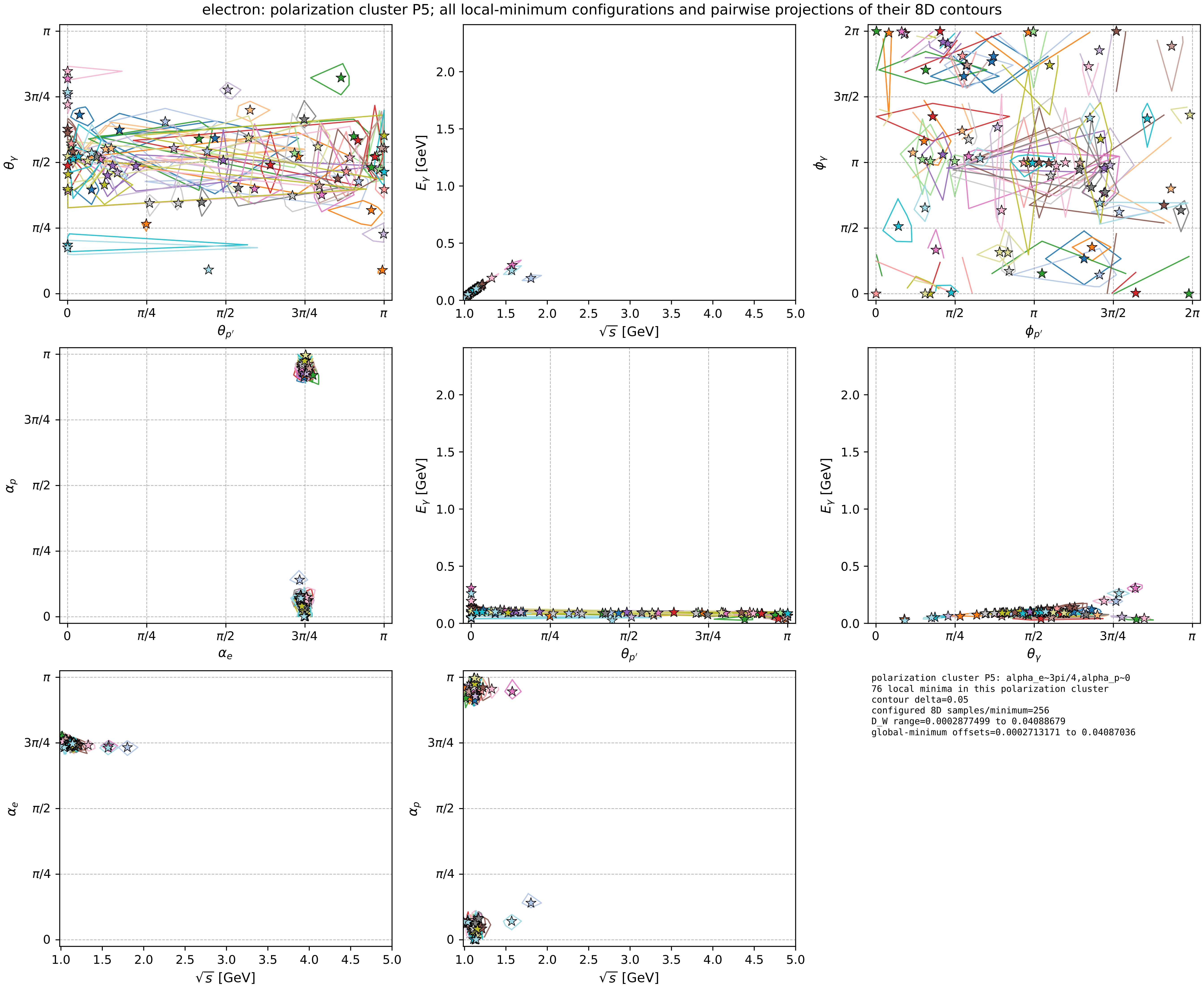
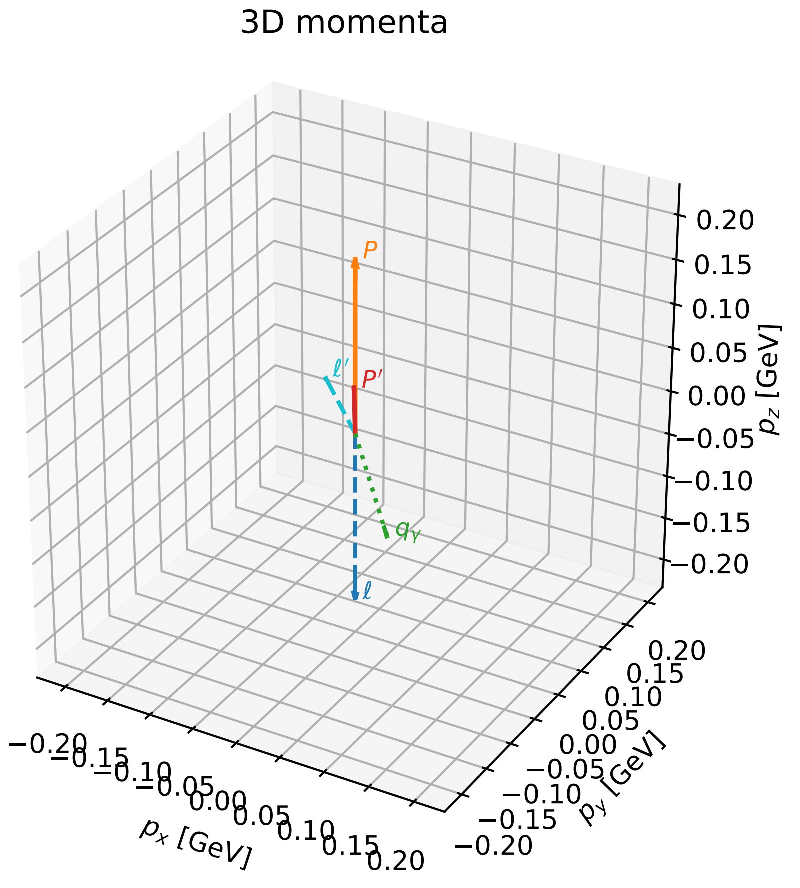


electron: polarization cluster P5; all local-minimum configurations and pairwise projections of their 8D contours
polarization cluster P5: alpha_e~3pi/4,alpha_p~0
76 local minima in this polarization cluster
contour delta=0.05
configured 8D samples/minimum=256
D_W range=0.0002877499 to 0.04088679
global-minimum offsets=0.0002713171 to 0.04087036

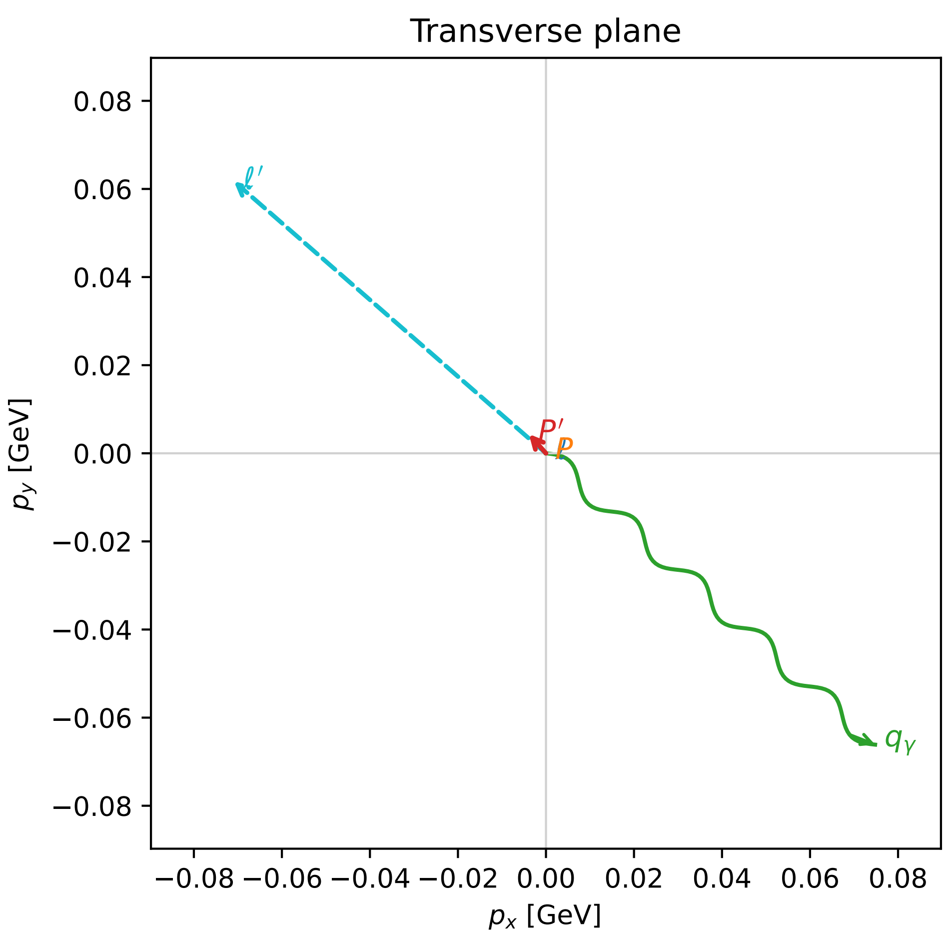
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

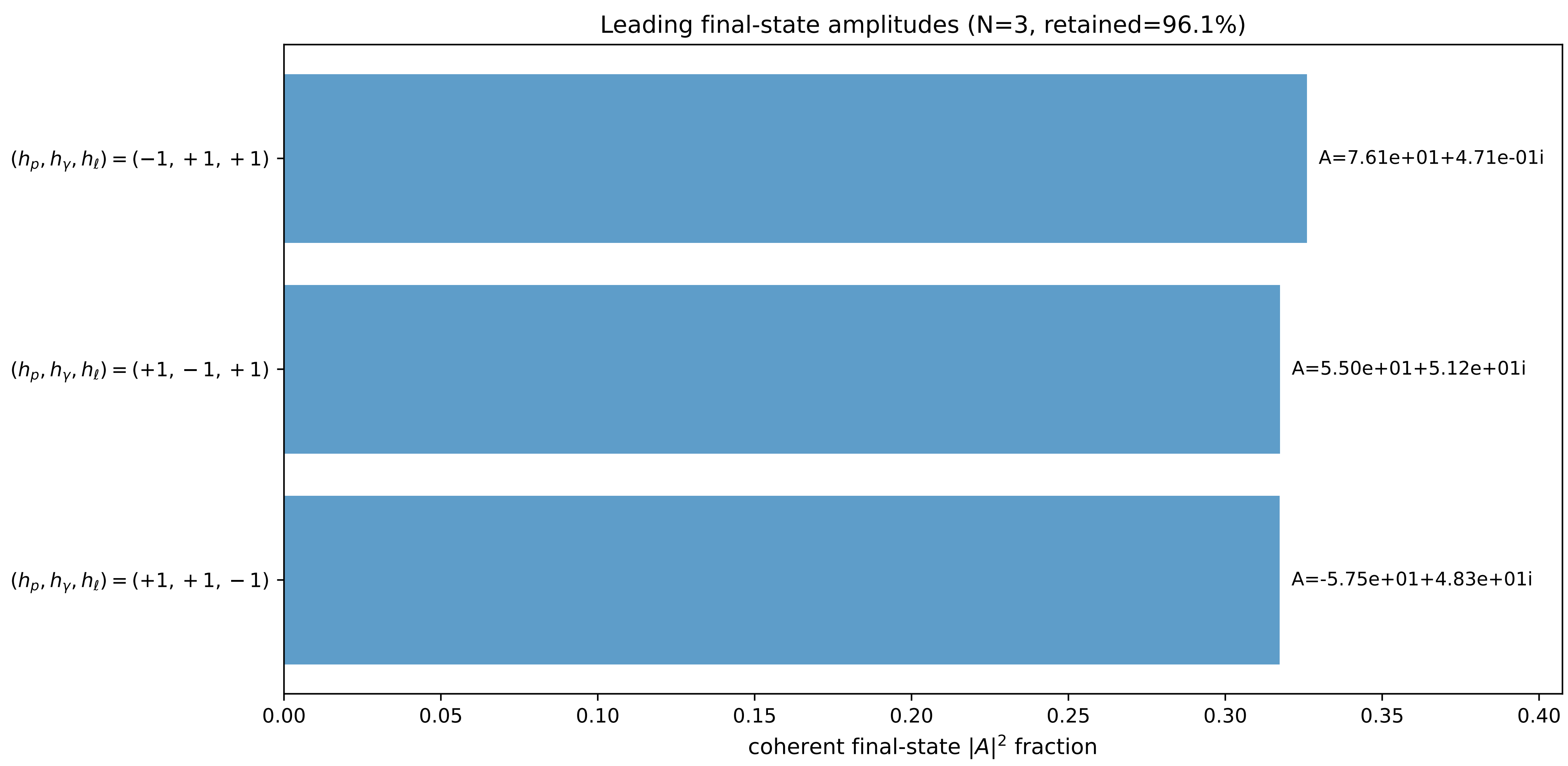



```
dw_mixing_angles_polarization_05_local_minimum_113  D_W=0.00028775
region: polarization_cluster_05_local_minimum_113
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3605141 rad
incoming lepton: -0.710155|+> +0.704046|->
proton mixing angle: alpha_p=3.009653 rad
incoming proton: -0.991309|+> +0.131557|->
```

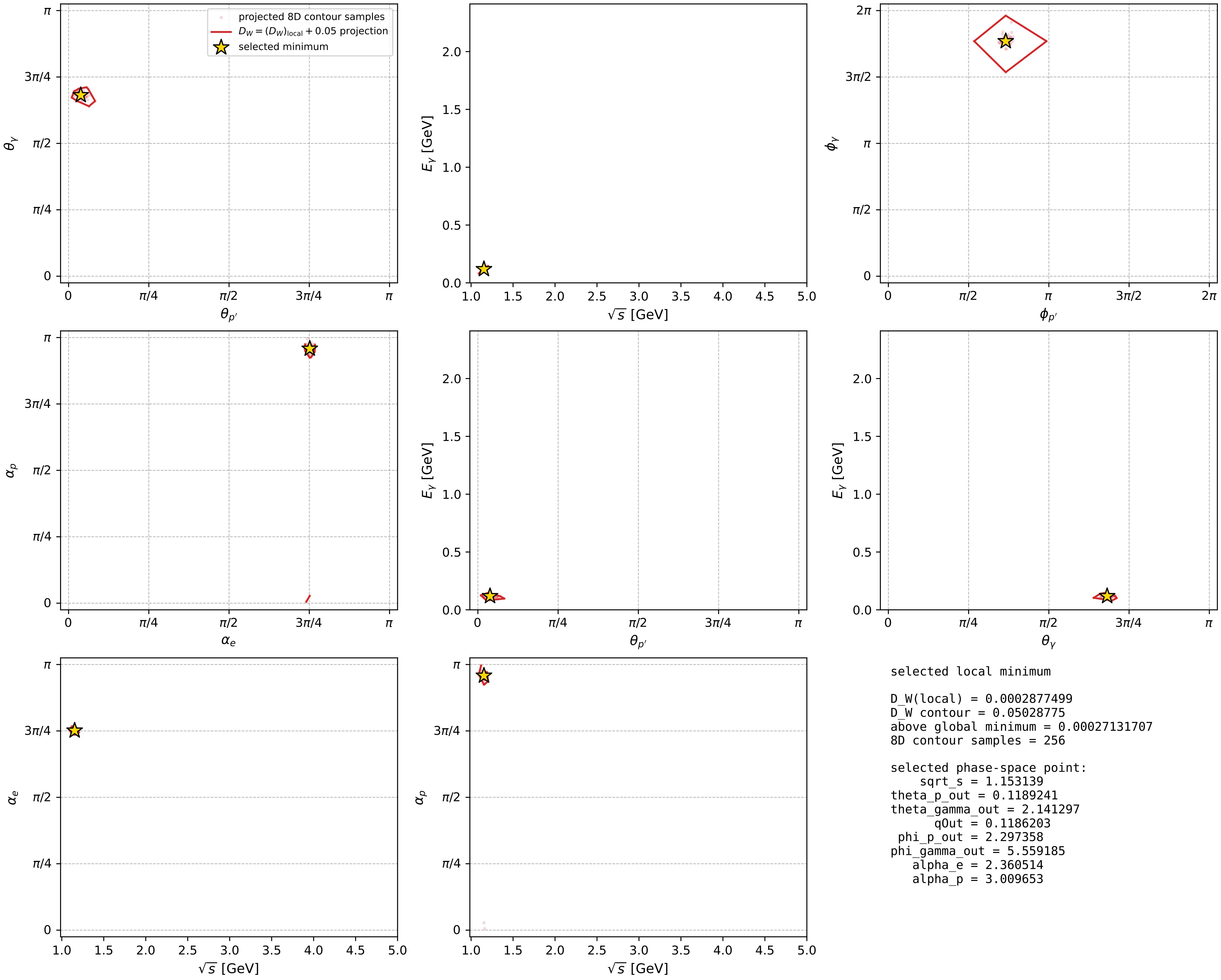
s=1.32973, $\sqrt{s}$=1.15314, $|\vec{P}|$=0.195069, $|\vec{P}'|$=0.0493626
$\theta_{p'}$=0.118924, θ_γ=2.1413, $\phi_{p'}$=2.29736, ϕ_γ=5.55919
E_γ=0.11862, $\theta_{\ell'}$=1.4121, $\phi_{\ell'}$=2.42506
Q^2=0.0430194, x_B=0.157407, t=-0.0210141, W^2=1.11012, y=0.607488

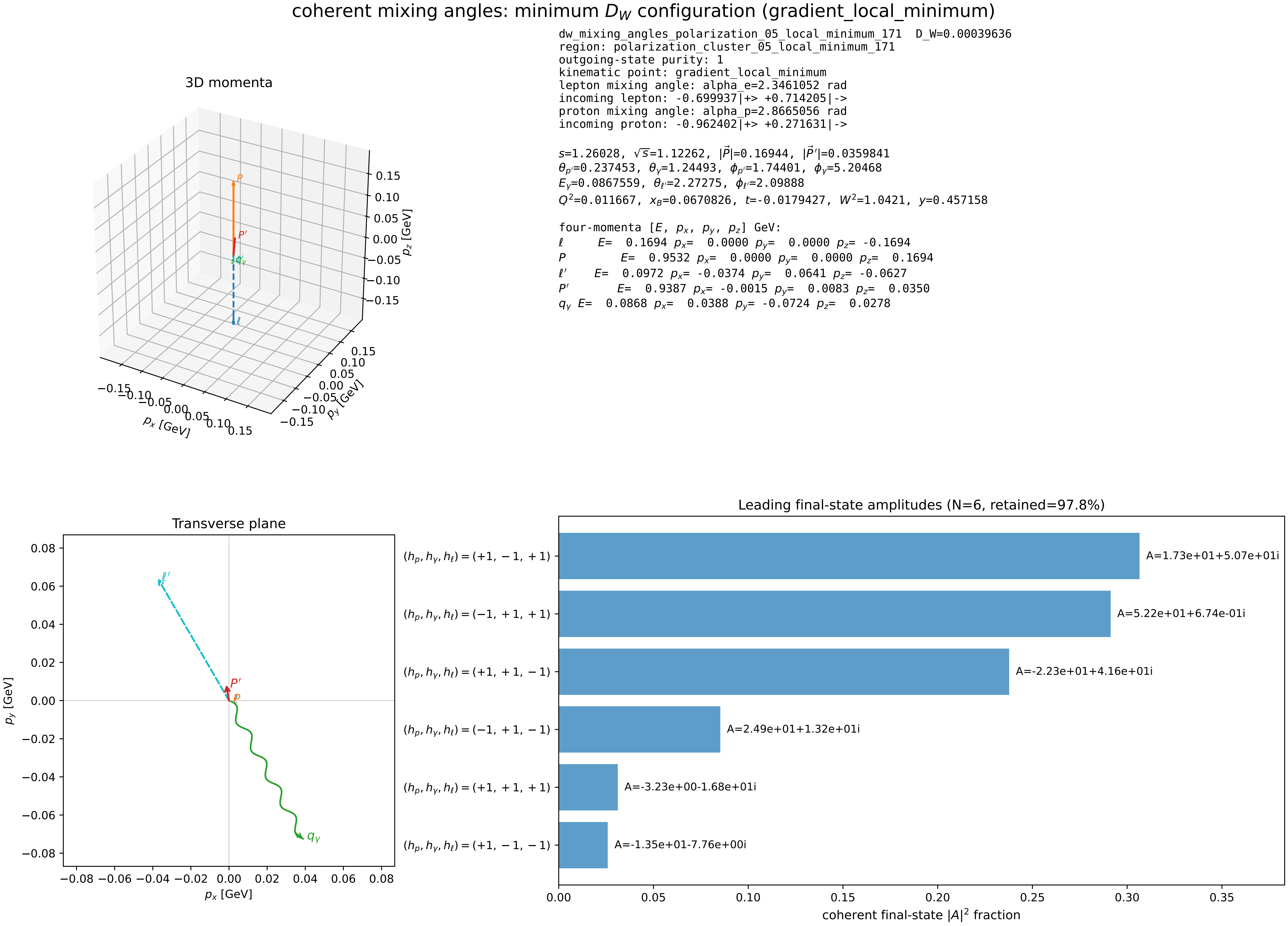
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1951 p_x= 0.0000 p_y= 0.0000 p_z= -0.1951
P E= 0.9581 p_x= 0.0000 p_y= 0.0000 p_z= 0.1951
ℓ' E= 0.0952 p_x= -0.0709 p_y= 0.0618 p_z= 0.0150
P' E= 0.9393 p_x= -0.0039 p_y= 0.0044 p_z= 0.0490
q_γ E= 0.1186 p_x= 0.0748 p_y= -0.0661 p_z= -0.0641



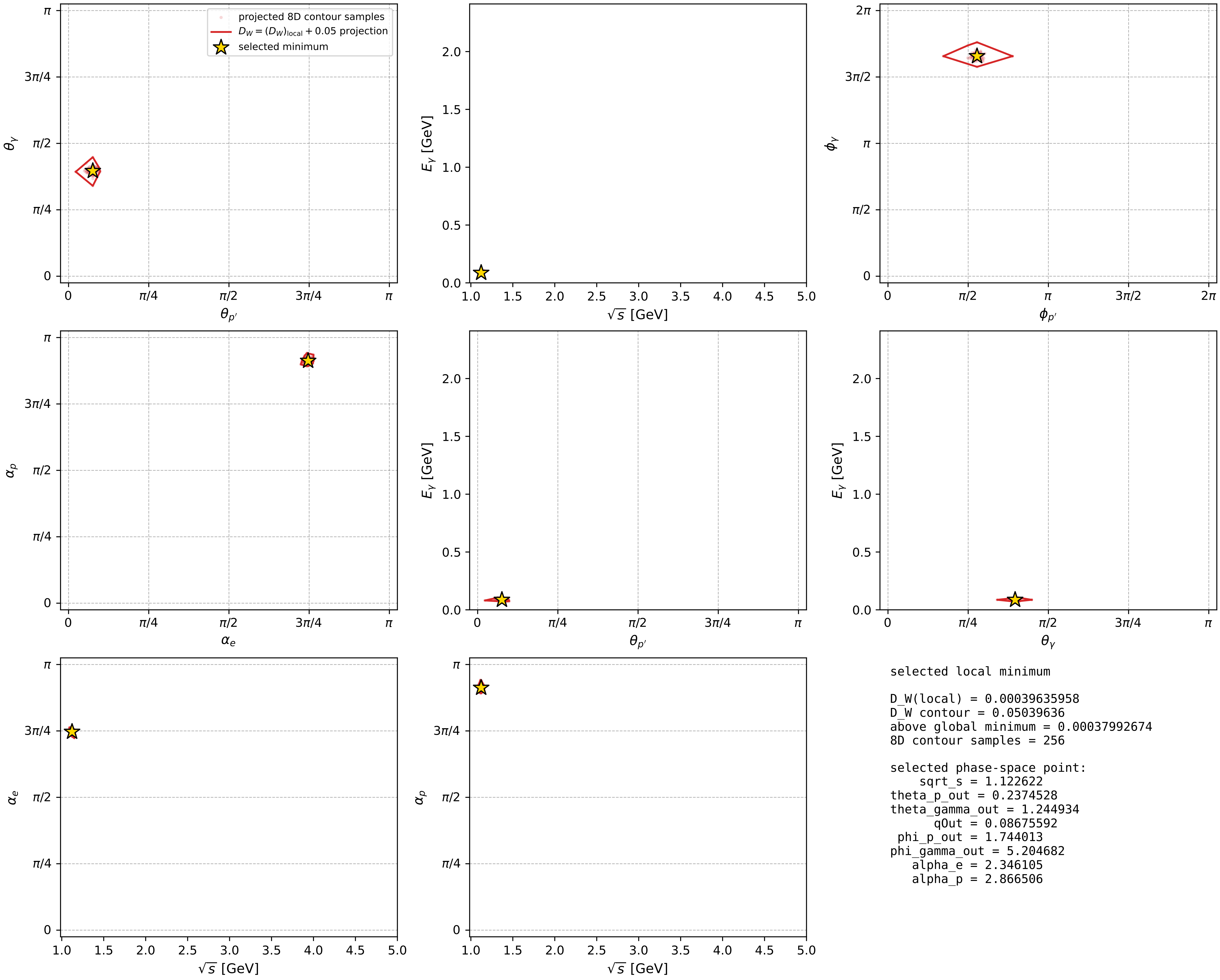


electron: polarization cluster P5, local minimum 113; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



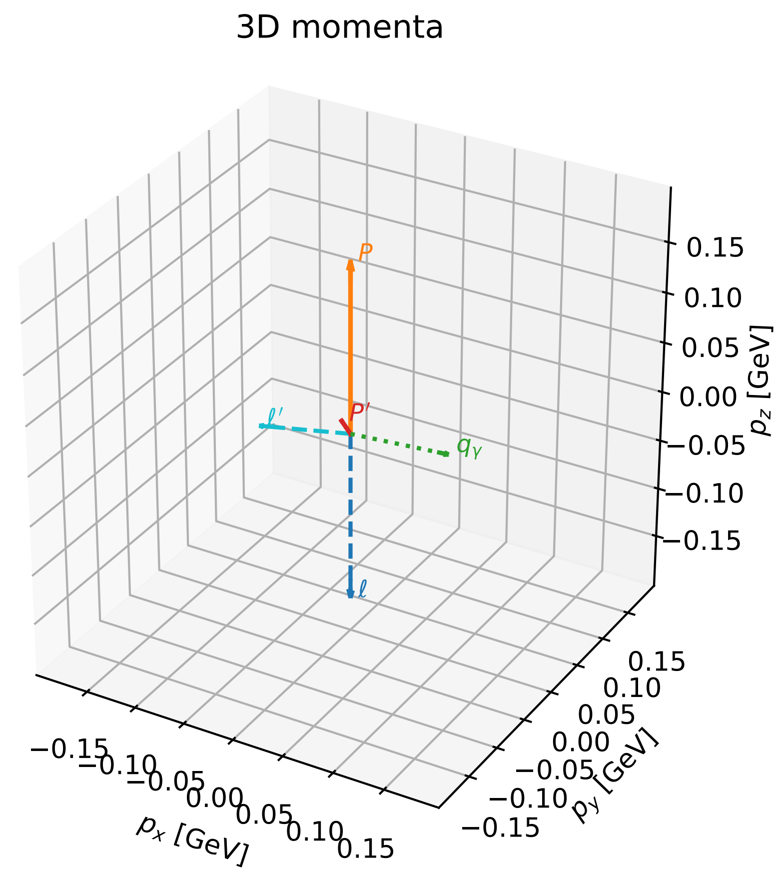

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_171 D_W=0.00039636
region: polarization_cluster_05_local_minimum_171
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3461052 rad
incoming lepton: -0.699937|+> +0.714205|->
proton mixing angle: alpha_p=2.8665056 rad
incoming proton: -0.962402|+> +0.271631|->
s=1.26028, $\sqrt{s}$=1.12262, $|\vec{P}|$=0.16944, $|\vec{P}'|$=0.0359841
$\theta_{p'}$=0.237453, θ_γ=1.24493, $\phi_{p'}$=1.74401, ϕ_γ=5.20468
E_γ=0.0867559, $\theta_{\ell'}$=2.27275, $\phi_{\ell'}$=2.09888
Q^2=0.011667, x_B=0.0670826, t=-0.0179427, W^2=1.0421, y=0.457158
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1694 p_x= 0.0000 p_y= 0.0000 p_z= -0.1694
P E= 0.9532 p_x= 0.0000 p_y= 0.0000 p_z= 0.1694
ℓ' E= 0.0972 p_x= -0.0374 p_y= 0.0641 p_z= -0.0627
P' E= 0.9387 p_x= -0.0015 p_y= 0.0083 p_z= 0.0350
q_γ E= 0.0868 p_x= 0.0388 p_y= -0.0724 p_z= 0.0278
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
ℓ'
P'
P
q_γ
ℓ
Leading final-state amplitudes (N=6, retained=97.8%)
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=1.73e+01+5.07e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=5.22e+01+6.74e-01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-2.23e+01+4.16e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=2.49e+01+1.32e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-3.23e+00-1.68e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-1.35e+01-7.76e+00i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P5, local minimum 171; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour




```
selected local minimum

D_W(local) = 0.00039635958
D_W contour = 0.05039636
above global minimum = 0.00037992674
8D contour samples = 256

selected phase-space point:
    sqrt_s = 1.122622
theta_p_out = 0.2374528
theta_gamma_out = 1.244934
      qOut = 0.08675592
 phi_p_out = 1.744013
phi_gamma_out = 5.204682
   alpha_e = 2.346105
   alpha_p = 2.866506
```

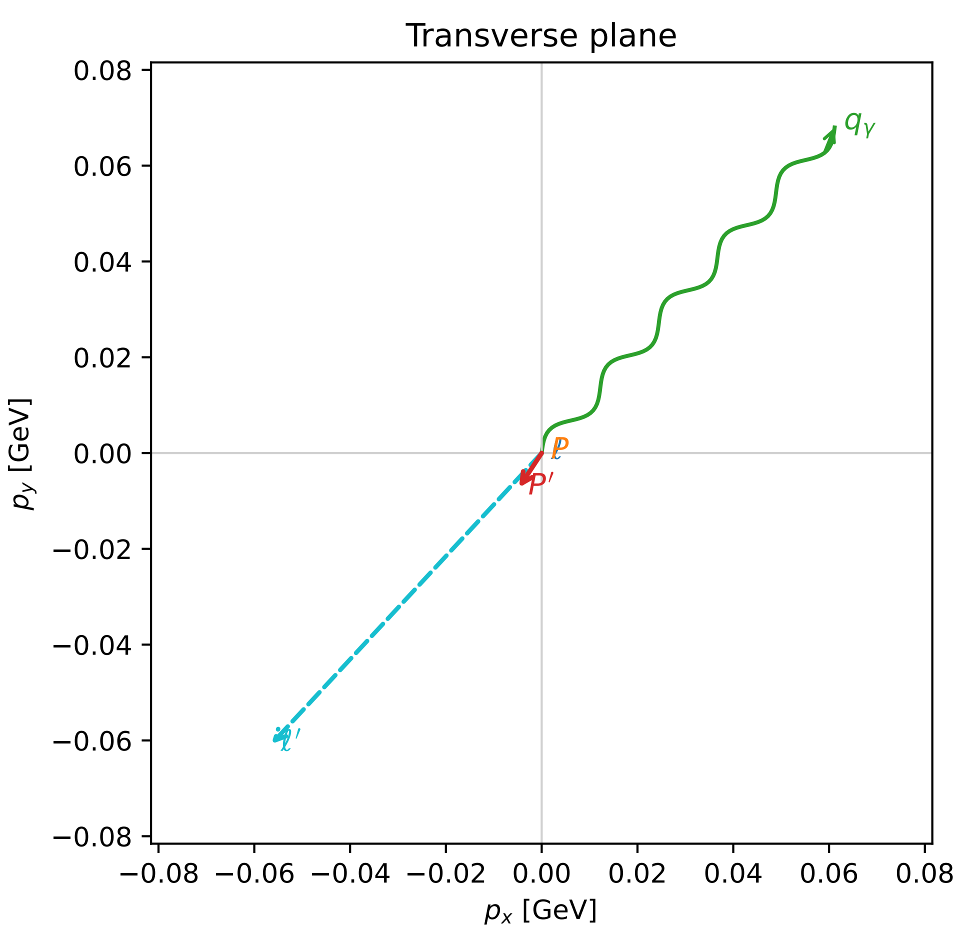
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

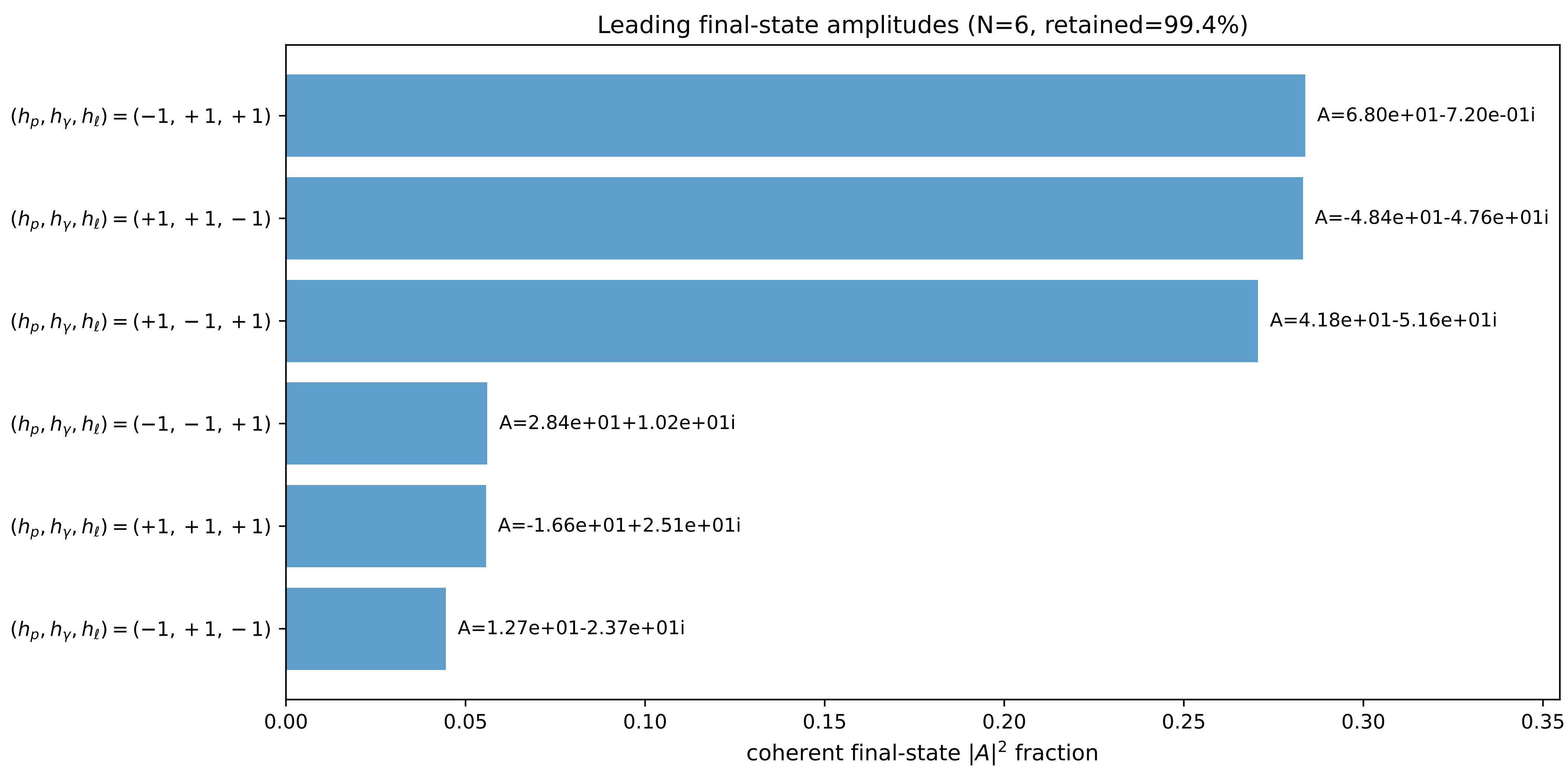


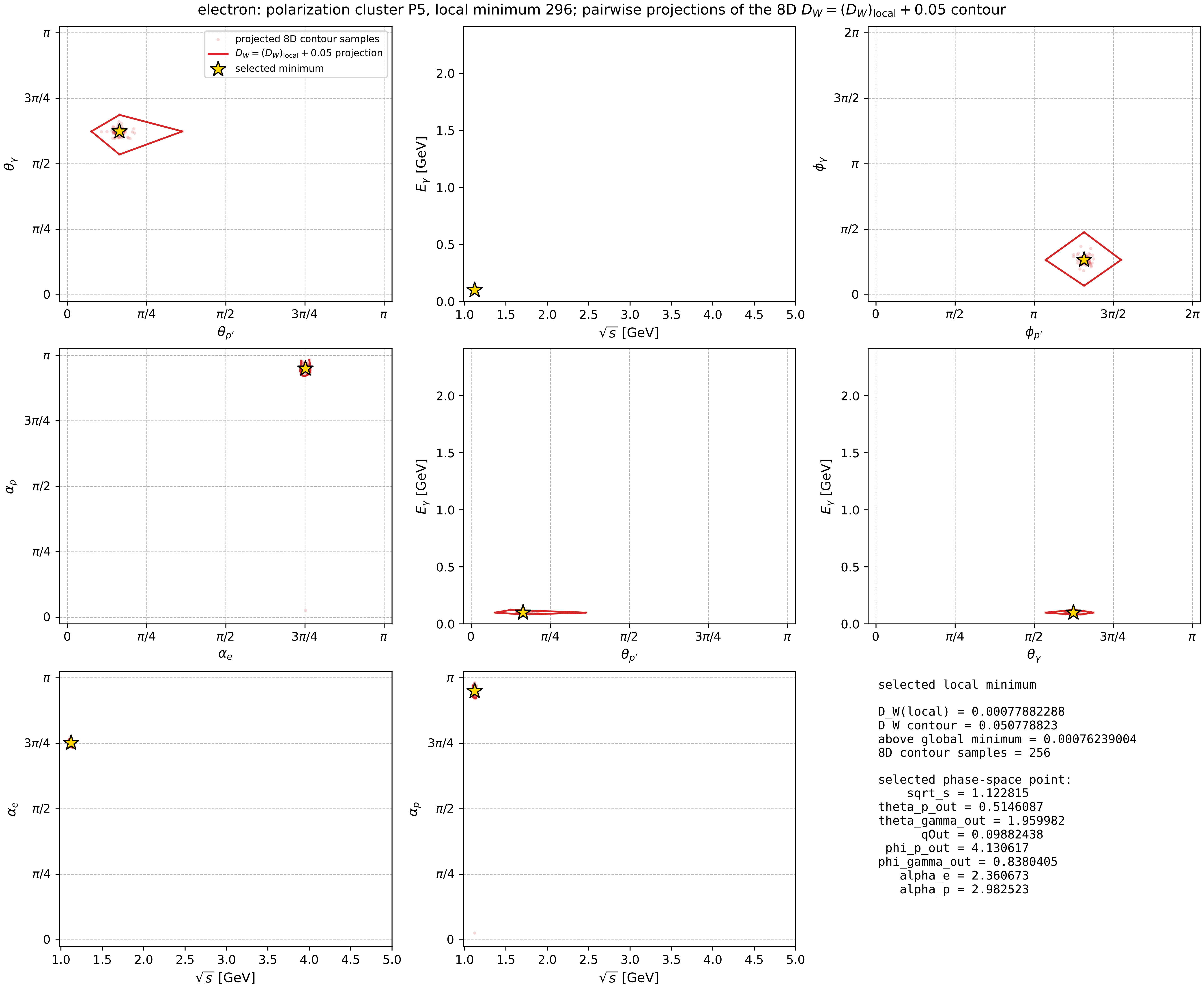

```
dw_mixing_angles_polarization_05_local_minimum_296  D_W=0.000778823
region: polarization_cluster_05_local_minimum_296
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3606731 rad
incoming lepton: -0.710267|+> +0.703933|->
proton mixing angle: alpha_p=2.9825229 rad
incoming proton: -0.987375|+> +0.1584|->
```

s=1.26071, $\sqrt{s}$=1.12281, $|\vec{P}|$=0.169604, $|\vec{P}'|$=0.0175589
$\theta_{p'}$=0.514609, θ_γ=1.95998, $\phi_{p'}$=4.13062, ϕ_γ=0.838041
E_γ=0.0988244, $\theta_{\ell'}$=1.309, $\phi_{\ell'}$=3.96395
Q^2=0.0366472, x_B=0.163034, t=-0.0236627, W^2=1.06798, y=0.590185

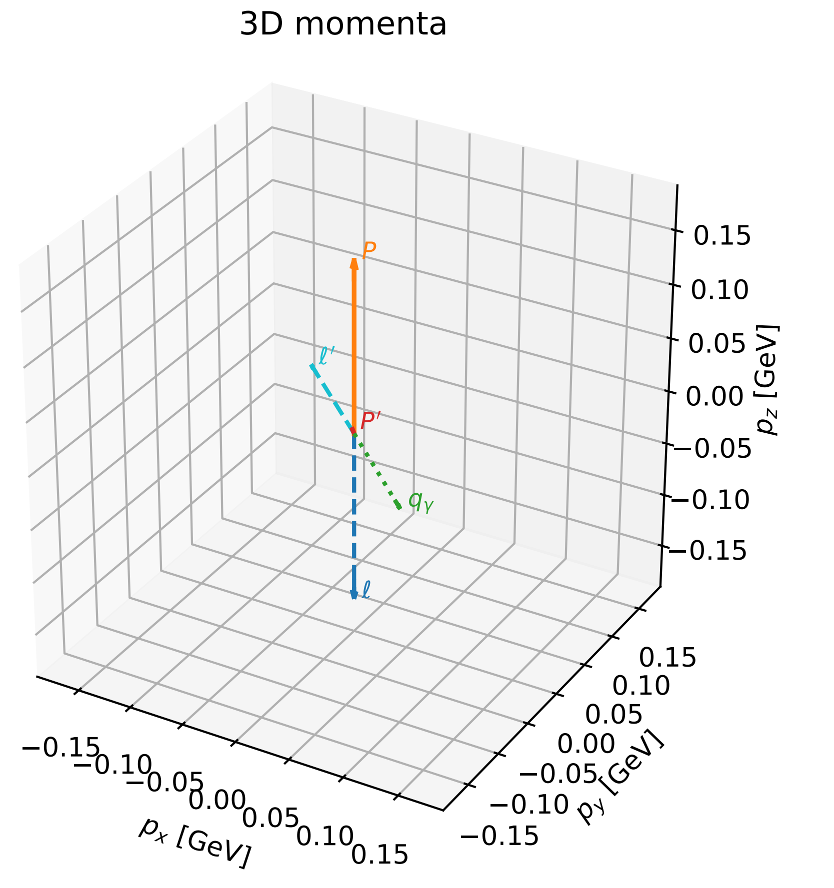
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1696 p_x= 0.0000 p_y= 0.0000 p_z= -0.1696
P E= 0.9532 p_x= 0.0000 p_y= 0.0000 p_z= 0.1696
ℓ' E= 0.0858 p_x= -0.0564 p_y= -0.0607 p_z= 0.0222
P' E= 0.9382 p_x= -0.0047 p_y= -0.0072 p_z= 0.0153
q_γ E= 0.0988 p_x= 0.0612 p_y= 0.0680 p_z= -0.0375






electron: polarization cluster P5, local minimum 296; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.00077882288
D_W contour = 0.050778823
above global minimum = 0.00076239004
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.122815
theta_p_out = 0.5146087
theta_gamma_out = 1.959982
qOut = 0.09882438
phi_p_out = 4.130617
phi_gamma_out = 0.8380405
alpha_e = 2.360673
alpha_p = 2.982523

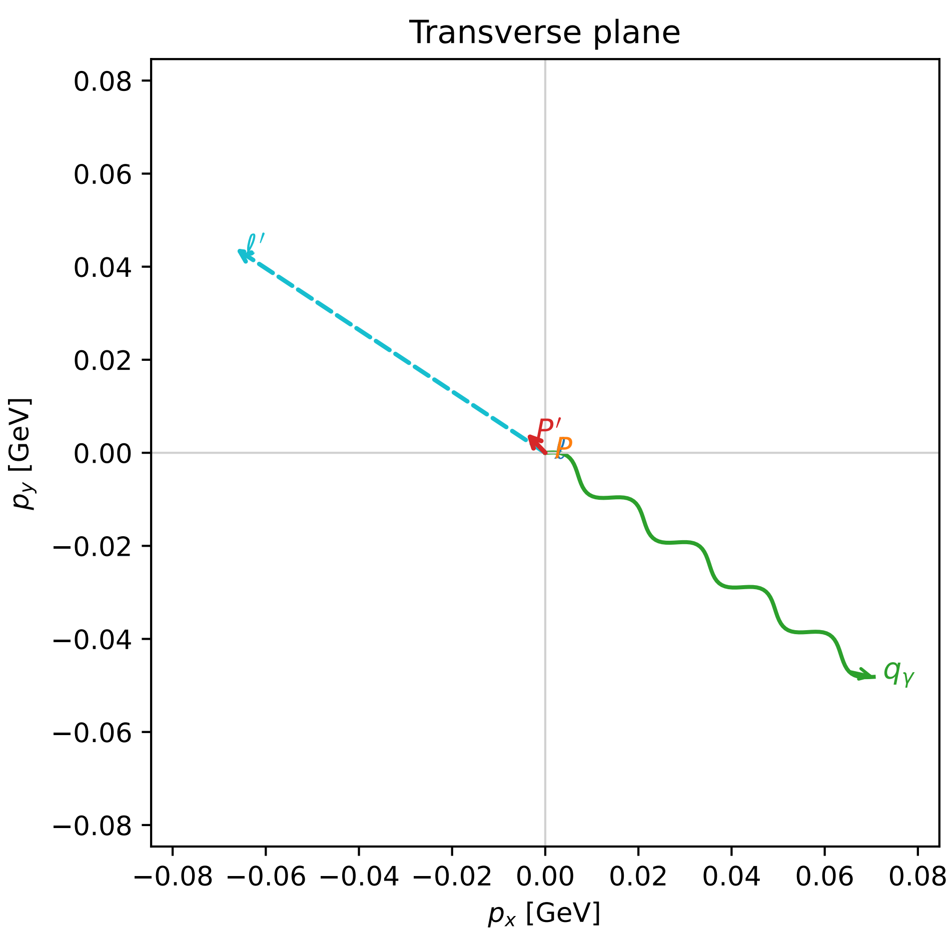
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

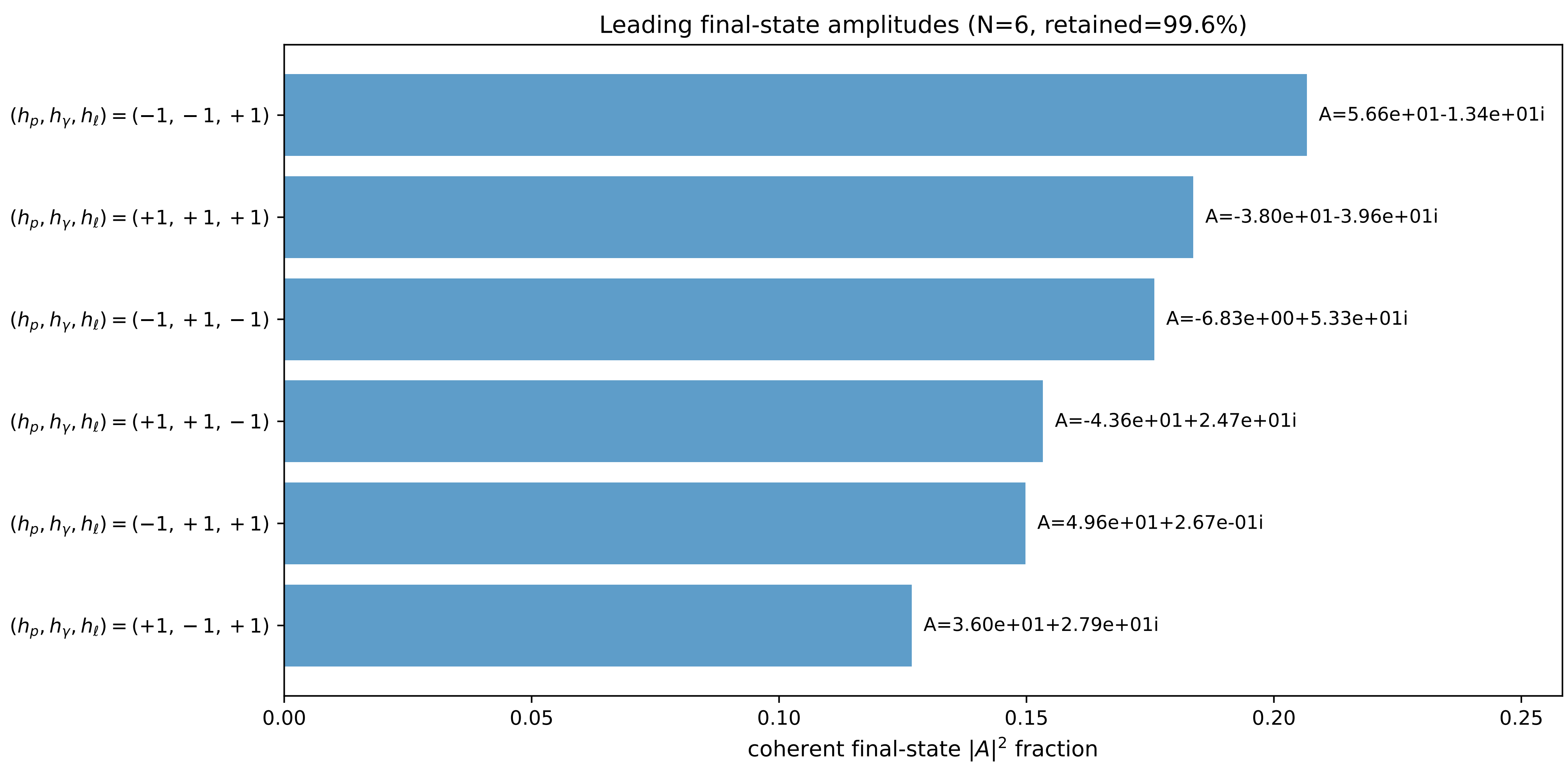


dw_mixing_angles_polarization_05_local_minimum_305 D_W=0.000802871
region: polarization_cluster_05_local_minimum_305
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3658533 rad
incoming lepton: -0.713904|+> +0.700244|->
proton mixing angle: alpha_p=3.0005656 rad
incoming proton: -0.990072|+> +0.14056|->

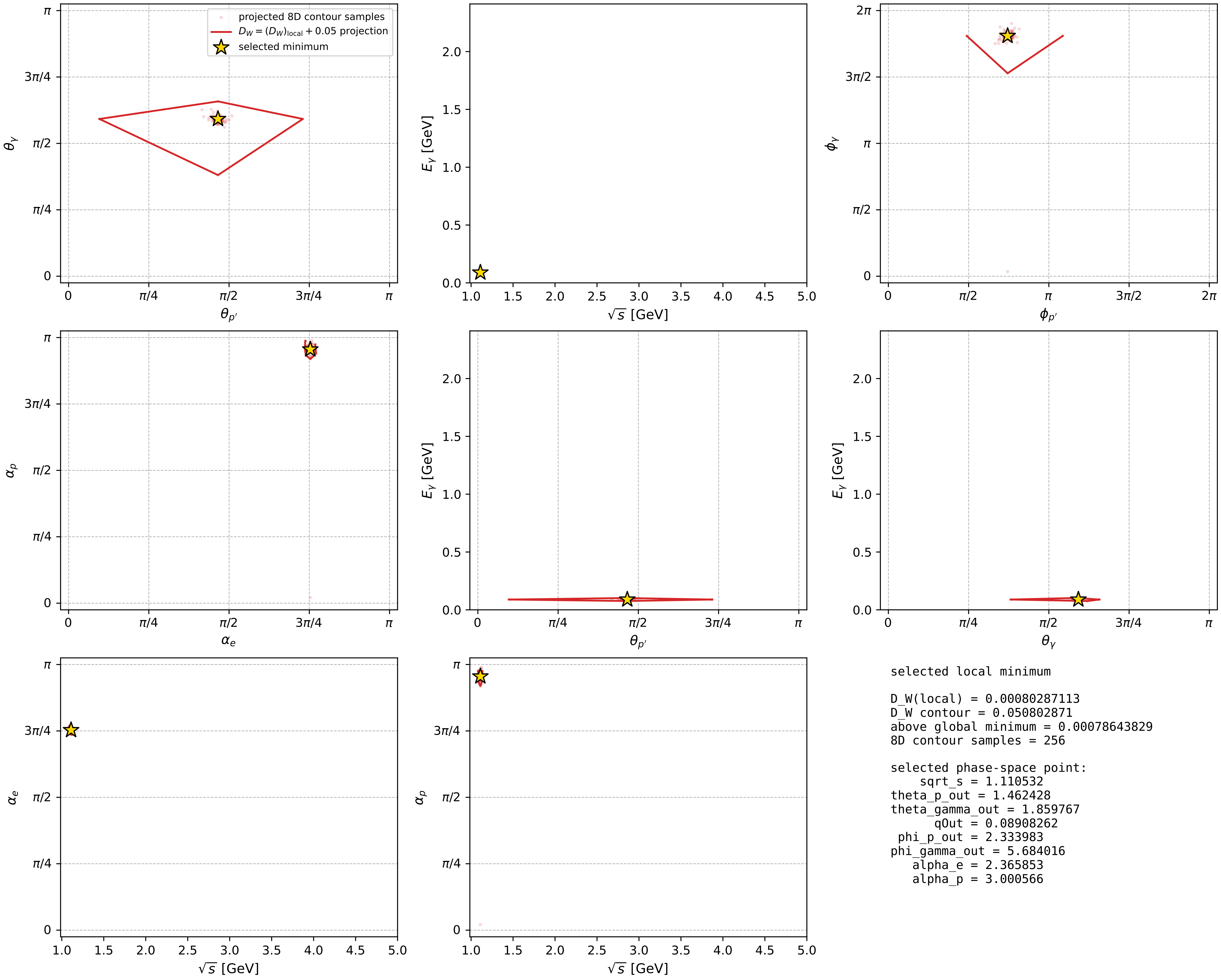
s=1.23328, $\sqrt{s}$=1.11053, $|\vec{P}|$=0.159129, $|\vec{P}'|$=0.00588583
$\theta_{p'}$=1.46243, θ_γ=1.85977, $\phi_{p'}$=2.33398, ϕ_γ=5.68402
E_γ=0.0890826, $\theta_{\ell'}$=1.26962, $\phi_{\ell'}$=2.55762
Q^2=0.0344285, x_B=0.169966, t=-0.0249749, W^2=1.04798, y=0.573118

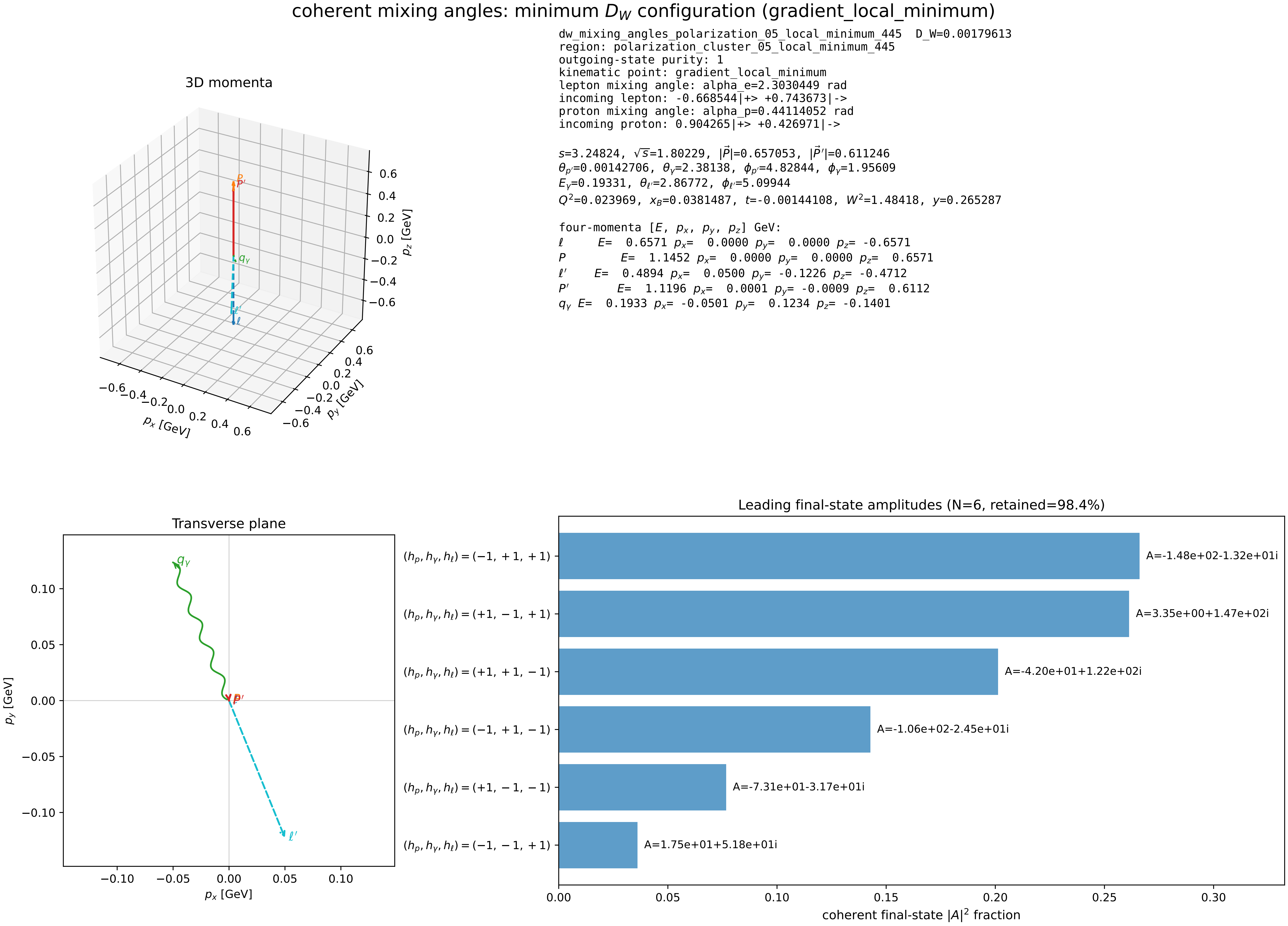
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1591 p_x= 0.0000 p_y= 0.0000 p_z= -0.1591
P E= 0.9514 p_x= 0.0000 p_y= 0.0000 p_z= 0.1591
ℓ' E= 0.0834 p_x= -0.0665 p_y= 0.0439 p_z= 0.0247
P' E= 0.9380 p_x= -0.0040 p_y= 0.0042 p_z= 0.0006
q_γ E= 0.0891 p_x= 0.0705 p_y= -0.0482 p_z= -0.0254



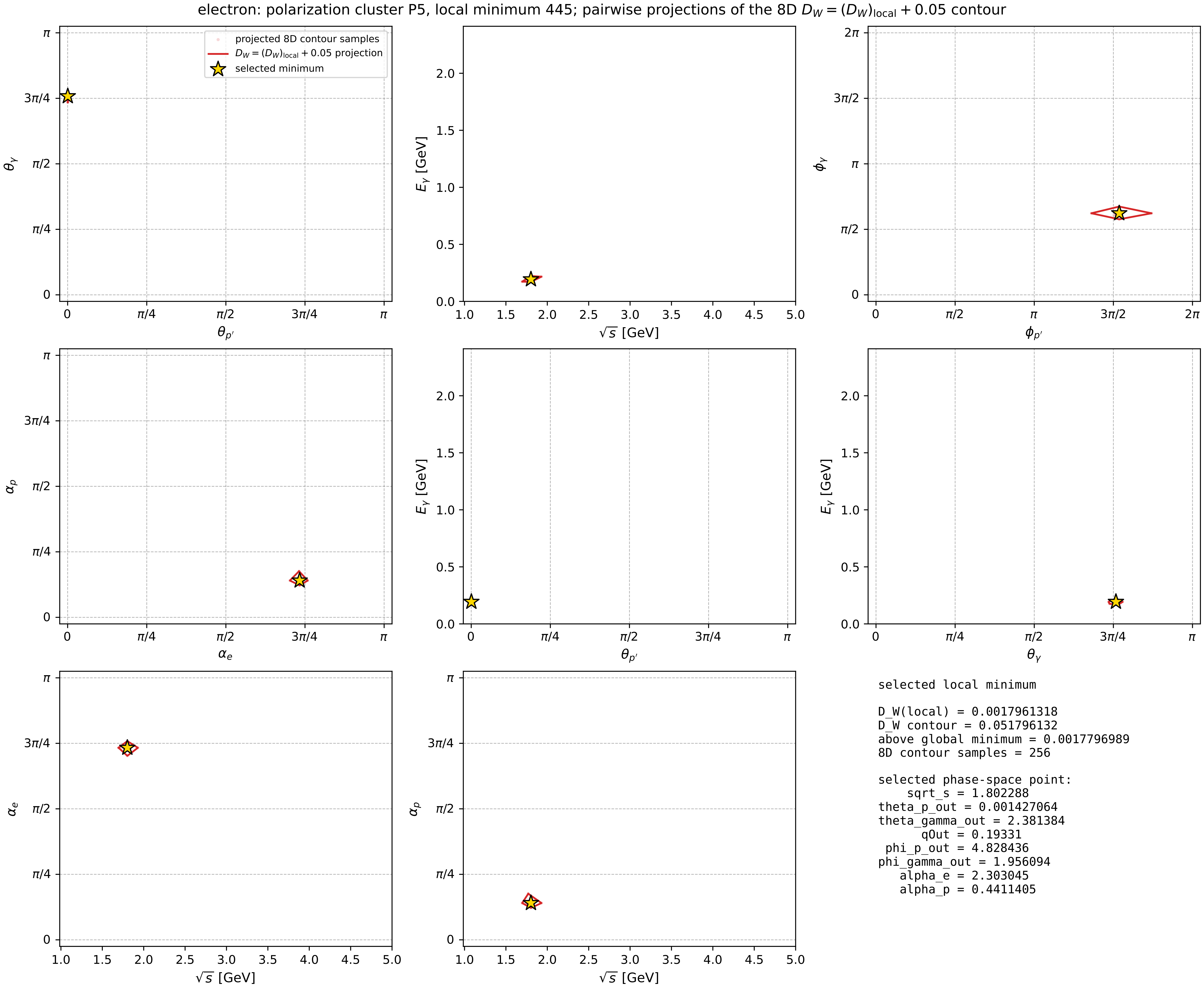


electron: polarization cluster P5, local minimum 305; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



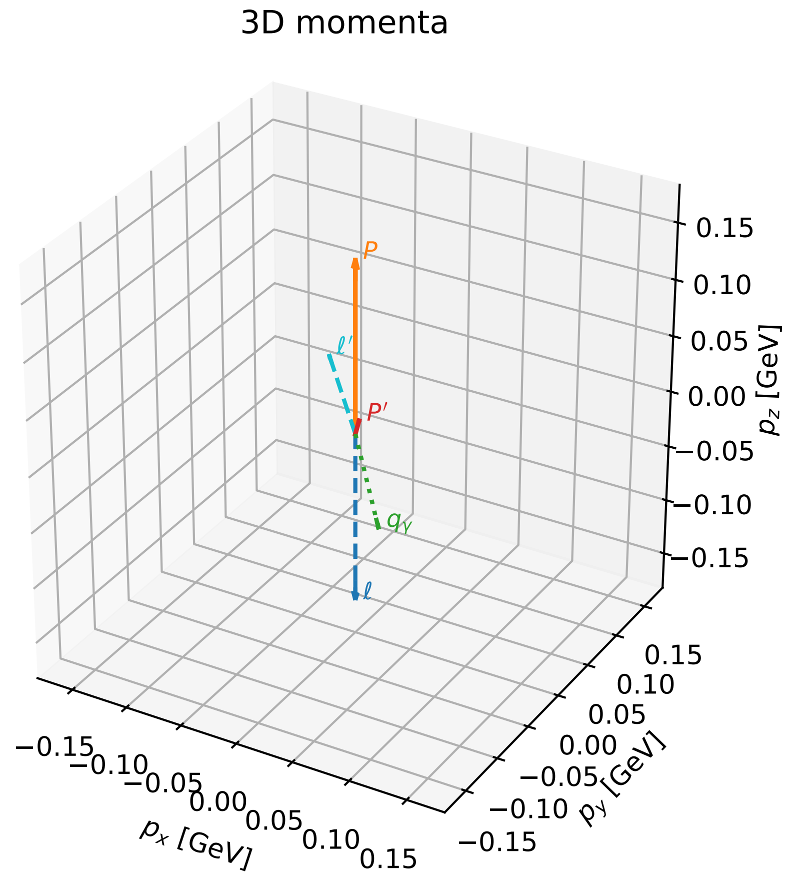


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_445 D_W=0.00179613
region: polarization_cluster_05_local_minimum_445
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3030449 rad
incoming lepton: -0.668544|+> +0.743673|->
proton mixing angle: alpha_p=0.44114052 rad
incoming proton: 0.904265|+> +0.426971|->
s=3.24824, $\sqrt{s}$=1.80229, $|\vec{P}|$=0.657053, $|\vec{P}'|$=0.611246
$\theta_{p'}$=0.00142706, θ_γ=2.38138, $\phi_{p'}$=4.82844, ϕ_γ=1.95609
E_γ=0.19331, $\theta_{\ell'}$=2.86772, $\phi_{\ell'}$=5.09944
Q^2=0.023969, x_B=0.0381487, t=-0.00144108, W^2=1.48418, y=0.265287
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.6571 p_x= 0.0000 p_y= 0.0000 p_z= -0.6571
P E= 1.1452 p_x= 0.0000 p_y= 0.0000 p_z= 0.6571
ℓ' E= 0.4894 p_x= 0.0500 p_y= -0.1226 p_z= -0.4712
P' E= 1.1196 p_x= 0.0001 p_y= -0.0009 p_z= 0.6112
q_γ E= 0.1933 p_x= -0.0501 p_y= 0.1234 p_z= -0.1401
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P'
q_γ
ℓ'
ℓ
Transverse plane
p_x [GeV]
p_y [GeV]
q_γ
P'
ℓ'
Leading final-state amplitudes (N=6, retained=98.4%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-1.48e+02-1.32e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=3.35e+00+1.47e+02i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-4.20e+01+1.22e+02i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-1.06e+02-2.45e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, -1)$ A=-7.31e+01-3.17e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=1.75e+01+5.18e+01i
coherent final-state $|A|^2$ fraction



electron: polarization cluster P5, local minimum 445; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0017961318
D_W contour = 0.051796132
above global minimum = 0.0017796989
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.802288
theta_p_out = 0.001427064
theta_gamma_out = 2.381384
qOut = 0.19331
phi_p_out = 4.828436
phi_gamma_out = 1.956094
alpha_e = 2.303045
alpha_p = 0.4411405

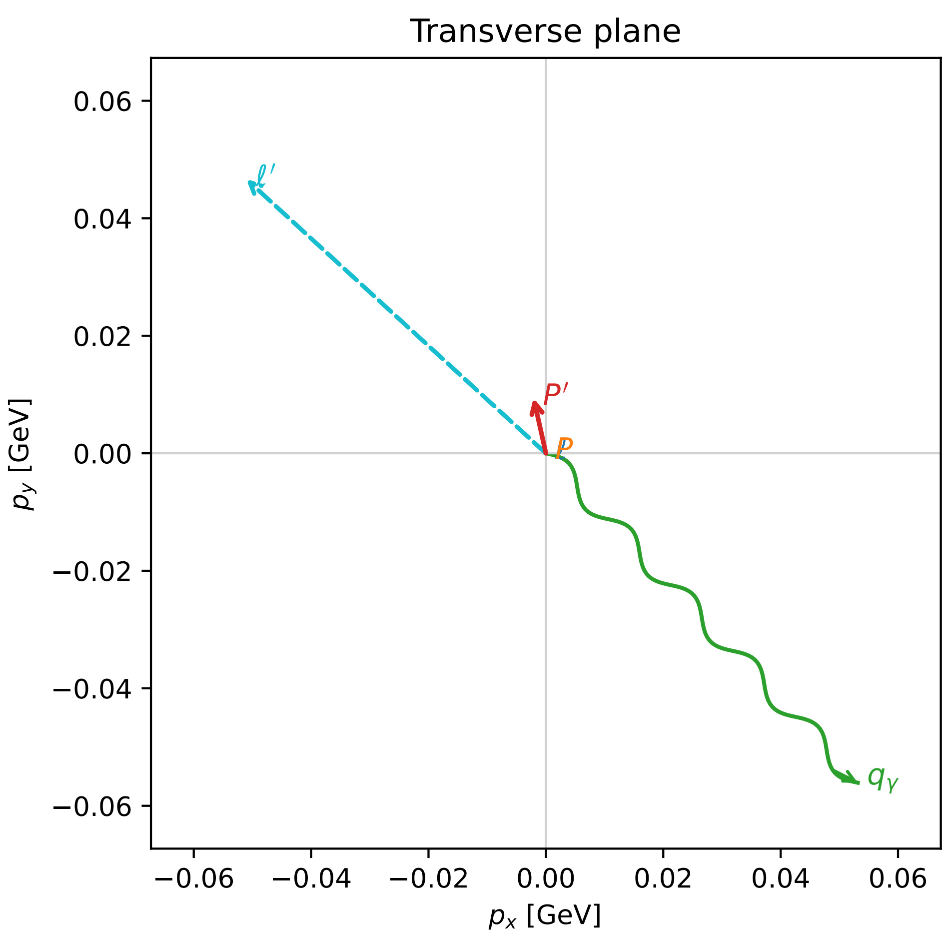
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

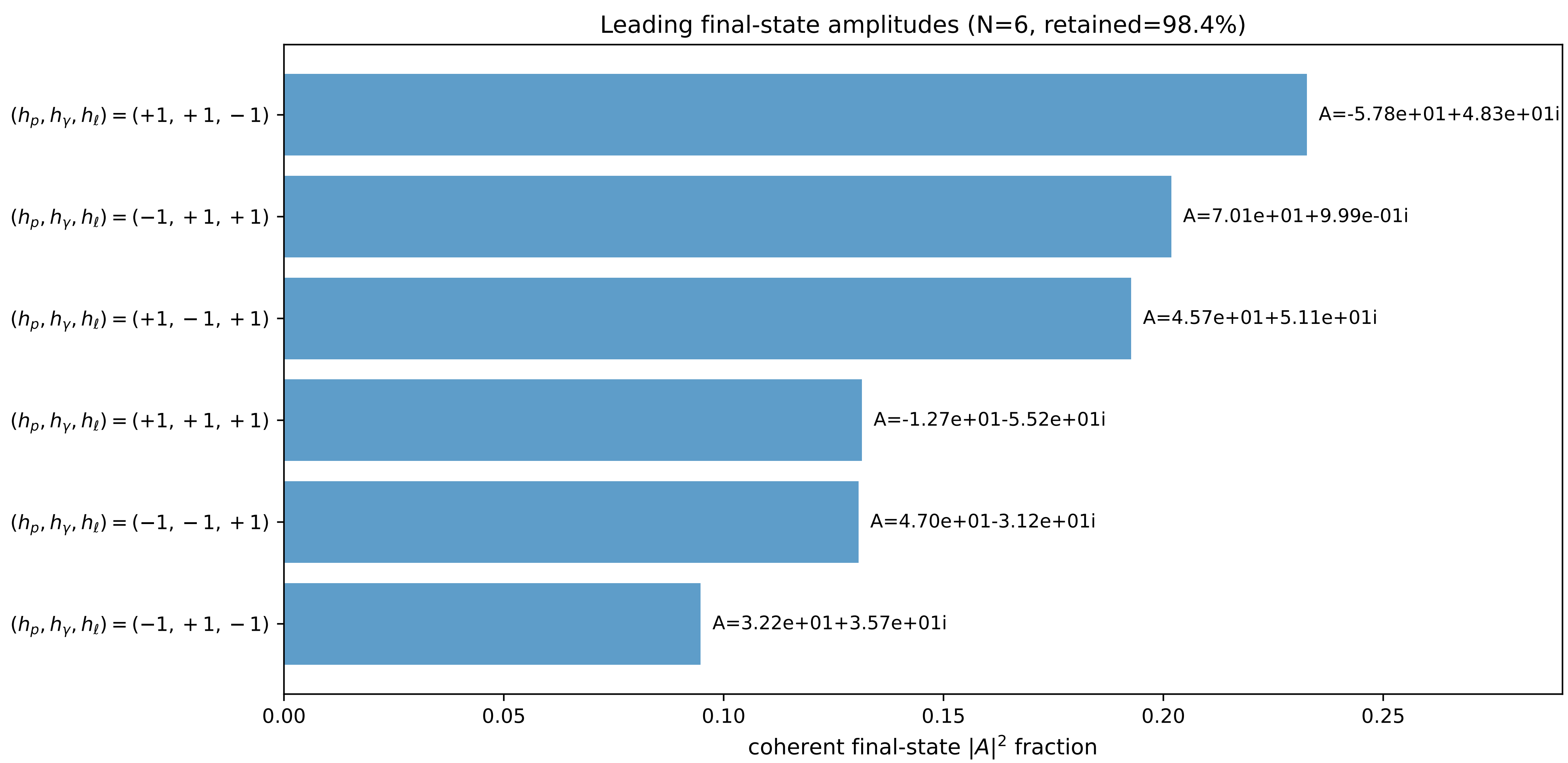


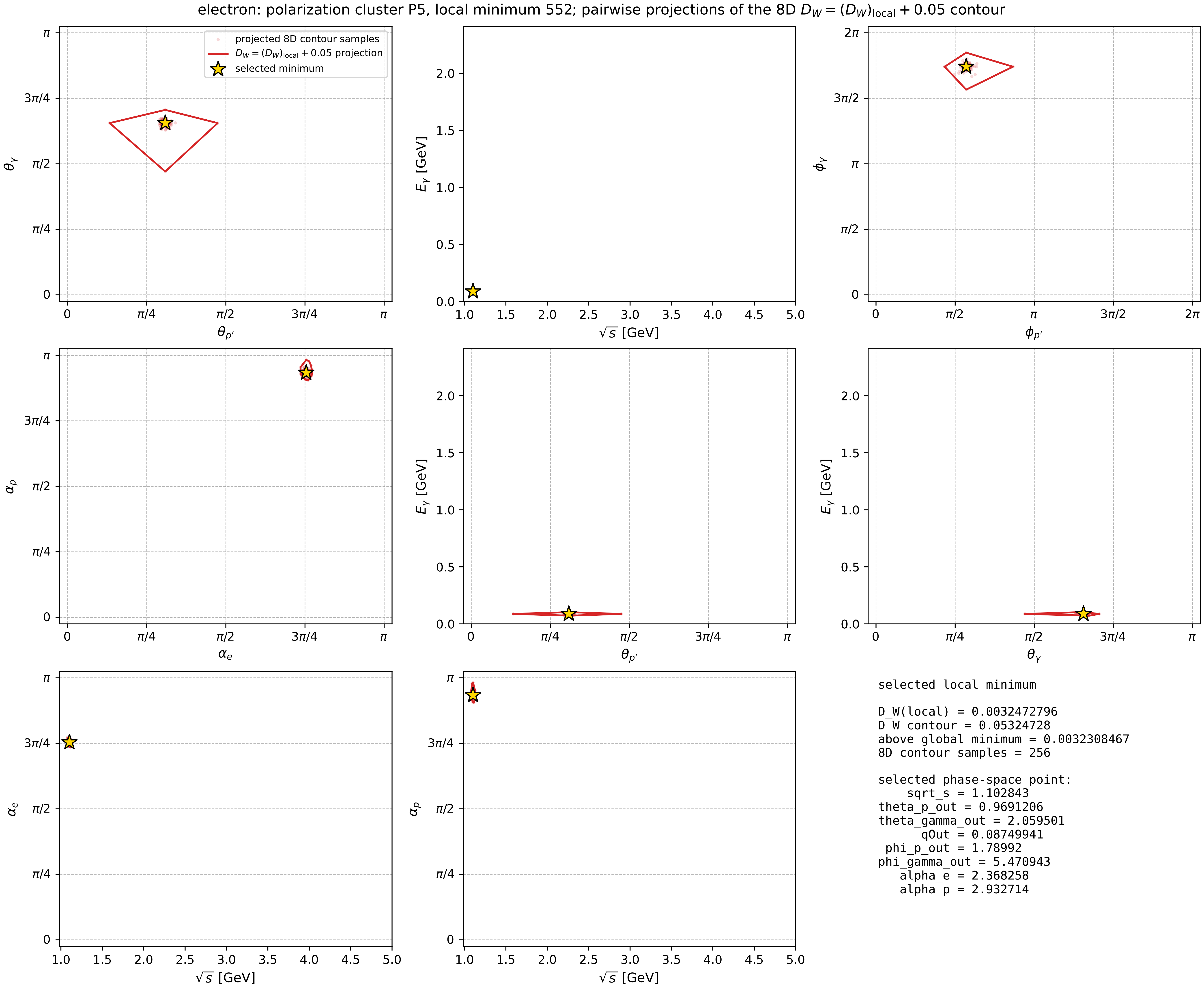
dw_mixing_angles_polarization_05_local_minimum_552 D_W=0.00324728
region: polarization_cluster_05_local_minimum_552
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.368258 rad
incoming lepton: -0.715585|+> +0.698525|->
proton mixing angle: alpha_p=2.9327138 rad
incoming proton: -0.978264|+> +0.207363|->

s=1.21626, $\sqrt{s}$=1.10284, $|\vec{P}|$=0.152523, $|\vec{P}'|$=0.0116976
$\theta_{p'}$=0.969121, θ_γ=2.0595, $\phi_{p'}$=1.78992, ϕ_γ=5.47094
E_γ=0.0874994, $\theta_{\ell'}$=1.10856, $\phi_{\ell'}$=2.40103
Q^2=0.0340822, x_B=0.170354, t=-0.0212303, W^2=1.04583, y=0.594693

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1525 p_x= 0.0000 p_y= 0.0000 p_z= -0.1525
P E= 0.9503 p_x= 0.0000 p_y= 0.0000 p_z= 0.1525
ℓ' E= 0.0773 p_x= -0.0510 p_y= 0.0467 p_z= 0.0345
P' E= 0.9381 p_x= -0.0021 p_y= 0.0094 p_z= 0.0066
q_γ E= 0.0875 p_x= 0.0531 p_y= -0.0561 p_z= -0.0411

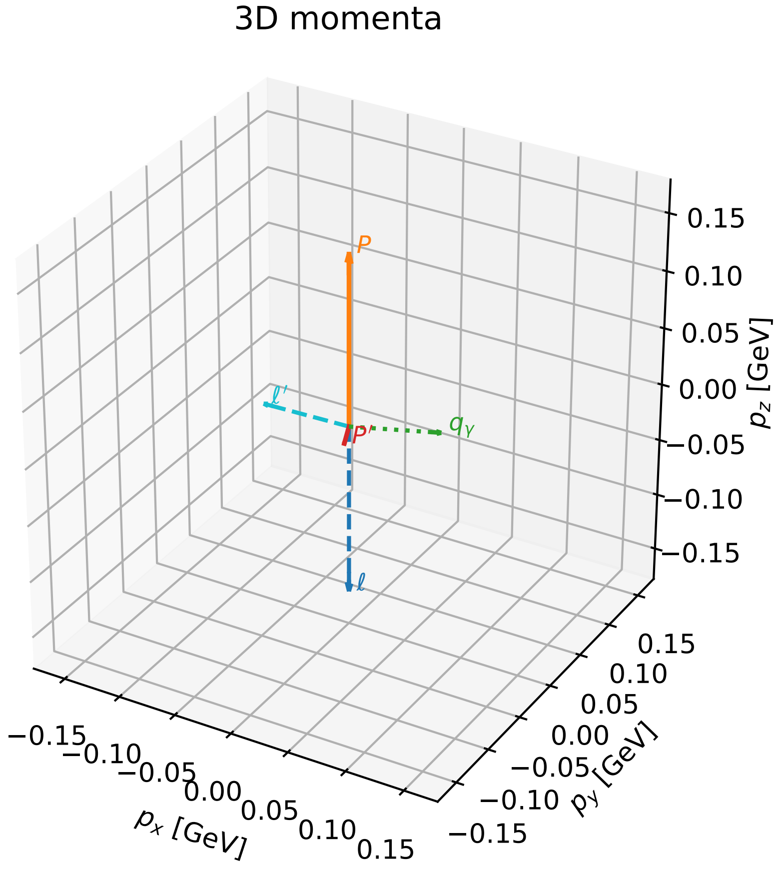






electron: polarization cluster P5, local minimum 552; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0032472796
D_W contour = 0.05324728
above global minimum = 0.0032308467
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.102843
theta_p_out = 0.9691206
theta_gamma_out = 2.059501
qOut = 0.08749941
phi_p_out = 1.78992
phi_gamma_out = 5.470943
alpha_e = 2.368258
alpha_p = 2.932714

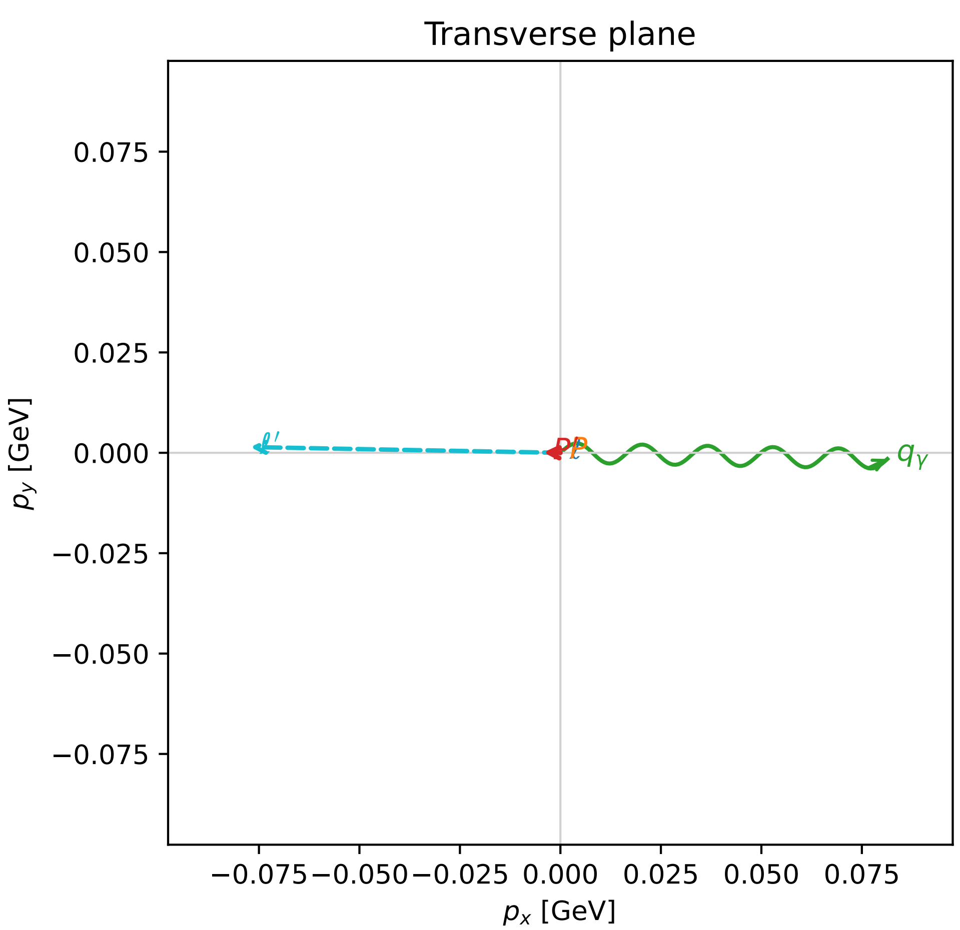
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

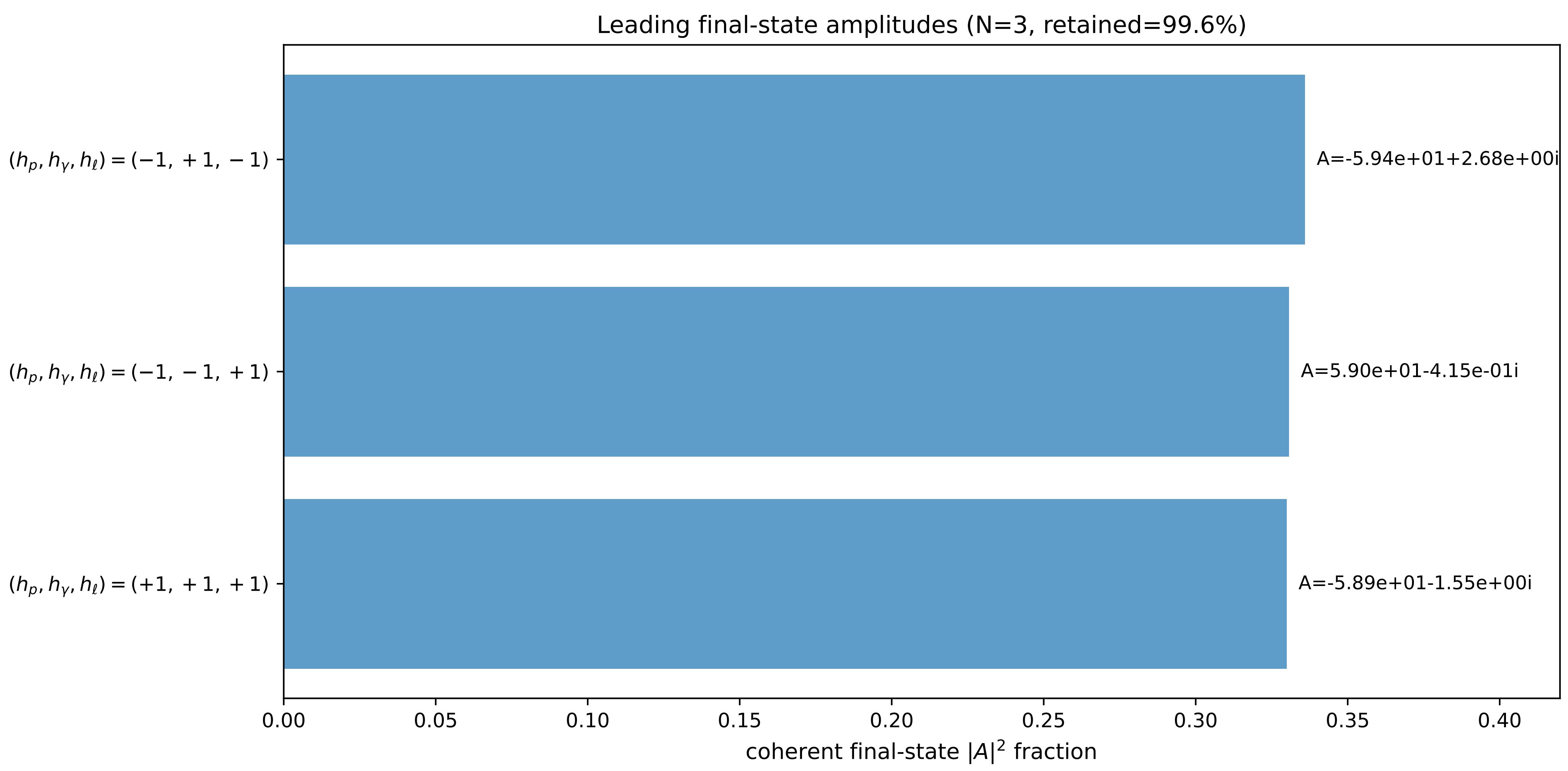


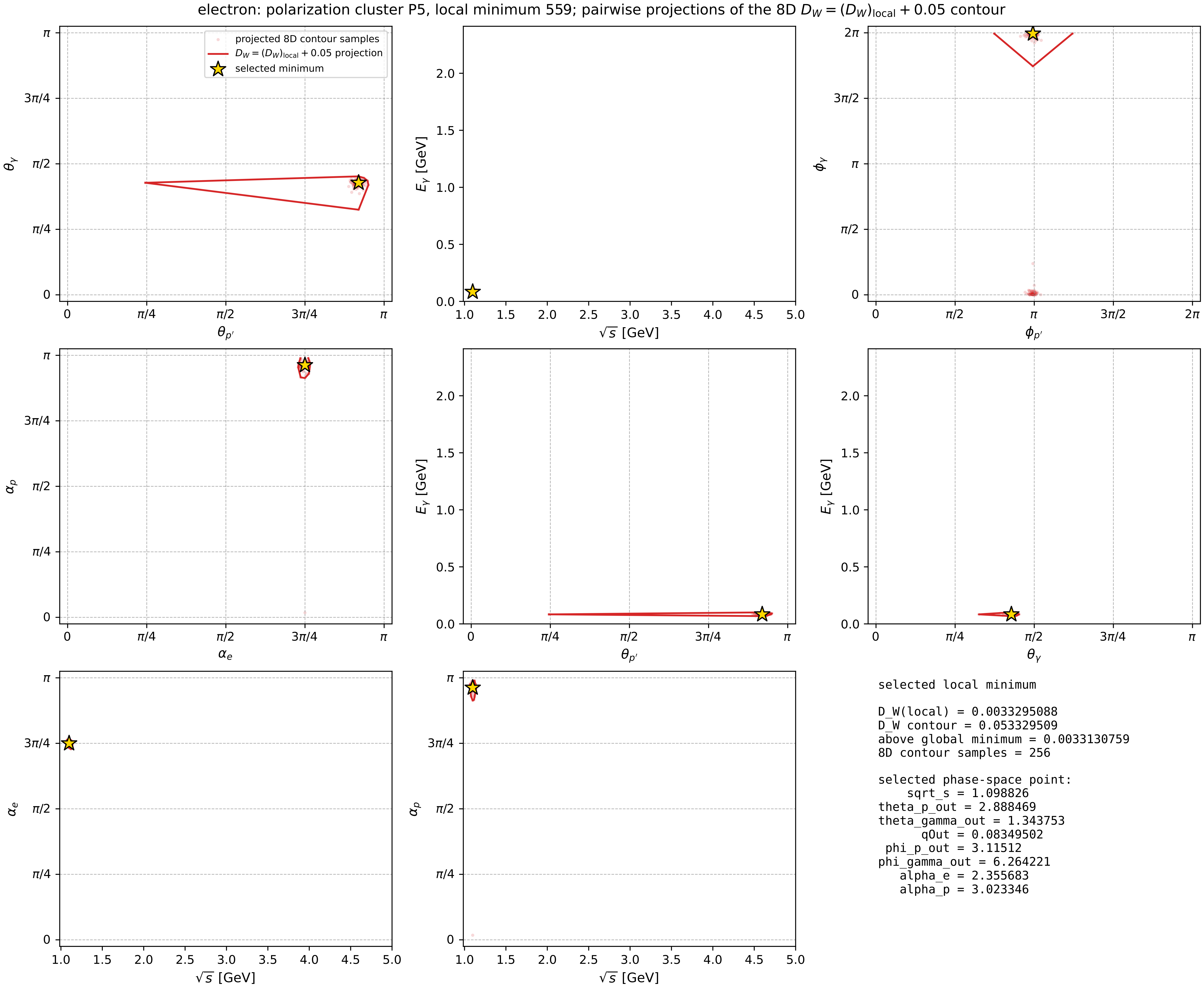

```
dw_mixing_angles_polarization_05_local_minimum_559  D_W=0.00332951
region: polarization_cluster_05_local_minimum_559
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3556829 rad
incoming lepton: -0.706745|+> +0.707468|->
proton mixing angle: alpha_p=3.0233463 rad
incoming proton: -0.993017|+> +0.117971|->
```

s=1.20742, $\sqrt{s}$=1.09883, $|\vec{P}|$=0.149056, $|\vec{P}'|$=0.01683
$\theta_{p'}$=2.88847, θ_γ=1.34375, $\phi_{p'}$=3.11512, ϕ_γ=6.26422
E_γ=0.083495, $\theta_{\ell'}$=1.60321, $\phi_{\ell'}$=3.12304
Q^2=0.0222623, x_B=0.123527, t=-0.0272232, W^2=1.0378, y=0.550173

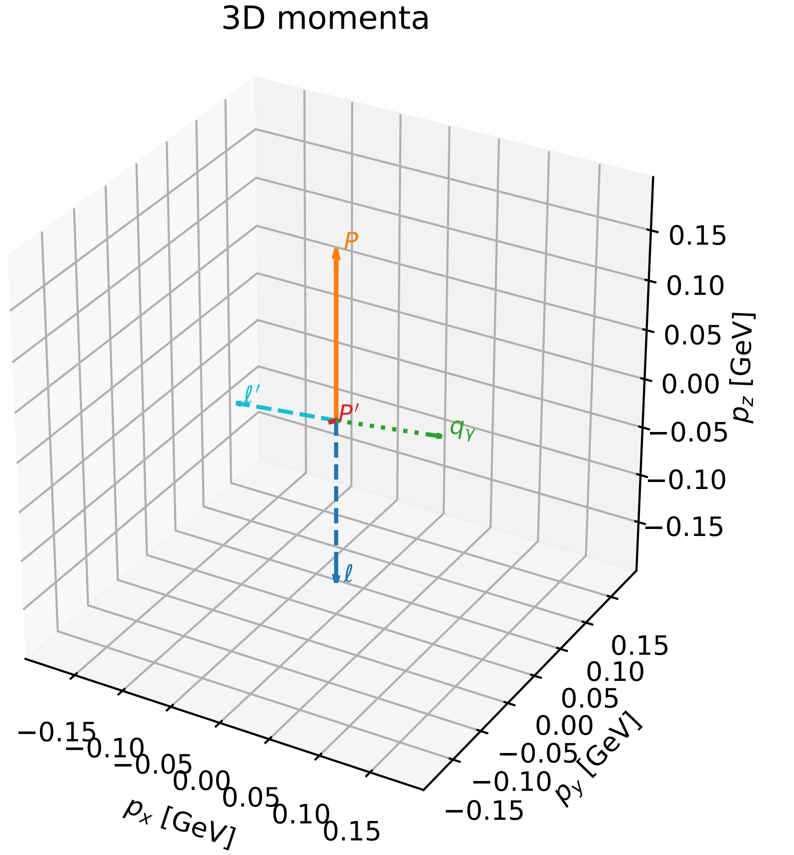
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1491 p_x= 0.0000 p_y= 0.0000 p_z= -0.1491
P E= 0.9498 p_x= 0.0000 p_y= 0.0000 p_z= 0.1491
ℓ' E= 0.0772 p_x= -0.0771 p_y= 0.0014 p_z= -0.0025
P' E= 0.9382 p_x= -0.0042 p_y= 0.0001 p_z= -0.0163
q_γ E= 0.0835 p_x= 0.0813 p_y= -0.0015 p_z= 0.0188






electron: polarization cluster P5, local minimum 559; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0033295088
D_W contour = 0.053329509
above global minimum = 0.0033130759
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.098826
theta_p_out = 2.888469
theta_gamma_out = 1.343753
qOut = 0.08349502
phi_p_out = 3.11512
phi_gamma_out = 6.264221
alpha_e = 2.355683
alpha_p = 3.023346

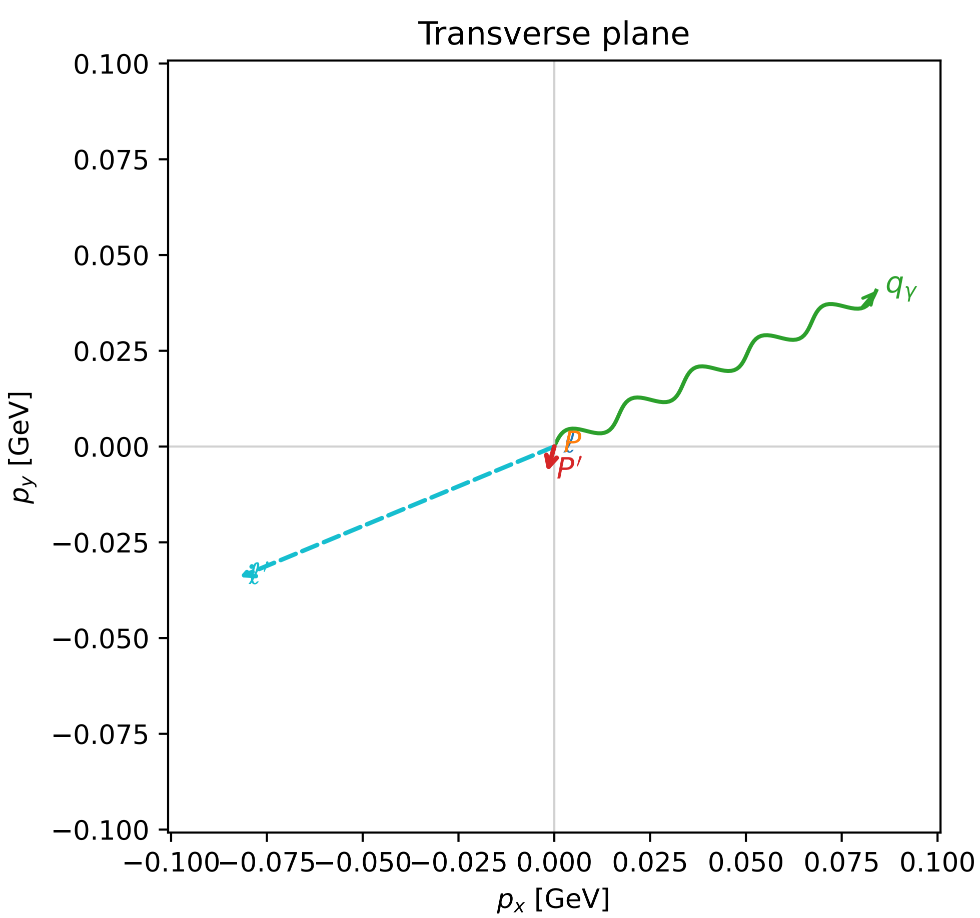
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

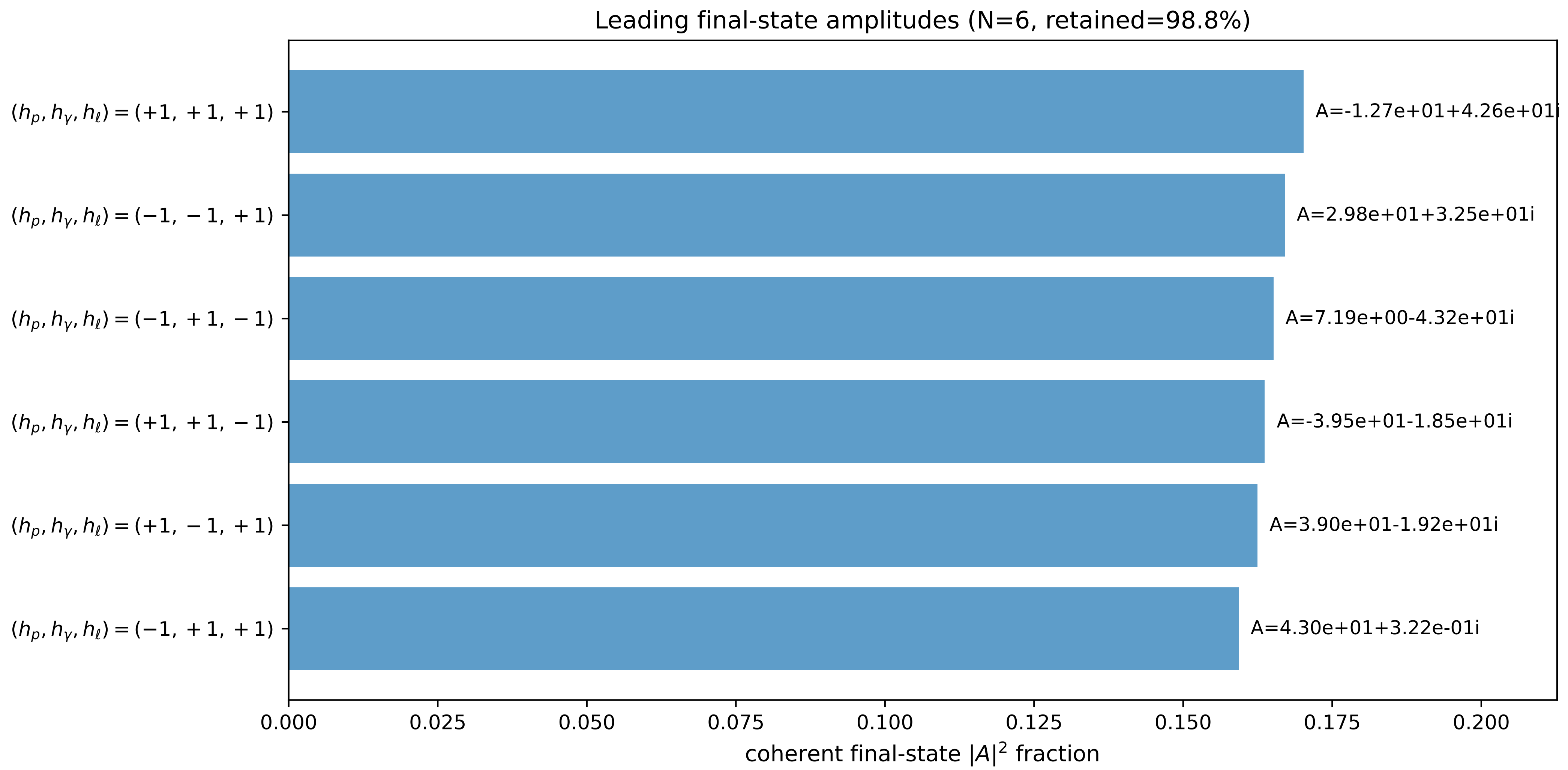


dw_mixing_angles_polarization_05_local_minimum_568 D_W=0.00349483
region: polarization_cluster_05_local_minimum_568
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3612194 rad
incoming lepton: -0.710651|+> +0.703545|->
proton mixing angle: alpha_p=2.944649 rad
incoming proton: -0.980669|+> +0.195673|->

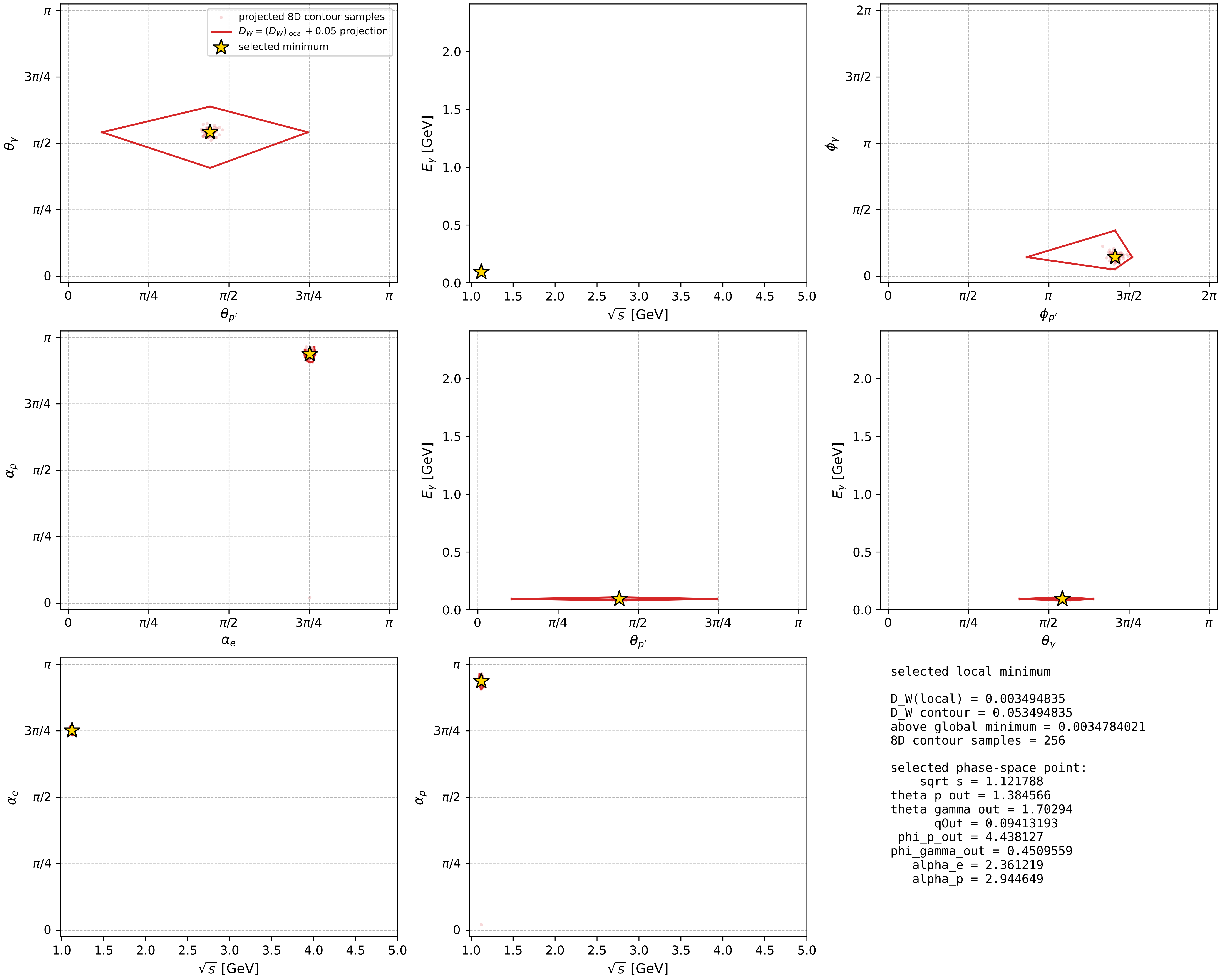
s=1.25841, $\sqrt{s}$=1.12179, $|\vec{P}|$=0.168732, $|\vec{P}'|$=0.0069356
$\theta_{p'}$=1.38457, θ_γ=1.70294, $\phi_{p'}$=4.43813, ϕ_γ=0.450956
E_γ=0.0941319, $\theta_{\ell'}$=1.44642, $\phi_{\ell'}$=3.53517
Q^2=0.0339989, x_B=0.160773, t=-0.0278593, W^2=1.05732, y=0.558612

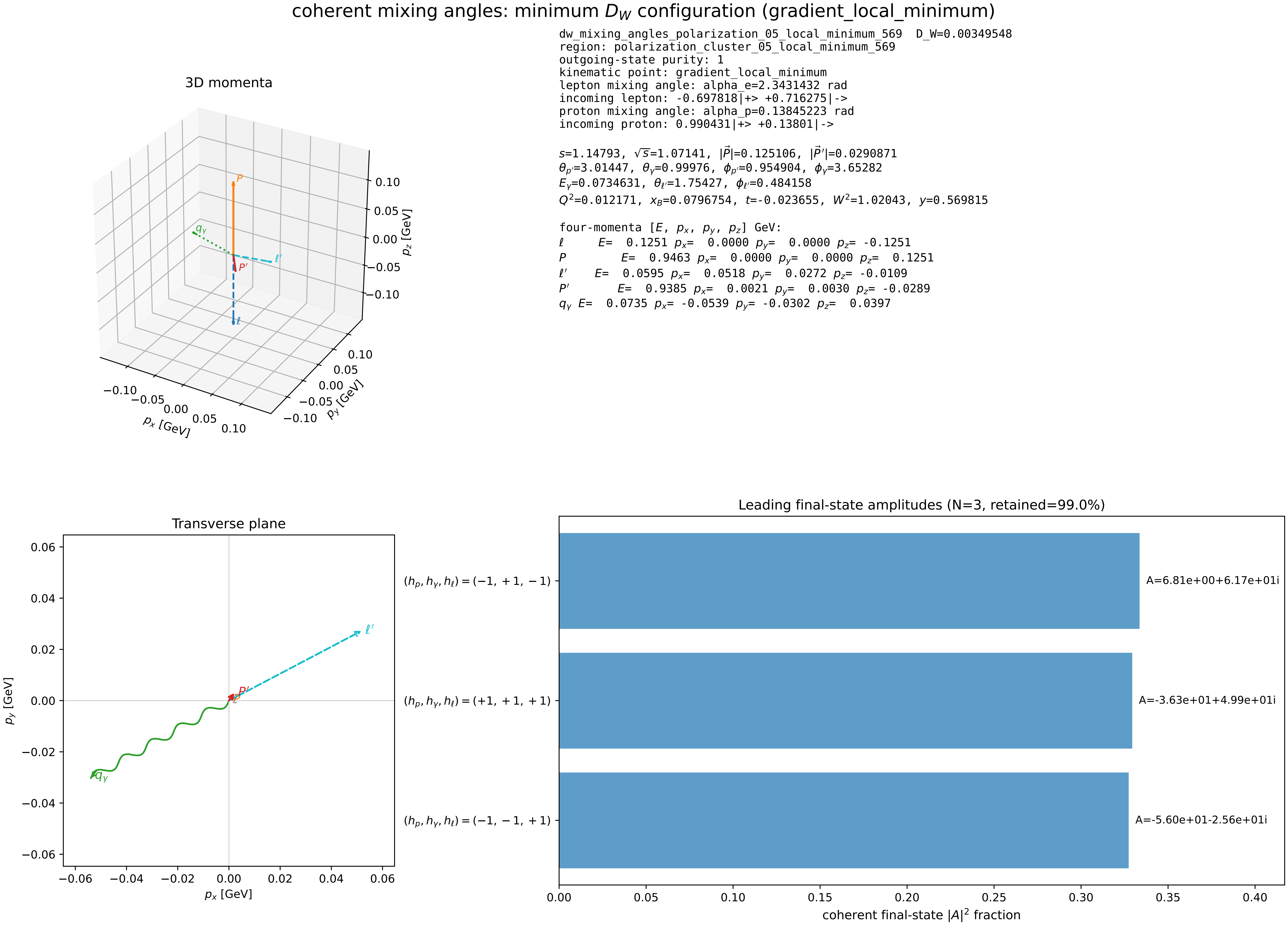
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1687 p_x= 0.0000 p_y= 0.0000 p_z= -0.1687
P E= 0.9531 p_x= 0.0000 p_y= 0.0000 p_z= 0.1687
ℓ' E= 0.0896 p_x= -0.0821 p_y= -0.0341 p_z= 0.0111
P' E= 0.9380 p_x= -0.0018 p_y= -0.0066 p_z= 0.0013
q_γ E= 0.0941 p_x= 0.0840 p_y= 0.0407 p_z= -0.0124

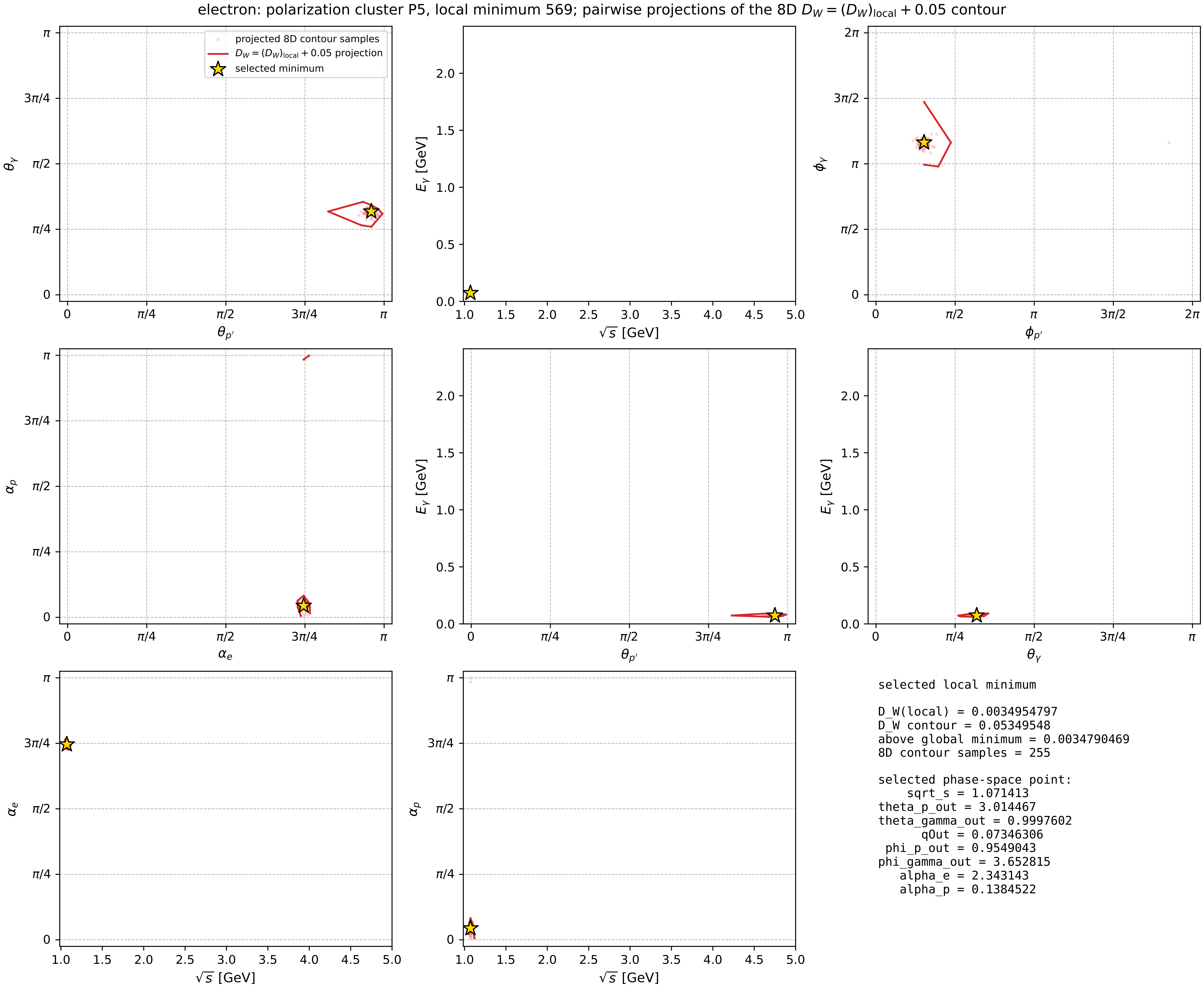




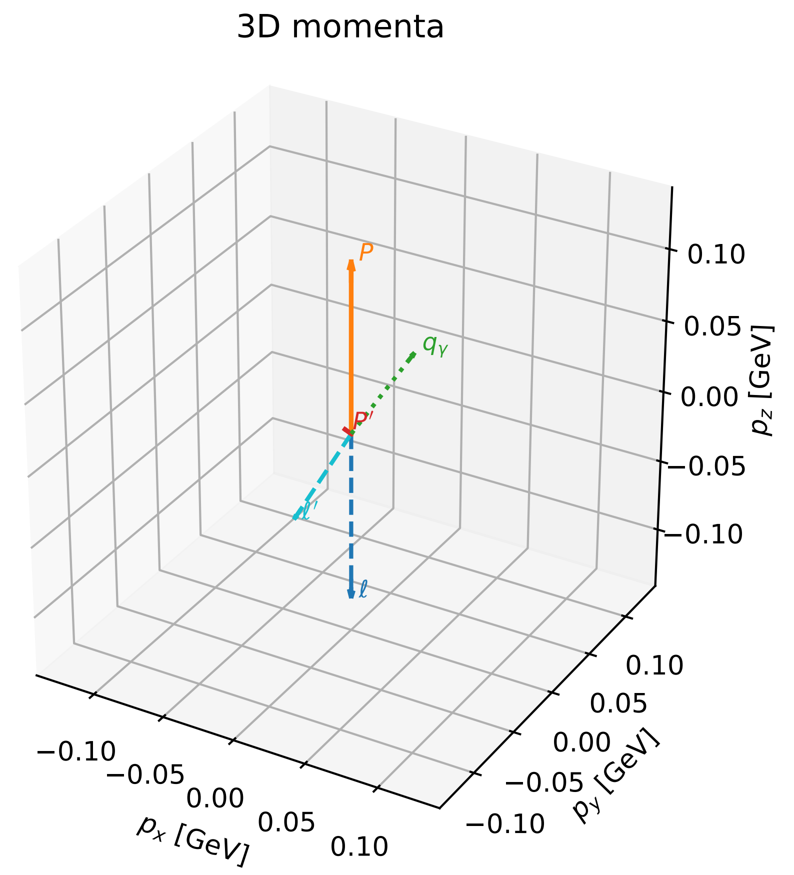
electron: polarization cluster P5, local minimum 568; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour




coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
P
q_γ
ℓ'
P'
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
dw_mixing_angles_polarization_05_local_minimum_569 D_W=0.00349548
region: polarization_cluster_05_local_minimum_569
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3431432 rad
incoming lepton: -0.697818|+> +0.716275|->
proton mixing angle: alpha_p=0.13845223 rad
incoming proton: 0.990431|+> +0.13801|->
s=1.14793, $\sqrt{s}$=1.07141, $|\vec{P}|$=0.125106, $|\vec{P}'|$=0.0290871
$\theta_{p'}$=3.01447, θ_γ=0.99976, $\phi_{p'}$=0.954904, ϕ_γ=3.65282
E_γ=0.0734631, $\theta_{\ell'}$=1.75427, $\phi_{\ell'}$=0.484158
Q^2=0.012171, x_B=0.0796754, t=-0.023655, W^2=1.02043, y=0.569815
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1251 p_x= 0.0000 p_y= 0.0000 p_z= -0.1251
P E= 0.9463 p_x= 0.0000 p_y= 0.0000 p_z= 0.1251
ℓ' E= 0.0595 p_x= 0.0518 p_y= 0.0272 p_z= -0.0109
P' E= 0.9385 p_x= 0.0021 p_y= 0.0030 p_z= -0.0289
q_γ E= 0.0735 p_x= -0.0539 p_y= -0.0302 p_z= 0.0397
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=3, retained=99.0%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=6.81e+00+6.17e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$
A=-3.63e+01+4.99e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=-5.60e+01-2.56e+01i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P5, local minimum 569; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0034954797
D_W contour = 0.05349548
above global minimum = 0.0034790469
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.071413
theta_p_out = 3.014467
theta_gamma_out = 0.9997602
qOut = 0.07346306
phi_p_out = 0.9549043
phi_gamma_out = 3.652815
alpha_e = 2.343143
alpha_p = 0.1384522

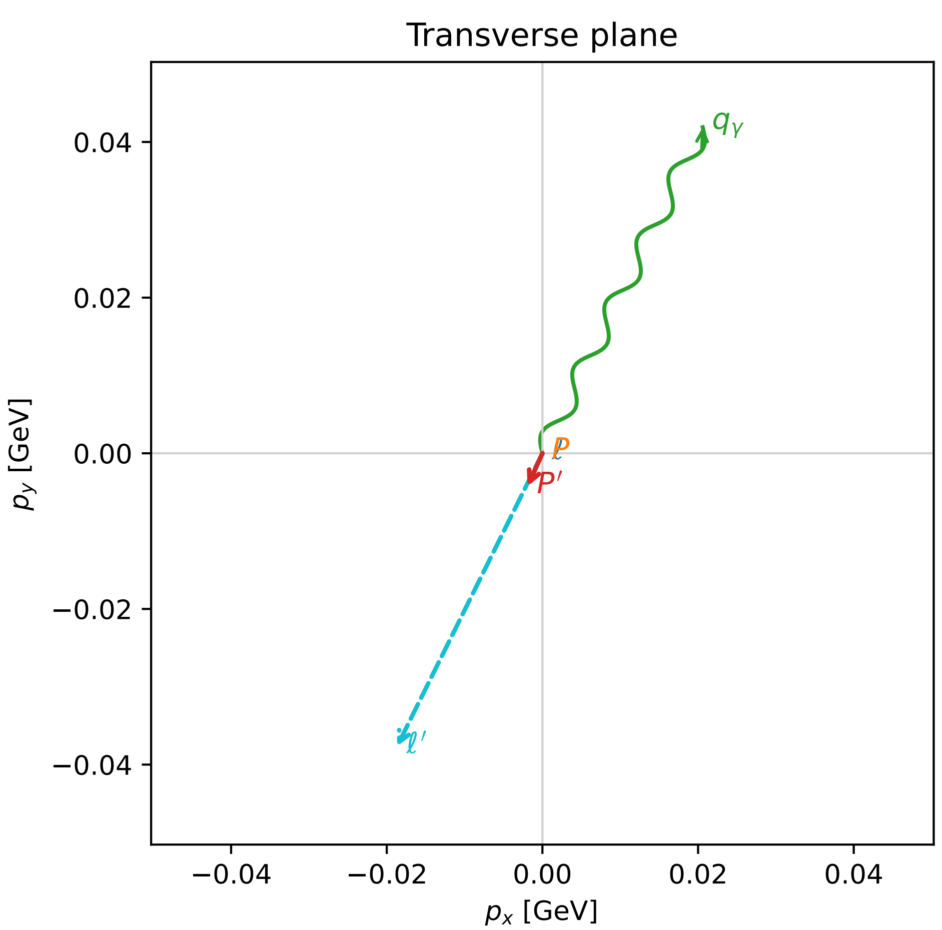
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

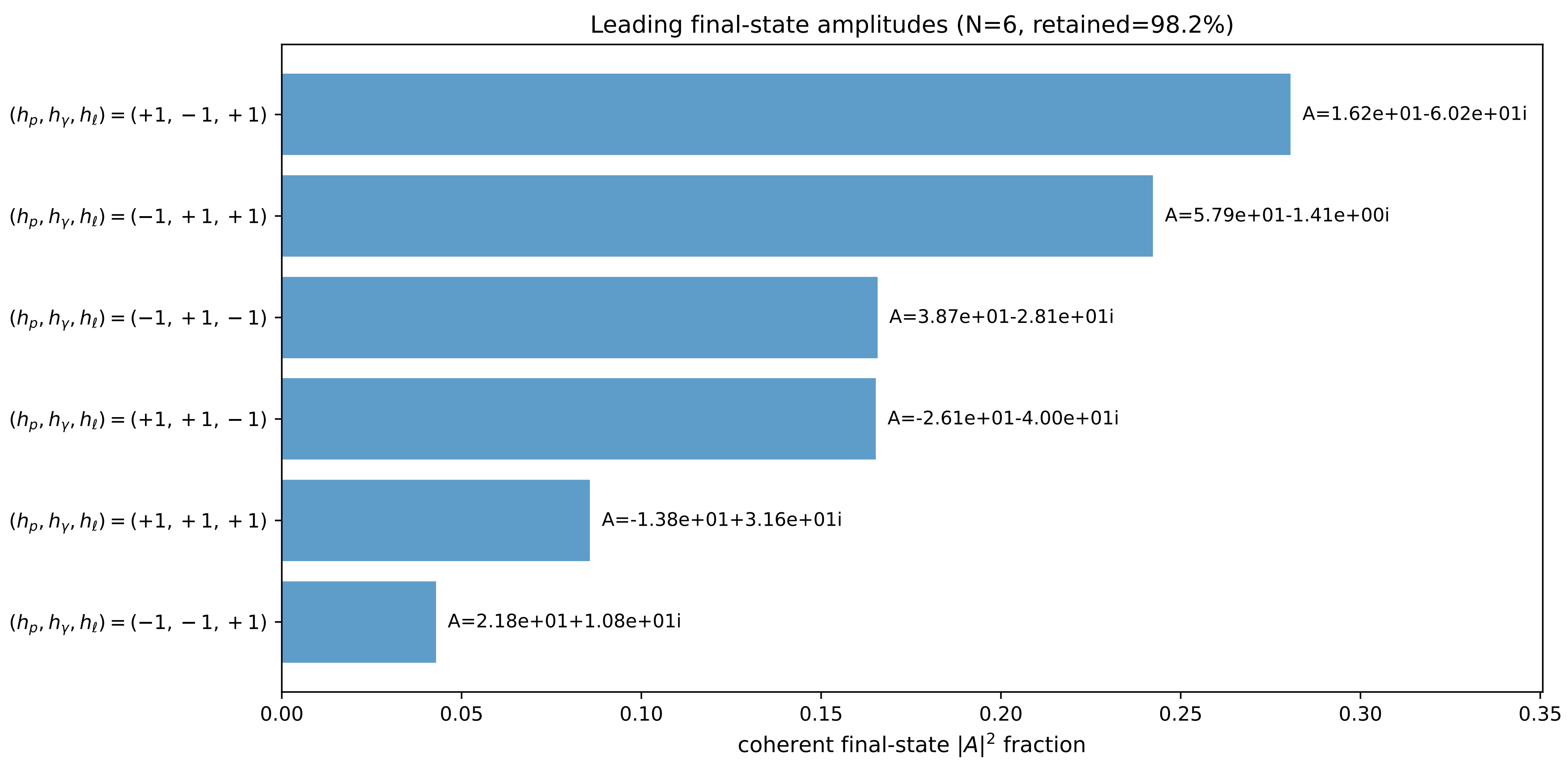


dw_mixing_angles_polarization_05_local_minimum_594 D_W=0.00399973
region: polarization_cluster_05_local_minimum_594
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3510222 rad
incoming lepton: -0.70344|+> +0.710755|->
proton mixing angle: alpha_p=2.8929763 rad
incoming proton: -0.969254|+> +0.246063|->

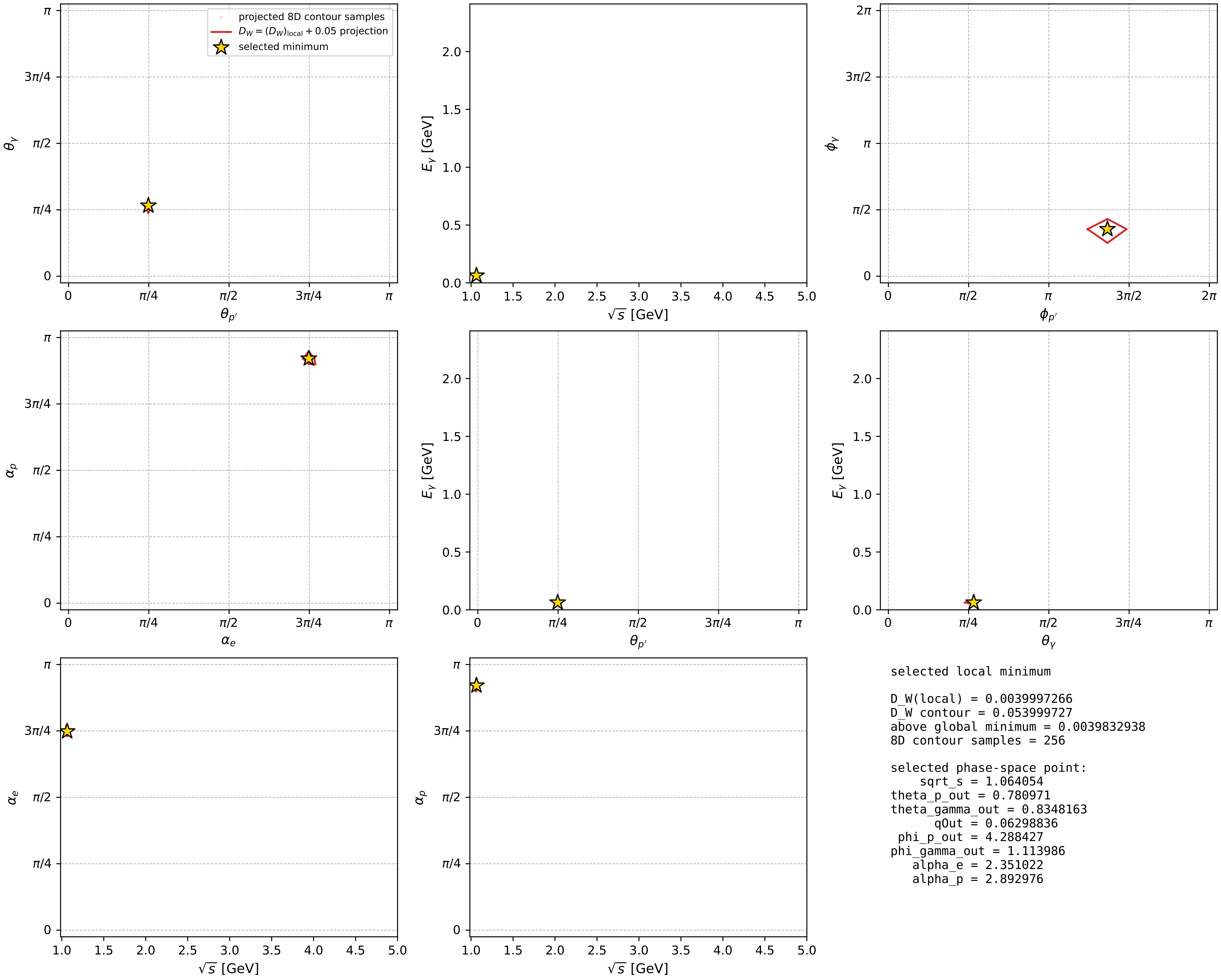
s=1.13221, $\sqrt{s}$=1.06405, $|\vec{P}|$=0.118586, $|\vec{P}'|$=0.00661233
$\theta_{p'}$=0.780971, θ_γ=0.834816, $\phi_{p'}$=4.28843, ϕ_γ=1.11399
E_γ=0.0629884, $\theta_{\ell'}$=2.41172, $\phi_{\ell'}$=4.25194
Q^2=0.00380877, x_B=0.0312156, t=-0.0129372, W^2=0.99805, y=0.483485

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1186 p_x= 0.0000 p_y= 0.0000 p_z= -0.1186
P E= 0.9455 p_x= 0.0000 p_y= 0.0000 p_z= 0.1186
ℓ' E= 0.0630 p_x= -0.0187 p_y= -0.0377 p_z= -0.0470
P' E= 0.9380 p_x= -0.0019 p_y= -0.0042 p_z= 0.0047
q_γ E= 0.0630 p_x= 0.0206 p_y= 0.0419 p_z= 0.0423

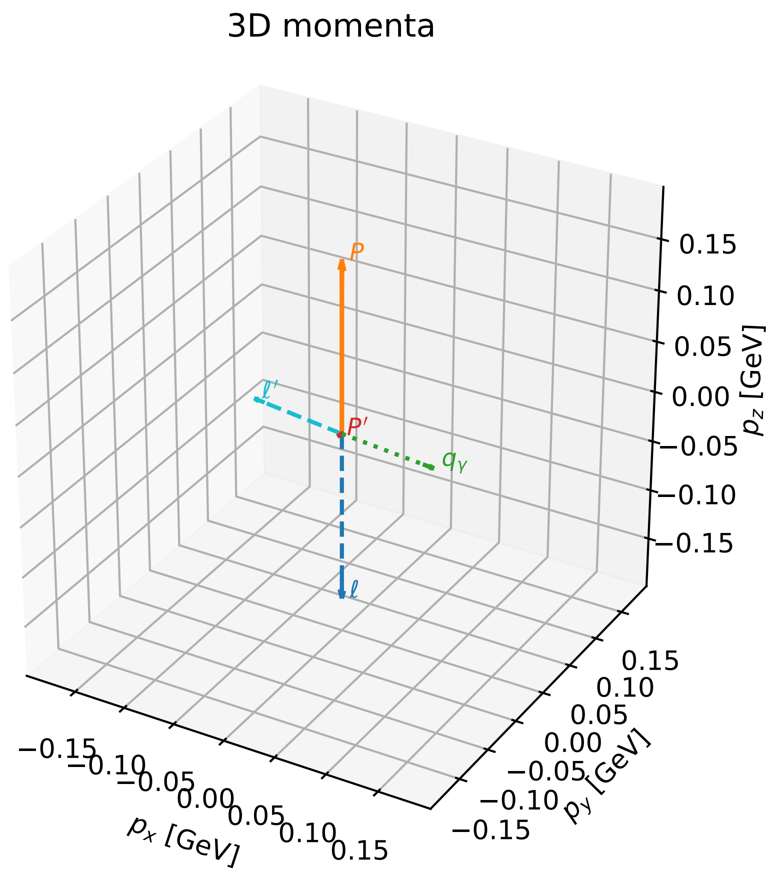




electron: polarization cluster P5, local minimum 594; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



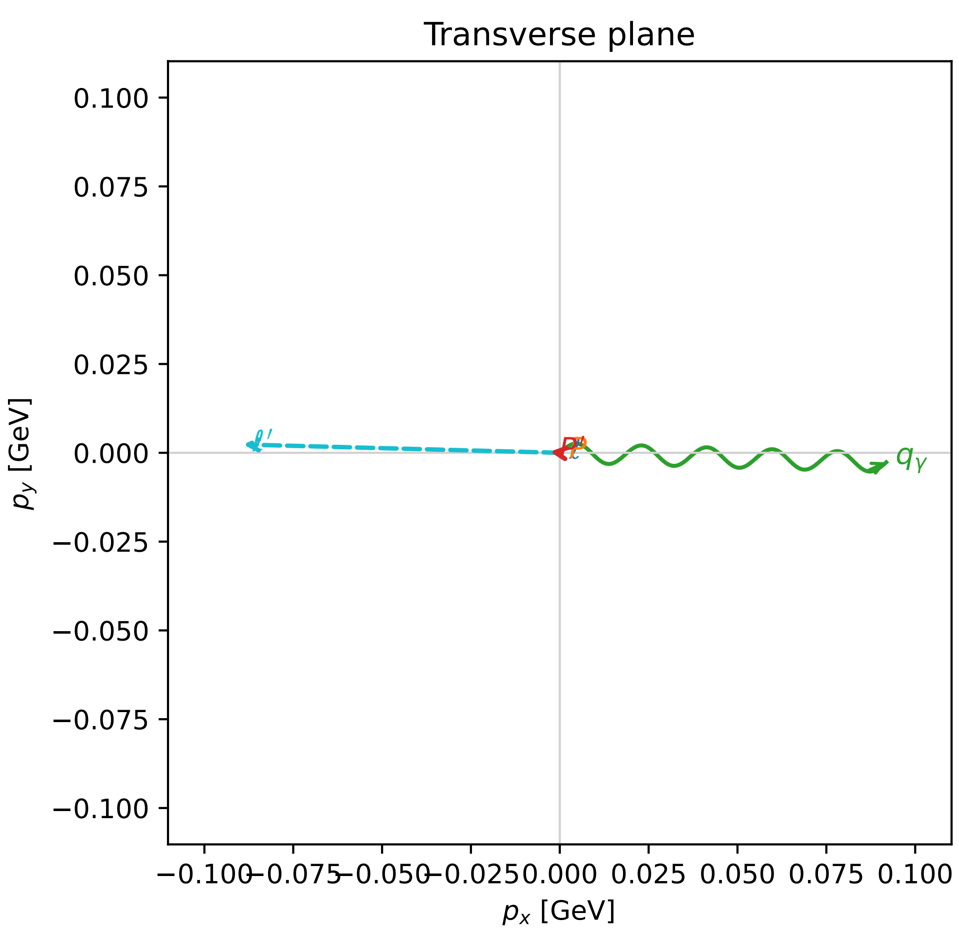
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

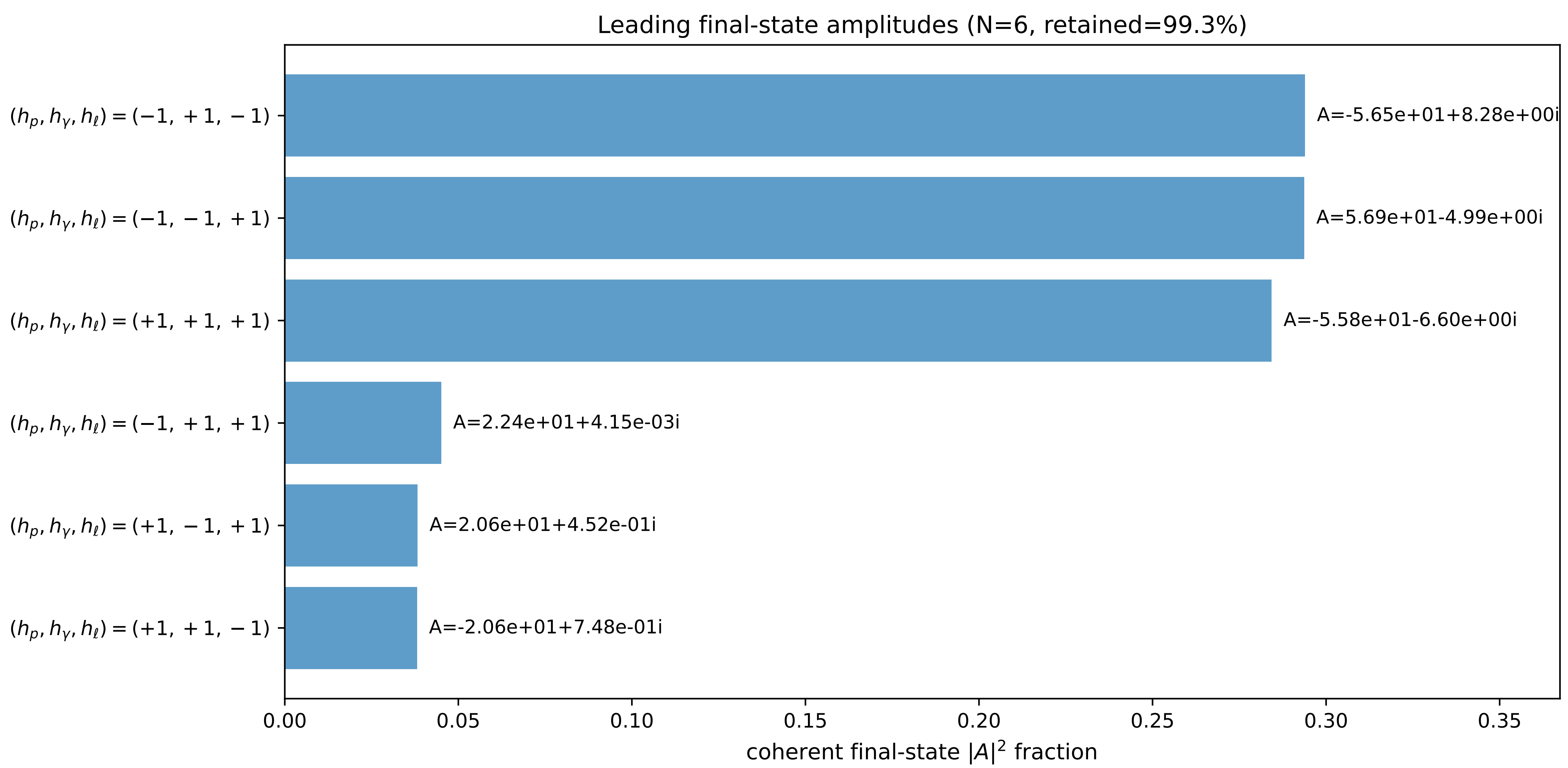


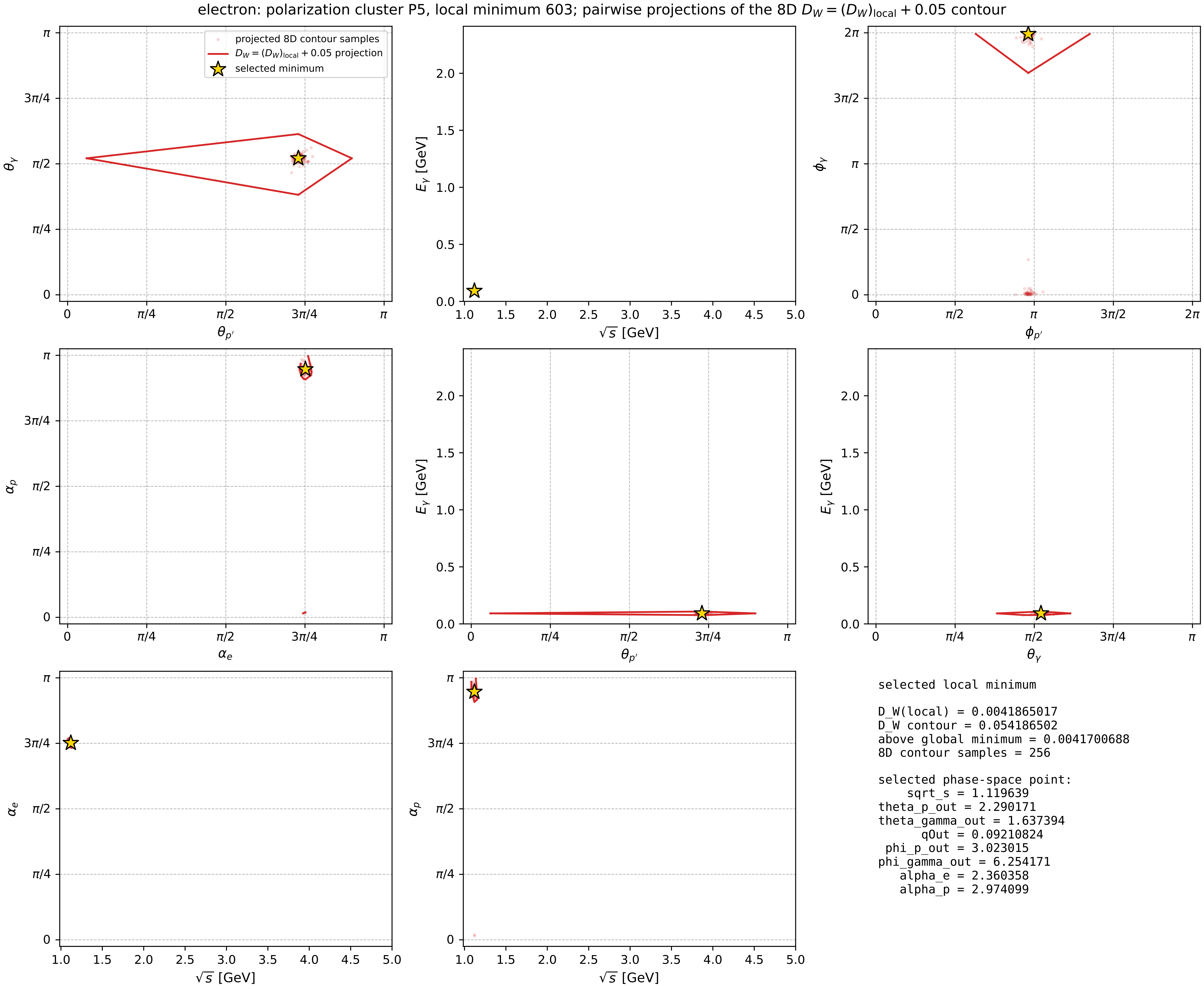
dw_mixing_angles_polarization_05_local_minimum_603 D_W=0.0041865
region: polarization_cluster_05_local_minimum_603
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3603579 rad
incoming lepton: -0.710045|+> +0.704157|->
proton mixing angle: alpha_p=2.9740991 rad
incoming proton: -0.986006|+> +0.166711|->

s=1.25359, $\sqrt{s}$=1.11964, $|\vec{P}|$=0.166905, $|\vec{P}'|$=0.00373106
$\theta_{p'}$=2.29017, θ_γ=1.63739, $\phi_{p'}$=3.02302, ϕ_γ=6.25417
E_γ=0.0921082, $\theta_{\ell'}$=1.47472, $\phi_{\ell'}$=3.1154
Q^2=0.0327502, x_B=0.158958, t=-0.0284749, W^2=1.05312, y=0.551255

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1669 p_x= 0.0000 p_y= 0.0000 p_z= -0.1669
P E= 0.9527 p_x= 0.0000 p_y= 0.0000 p_z= 0.1669
ℓ' E= 0.0895 p_x= -0.0891 p_y= 0.0023 p_z= 0.0086
P' E= 0.9380 p_x= -0.0028 p_y= 0.0003 p_z= -0.0025
q_γ E= 0.0921 p_x= 0.0919 p_y= -0.0027 p_z= -0.0061

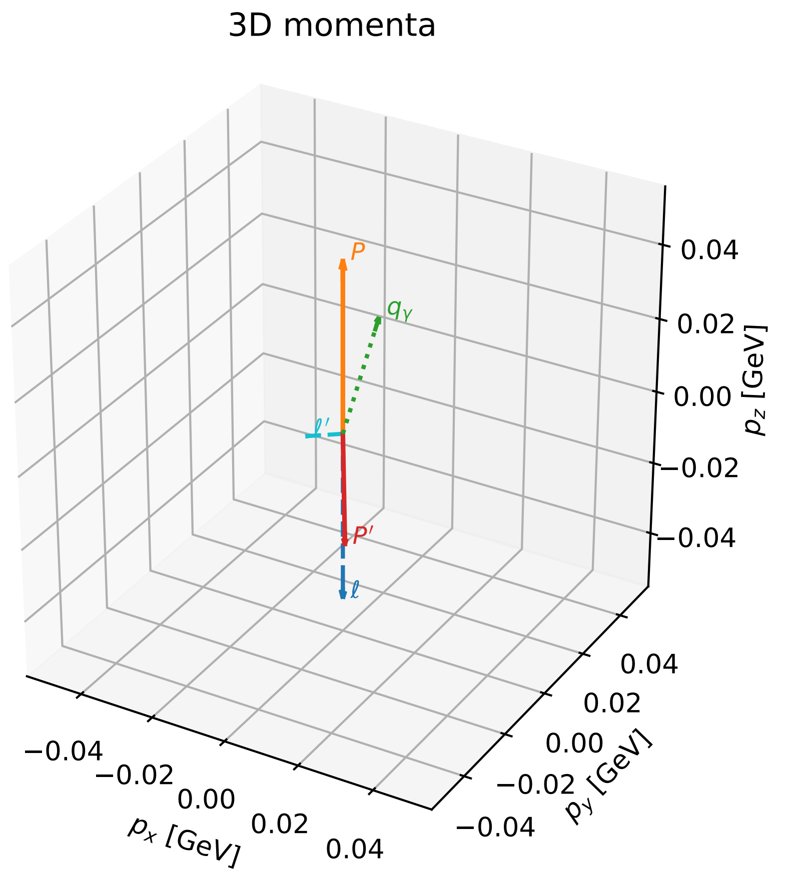






electron: polarization cluster P5, local minimum 603; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.0041865017
D_W contour = 0.054186502
above global minimum = 0.0041700688
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.119639
theta_p_out = 2.290171
theta_gamma_out = 1.637394
qOut = 0.09210824
phi_p_out = 3.023015
phi_gamma_out = 6.254171
alpha_e = 2.360358
alpha_p = 2.974099

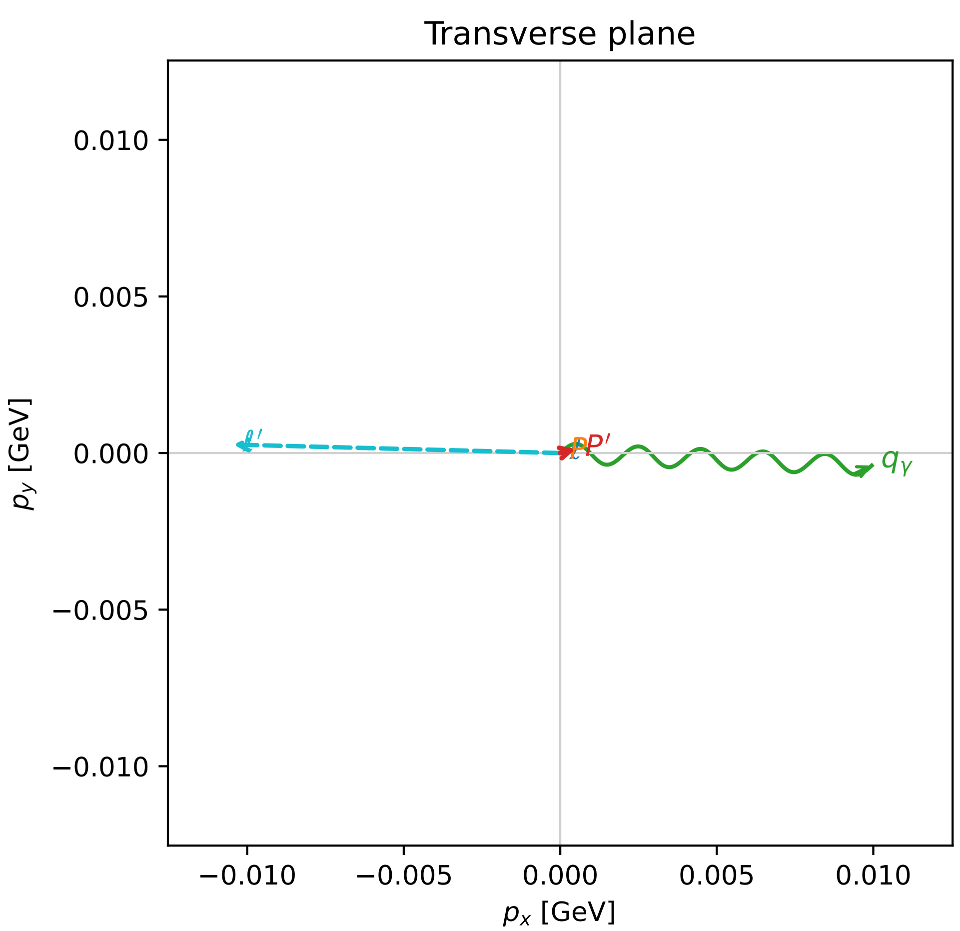
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

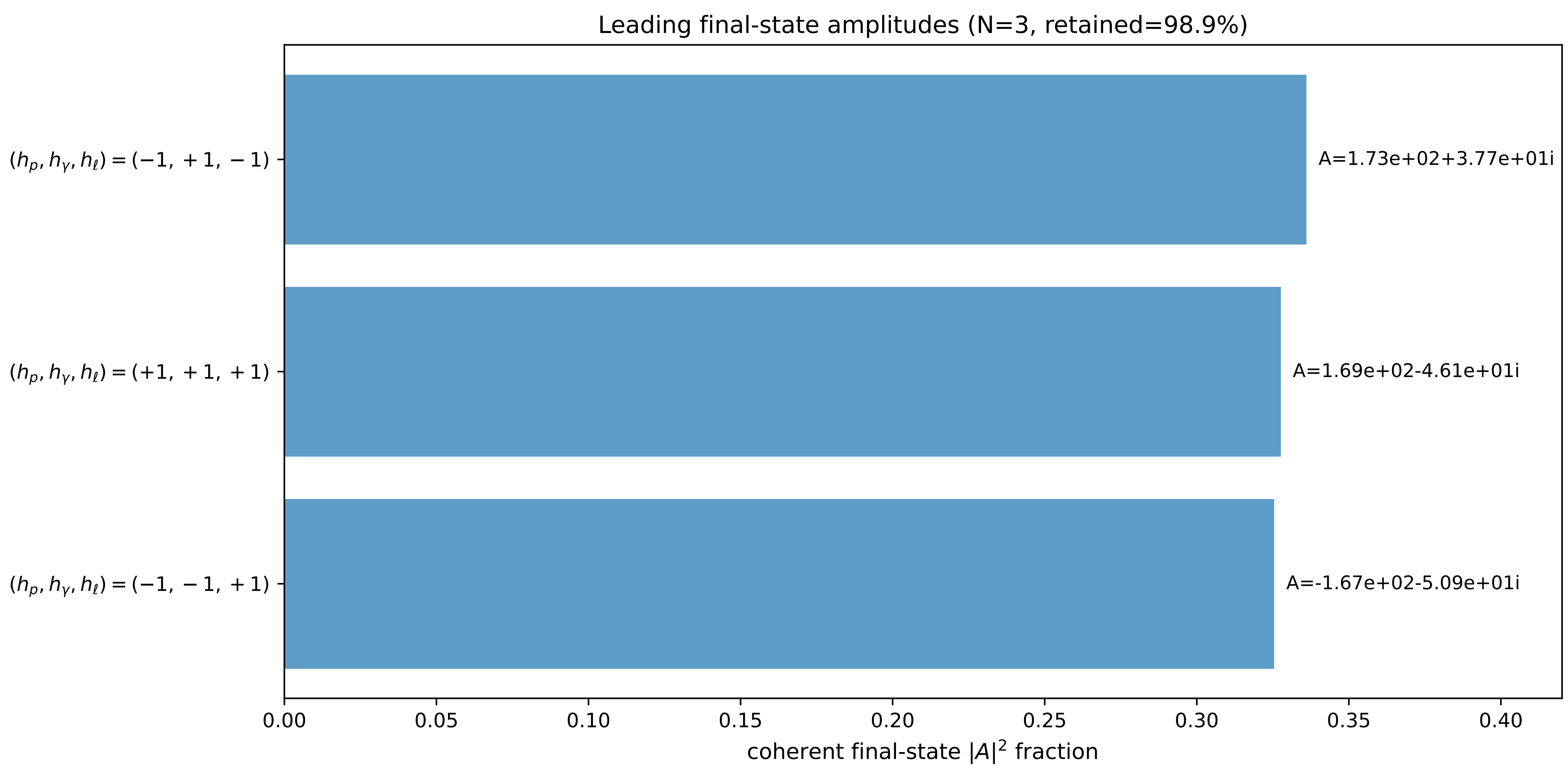


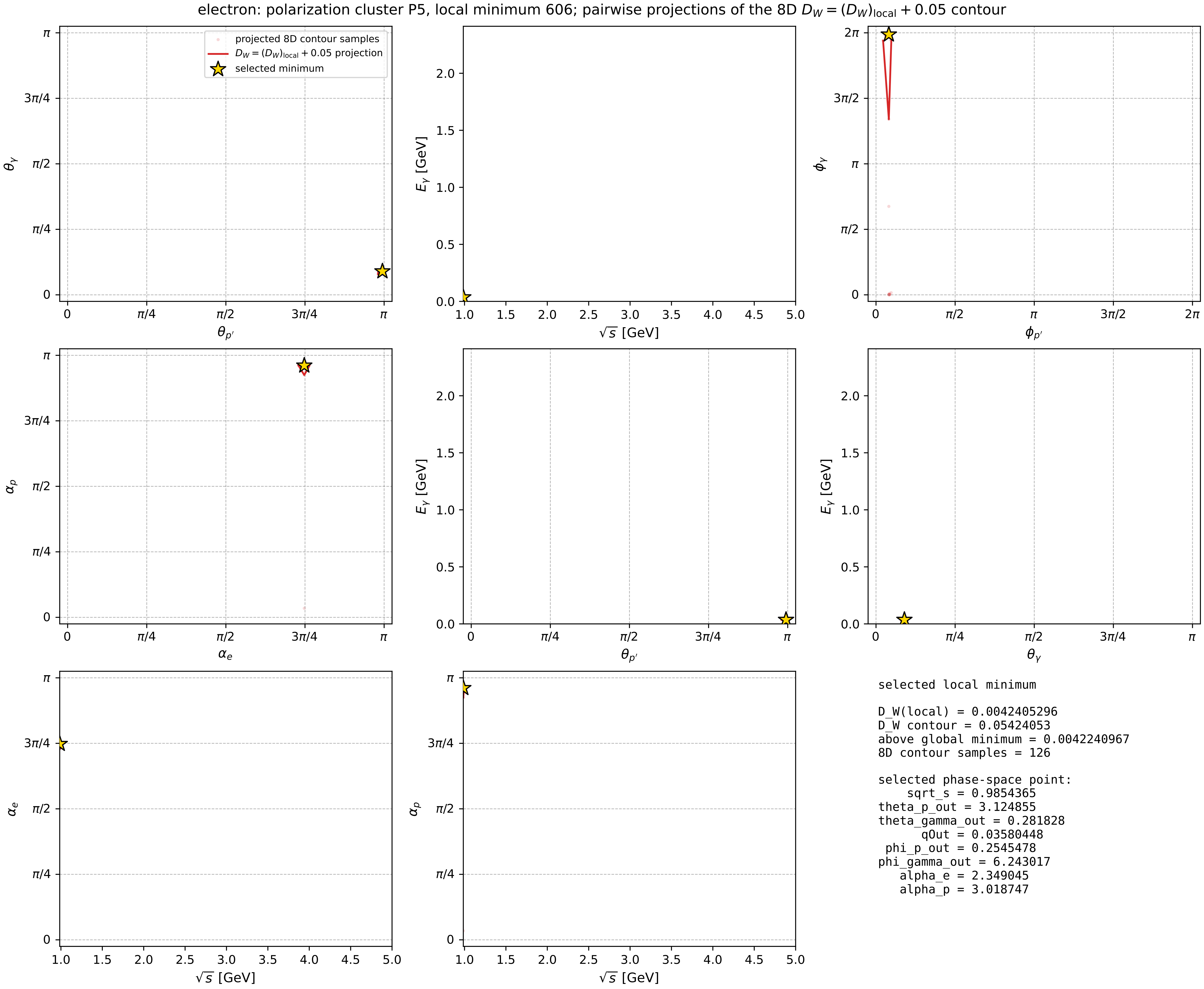
dw_mixing_angles_polarization_05_local_minimum_606 D_W=0.00424053
region: polarization_cluster_05_local_minimum_606
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3490452 rad
incoming lepton: -0.702033|+> +0.712144|->
proton mixing angle: alpha_p=3.0187472 rad
incoming proton: -0.992464|+> +0.122537|->

s=0.971085, $\sqrt{s}$=0.985437, $|\vec{P}|$=0.0462921, $|\vec{P}'|$=0.0305854
$\theta_{p'}$=3.12485, θ_γ=0.281828, $\phi_{p'}$=0.254548, ϕ_γ=6.24302
E_γ=0.0358045, $\theta_{\ell'}$=1.92053, $\phi_{\ell'}$=3.11566
Q^2=0.000677509, x_B=0.009682, t=-0.00590934, W^2=0.949143, y=0.766938

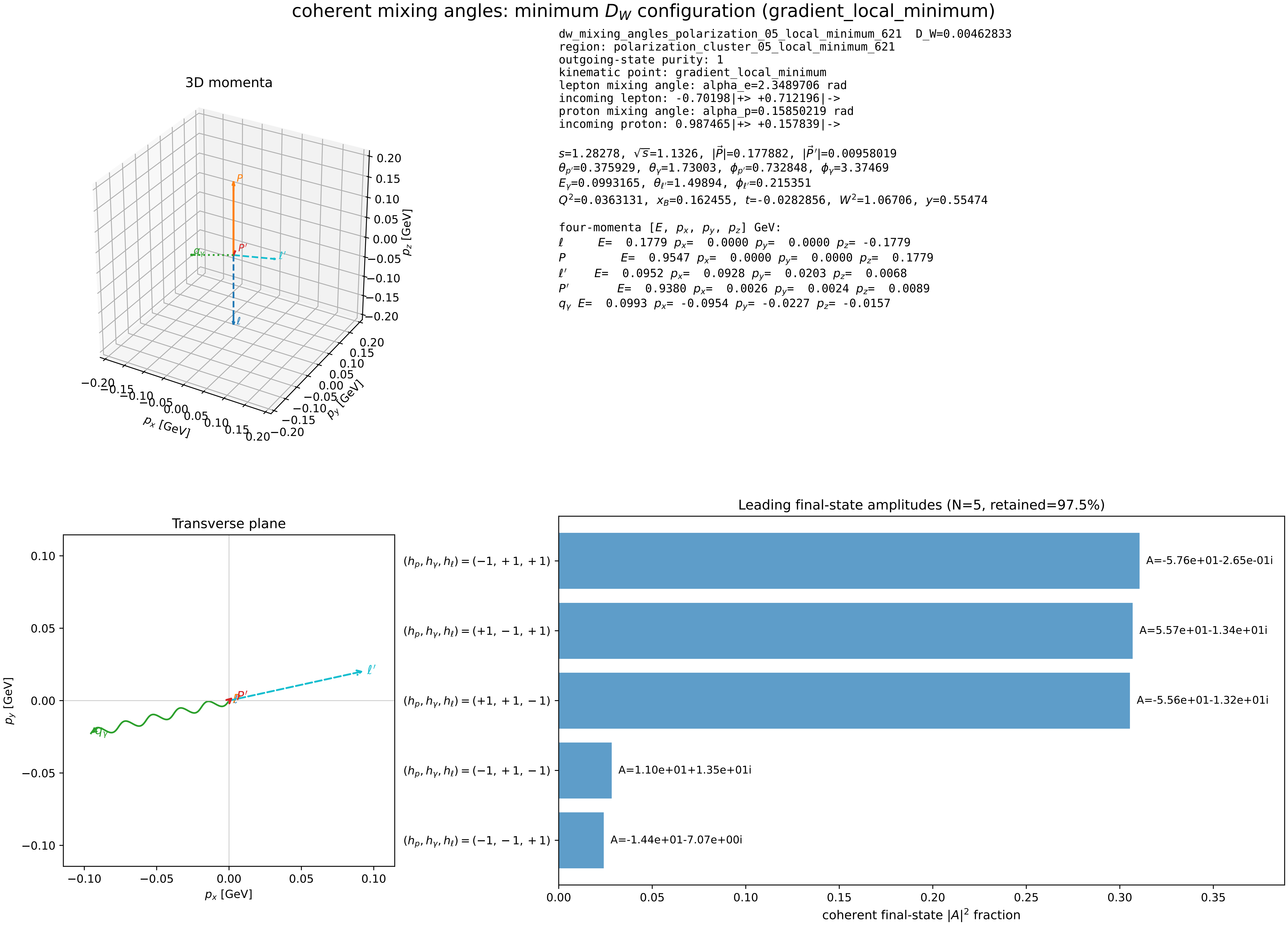
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0463 p_x= 0.0000 p_y= 0.0000 p_z= -0.0463
P E= 0.9391 p_x= 0.0000 p_y= 0.0000 p_z= 0.0463
ℓ' E= 0.0111 p_x= -0.0104 p_y= 0.0003 p_z= -0.0038
P' E= 0.9385 p_x= 0.0005 p_y= 0.0001 p_z= -0.0306
q_γ E= 0.0358 p_x= 0.0099 p_y= -0.0004 p_z= 0.0344

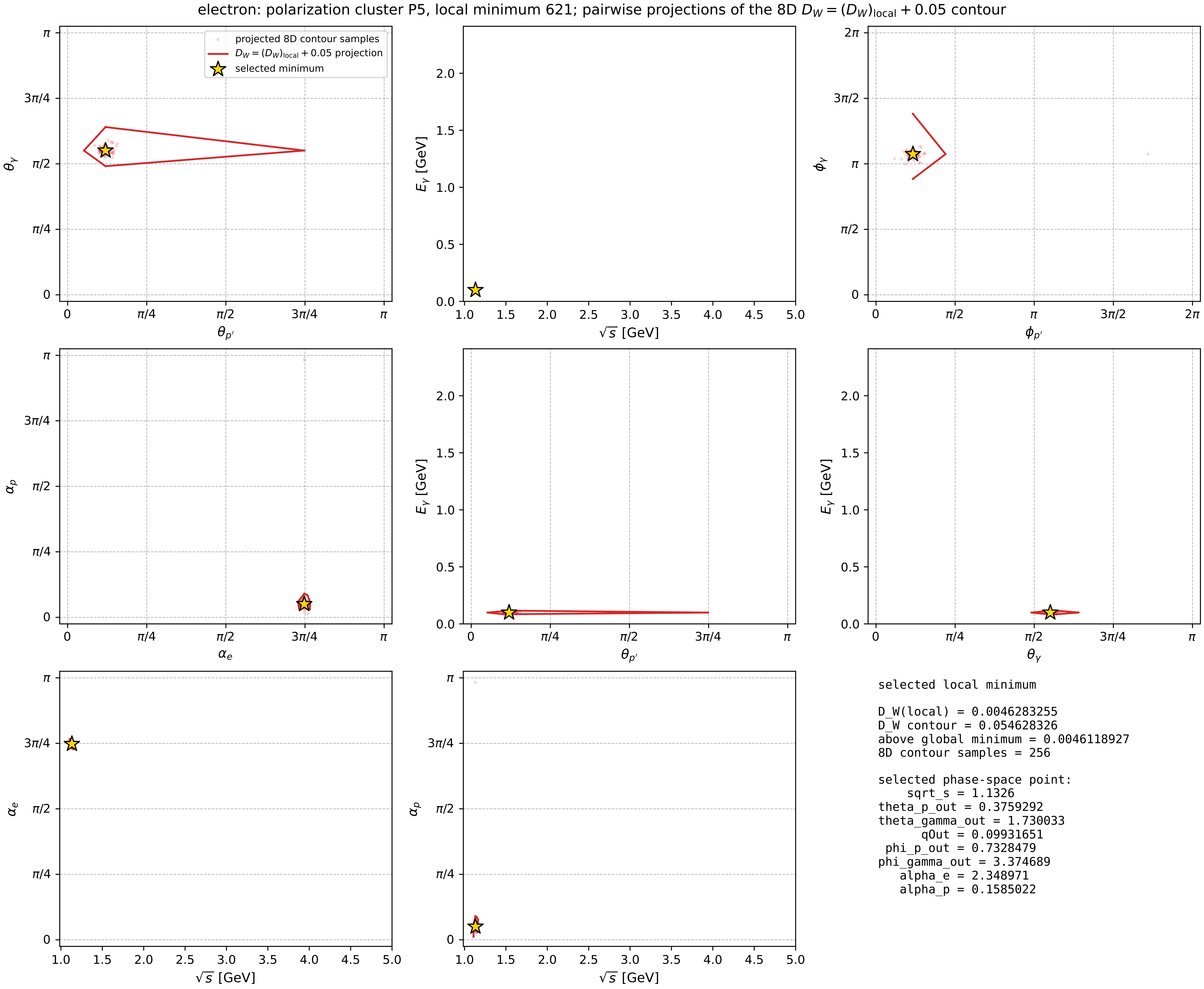


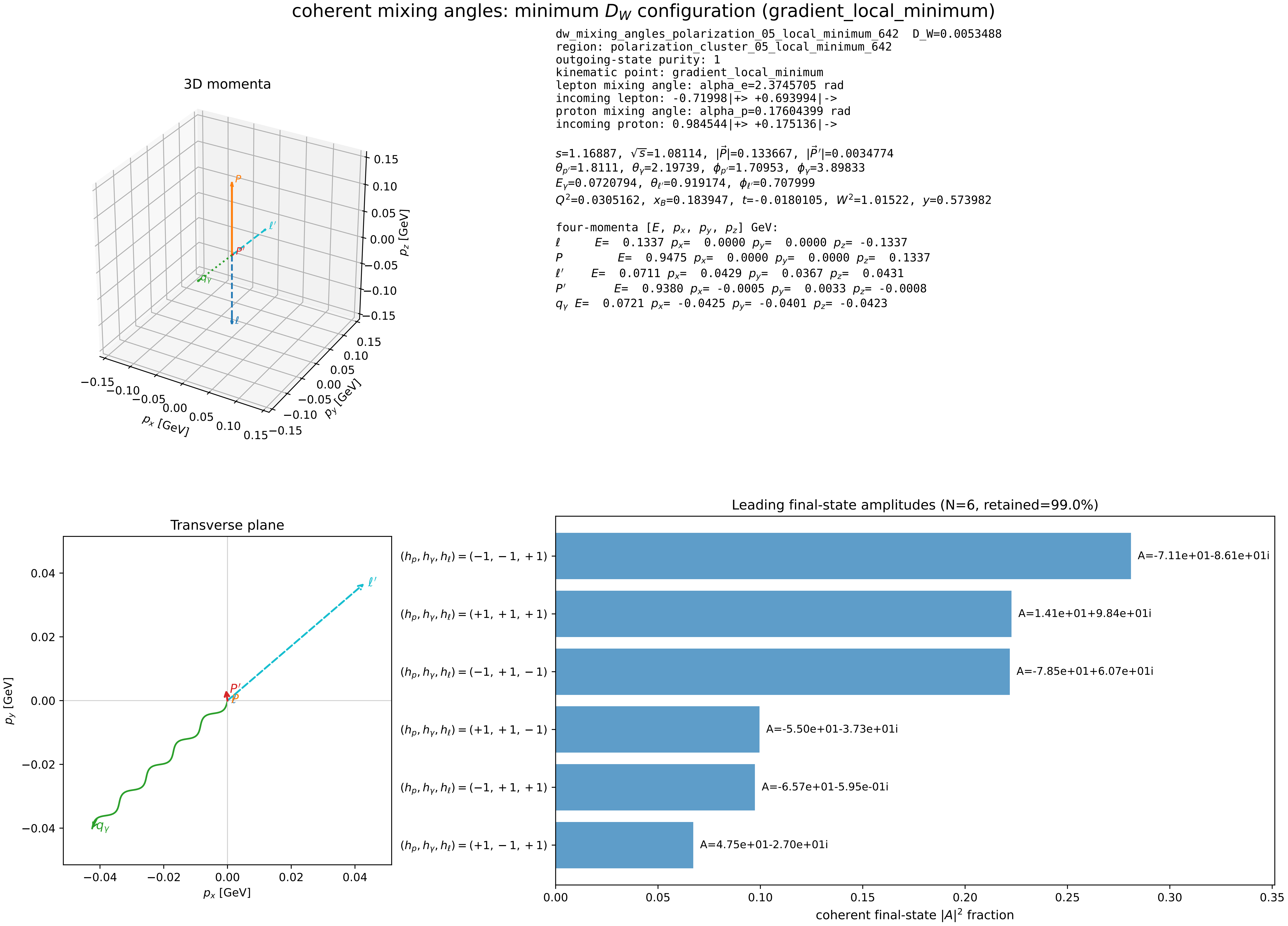




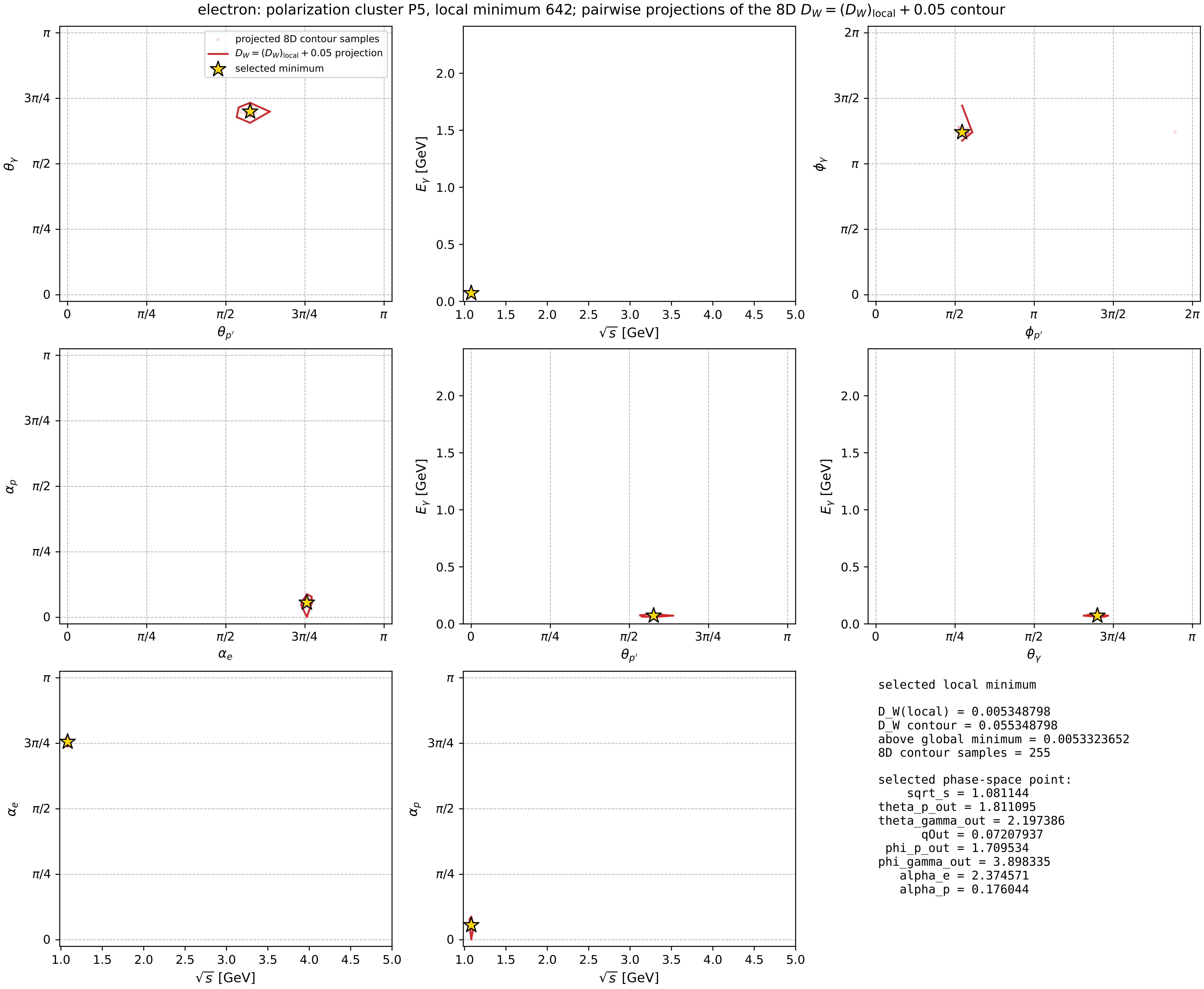
electron: polarization cluster P5, local minimum 606; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
E_γ [GeV]
$\theta_{p'}$
E_γ [GeV]
θ_γ
α_e
$\sqrt{s}$ [GeV]
α_p
$\sqrt{s}$ [GeV]
selected local minimum
D_W(local) = 0.0042405296
D_W contour = 0.05424053
above global minimum = 0.0042240967
8D contour samples = 126
selected phase-space point:
sqrt_s = 0.9854365
theta_p_out = 3.124855
theta_gamma_out = 0.281828
qOut = 0.03580448
phi_p_out = 0.2545478
phi_gamma_out = 6.243017
alpha_e = 2.349045
alpha_p = 3.018747


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_621 D_W=0.00462833
region: polarization_cluster_05_local_minimum_621
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3489706 rad
incoming lepton: -0.70198|+> +0.712196|->
proton mixing angle: alpha_p=0.15850219 rad
incoming proton: 0.987465|+> +0.157839|->
s=1.28278, $\sqrt{s}$=1.1326, $|\vec{P}|$=0.177882, $|\vec{P}'|$=0.00958019
$\theta_{p'}$=0.375929, θ_γ=1.73003, $\phi_{p'}$=0.732848, ϕ_γ=3.37469
E_γ=0.0993165, $\theta_{\ell'}$=1.49894, $\phi_{\ell'}$=0.215351
Q^2=0.0363131, x_B=0.162455, t=-0.0282856, W^2=1.06706, y=0.55474
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1779 p_x= 0.0000 p_y= 0.0000 p_z= -0.1779
P E= 0.9547 p_x= 0.0000 p_y= 0.0000 p_z= 0.1779
ℓ' E= 0.0952 p_x= 0.0928 p_y= 0.0203 p_z= 0.0068
P' E= 0.9380 p_x= 0.0026 p_y= 0.0024 p_z= 0.0089
q_γ E= 0.0993 p_x= -0.0954 p_y= -0.0227 p_z= -0.0157
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=5, retained=97.5%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-5.76e+01-2.65e-01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=5.57e+01-1.34e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-5.56e+01-1.32e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=1.10e+01+1.35e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-1.44e+01-7.07e+00i
coherent final-state $|A|^2$ fraction

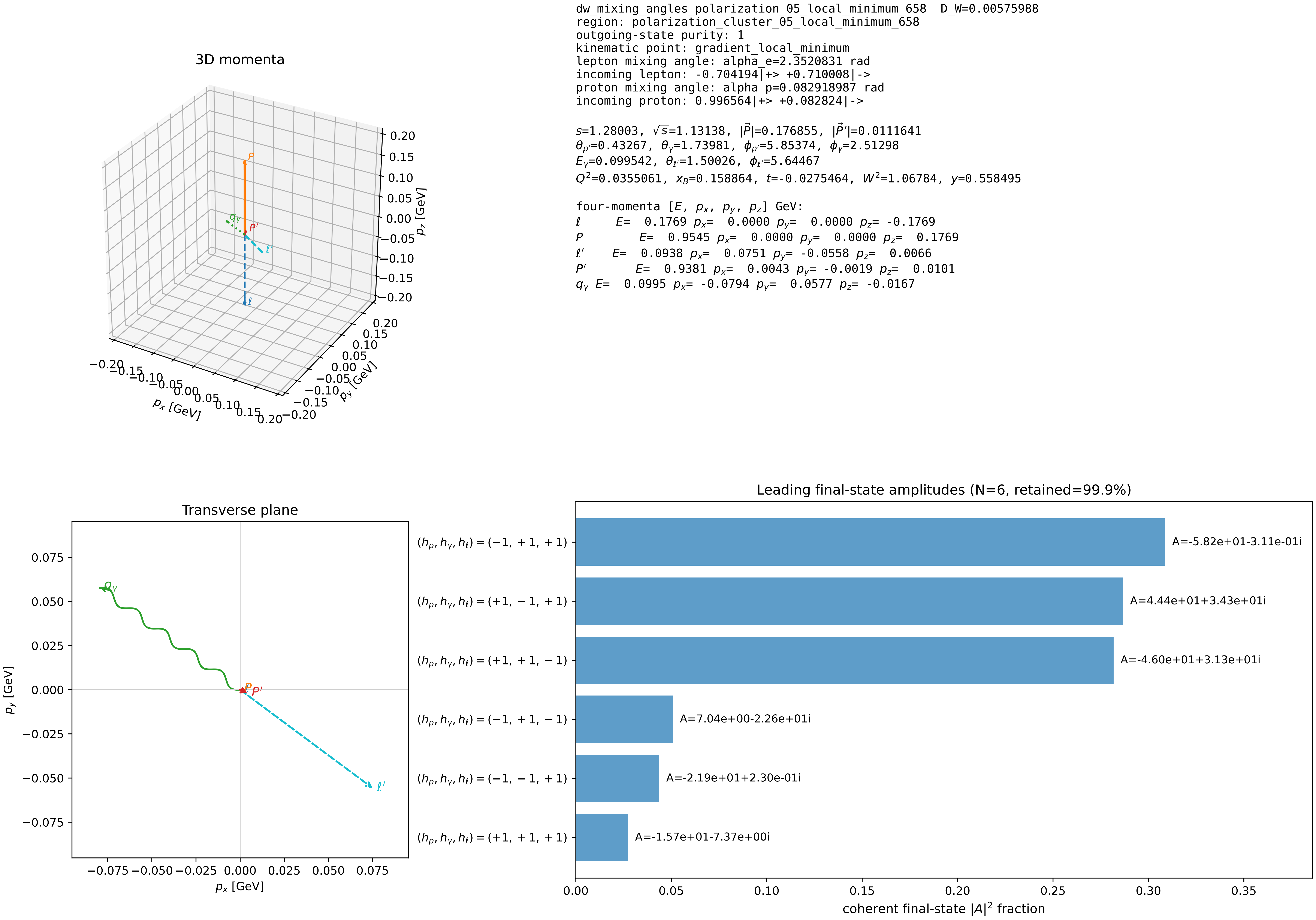

electron: polarization cluster P5, local minimum 621; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0046283255
D_W contour = 0.054628326
above global minimum = 0.0046118927
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.1326
theta_p_out = 0.3759292
theta_gamma_out = 1.730033
qOut = 0.09931651
phi_p_out = 0.7328479
phi_gamma_out = 3.374689
alpha_e = 2.348971
alpha_p = 0.1585022



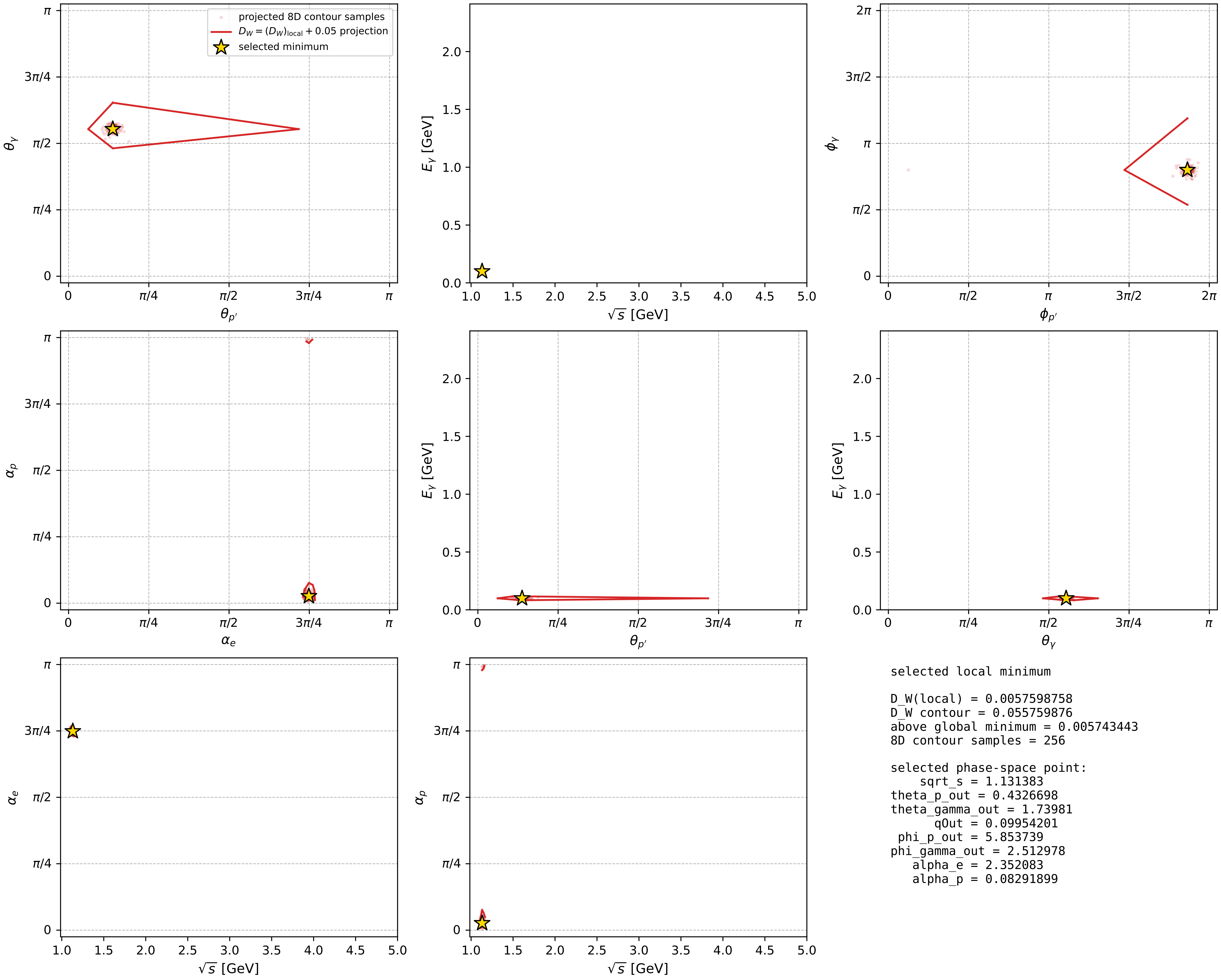
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
P
ℓ'
P'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
dw_mixing_angles_polarization_05_local_minimum_642 D_W=0.0053488
region: polarization_cluster_05_local_minimum_642
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3745705 rad
incoming lepton: -0.71998|+> +0.693994|->
proton mixing angle: alpha_p=0.17604399 rad
incoming proton: 0.984544|+> +0.175136|->
s=1.16887, $\sqrt{s}$=1.08114, $|\vec{P}|$=0.133667, $|\vec{P}'|$=0.0034774
$\theta_{p'}$=1.8111, θ_γ=2.19739, $\phi_{p'}$=1.70953, ϕ_γ=3.89833
E_γ=0.0720794, $\theta_{\ell'}$=0.919174, $\phi_{\ell'}$=0.707999
Q^2=0.0305162, x_B=0.183947, t=-0.0180105, W^2=1.01522, y=0.573982
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1337 p_x= 0.0000 p_y= 0.0000 p_z= -0.1337
P E= 0.9475 p_x= 0.0000 p_y= 0.0000 p_z= 0.1337
ℓ' E= 0.0711 p_x= 0.0429 p_y= 0.0367 p_z= 0.0431
P' E= 0.9380 p_x= -0.0005 p_y= 0.0033 p_z= -0.0008
q_γ E= 0.0721 p_x= -0.0425 p_y= -0.0401 p_z= -0.0423
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=99.0%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-7.11e+01-8.61e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=1.41e+01+9.84e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-7.85e+01+6.07e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-5.50e+01-3.73e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-6.57e+01-5.95e-01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=4.75e+01-2.70e+01i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P5, local minimum 642; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.005348798
D_W contour = 0.055348798
above global minimum = 0.0053323652
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.081144
theta_p_out = 1.811095
theta_gamma_out = 2.197386
qOut = 0.07207937
phi_p_out = 1.709534
phi_gamma_out = 3.898335
alpha_e = 2.374571
alpha_p = 0.176044

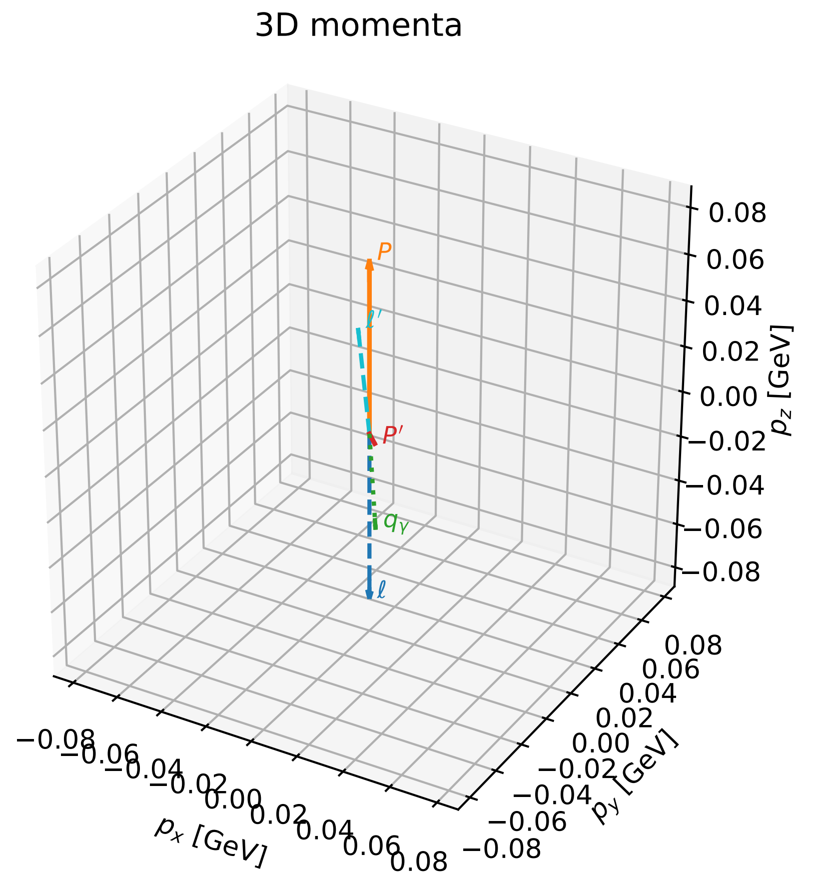
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)



electron: polarization cluster P5, local minimum 658; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



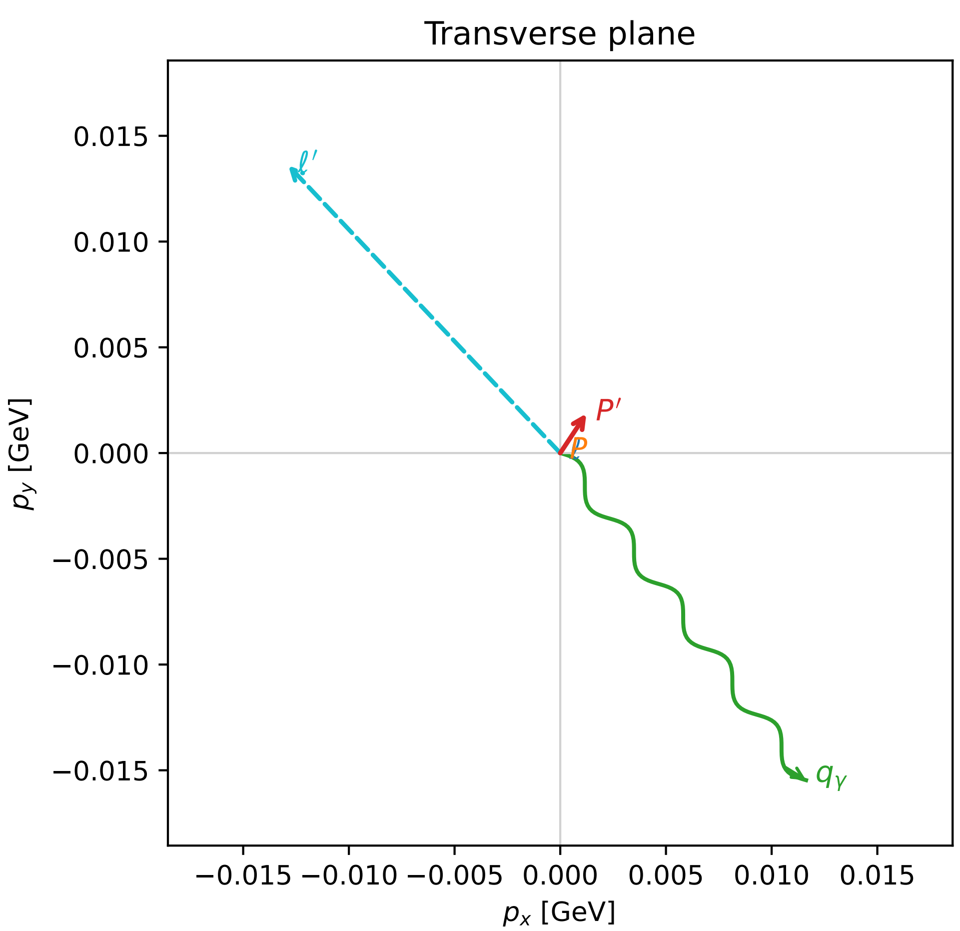
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

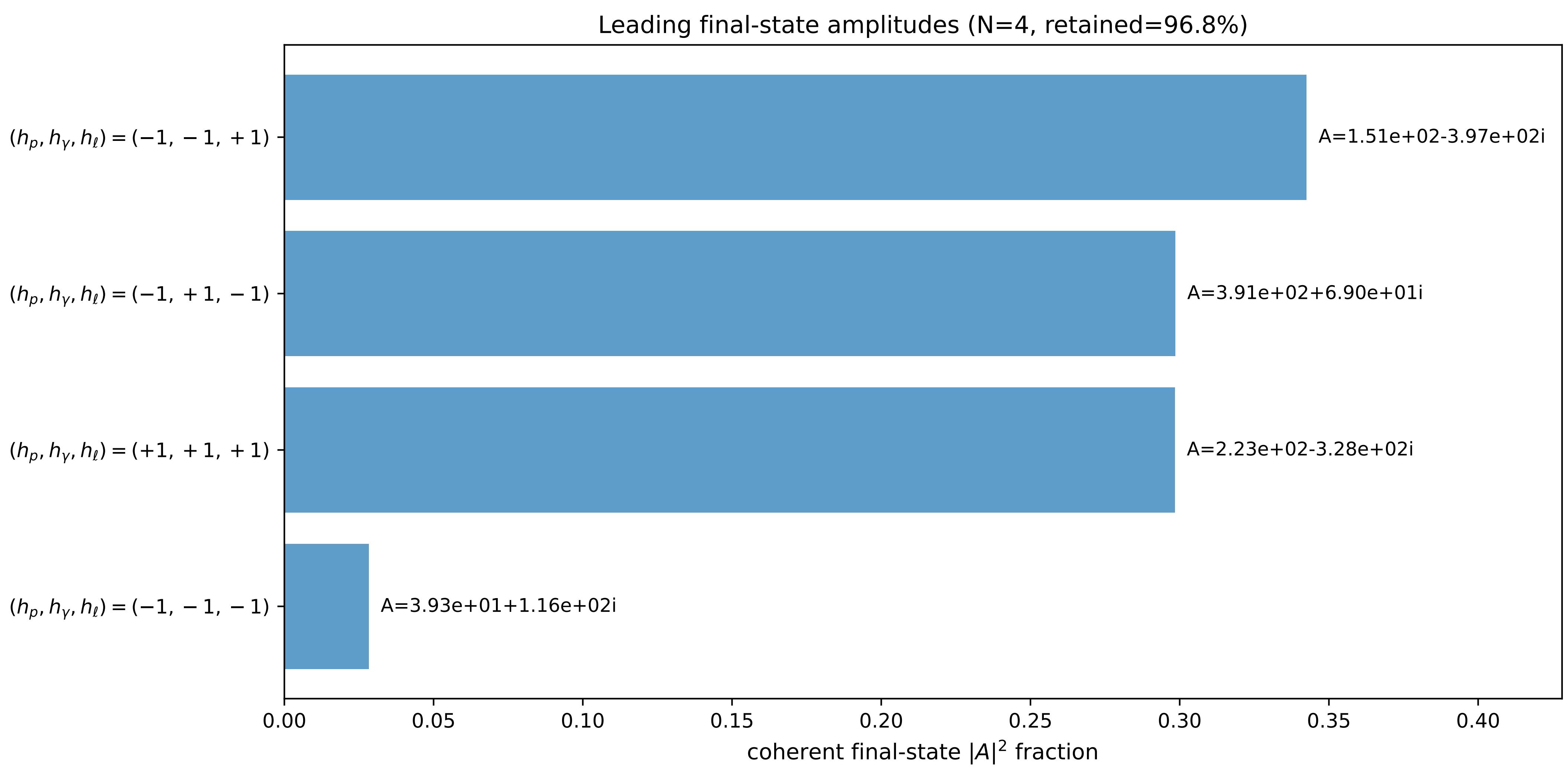


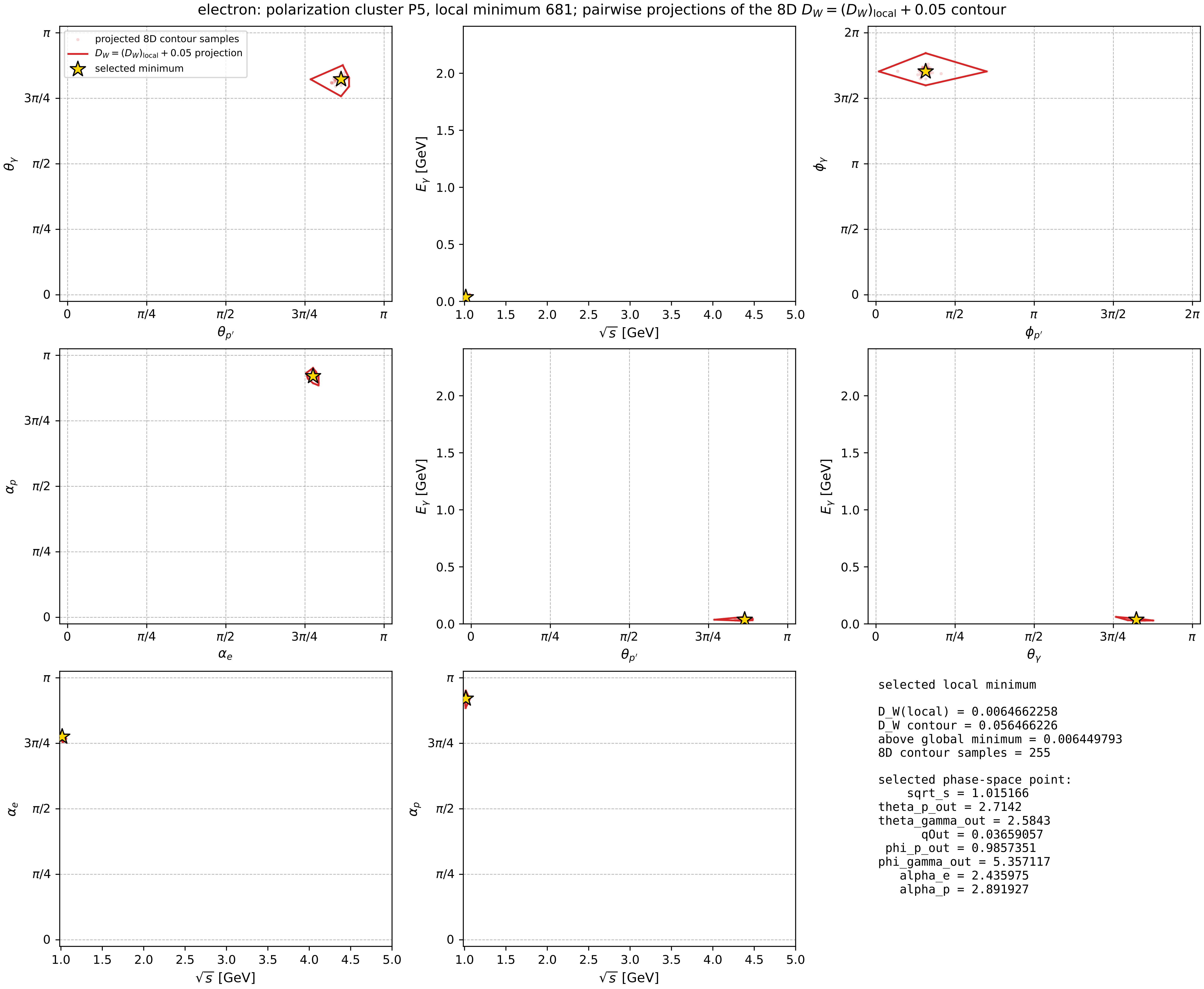
dw_mixing_angles_polarization_05_local_minimum_681 D_W=0.00646623
region: polarization_cluster_05_local_minimum_681
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.435975 rad
incoming lepton: -0.761211|+> +0.648504|->
proton mixing angle: alpha_p=2.8919268 rad
incoming proton: -0.968995|+> +0.24708|->

s=1.03056, $\sqrt{s}$=1.01517, $|\vec{P}|$=0.0742313, $|\vec{P}'|$=0.00540877
$\theta_{p'}$=2.7142, θ_γ=2.5843, $\phi_{p'}$=0.985735, ϕ_γ=5.35712
E_γ=0.0365906, $\theta_{\ell'}$=0.479838, $\phi_{\ell'}$=2.32862
Q^2=0.0113623, x_B=0.142509, t=-0.0062618, W^2=0.948212, y=0.529009

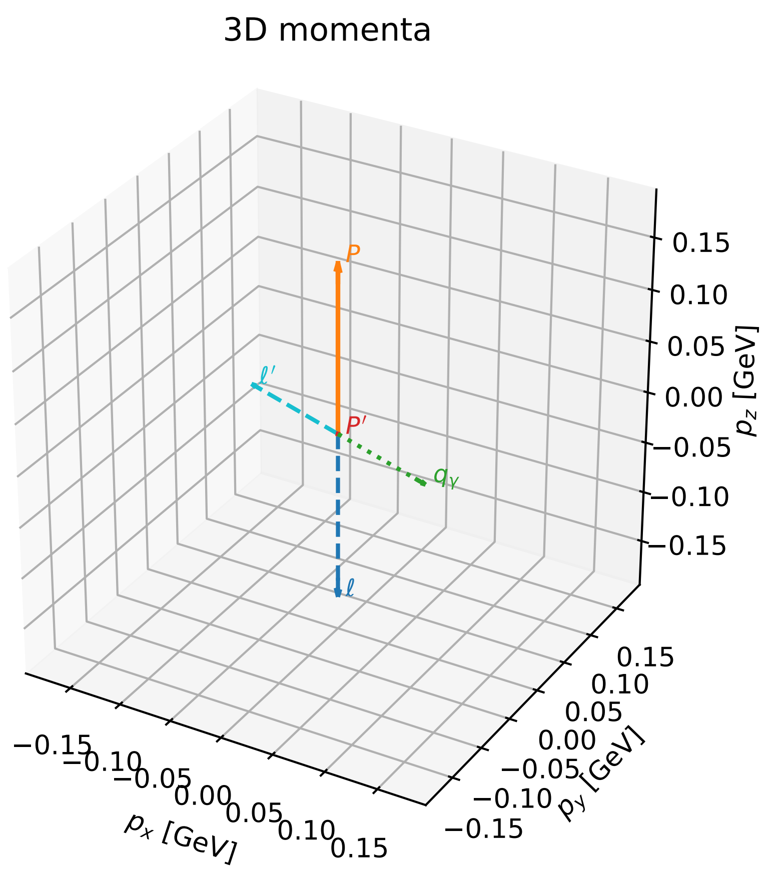
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0742 p_x= 0.0000 p_y= 0.0000 p_z= -0.0742
P E= 0.9409 p_x= 0.0000 p_y= 0.0000 p_z= 0.0742
ℓ' E= 0.0406 p_x= -0.0129 p_y= 0.0136 p_z= 0.0360
P' E= 0.9380 p_x= 0.0012 p_y= 0.0019 p_z= -0.0049
q_γ E= 0.0366 p_x= 0.0116 p_y= -0.0155 p_z= -0.0311






electron: polarization cluster P5, local minimum 681; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0064662258
D_W contour = 0.056466226
above global minimum = 0.006449793
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.015166
theta_p_out = 2.7142
theta_gamma_out = 2.5843
qOut = 0.03659057
phi_p_out = 0.9857351
phi_gamma_out = 5.357117
alpha_e = 2.435975
alpha_p = 2.891927

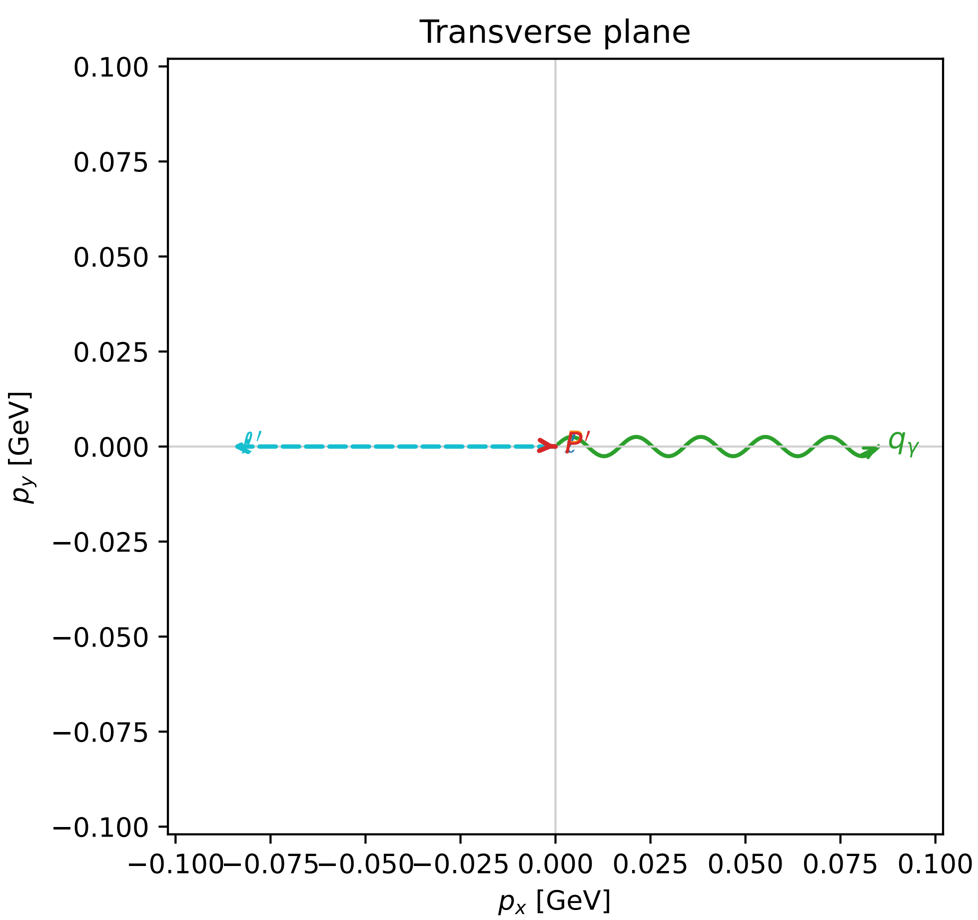
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

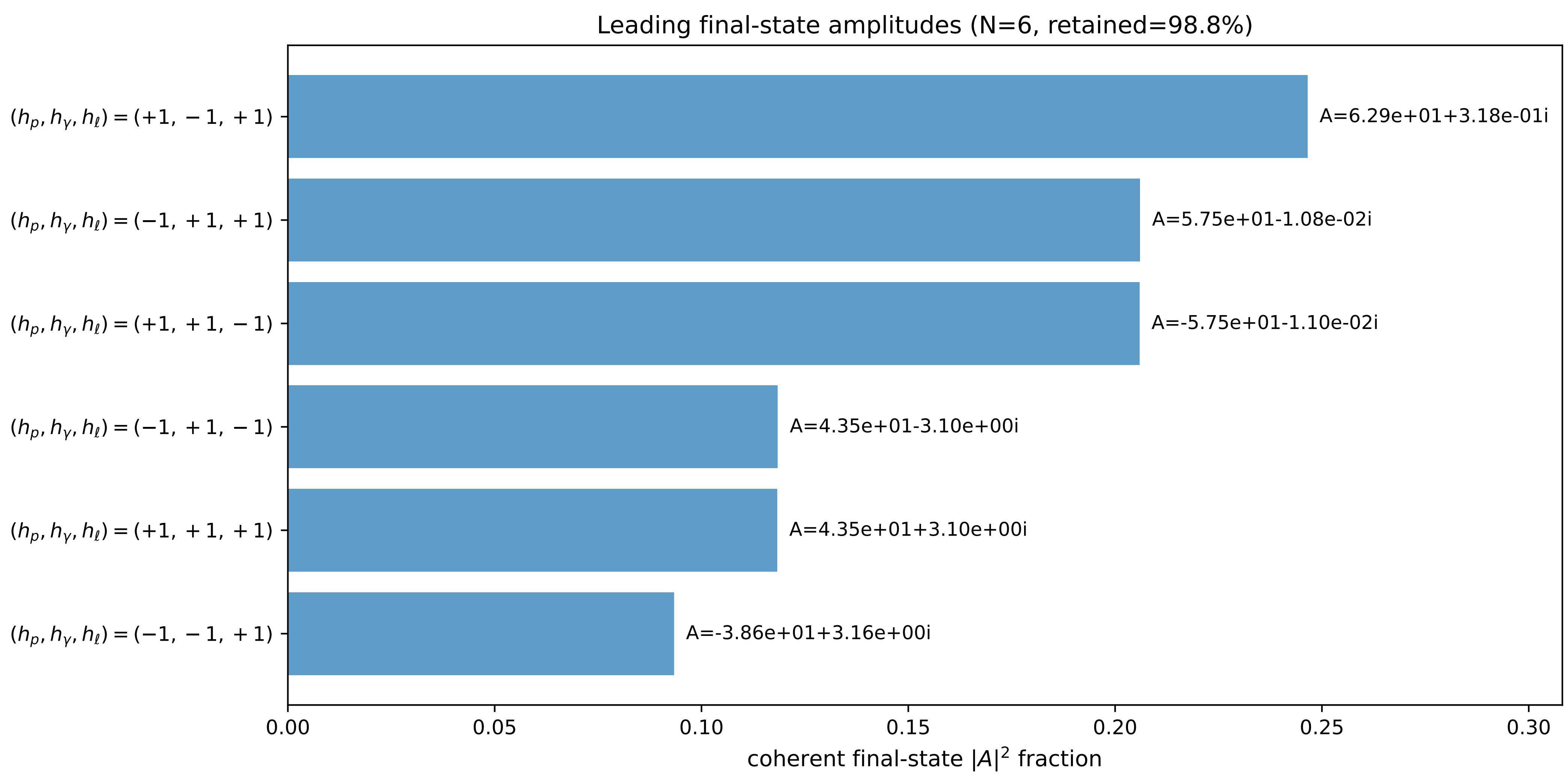


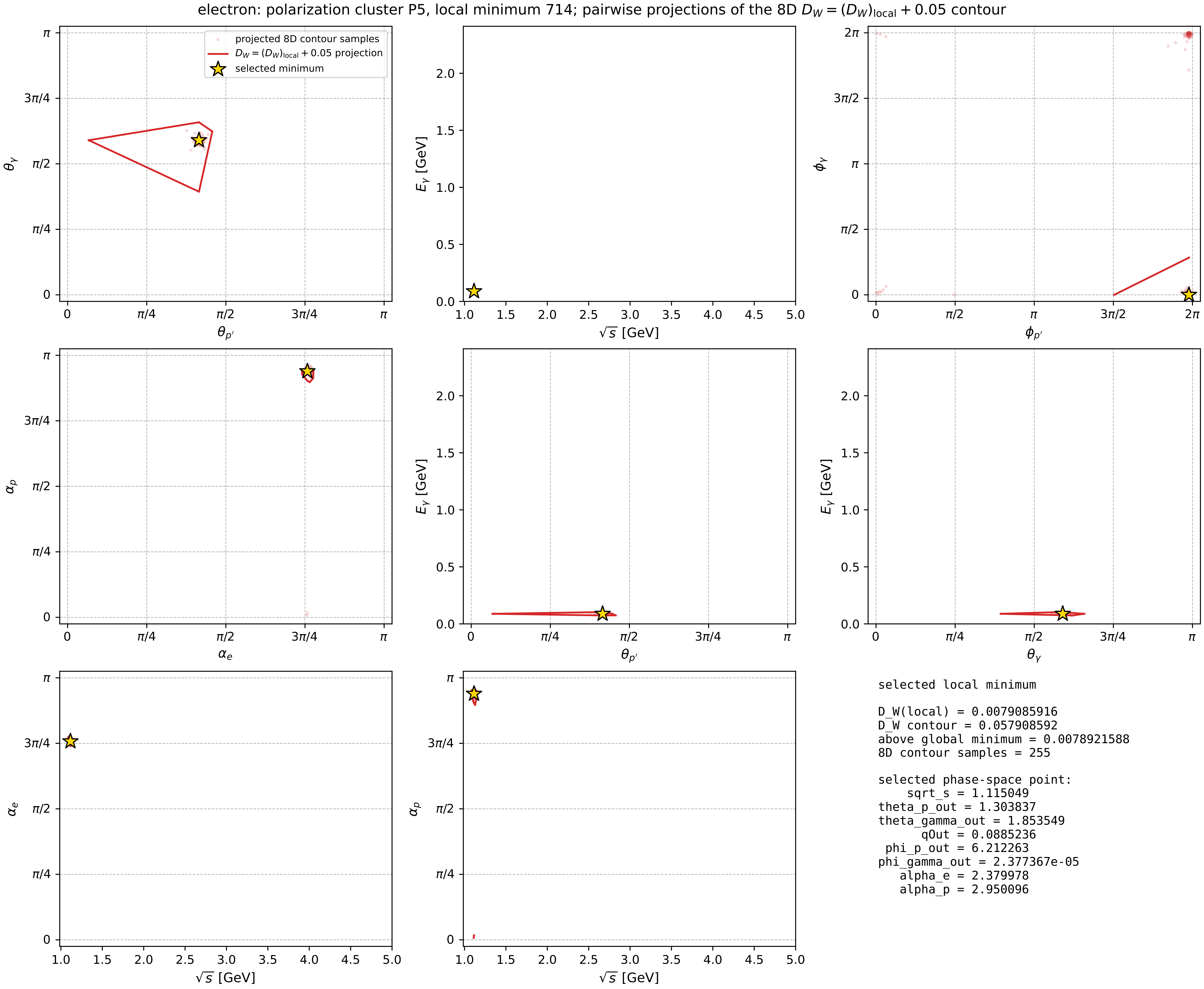
dw_mixing_angles_polarization_05_local_minimum_714 D_W=0.00790859
region: polarization_cluster_05_local_minimum_714
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3799775 rad
incoming lepton: -0.723722|+> +0.690091|->
proton mixing angle: alpha_p=2.9500959 rad
incoming proton: -0.98172|+> +0.190328|->

s=1.24333, $\sqrt{s}$=1.11505, $|\vec{P}|$=0.162992, $|\vec{P}'|$=5.8715e-07
$\theta_{p'}$=1.30384, θ_γ=1.85355, $\phi_{p'}$=6.21226, ϕ_γ=2.37737e-05
E_γ=0.0885236, $\theta_{\ell'}$=1.28805, $\phi_{\ell'}$=3.14162
Q^2=0.0369088, x_B=0.181835, t=-0.0263689, W^2=1.04591, y=0.558416

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1630 p_x= 0.0000 p_y= 0.0000 p_z= -0.1630
P E= 0.9521 p_x= 0.0000 p_y= 0.0000 p_z= 0.1630
ℓ' E= 0.0885 p_x= -0.0850 p_y= -0.0000 p_z= 0.0247
P' E= 0.9380 p_x= 0.0000 p_y= -0.0000 p_z= 0.0000
q_γ E= 0.0885 p_x= 0.0850 p_y= 0.0000 p_z= -0.0247







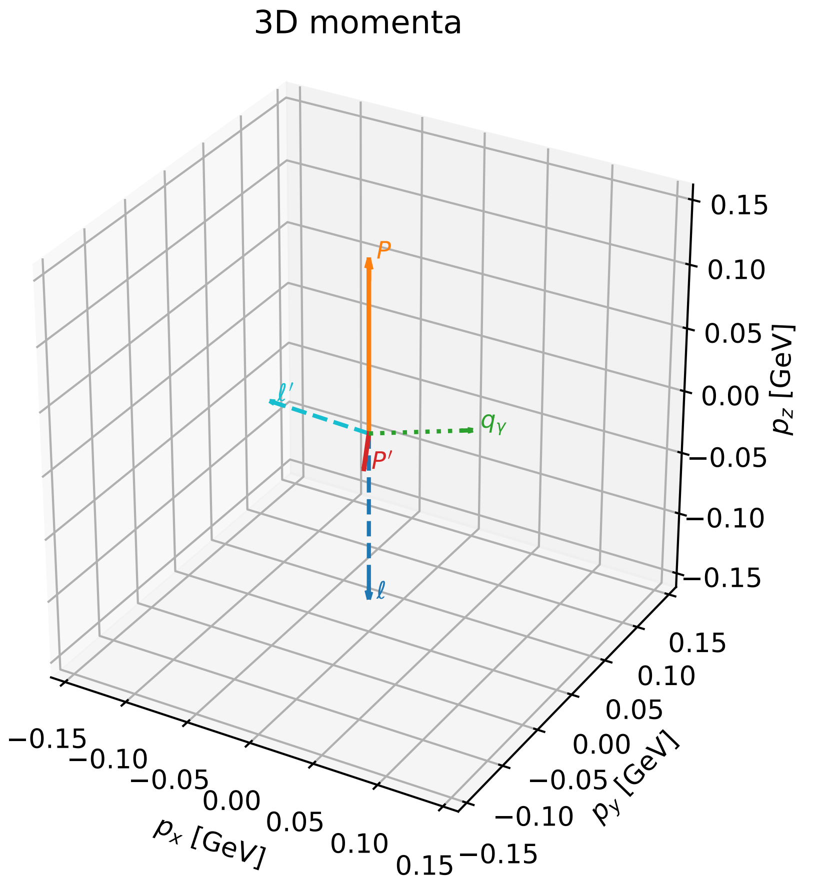
electron: polarization cluster P5, local minimum 714; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.0079085916
D_W contour = 0.057908592
above global minimum = 0.0078921588
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.115049
theta_p_out = 1.303837
theta_gamma_out = 1.853549
qOut = 0.0885236
phi_p_out = 6.212263
phi_gamma_out = 2.377367e-05
alpha_e = 2.379978
alpha_p = 2.950096

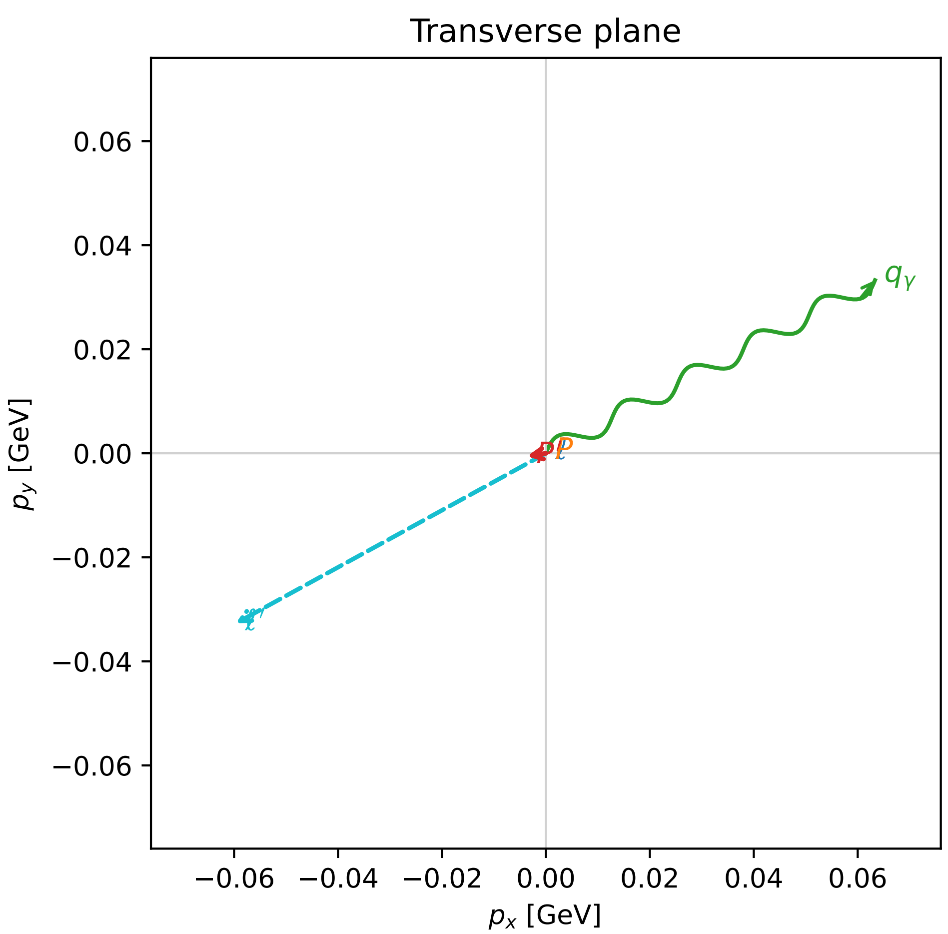
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

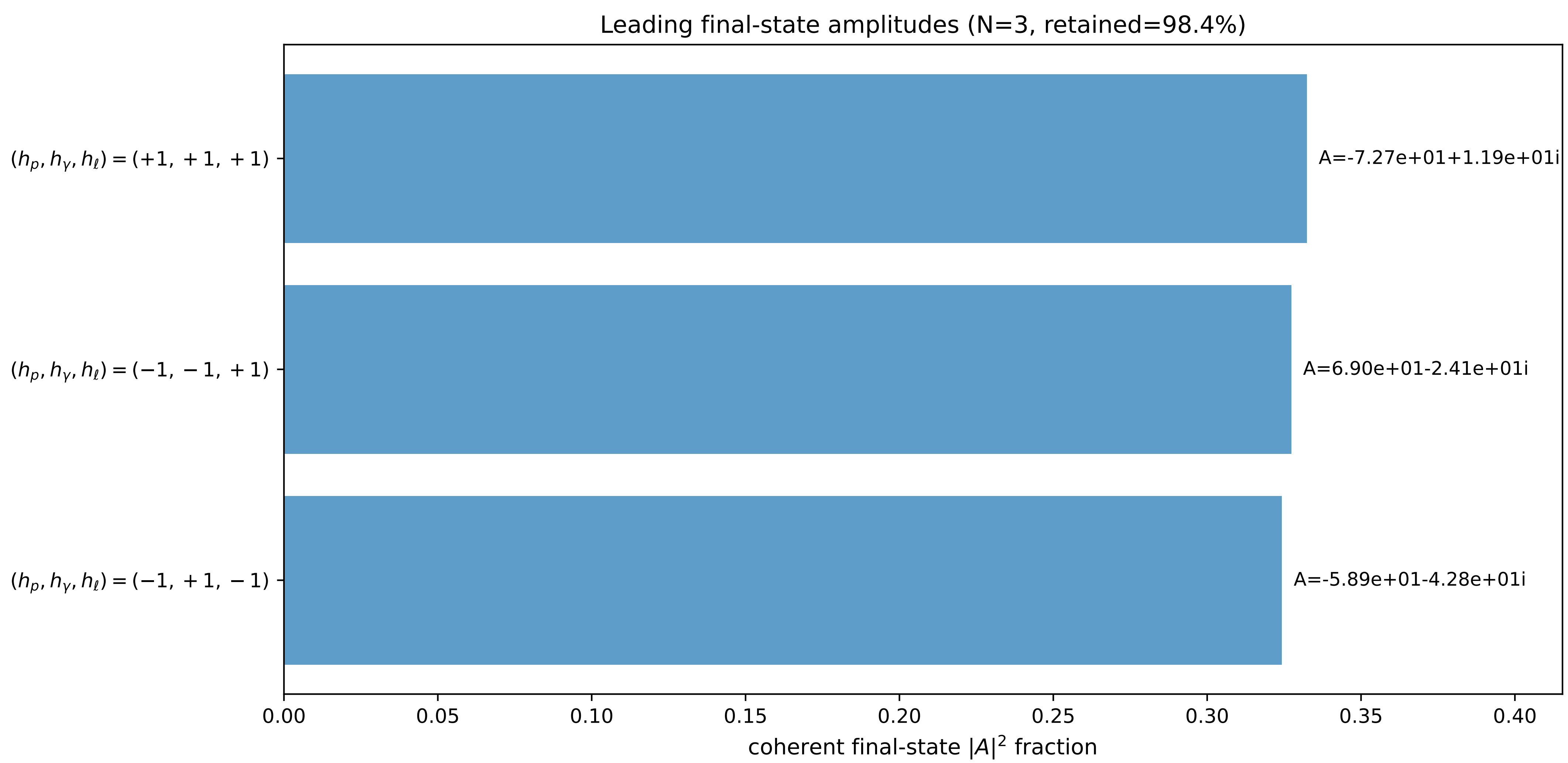
dw_mixing_angles_polarization_05_local_minimum_728 D_W=0.00832553
region: polarization_cluster_05_local_minimum_728
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3656503 rad
incoming lepton: -0.713761|+> +0.700389|->
proton mixing angle: alpha_p=3.0654637 rad
incoming proton: -0.997104|+> +0.0760554|->

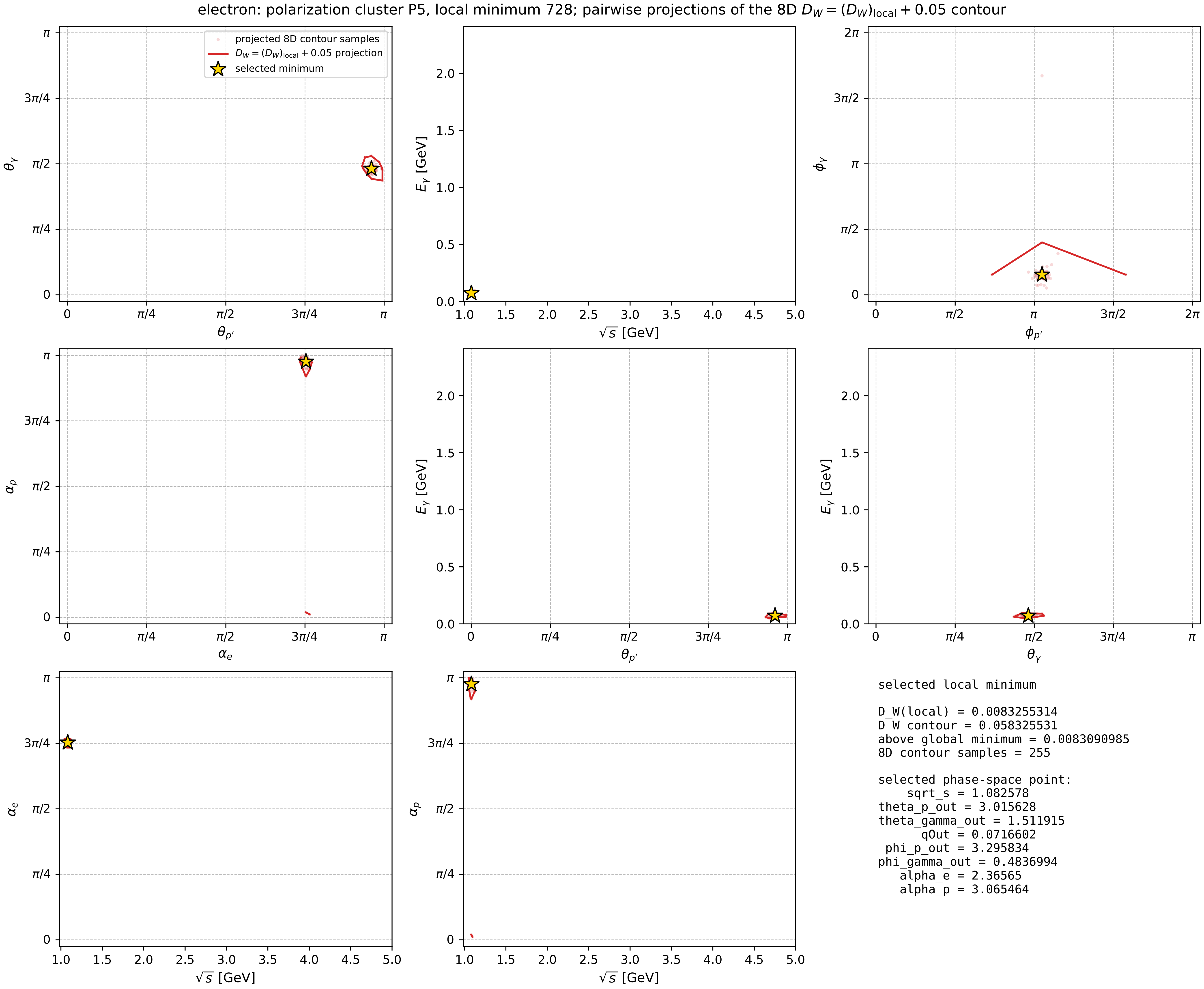
s=1.17197, $\sqrt{s}$=1.08258, $|\vec{P}|$=0.134923, $|\vec{P}'|$=0.0293
$\theta_{p'}$=3.01563, θ_γ=1.51191, $\phi_{p'}$=3.29583, ϕ_γ=0.483699
E_γ=0.0716602, $\theta_{\ell'}$=1.22072, $\phi_{\ell'}$=3.64279
Q^2=0.0262585, x_B=0.162589, t=-0.0268219, W^2=1.01509, y=0.552843

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1349 p_x= 0.0000 p_y= 0.0000 p_z= -0.1349
P E= 0.9477 p_x= 0.0000 p_y= 0.0000 p_z= 0.1349
ℓ' E= 0.0725 p_x= -0.0597 p_y= -0.0327 p_z= 0.0249
P' E= 0.9385 p_x= -0.0036 p_y= -0.0006 p_z= -0.0291
q_γ E= 0.0717 p_x= 0.0633 p_y= 0.0333 p_z= 0.0042

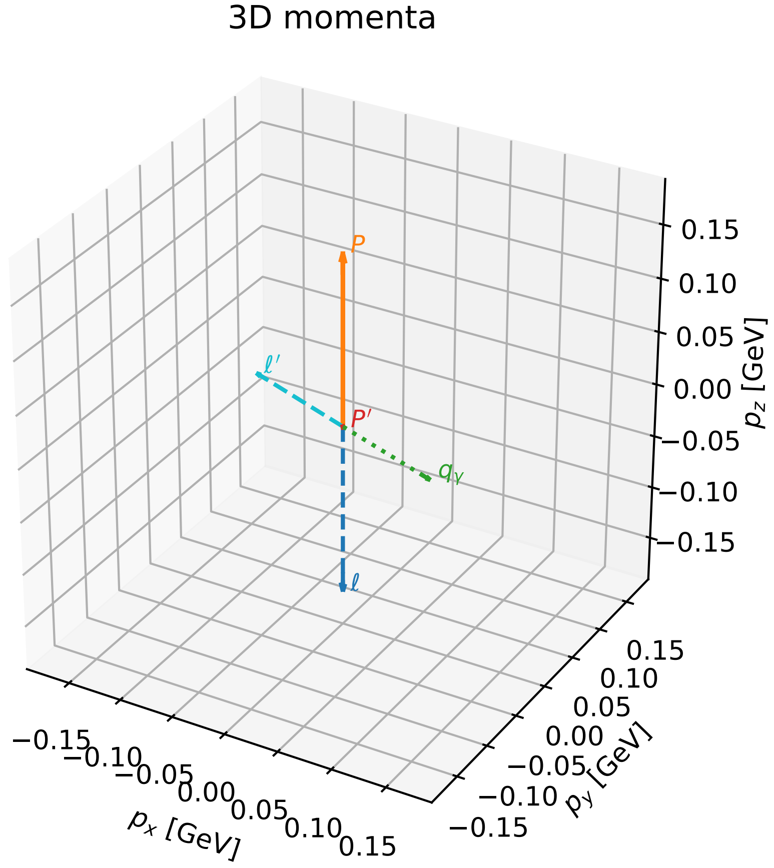







electron: polarization cluster P5, local minimum 728; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.0083255314
D_W contour = 0.058325531
above global minimum = 0.0083090985
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.082578
theta_p_out = 3.015628
theta_gamma_out = 1.511915
qOut = 0.0716602
phi_p_out = 3.295834
phi_gamma_out = 0.4836994
alpha_e = 2.36565
alpha_p = 3.065464

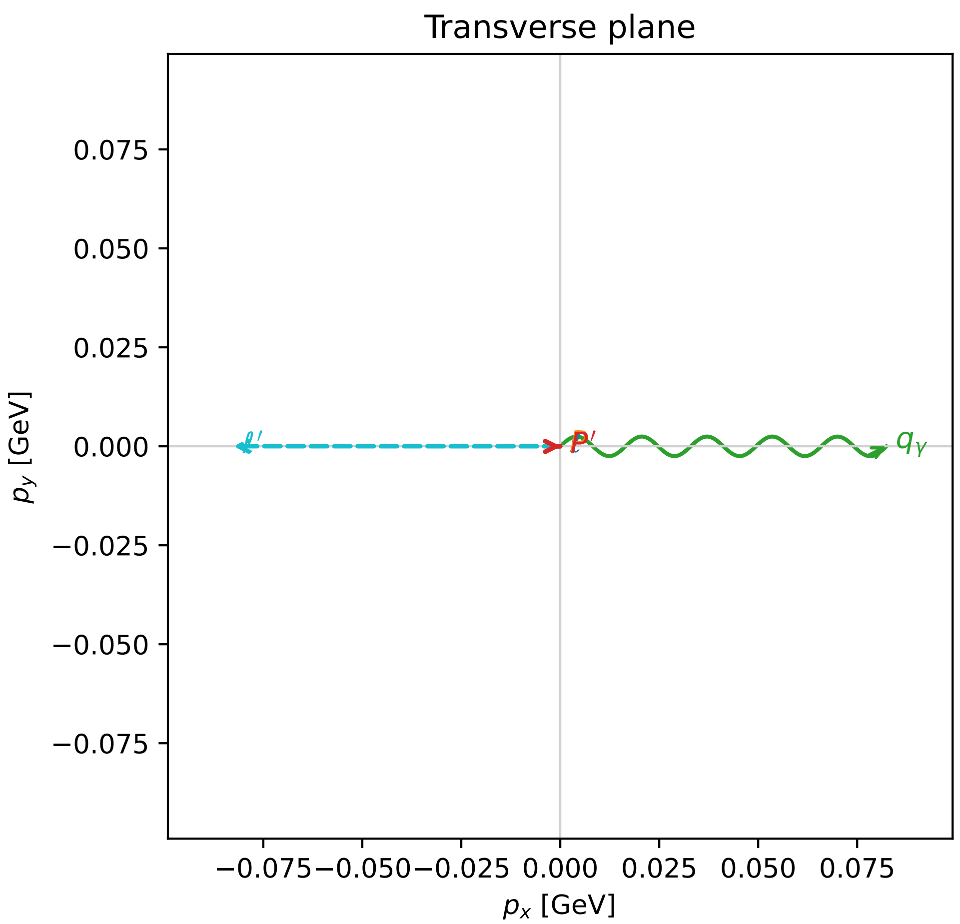
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

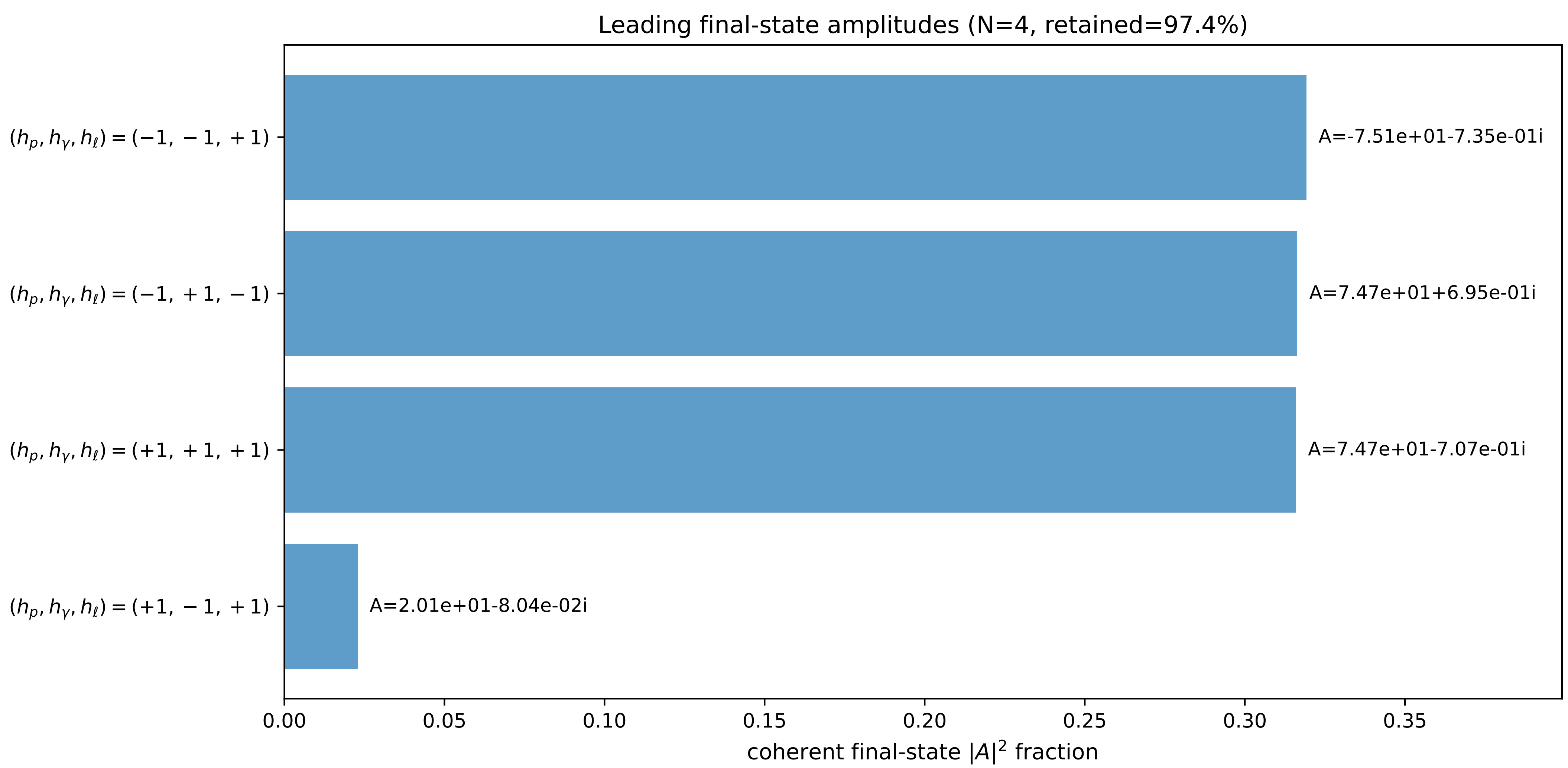


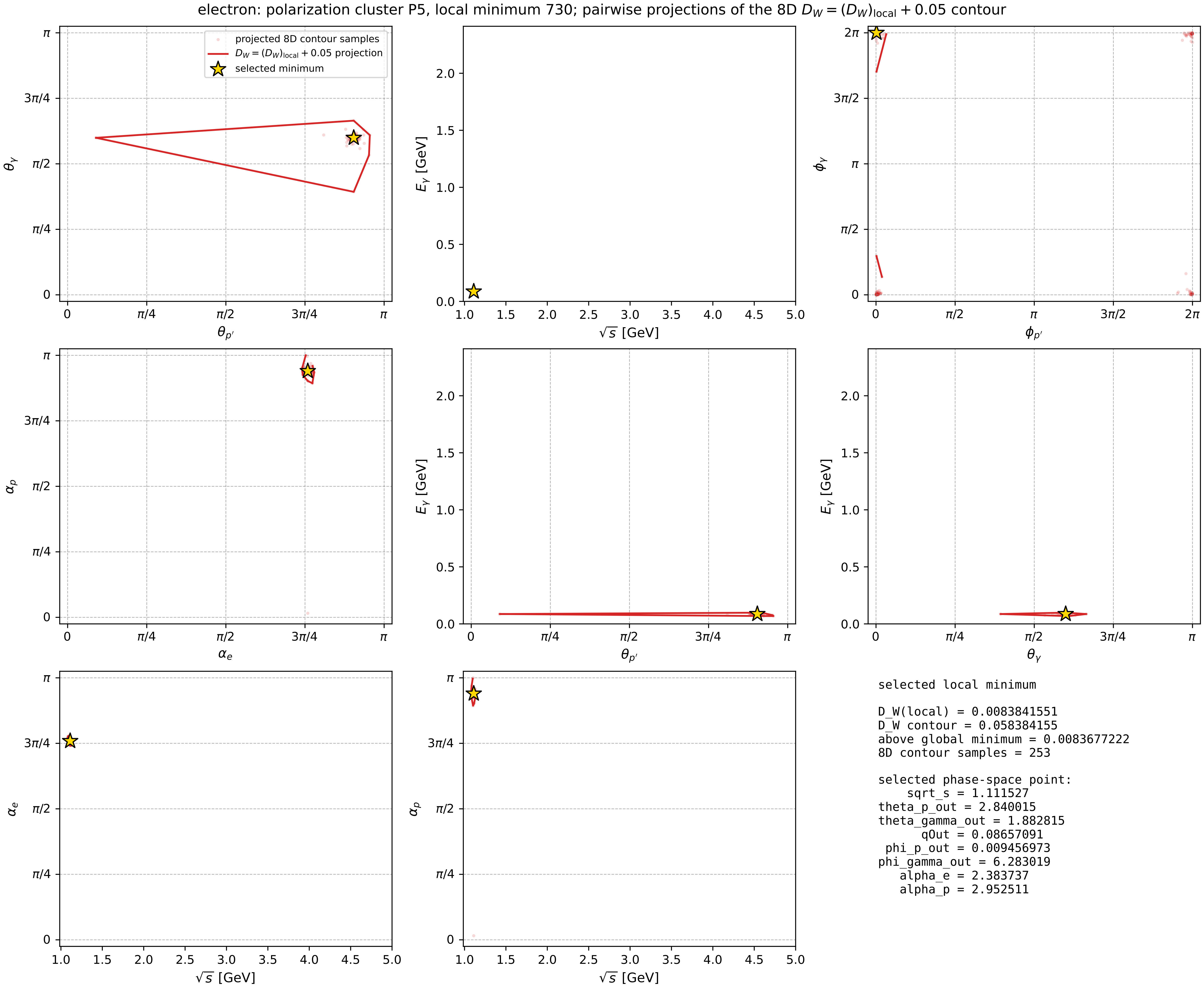
dw_mixing_angles_polarization_05_local_minimum_730 D_W=0.00838416
region: polarization_cluster_05_local_minimum_730
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3837369 rad
incoming lepton: -0.726312|+> +0.687366|->
proton mixing angle: alpha_p=2.9525109 rad
incoming proton: -0.982177|+> +0.187957|->

s=1.23549, $\sqrt{s}$=1.11153, $|\vec{P}|$=0.159981, $|\vec{P}'|$=0.00066366
$\theta_{p'}$=2.84002, θ_γ=1.88282, $\phi_{p'}$=0.00945697, ϕ_γ=6.28302
E_γ=0.0865709, $\theta_{\ell'}$=1.25254, $\phi_{\ell'}$=3.14145
Q^2=0.0365284, x_B=0.18368, t=-0.0256138, W^2=1.04218, y=0.559173

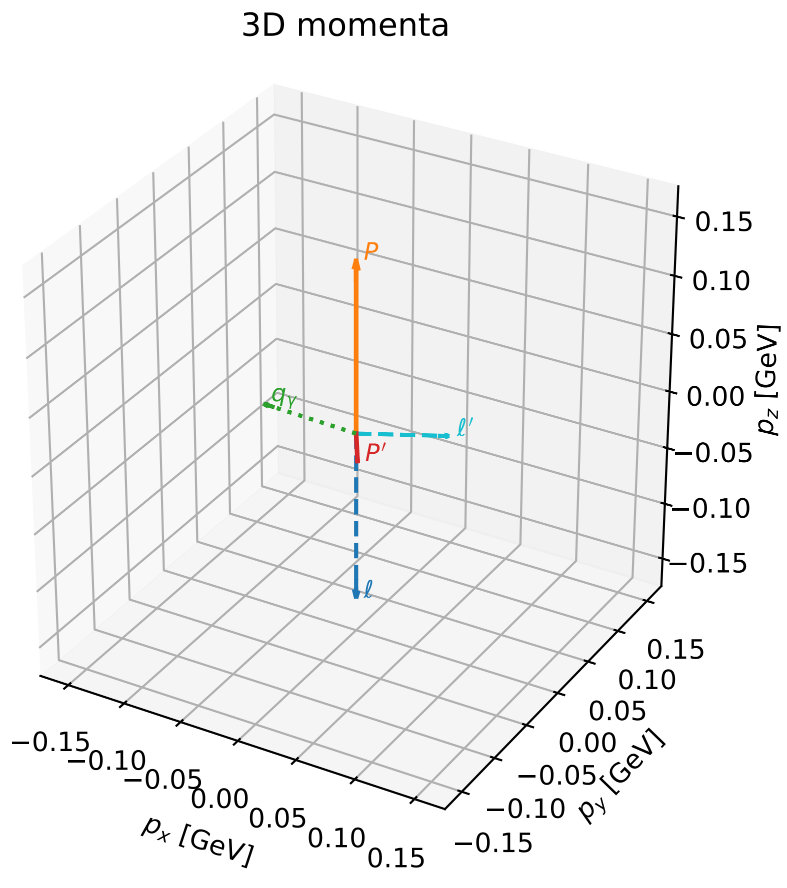
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1600 p_x= 0.0000 p_y= 0.0000 p_z= -0.1600
P E= 0.9515 p_x= 0.0000 p_y= 0.0000 p_z= 0.1600
ℓ' E= 0.0870 p_x= -0.0826 p_y= 0.0000 p_z= 0.0272
P' E= 0.9380 p_x= 0.0002 p_y= 0.0000 p_z= -0.0006
q_γ E= 0.0866 p_x= 0.0824 p_y= -0.0000 p_z= -0.0266






electron: polarization cluster P5, local minimum 730; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.0083841551
D_W contour = 0.058384155
above global minimum = 0.0083677222
8D contour samples = 253
selected phase-space point:
sqrt_s = 1.111527
theta_p_out = 2.840015
theta_gamma_out = 1.882815
qOut = 0.08657091
phi_p_out = 0.009456973
phi_gamma_out = 6.283019
alpha_e = 2.383737
alpha_p = 2.952511

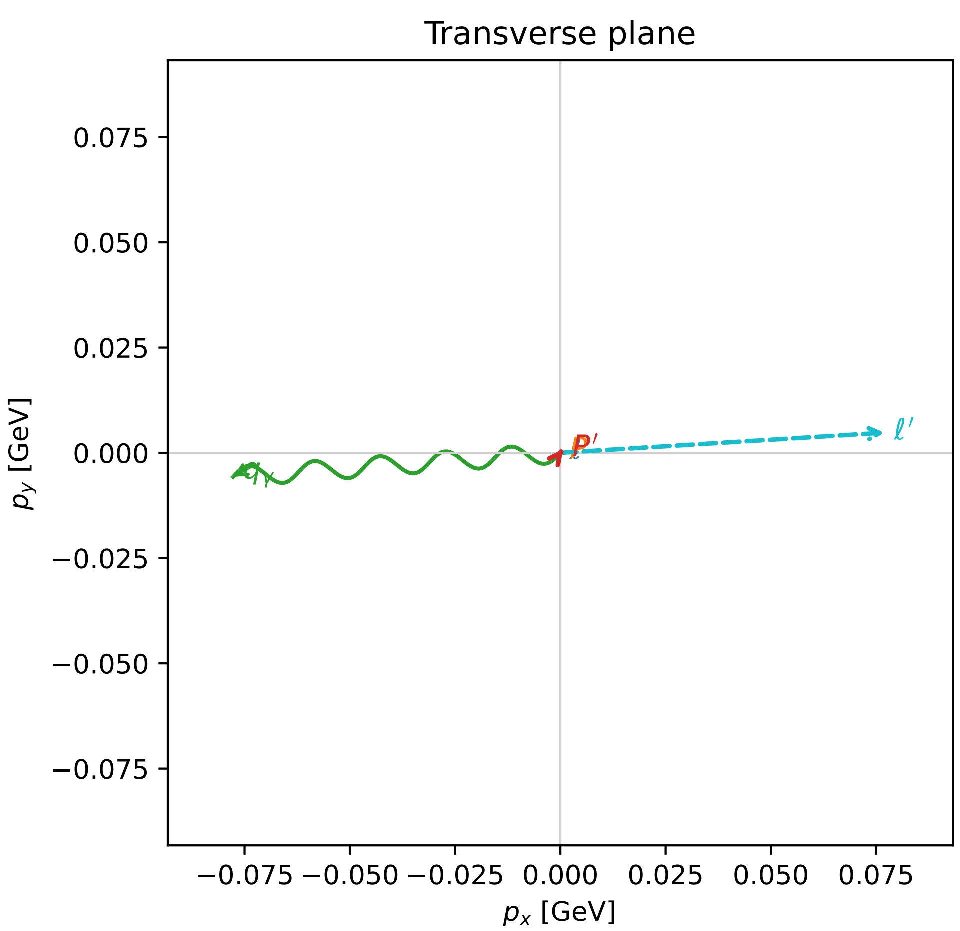
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

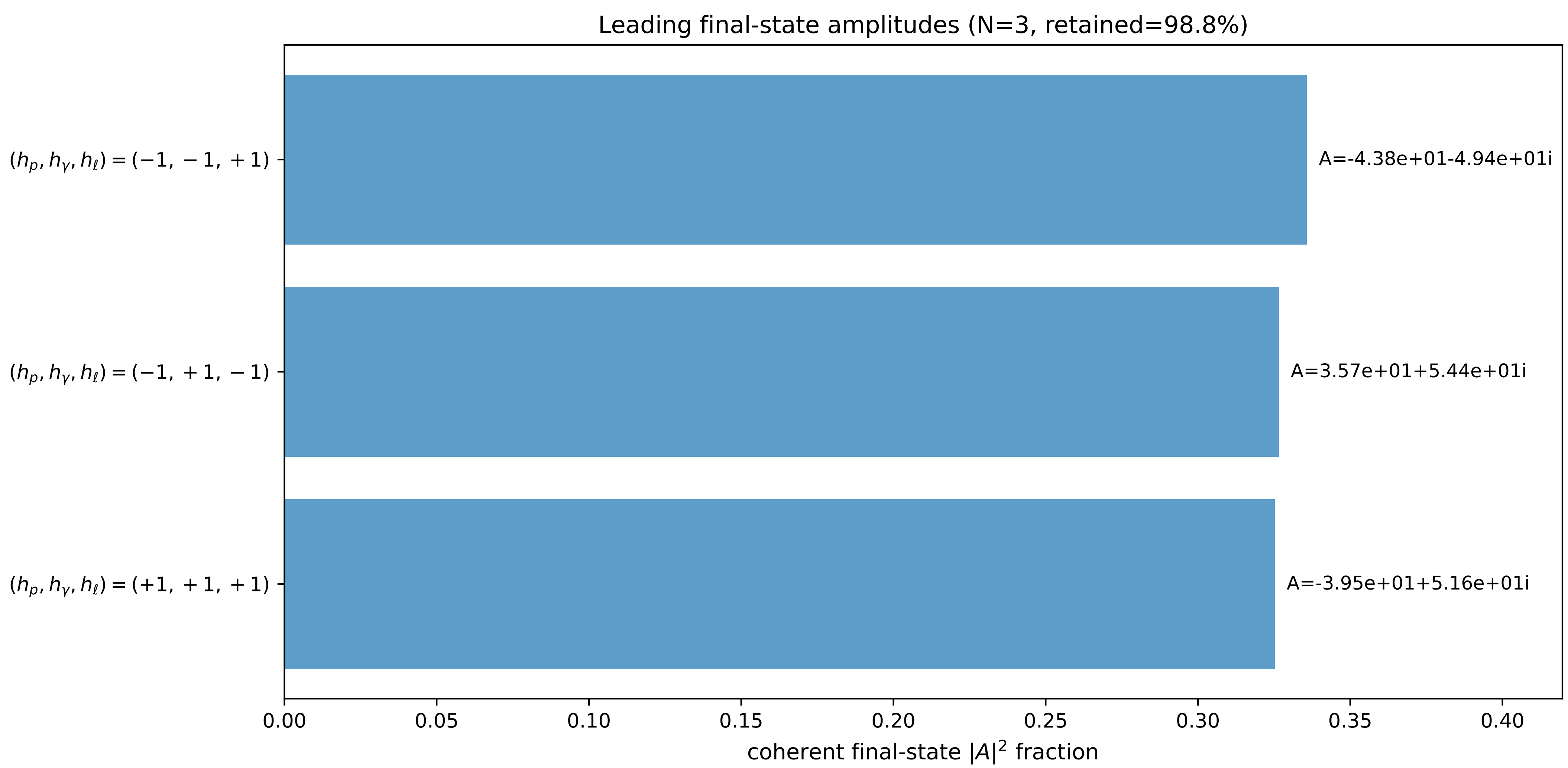


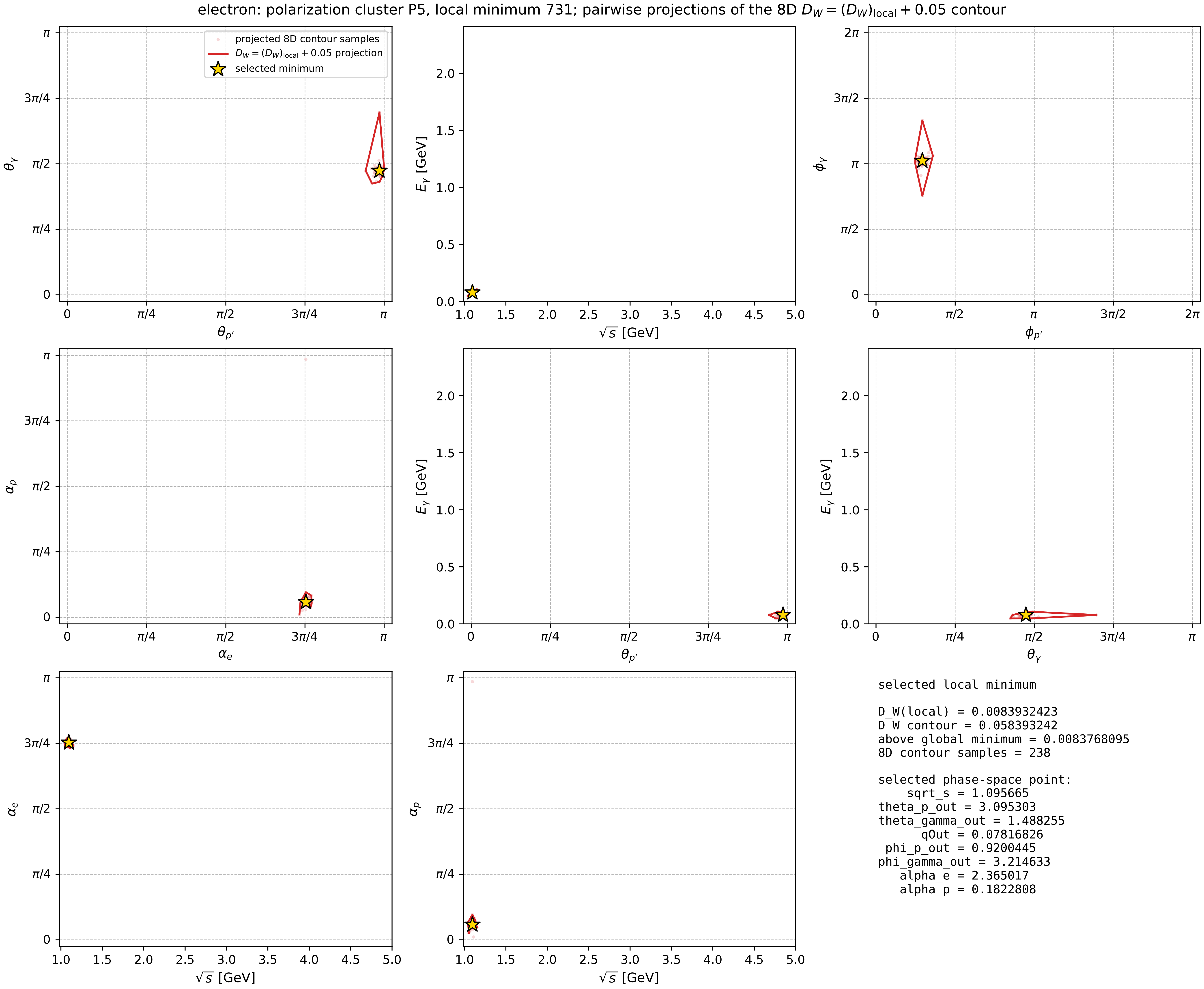
dw_mixing_angles_polarization_05_local_minimum_731 D_W=0.00839324
region: polarization_cluster_05_local_minimum_731
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3650174 rad
incoming lepton: -0.713318|+> +0.700841|->
proton mixing angle: alpha_p=0.18228078 rad
incoming proton: 0.983433|+> +0.181273|->

s=1.20048, $\sqrt{s}$=1.09567, $|\vec{P}|$=0.146321, $|\vec{P}'|$=0.024235
$\theta_{p'}$=3.0953, θ_γ=1.48825, $\phi_{p'}$=0.920045, ϕ_γ=3.21463
E_γ=0.0781683, $\theta_{\ell'}$=1.34453, $\phi_{\ell'}$=0.0621503
Q^2=0.0283707, x_B=0.161665, t=-0.0289599, W^2=1.02696, y=0.547318

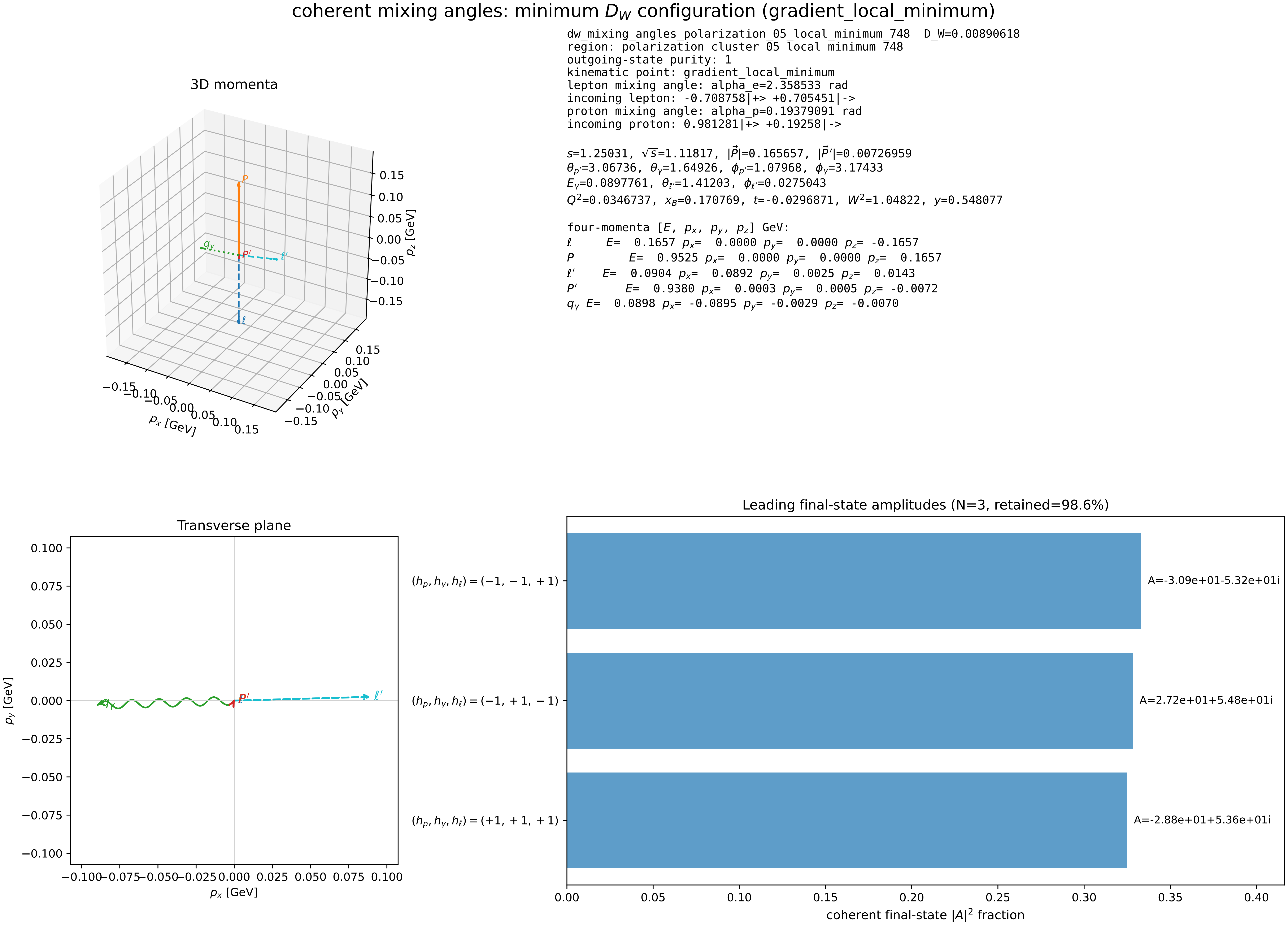
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1463 p_x= 0.0000 p_y= 0.0000 p_z= -0.1463
P E= 0.9493 p_x= 0.0000 p_y= 0.0000 p_z= 0.1463
ℓ' E= 0.0792 p_x= 0.0770 p_y= 0.0048 p_z= 0.0178
P' E= 0.9383 p_x= 0.0007 p_y= 0.0009 p_z= -0.0242
q_γ E= 0.0782 p_x= -0.0777 p_y= -0.0057 p_z= 0.0064

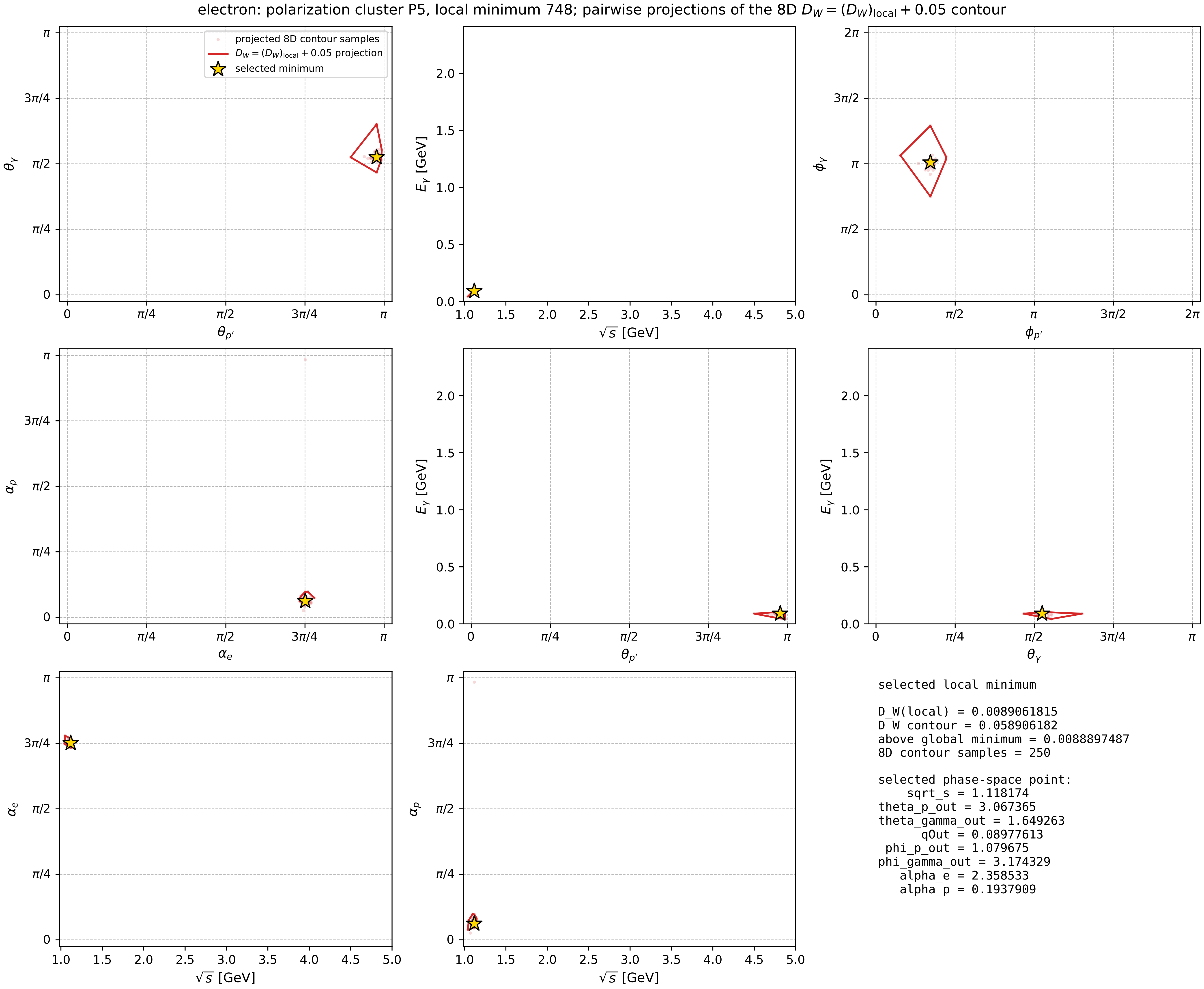




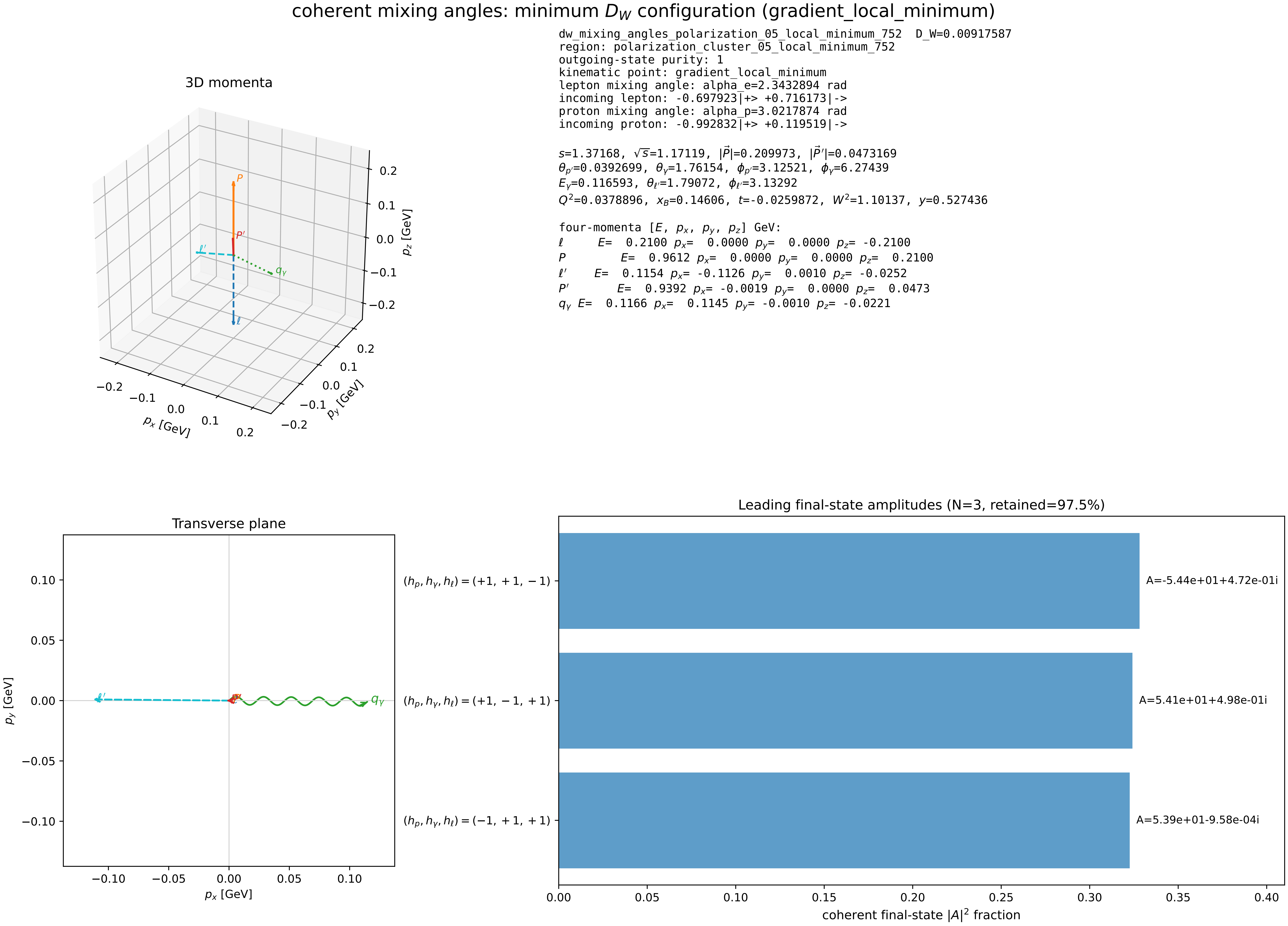


electron: polarization cluster P5, local minimum 731; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.0083932423
D_W contour = 0.058393242
above global minimum = 0.0083768095
8D contour samples = 238
selected phase-space point:
sqrt_s = 1.095665
theta_p_out = 3.095303
theta_gamma_out = 1.488255
qOut = 0.07816826
phi_p_out = 0.9200445
phi_gamma_out = 3.214633
alpha_e = 2.365017
alpha_p = 0.1822808

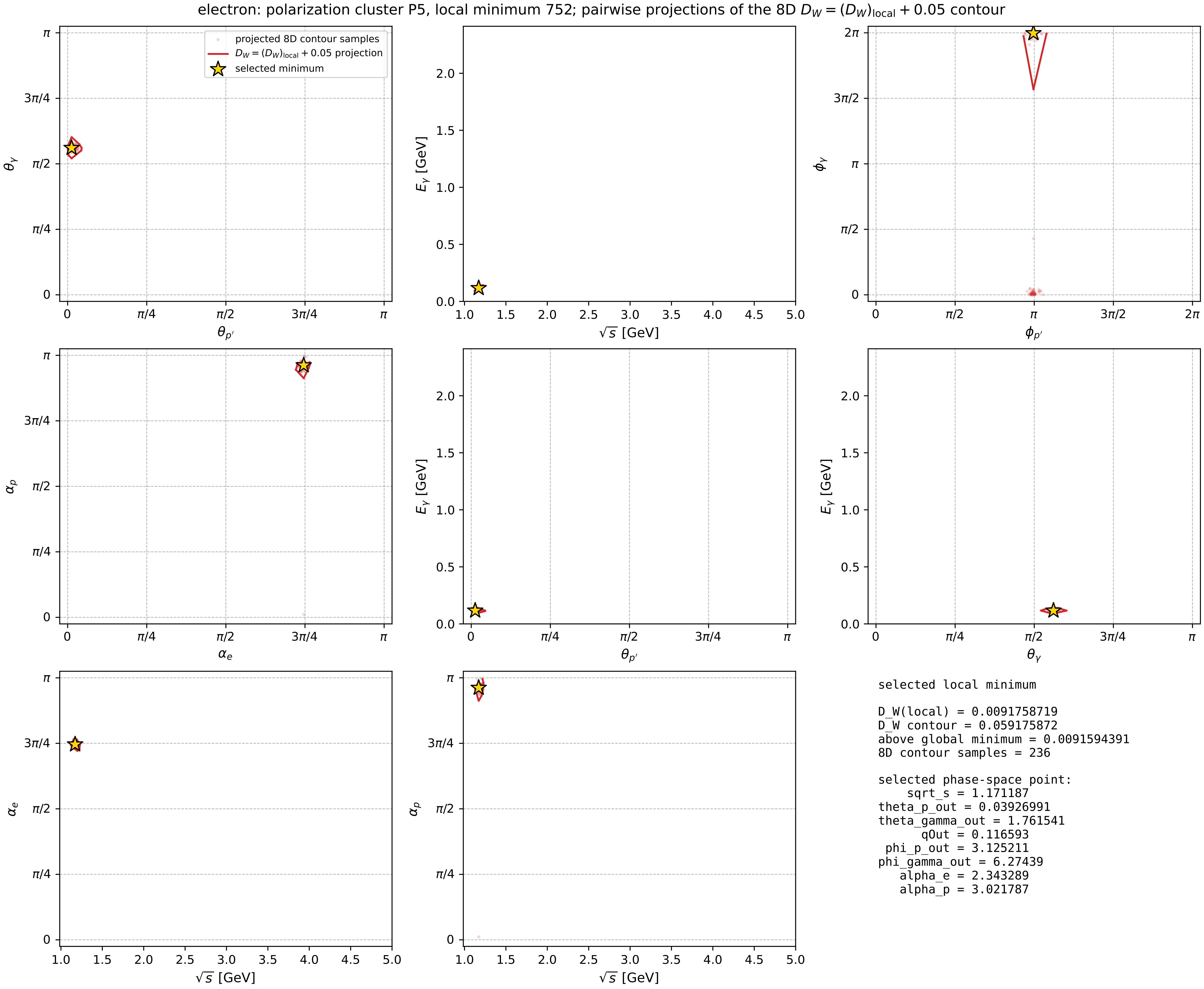

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_748 D_W=0.00890618
region: polarization_cluster_05_local_minimum_748
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.358533 rad
incoming lepton: -0.708758|+> +0.705451|->
proton mixing angle: alpha_p=0.19379091 rad
incoming proton: 0.981281|+> +0.19258|->
s=1.25031, $\sqrt{s}$=1.11817, $|\vec{P}|$=0.165657, $|\vec{P}'|$=0.00726959
$\theta_{p'}$=3.06736, θ_γ=1.64926, $\phi_{p'}$=1.07968, ϕ_γ=3.17433
E_γ=0.0897761, $\theta_{\ell'}$=1.41203, $\phi_{\ell'}$=0.0275043
Q^2=0.0346737, x_B=0.170769, t=-0.0296871, W^2=1.04822, y=0.548077
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1657 p_x= 0.0000 p_y= 0.0000 p_z= -0.1657
P E= 0.9525 p_x= 0.0000 p_y= 0.0000 p_z= 0.1657
ℓ' E= 0.0904 p_x= 0.0892 p_y= 0.0025 p_z= 0.0143
P' E= 0.9380 p_x= 0.0003 p_y= 0.0005 p_z= -0.0072
q_γ E= 0.0898 p_x= -0.0895 p_y= -0.0029 p_z= -0.0070
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=3, retained=98.6%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=-3.09e+01-5.32e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=2.72e+01+5.48e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$
A=-2.88e+01+5.36e+01i
coherent final-state $|A|^2$ fraction



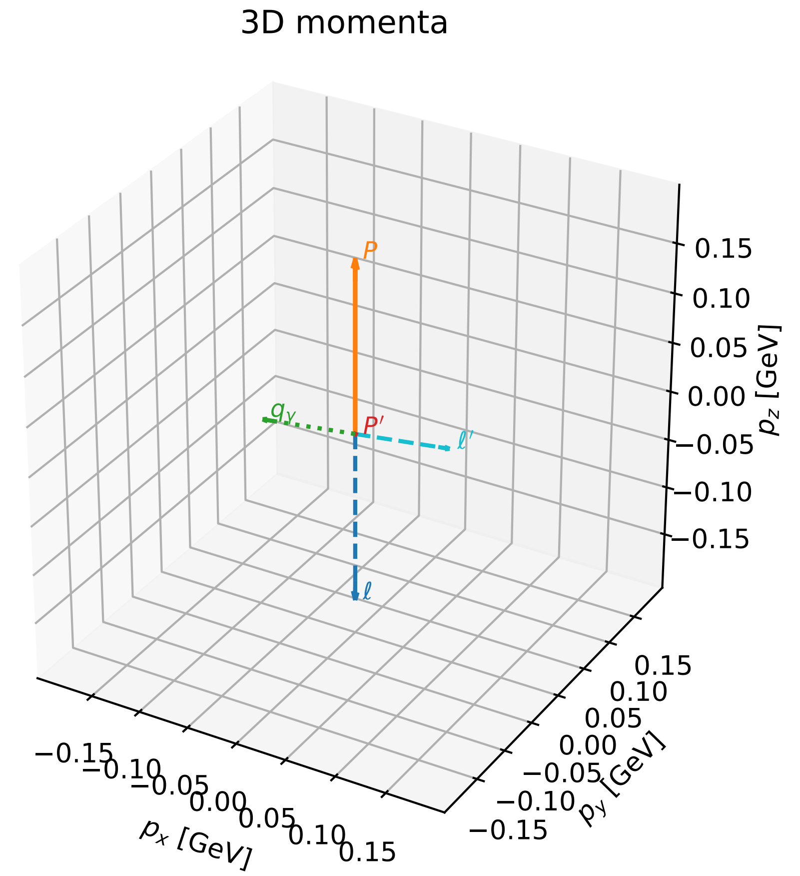
electron: polarization cluster P5, local minimum 748; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0089061815
D_W contour = 0.058906182
above global minimum = 0.0088897487
8D contour samples = 250
selected phase-space point:
sqrt_s = 1.118174
theta_p_out = 3.067365
theta_gamma_out = 1.649263
qOut = 0.08977613
phi_p_out = 1.079675
phi_gamma_out = 3.174329
alpha_e = 2.358533
alpha_p = 0.1937909



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
P'
ℓ'
q_γ
ℓ
dw_mixing_angles_polarization_05_local_minimum_752 D_W=0.00917587
region: polarization_cluster_05_local_minimum_752
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3432894 rad
incoming lepton: -0.697923|+> +0.716173|->
proton mixing angle: alpha_p=3.0217874 rad
incoming proton: -0.992832|+> +0.119519|->
s=1.37168, $\sqrt{s}$=1.17119, $|\vec{P}|$=0.209973, $|\vec{P}'|$=0.0473169
$\theta_{p'}$=0.0392699, θ_γ=1.76154, $\phi_{p'}$=3.12521, ϕ_γ=6.27439
E_γ=0.116593, $\theta_{\ell'}$=1.79072, $\phi_{\ell'}$=3.13292
Q^2=0.0378896, x_B=0.14606, t=-0.0259872, W^2=1.10137, y=0.527436
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2100 p_x= 0.0000 p_y= 0.0000 p_z= -0.2100
P E= 0.9612 p_x= 0.0000 p_y= 0.0000 p_z= 0.2100
ℓ' E= 0.1154 p_x= -0.1126 p_y= 0.0010 p_z= -0.0252
P' E= 0.9392 p_x= -0.0019 p_y= 0.0000 p_z= 0.0473
q_γ E= 0.1166 p_x= 0.1145 p_y= -0.0010 p_z= -0.0221
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=3, retained=97.5%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=-5.44e+01+4.72e-01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=5.41e+01+4.98e-01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
A=5.39e+01-9.58e-04i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P5, local minimum 752; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.0091758719
D_W contour = 0.059175872
above global minimum = 0.0091594391
8D contour samples = 236
selected phase-space point:
sqrt_s = 1.171187
theta_p_out = 0.03926991
theta_gamma_out = 1.761541
qOut = 0.116593
phi_p_out = 3.125211
phi_gamma_out = 6.27439
alpha_e = 2.343289
alpha_p = 3.021787

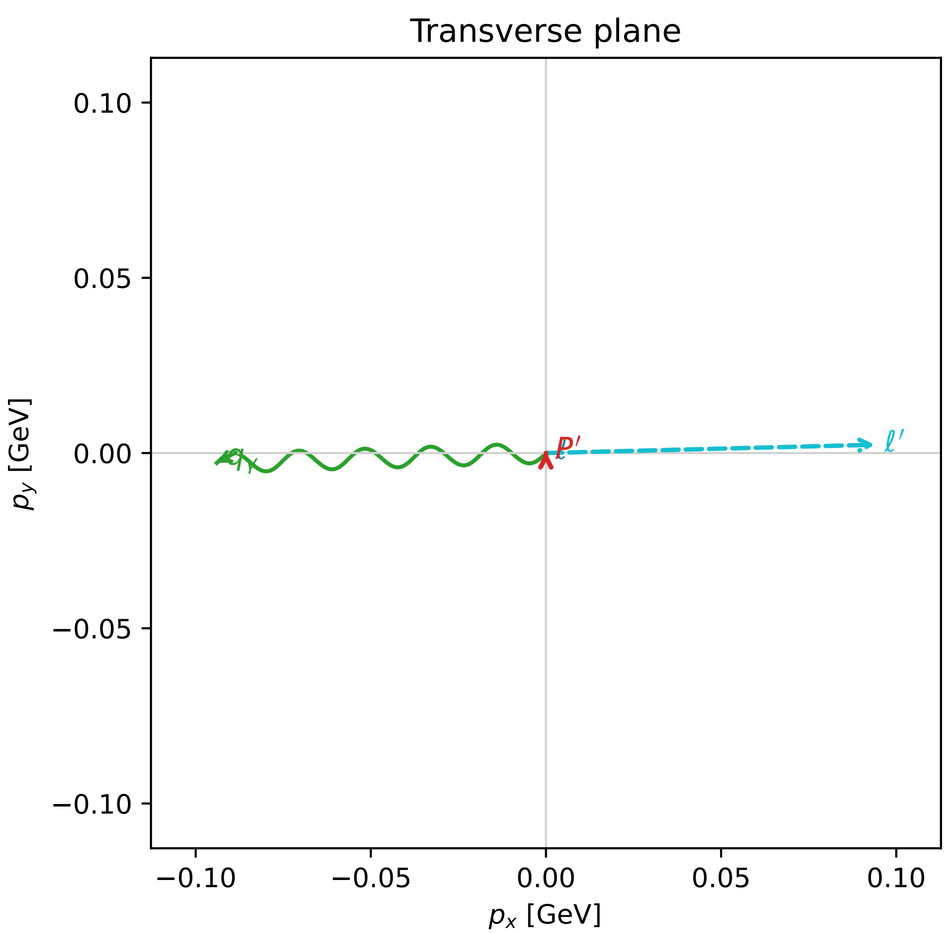
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

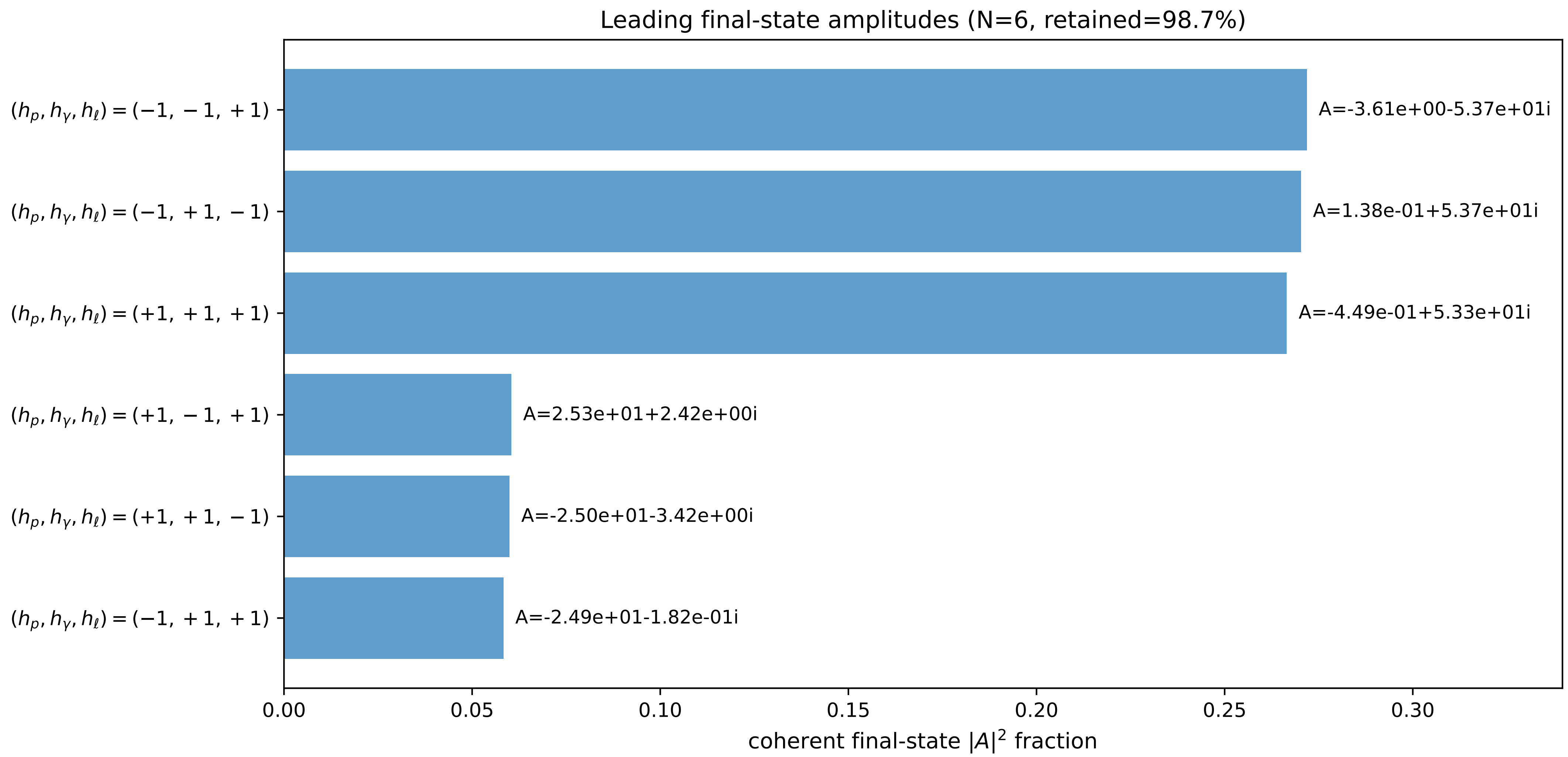


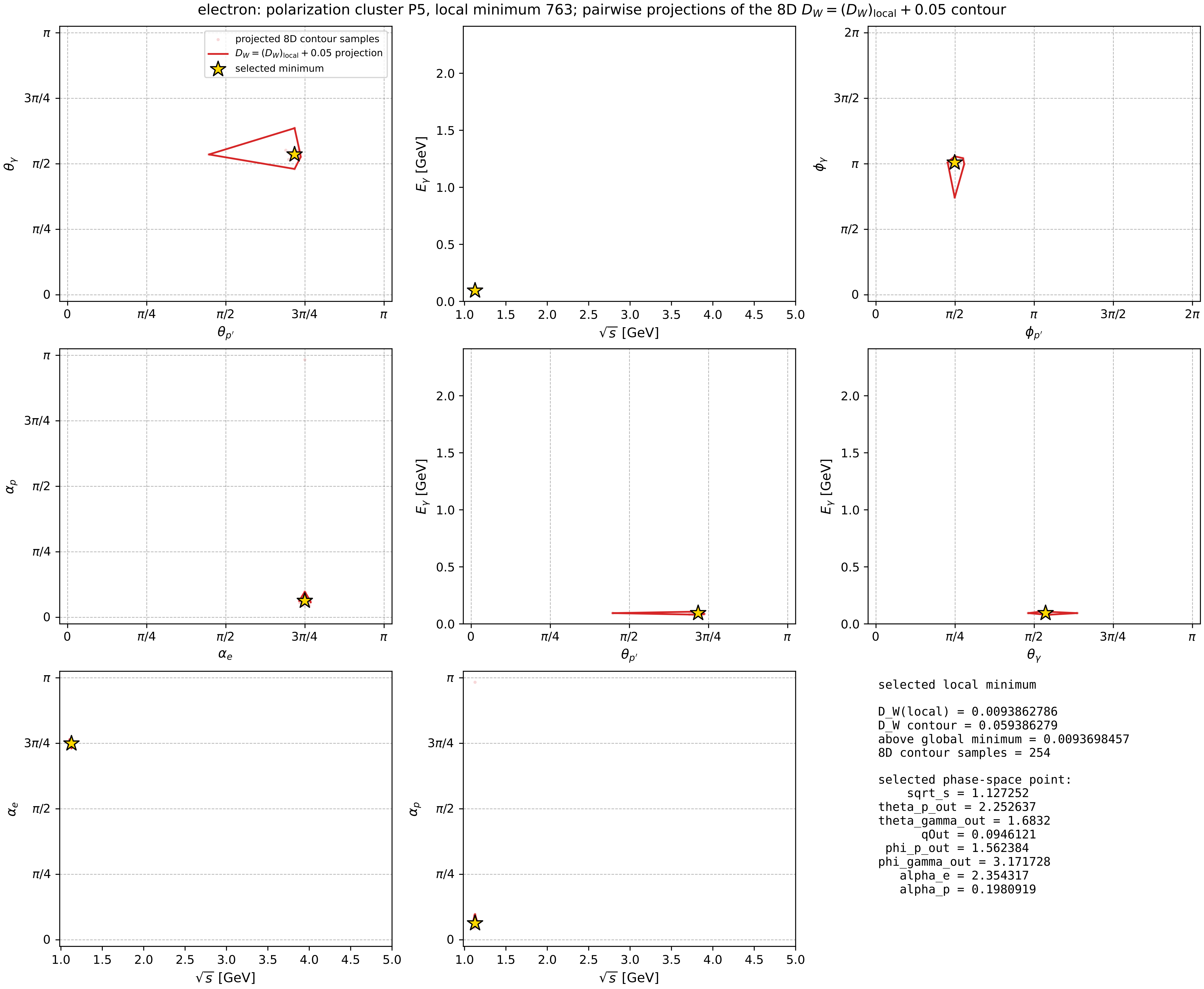
dw_mixing_angles_polarization_05_local_minimum_763 D_W=0.00938628
region: polarization_cluster_05_local_minimum_763
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3543166 rad
incoming lepton: -0.705778|+> +0.708433|->
proton mixing angle: alpha_p=0.19809192 rad
incoming proton: 0.980444|+> +0.196799|->

s=1.2707, $\sqrt{s}$=1.12725, $|\vec{P}|$=0.173365, $|\vec{P}'|$=0.000596262
$\theta_{p'}$=2.25264, θ_γ=1.6832, $\phi_{p'}$=1.56238, ϕ_γ=3.17173
E_γ=0.0946121, $\theta_{\ell'}$=1.45443, $\phi_{\ell'}$=0.0252137
Q^2=0.0366239, x_B=0.171051, t=-0.0299336, W^2=1.05733, y=0.547806

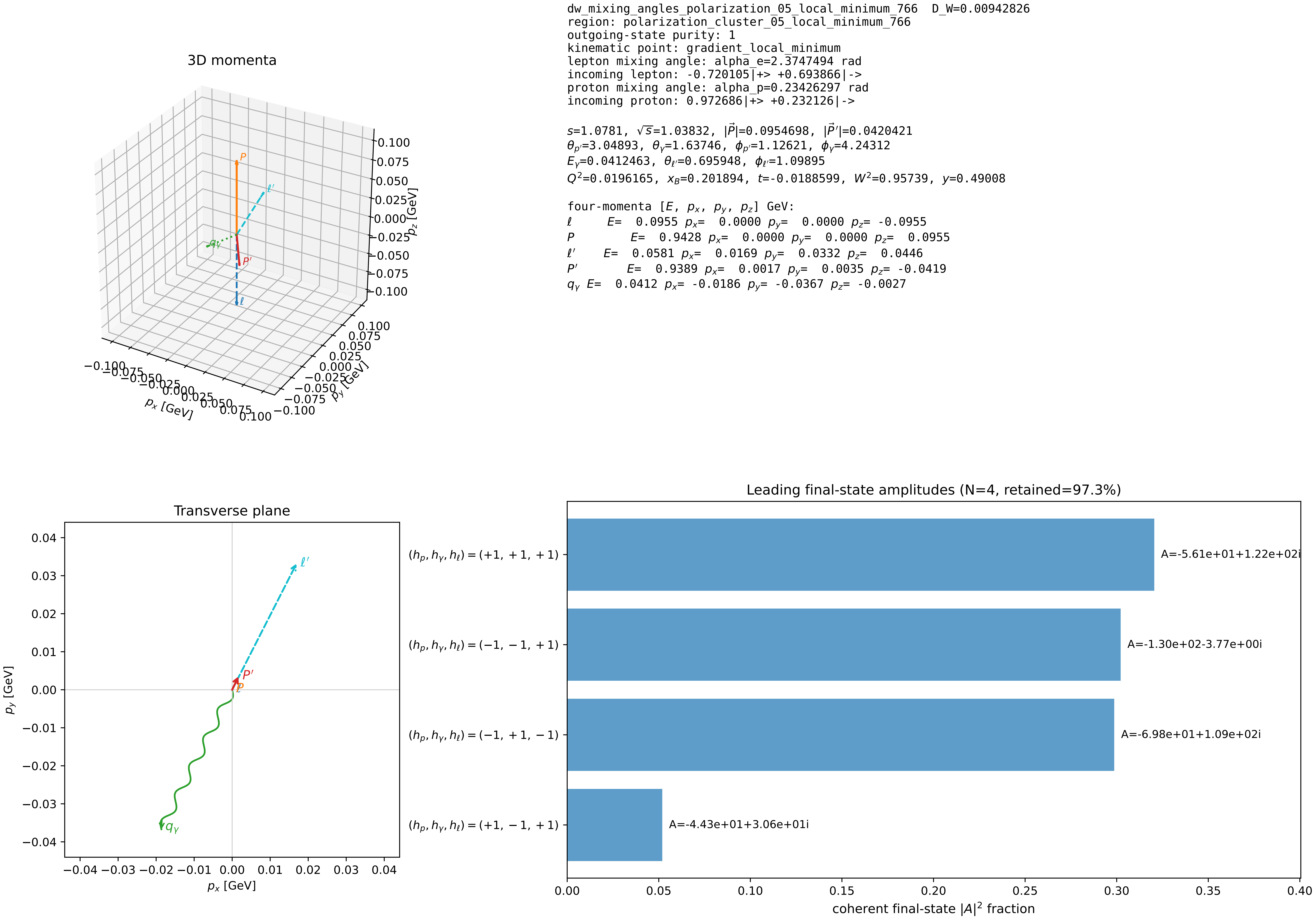
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1734 p_x= 0.0000 p_y= 0.0000 p_z= -0.1734
P E= 0.9539 p_x= 0.0000 p_y= 0.0000 p_z= 0.1734
ℓ' E= 0.0946 p_x= 0.0940 p_y= 0.0024 p_z= 0.0110
P' E= 0.9380 p_x= 0.0000 p_y= 0.0005 p_z= -0.0004
q_γ E= 0.0946 p_x= -0.0940 p_y= -0.0028 p_z= -0.0106

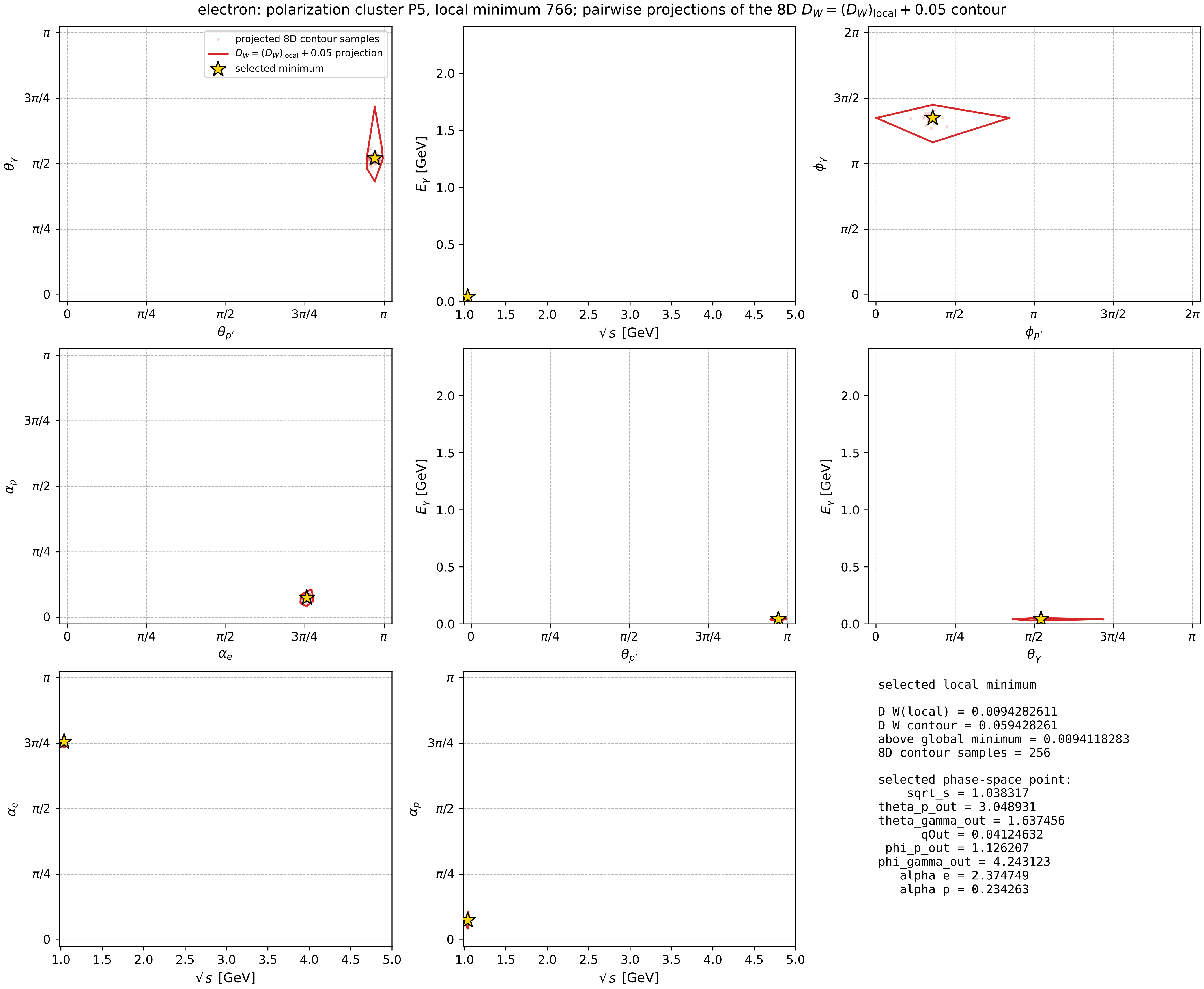




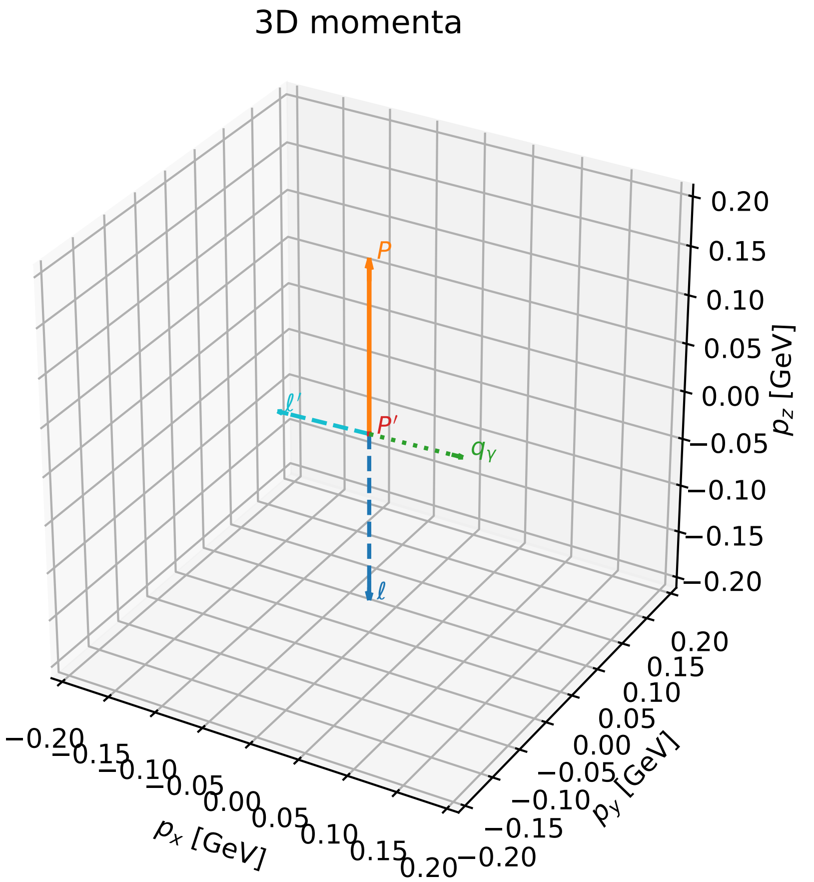

electron: polarization cluster P5, local minimum 763; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0093862786
D_W contour = 0.059386279
above global minimum = 0.0093698457
8D contour samples = 254
selected phase-space point:
sqrt_s = 1.127252
theta_p_out = 2.252637
theta_gamma_out = 1.6832
qOut = 0.0946121
phi_p_out = 1.562384
phi_gamma_out = 3.171728
alpha_e = 2.354317
alpha_p = 0.1980919

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)




electron: polarization cluster P5, local minimum 766; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.0094282611
D_W contour = 0.059428261
above global minimum = 0.0094118283
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.038317
theta_p_out = 3.048931
theta_gamma_out = 1.637456
qOut = 0.04124632
phi_p_out = 1.126207
phi_gamma_out = 4.243123
alpha_e = 2.374749
alpha_p = 0.234263

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

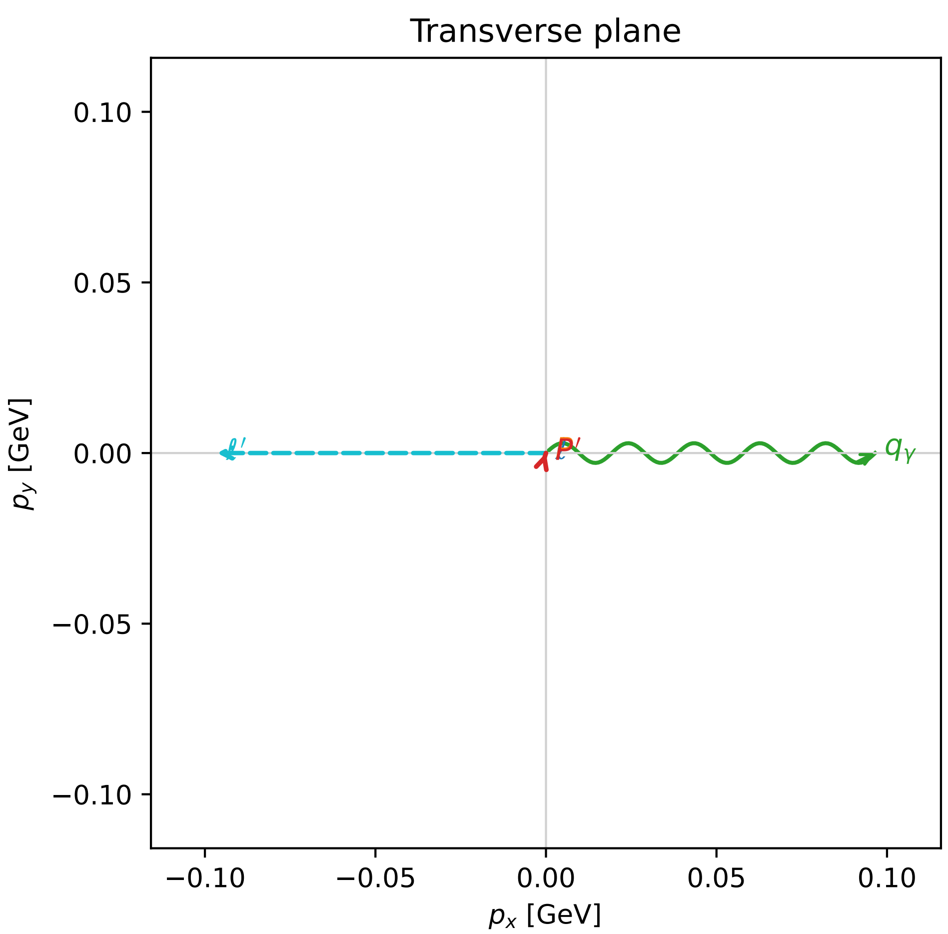


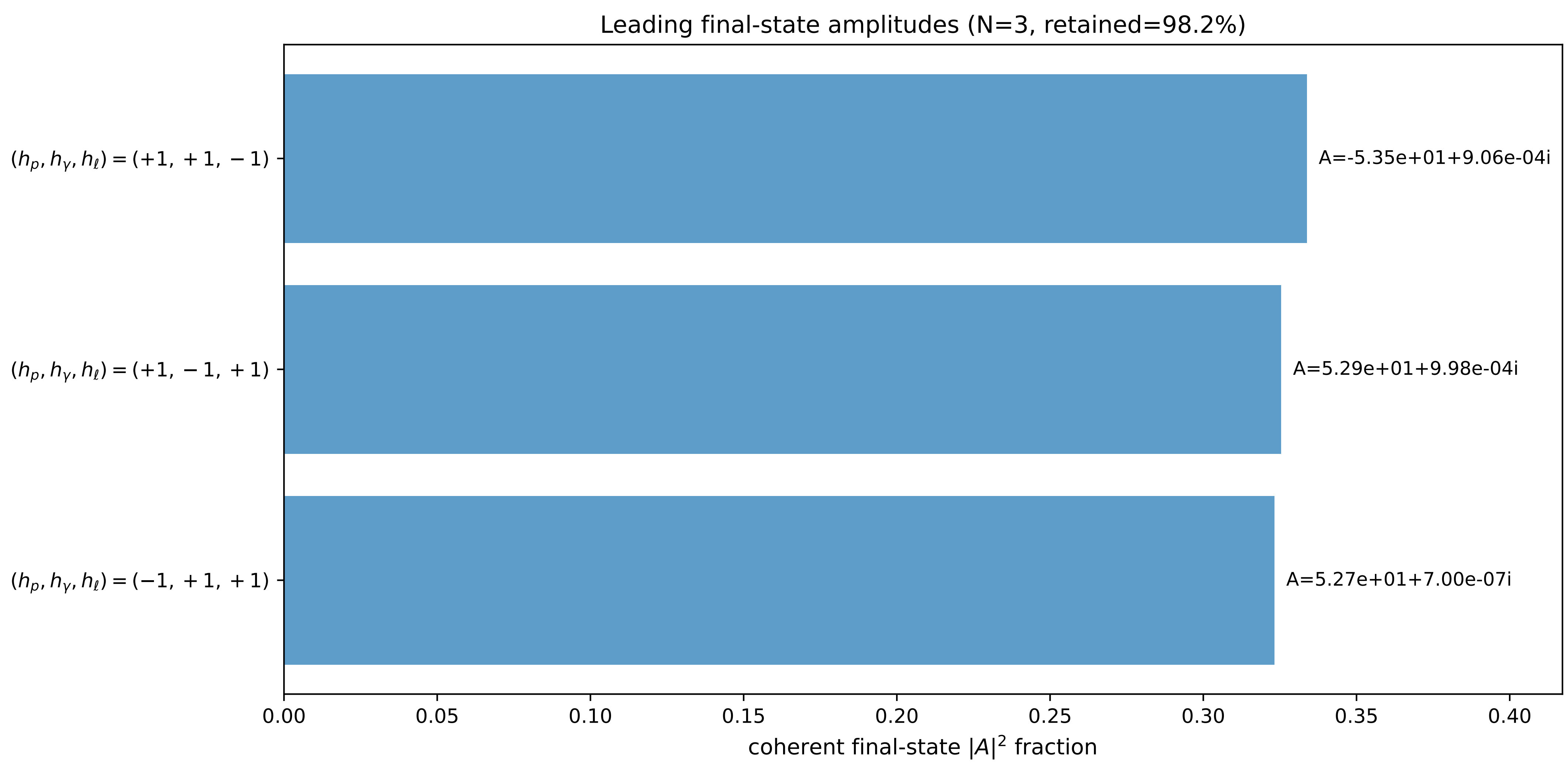
dw_mixing_angles_polarization_05_local_minimum_776 D_W=0.00968106
region: polarization_cluster_05_local_minimum_776
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3464431 rad
incoming lepton: -0.700178|+> +0.713968|->
proton mixing angle: alpha_p=2.9353127 rad
incoming proton: -0.9788|+> +0.20482|->

s=1.27959, $\sqrt{s}$=1.13119, $|\vec{P}|$=0.176693, $|\vec{P}'|$=0.000242767
$\theta_{p'}$=8.30818e-07, θ_γ=1.53179, $\phi_{p'}$=1.27106, ϕ_γ=6.28317
E_γ=0.09659, $\theta_{\ell'}$=1.61231, $\phi_{\ell'}$=3.14157
Q^2=0.0327203, x_B=0.152955, t=-0.0308627, W^2=1.06104, y=0.535139

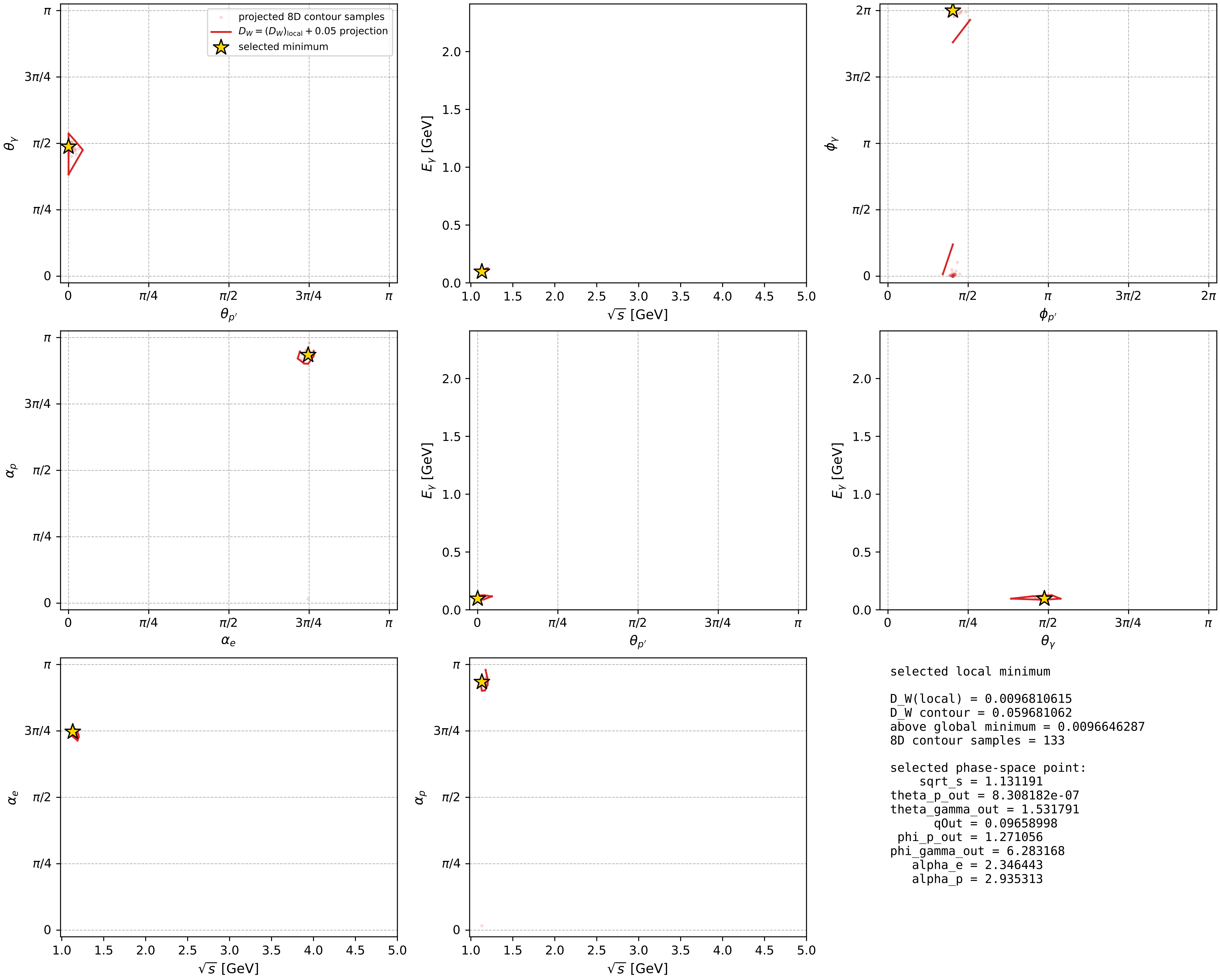
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 0.1767	p_x= 0.0000	p_y= 0.0000	p_z= -0.1767
P	E= 0.9545	p_x= 0.0000	p_y= 0.0000	p_z= 0.1767
ℓ'	E= 0.0966	p_x= -0.0965	p_y= 0.0000	p_z= -0.0040
P'	E= 0.9380	p_x= 0.0000	p_y= 0.0000	p_z= 0.0002
q_γ	E= 0.0966	p_x= 0.0965	p_y= -0.0000	p_z= 0.0038

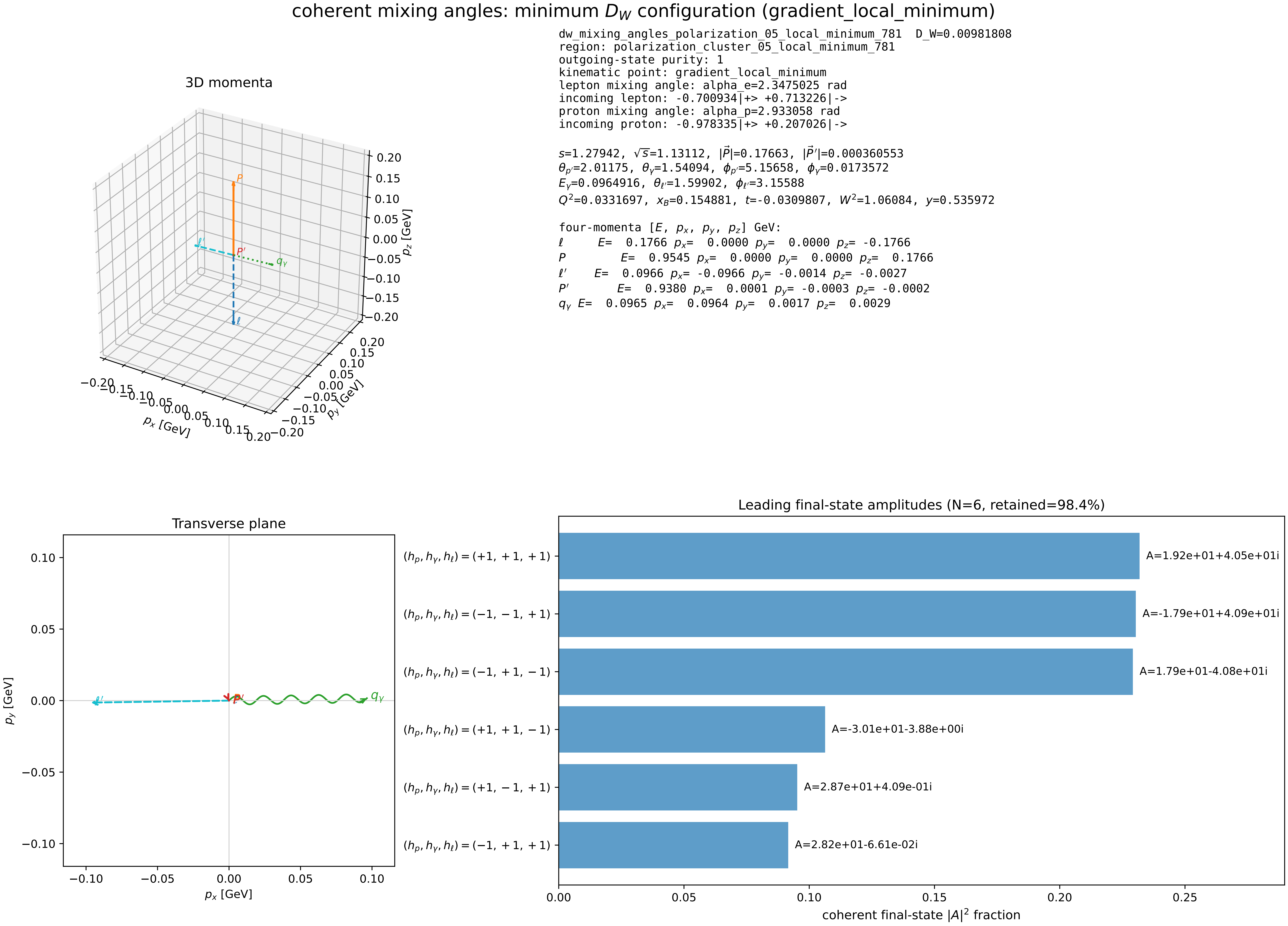




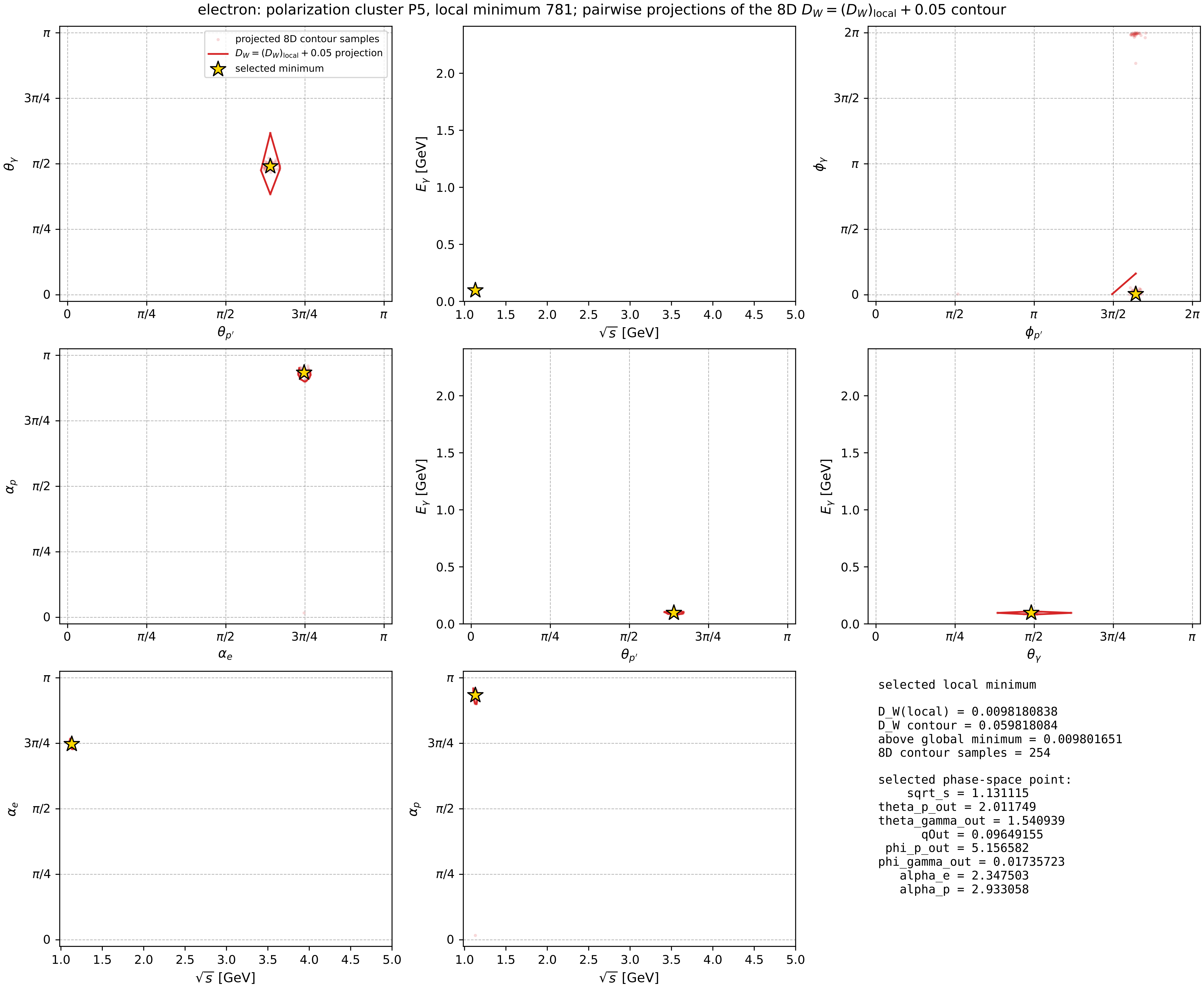
electron: polarization cluster P5, local minimum 776; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



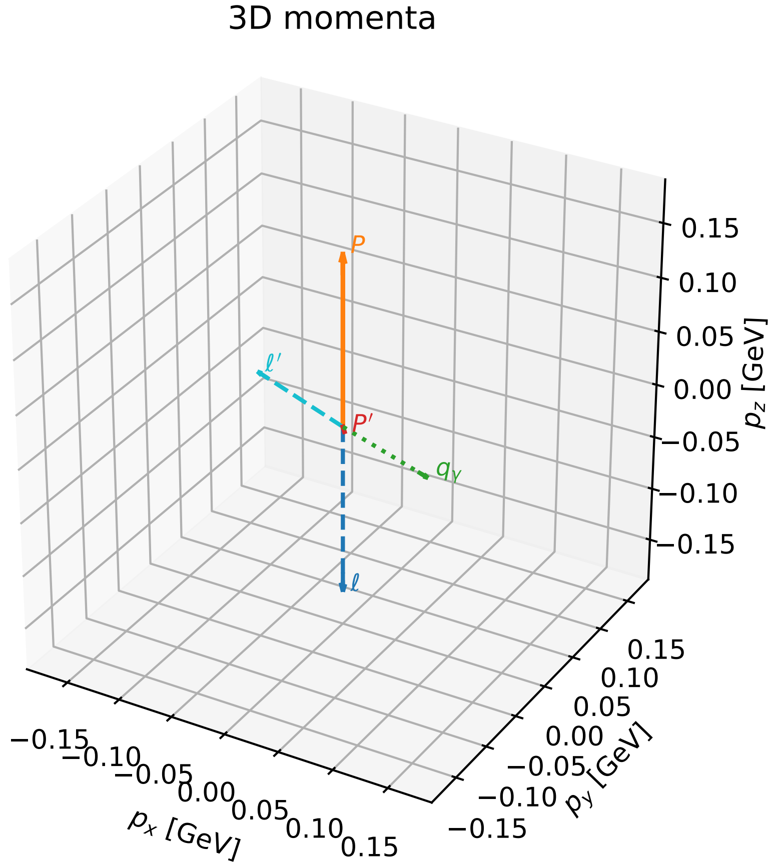

```
selected local minimum

D_W(local) = 0.0096810615
D_W contour = 0.059681062
above global minimum = 0.0096646287
8D contour samples = 133

selected phase-space point:
    sqrt_s = 1.131191
theta_p_out = 8.308182e-07
theta_gamma_out = 1.531791
       qOut = 0.09658998
  phi_p_out = 1.271056
phi_gamma_out = 6.283168
    alpha_e = 2.346443
    alpha_p = 2.935313
```



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_781 D_W=0.00981808
region: polarization_cluster_05_local_minimum_781
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3475025 rad
incoming lepton: -0.700934|+> +0.713226|->
proton mixing angle: alpha_p=2.933058 rad
incoming proton: -0.978335|+> +0.207026|->
s=1.27942, $\sqrt{s}$=1.13112, $|\vec{P}|$=0.17663, $|\vec{P}'|$=0.000360553
$\theta_{p'}$=2.01175, θ_γ=1.54094, $\phi_{p'}$=5.15658, ϕ_γ=0.0173572
E_γ=0.0964916, $\theta_{\ell'}$=1.59902, $\phi_{\ell'}$=3.15588
Q^2=0.0331697, x_B=0.154881, t=-0.0309807, W^2=1.06084, y=0.535972
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1766 p_x= 0.0000 p_y= 0.0000 p_z= -0.1766
P E= 0.9545 p_x= 0.0000 p_y= 0.0000 p_z= 0.1766
ℓ' E= 0.0966 p_x= -0.0966 p_y= -0.0014 p_z= -0.0027
P' E= 0.9380 p_x= 0.0001 p_y= -0.0003 p_z= -0.0002
q_γ E= 0.0965 p_x= 0.0964 p_y= 0.0017 p_z= 0.0029
3D momenta
P
ℓ'
P'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
−0.20 −0.15 −0.10 −0.05 0.00 0.05 0.10 0.15 0.20
Transverse plane
p_x [GeV]
p_y [GeV]
−0.10 −0.05 0.00 0.05 0.10
Leading final-state amplitudes (N=6, retained=98.4%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=1.92e+01+4.05e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-1.79e+01+4.09e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=1.79e+01-4.08e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-3.01e+01-3.88e+00i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=2.87e+01+4.09e-01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=2.82e+01-6.61e-02i
0.00 0.05 0.10 0.15 0.20 0.25
coherent final-state $|A|^2$ fraction


electron: polarization cluster P5, local minimum 781; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.0098180838
D_W contour = 0.059818084
above global minimum = 0.009801651
8D contour samples = 254
selected phase-space point:
sqrt_s = 1.131115
theta_p_out = 2.011749
theta_gamma_out = 1.540939
qOut = 0.09649155
phi_p_out = 5.156582
phi_gamma_out = 0.01735723
alpha_e = 2.347503
alpha_p = 2.933058

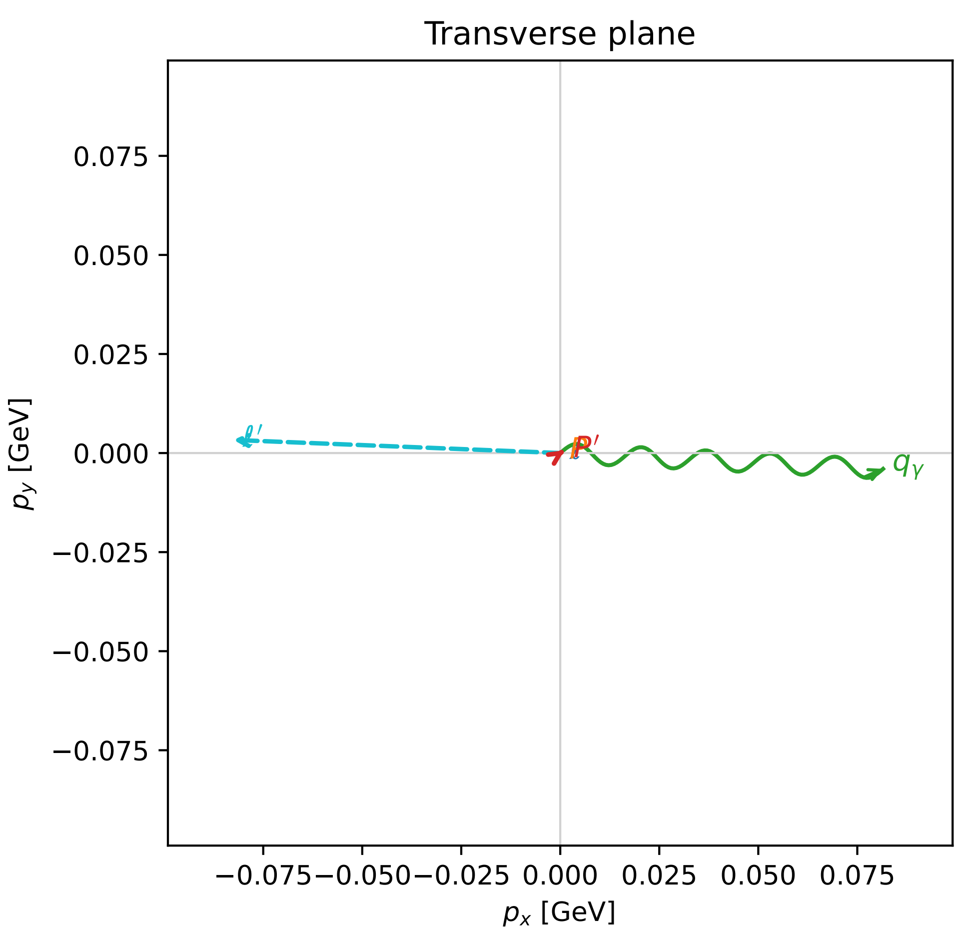
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

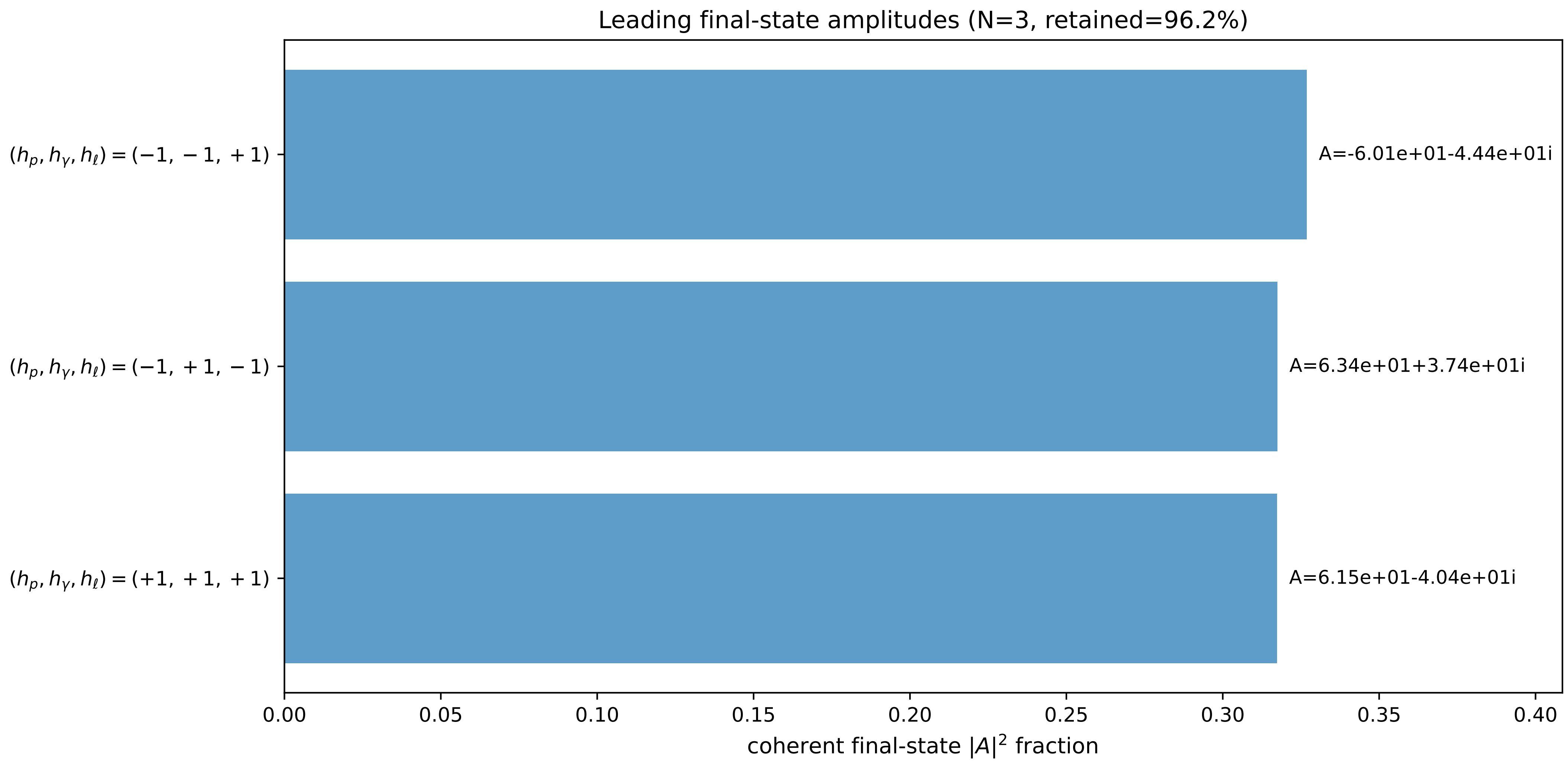


dw_mixing_angles_polarization_05_local_minimum_793 D_W=0.0102042
region: polarization_cluster_05_local_minimum_793
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3866647 rad
incoming lepton: -0.728321|+> +0.685236|->
proton mixing angle: alpha_p=2.9399128 rad
incoming proton: -0.979731|+> +0.200315|->

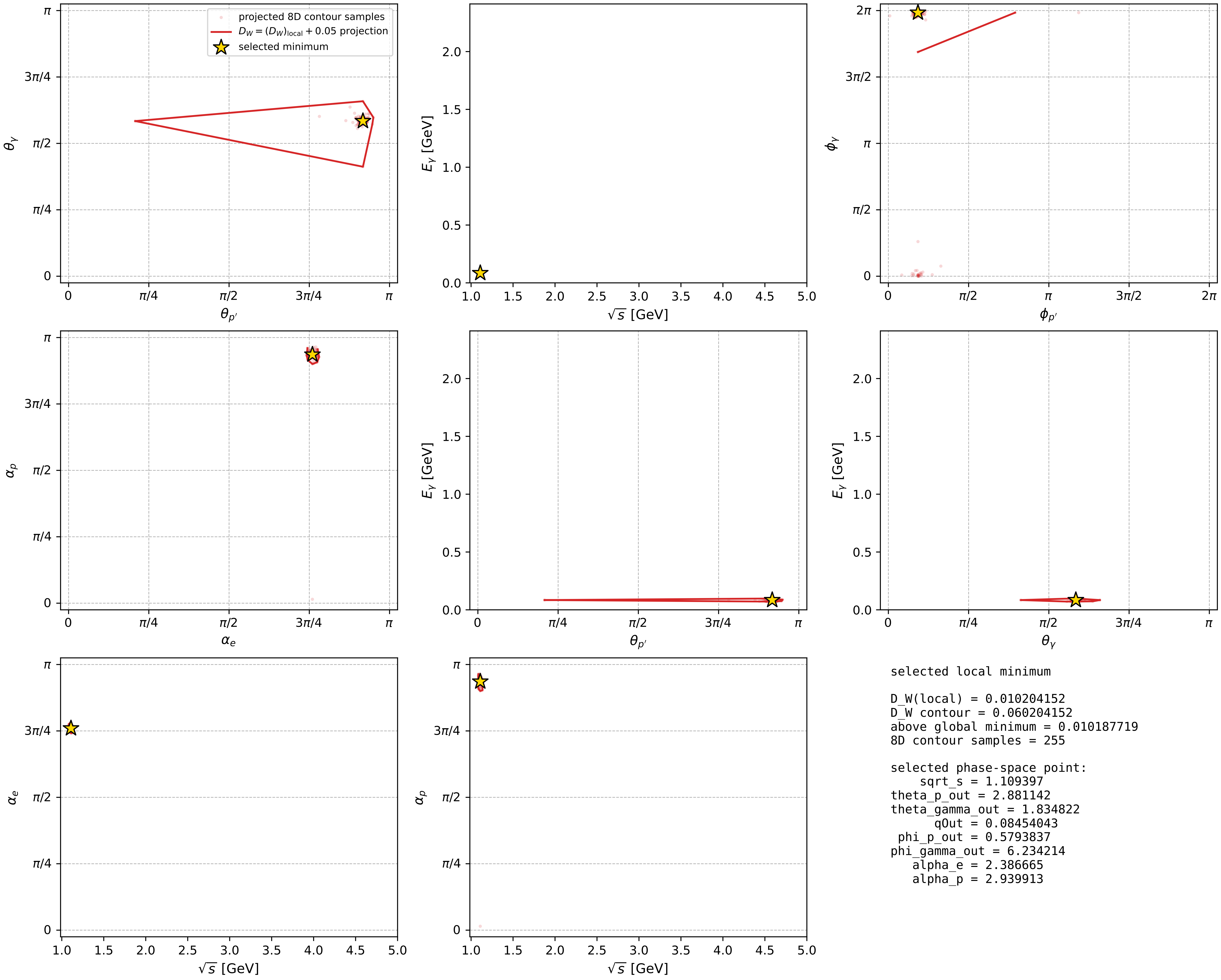
s=1.23076, $\sqrt{s}$=1.1094, $|\vec{P}|$=0.158156, $|\vec{P}'|$=0.00484159
$\theta_{p'}$=2.88114, θ_γ=1.83482, $\phi_{p'}$=0.579384, ϕ_γ=6.23421
E_γ=0.0845404, $\theta_{\ell'}$=1.25779, $\phi_{\ell'}$=3.10149
Q^2=0.0359278, x_B=0.185045, t=-0.0263416, W^2=1.03807, y=0.553284

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1582 p_x= 0.0000 p_y= 0.0000 p_z= -0.1582
P E= 0.9512 p_x= 0.0000 p_y= 0.0000 p_z= 0.1582
ℓ' E= 0.0868 p_x= -0.0826 p_y= 0.0033 p_z= 0.0267
P' E= 0.9380 p_x= 0.0010 p_y= 0.0007 p_z= -0.0047
q_γ E= 0.0845 p_x= 0.0815 p_y= -0.0040 p_z= -0.0221

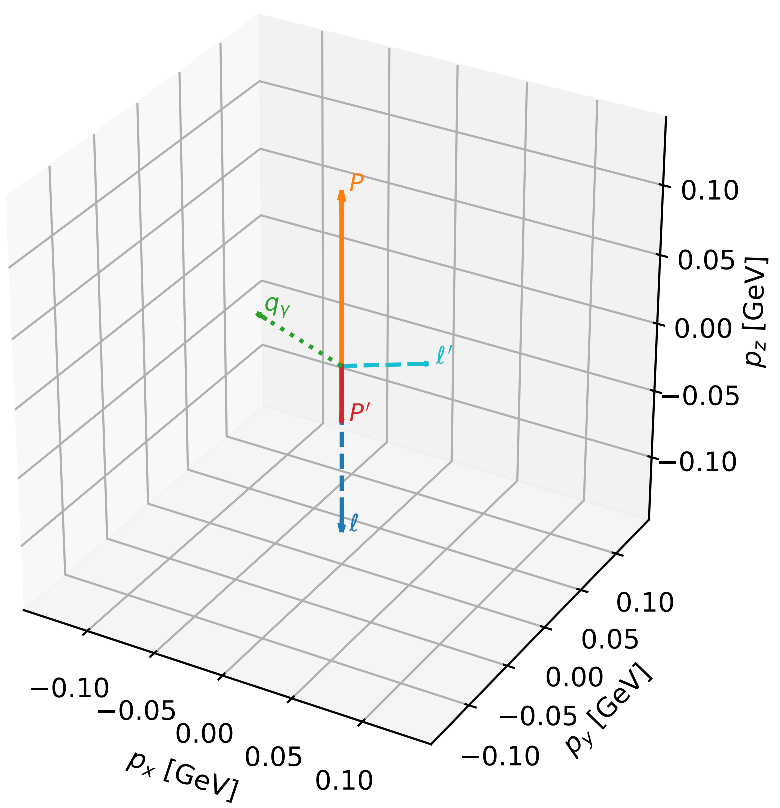




electron: polarization cluster P5, local minimum 793; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



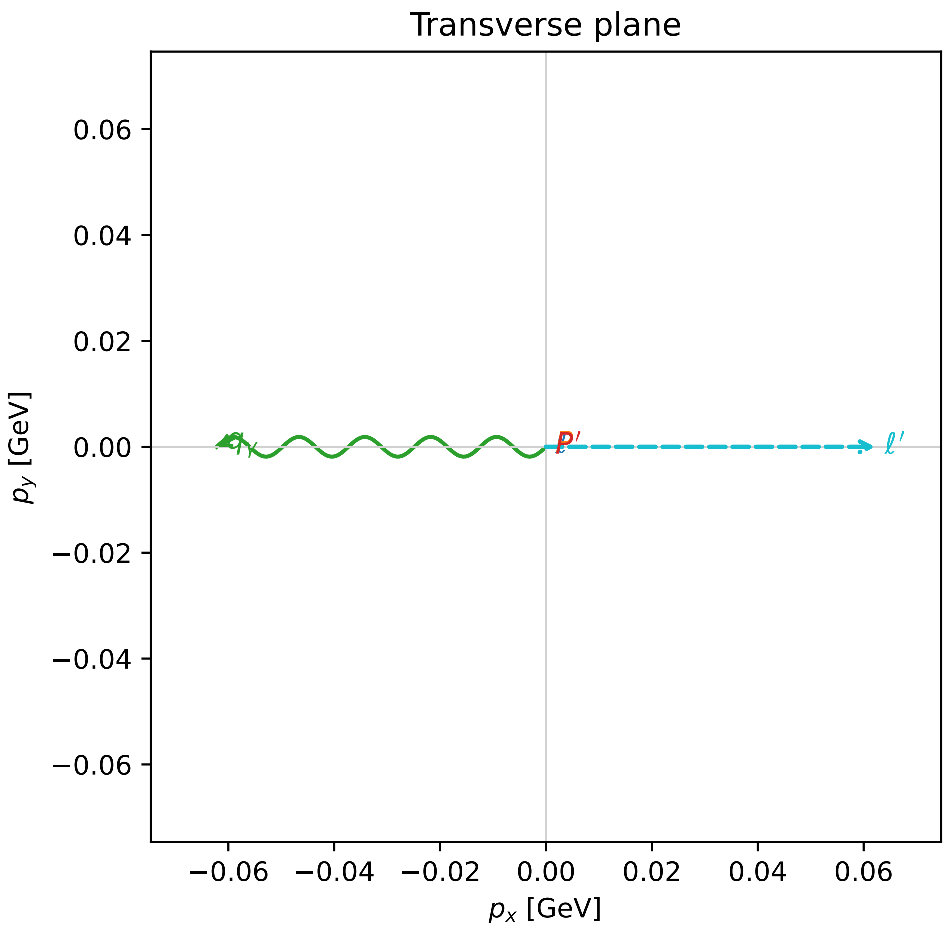
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

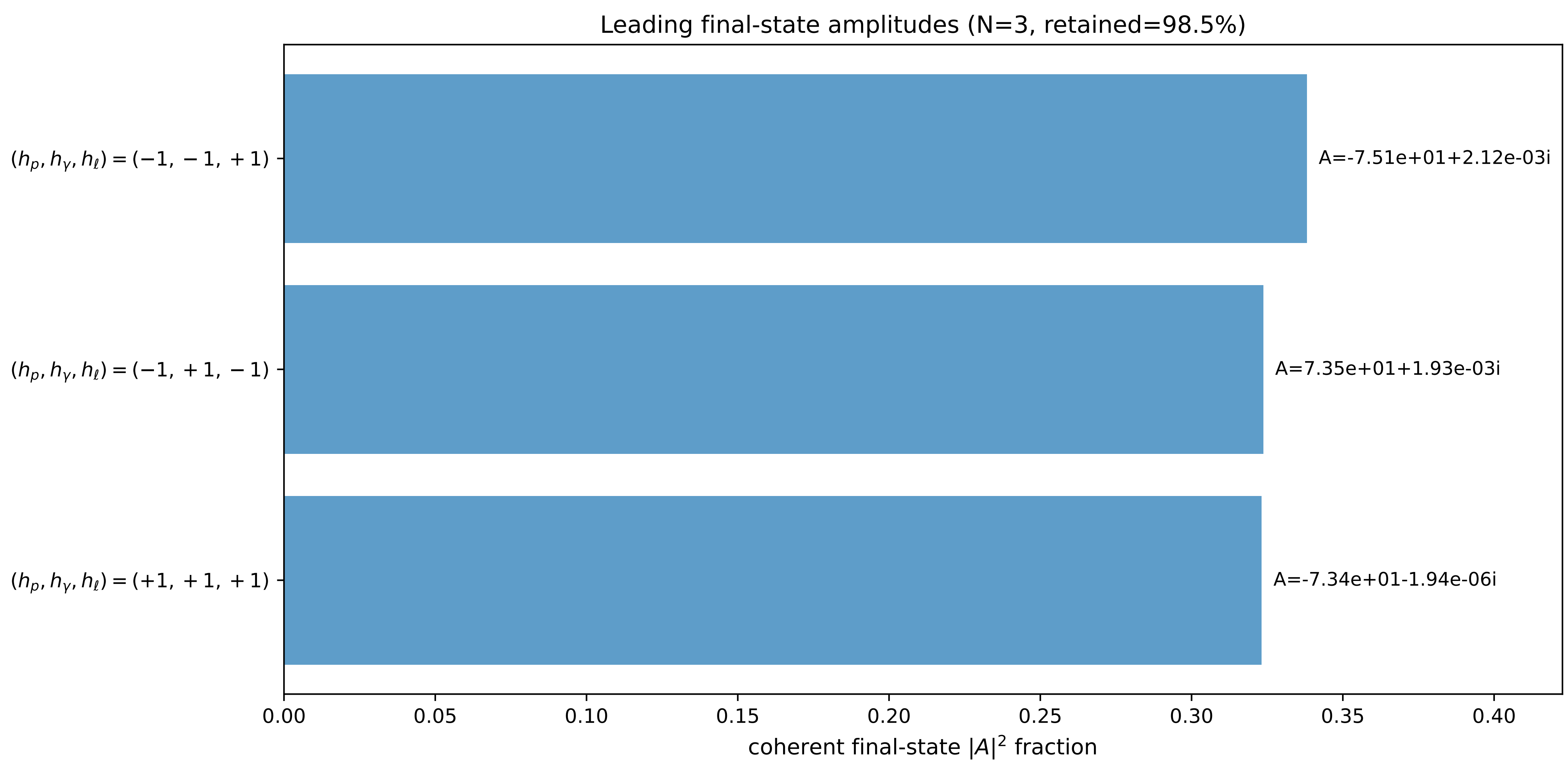


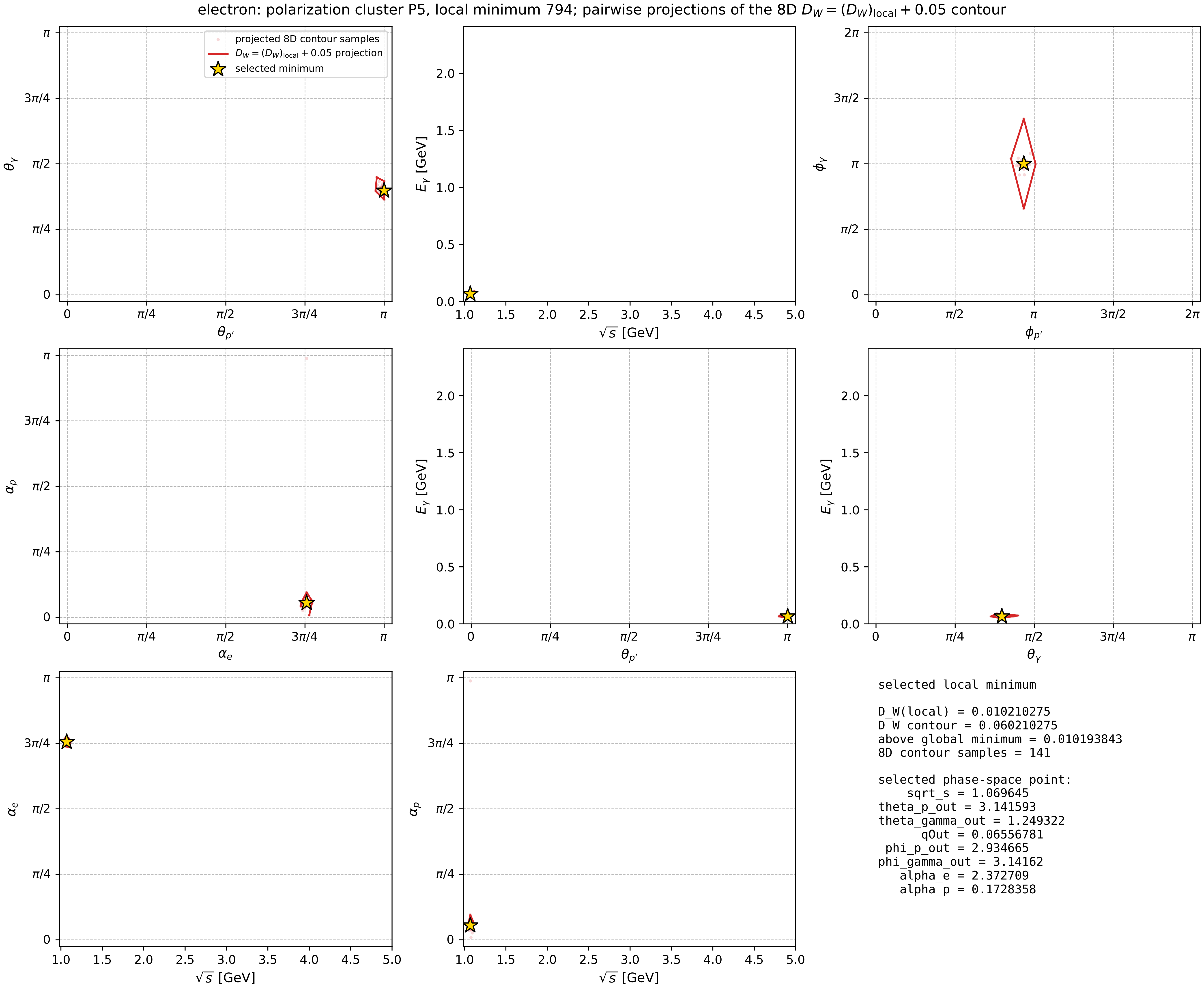
dw_mixing_angles_polarization_05_local_minimum_794 D_W=0.0102103
region: polarization_cluster_05_local_minimum_794
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.372709 rad
incoming lepton: -0.718687|+> +0.695333|->
proton mixing angle: alpha_p=0.17283578 rad
incoming proton: 0.985101|+> +0.171977|->

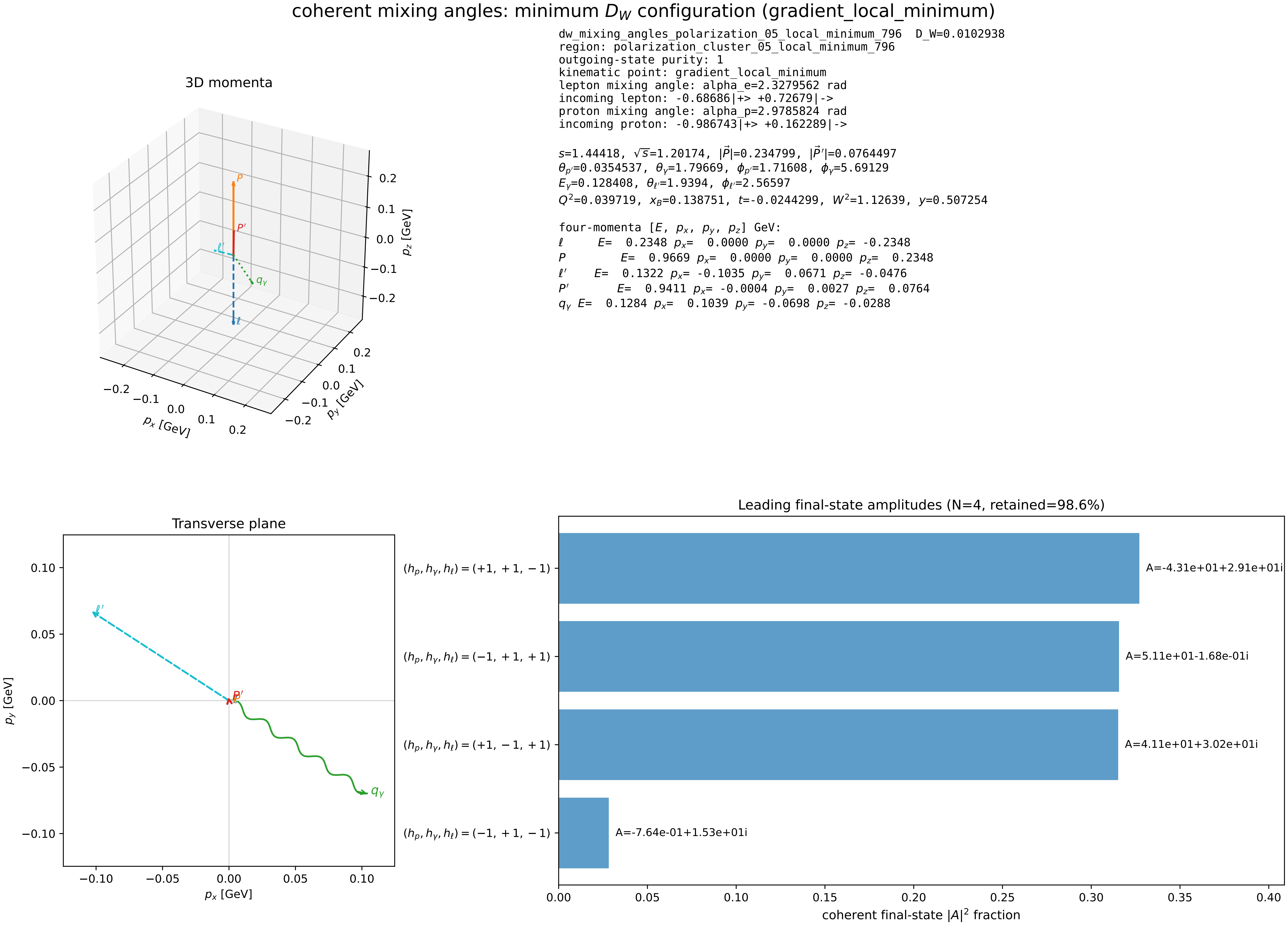
s=1.14414, $\sqrt{s}$=1.06964, $|\vec{P}|$=0.123543, $|\vec{P}'|$=0.0402749
$\theta_{p'}$=3.14159, θ_γ=1.24932, $\phi_{p'}$=2.93467, ϕ_γ=3.14162
E_γ=0.0655678, $\theta_{\ell'}$=1.26619, $\phi_{\ell'}$=2.75644e-05
Q^2=0.0209452, x_B=0.143724, t=-0.0267839, W^2=1.00463, y=0.551399

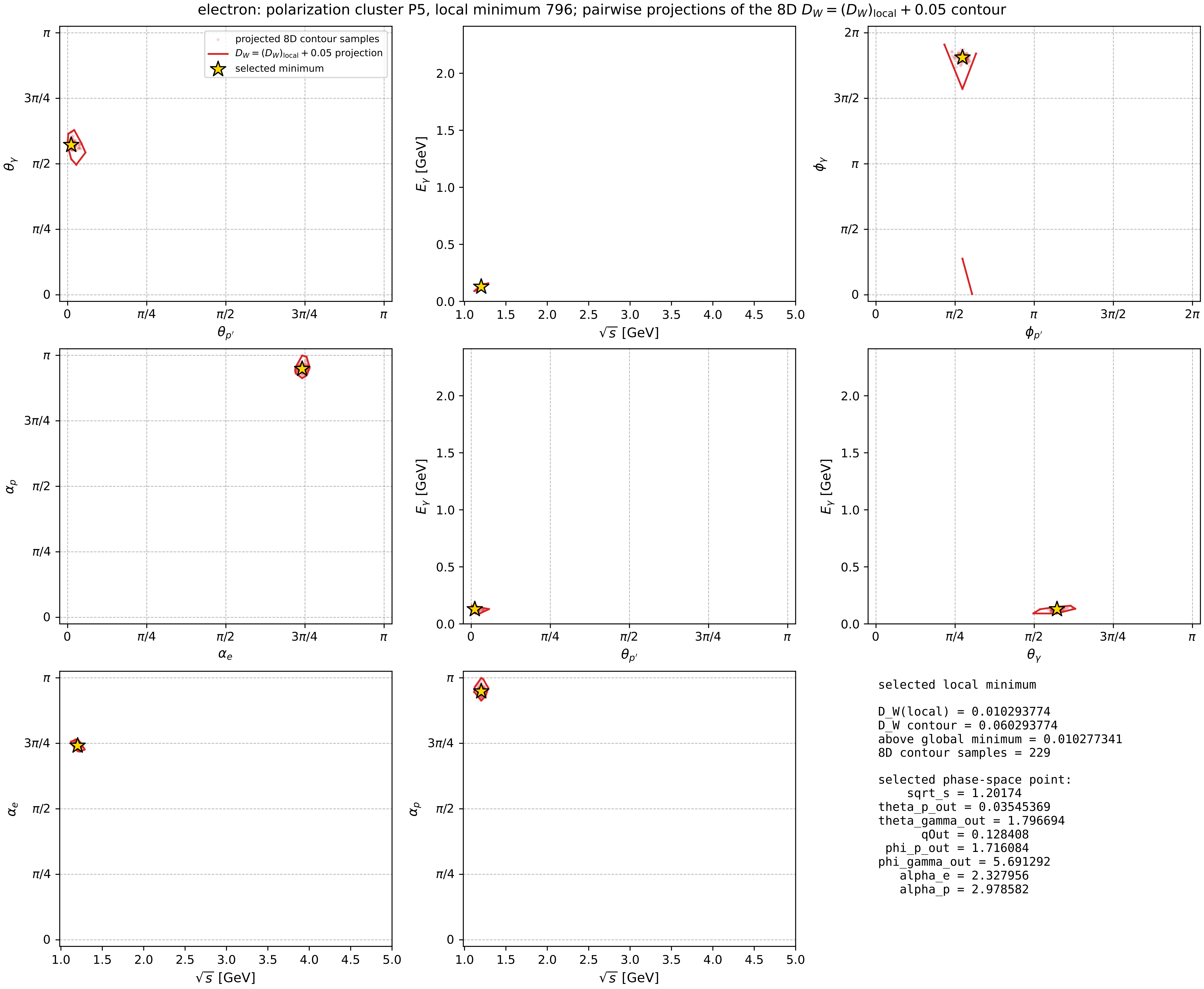
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1235 p_x= 0.0000 p_y= 0.0000 p_z= -0.1235
P E= 0.9461 p_x= 0.0000 p_y= 0.0000 p_z= 0.1235
ℓ' E= 0.0652 p_x= 0.0622 p_y= 0.0000 p_z= 0.0196
P' E= 0.9389 p_x= -0.0000 p_y= 0.0000 p_z= -0.0403
q_γ E= 0.0656 p_x= -0.0622 p_y= -0.0000 p_z= 0.0207

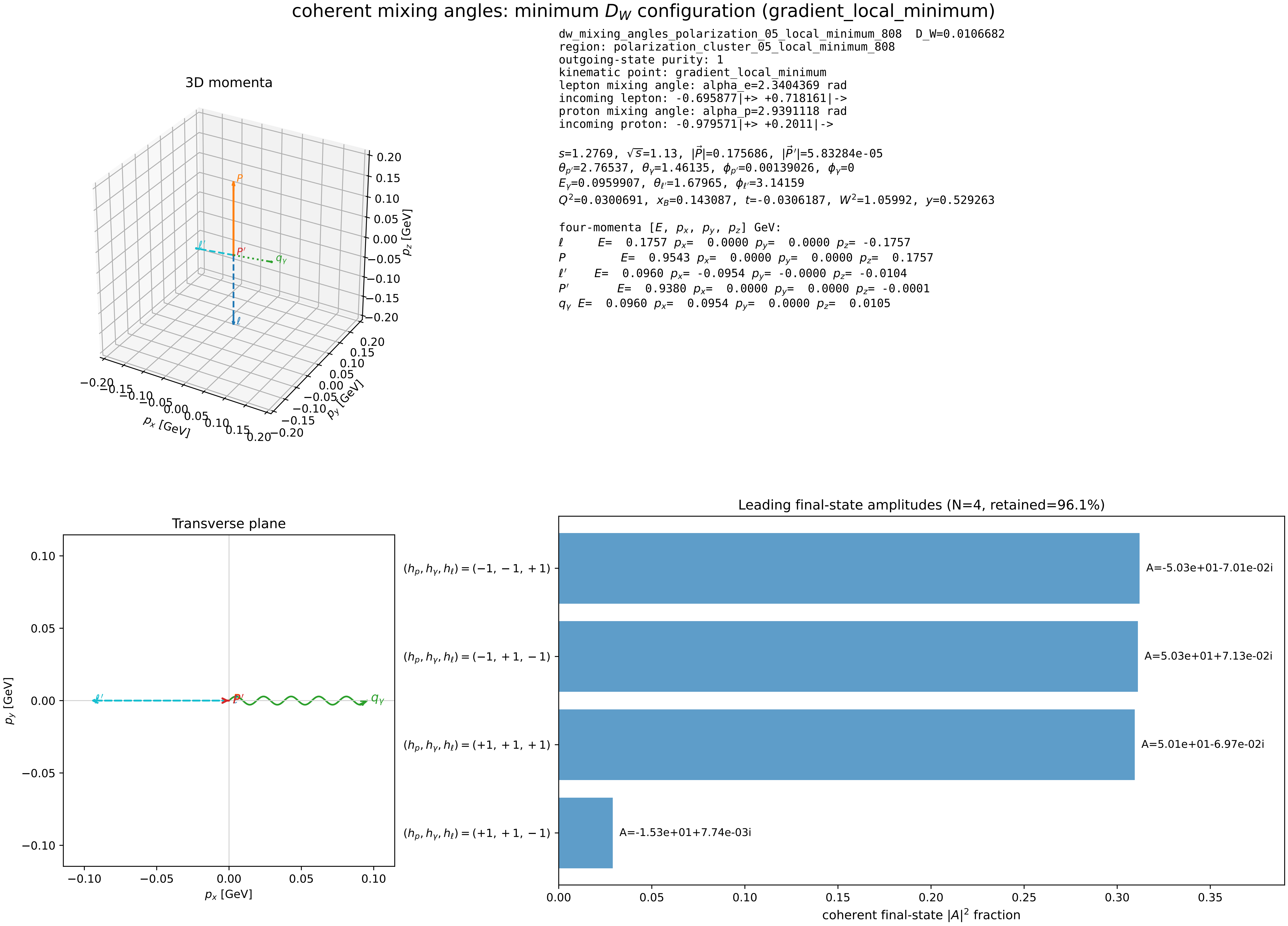




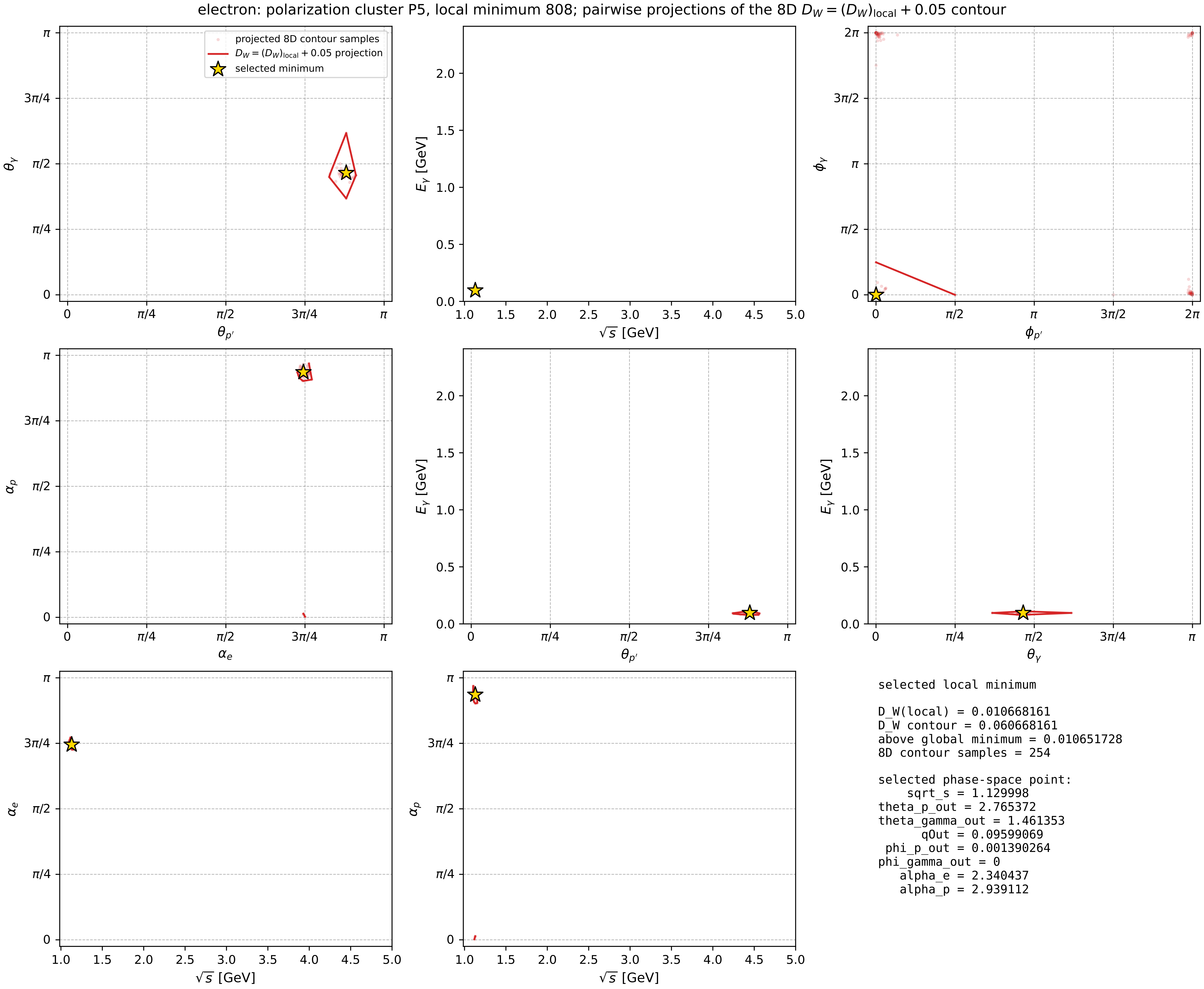

electron: polarization cluster P5, local minimum 794; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.010210275
D_W contour = 0.060210275
above global minimum = 0.010193843
8D contour samples = 141
selected phase-space point:
sqrt_s = 1.069645
theta_p_out = 3.141593
theta_gamma_out = 1.249322
qOut = 0.06556781
phi_p_out = 2.934665
phi_gamma_out = 3.14162
alpha_e = 2.372709
alpha_p = 0.1728358


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_796 D_W=0.0102938
region: polarization_cluster_05_local_minimum_796
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3279562 rad
incoming lepton: -0.68686|+> +0.72679|->
proton mixing angle: alpha_p=2.9785824 rad
incoming proton: -0.986743|+> +0.162289|->
s=1.44418, $\sqrt{s}$=1.20174, $|\vec{P}|$=0.234799, $|\vec{P}'|$=0.0764497
$\theta_{p'}$=0.0354537, θ_γ=1.79669, $\phi_{p'}$=1.71608, ϕ_γ=5.69129
E_γ=0.128408, $\theta_{\ell'}$=1.9394, $\phi_{\ell'}$=2.56597
Q^2=0.039719, x_B=0.138751, t=-0.0244299, W^2=1.12639, y=0.507254
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2348 p_x= 0.0000 p_y= 0.0000 p_z= -0.2348
P E= 0.9669 p_x= 0.0000 p_y= 0.0000 p_z= 0.2348
ℓ' E= 0.1322 p_x= -0.1035 p_y= 0.0671 p_z= -0.0476
P' E= 0.9411 p_x= -0.0004 p_y= 0.0027 p_z= 0.0764
q_γ E= 0.1284 p_x= 0.1039 p_y= -0.0698 p_z= -0.0288
3D momenta
P
P'
ℓ'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
−0.2
−0.1
0.0
0.1
0.2
Transverse plane
ℓ'
P'
q_γ
p_x [GeV]
p_y [GeV]
−0.10
−0.05
0.00
0.05
0.10
Leading final-state amplitudes (N=4, retained=98.6%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=-4.31e+01+2.91e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
A=5.11e+01-1.68e-01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=4.11e+01+3.02e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=-7.64e-01+1.53e+01i
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
coherent final-state $|A|^2$ fraction


electron: polarization cluster P5, local minimum 796; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.010293774
D_W contour = 0.060293774
above global minimum = 0.010277341
8D contour samples = 229
selected phase-space point:
sqrt_s = 1.20174
theta_p_out = 0.03545369
theta_gamma_out = 1.796694
qOut = 0.128408
phi_p_out = 1.716084
phi_gamma_out = 5.691292
alpha_e = 2.327956
alpha_p = 2.978582



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
dw_mixing_angles_polarization_05_local_minimum_808 D_W=0.0106682
region: polarization_cluster_05_local_minimum_808
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3404369 rad
incoming lepton: -0.695877|+> +0.718161|->
proton mixing angle: alpha_p=2.9391118 rad
incoming proton: -0.979571|+> +0.2011|->
s=1.2769, $\sqrt{s}$=1.13, $|\vec{P}|$=0.175686, $|\vec{P}'|$=5.83284e-05
$\theta_{p'}$=2.76537, θ_γ=1.46135, $\phi_{p'}$=0.00139026, ϕ_γ=0
E_γ=0.0959907, $\theta_{\ell'}$=1.67965, $\phi_{\ell'}$=3.14159
Q^2=0.0300691, x_B=0.143087, t=-0.0306187, W^2=1.05992, y=0.529263
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1757 p_x= 0.0000 p_y= 0.0000 p_z= -0.1757
P E= 0.9543 p_x= 0.0000 p_y= 0.0000 p_z= 0.1757
ℓ' E= 0.0960 p_x= -0.0954 p_y= -0.0000 p_z= -0.0104
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= -0.0001
q_γ E= 0.0960 p_x= 0.0954 p_y= 0.0000 p_z= 0.0105
P
P'
ℓ'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
Leading final-state amplitudes (N=4, retained=96.1%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=-5.03e+01-7.01e-02i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=5.03e+01+7.13e-02i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$
A=5.01e+01-6.97e-02i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=-1.53e+01+7.74e-03i
coherent final-state $|A|^2$ fraction

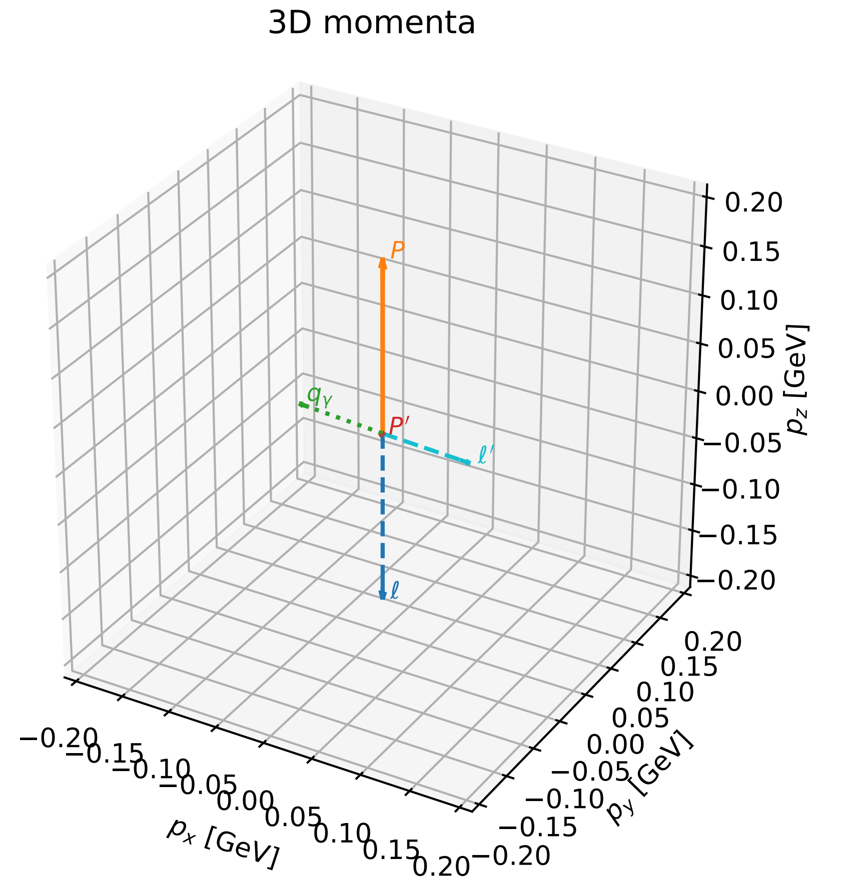

electron: polarization cluster P5, local minimum 808; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.010668161
D_W contour = 0.060668161
above global minimum = 0.010651728
8D contour samples = 254
selected phase-space point:
sqrt_s = 1.129998
theta_p_out = 2.765372
theta_gamma_out = 1.461353
qOut = 0.09599069
phi_p_out = 0.001390264
phi_gamma_out = 0
alpha_e = 2.340437
alpha_p = 2.939112

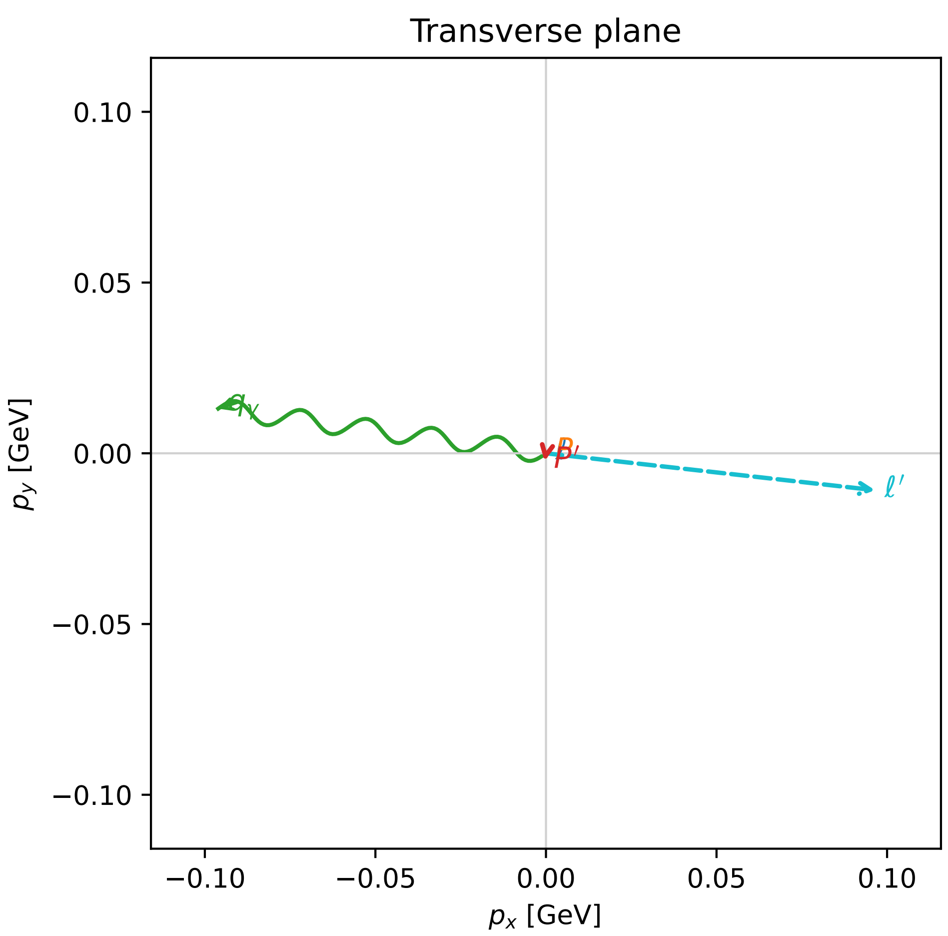
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

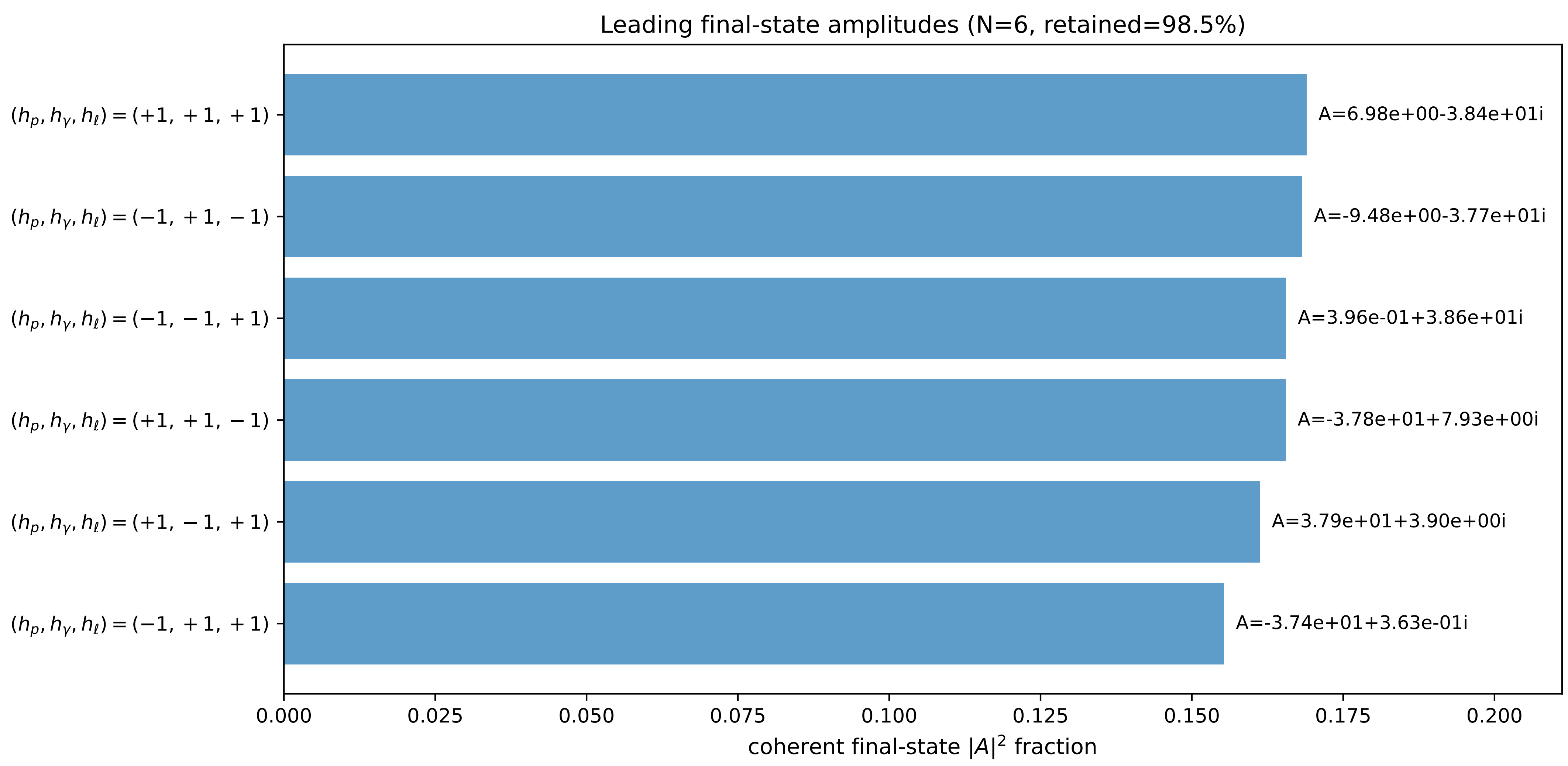
```
dw_mixing_angles_polarization_05_local_minimum_810  D_W=0.0107718
region: polarization_cluster_05_local_minimum_810
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3445458 rad
incoming lepton: -0.698822|+> +0.715295|->
proton mixing angle: alpha_p=0.20679792 rad
incoming proton: 0.978693|+> +0.205327|->
```

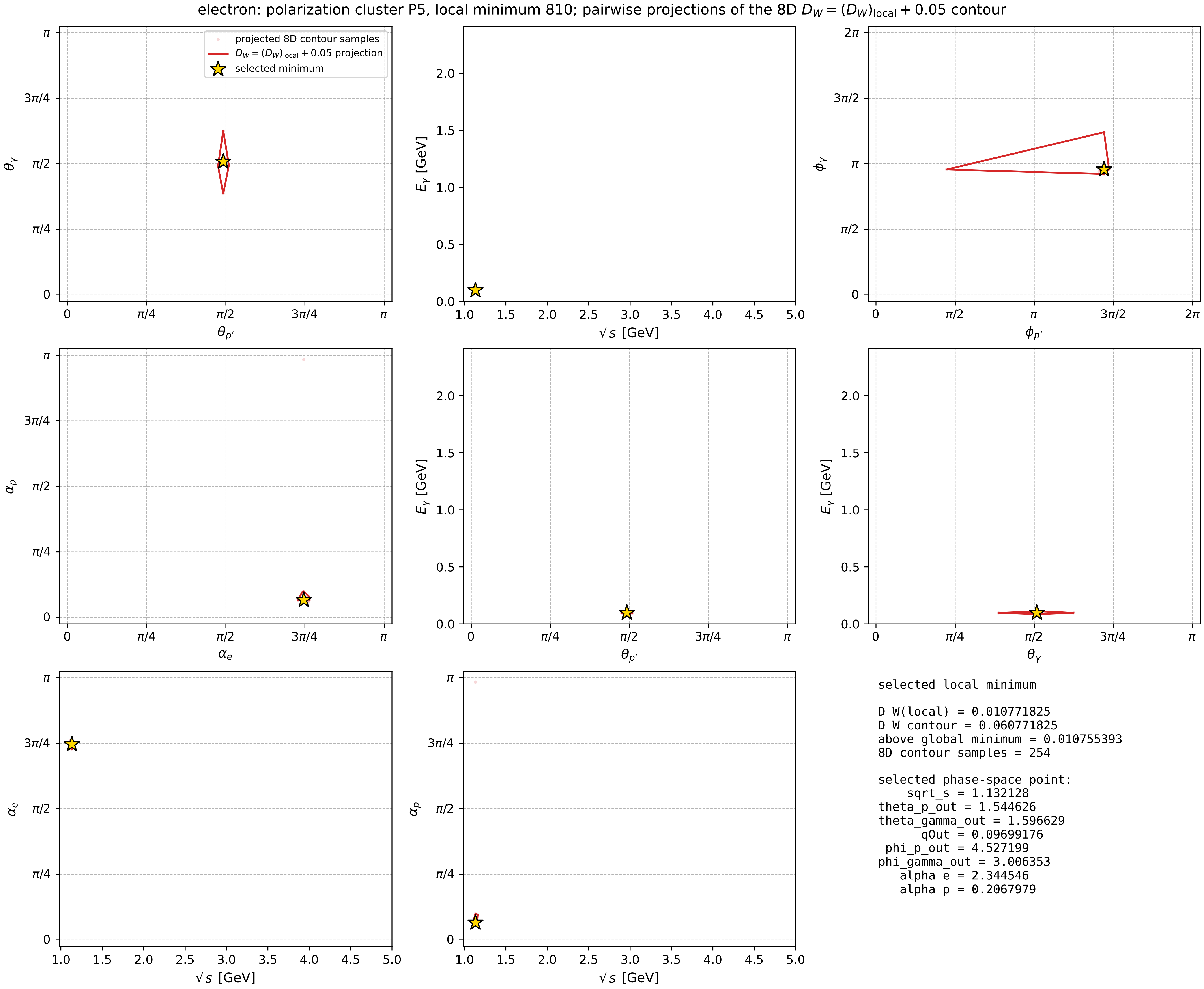
s=1.28171, $\sqrt{s}$=1.13213, $|\vec{P}|$=0.177484, $|\vec{P}'|$=0.00230333
$\theta_{p'}$=1.54463, θ_γ=1.59663, $\phi_{p'}$=4.5272, ϕ_γ=3.00635
E_γ=0.0969918, $\theta_{\ell'}$=1.54562, $\phi_{\ell'}$=6.17163
Q^2=0.0353469, x_B=0.162677, t=-0.0312075, W^2=1.06178, y=0.540677

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1775 p_x= 0.0000 p_y= 0.0000 p_z= -0.1775
P E= 0.9546 p_x= 0.0000 p_y= 0.0000 p_z= 0.1775
ℓ' E= 0.0971 p_x= 0.0965 p_y= -0.0108 p_z= 0.0024
P' E= 0.9380 p_x= -0.0004 p_y= -0.0023 p_z= 0.0001
q_γ E= 0.0970 p_x= -0.0961 p_y= 0.0131 p_z= -0.0025

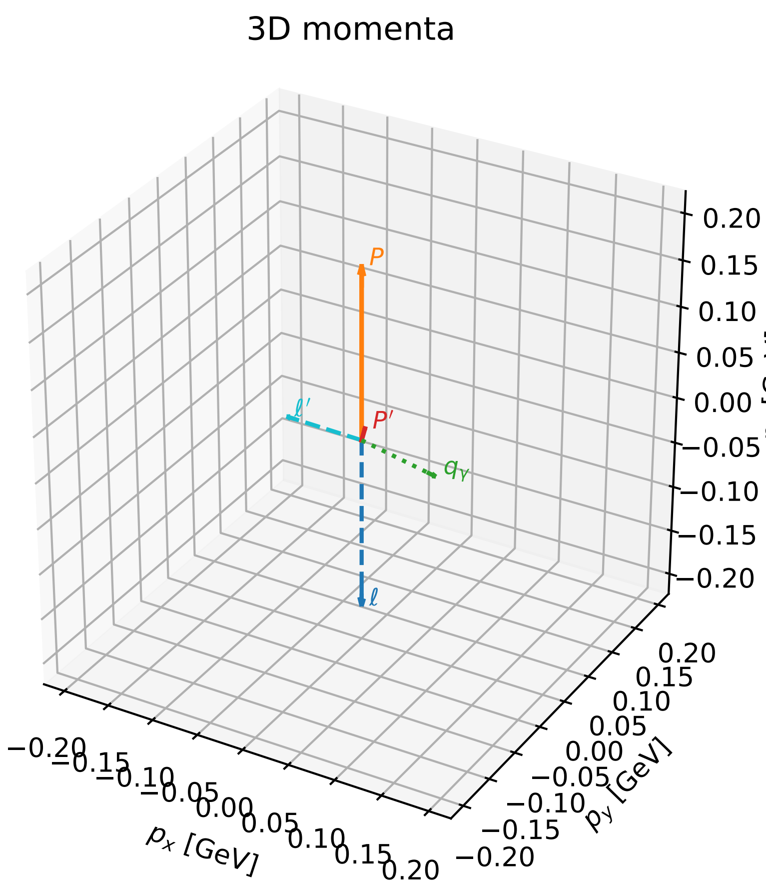







electron: polarization cluster P5, local minimum 810; pairwise projections of the 8D $D_W = (D_W)_{\rm local} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\rm local} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
θ_γ
selected local minimum
D_W(local) = 0.010771825
D_W contour = 0.060771825
above global minimum = 0.010755393
8D contour samples = 254
selected phase-space point:
sqrt_s = 1.132128
theta_p_out = 1.544626
theta_gamma_out = 1.596629
qOut = 0.09699176
phi_p_out = 4.527199
phi_gamma_out = 3.006353
alpha_e = 2.344546
alpha_p = 0.2067979

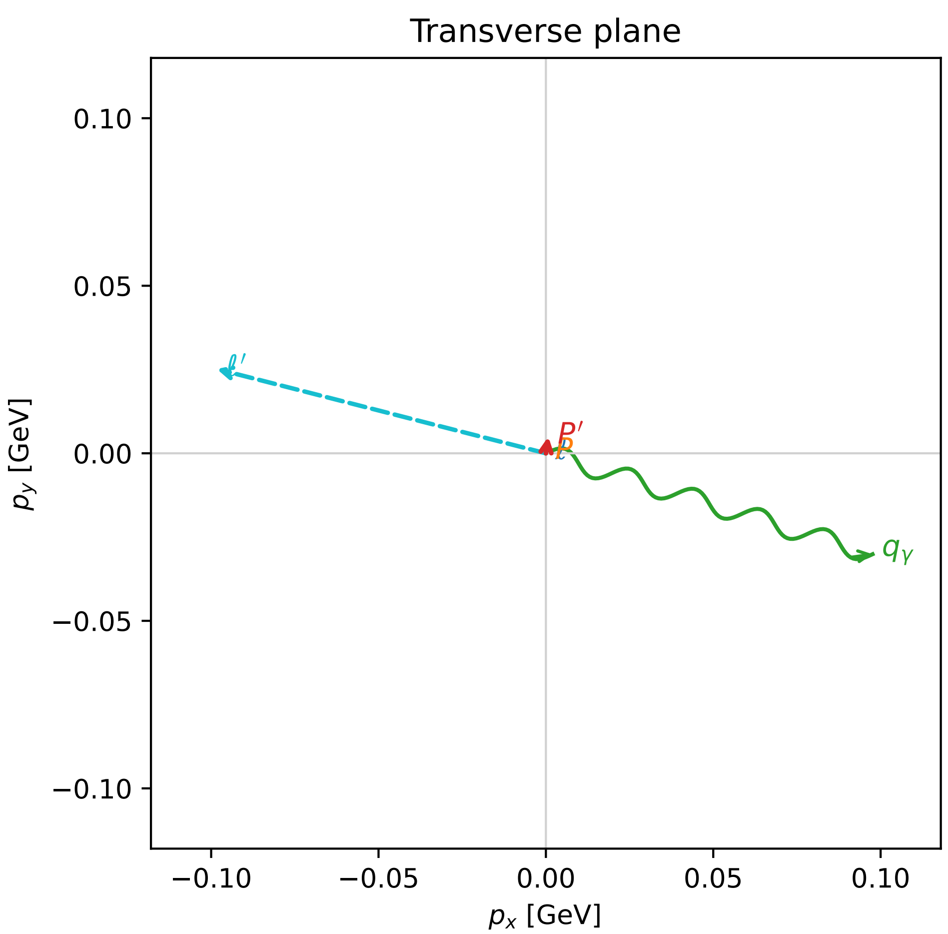
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

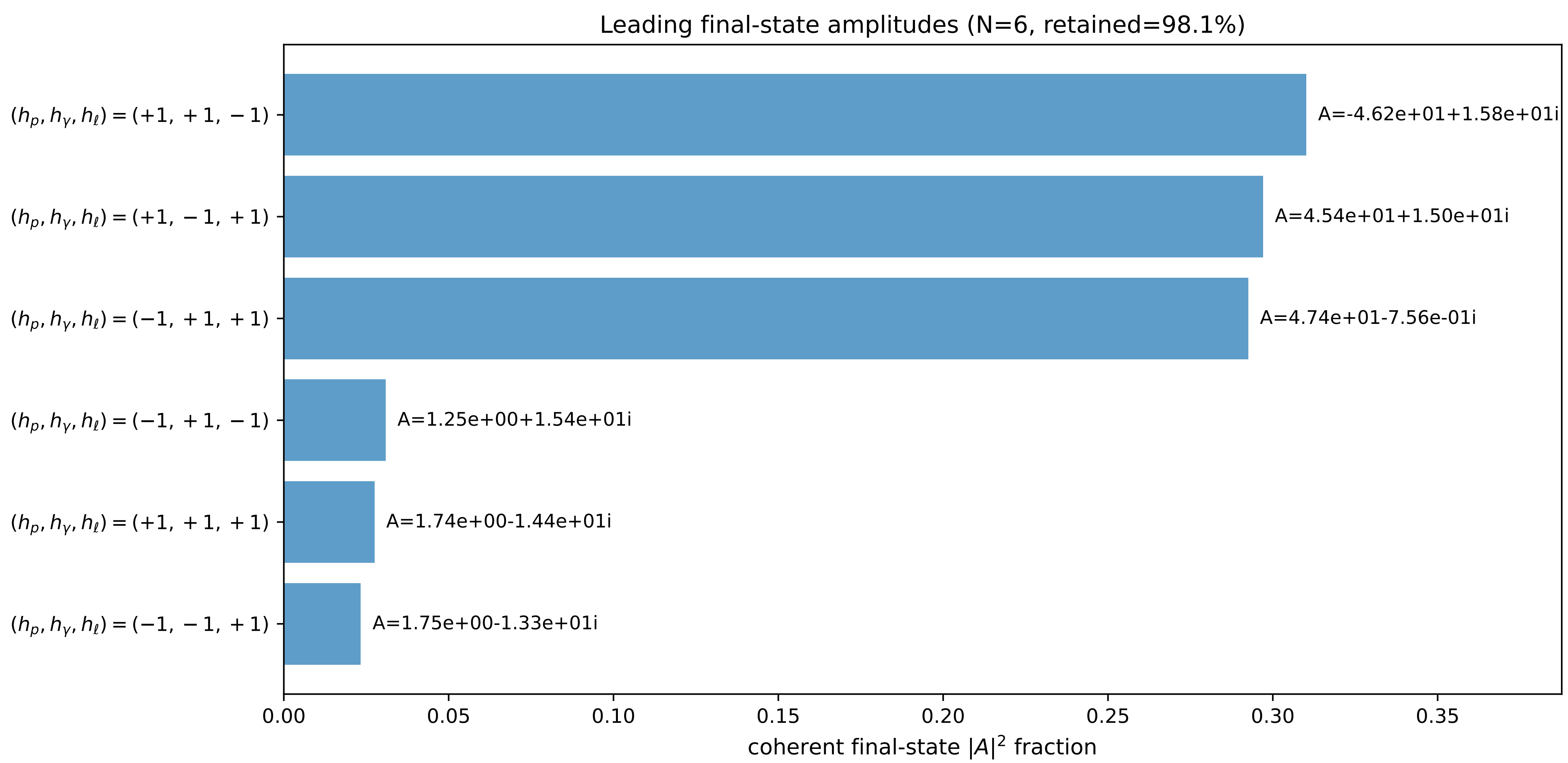


dw_mixing_angles_polarization_05_local_minimum_830 D_W=0.0113965
region: polarization_cluster_05_local_minimum_830
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3338616 rad
incoming lepton: -0.69114|+> +0.722721|->
proton mixing angle: alpha_p=2.9267632 rad
incoming proton: -0.977013|+> +0.213181|->

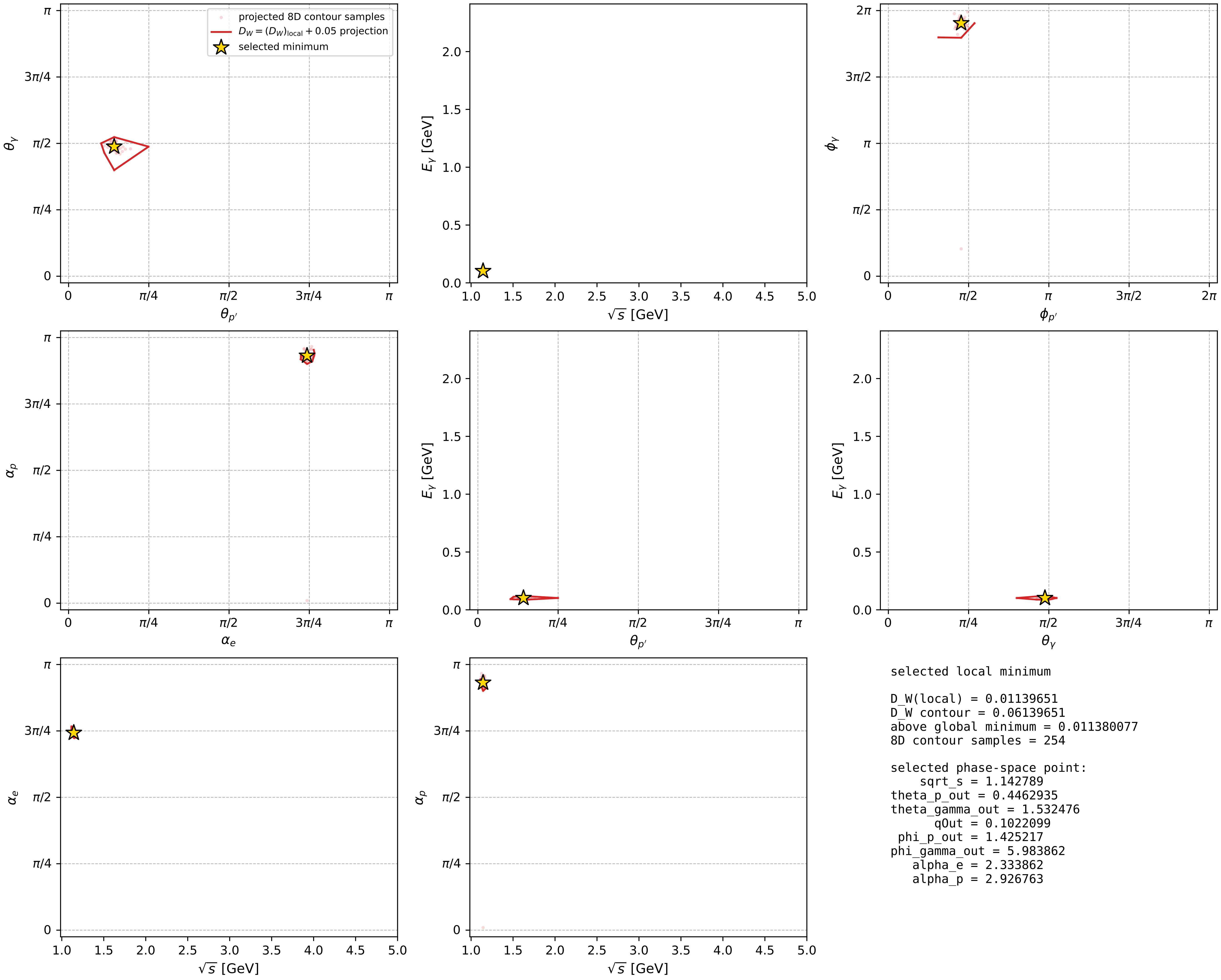
s=1.30597, $\sqrt{s}$=1.14279, $|\vec{P}|$=0.186439, $|\vec{P}'|$=0.0116342
$\theta_{p'}$=0.446293, θ_γ=1.53248, $\phi_{p'}$=1.42522, ϕ_γ=5.98386
E_γ=0.10221, $\theta_{\ell'}$=1.71185, $\phi_{\ell'}$=2.89118
Q^2=0.0328487, x_B=0.146199, t=-0.0306475, W^2=1.07168, y=0.527277

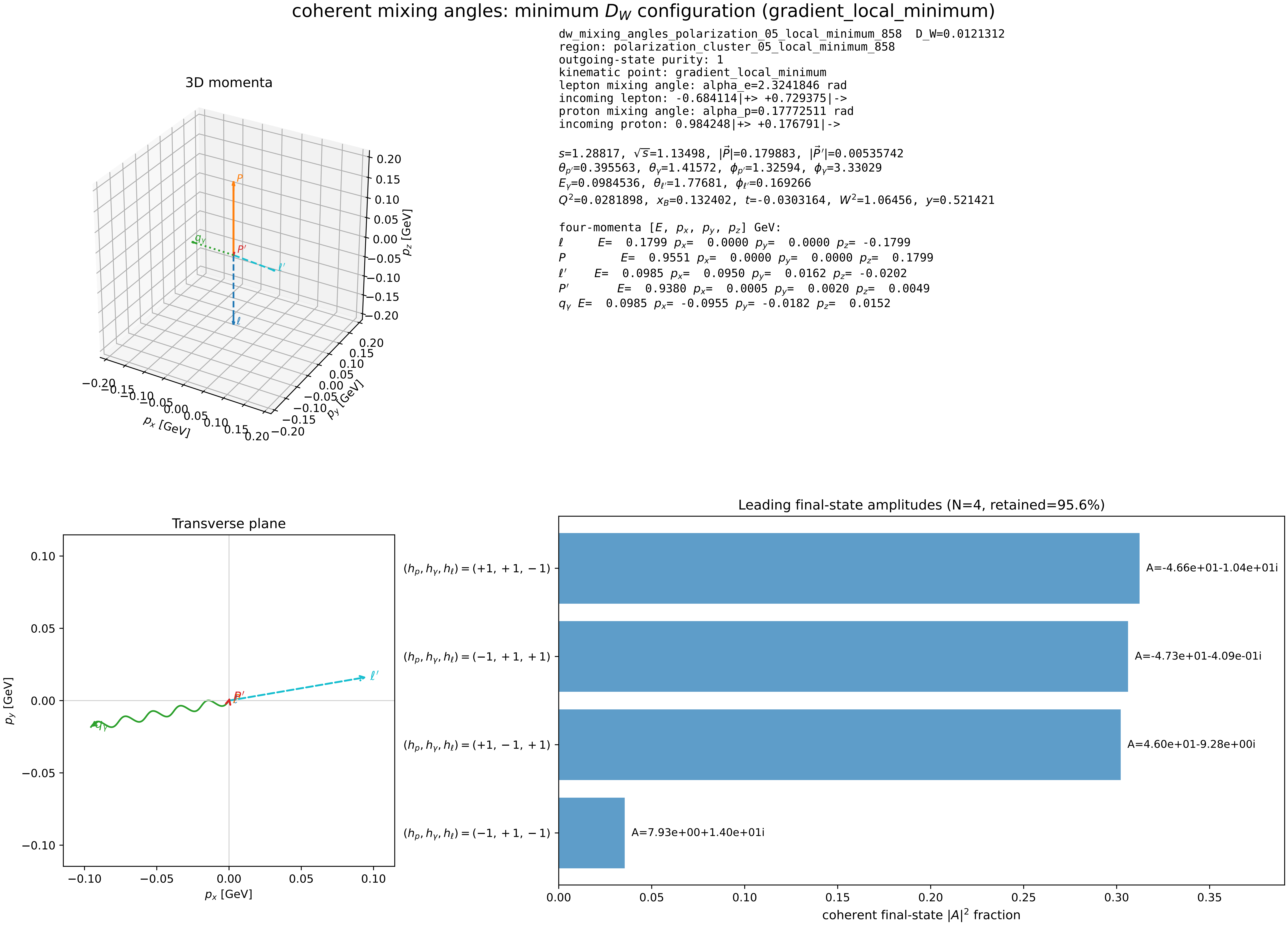
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1864 p_x= 0.0000 p_y= 0.0000 p_z= -0.1864
P E= 0.9563 p_x= 0.0000 p_y= 0.0000 p_z= 0.1864
ℓ' E= 0.1025 p_x= -0.0983 p_y= 0.0251 p_z= -0.0144
P' E= 0.9381 p_x= 0.0007 p_y= 0.0050 p_z= 0.0105
q_γ E= 0.1022 p_x= 0.0976 p_y= -0.0301 p_z= 0.0039



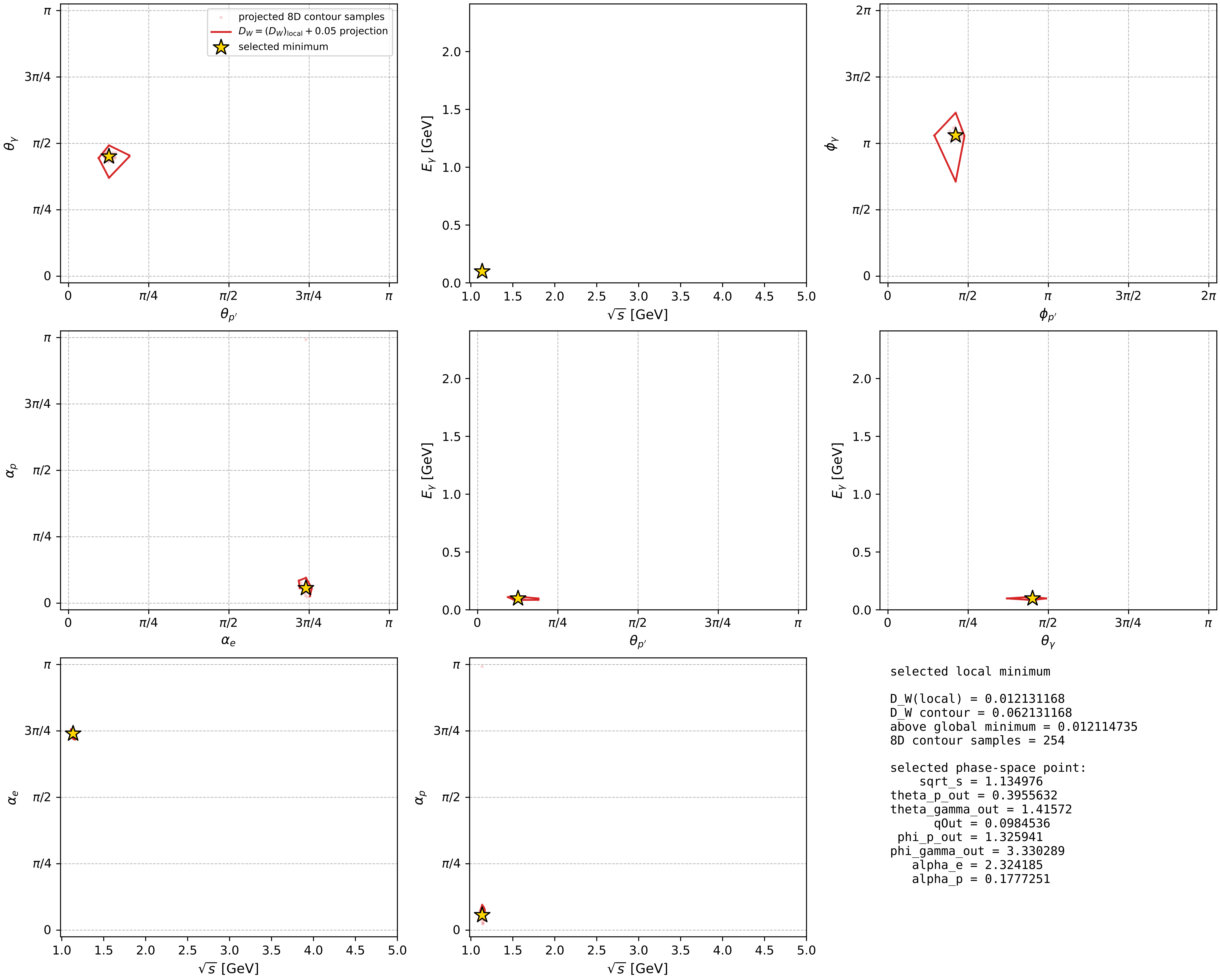


electron: polarization cluster P5, local minimum 830; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



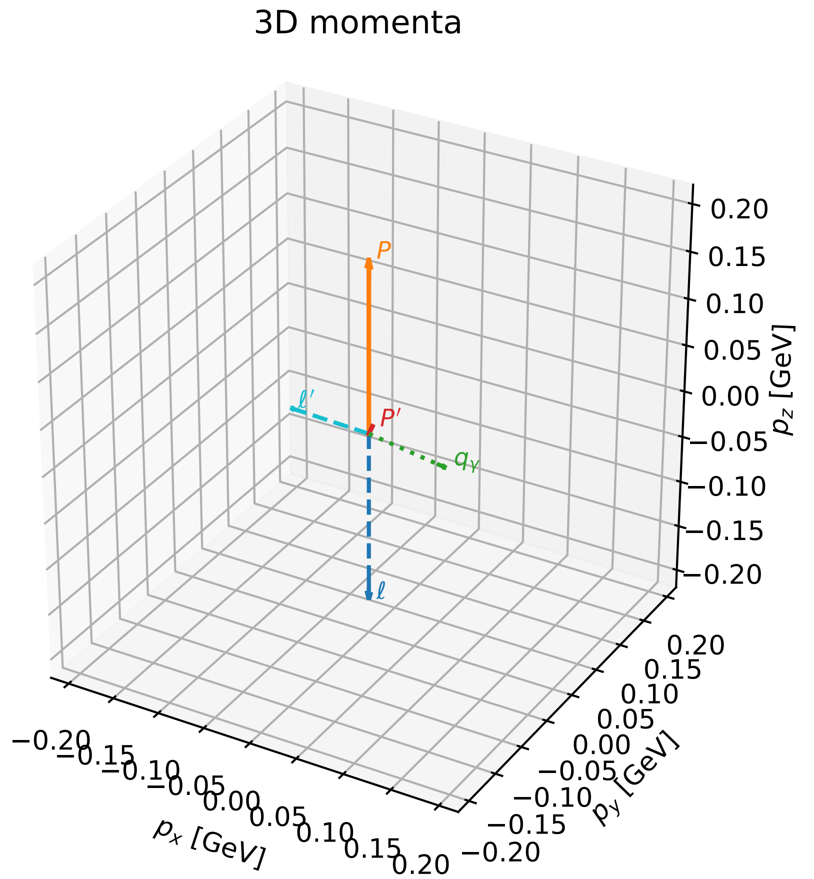

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_858 D_W=0.0121312
region: polarization_cluster_05_local_minimum_858
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3241846 rad
incoming lepton: -0.684114|+> +0.729375|->
proton mixing angle: alpha_p=0.17772511 rad
incoming proton: 0.984248|+> +0.176791|->
s=1.28817, $\sqrt{s}$=1.13498, $|\vec{P}|$=0.179883, $|\vec{P}'|$=0.00535742
$\theta_{p'}$=0.395563, θ_γ=1.41572, $\phi_{p'}$=1.32594, ϕ_γ=3.33029
E_γ=0.0984536, $\theta_{\ell'}$=1.77681, $\phi_{\ell'}$=0.169266
Q^2=0.0281898, x_B=0.132402, t=-0.0303164, W^2=1.06456, y=0.521421
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1799 p_x= 0.0000 p_y= 0.0000 p_z= -0.1799
P E= 0.9551 p_x= 0.0000 p_y= 0.0000 p_z= 0.1799
ℓ' E= 0.0985 p_x= 0.0950 p_y= 0.0162 p_z= -0.0202
P' E= 0.9380 p_x= 0.0005 p_y= 0.0020 p_z= 0.0049
q_γ E= 0.0985 p_x= -0.0955 p_y= -0.0182 p_z= 0.0152
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=4, retained=95.6%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-4.66e+01-1.04e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-4.73e+01-4.09e-01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=4.60e+01-9.28e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=7.93e+00+1.40e+01i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P5, local minimum 858; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour




```
selected local minimum

D_W(local) = 0.012131168
D_W contour = 0.062131168
above global minimum = 0.012114735
8D contour samples = 254

selected phase-space point:
    sqrt_s = 1.134976
theta_p_out = 0.3955632
theta_gamma_out = 1.41572
      qOut = 0.0984536
 phi_p_out = 1.325941
phi_gamma_out = 3.330289
   alpha_e = 2.324185
   alpha_p = 0.1777251
```

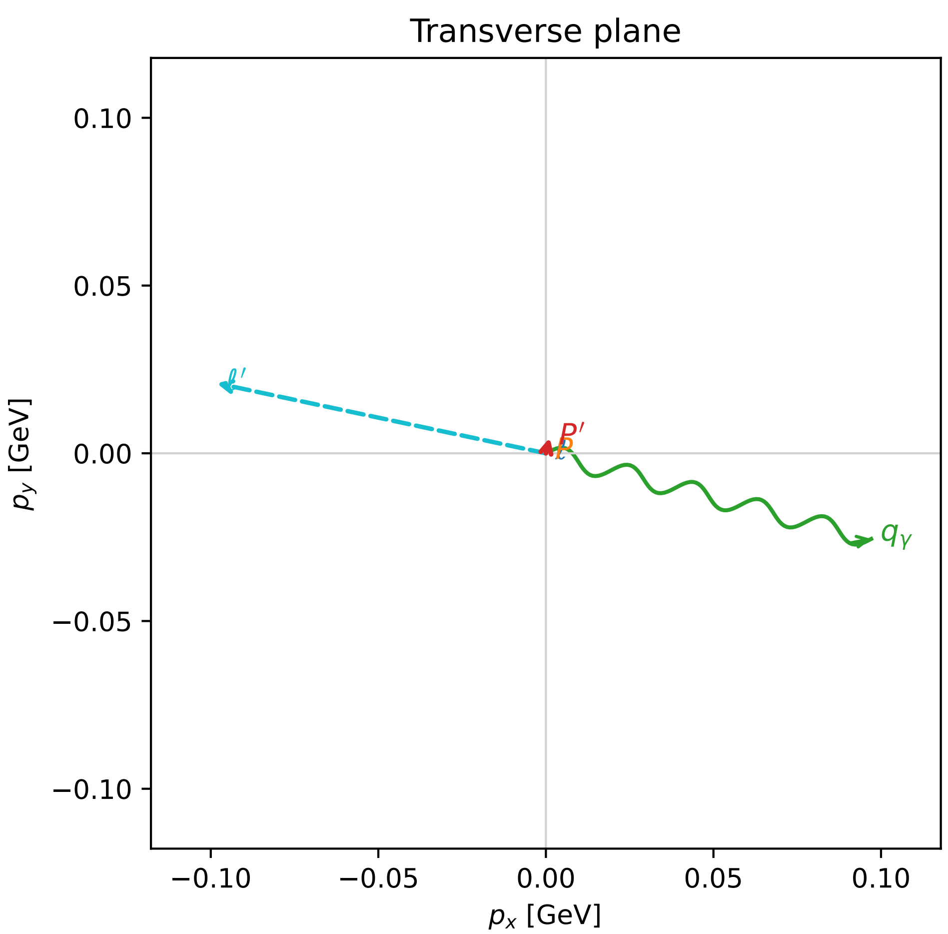
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

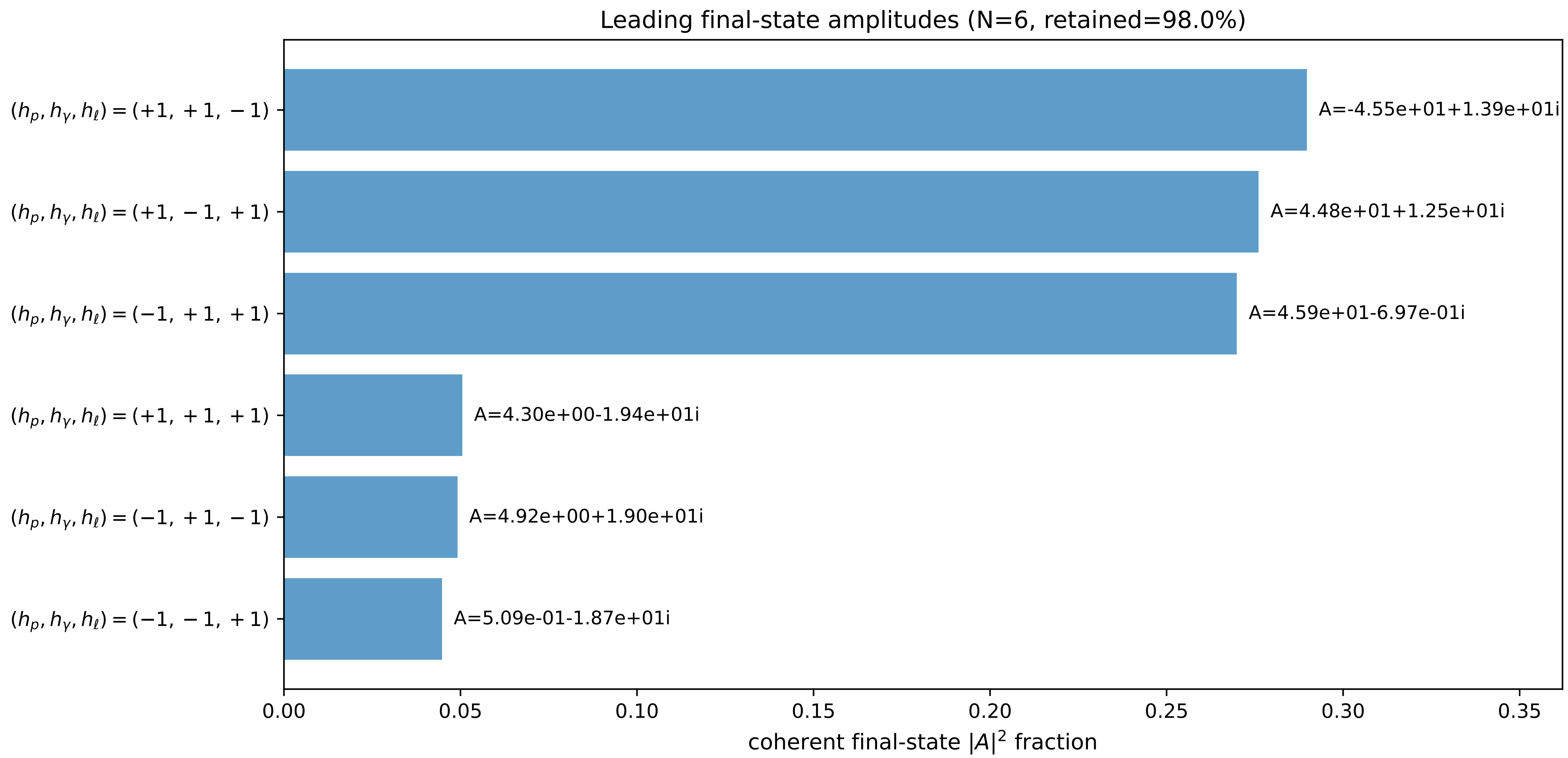


dw_mixing_angles_polarization_05_local_minimum_865 D_W=0.0122529
region: polarization_cluster_05_local_minimum_865
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.337102 rad
incoming lepton: -0.693478|+> +0.720477|->
proton mixing angle: alpha_p=2.9207102 rad
incoming proton: -0.975704|+> +0.219091|->

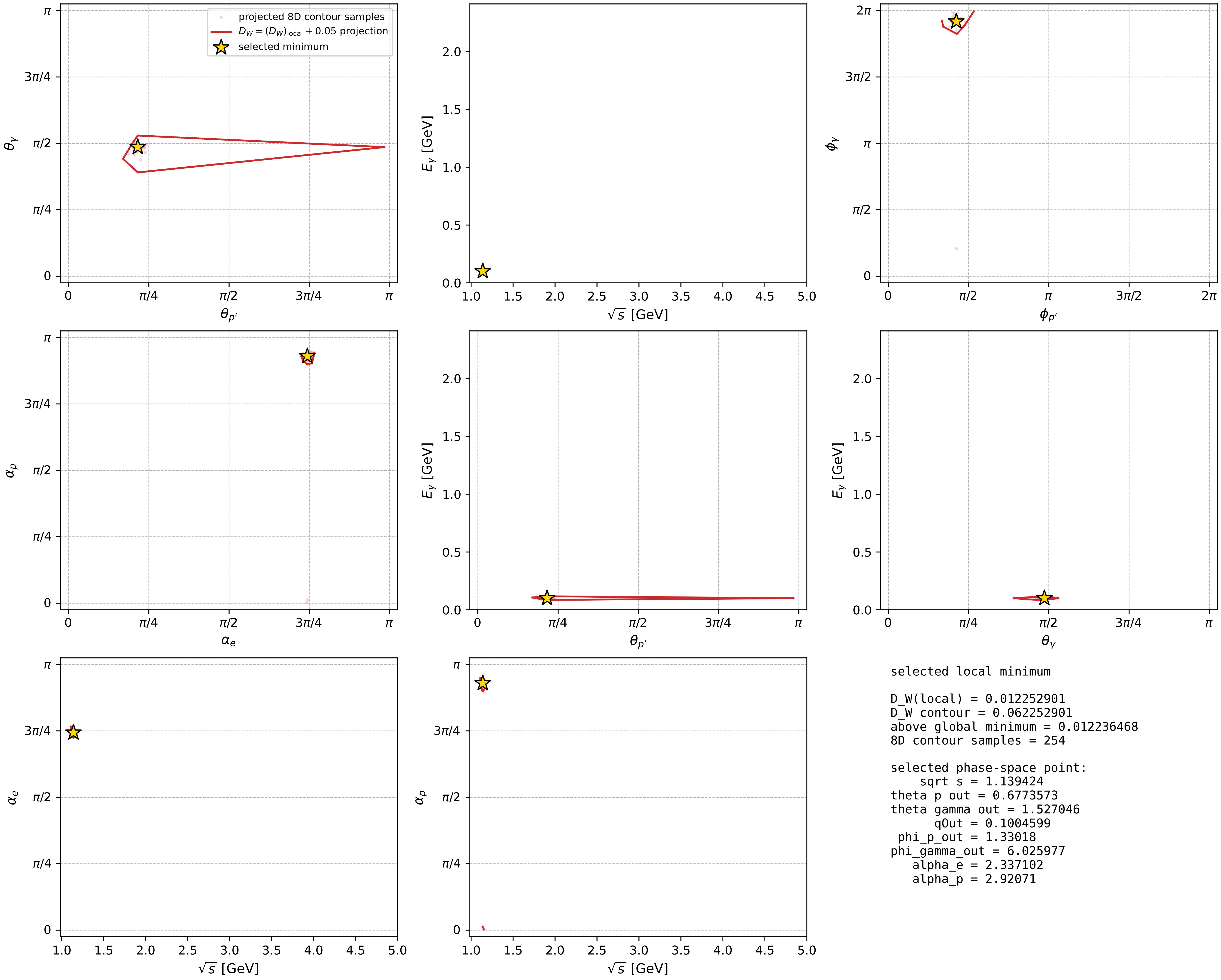
s=1.29829, $\sqrt{s}$=1.13942, $|\vec{P}|$=0.18362, $|\vec{P}'|$=0.00765613
$\theta_{p'}$=0.677357, θ_γ=1.52705, $\phi_{p'}$=1.33018, ϕ_γ=6.02598
E_γ=0.10046, $\theta_{\ell'}$=1.67362, $\phi_{\ell'}$=2.93219
Q^2=0.0332619, x_B=0.150035, t=-0.0312682, W^2=1.06828, y=0.529807

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1836 p_x= 0.0000 p_y= 0.0000 p_z= -0.1836
P E= 0.9558 p_x= 0.0000 p_y= 0.0000 p_z= 0.1836
ℓ' E= 0.1009 p_x= -0.0982 p_y= 0.0209 p_z= -0.0104
P' E= 0.9380 p_x= 0.0011 p_y= 0.0047 p_z= 0.0060
q_γ E= 0.1005 p_x= 0.0971 p_y= -0.0255 p_z= 0.0044

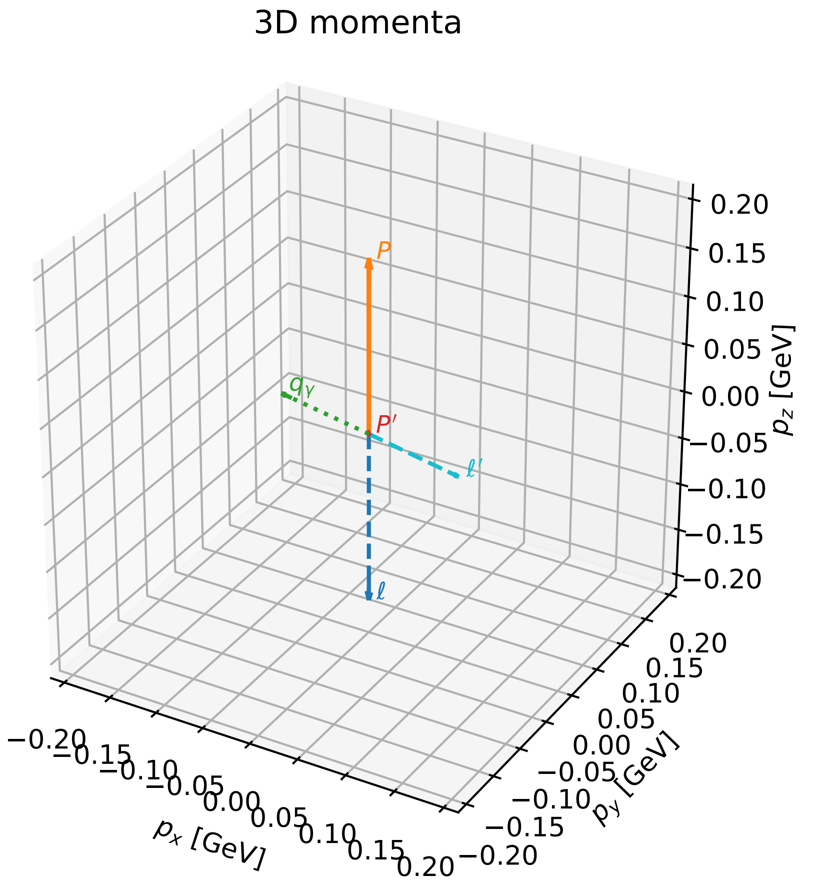




electron: polarization cluster P5, local minimum 865; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



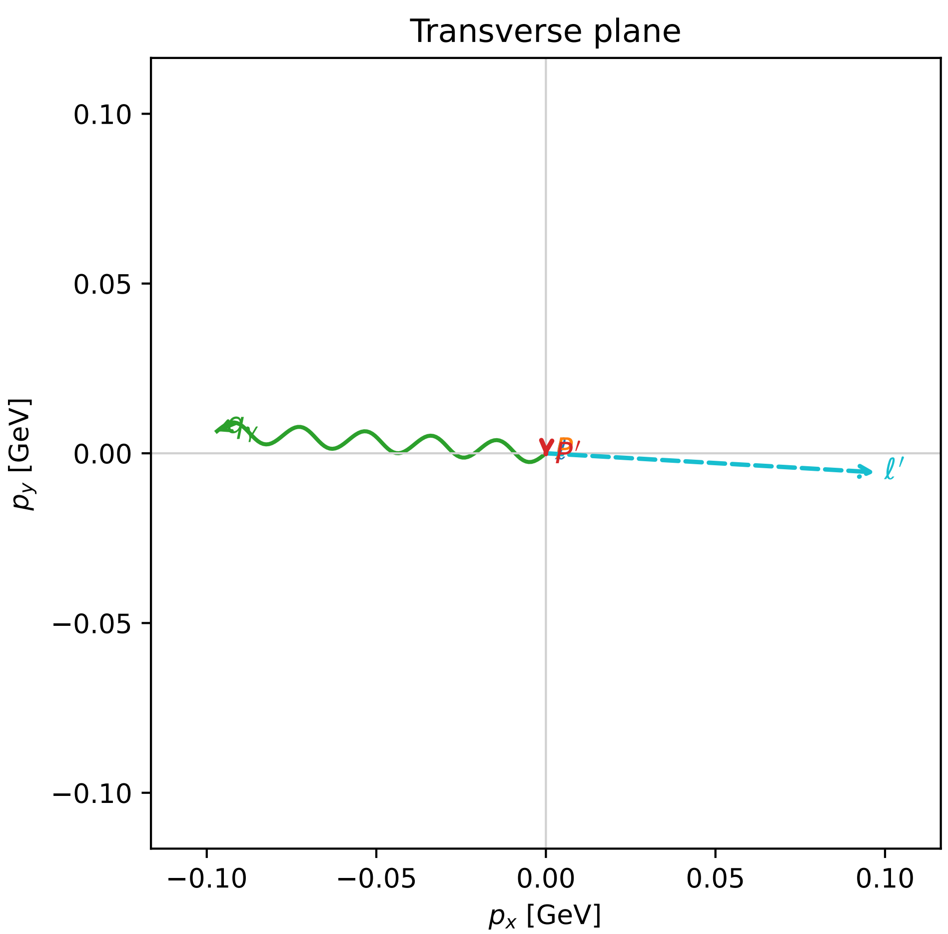
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

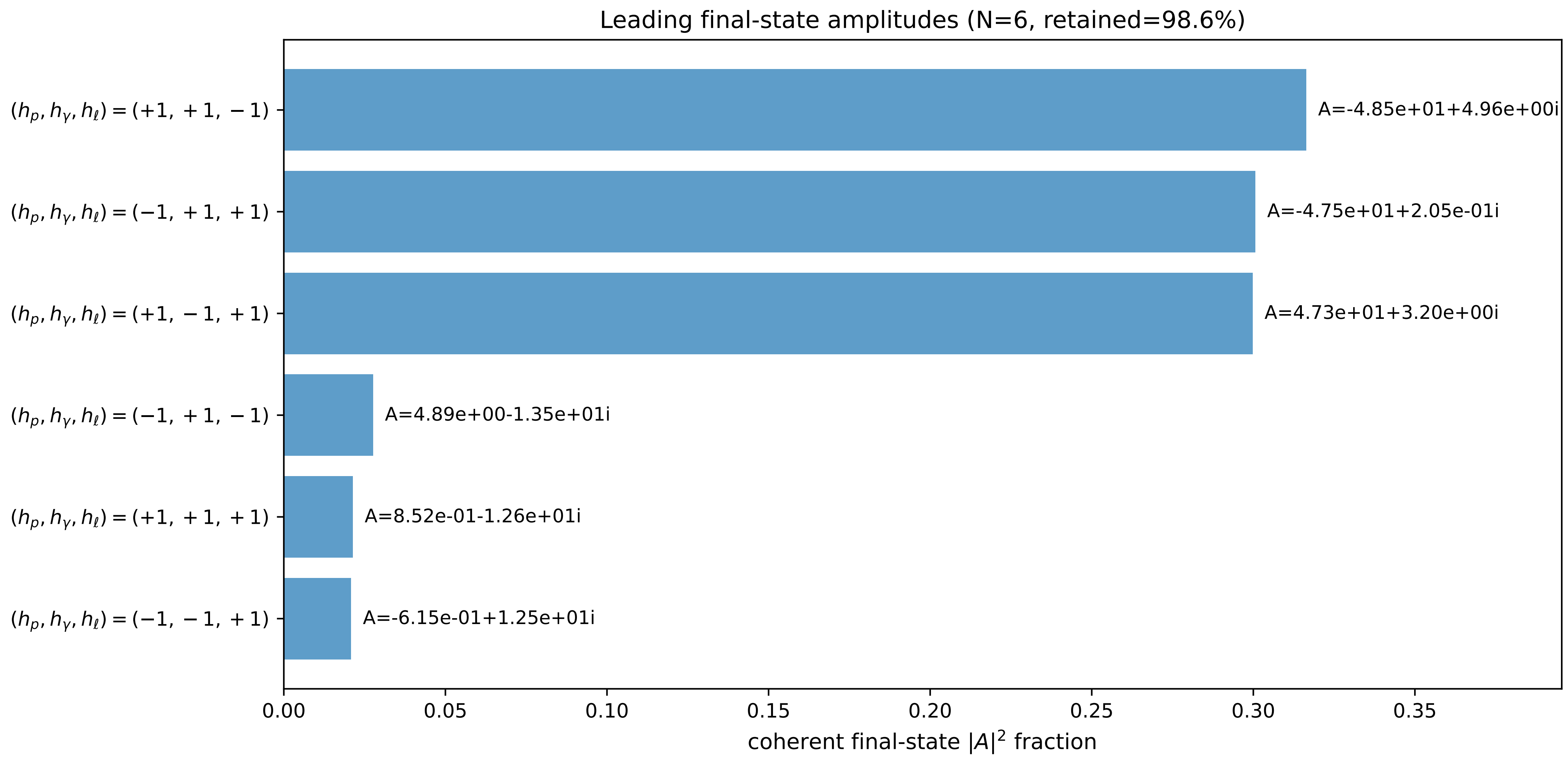


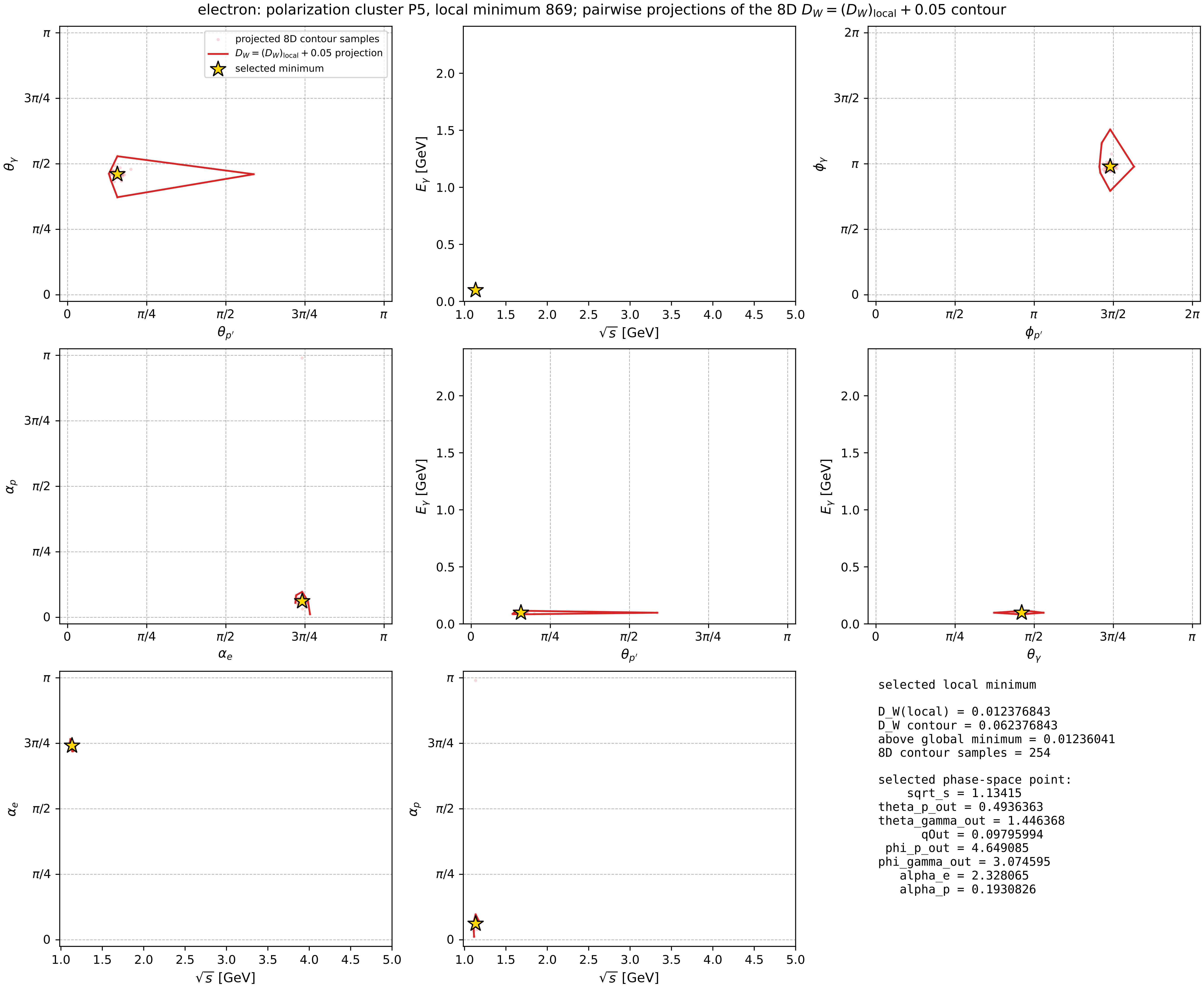
dw_mixing_angles_polarization_05_local_minimum_869 D_W=0.0123768
region: polarization_cluster_05_local_minimum_869
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3280646 rad
incoming lepton: -0.686939|+> +0.726715|->
proton mixing angle: alpha_p=0.19308263 rad
incoming proton: 0.981417|+> +0.191885|->

s=1.2863, $\sqrt{s}$=1.13415, $|\vec{P}|$=0.179187, $|\vec{P}'|$=0.00193206
$\theta_{p'}$=0.493636, θ_γ=1.44637, $\phi_{p'}$=4.64909, ϕ_γ=3.07459
E_γ=0.0979599, $\theta_{\ell'}$=1.71242, $\phi_{\ell'}$=6.22561
Q^2=0.030221, x_B=0.14125, t=-0.0312144, W^2=1.06358, y=0.526394

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1792 p_x= 0.0000 p_y= 0.0000 p_z= -0.1792
P E= 0.9550 p_x= 0.0000 p_y= 0.0000 p_z= 0.1792
ℓ' E= 0.0982 p_x= 0.0970 p_y= -0.0056 p_z= -0.0139
P' E= 0.9380 p_x= -0.0001 p_y= -0.0009 p_z= 0.0017
q_γ E= 0.0980 p_x= -0.0970 p_y= 0.0065 p_z= 0.0122

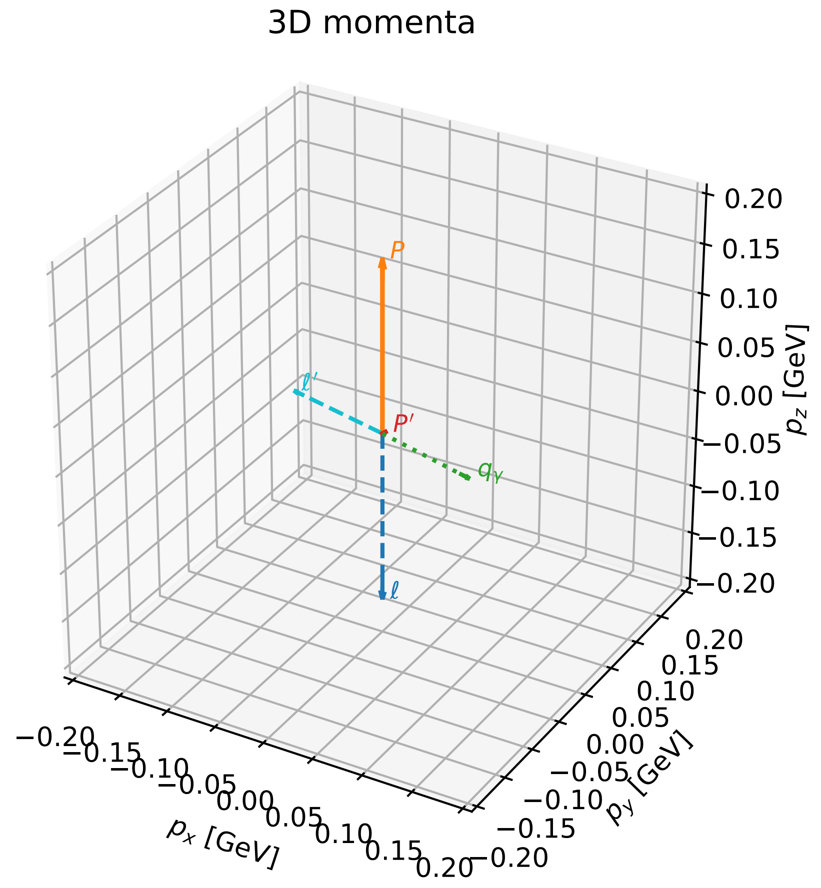






electron: polarization cluster P5, local minimum 869; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.012376843
D_W contour = 0.062376843
above global minimum = 0.01236041
8D contour samples = 254
selected phase-space point:
sqrt_s = 1.13415
theta_p_out = 0.4936363
theta_gamma_out = 1.446368
qOut = 0.09795994
phi_p_out = 4.649085
phi_gamma_out = 3.074595
alpha_e = 2.328065
alpha_p = 0.1930826

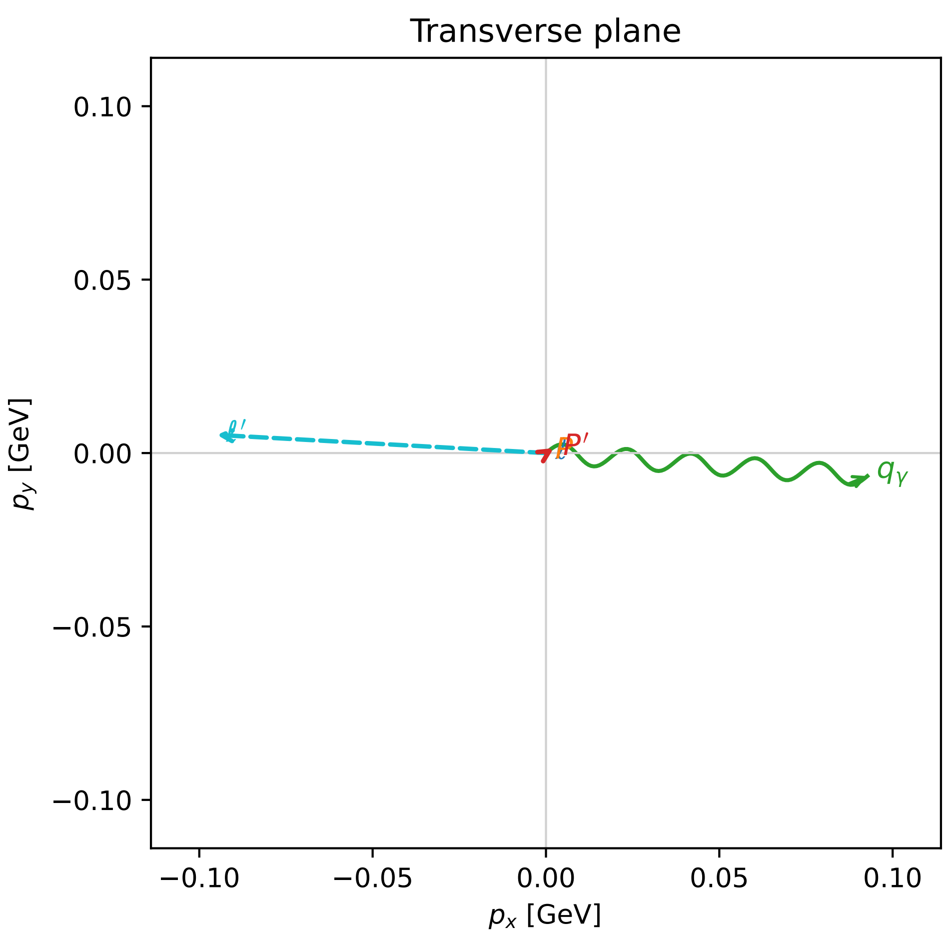
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

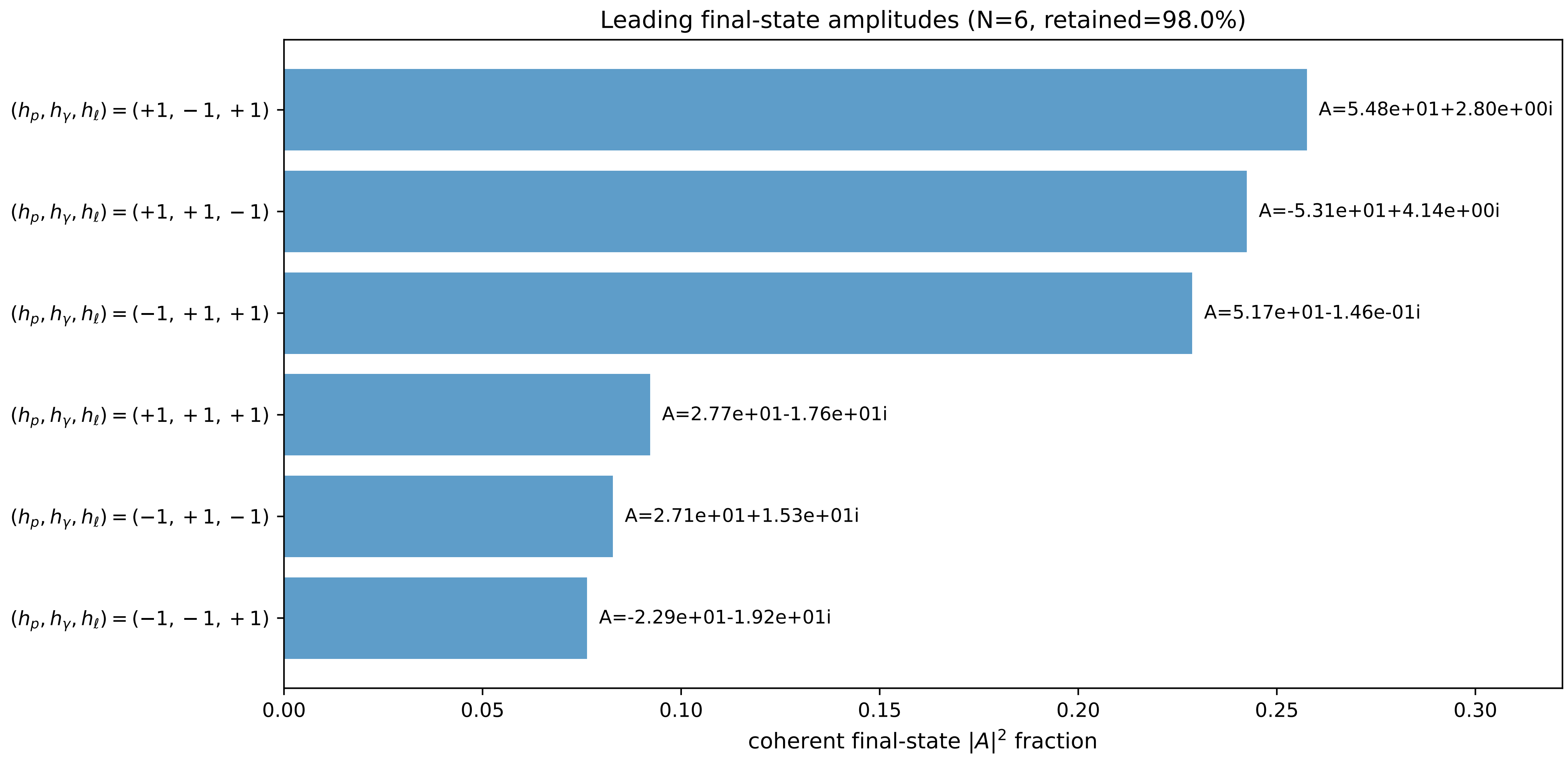


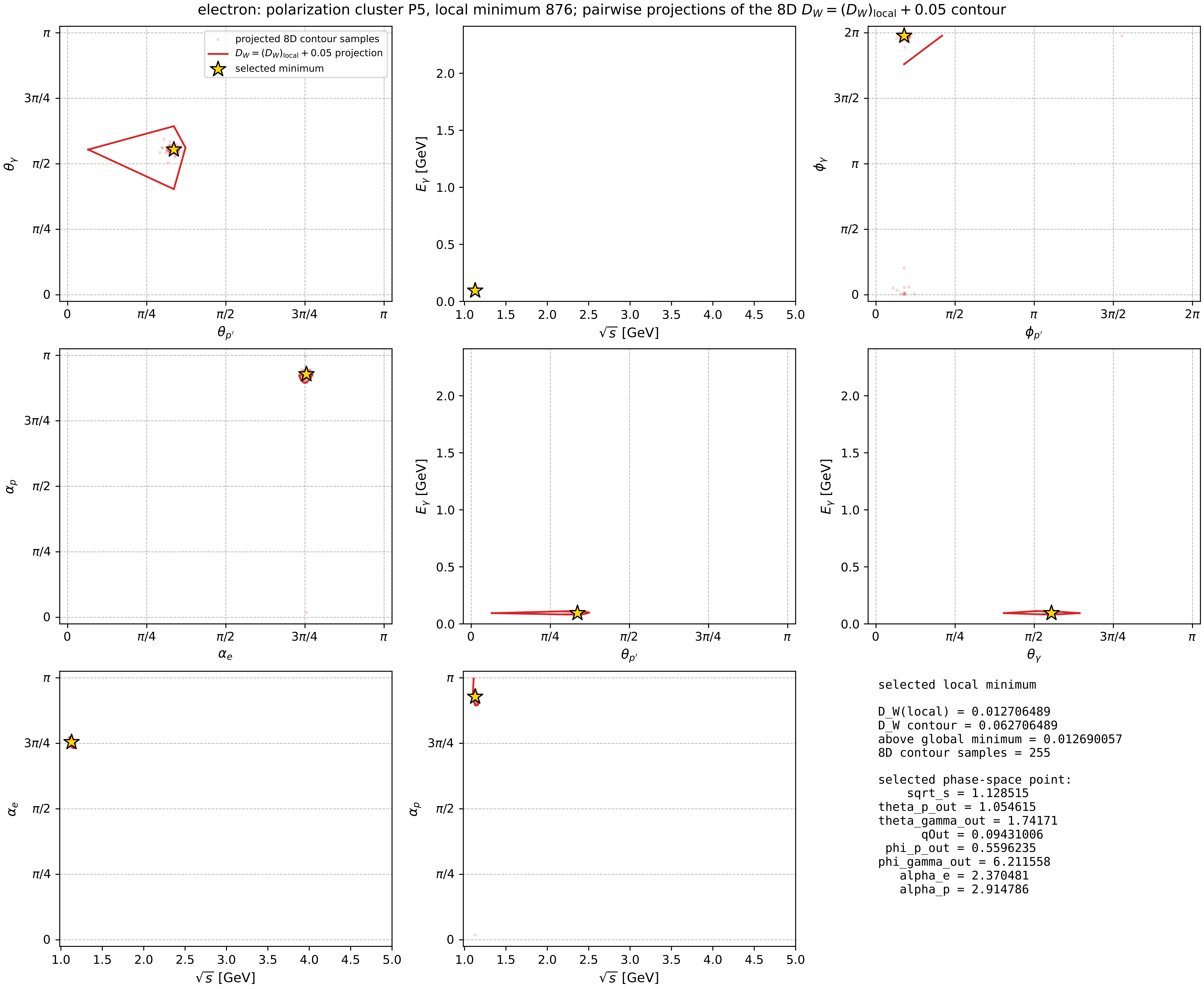
dw_mixing_angles_polarization_05_local_minimum_876 D_W=0.0127065
region: polarization_cluster_05_local_minimum_876
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3704813 rad
incoming lepton: -0.717137|+> +0.696933|->
proton mixing angle: alpha_p=2.9147855 rad
incoming proton: -0.974389|+> +0.224868|->

s=1.27355, $\sqrt{s}$=1.12852, $|\vec{P}|$=0.174433, $|\vec{P}'|$=0.00305684
$\theta_{p'}$=1.05461, θ_γ=1.74171, $\phi_{p'}$=0.559624, ϕ_γ=6.21156
E_γ=0.0943101, $\theta_{\ell'}$=1.41916, $\phi_{\ell'}$=3.08647
Q^2=0.0386304, x_B=0.179504, t=-0.0296516, W^2=1.05642, y=0.546622

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1744 p_x= 0.0000 p_y= 0.0000 p_z= -0.1744
P E= 0.9541 p_x= 0.0000 p_y= 0.0000 p_z= 0.1744
ℓ' E= 0.0962 p_x= -0.0950 p_y= 0.0052 p_z= 0.0145
P' E= 0.9380 p_x= 0.0023 p_y= 0.0014 p_z= 0.0015
q_γ E= 0.0943 p_x= 0.0927 p_y= -0.0067 p_z= -0.0160





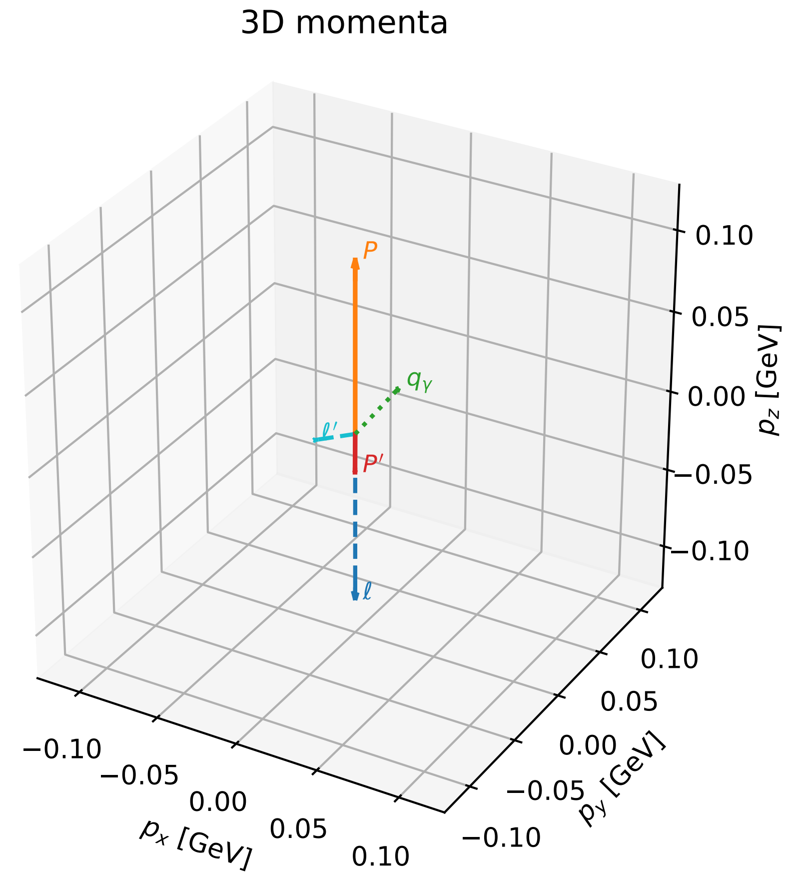

electron: polarization cluster P5, local minimum 876; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.012706489
D_W contour = 0.062706489
above global minimum = 0.012690057
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.128515
theta_p_out = 1.054615
theta_gamma_out = 1.74171
qOut = 0.09431006
phi_p_out = 0.5596235
phi_gamma_out = 6.211558
alpha_e = 2.370481
alpha_p = 2.914786

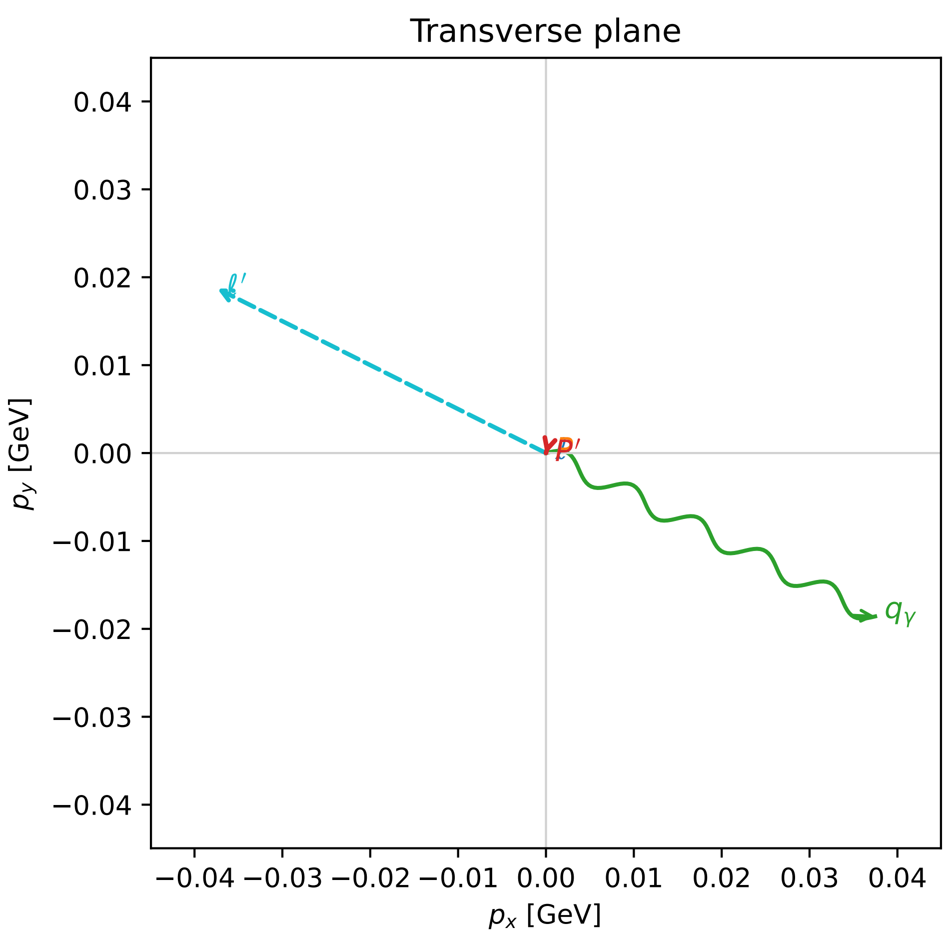
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

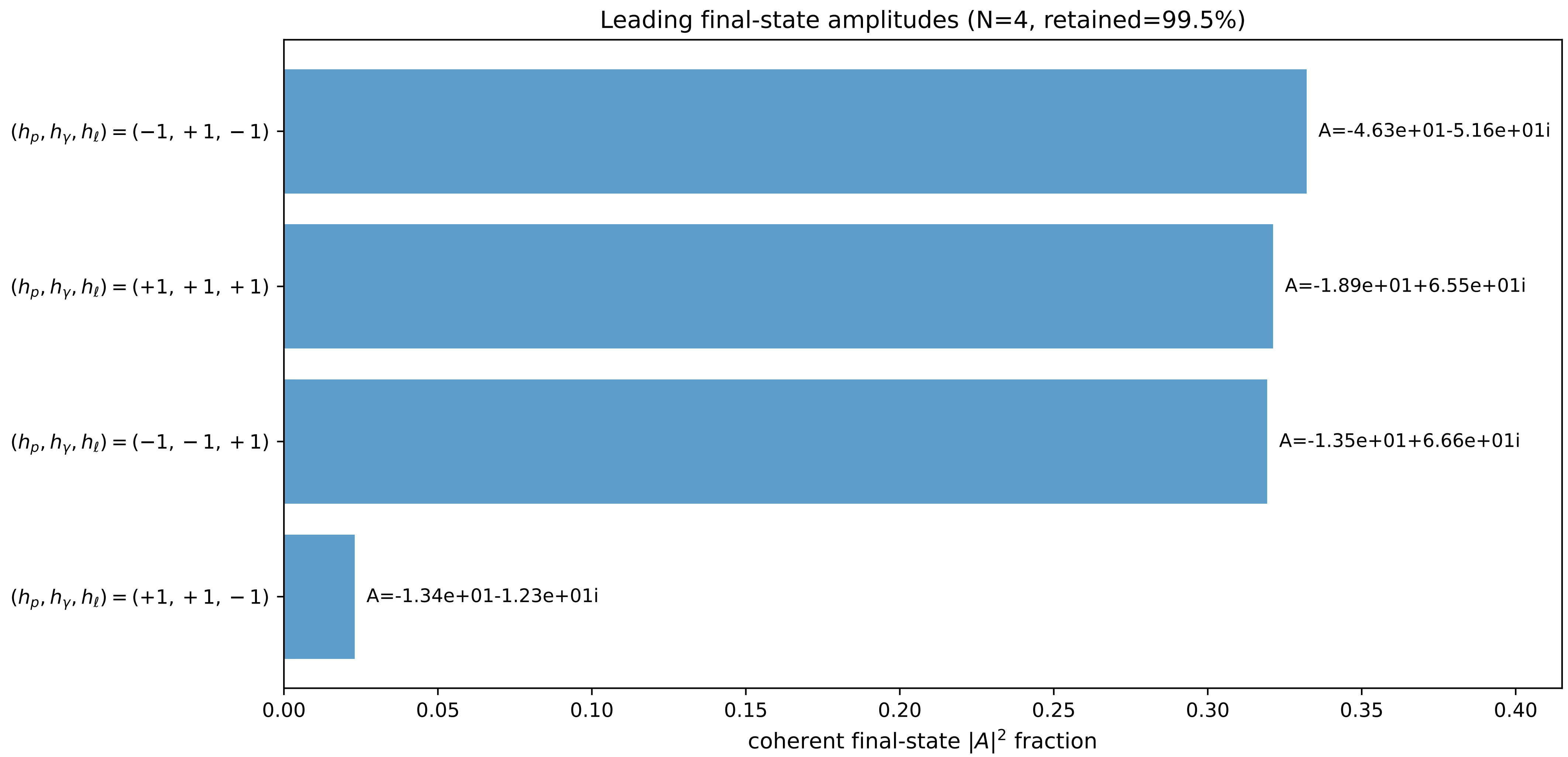
dw_mixing_angles_polarization_05_local_minimum_898 D_W=0.0135939
region: polarization_cluster_05_local_minimum_898
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3381561 rad
incoming lepton: -0.694237|+> +0.719746|->
proton mixing angle: alpha_p=3.0405209 rad
incoming proton: -0.994897|+> +0.1009|->

s=1.10334, $\sqrt{s}$=1.0504, $|\vec{P}|$=0.106383, $|\vec{P}'|$=0.0239989
$\theta_{p'}$=3.13575, θ_γ=0.715162, $\phi_{p'}$=4.43498, ϕ_γ=5.82273
E_γ=0.0637743, $\theta_{\ell'}$=2.09423, $\phi_{\ell'}$=2.67785
Q^2=0.00514141, x_B=0.0404422, t=-0.0169669, W^2=1.00183, y=0.568836

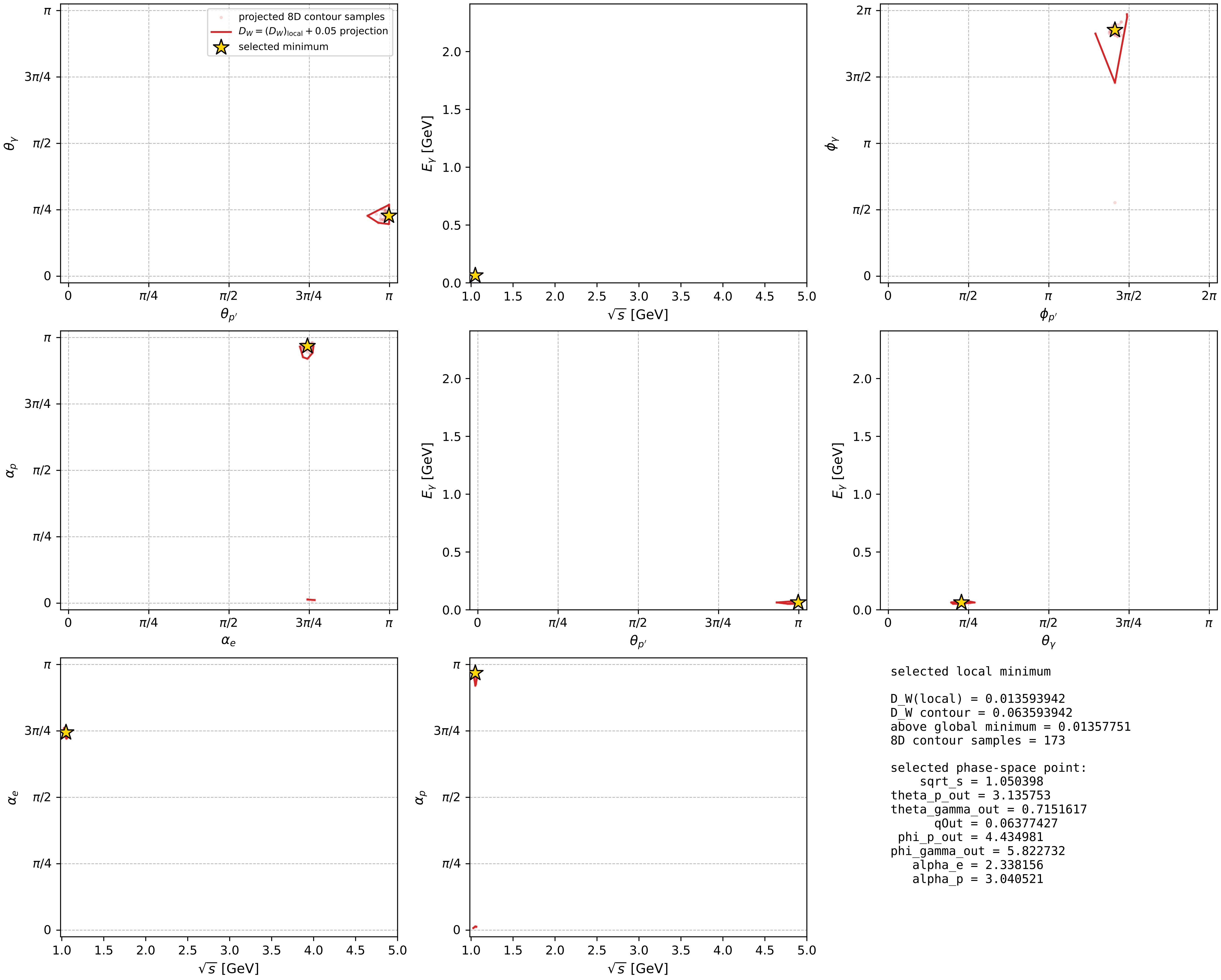
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1064 p_x= 0.0000 p_y= 0.0000 p_z= -0.1064
P E= 0.9440 p_x= 0.0000 p_y= 0.0000 p_z= 0.1064
ℓ' E= 0.0483 p_x= -0.0374 p_y= 0.0187 p_z= -0.0242
P' E= 0.9383 p_x= -0.0000 p_y= -0.0001 p_z= -0.0240
q_γ E= 0.0638 p_x= 0.0375 p_y= -0.0186 p_z= 0.0481

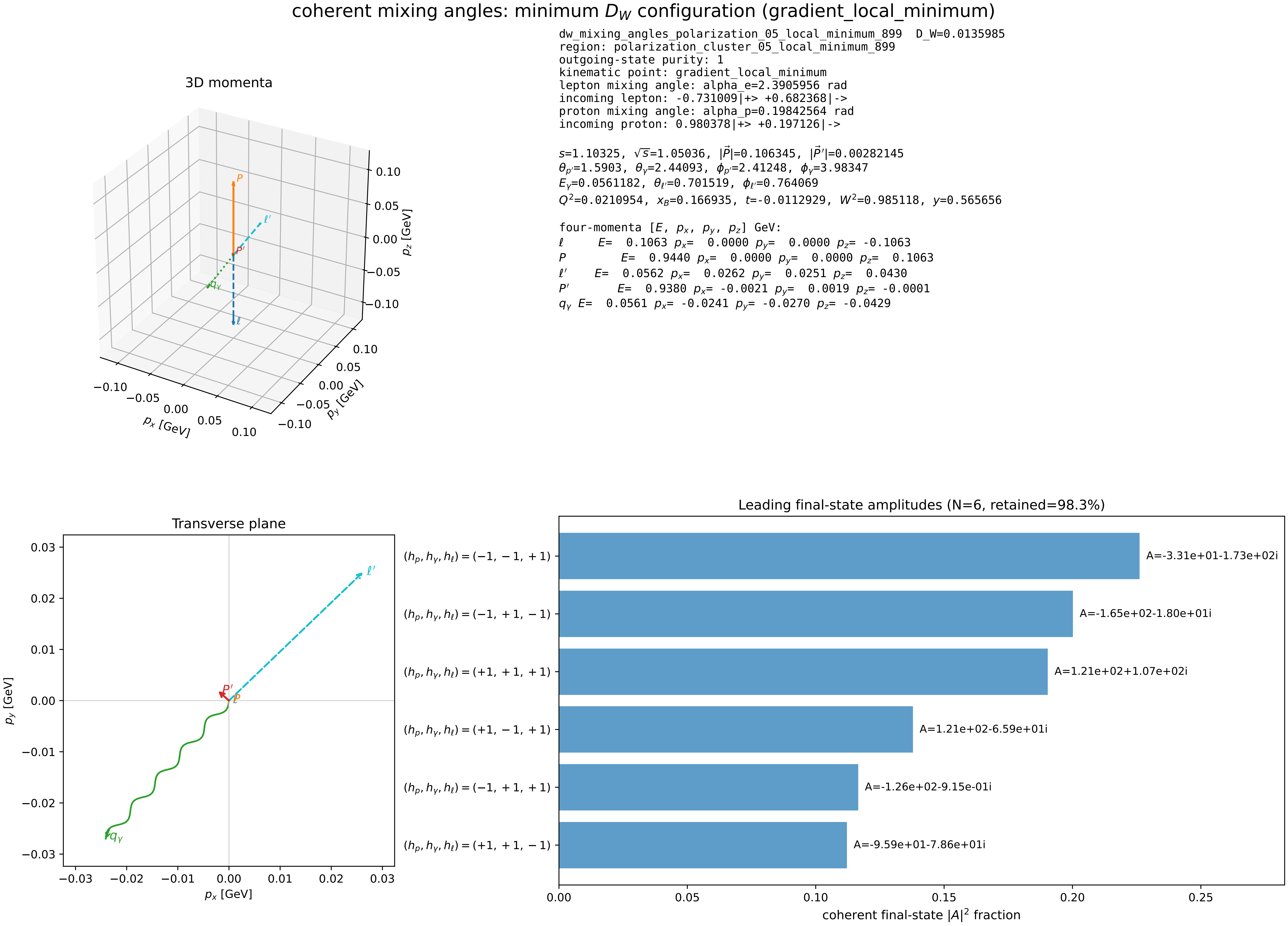




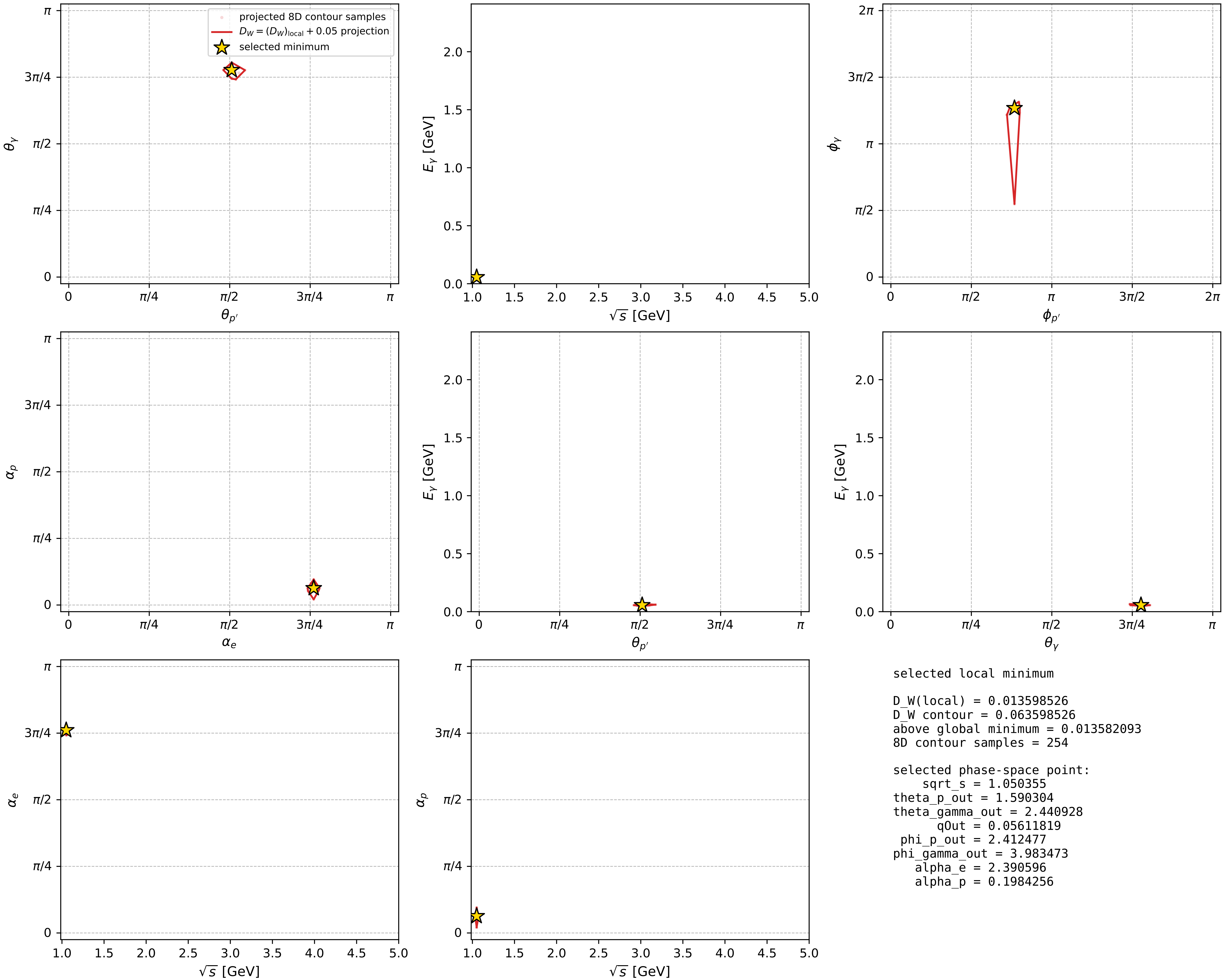


electron: polarization cluster P5, local minimum 898; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour




coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_899 D_W=0.0135985
region: polarization_cluster_05_local_minimum_899
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3905956 rad
incoming lepton: -0.731009|+> +0.682368|->
proton mixing angle: alpha_p=0.19842564 rad
incoming proton: 0.980378|+> +0.197126|->
s=1.10325, $\sqrt{s}$=1.05036, $|\vec{P}|$=0.106345, $|\vec{P}'|$=0.00282145
$\theta_{p'}$=1.5903, θ_γ=2.44093, $\phi_{p'}$=2.41248, ϕ_γ=3.98347
E_γ=0.0561182, $\theta_{\ell'}$=0.701519, $\phi_{\ell'}$=0.764069
Q^2=0.0210954, x_B=0.166935, t=-0.0112929, W^2=0.985118, y=0.565656
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1063 p_x= 0.0000 p_y= 0.0000 p_z= -0.1063
P E= 0.9440 p_x= 0.0000 p_y= 0.0000 p_z= 0.1063
ℓ' E= 0.0562 p_x= 0.0262 p_y= 0.0251 p_z= 0.0430
P' E= 0.9380 p_x= -0.0021 p_y= 0.0019 p_z= -0.0001
q_γ E= 0.0561 p_x= -0.0241 p_y= -0.0270 p_z= -0.0429
3D momenta
P
ℓ'
P'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=98.3%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-3.31e+01-1.73e+02i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-1.65e+02-1.80e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=1.21e+02+1.07e+02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=1.21e+02-6.59e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-1.26e+02-9.15e-01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-9.59e+01-7.86e+01i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P5, local minimum 899; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



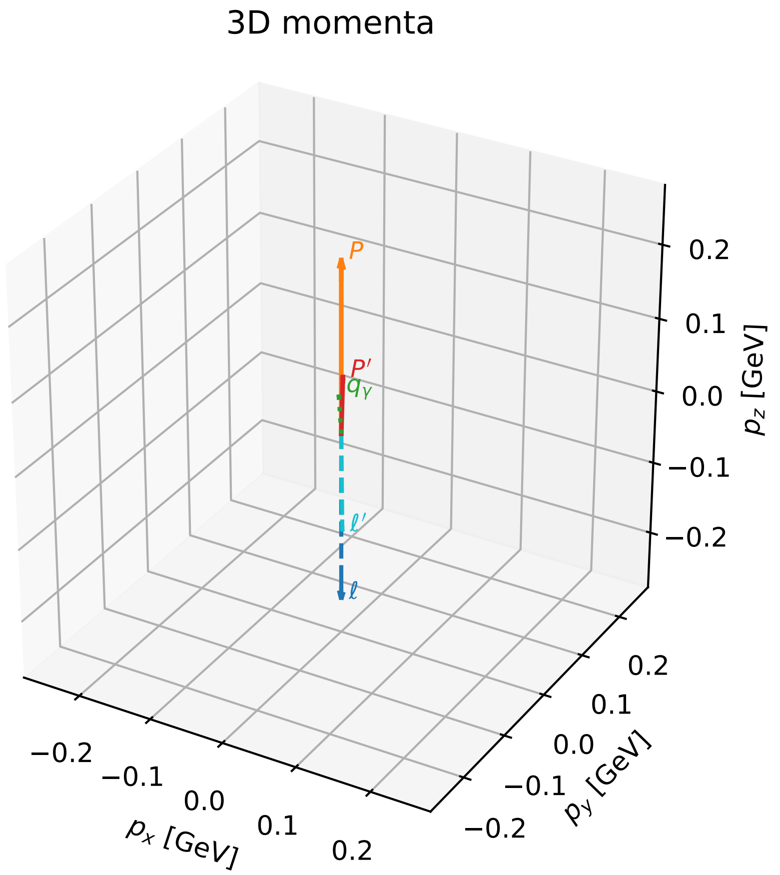

```
selected local minimum

D_W(local) = 0.013598526
D_W contour = 0.063598526
above global minimum = 0.013582093
8D contour samples = 254

selected phase-space point:
    sqrt_s = 1.050355
theta_p_out = 1.590304
theta_gamma_out = 2.440928
      qOut = 0.05611819
 phi_p_out = 2.412477
phi_gamma_out = 3.983473
   alpha_e = 2.390596
   alpha_p = 0.1984256
```

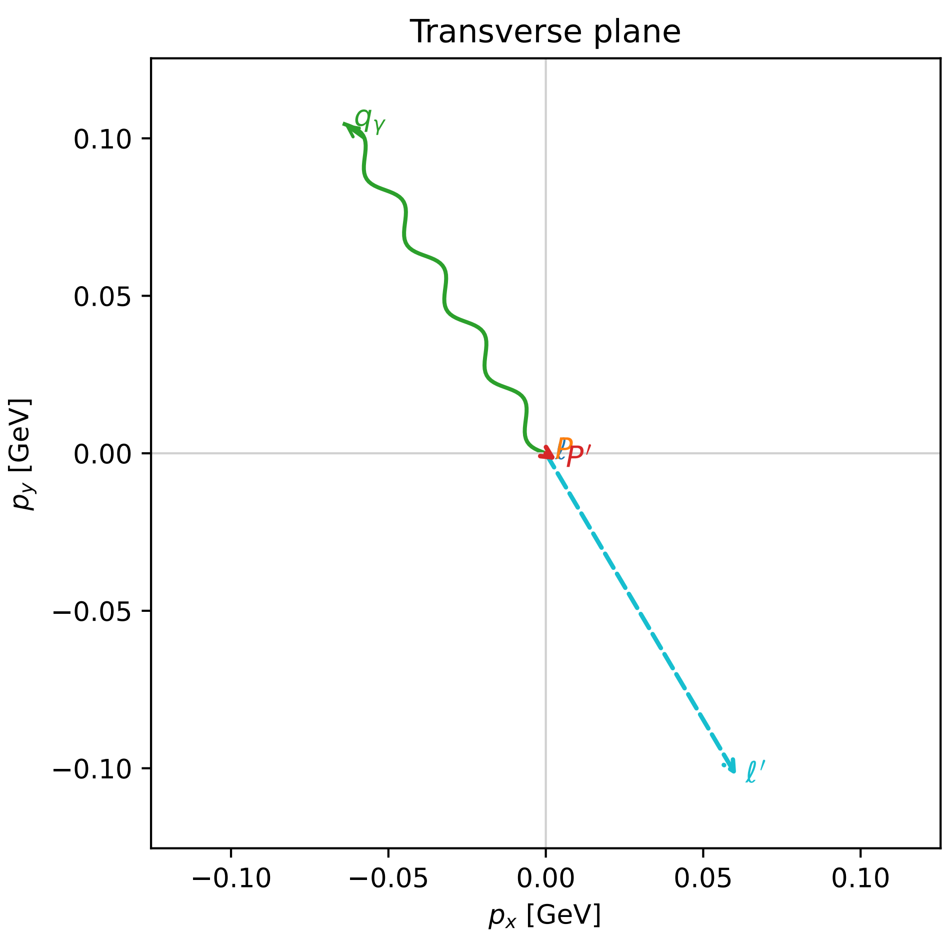
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

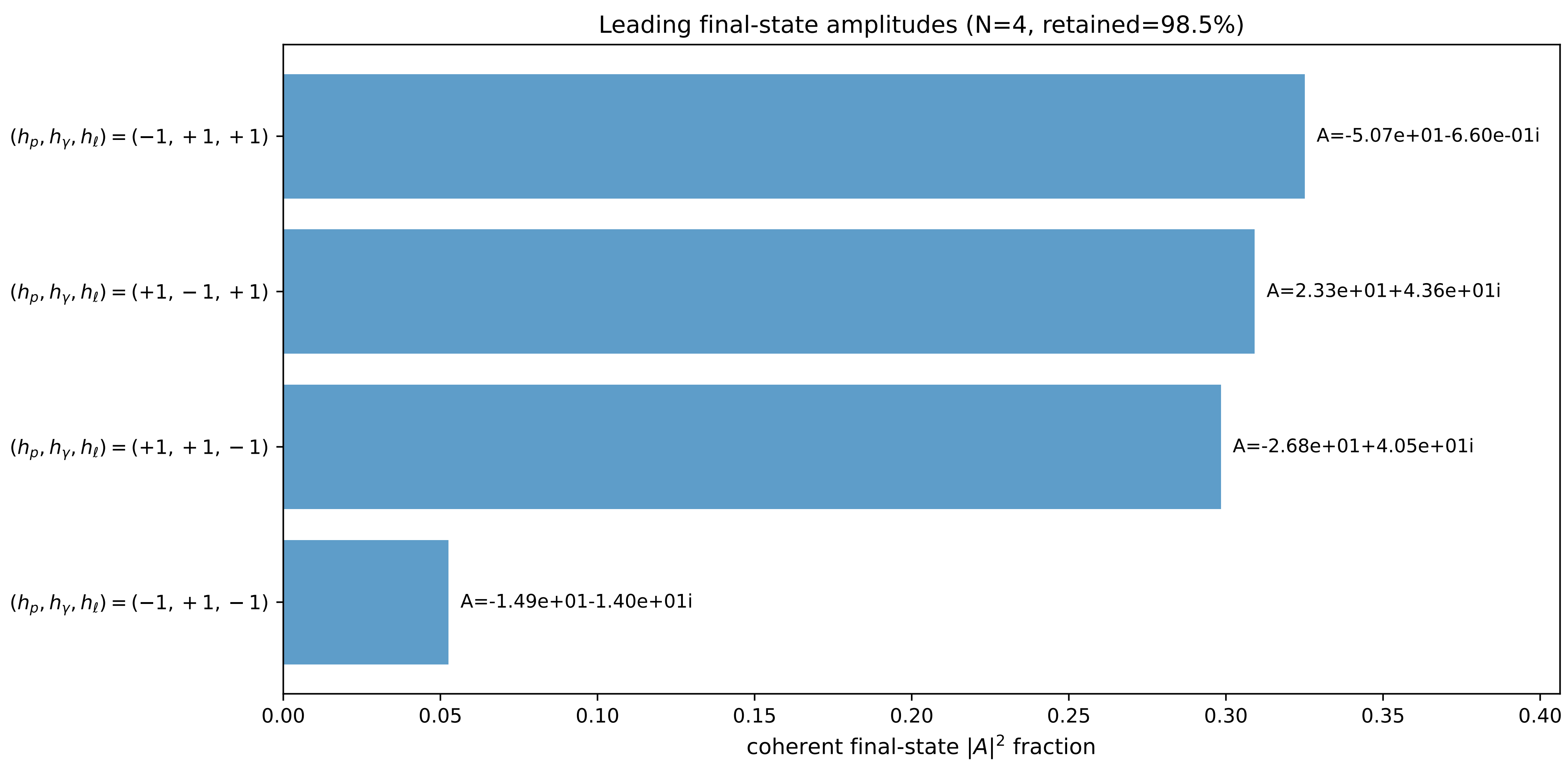
dw_mixing_angles_polarization_05_local_minimum_905 D_W=0.013785
region: polarization_cluster_05_local_minimum_905
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3400151 rad
incoming lepton: -0.695574|+> +0.718454|->
proton mixing angle: alpha_p=0.16911362 rad
incoming proton: 0.985734|+> +0.168309|->

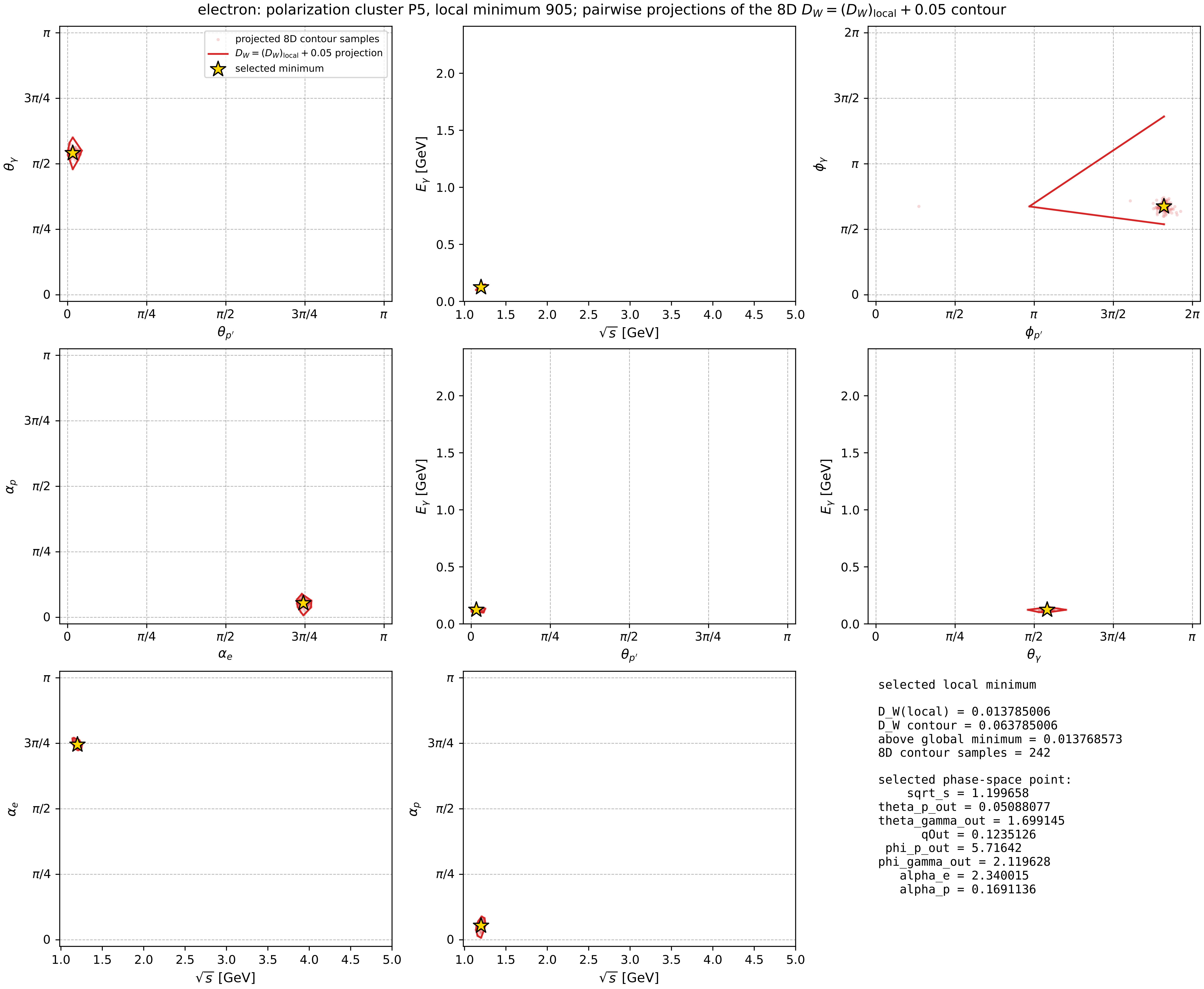
s=1.43918, $\sqrt{s}$=1.19966, $|\vec{P}|$=0.233122, $|\vec{P}'|$=0.0794312
$\theta_{p'}$=0.0508808, θ_γ=1.69914, $\phi_{p'}$=5.71642, ϕ_γ=2.11963
E_γ=0.123513, $\theta_{\ell'}$=2.06151, $\phi_{\ell'}$=5.24628
Q^2=0.0332282, x_B=0.12345, t=-0.0230349, W^2=1.11578, y=0.481223

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2331 p_x= 0.0000 p_y= 0.0000 p_z= -0.2331
P E= 0.9665 p_x= 0.0000 p_y= 0.0000 p_z= 0.2331
ℓ' E= 0.1348 p_x= 0.0605 p_y= -0.1023 p_z= -0.0635
P' E= 0.9414 p_x= 0.0034 p_y= -0.0022 p_z= 0.0793
q_γ E= 0.1235 p_x= -0.0639 p_y= 0.1045 p_z= -0.0158



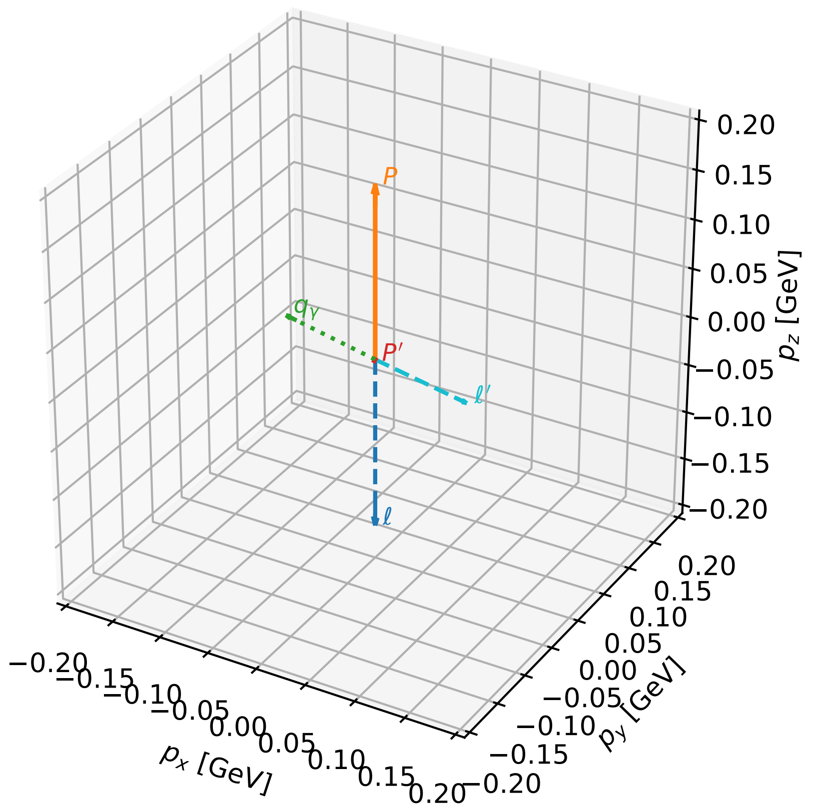





electron: polarization cluster P5, local minimum 905; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.013785006
D_W contour = 0.063785006
above global minimum = 0.013768573
8D contour samples = 242
selected phase-space point:
sqrt_s = 1.199658
theta_p_out = 0.05088077
theta_gamma_out = 1.699145
qOut = 0.1235126
phi_p_out = 5.71642
phi_gamma_out = 2.119628
alpha_e = 2.340015
alpha_p = 0.1691136

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

3D momenta

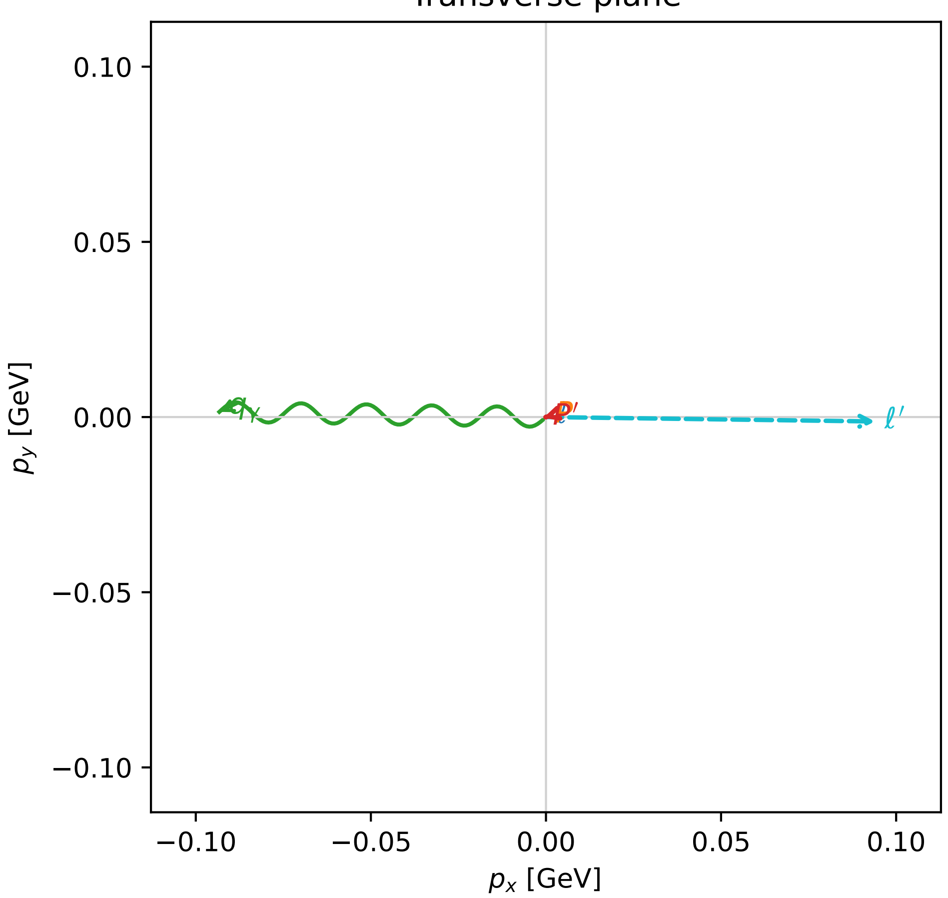



```
dw_mixing_angles_polarization_05_local_minimum_917  D_W=0.0144442
region: polarization_cluster_05_local_minimum_917
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3270893 rad
incoming lepton: -0.68623|+> +0.727385|->
proton mixing angle: alpha_p=0.19706821 rad
incoming proton: 0.980645|+> +0.195795|->
```

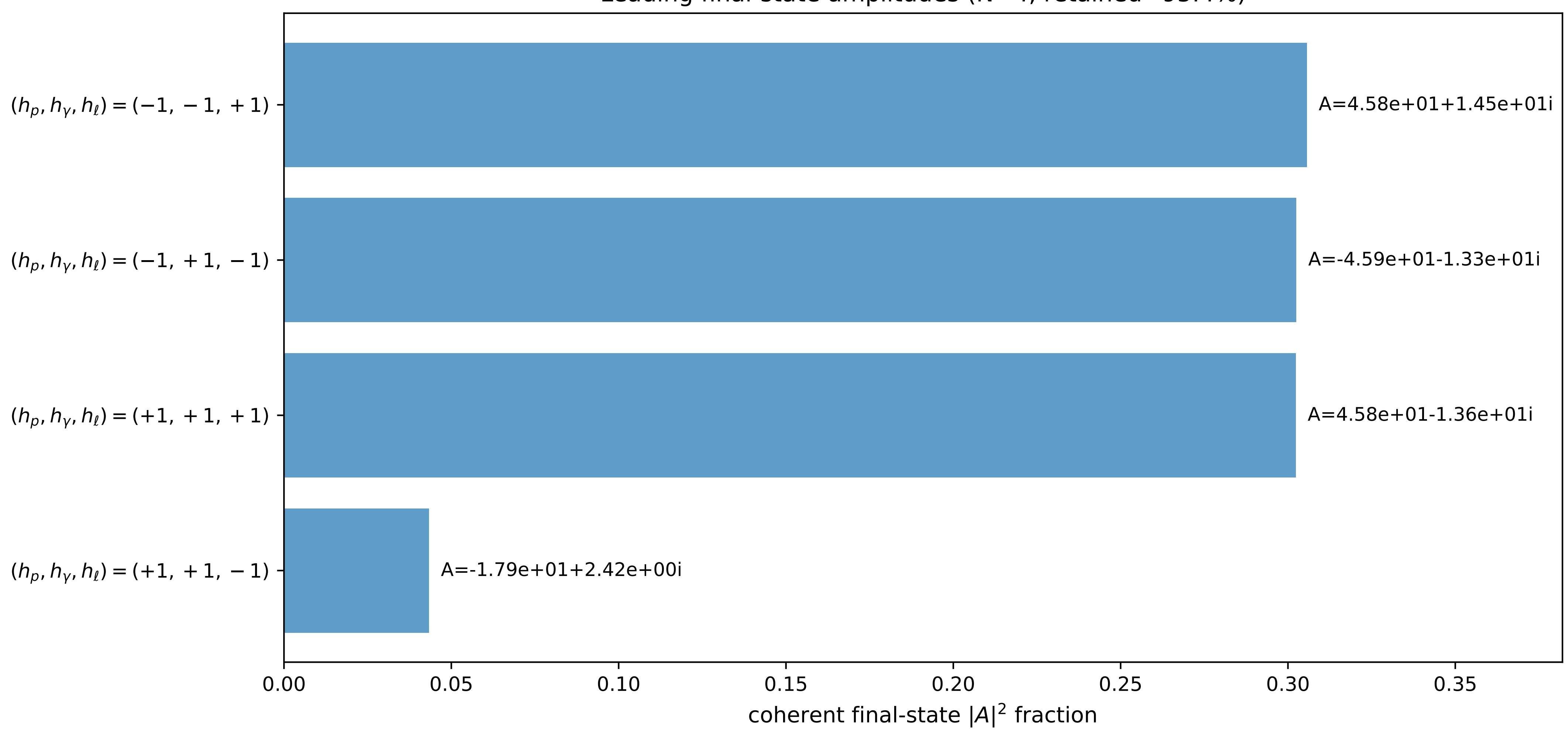
s=1.2736, $\sqrt{s}$=1.12854, $|\vec{P}|$=0.174454, $|\vec{P}'|$=0.00170364
$\theta_{p'}$=2.68144, θ_γ=1.37769, $\phi_{p'}$=3.43006, ϕ_γ=3.12576
E_γ=0.0950533, $\theta_{\ell'}$=1.74676, $\phi_{\ell'}$=6.26976
Q^2=0.0274829, x_B=0.133591, t=-0.0307109, W^2=1.05809, y=0.522465

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1745 p_x= 0.0000 p_y= 0.0000 p_z= -0.1745
P E= 0.9541 p_x= 0.0000 p_y= 0.0000 p_z= 0.1745
ℓ' E= 0.0955 p_x= 0.0940 p_y= -0.0013 p_z= -0.0167
P' E= 0.9380 p_x= -0.0007 p_y= -0.0002 p_z= -0.0015
q_γ E= 0.0951 p_x= -0.0933 p_y= 0.0015 p_z= 0.0182

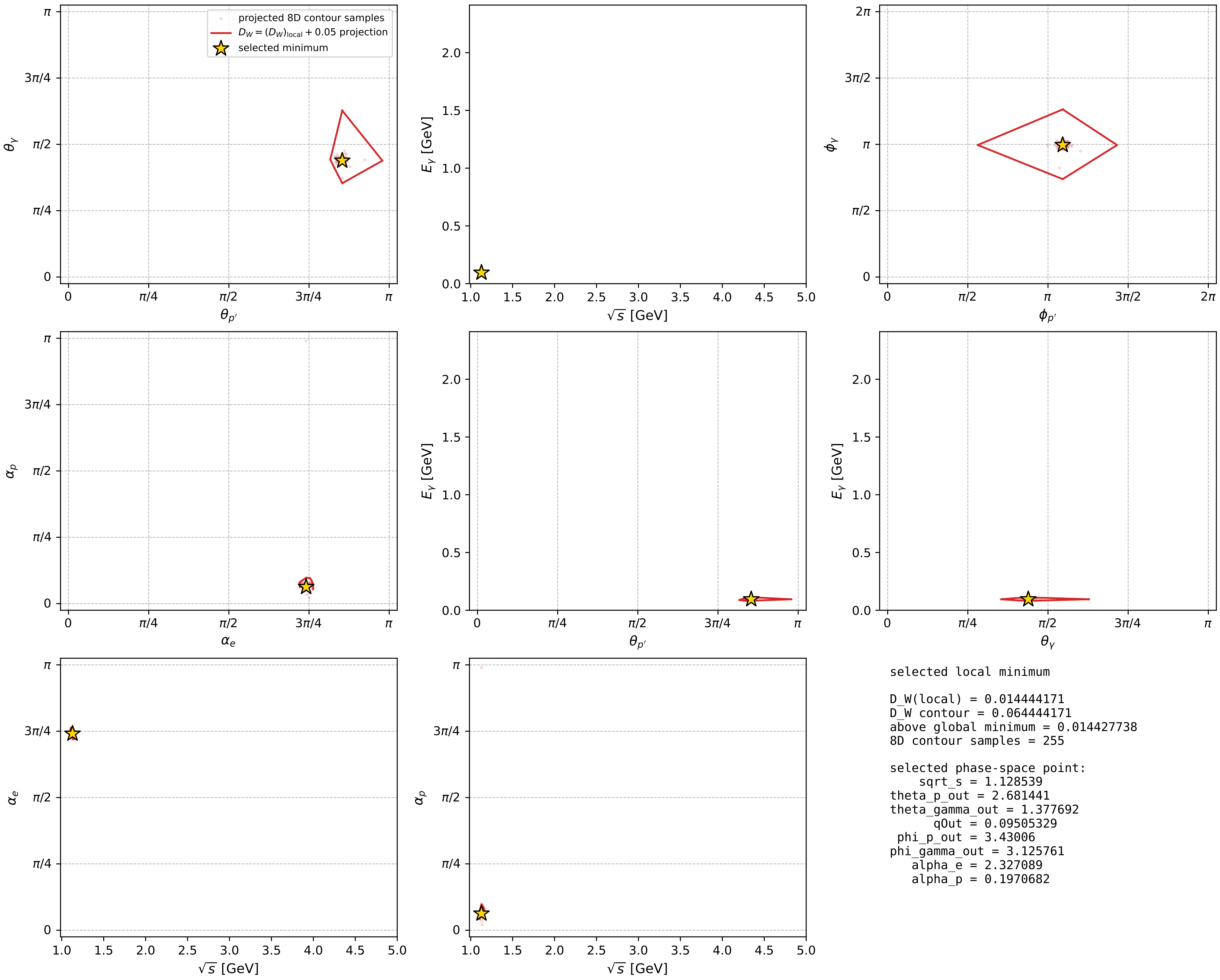
Transverse plane

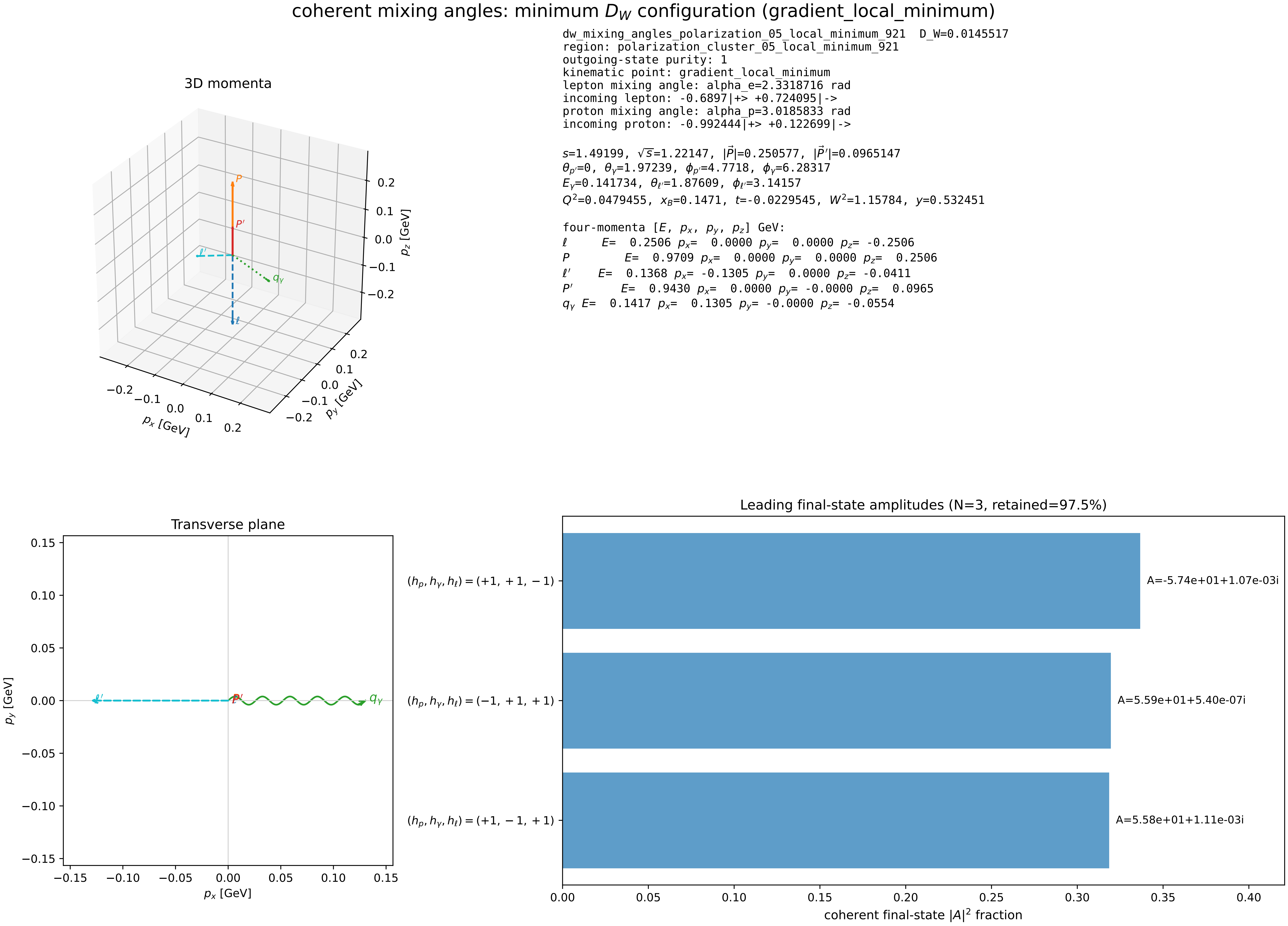


Leading final-state amplitudes (N=4, retained=95.4%)

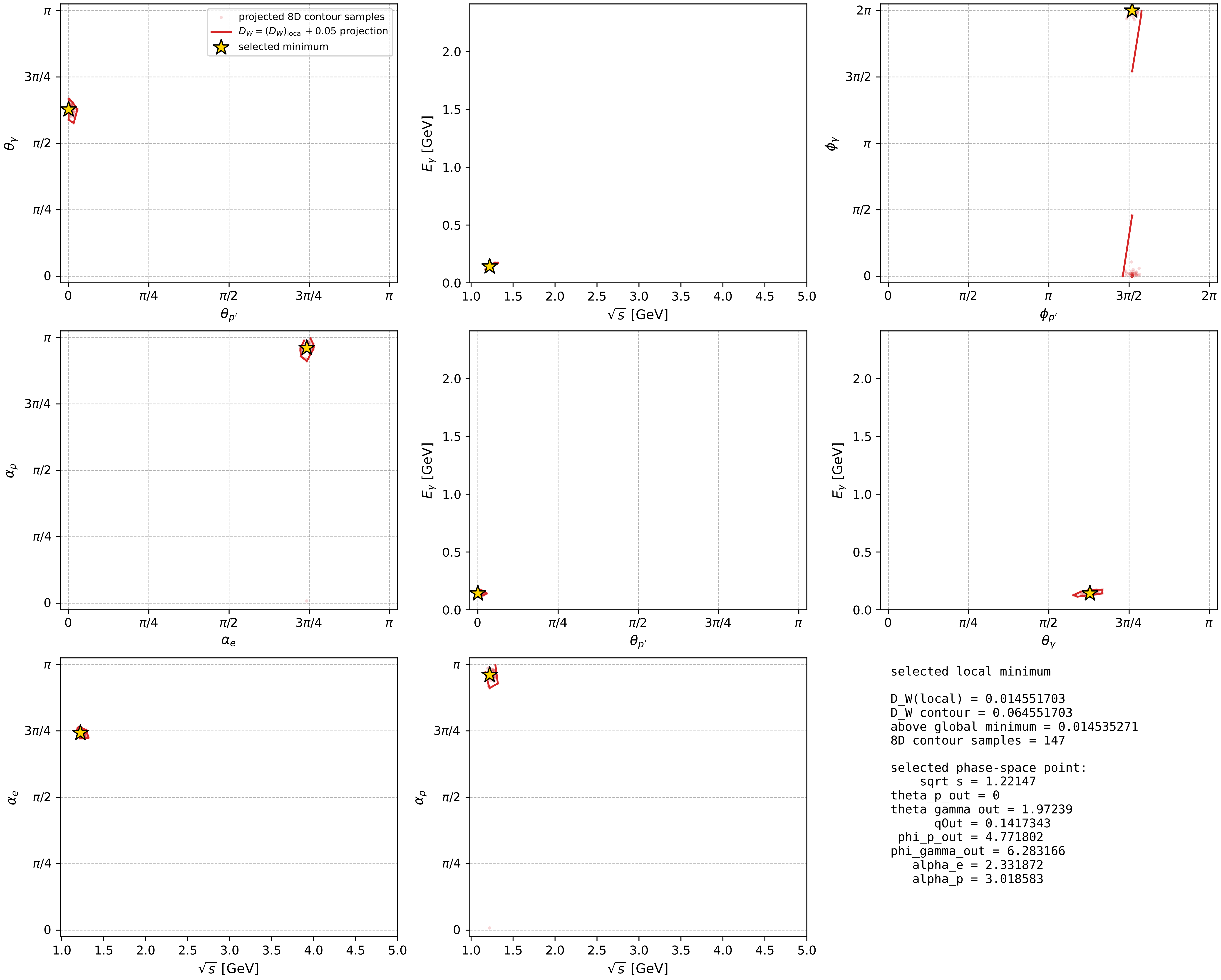


electron: polarization cluster P5, local minimum 917; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour

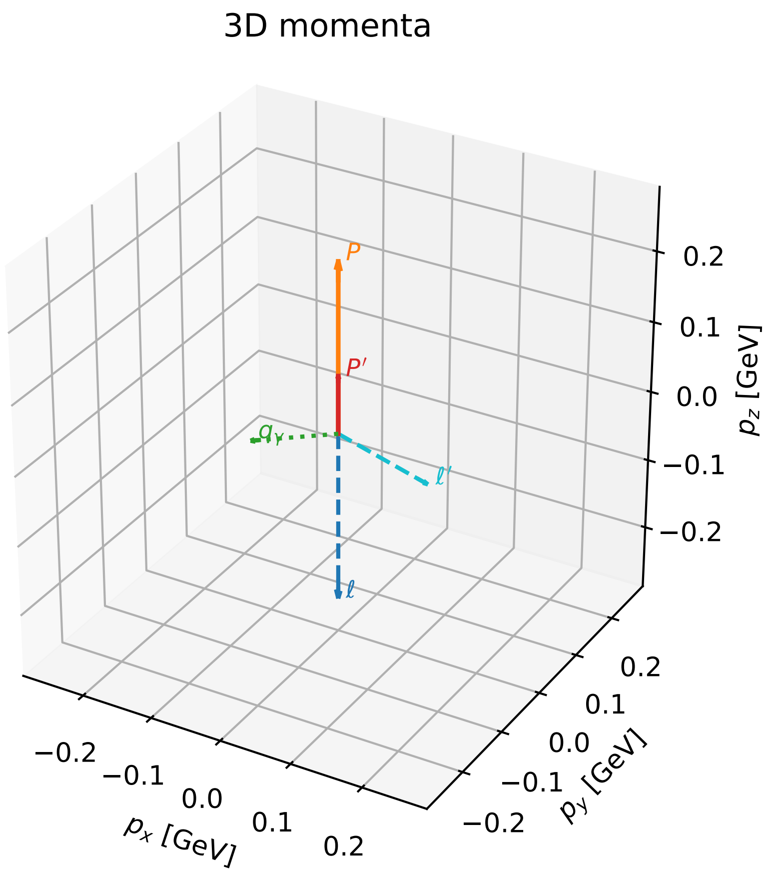



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_921 D_W=0.0145517
region: polarization_cluster_05_local_minimum_921
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3318716 rad
incoming lepton: -0.6897|+> +0.724095|->
proton mixing angle: alpha_p=3.0185833 rad
incoming proton: -0.992444|+> +0.122699|->
s=1.49199, $\sqrt{s}$=1.22147, $|\vec{P}|$=0.250577, $|\vec{P}'|$=0.0965147
$\theta_{p'}$=0, θ_γ=1.97239, $\phi_{p'}$=4.7718, ϕ_γ=6.28317
E_γ=0.141734, $\theta_{\ell'}$=1.87609, $\phi_{\ell'}$=3.14157
Q^2=0.0479455, x_B=0.1471, t=-0.0229545, W^2=1.15784, y=0.532451
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2506 p_x= 0.0000 p_y= 0.0000 p_z= -0.2506
P E= 0.9709 p_x= 0.0000 p_y= 0.0000 p_z= 0.2506
ℓ' E= 0.1368 p_x= -0.1305 p_y= 0.0000 p_z= -0.0411
P' E= 0.9430 p_x= 0.0000 p_y= -0.0000 p_z= 0.0965
q_γ E= 0.1417 p_x= 0.1305 p_y= -0.0000 p_z= -0.0554
3D momenta
P
P'
ℓ'
q_γ
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
ℓ'
P'
q_γ
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=3, retained=97.5%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-5.74e+01+1.07e-03i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=5.59e+01+5.40e-07i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=5.58e+01+1.11e-03i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P5, local minimum 921; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



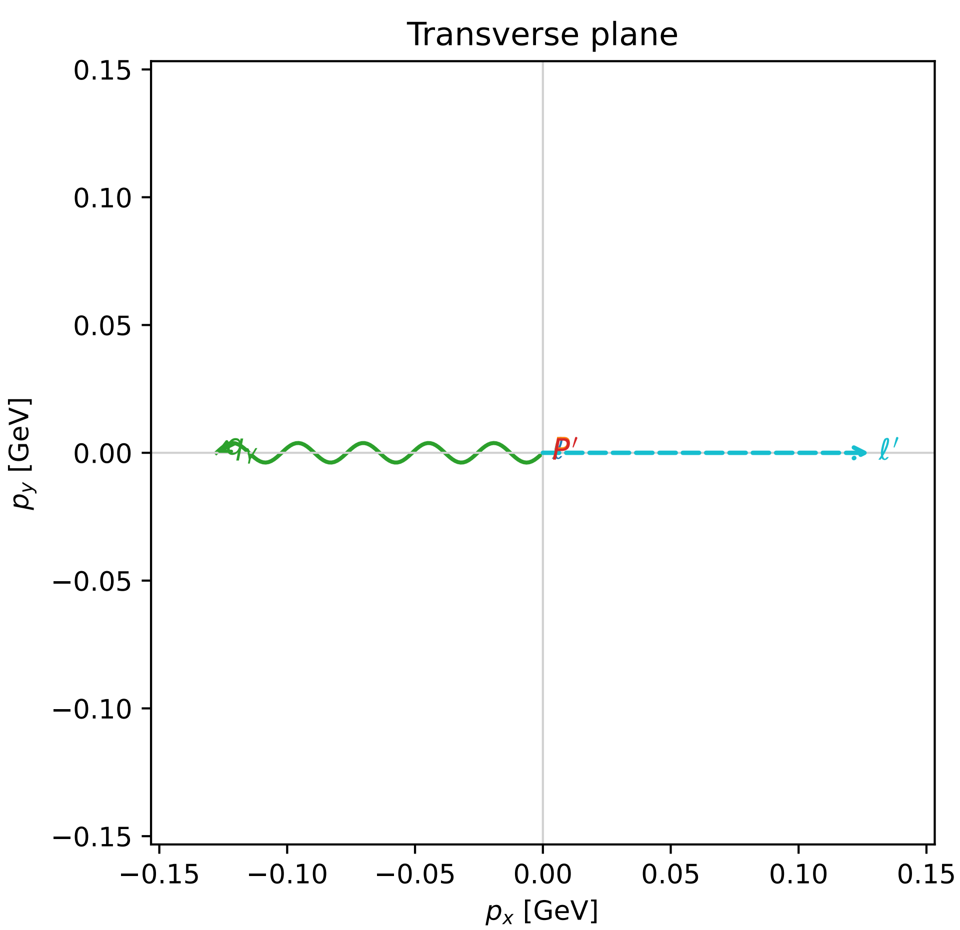
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

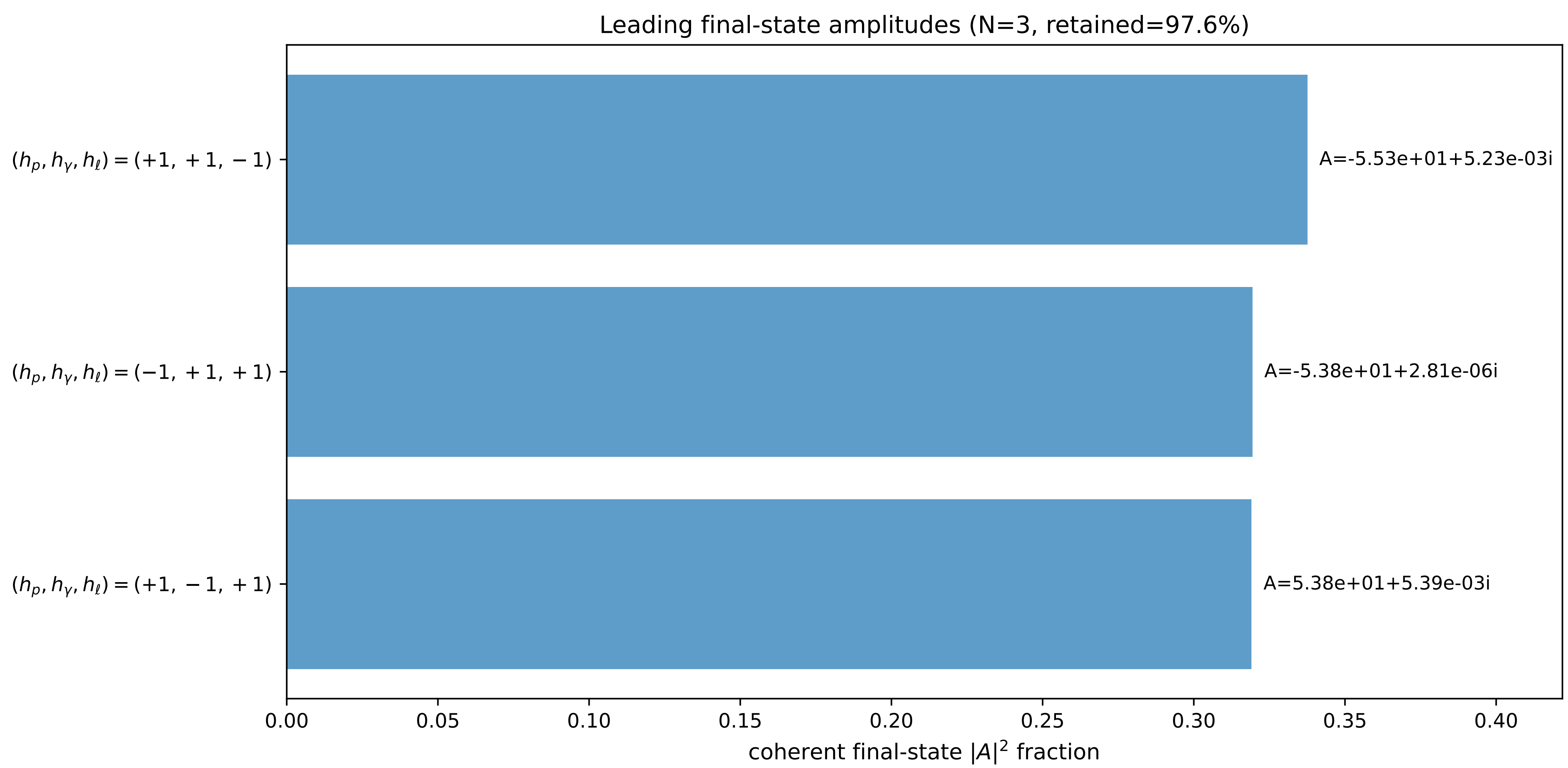


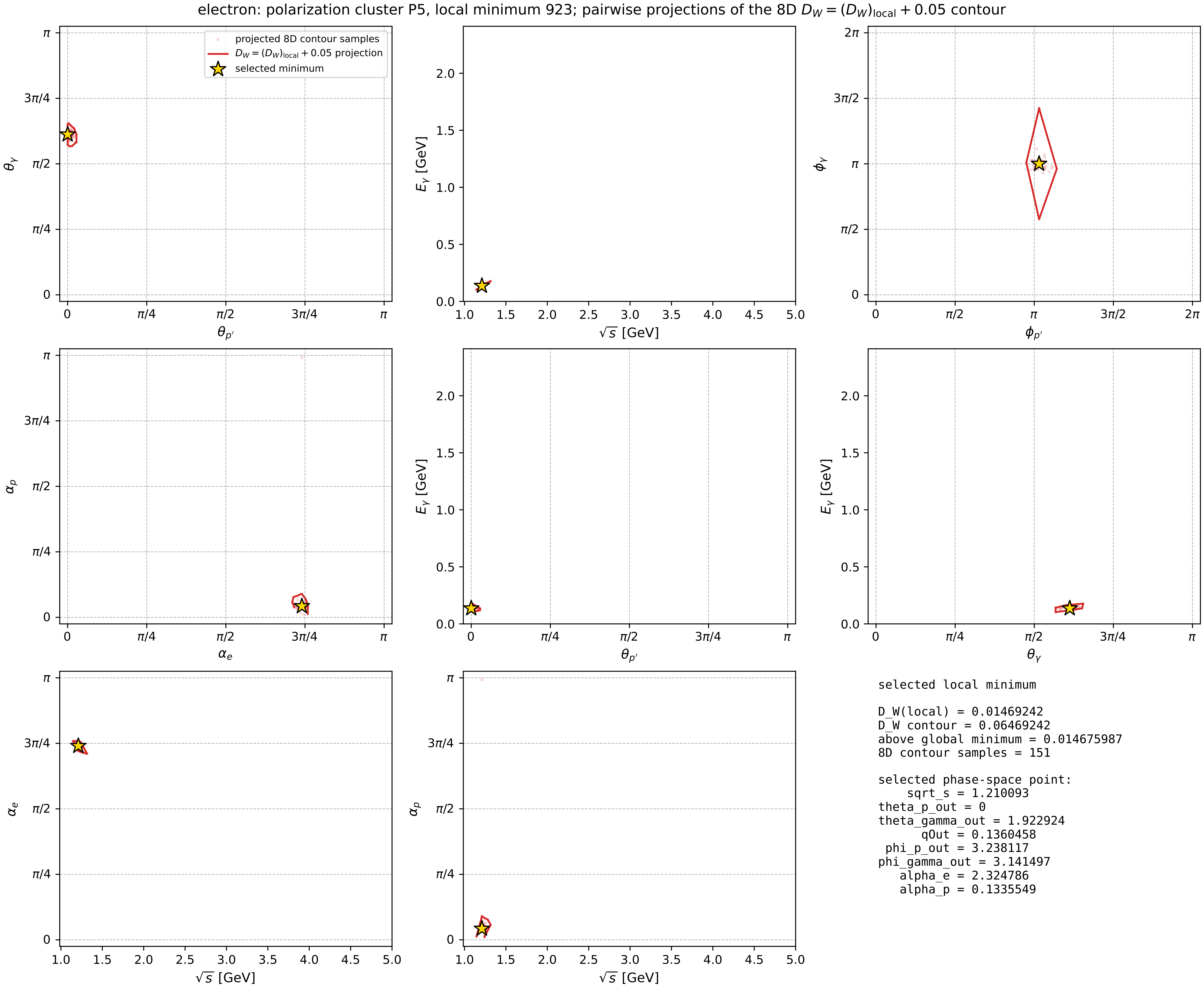

```
dw_mixing_angles_polarization_05_local_minimum_923  D_W=0.0146924
region: polarization_cluster_05_local_minimum_923
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.324786 rad
incoming lepton: -0.684552|+> +0.728964|->
proton mixing angle: alpha_p=0.13355488 rad
incoming proton: 0.991095|+> +0.133158|->
```

s=1.46432, $\sqrt{s}$=1.21009, $|\vec{P}|$=0.241502, $|\vec{P}'|$=0.0821138
$\theta_{p'}$=0, θ_γ=1.92292, $\phi_{p'}$=3.23812, ϕ_γ=3.1415
E_γ=0.136046, $\theta_{\ell'}$=1.83971, $\phi_{\ell'}$=6.28309
Q^2=0.0469803, x_B=0.151118, t=-0.0246754, W^2=1.14375, y=0.531898

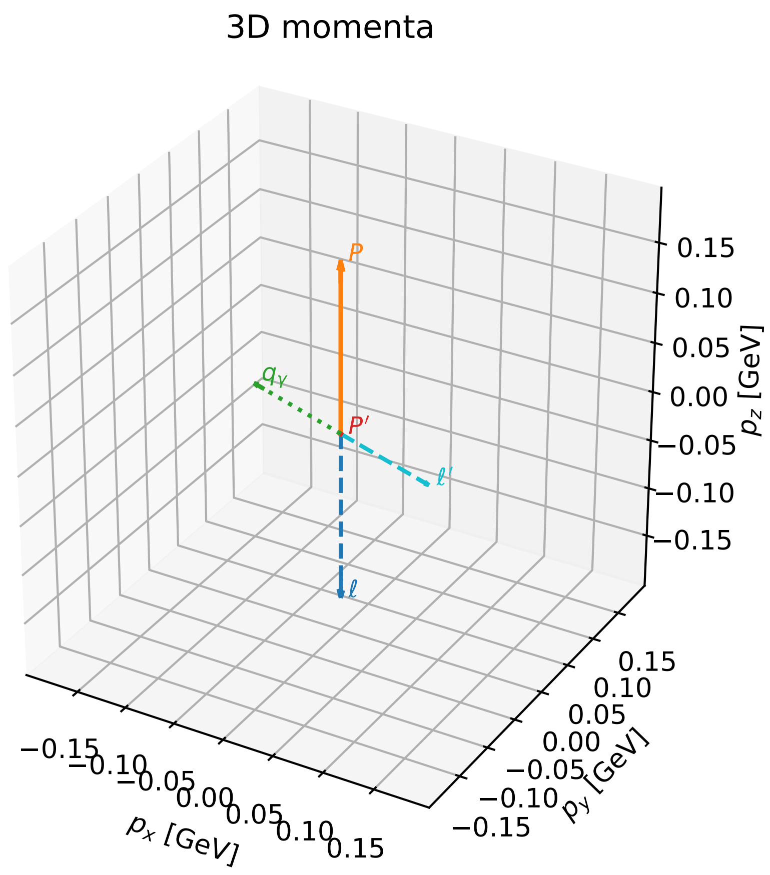
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2415 p_x= 0.0000 p_y= 0.0000 p_z= -0.2415
P E= 0.9686 p_x= 0.0000 p_y= 0.0000 p_z= 0.2415
ℓ' E= 0.1325 p_x= 0.1277 p_y= -0.0000 p_z= -0.0352
P' E= 0.9416 p_x= -0.0000 p_y= -0.0000 p_z= 0.0821
q_γ E= 0.1360 p_x= -0.1277 p_y= 0.0000 p_z= -0.0469






electron: polarization cluster P5, local minimum 923; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.01469242
D_W contour = 0.06469242
above global minimum = 0.014675987
8D contour samples = 151
selected phase-space point:
sqrt_s = 1.210093
theta_p_out = 0
theta_gamma_out = 1.922924
qOut = 0.1360458
phi_p_out = 3.238117
phi_gamma_out = 3.141497
alpha_e = 2.324786
alpha_p = 0.1335549

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)




```
dw_mixing_angles_polarization_05_local_minimum_927  D_W=0.014801
region: polarization_cluster_05_local_minimum_927
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.323091 rad
incoming lepton: -0.683316|+> +0.730123|->
proton mixing angle: alpha_p=0.16705003 rad
incoming proton: 0.98608|+> +0.166274|->
```

s=1.26194, $\sqrt{s}$=1.12336, $|\vec{P}|$=0.170068, $|\vec{P}'|$=0.00017414

$\theta_{p'}$=2.49932, θ_γ=1.28982, $\phi_{p'}$=4.04792, ϕ_γ=3.13144

E_γ=0.0926688, $\theta_{\ell'}$=1.85013, $\phi_{\ell'}$=6.27396

Q^2=0.022835, x_B=0.116104, t=-0.0287367, W^2=1.05369, y=0.514732

four-momenta [E, p_x, p_y, p_z] GeV:

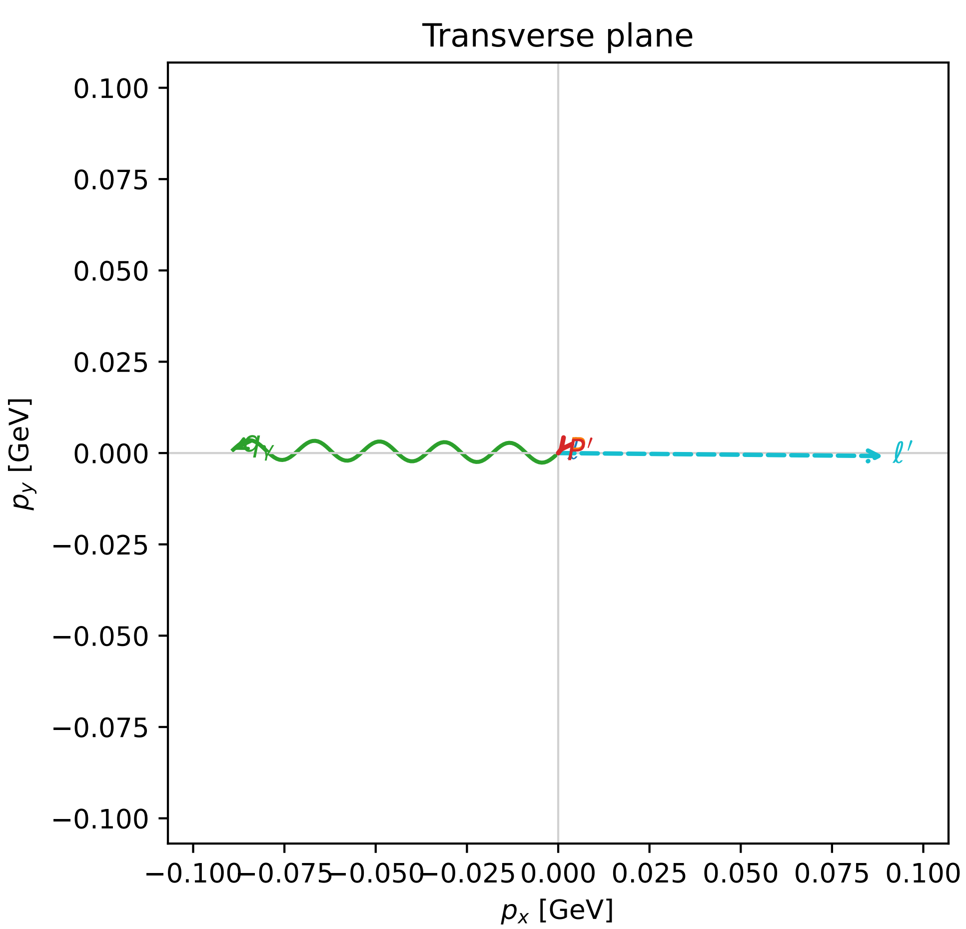
ℓ E= 0.1701 p_x= 0.0000 p_y= 0.0000 p_z= -0.1701

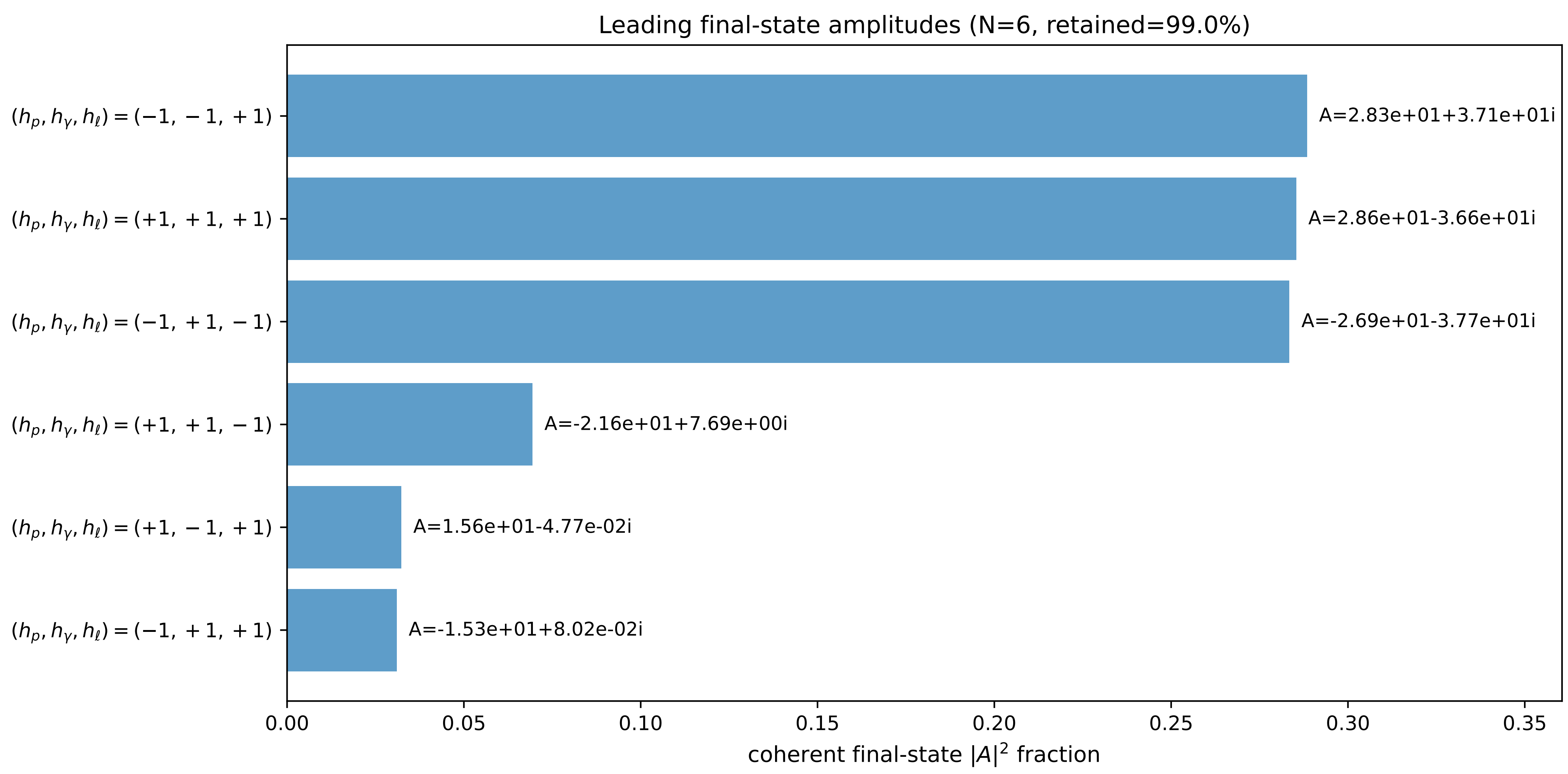
P E= 0.9533 p_x= 0.0000 p_y= 0.0000 p_z= 0.1701

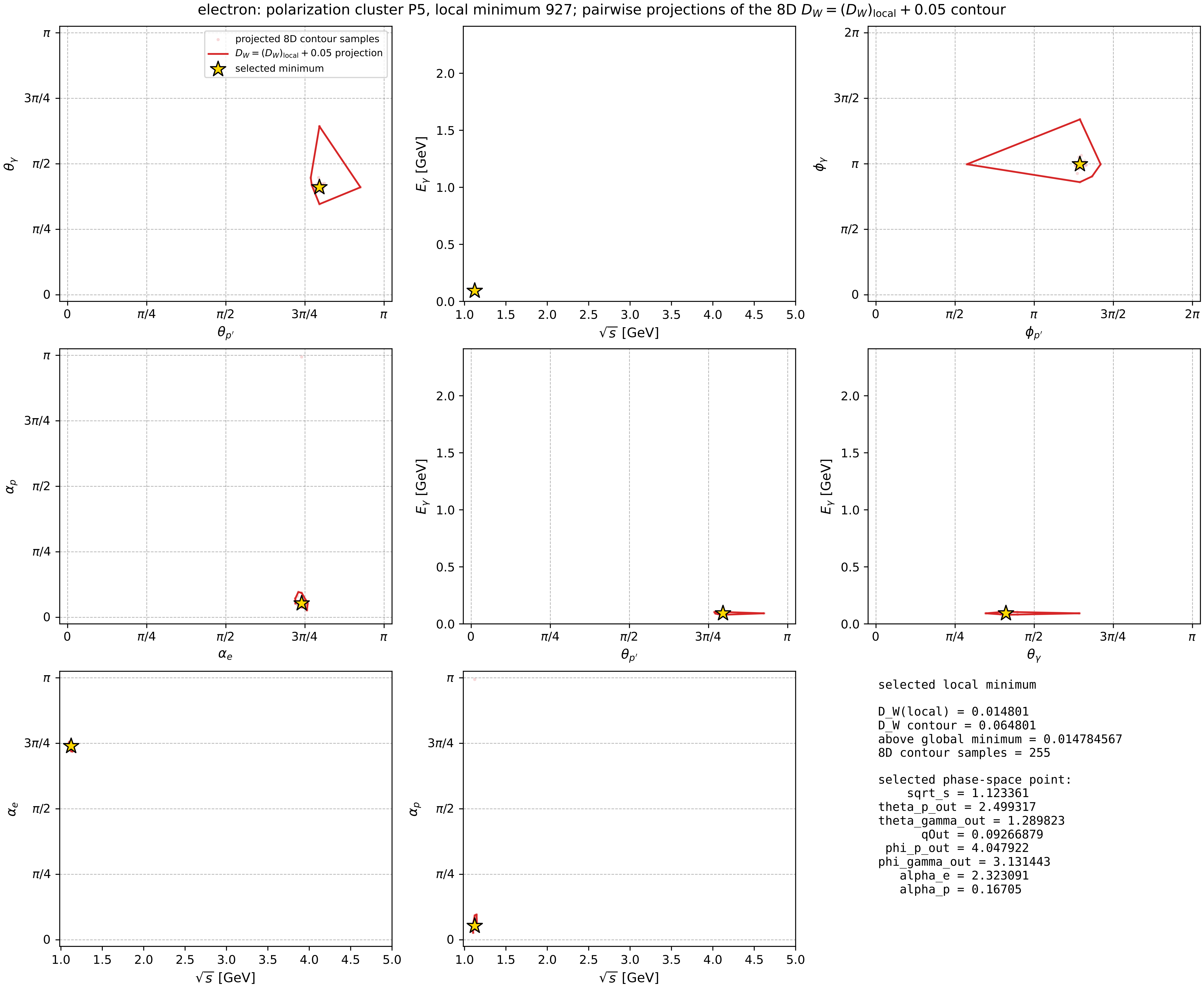
ℓ' E= 0.0927 p_x= 0.0891 p_y= -0.0008 p_z= -0.0256

P' E= 0.9380 p_x= -0.0001 p_y= -0.0001 p_z= -0.0001

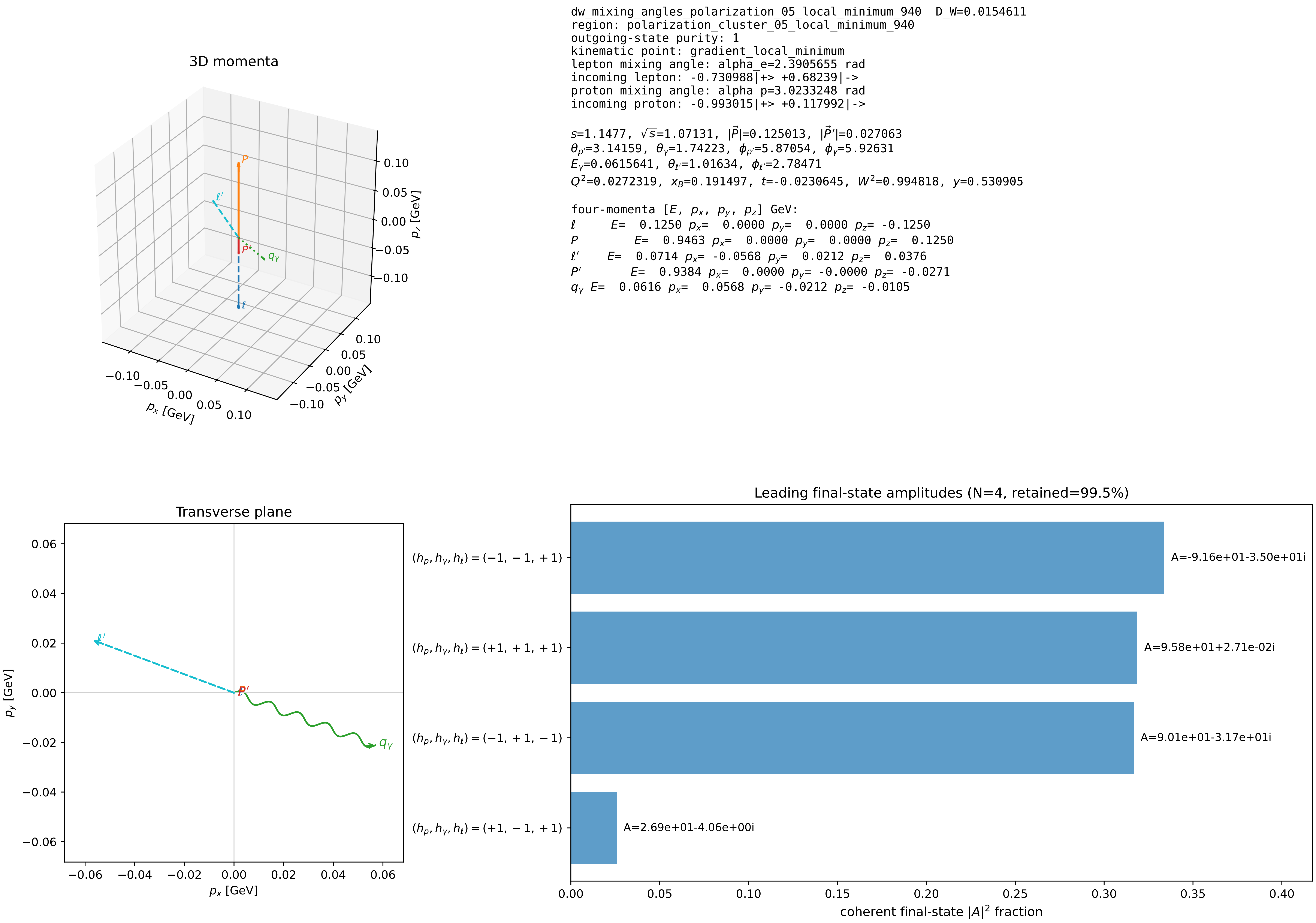
q_γ E= 0.0927 p_x= -0.0890 p_y= 0.0009 p_z= 0.0257



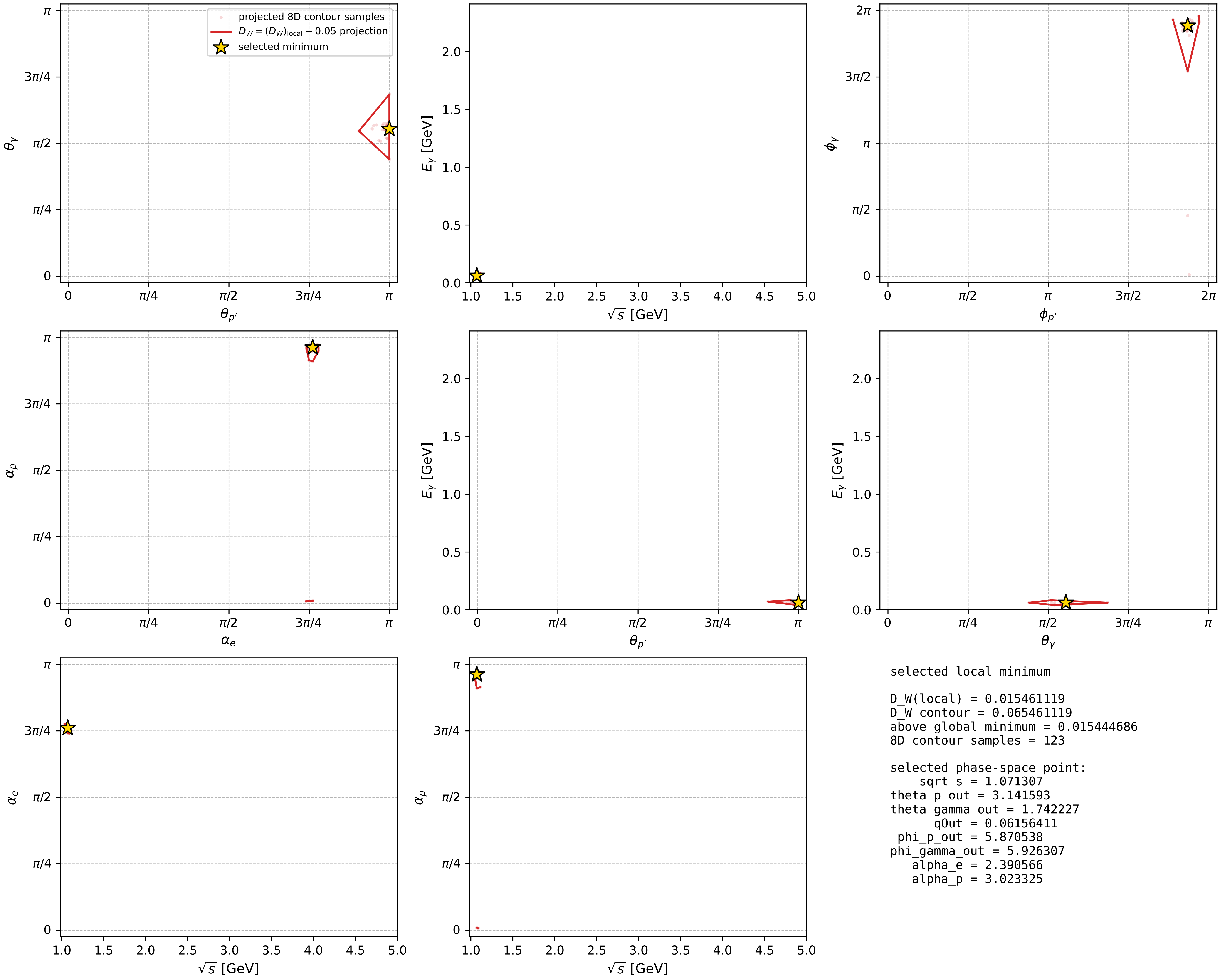



electron: polarization cluster P5, local minimum 927; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.014801
D_W contour = 0.064801
above global minimum = 0.014784567
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.123361
theta_p_out = 2.499317
theta_gamma_out = 1.289823
qOut = 0.09266879
phi_p_out = 4.047922
phi_gamma_out = 3.131443
alpha_e = 2.323091
alpha_p = 0.16705

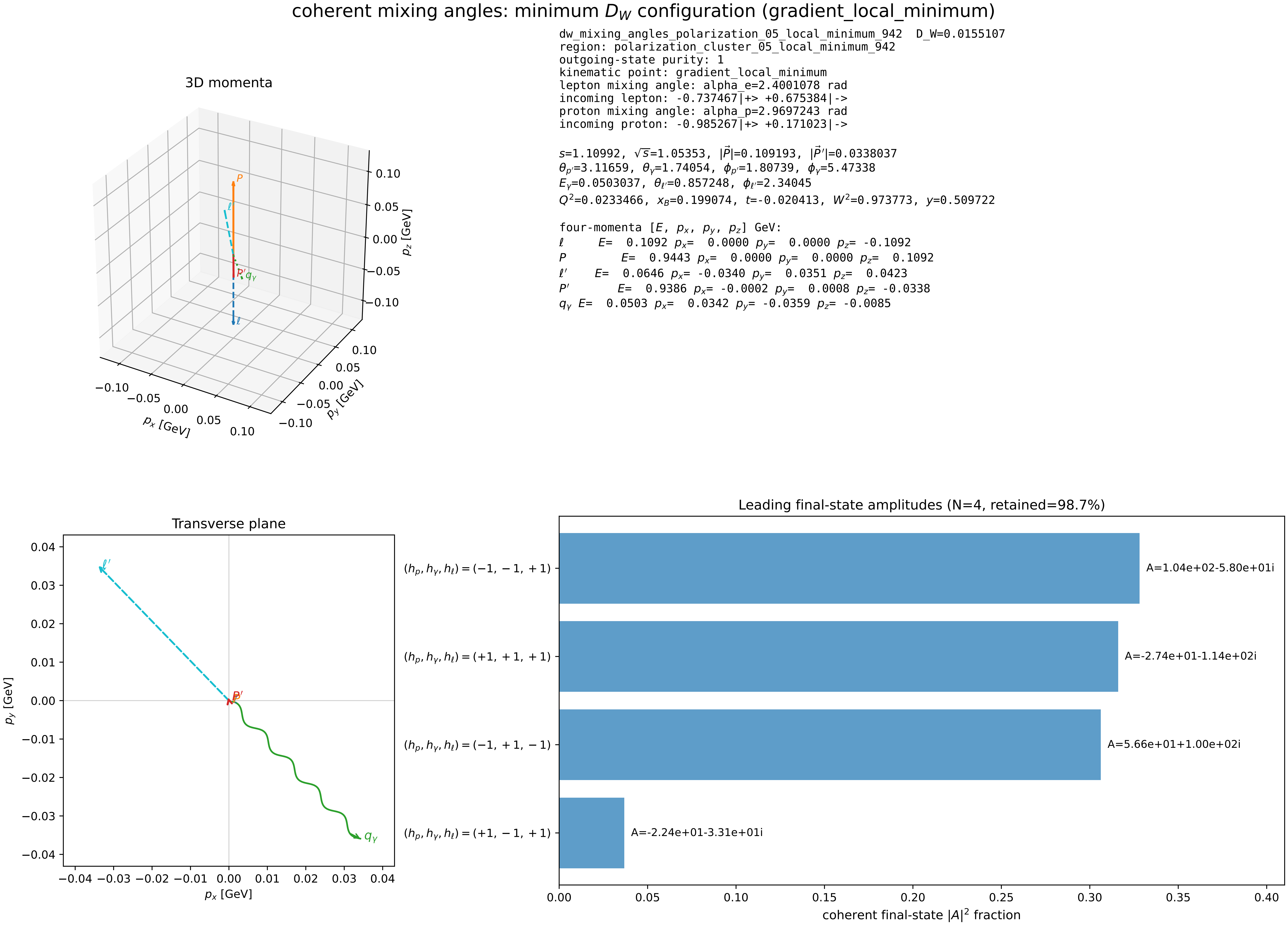
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

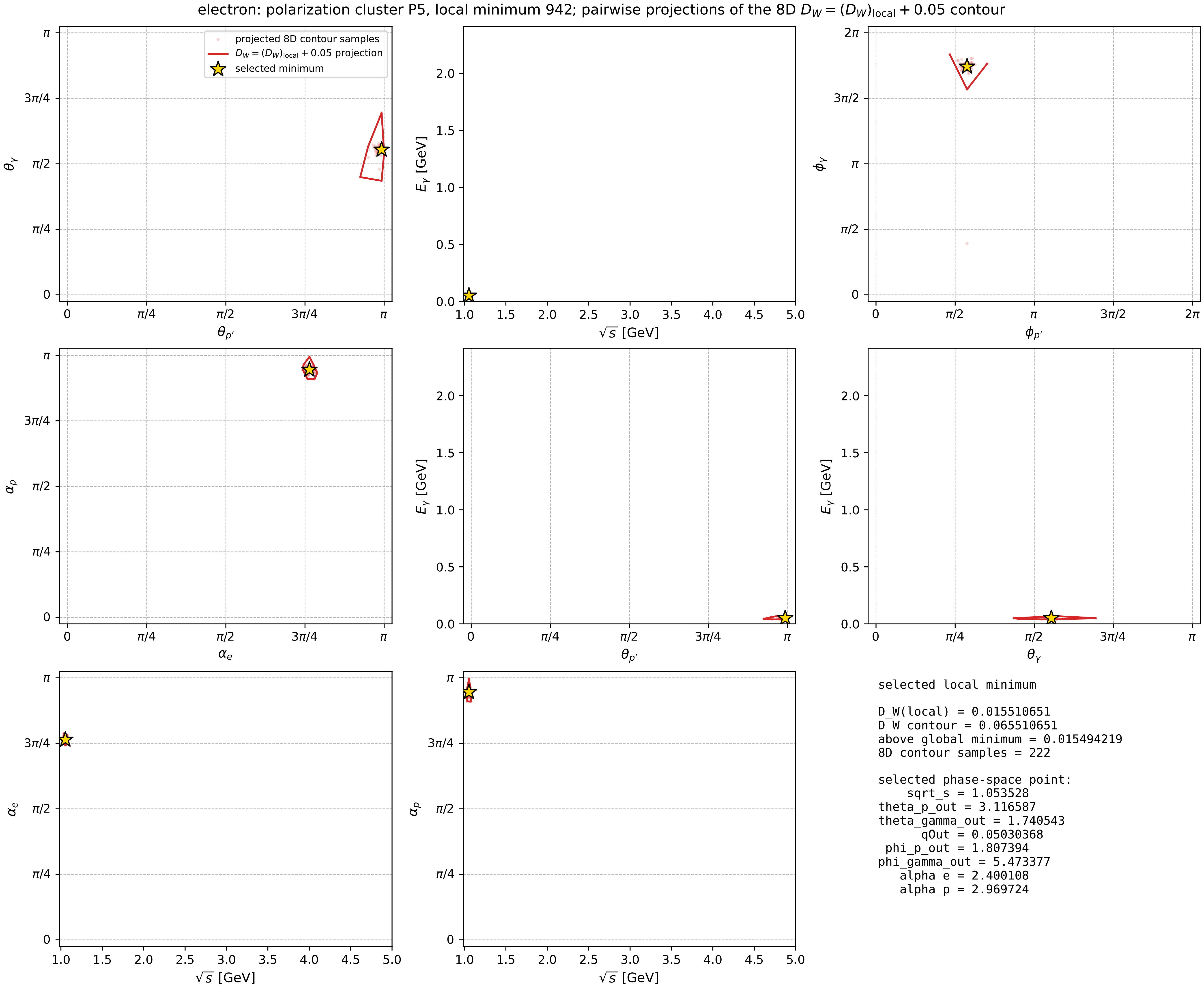


electron: polarization cluster P5, local minimum 940; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour

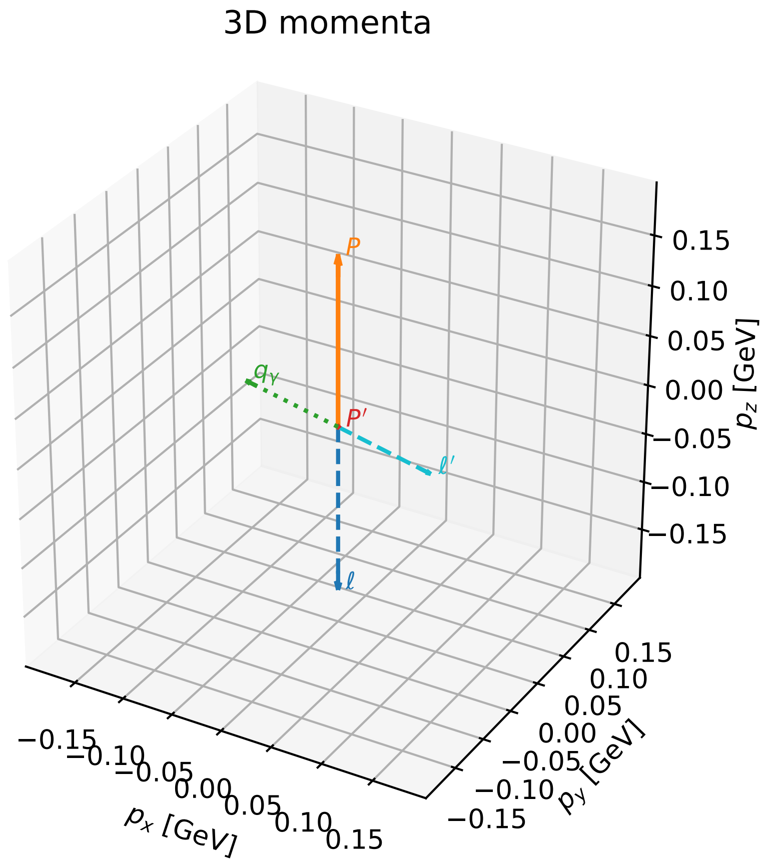



```
selected local minimum

D_W(local) = 0.015461119
D_W contour = 0.065461119
above global minimum = 0.015444686
8D contour samples = 123

selected phase-space point:
    sqrt_s = 1.071307
theta_p_out = 3.141593
theta_gamma_out = 1.742227
       qOut = 0.06156411
 phi_p_out = 5.870538
phi_gamma_out = 5.926307
   alpha_e = 2.390566
   alpha_p = 3.023325
```


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_942 D_W=0.0155107
region: polarization_cluster_05_local_minimum_942
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.4001078 rad
incoming lepton: -0.737467|+> +0.675384|->
proton mixing angle: alpha_p=2.9697243 rad
incoming proton: -0.985267|+> +0.171023|->
s=1.10992, $\sqrt{s}$=1.05353, $|\vec{P}|$=0.109193, $|\vec{P}'|$=0.0338037
$\theta_{p'}$=3.11659, θ_γ=1.74054, $\phi_{p'}$=1.80739, ϕ_γ=5.47338
E_γ=0.0503037, $\theta_{\ell'}$=0.857248, $\phi_{\ell'}$=2.34045
Q^2=0.0233466, x_B=0.199074, t=-0.020413, W^2=0.973773, y=0.509722
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1092 p_x= 0.0000 p_y= 0.0000 p_z= -0.1092
P E= 0.9443 p_x= 0.0000 p_y= 0.0000 p_z= 0.1092
ℓ' E= 0.0646 p_x= -0.0340 p_y= 0.0351 p_z= 0.0423
P' E= 0.9386 p_x= -0.0002 p_y= 0.0008 p_z= -0.0338
q_γ E= 0.0503 p_x= 0.0342 p_y= -0.0359 p_z= -0.0085
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
ℓ'
P'
q_γ
Leading final-state amplitudes (N=4, retained=98.7%)
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$
A=1.04e+02-5.80e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$
A=-2.74e+01-1.14e+02i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=5.66e+01+1.00e+02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=-2.24e+01-3.31e+01i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P5, local minimum 942; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
0
$\pi/4$
$\pi/2$
$3\pi/4$
π
$3\pi/2$
2π
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
selected local minimum
D_W(local) = 0.015510651
D_W contour = 0.065510651
above global minimum = 0.015494219
8D contour samples = 222
selected phase-space point:
sqrt_s = 1.053528
theta_p_out = 3.116587
theta_gamma_out = 1.740543
qOut = 0.05030368
phi_p_out = 1.807394
phi_gamma_out = 5.473377
alpha_e = 2.400108
alpha_p = 2.969724

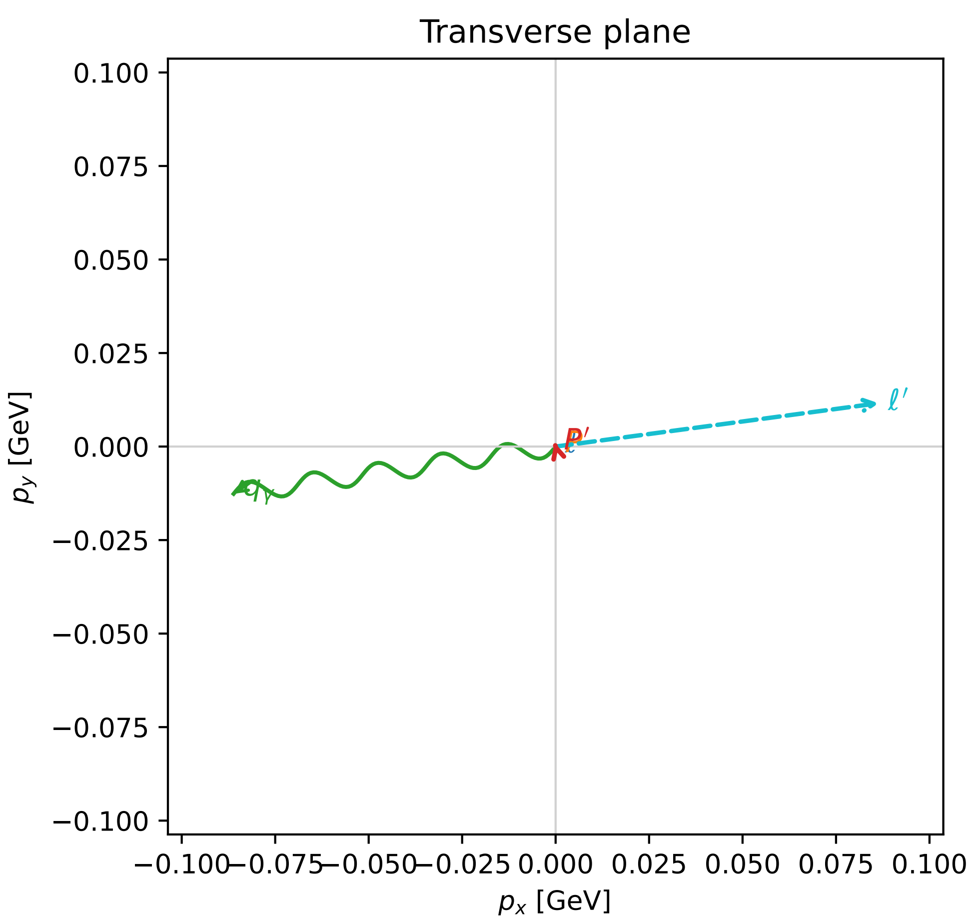
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

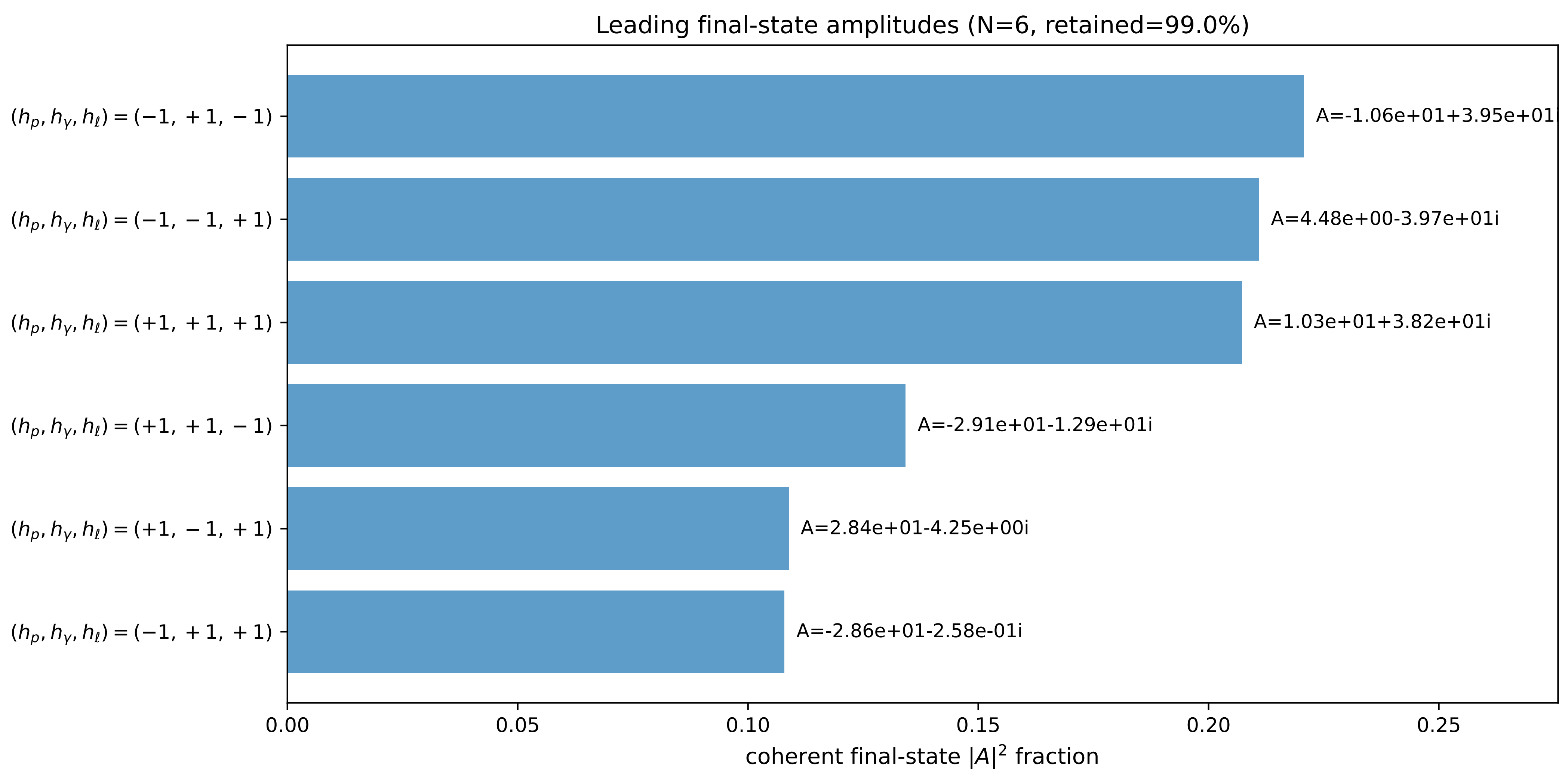


dw_mixing_angles_polarization_05_local_minimum_945 D_W=0.0156741
region: polarization_cluster_05_local_minimum_945
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3212639 rad
incoming lepton: -0.681981|+> +0.73137|->
proton mixing angle: alpha_p=0.16486539 rad
incoming proton: 0.98644|+> +0.16412|->

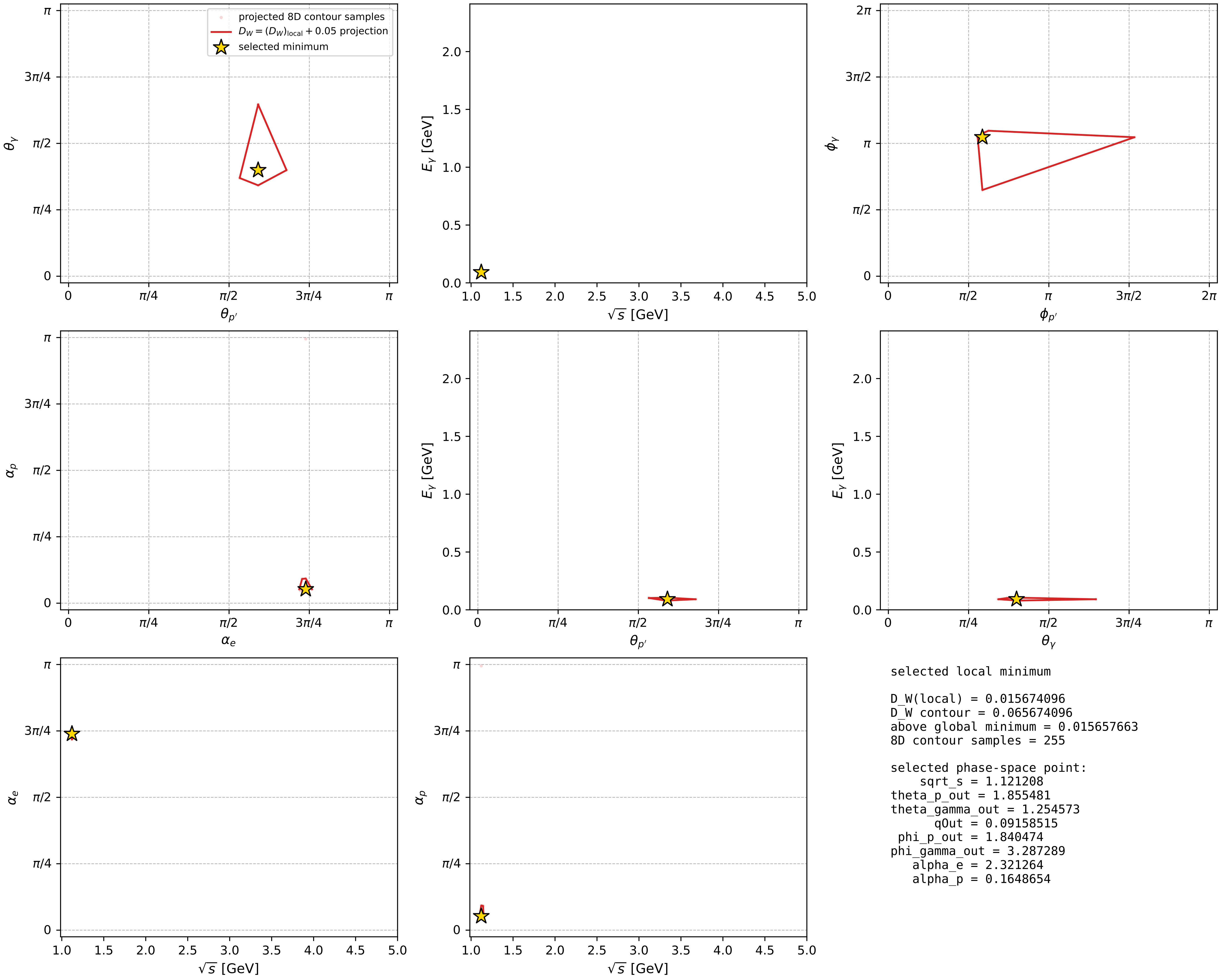
s=1.25711, $\sqrt{s}$=1.12121, $|\vec{P}|$=0.168239, $|\vec{P}'|$=0.00113839
$\theta_{p'}$=1.85548, θ_γ=1.25457, $\phi_{p'}$=1.84047, ϕ_γ=3.28729
E_γ=0.0915852, $\theta_{\ell'}$=1.88322, $\phi_{\ell'}$=0.133261
Q^2=0.021353, x_B=0.110545, t=-0.0281894, W^2=1.05165, y=0.512006

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1682 p_x= 0.0000 p_y= 0.0000 p_z= -0.1682
P E= 0.9530 p_x= 0.0000 p_y= 0.0000 p_z= 0.1682
ℓ' E= 0.0916 p_x= 0.0864 p_y= 0.0116 p_z= -0.0282
P' E= 0.9380 p_x= -0.0003 p_y= 0.0011 p_z= -0.0003
q_γ E= 0.0916 p_x= -0.0861 p_y= -0.0126 p_z= 0.0285



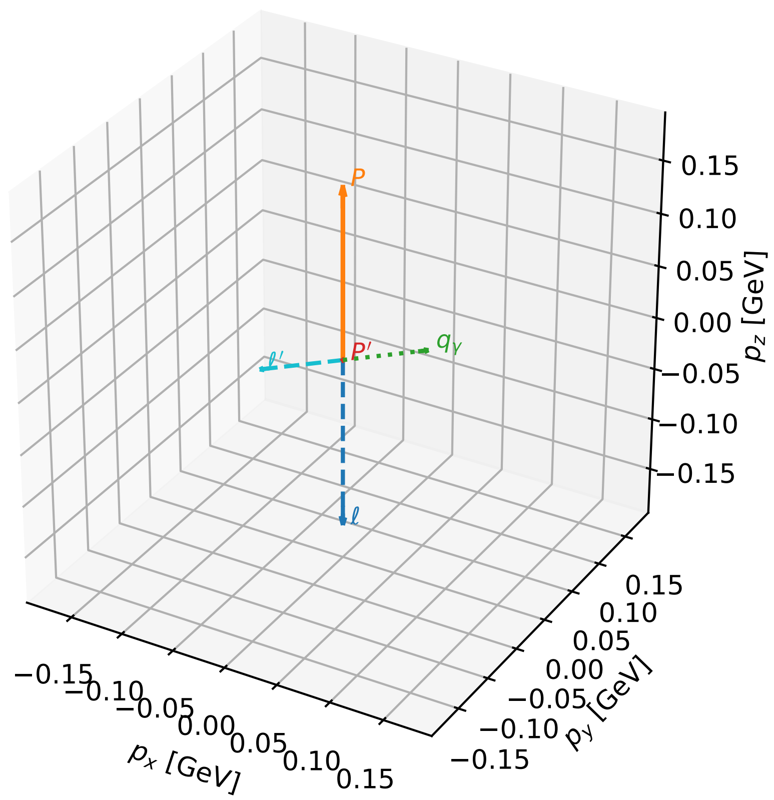


electron: polarization cluster P5, local minimum 945; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

3D momenta

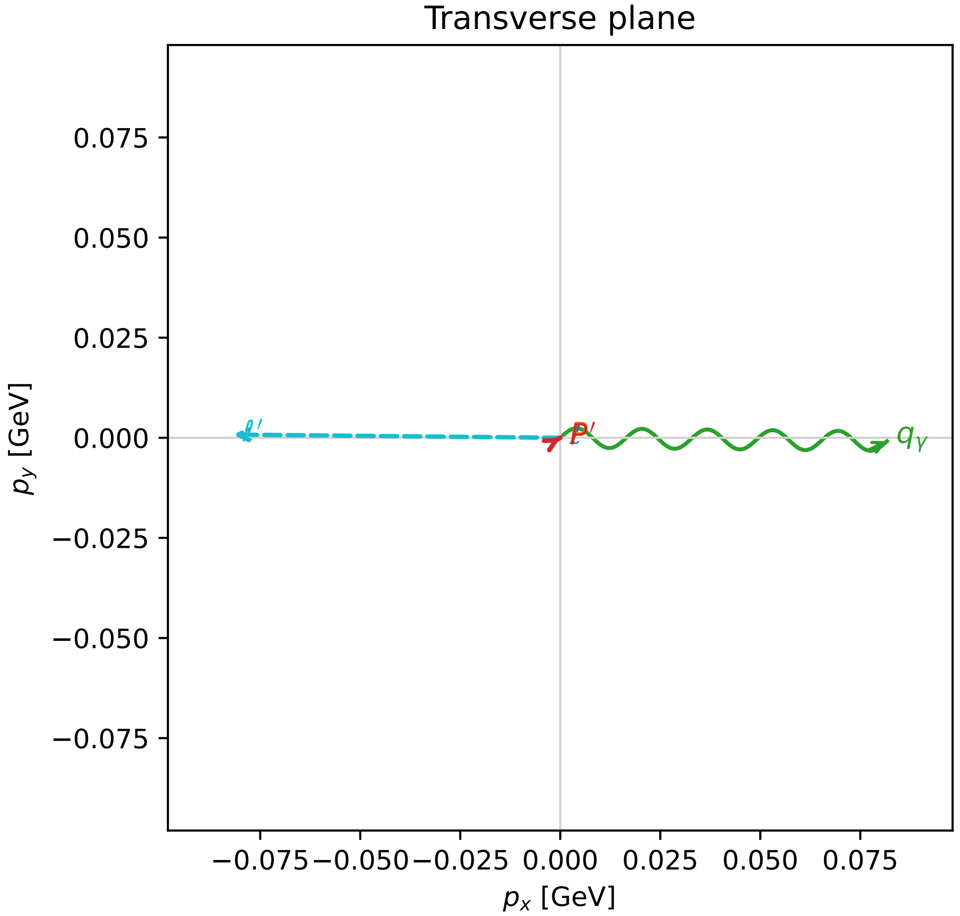



```
dw_mixing_angles_polarization_05_local_minimum_952  D_W=0.0160524
region: polarization_cluster_05_local_minimum_952
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3266746 rad
incoming lepton: -0.685928|+> +0.727669|->
proton mixing angle: alpha_p=2.9980848 rad
incoming proton: -0.98972|+> +0.143016|->
```

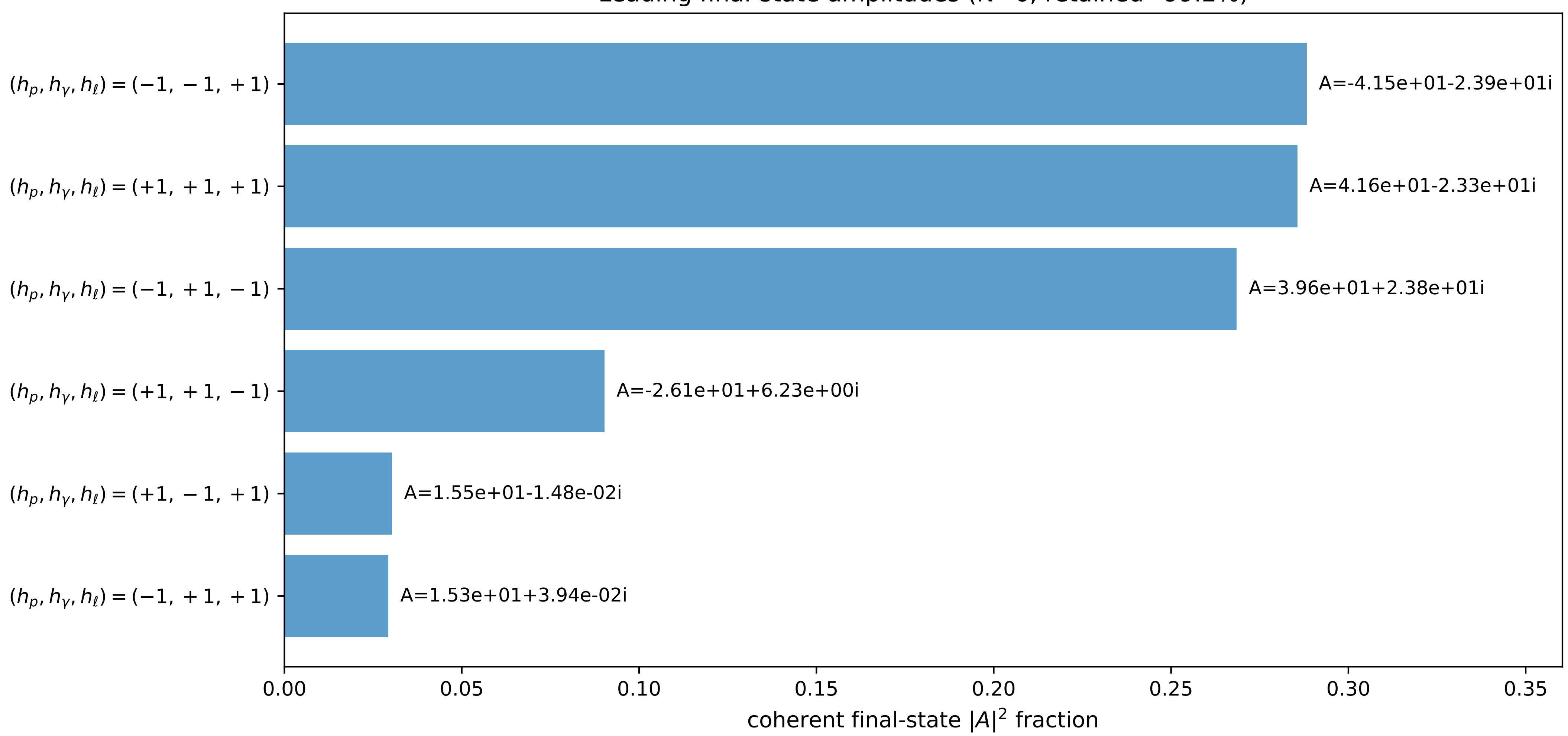
s=1.24192, $\sqrt{s}$=1.11442, $|\vec{P}|$=0.162451, $|\vec{P}'|$=0.000137254
$\theta_{p'}$=2.52225, θ_γ=1.1836, $\phi_{p'}$=0.509658, ϕ_γ=6.27294
E_γ=0.0881959, $\theta_{\ell'}$=1.95653, $\phi_{\ell'}$=3.13183
Q^2=0.0178785, x_B=0.0975208, t=-0.0262317, W^2=1.0453, y=0.506329

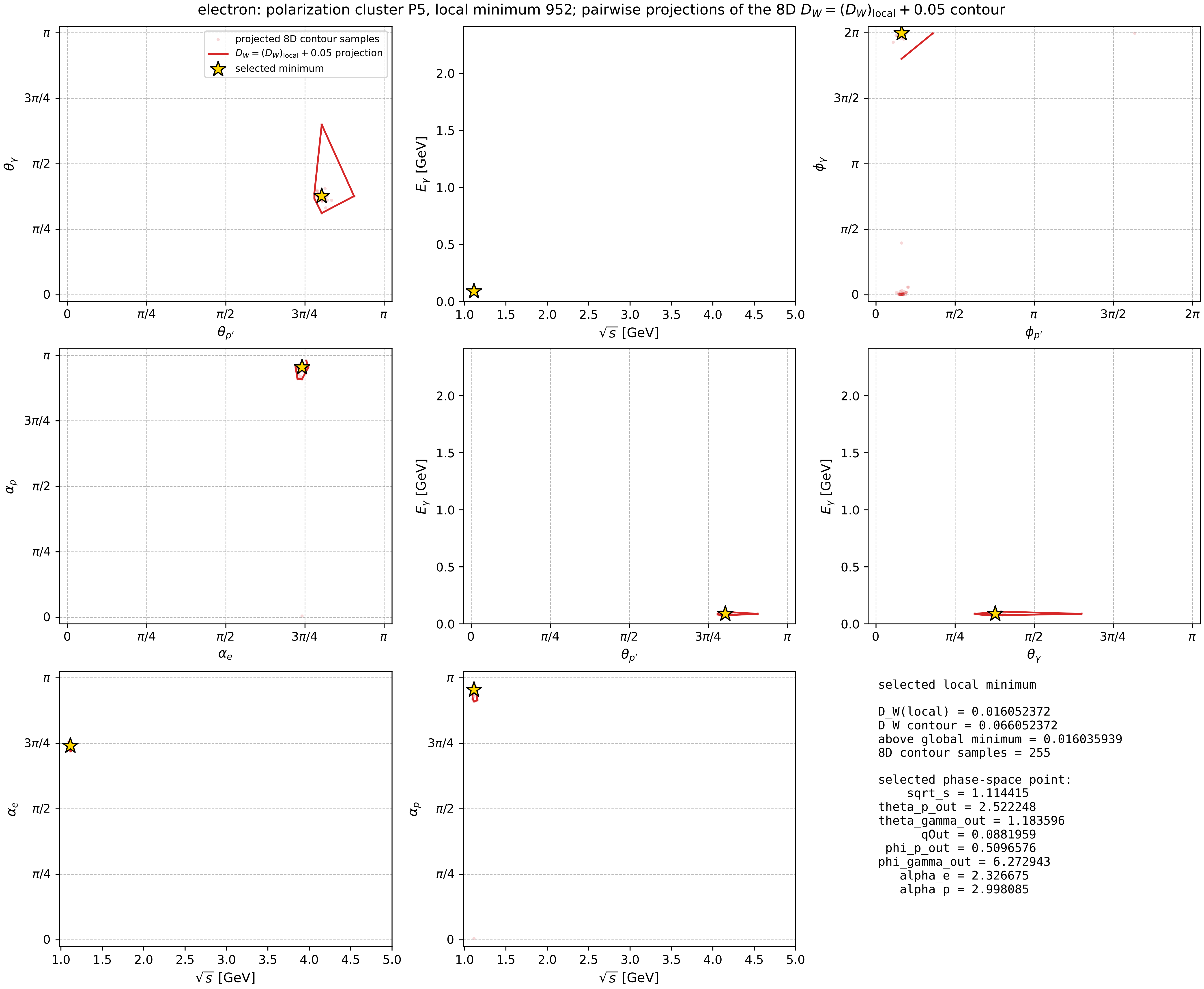
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1625 p_x= 0.0000 p_y= 0.0000 p_z= -0.1625
P E= 0.9520 p_x= 0.0000 p_y= 0.0000 p_z= 0.1625
ℓ' E= 0.0882 p_x= -0.0817 p_y= 0.0008 p_z= -0.0332
P' E= 0.9380 p_x= 0.0001 p_y= 0.0000 p_z= -0.0001
q_γ E= 0.0882 p_x= 0.0817 p_y= -0.0008 p_z= 0.0333

Transverse plane



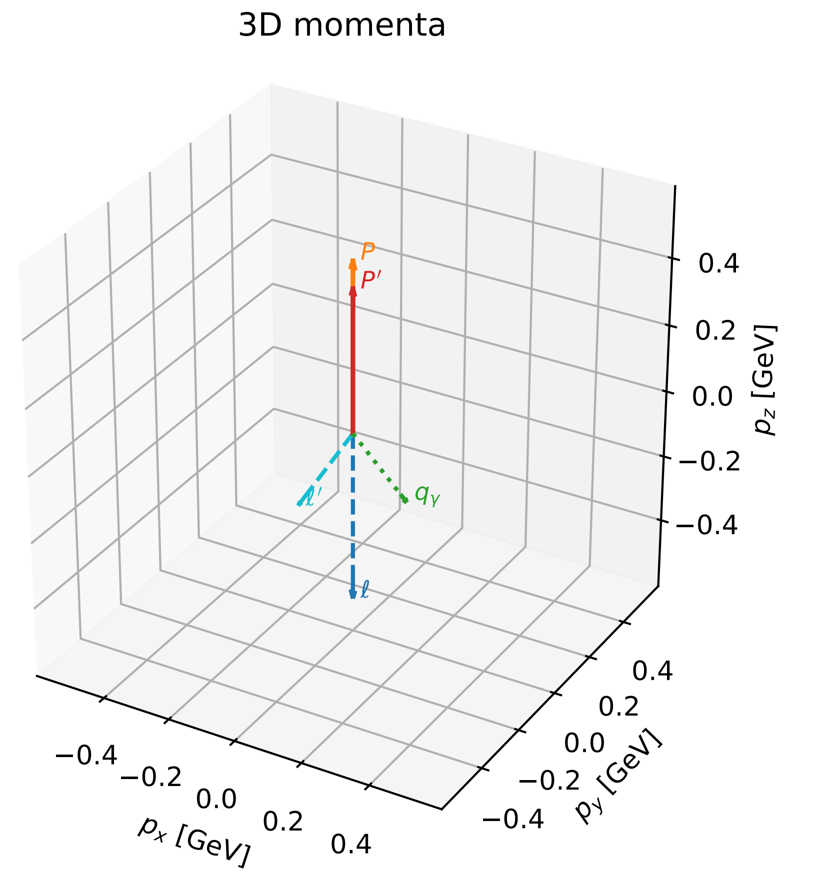
Leading final-state amplitudes (N=6, retained=99.2%)





electron: polarization cluster P5, local minimum 952; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.016052372
D_W contour = 0.066052372
above global minimum = 0.016035939
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.114415
theta_p_out = 2.522248
theta_gamma_out = 1.183596
qOut = 0.0881959
phi_p_out = 0.5096576
phi_gamma_out = 6.272943
alpha_e = 2.326675
alpha_p = 2.998085

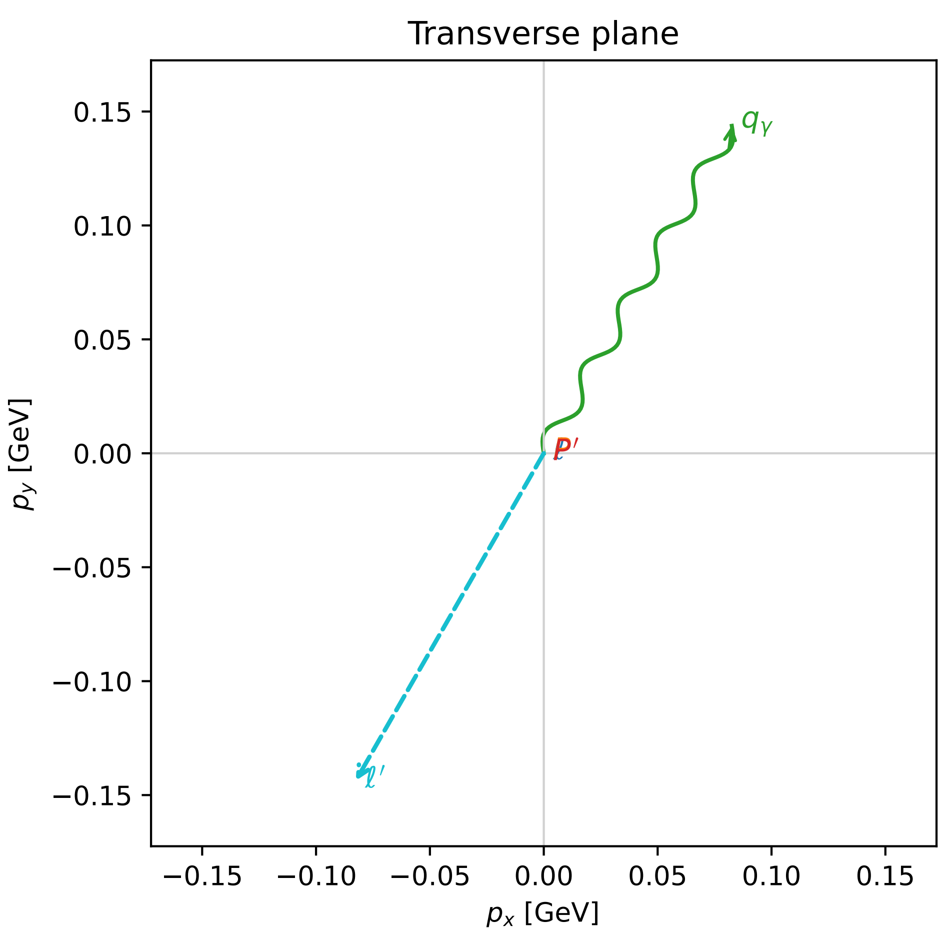
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

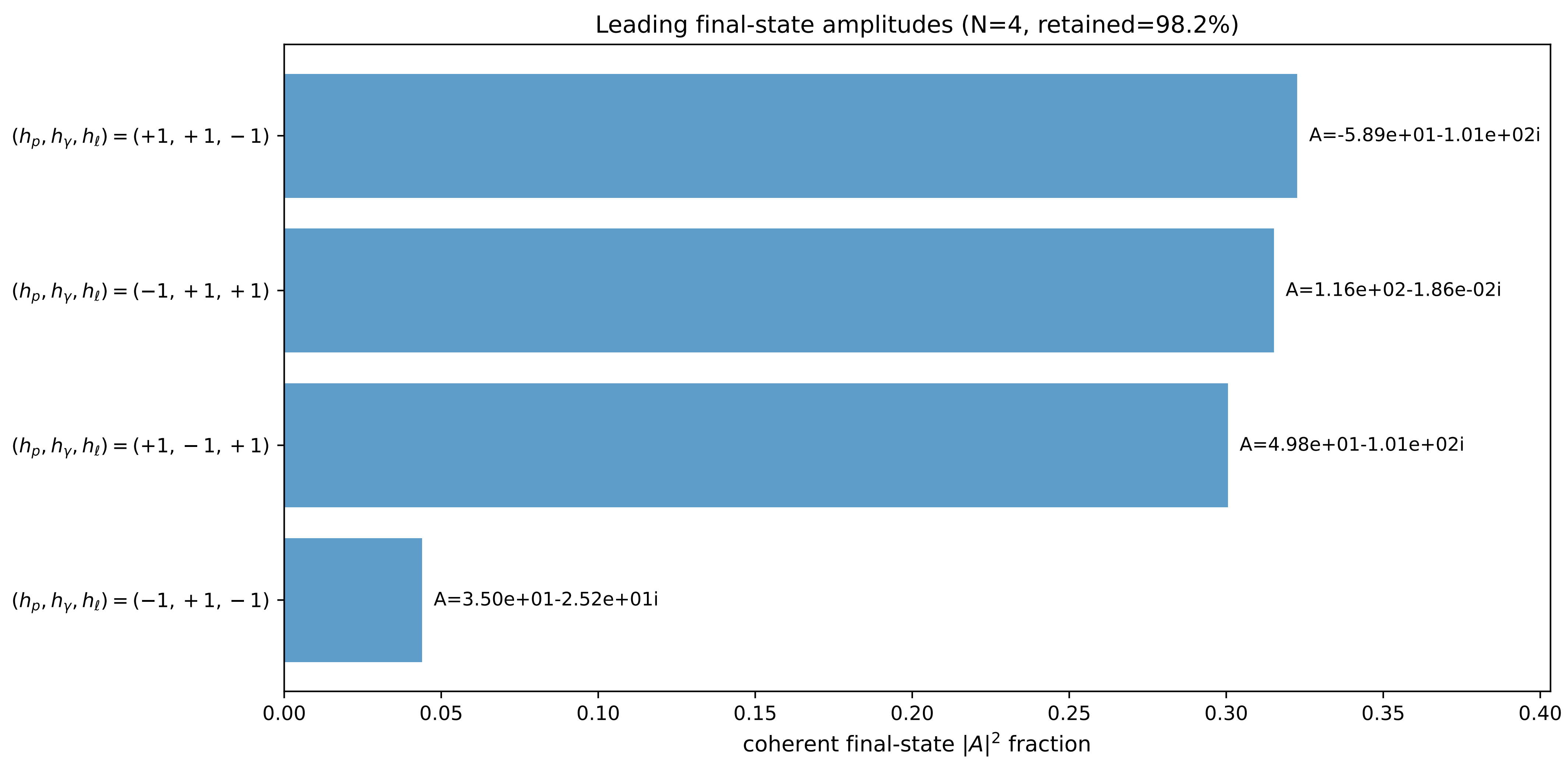


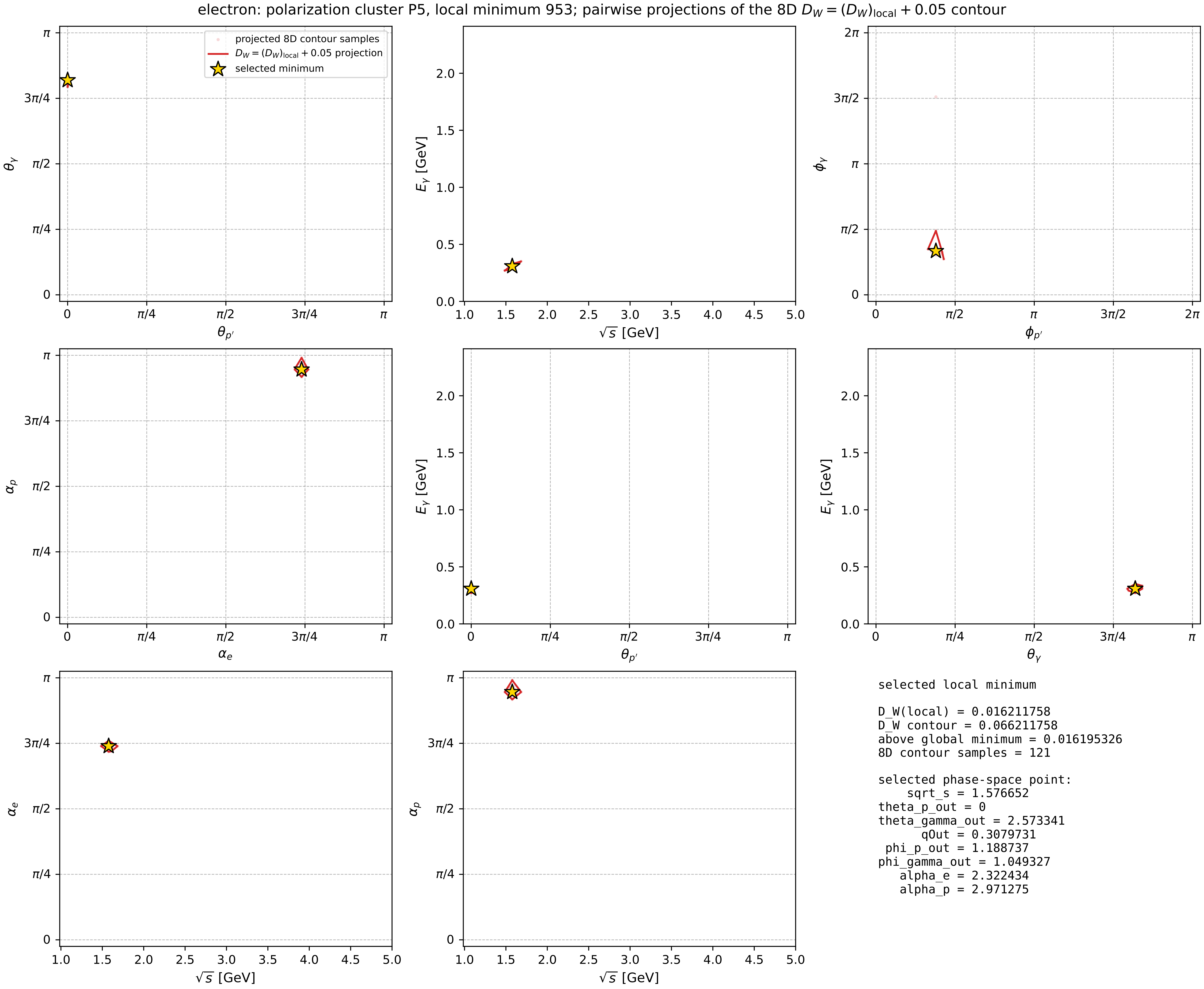
dw_mixing_angles_polarization_05_local_minimum_953 D_W=0.0162118
region: polarization_cluster_05_local_minimum_953
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3224337 rad
incoming lepton: -0.682836|+> +0.730572|->
proton mixing angle: alpha_p=2.9712753 rad
incoming proton: -0.985531|+> +0.169495|->

s=2.48583, $\sqrt{s}$=1.57665, $|\vec{P}|$=0.509303, $|\vec{P}'|$=0.429193
$\theta_{p'}$=0, θ_γ=2.57334, $\phi_{p'}$=1.18874, ϕ_γ=1.04933
E_γ=0.307973, $\theta_{\ell'}$=2.36777, $\phi_{\ell'}$=4.19092
Q^2=0.0687868, x_B=0.0742063, t=-0.00513449, W^2=1.73802, y=0.577194

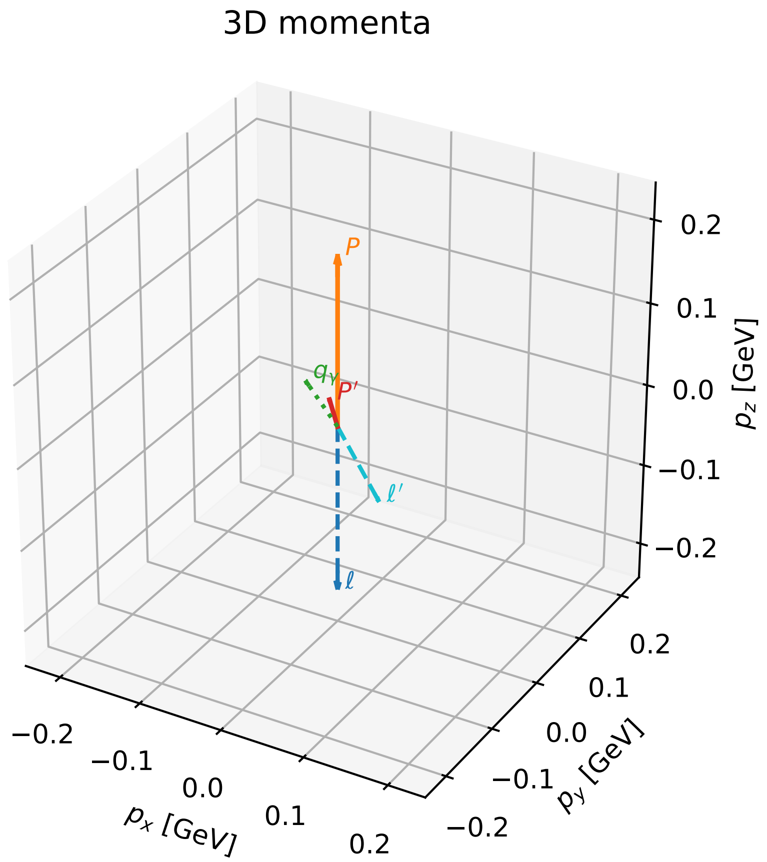
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.5093 p_x= 0.0000 p_y= 0.0000 p_z= -0.5093
P E= 1.0673 p_x= 0.0000 p_y= 0.0000 p_z= 0.5093
ℓ' E= 0.2372 p_x= -0.0826 p_y= -0.1437 p_z= -0.1696
P' E= 1.0315 p_x= 0.0000 p_y= 0.0000 p_z= 0.4292
q_γ E= 0.3080 p_x= 0.0826 p_y= 0.1437 p_z= -0.2596






electron: polarization cluster P5, local minimum 953; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.016211758
D_W contour = 0.066211758
above global minimum = 0.016195326
8D contour samples = 121
selected phase-space point:
sqrt_s = 1.576652
theta_p_out = 0
theta_gamma_out = 2.573341
qOut = 0.3079731
phi_p_out = 1.188737
phi_gamma_out = 1.049327
alpha_e = 2.322434
alpha_p = 2.971275

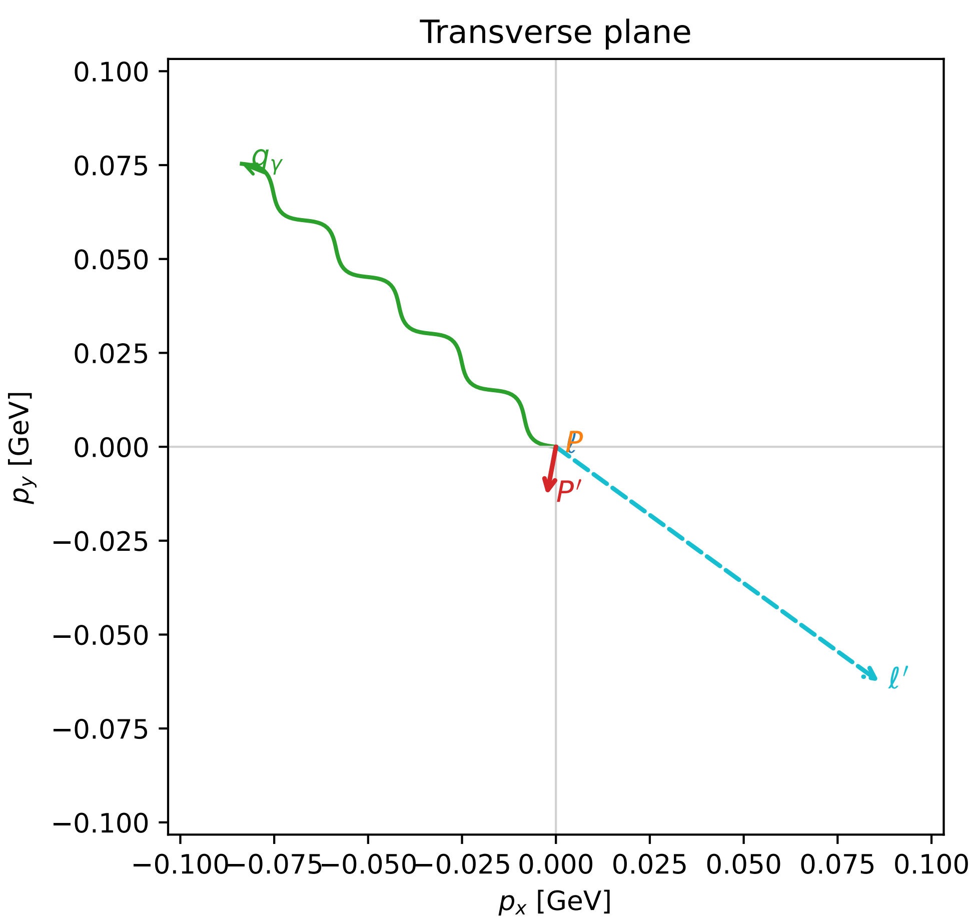
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

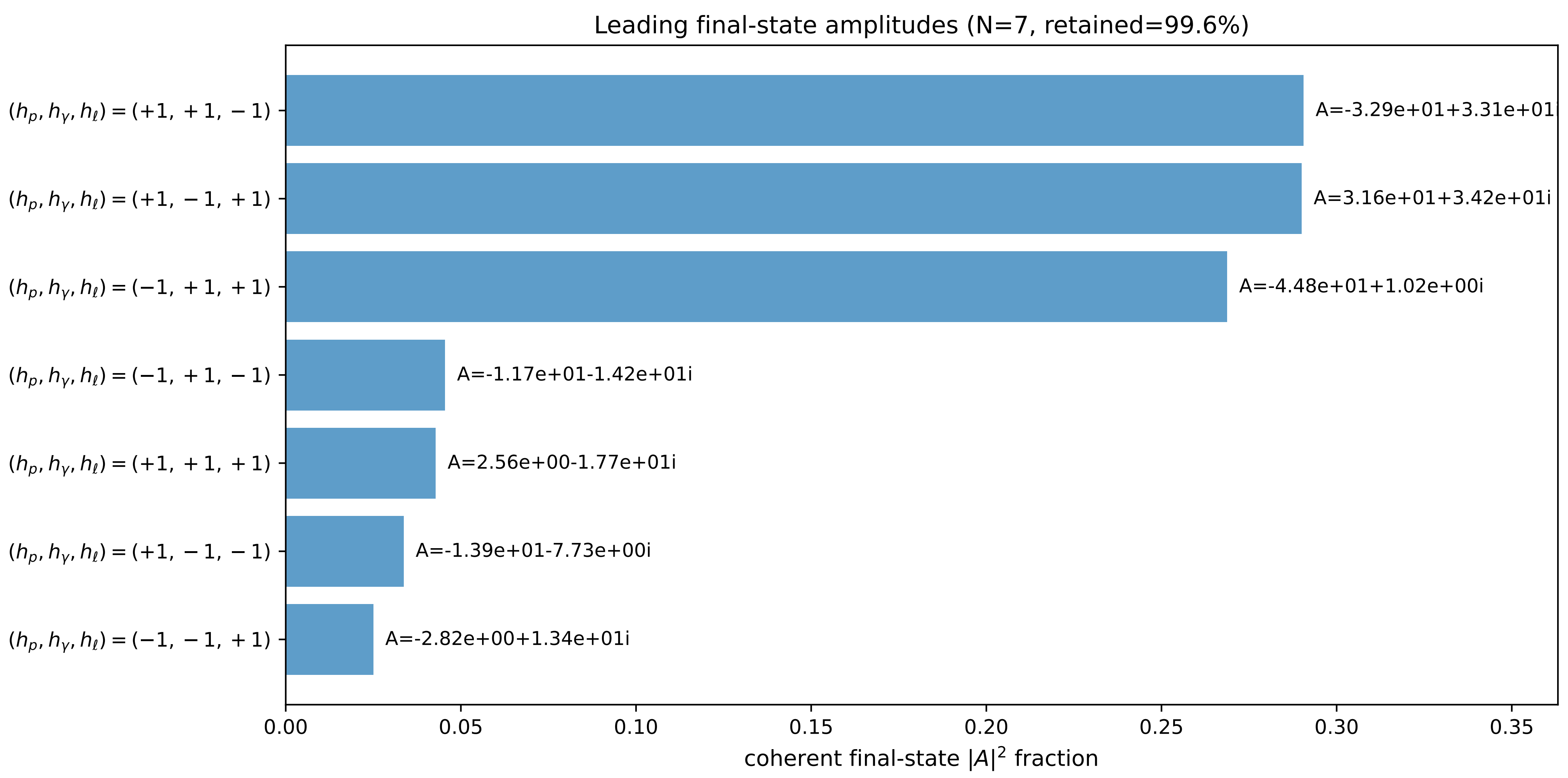



```
dw_mixing_angles_polarization_05_local_minimum_954  D_W=0.0163484
region: polarization_cluster_05_local_minimum_954
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3149515 rad
incoming lepton: -0.677351|+> +0.73566|->
proton mixing angle: alpha_p=0.26048191 rad
incoming proton: 0.966266|+> +0.257546|->
```

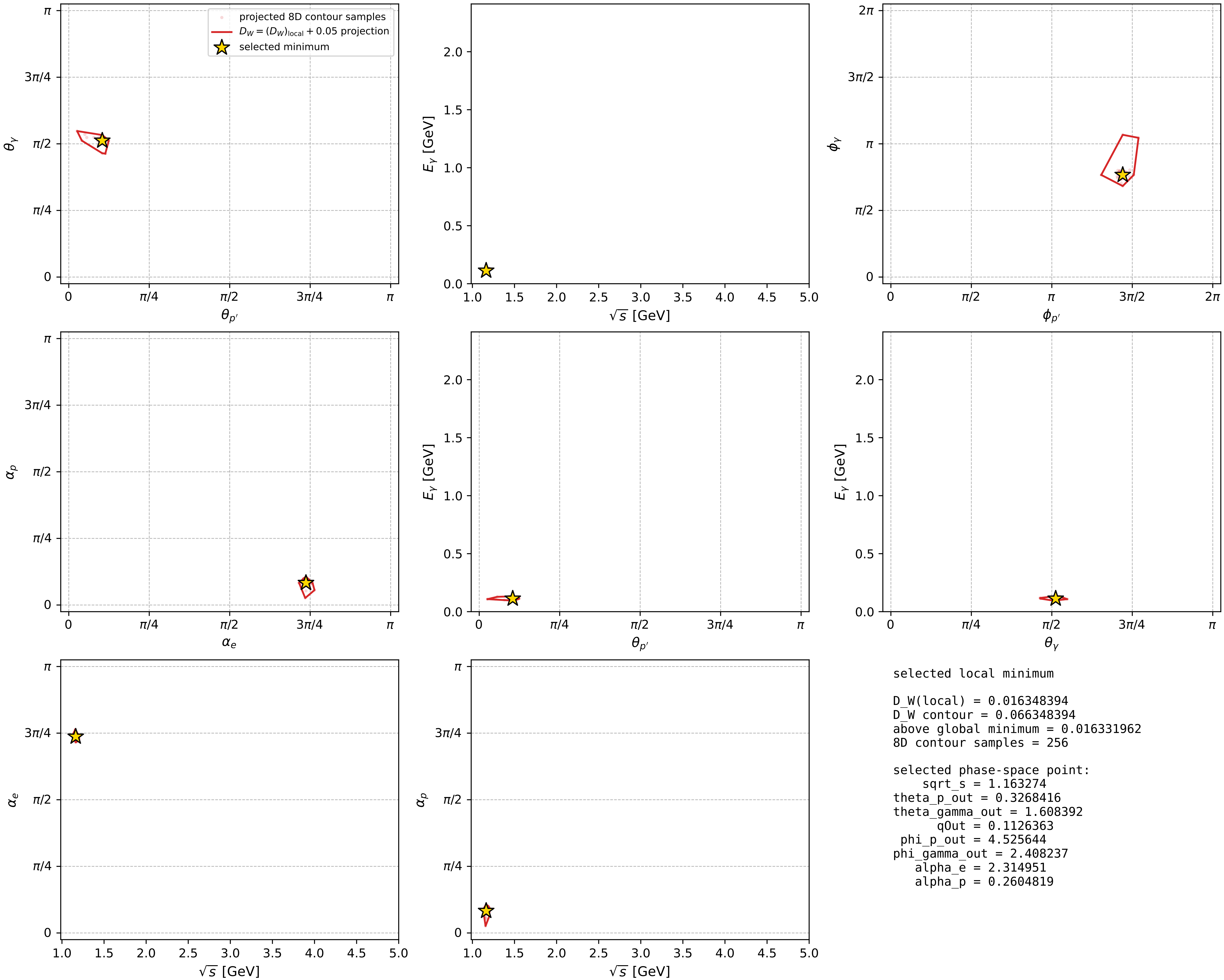
s=1.35321, $\sqrt{s}$=1.16327, $|\vec{P}|$=0.203461, $|\vec{P}'|$=0.0407186
$\theta_{p'}$=0.326842, θ_γ=1.60839, $\phi_{p'}$=4.52564, ϕ_γ=2.40824
E_γ=0.112636, $\theta_{\ell'}$=1.88304, $\phi_{\ell'}$=5.65504
Q^2=0.0315054, x_B=0.128664, t=-0.0269241, W^2=1.0932, y=0.517292

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2035 p_x= 0.0000 p_y= 0.0000 p_z= -0.2035
P E= 0.9598 p_x= 0.0000 p_y= 0.0000 p_z= 0.2035
ℓ' E= 0.1118 p_x= 0.0860 p_y= -0.0625 p_z= -0.0343
P' E= 0.9389 p_x= -0.0024 p_y= -0.0128 p_z= 0.0386
q_γ E= 0.1126 p_x= -0.0836 p_y= 0.0753 p_z= -0.0042



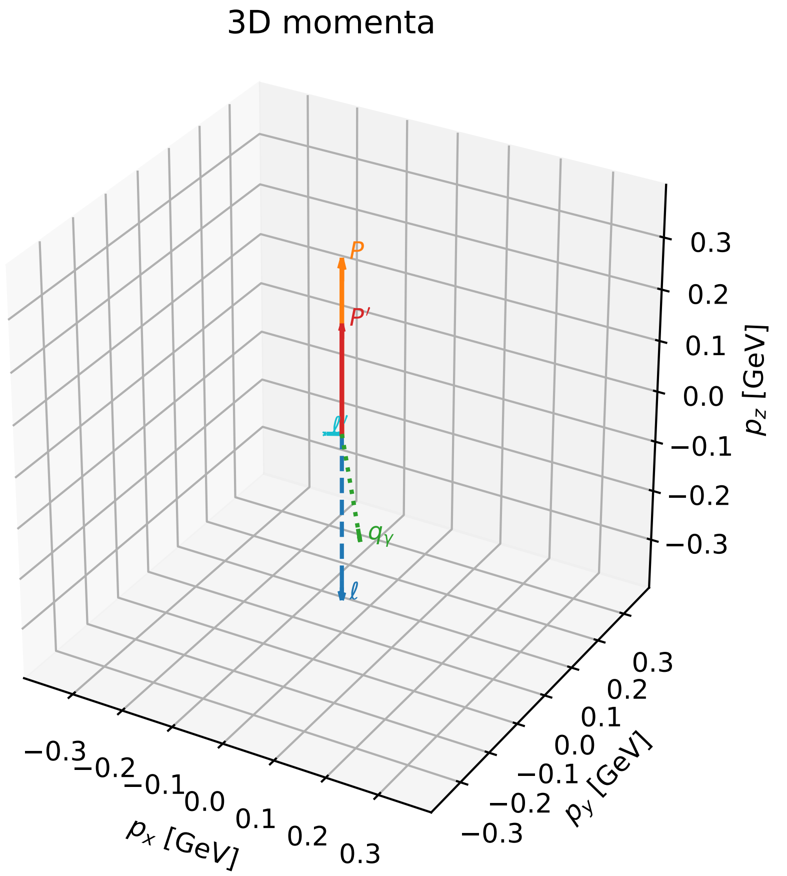


electron: polarization cluster P5, local minimum 954; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour




```
selected local minimum

D_W(local) = 0.016348394
D_W contour = 0.066348394
above global minimum = 0.016331962
8D contour samples = 256

selected phase-space point:
    sqrt_s = 1.163274
theta_p_out = 0.3268416
theta_gamma_out = 1.608392
       qOut = 0.1126363
 phi_p_out = 4.525644
phi_gamma_out = 2.408237
   alpha_e = 2.314951
   alpha_p = 0.2604819
```

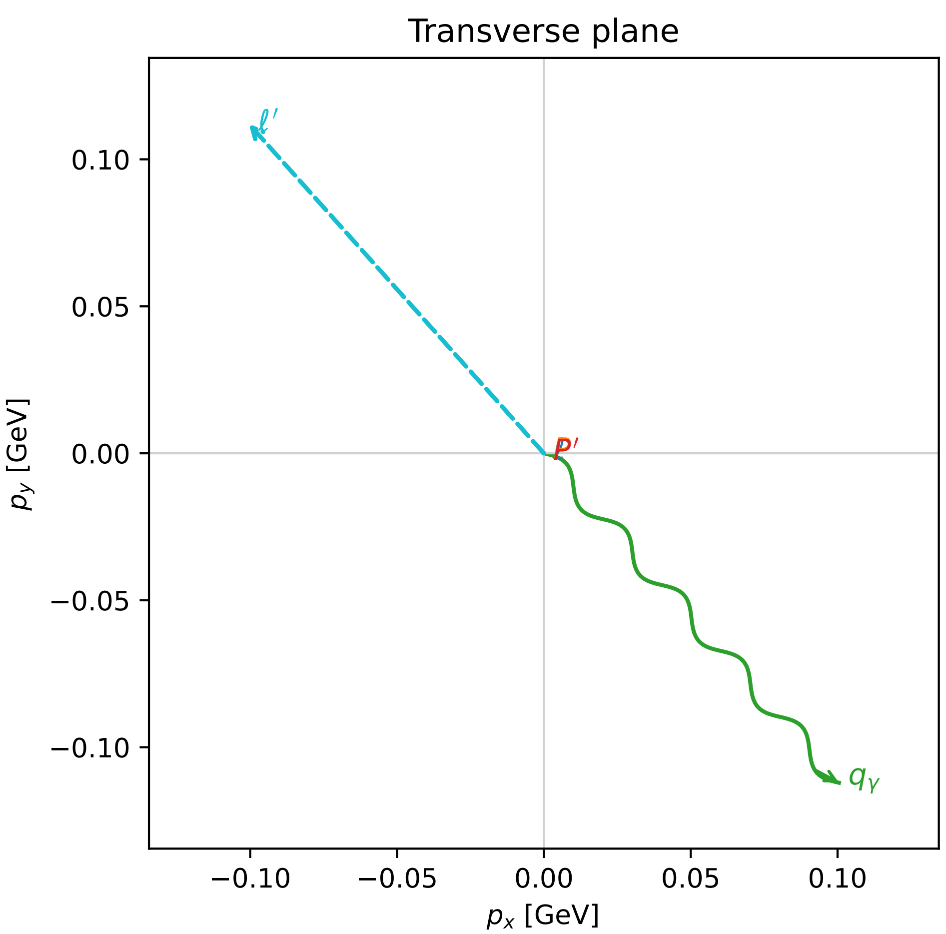
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

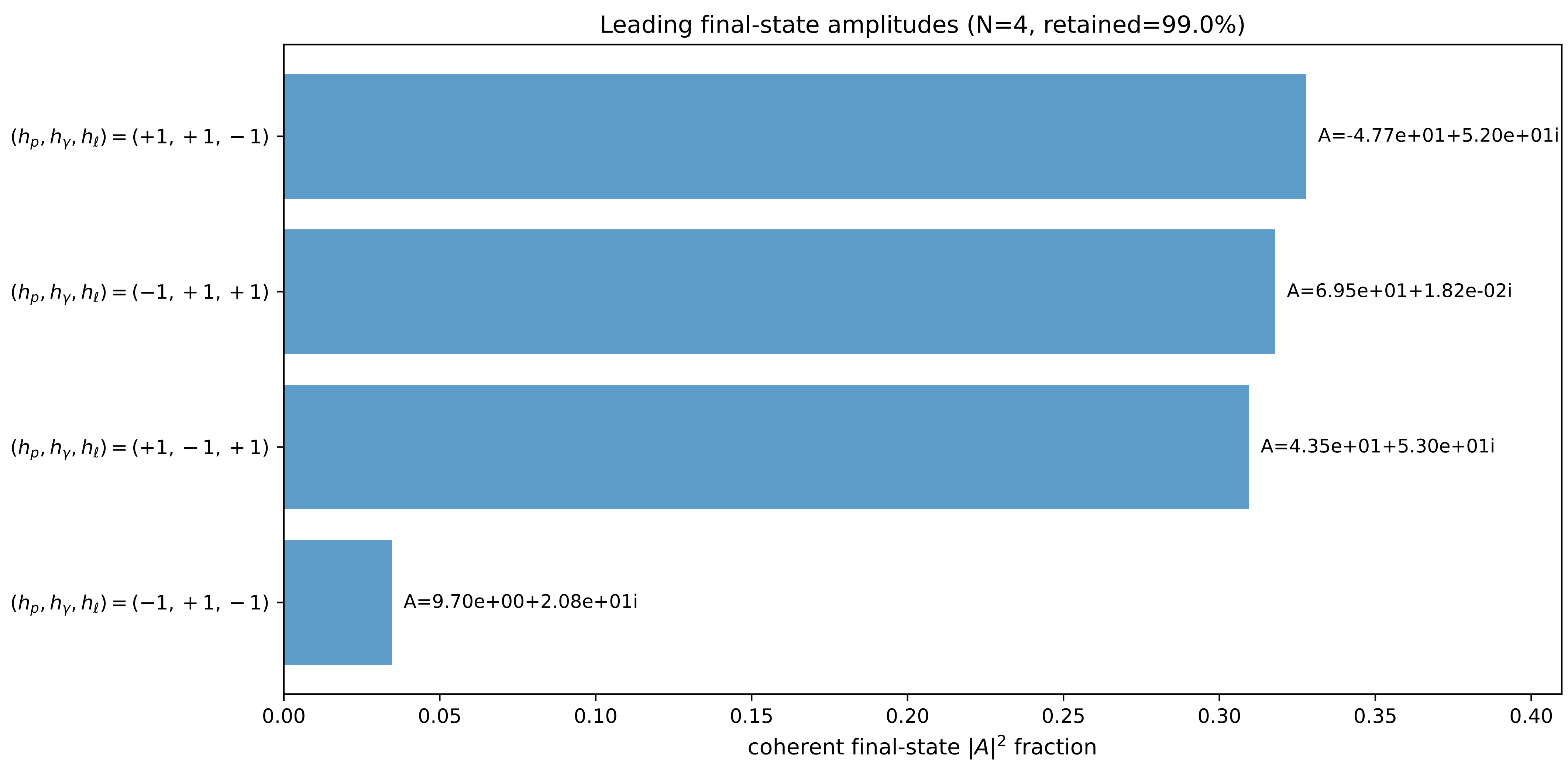


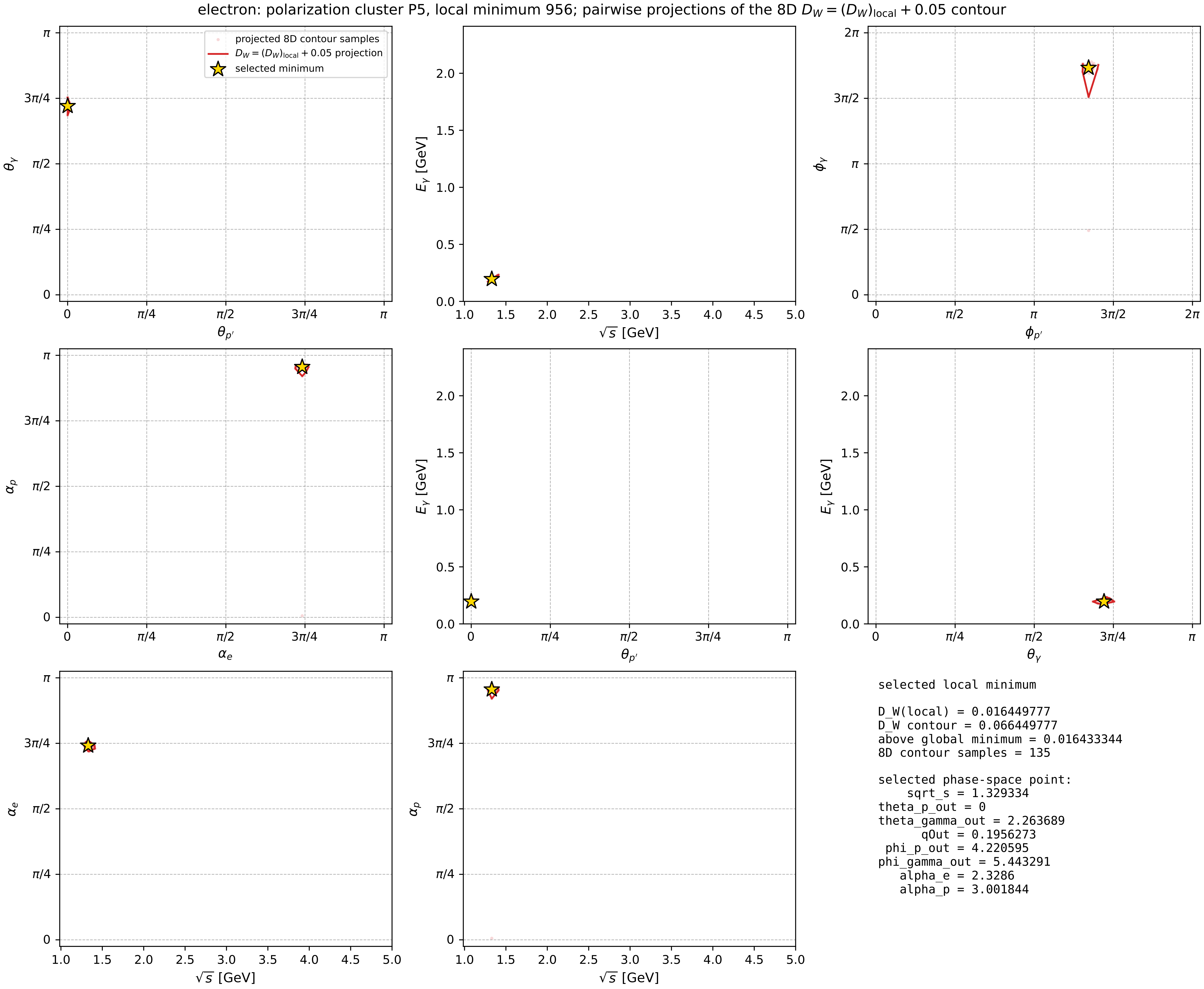
dw_mixing_angles_polarization_05_local_minimum_956 D_W=0.0164498
region: polarization_cluster_05_local_minimum_956
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3285997 rad
incoming lepton: -0.687328|+> +0.726348|->
proton mixing angle: alpha_p=3.0018437 rad
incoming proton: -0.990251|+> +0.139295|->

s=1.76713, $\sqrt{s}$=1.32933, $|\vec{P}|$=0.333732, $|\vec{P}'|$=0.209446
$\theta_{p'}$=0, θ_γ=2.26369, $\phi_{p'}$=4.22059, ϕ_γ=5.44329
E_γ=0.195627, $\theta_{\ell'}$=2.08228, $\phi_{\ell'}$=2.3017
Q^2=0.0588172, x_B=0.120726, t=-0.0142567, W^2=1.30822, y=0.549087

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.3337 p_x= 0.0000 p_y= 0.0000 p_z= -0.3337
P E= 0.9956 p_x= 0.0000 p_y= 0.0000 p_z= 0.3337
ℓ' E= 0.1726 p_x= -0.1005 p_y= 0.1121 p_z= -0.0845
P' E= 0.9611 p_x= -0.0000 p_y= -0.0000 p_z= 0.2094
q_γ E= 0.1956 p_x= 0.1005 p_y= -0.1121 p_z= -0.1250

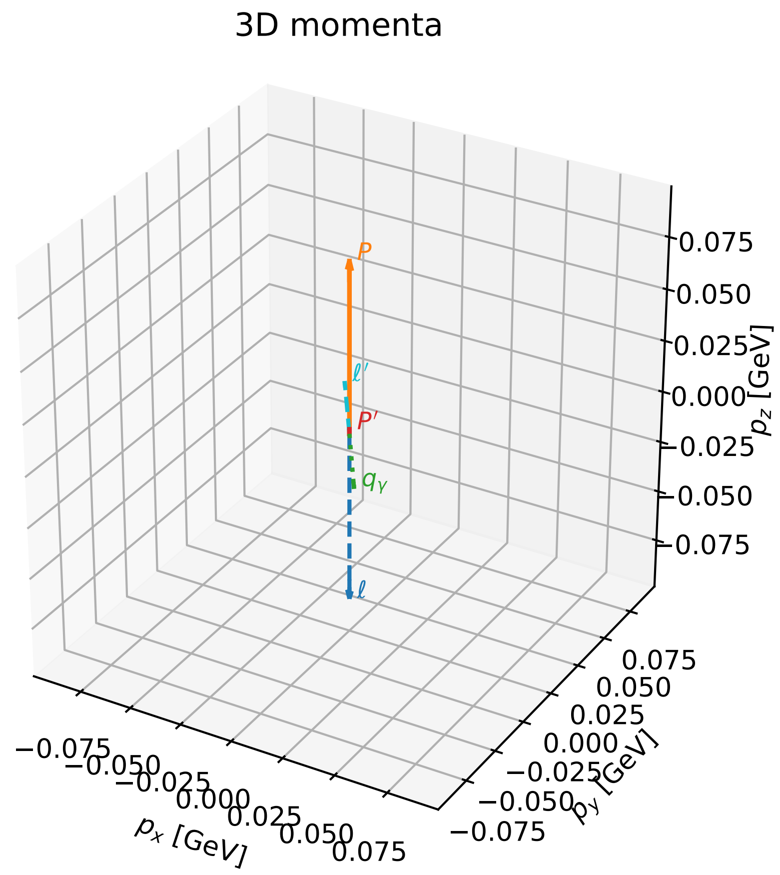






electron: polarization cluster P5, local minimum 956; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.016449777
D_W contour = 0.066449777
above global minimum = 0.016433344
8D contour samples = 135
selected phase-space point:
sqrt_s = 1.329334
theta_p_out = 0
theta_gamma_out = 2.263689
qOut = 0.1956273
phi_p_out = 4.220595
phi_gamma_out = 5.443291
alpha_e = 2.3286
alpha_p = 3.001844

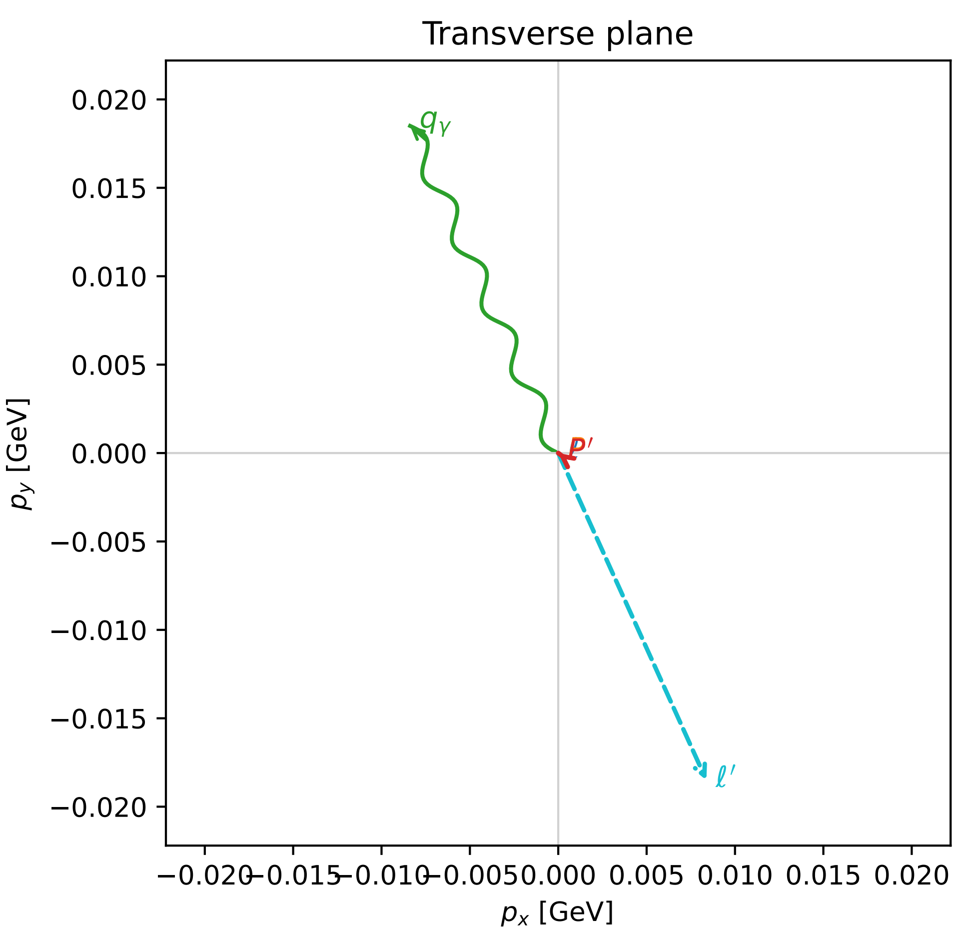
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

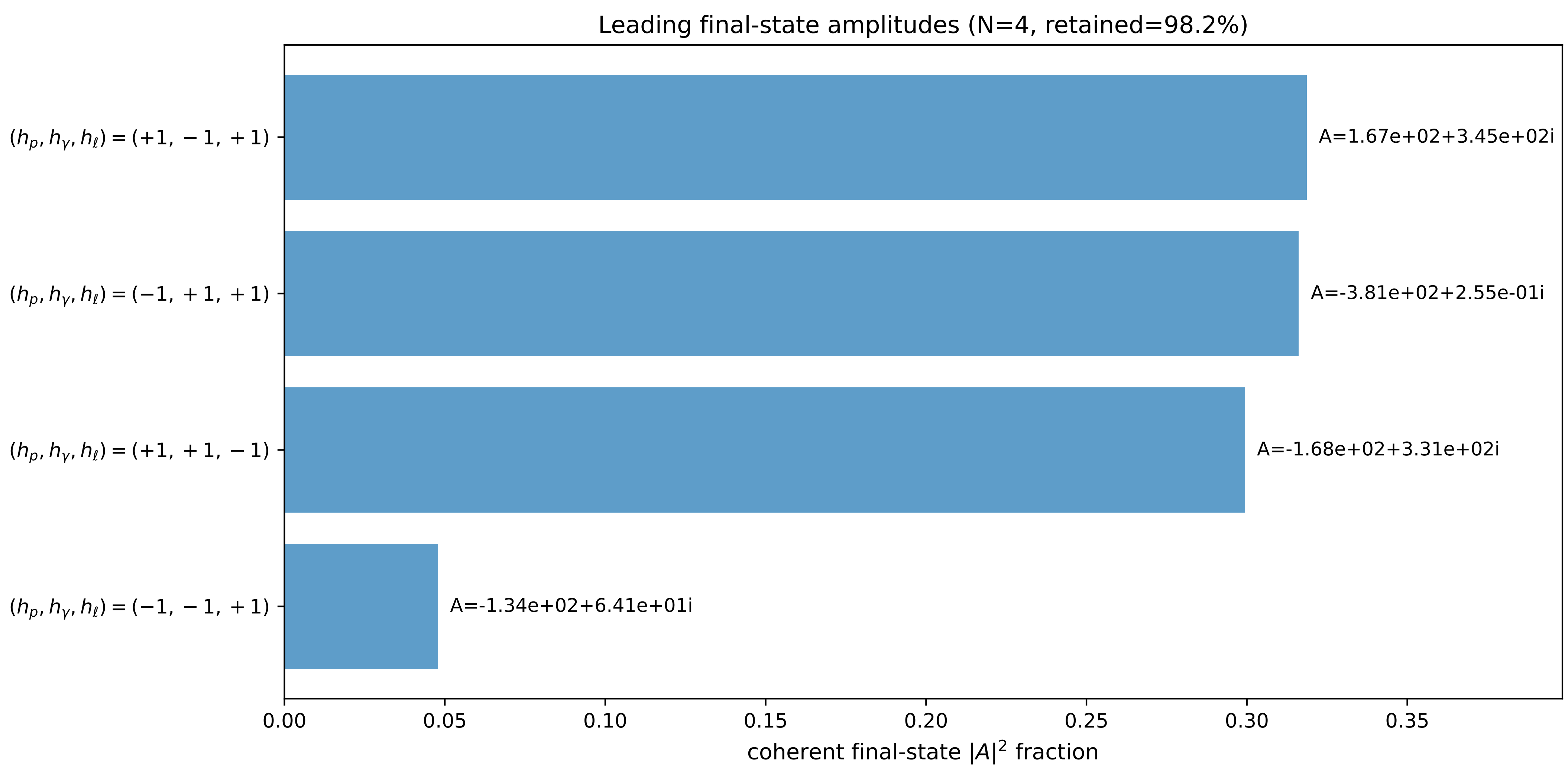



```
dw_mixing_angles_polarization_05_local_minimum_962  D_W=0.016629
region: polarization_cluster_05_local_minimum_962
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3731332 rad
incoming lepton: -0.718982|+> +0.695028|->
proton mixing angle: alpha_p=0.19089651 rad
incoming proton: 0.981835|+> +0.189739|->
```

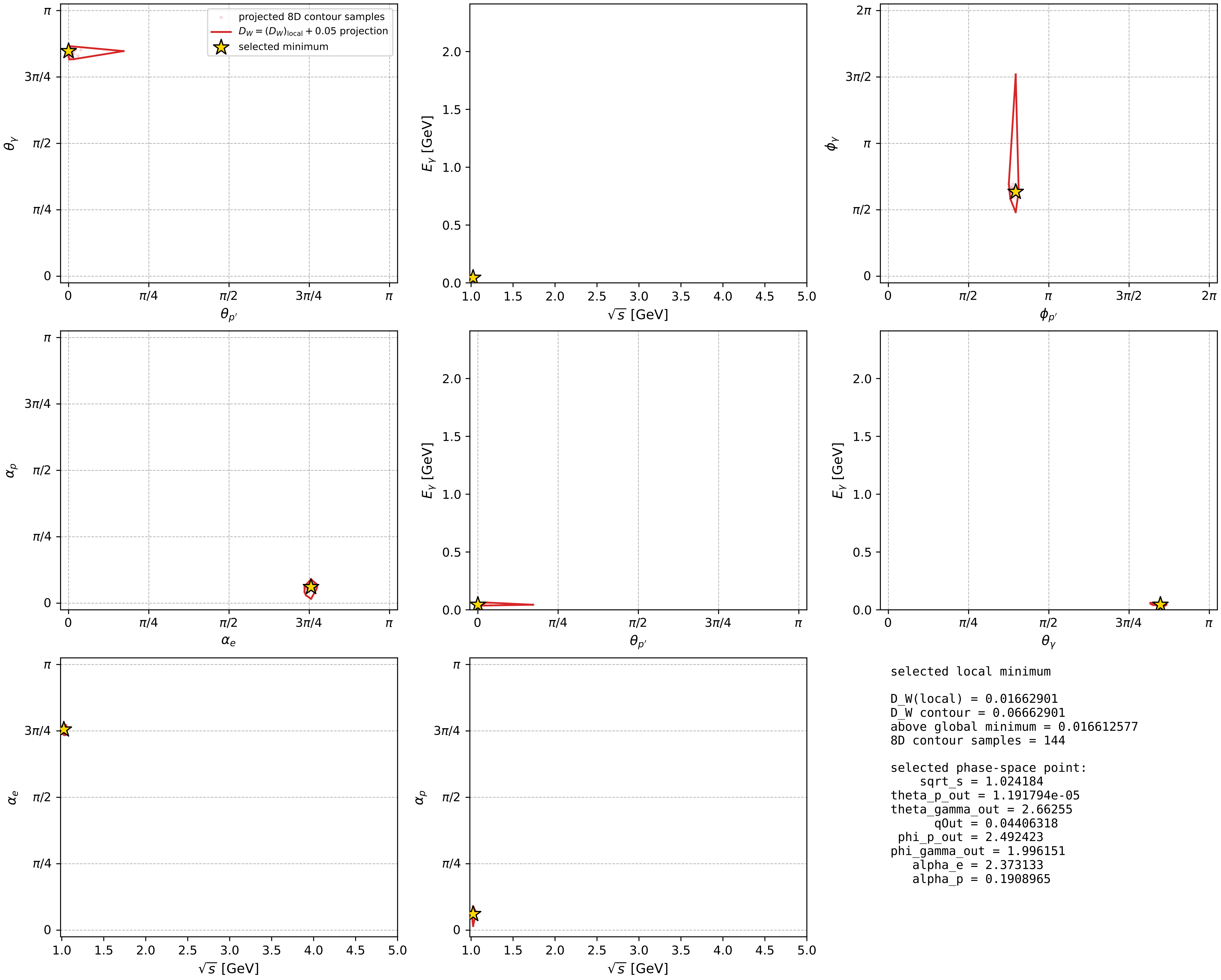
s=1.04895, $\sqrt{s}$=1.02418, $|\vec{P}|$=0.0825568, $|\vec{P}'|$=0.00220854
$\theta_{p'}$=1.19179e-05, θ_γ=2.66255, $\phi_{p'}$=2.49242, ϕ_γ=1.99615
E_γ=0.0440632, $\theta_{\ell'}$=0.503217, $\phi_{\ell'}$=5.13774
Q^2=0.0130458, x_B=0.136062, t=-0.00644271, W^2=0.962679, y=0.566977

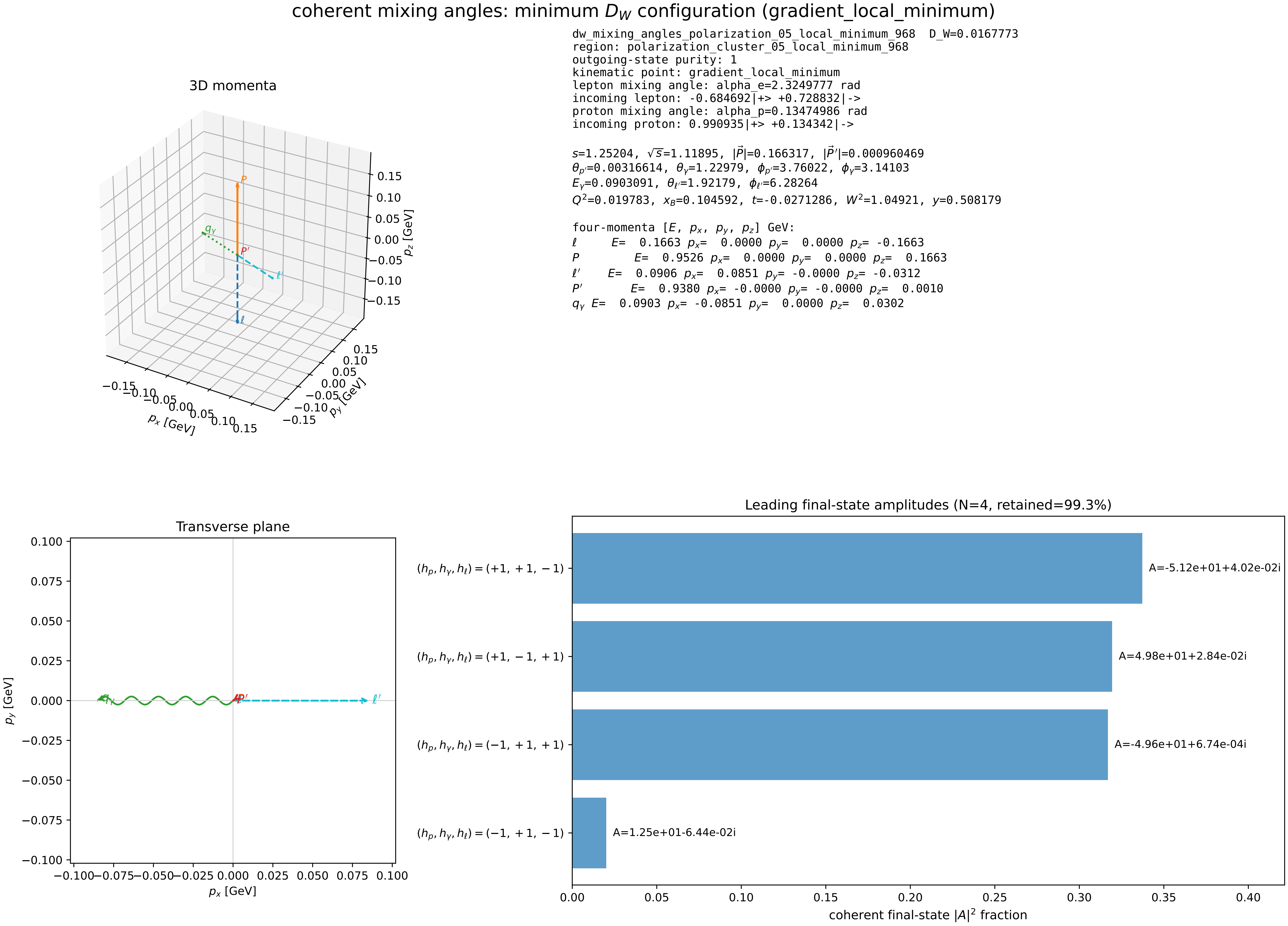
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0826 p_x= 0.0000 p_y= 0.0000 p_z= -0.0826
P E= 0.9416 p_x= 0.0000 p_y= 0.0000 p_z= 0.0826
ℓ' E= 0.0421 p_x= 0.0084 p_y= -0.0185 p_z= 0.0369
P' E= 0.9380 p_x= -0.0000 p_y= 0.0000 p_z= 0.0022
q_γ E= 0.0441 p_x= -0.0084 p_y= 0.0185 p_z= -0.0391





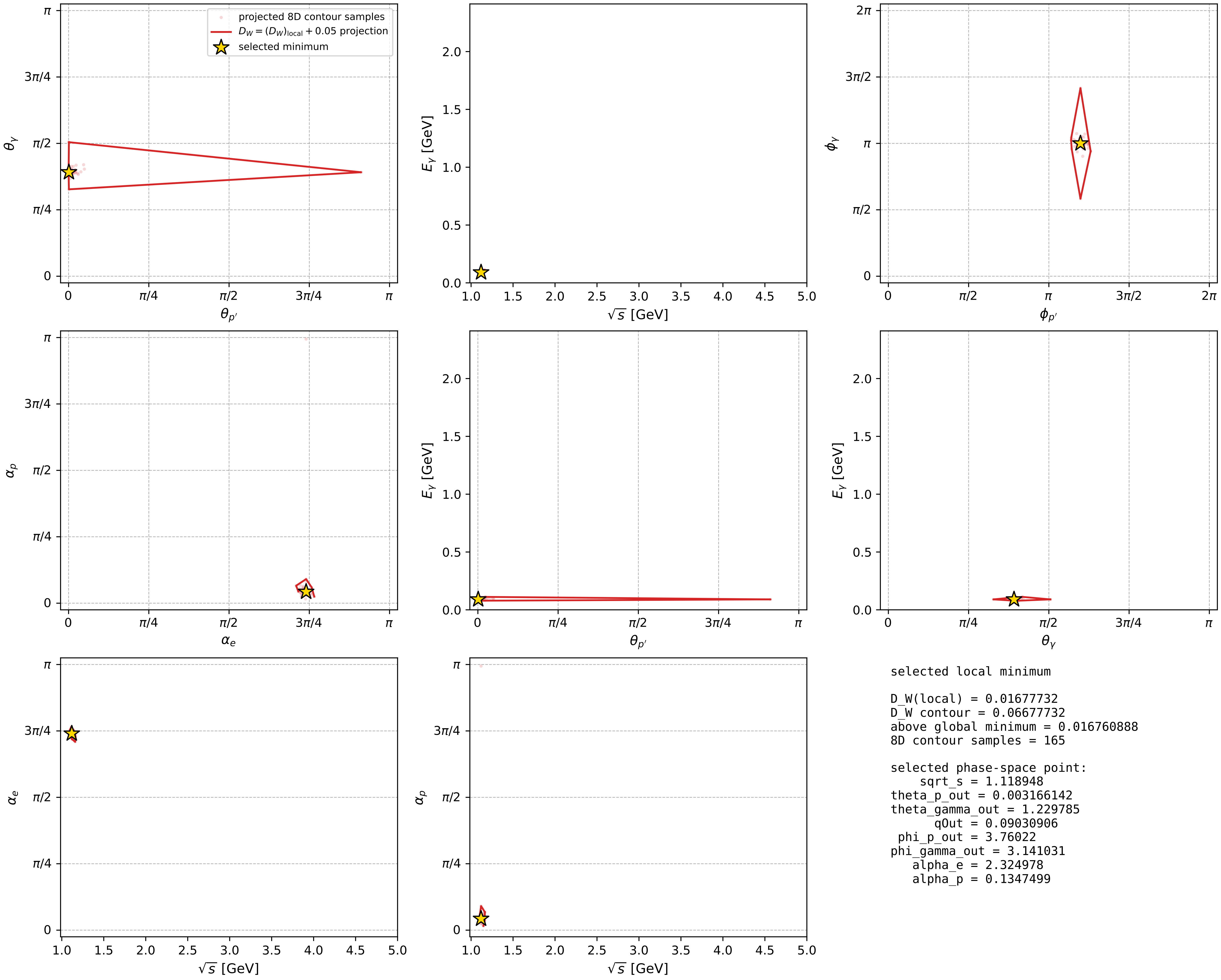
electron: polarization cluster P5, local minimum 962; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



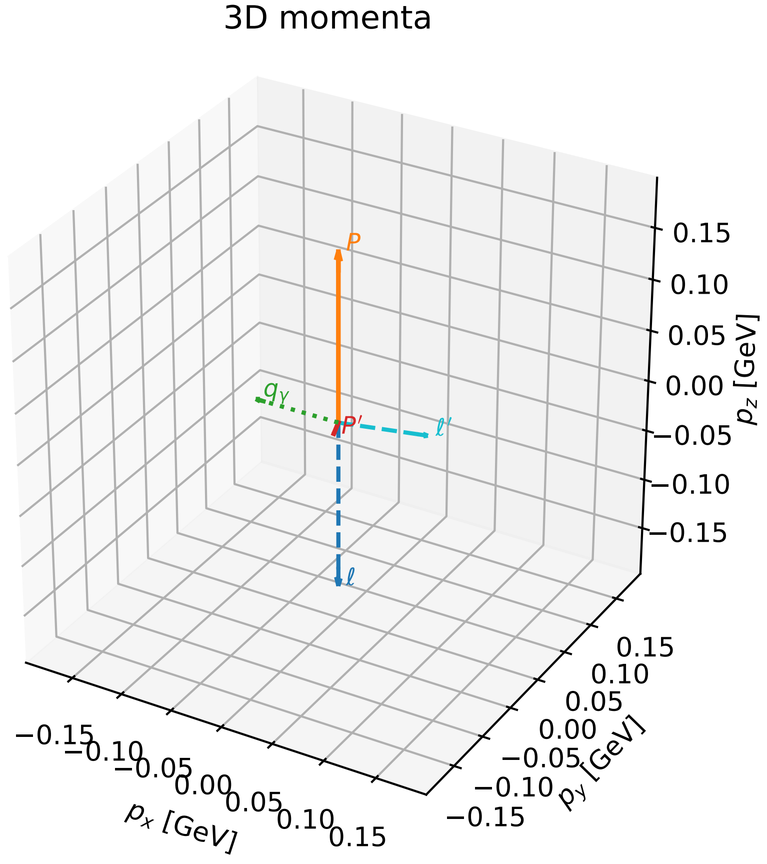


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_968 D_W=0.0167773
region: polarization_cluster_05_local_minimum_968
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3249777 rad
incoming lepton: -0.684692|+> +0.728832|->
proton mixing angle: alpha_p=0.13474986 rad
incoming proton: 0.990935|+> +0.134342|->
s=1.25204, $\sqrt{s}$=1.11895, $|\vec{P}|$=0.166317, $|\vec{P}'|$=0.000960469
$\theta_{p'}$=0.00316614, θ_γ=1.22979, $\phi_{p'}$=3.76022, ϕ_γ=3.14103
E_γ=0.0903091, $\theta_{\ell'}$=1.92179, $\phi_{\ell'}$=6.28264
Q^2=0.019783, x_B=0.104592, t=-0.0271286, W^2=1.04921, y=0.508179
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1663 p_x= 0.0000 p_y= 0.0000 p_z= -0.1663
P E= 0.9526 p_x= 0.0000 p_y= 0.0000 p_z= 0.1663
ℓ' E= 0.0906 p_x= 0.0851 p_y= -0.0000 p_z= -0.0312
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= 0.0010
q_γ E= 0.0903 p_x= -0.0851 p_y= 0.0000 p_z= 0.0302
3D momenta
P
q_γ
P'
ℓ'
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=4, retained=99.3%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-5.12e+01+4.02e-02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=4.98e+01+2.84e-02i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-4.96e+01+6.74e-04i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=1.25e+01-6.44e-02i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P5, local minimum 968; pairwise projections of the 8D $D_W = (D_W)_{\rm local} + 0.05$ contour



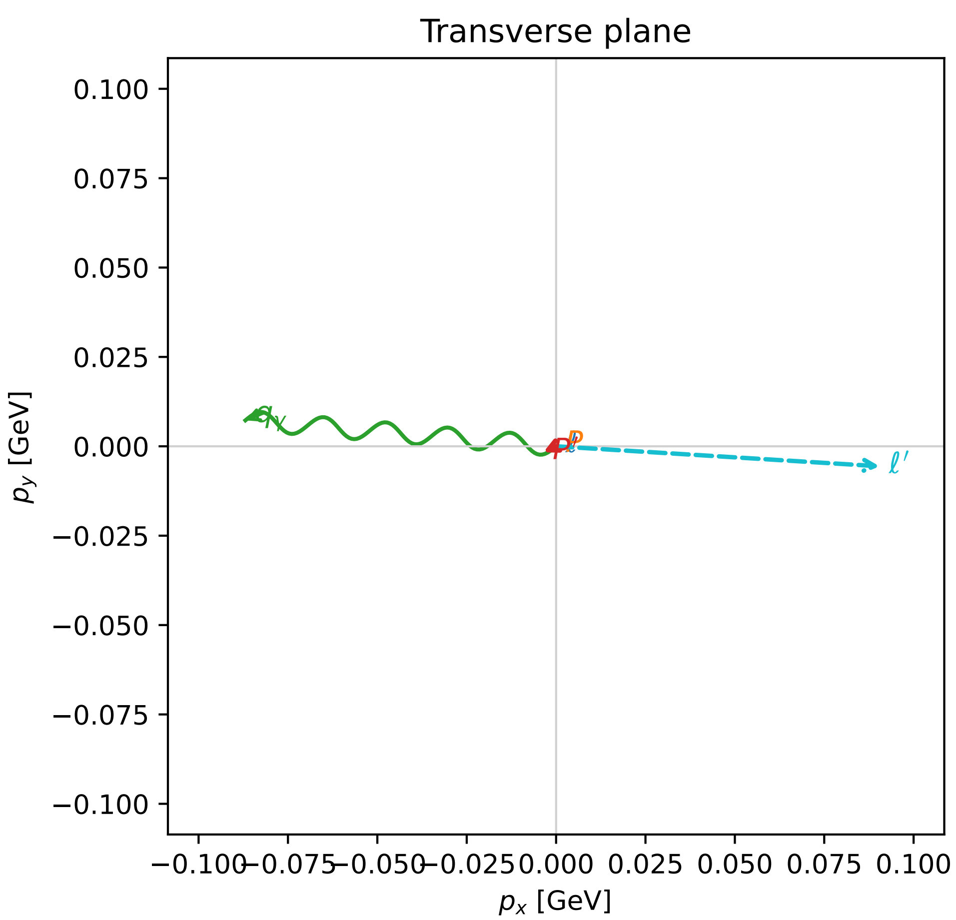
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

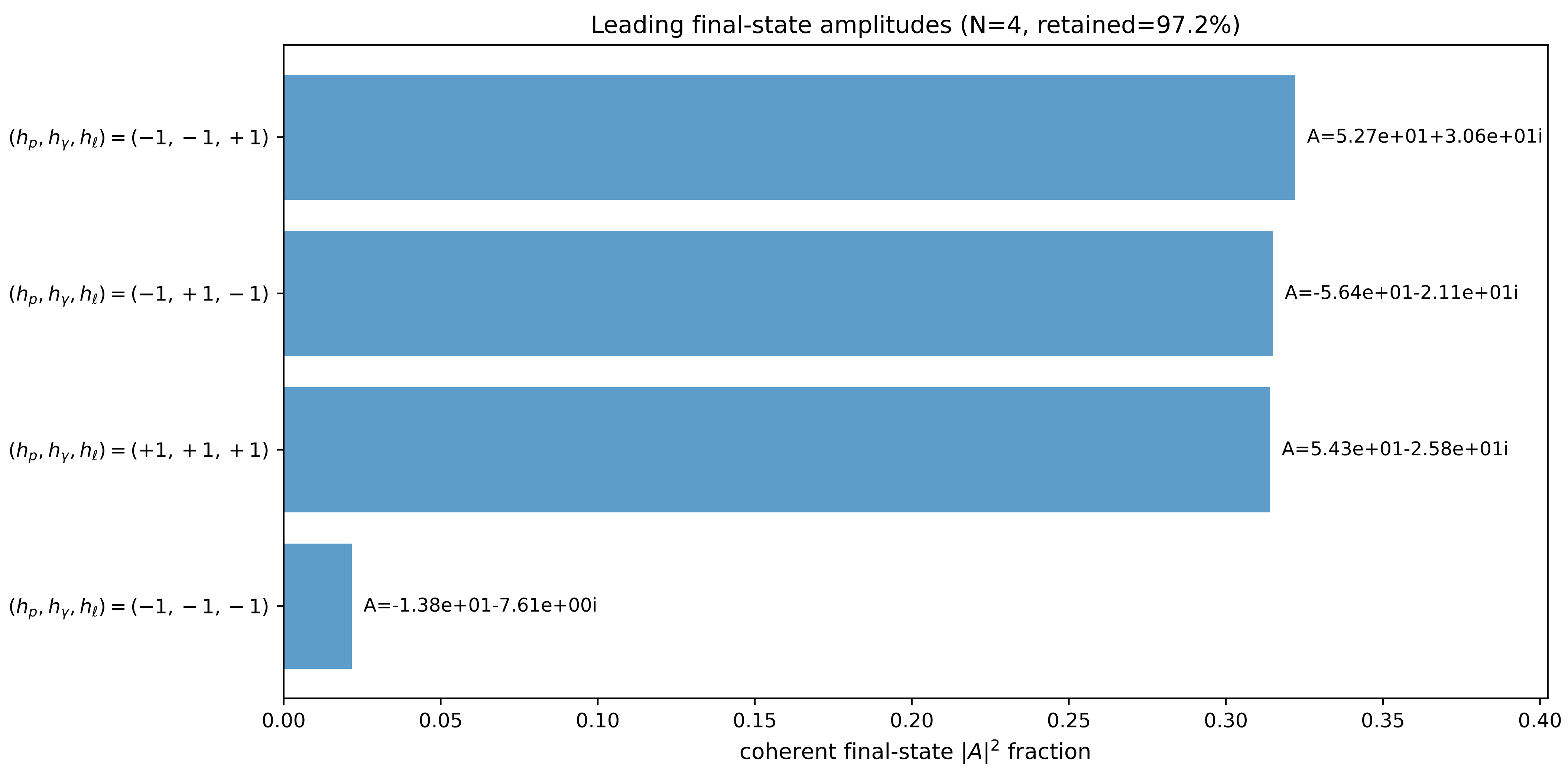



```
dw_mixing_angles_polarization_05_local_minimum_976  D_W=0.0172009
region: polarization_cluster_05_local_minimum_976
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3677104 rad
incoming lepton: -0.715203|+> +0.698917|->
proton mixing angle: alpha_p=0.23342827 rad
incoming proton: 0.972879|+> +0.231314|->
```

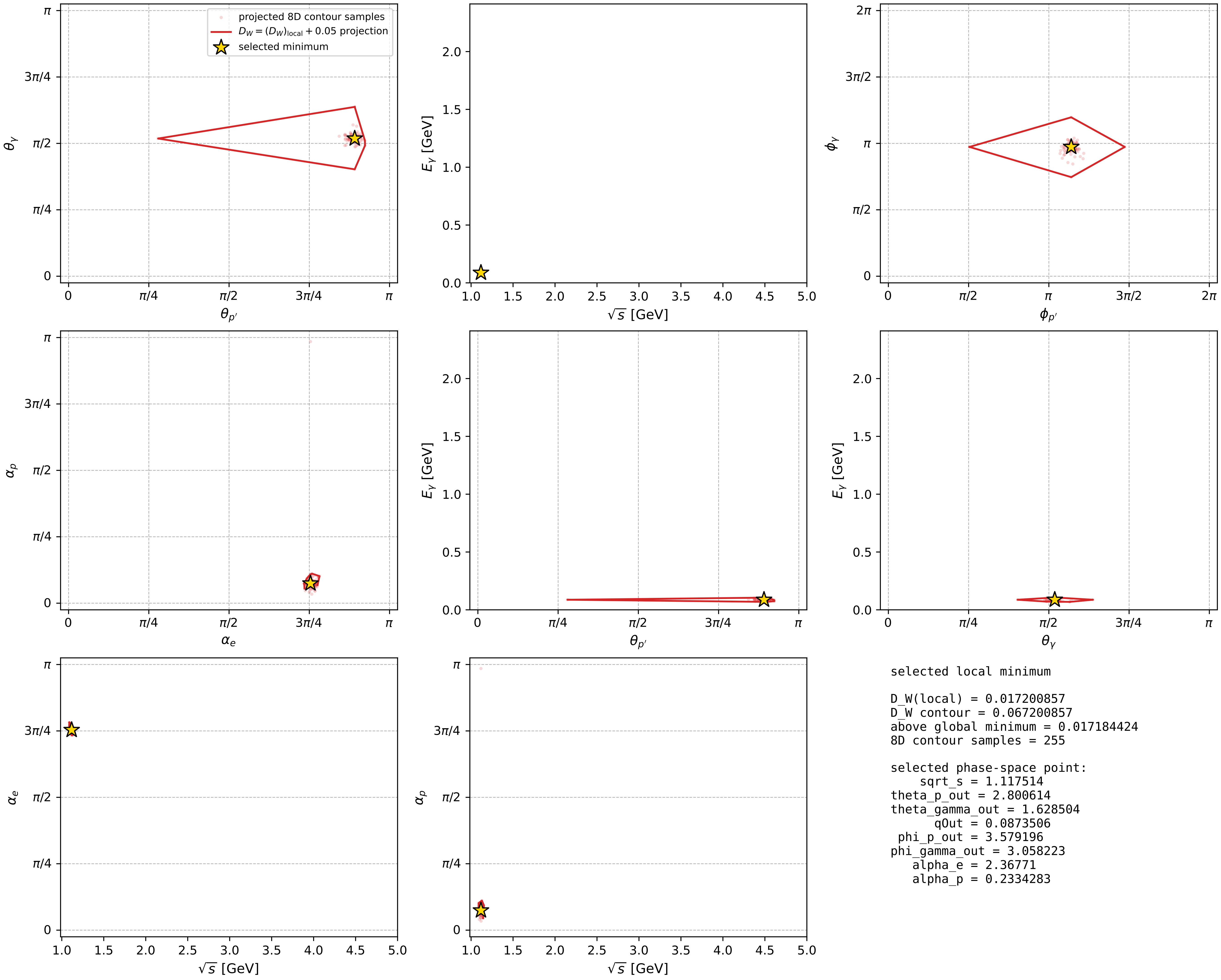
s=1.24884, $\sqrt{s}$=1.11751, $|\vec{P}|$=0.165095, $|\vec{P}'|$=0.011821
$\theta_{p'}$=2.80061, θ_γ=1.6285, $\phi_{p'}$=3.5792, ϕ_γ=3.05822
E_γ=0.0873506, $\theta_{\ell'}$=1.39419, $\phi_{\ell'}$=6.22152
Q^2=0.0357484, x_B=0.179712, t=-0.0308688, W^2=1.04302, y=0.539091

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1651 p_x= 0.0000 p_y= 0.0000 p_z= -0.1651
P E= 0.9524 p_x= 0.0000 p_y= 0.0000 p_z= 0.1651
ℓ' E= 0.0921 p_x= 0.0905 p_y= -0.0056 p_z= 0.0162
P' E= 0.9381 p_x= -0.0036 p_y= -0.0017 p_z= -0.0111
q_γ E= 0.0874 p_x= -0.0869 p_y= 0.0073 p_z= -0.0050

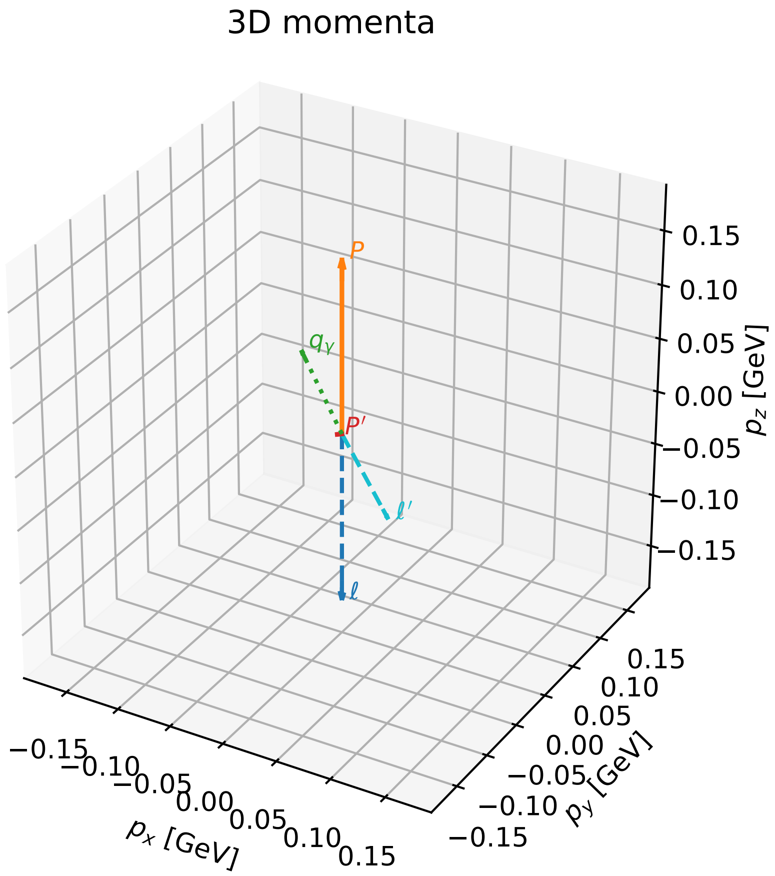




electron: polarization cluster P5, local minimum 976; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



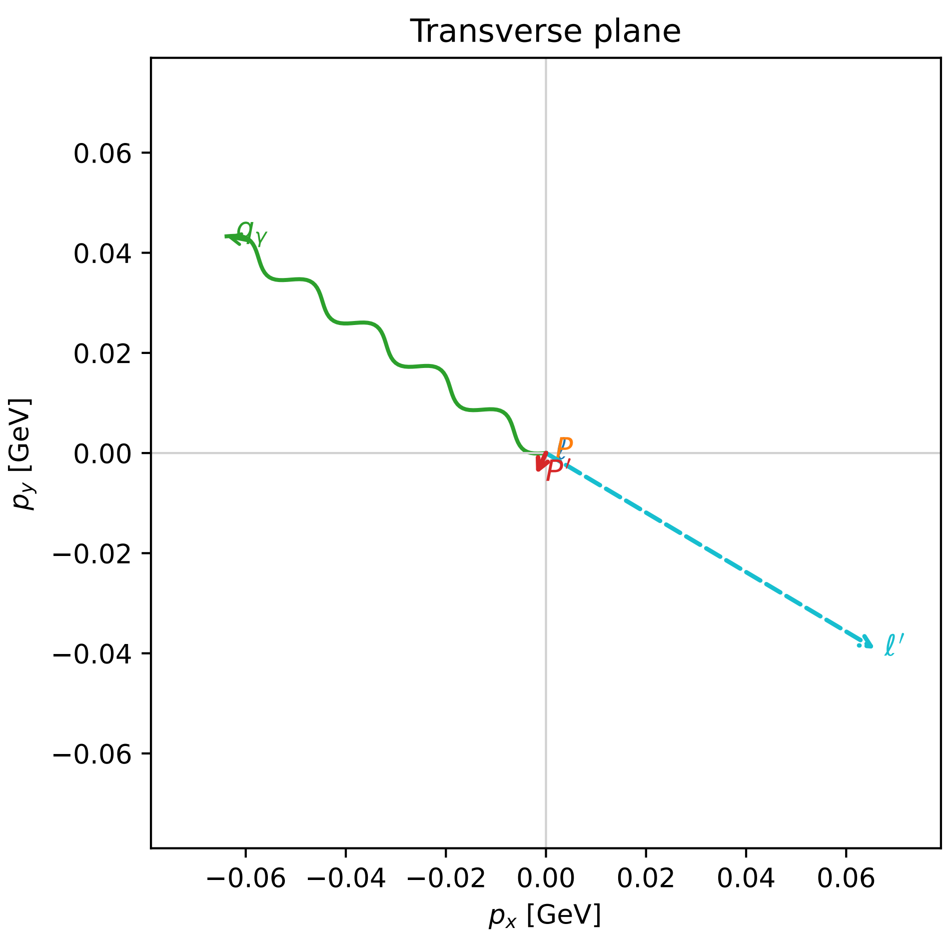
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

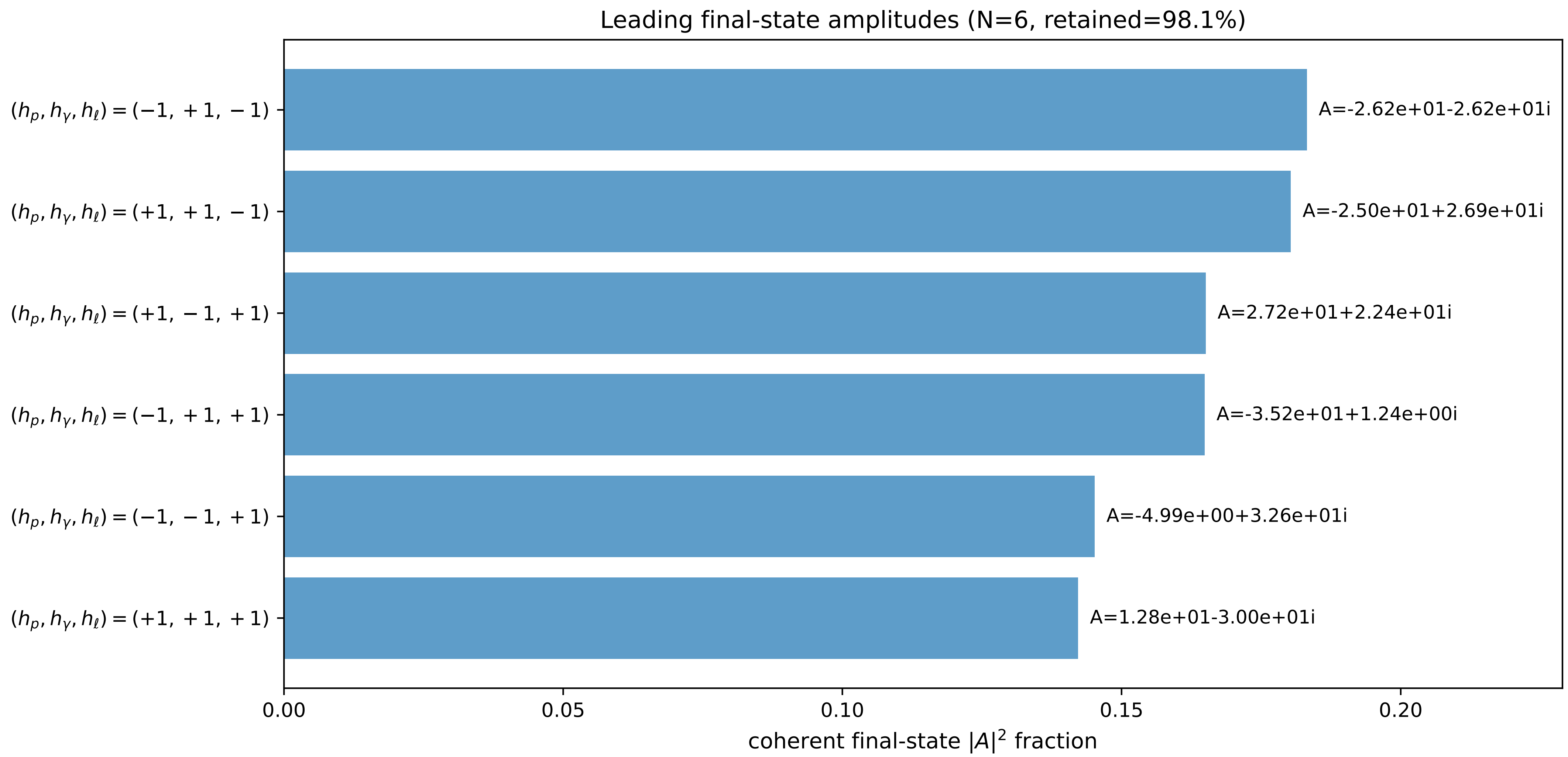


dw_mixing_angles_polarization_05_local_minimum_978 D_W=0.0172885
region: polarization_cluster_05_local_minimum_978
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3043221 rad
incoming lepton: -0.669493|+> +0.742819|->
proton mixing angle: alpha_p=0.19666807 rad
incoming proton: 0.980723|+> +0.195403|->

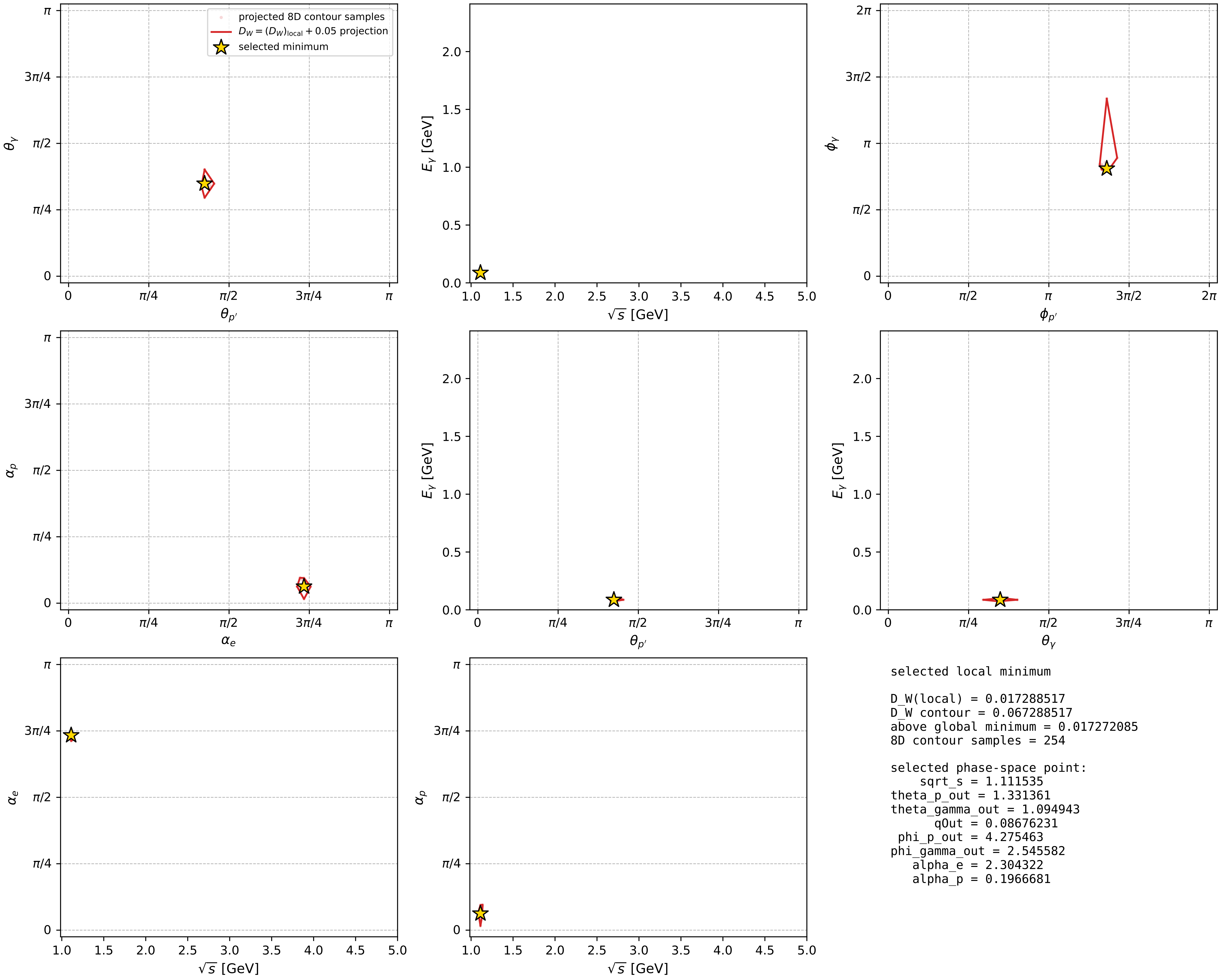
s=1.23551, $\sqrt{s}$=1.11153, $|\vec{P}|$=0.159988, $|\vec{P}'|$=0.00476439
$\theta_{p'}$=1.33136, θ_γ=1.09494, $\phi_{p'}$=4.27546, ϕ_γ=2.54558
E_γ=0.0867623, $\theta_{\ell'}$=2.06138, $\phi_{\ell'}$=5.74693
Q^2=0.0146817, x_B=0.0827264, t=-0.0250741, W^2=1.04264, y=0.498989

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1600 p_x= 0.0000 p_y= 0.0000 p_z= -0.1600
P E= 0.9515 p_x= 0.0000 p_y= 0.0000 p_z= 0.1600
ℓ' E= 0.0868 p_x= 0.0658 p_y= -0.0391 p_z= -0.0409
P' E= 0.9380 p_x= -0.0020 p_y= -0.0042 p_z= 0.0011
q_γ E= 0.0868 p_x= -0.0638 p_y= 0.0433 p_z= 0.0397

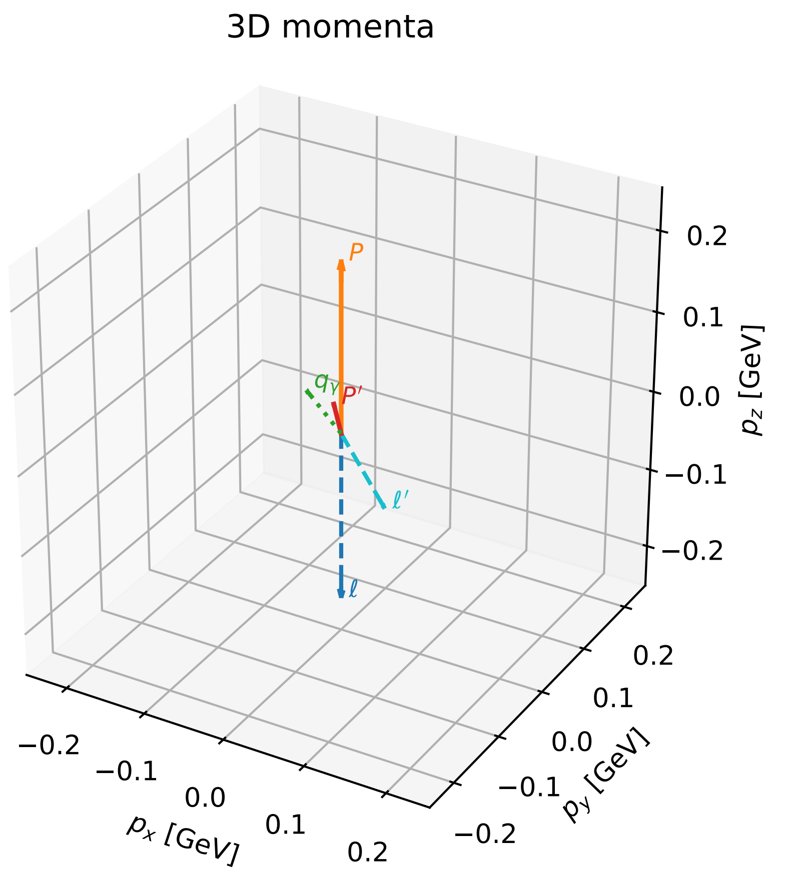




electron: polarization cluster P5, local minimum 978; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



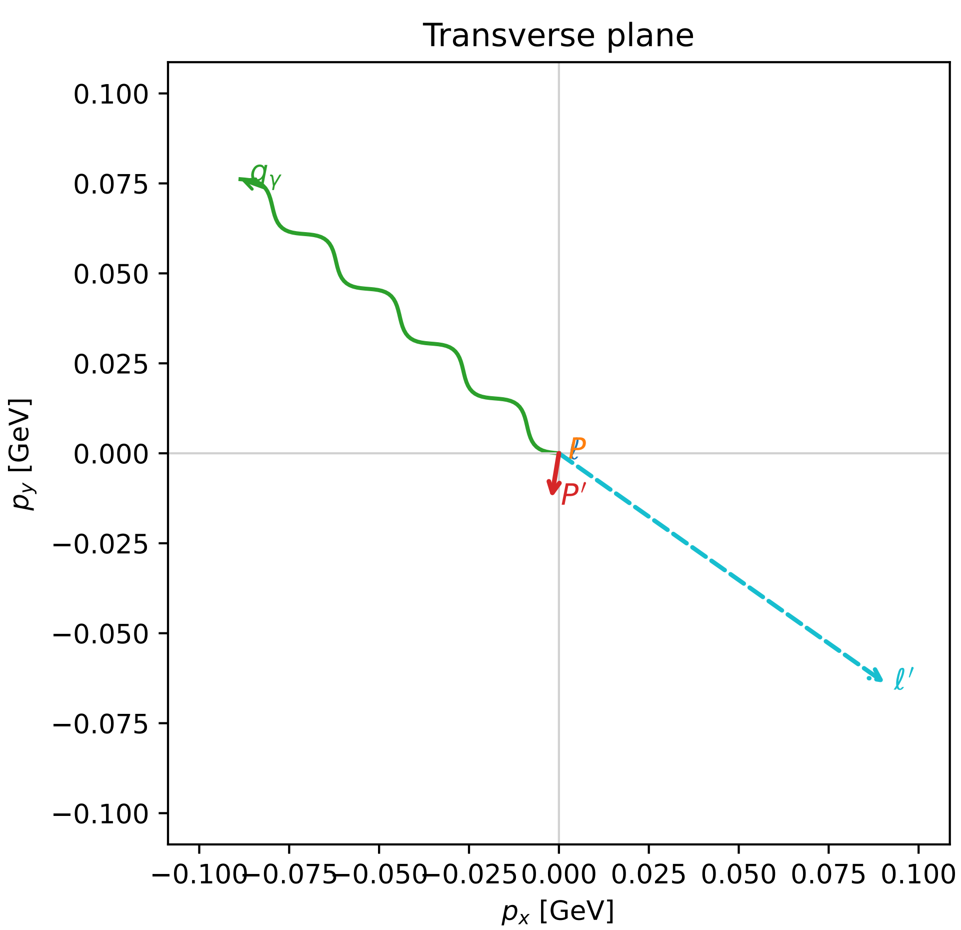
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

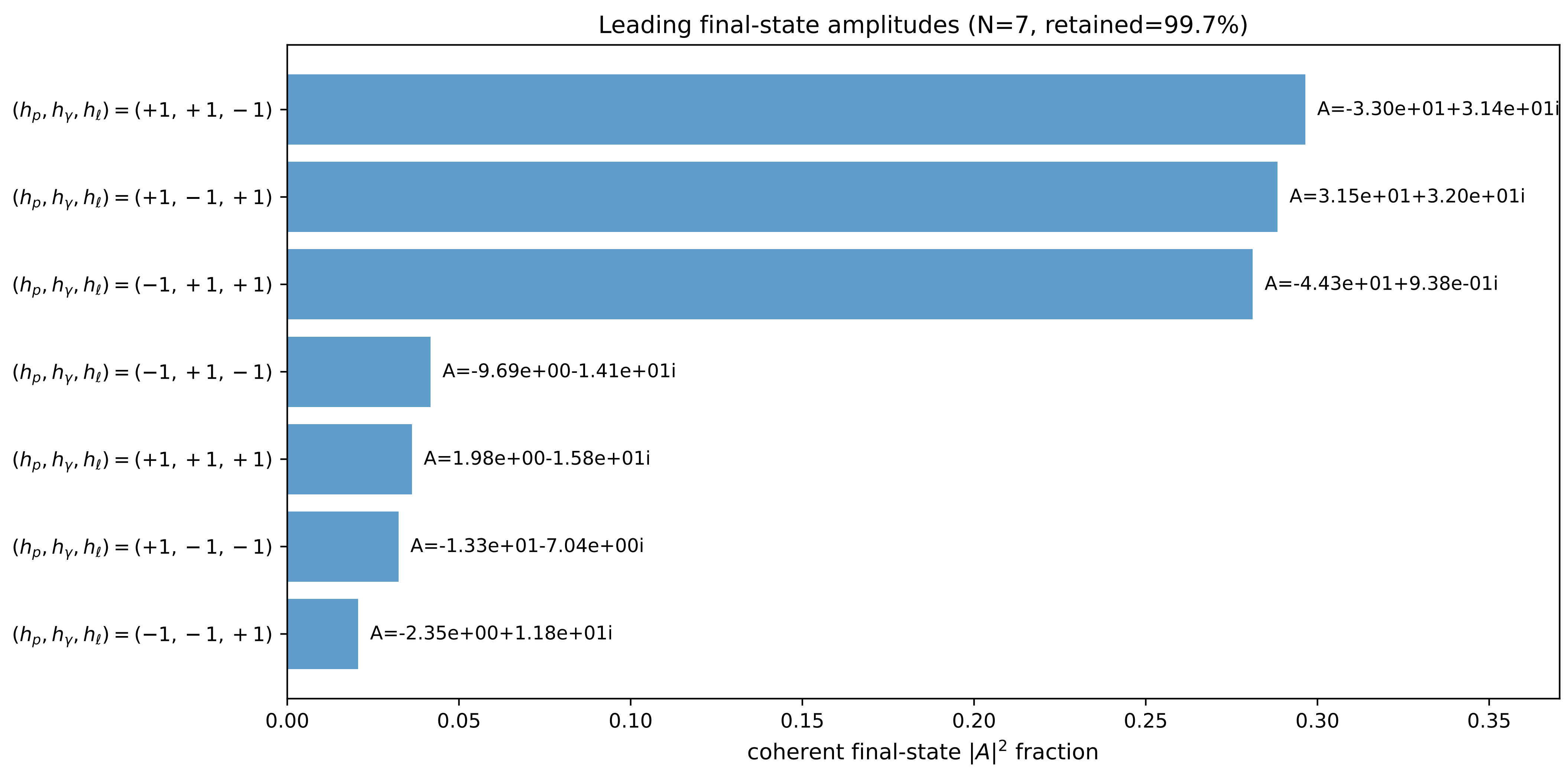


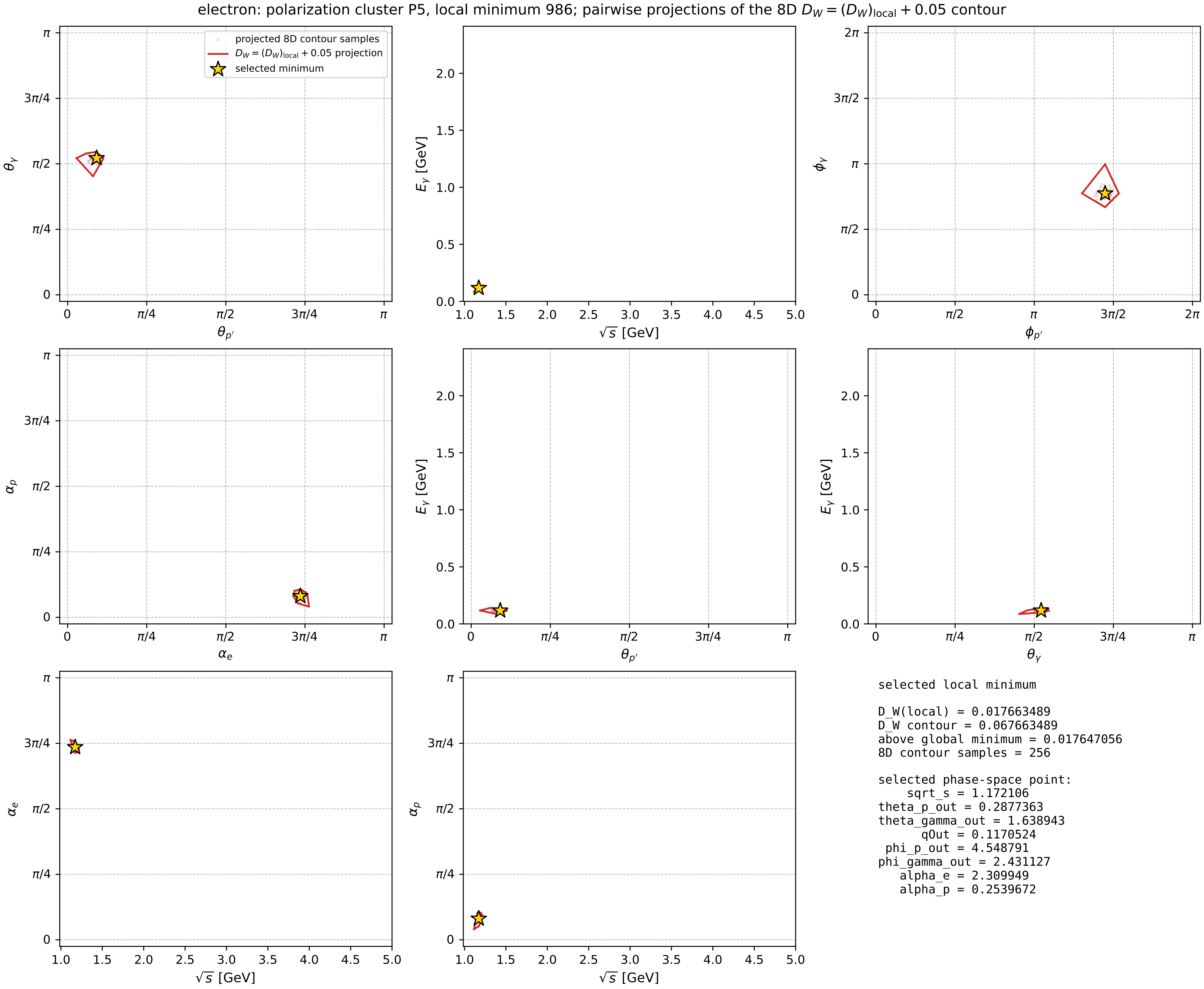
dw_mixing_angles_polarization_05_local_minimum_986 D_W=0.0176635
region: polarization_cluster_05_local_minimum_986
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3099488 rad
incoming lepton: -0.673662|+> +0.73904|->
proton mixing angle: alpha_p=0.25396717 rad
incoming proton: 0.967923|+> +0.251246|->

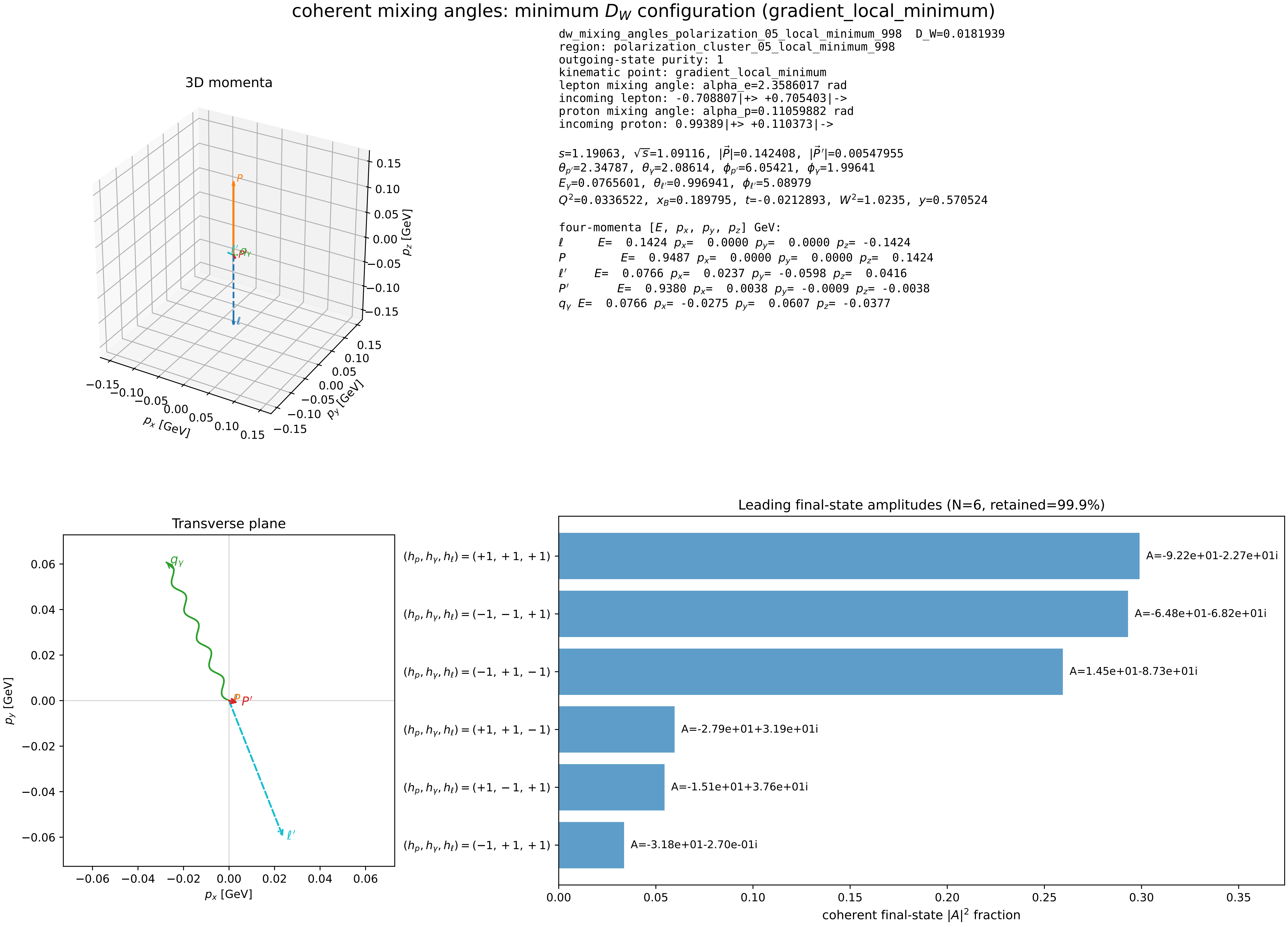
s=1.37383, $\sqrt{s}$=1.17211, $|\vec{P}|$=0.210727, $|\vec{P}'|$=0.0442587
$\theta_{p'}$=0.287736, θ_γ=1.63894, $\phi_{p'}$=4.54879, ϕ_γ=2.43113
E_γ=0.117052, $\theta_{\ell'}$=1.87247, $\phi_{\ell'}$=5.66972
Q^2=0.0343656, x_B=0.13403, t=-0.0279795, W^2=1.10188, y=0.519045

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.2107 p_x= 0.0000 p_y= 0.0000 p_z= -0.2107
P E= 0.9614 p_x= 0.0000 p_y= 0.0000 p_z= 0.2107
ℓ' E= 0.1160 p_x= 0.0906 p_y= -0.0638 p_z= -0.0345
P' E= 0.9390 p_x= -0.0020 p_y= -0.0124 p_z= 0.0424
q_γ E= 0.1171 p_x= -0.0885 p_y= 0.0762 p_z= -0.0080

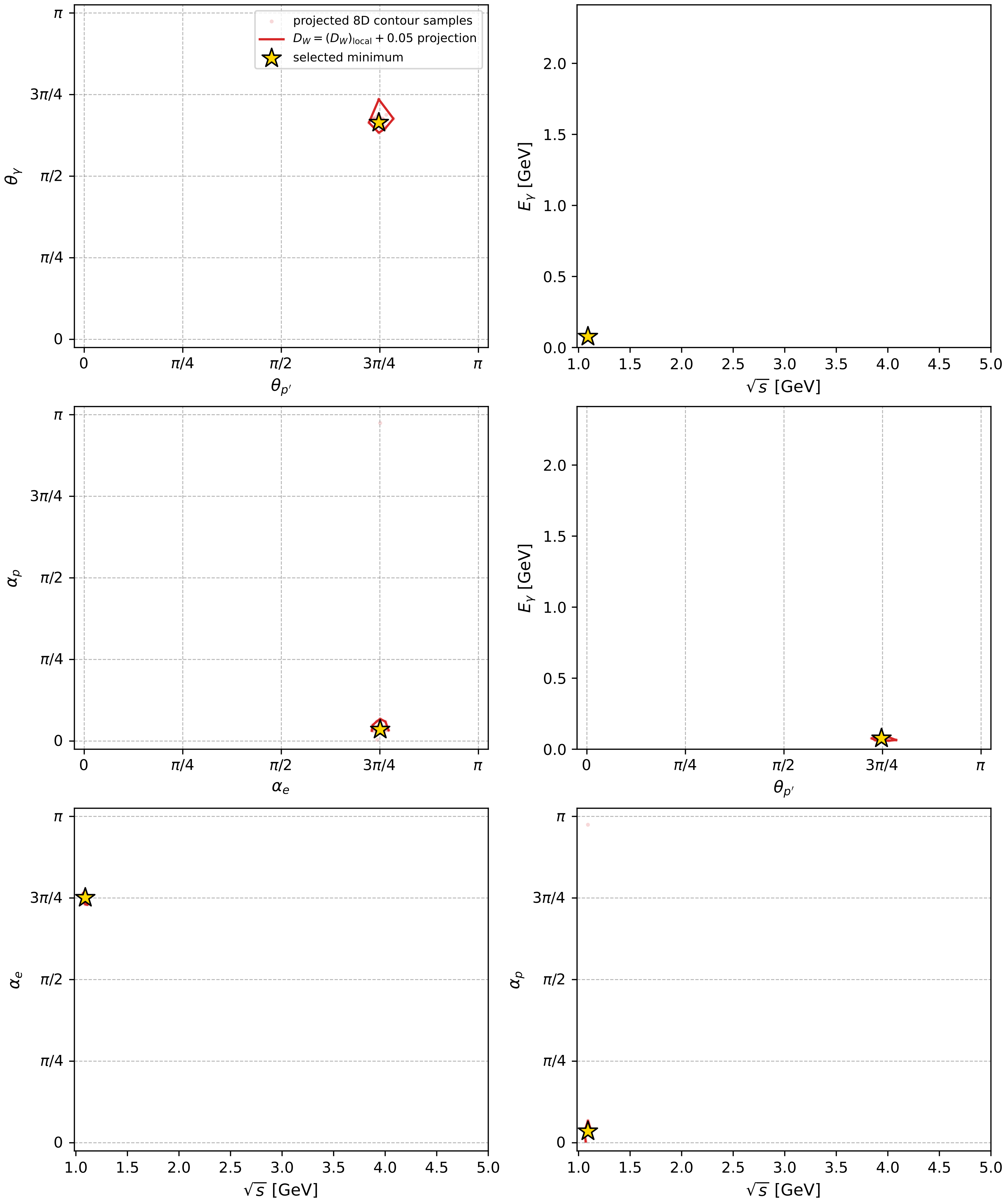





electron: polarization cluster P5, local minimum 986; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.017663489
D_W contour = 0.067663489
above global minimum = 0.017647056
8D contour samples = 256
selected phase-space point:
sqrt_s = 1.172106
theta_p_out = 0.2877363
theta_gamma_out = 1.638943
qOut = 0.1170524
phi_p_out = 4.548791
phi_gamma_out = 2.431127
alpha_e = 2.309949
alpha_p = 0.2539672

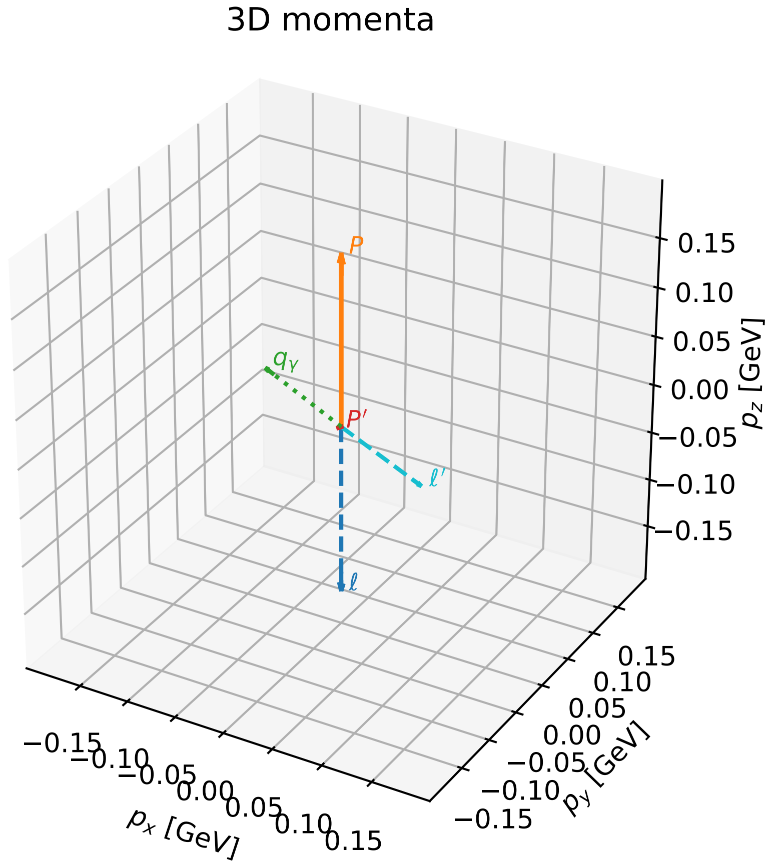

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_998 D_W=0.0181939
region: polarization_cluster_05_local_minimum_998
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3586017 rad
incoming lepton: -0.708807|+> +0.705403|->
proton mixing angle: alpha_p=0.11059882 rad
incoming proton: 0.99389|+> +0.110373|->
s=1.19063, $\sqrt{s}$=1.09116, $|\vec{P}|$=0.142408, $|\vec{P}'|$=0.00547955
$\theta_{p'}$=2.34787, θ_γ=2.08614, $\phi_{p'}$=6.05421, ϕ_γ=1.99641
E_γ=0.0765601, $\theta_{\ell'}$=0.996941, $\phi_{\ell'}$=5.08979
Q^2=0.0336522, x_B=0.189795, t=-0.0212893, W^2=1.0235, y=0.570524
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1424 p_x= 0.0000 p_y= 0.0000 p_z= -0.1424
P E= 0.9487 p_x= 0.0000 p_y= 0.0000 p_z= 0.1424
ℓ' E= 0.0766 p_x= 0.0237 p_y= -0.0598 p_z= 0.0416
P' E= 0.9380 p_x= 0.0038 p_y= -0.0009 p_z= -0.0038
q_γ E= 0.0766 p_x= -0.0275 p_y= 0.0607 p_z= -0.0377
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
q_γ
P'
ℓ'
Leading final-state amplitudes (N=6, retained=99.9%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-9.22e+01-2.27e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-6.48e+01-6.82e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=1.45e+01-8.73e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-2.79e+01+3.19e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-1.51e+01+3.76e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-3.18e+01-2.70e-01i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P5, local minimum 998; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour




```
selected local minimum

D_W(local) = 0.018193918
D_W contour = 0.068193918
above global minimum = 0.018177485
8D contour samples = 254

selected phase-space point:
    sqrt_s = 1.091158
theta_p_out = 2.347875
theta_gamma_out = 2.086135
      qOut = 0.07656009
 phi_p_out = 6.054212
phi_gamma_out = 1.996409
   alpha_e = 2.358602
   alpha_p = 0.1105988
```

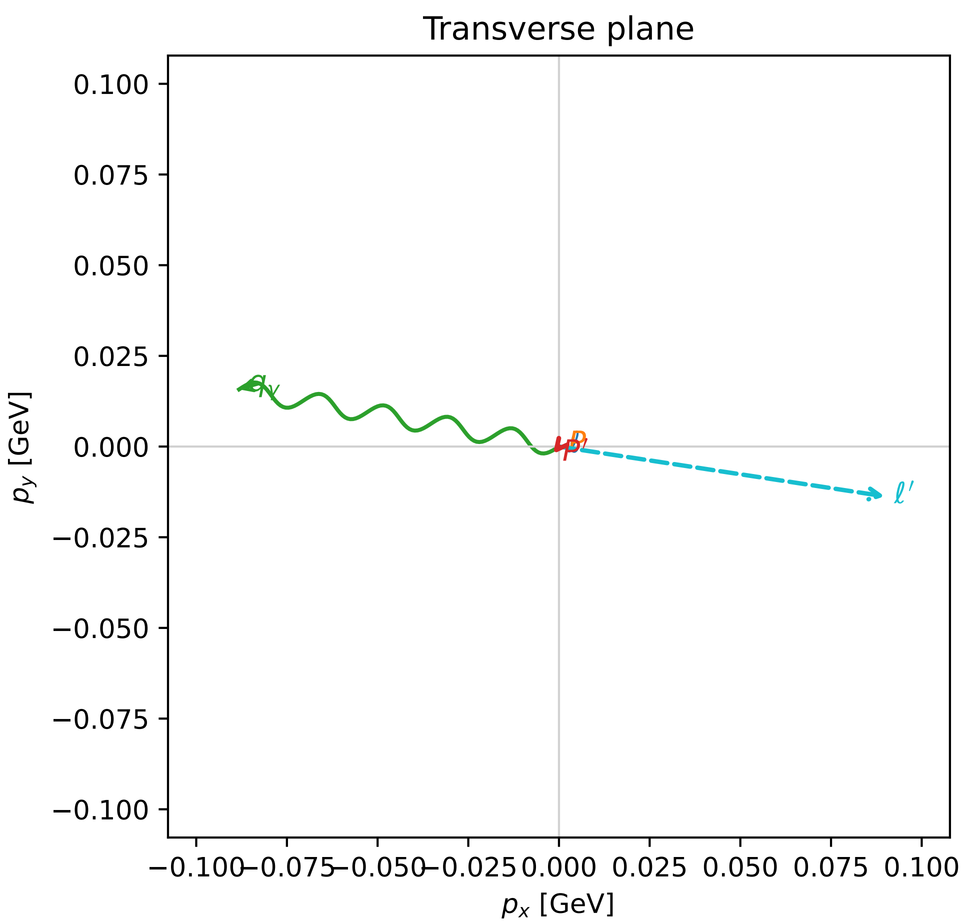
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

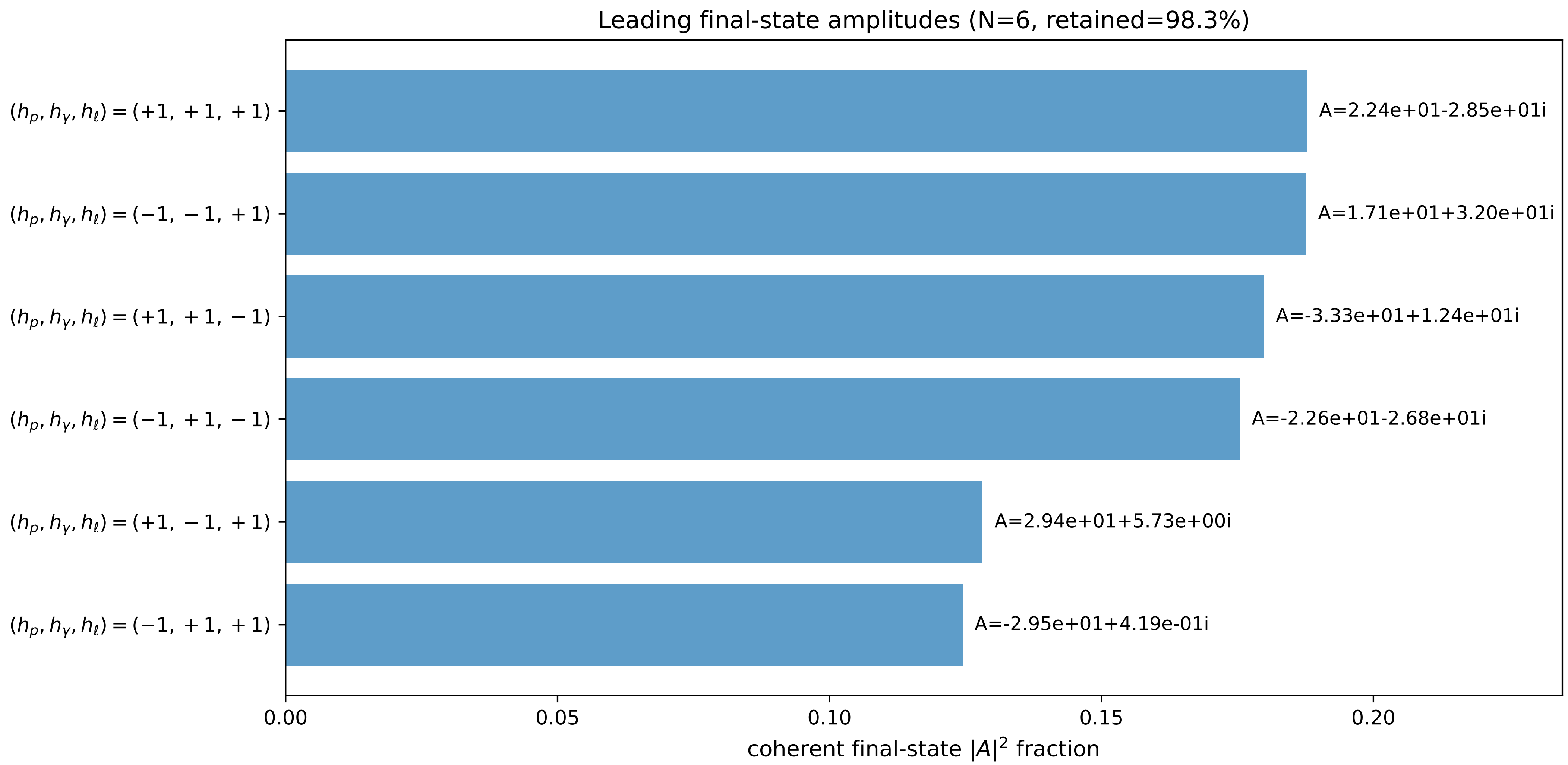


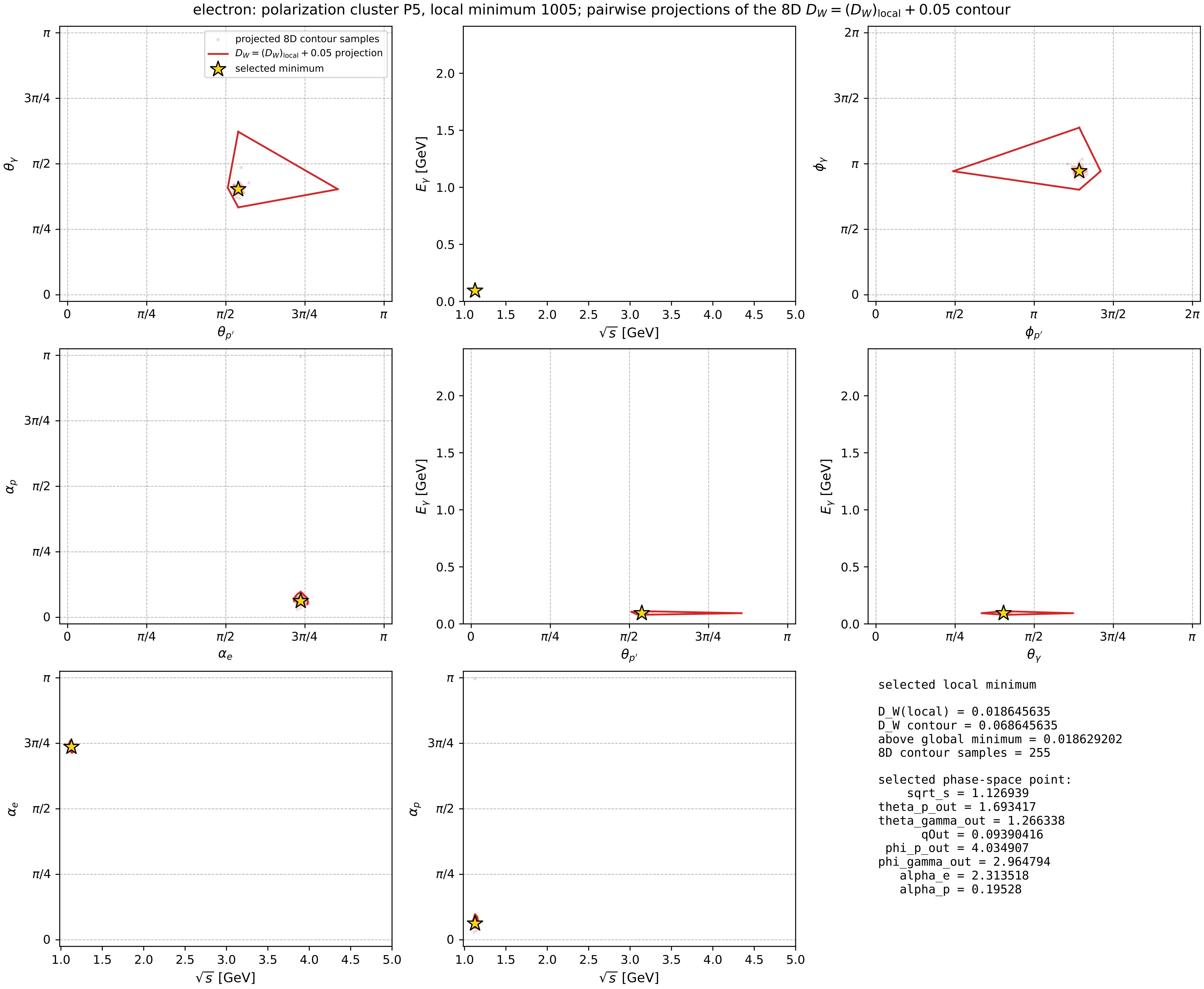
dw_mixing_angles_polarization_05_local_minimum_1005 D_W=0.0186456
region: polarization_cluster_05_local_minimum_1005
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3135185 rad
incoming lepton: -0.676296|+> +0.73663|->
proton mixing angle: alpha_p=0.19527999 rad
incoming proton: 0.980993|+> +0.194041|->

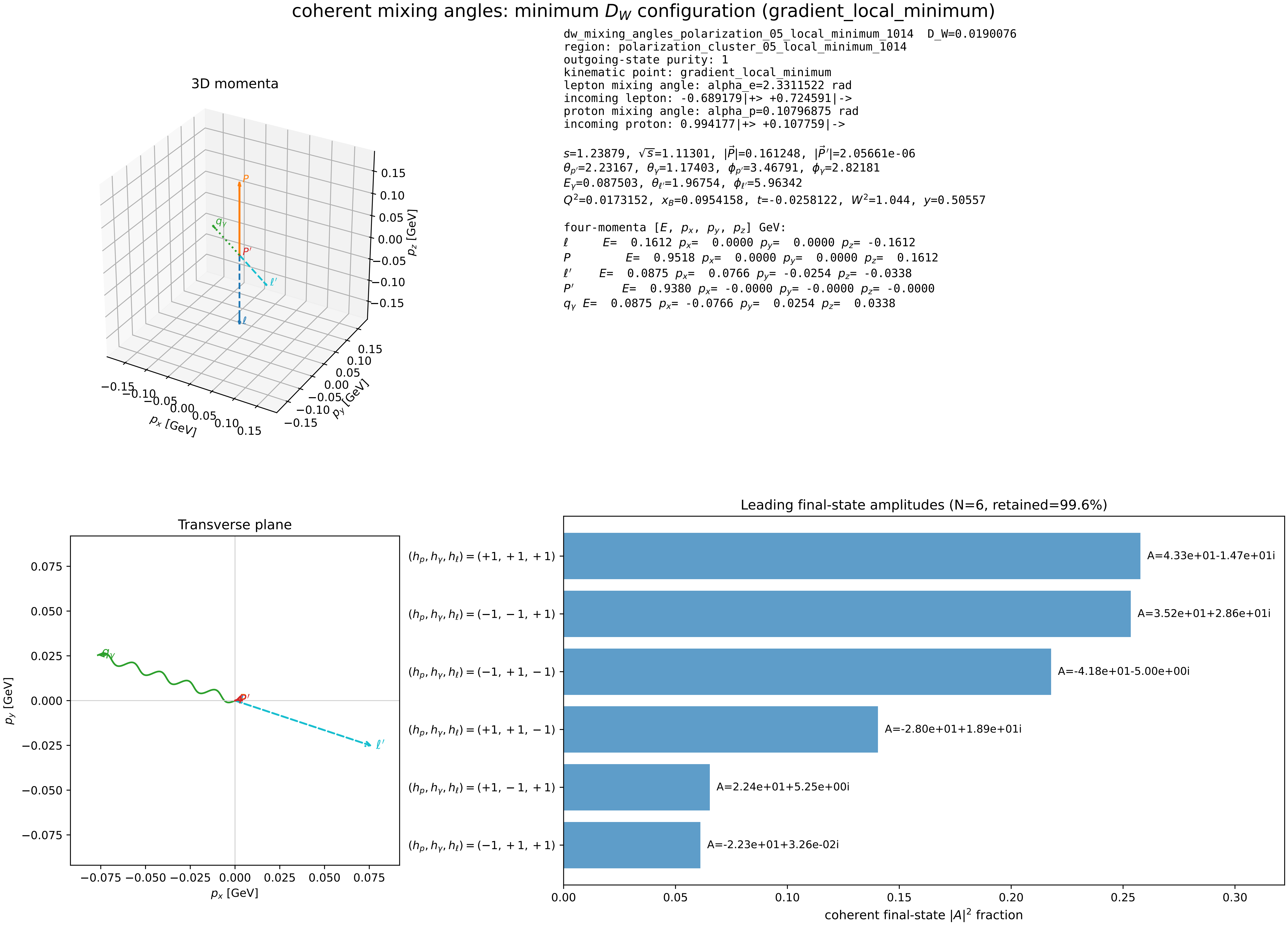
s=1.26999, $\sqrt{s}$=1.12694, $|\vec{P}|$=0.1731, $|\vec{P}'|$=0.00262309
$\theta_{p'}$=1.69342, θ_γ=1.26634, $\phi_{p'}$=4.03491, ϕ_γ=2.96479
E_γ=0.0939042, $\theta_{\ell'}$=1.868, $\phi_{\ell'}$=6.13152
Q^2=0.023265, x_B=0.116778, t=-0.0298309, W^2=1.0558, y=0.510637

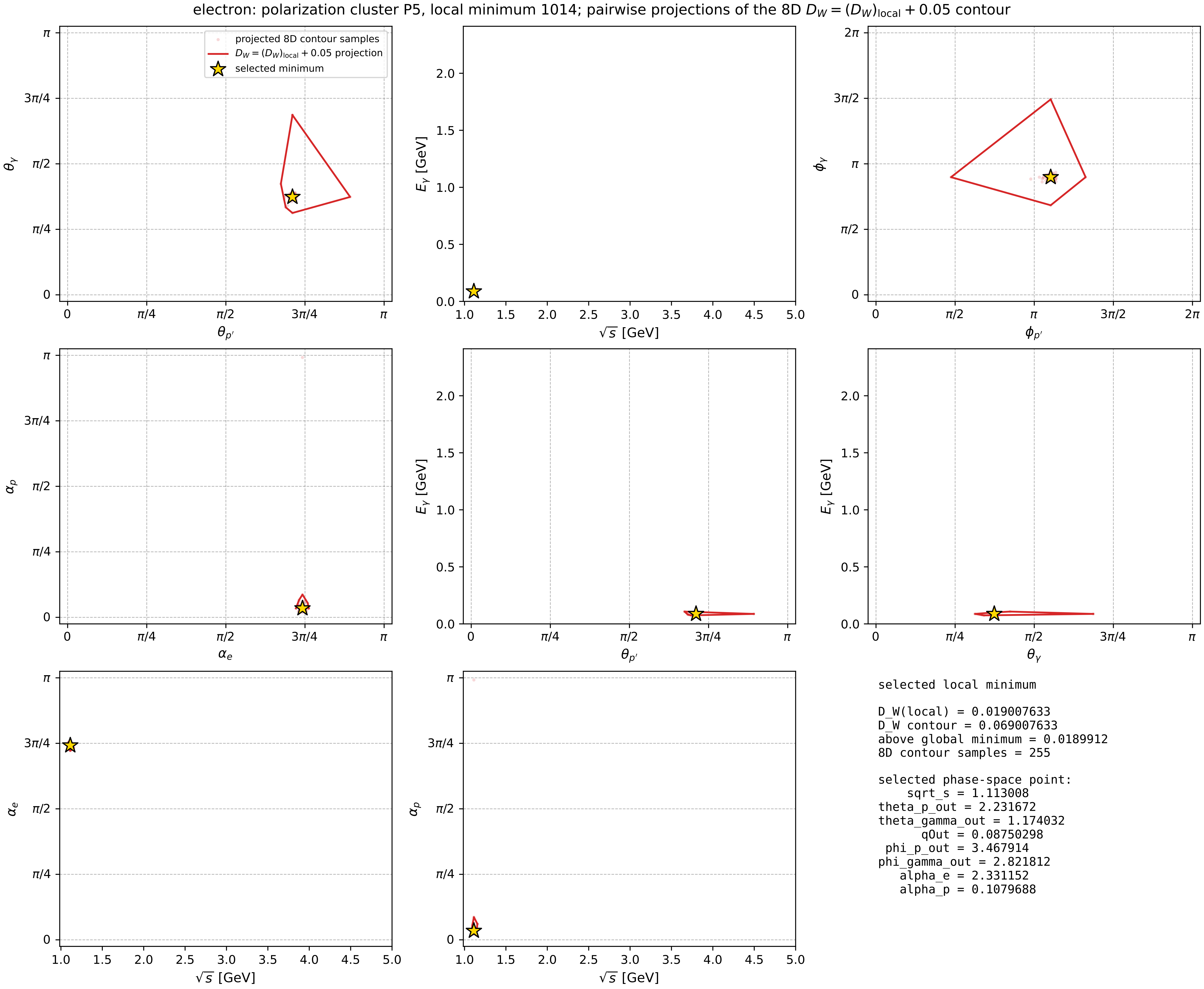
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1731 p_x= 0.0000 p_y= 0.0000 p_z= -0.1731
P E= 0.9538 p_x= 0.0000 p_y= 0.0000 p_z= 0.1731
ℓ' E= 0.0950 p_x= 0.0898 p_y= -0.0137 p_z= -0.0278
P' E= 0.9380 p_x= -0.0016 p_y= -0.0020 p_z= -0.0003
q_γ E= 0.0939 p_x= -0.0882 p_y= 0.0158 p_z= 0.0282

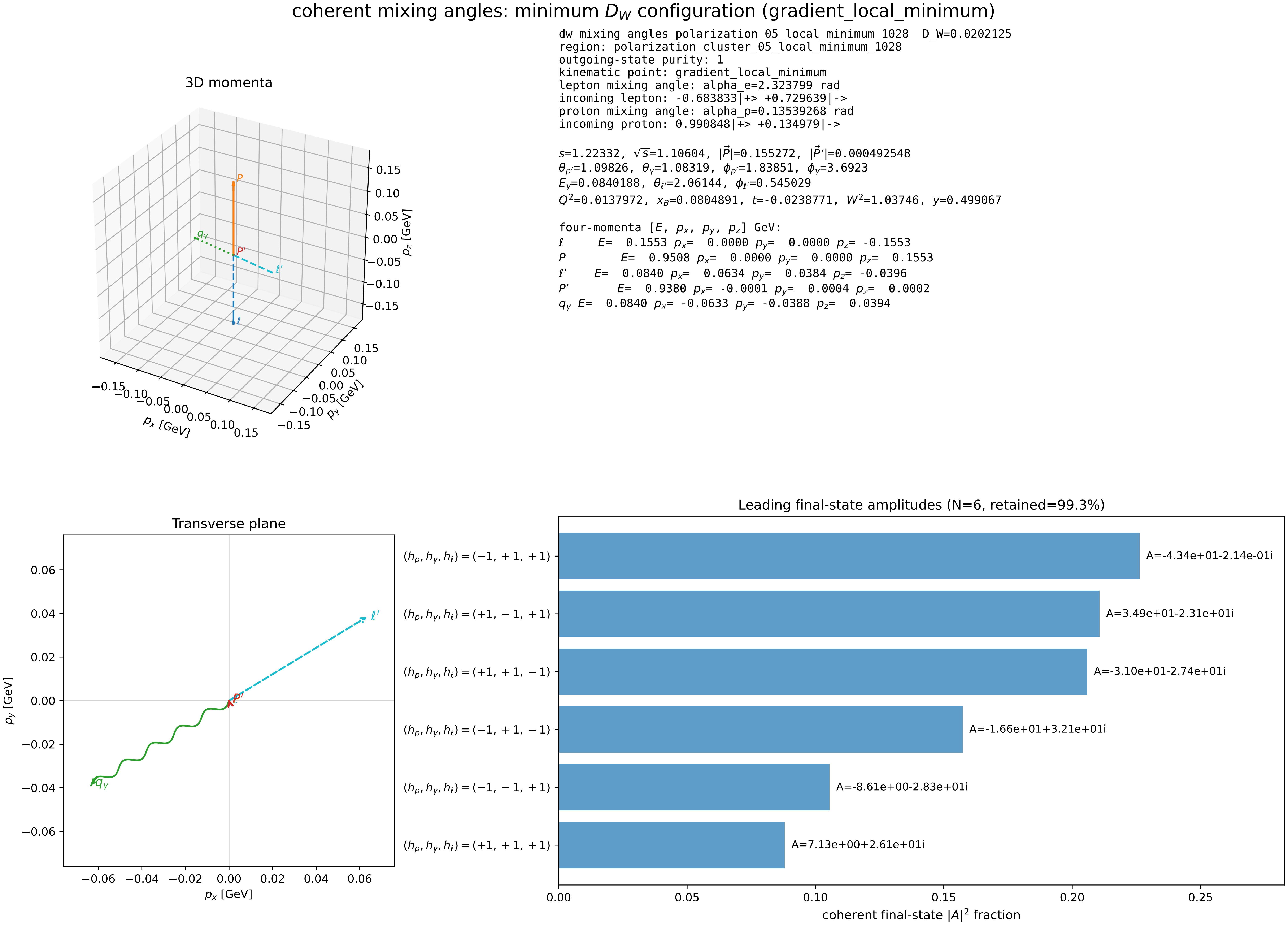




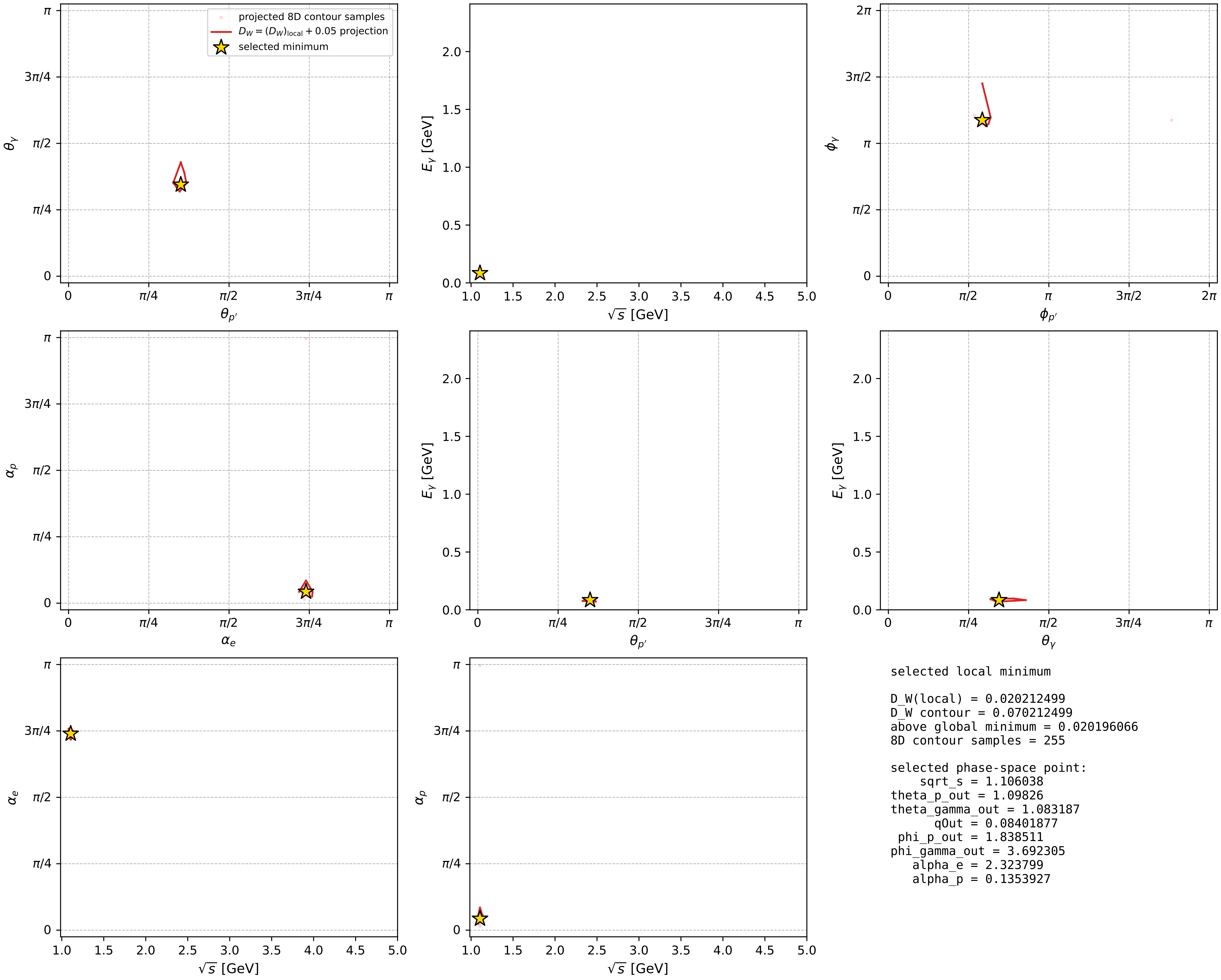

electron: polarization cluster P5, local minimum 1005; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
E_γ [GeV]
$\theta_{p'}$
E_γ [GeV]
θ_γ
α_e
$\sqrt{s}$ [GeV]
α_p
$\sqrt{s}$ [GeV]
selected local minimum
D_W(local) = 0.018645635
D_W contour = 0.068645635
above global minimum = 0.018629202
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.126939
theta_p_out = 1.693417
theta_gamma_out = 1.266338
qOut = 0.09390416
phi_p_out = 4.034907
phi_gamma_out = 2.964794
alpha_e = 2.313518
alpha_p = 0.19528

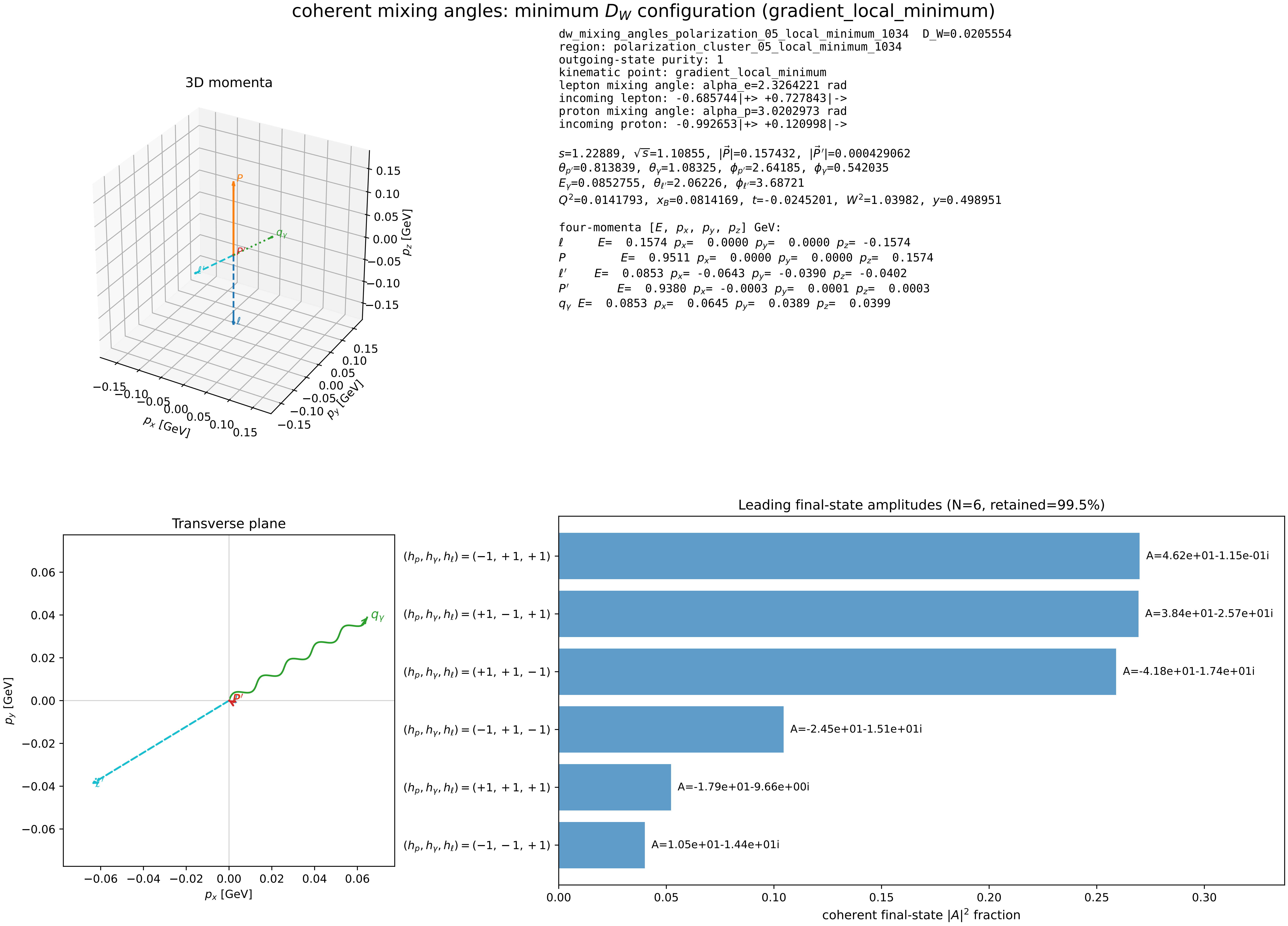

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_1014 D_W=0.0190076
region: polarization_cluster_05_local_minimum_1014
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3311522 rad
incoming lepton: -0.689179|+> +0.724591|->
proton mixing angle: alpha_p=0.10796875 rad
incoming proton: 0.994177|+> +0.107759|->
s=1.23879, $\sqrt{s}$=1.11301, $|\vec{P}|$=0.161248, $|\vec{P}'|$=2.05661e-06
$\theta_{p'}$=2.23167, θ_γ=1.17403, $\phi_{p'}$=3.46791, ϕ_γ=2.82181
E_γ=0.087503, $\theta_{\ell'}$=1.96754, $\phi_{\ell'}$=5.96342
Q^2=0.0173152, x_B=0.0954158, t=-0.0258122, W^2=1.044, y=0.50557
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1612 p_x= 0.0000 p_y= 0.0000 p_z= -0.1612
P E= 0.9518 p_x= 0.0000 p_y= 0.0000 p_z= 0.1612
ℓ' E= 0.0875 p_x= 0.0766 p_y= -0.0254 p_z= -0.0338
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= -0.0000
q_γ E= 0.0875 p_x= -0.0766 p_y= 0.0254 p_z= 0.0338
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=99.6%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=4.33e+01-1.47e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=3.52e+01+2.86e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-4.18e+01-5.00e+00i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-2.80e+01+1.89e+01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=2.24e+01+5.25e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-2.23e+01+3.26e-02i
coherent final-state $|A|^2$ fraction


electron: polarization cluster P5, local minimum 1014; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.019007633
D_W contour = 0.069007633
above global minimum = 0.0189912
8D contour samples = 255
selected phase-space point:
sqrt_s = 1.113008
theta_p_out = 2.231672
theta_gamma_out = 1.174032
qOut = 0.08750298
phi_p_out = 3.467914
phi_gamma_out = 2.821812
alpha_e = 2.331152
alpha_p = 0.1079688

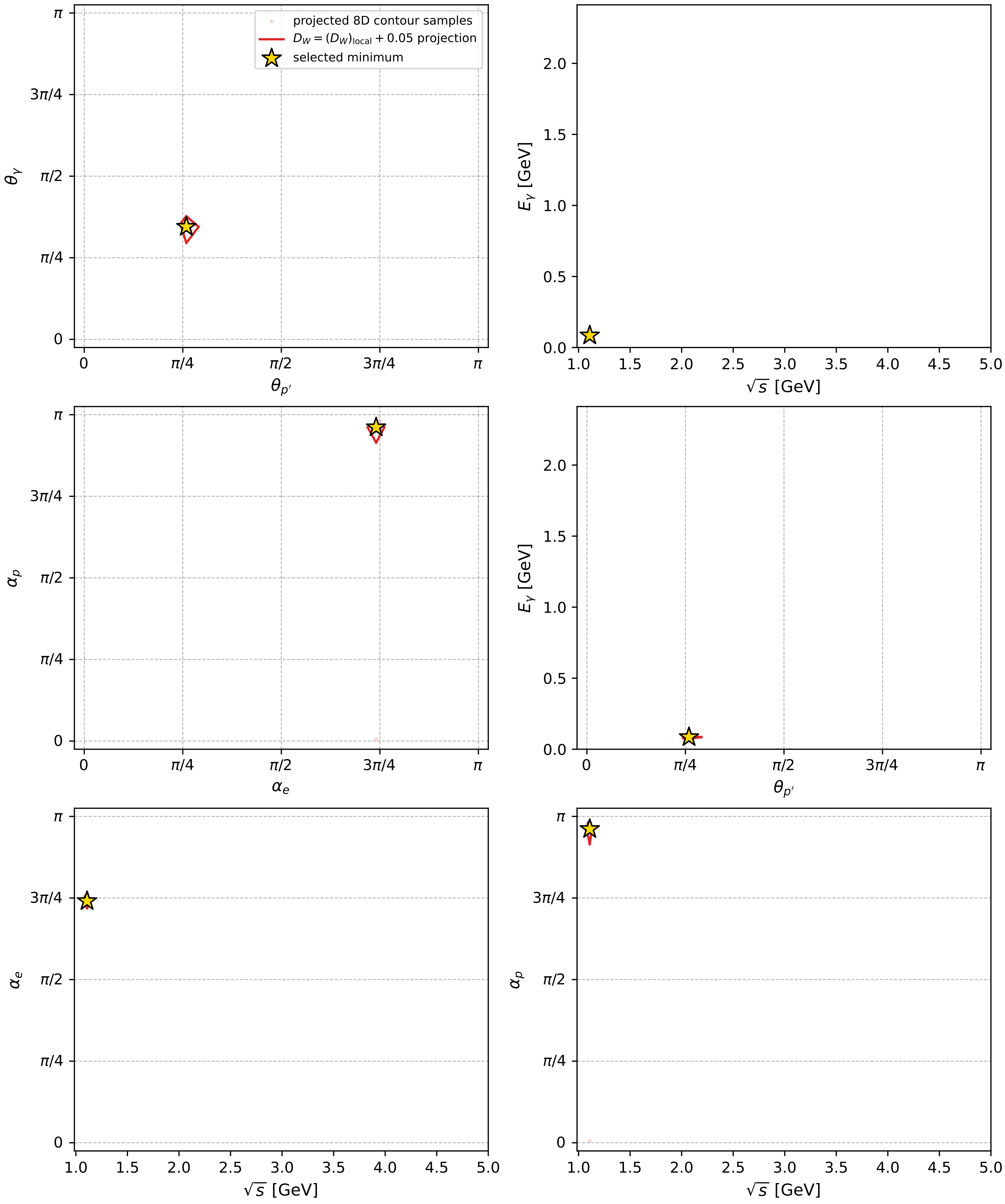

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
dw_mixing_angles_polarization_05_local_minimum_1028 D_W=0.0202125
region: polarization_cluster_05_local_minimum_1028
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.323799 rad
incoming lepton: -0.683833|+> +0.729639|->
proton mixing angle: alpha_p=0.13539268 rad
incoming proton: 0.990848|+> +0.134979|->
s=1.22332, $\sqrt{s}$=1.10604, $|\vec{P}|$=0.155272, $|\vec{P}'|$=0.000492548
$\theta_{p'}$=1.09826, θ_γ=1.08319, $\phi_{p'}$=1.83851, ϕ_γ=3.6923
E_γ=0.0840188, $\theta_{\ell'}$=2.06144, $\phi_{\ell'}$=0.545029
Q^2=0.0137972, x_B=0.0804891, t=-0.0238771, W^2=1.03746, y=0.499067
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1553 p_x= 0.0000 p_y= 0.0000 p_z= -0.1553
P E= 0.9508 p_x= 0.0000 p_y= 0.0000 p_z= 0.1553
ℓ' E= 0.0840 p_x= 0.0634 p_y= 0.0384 p_z= -0.0396
P' E= 0.9380 p_x= -0.0001 p_y= 0.0004 p_z= 0.0002
q_γ E= 0.0840 p_x= -0.0633 p_y= -0.0388 p_z= 0.0394
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=99.3%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-4.34e+01-2.14e-01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=3.49e+01-2.31e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-3.10e+01-2.74e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-1.66e+01+3.21e+01i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-8.61e+00-2.83e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=7.13e+00+2.61e+01i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P5, local minimum 1028; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



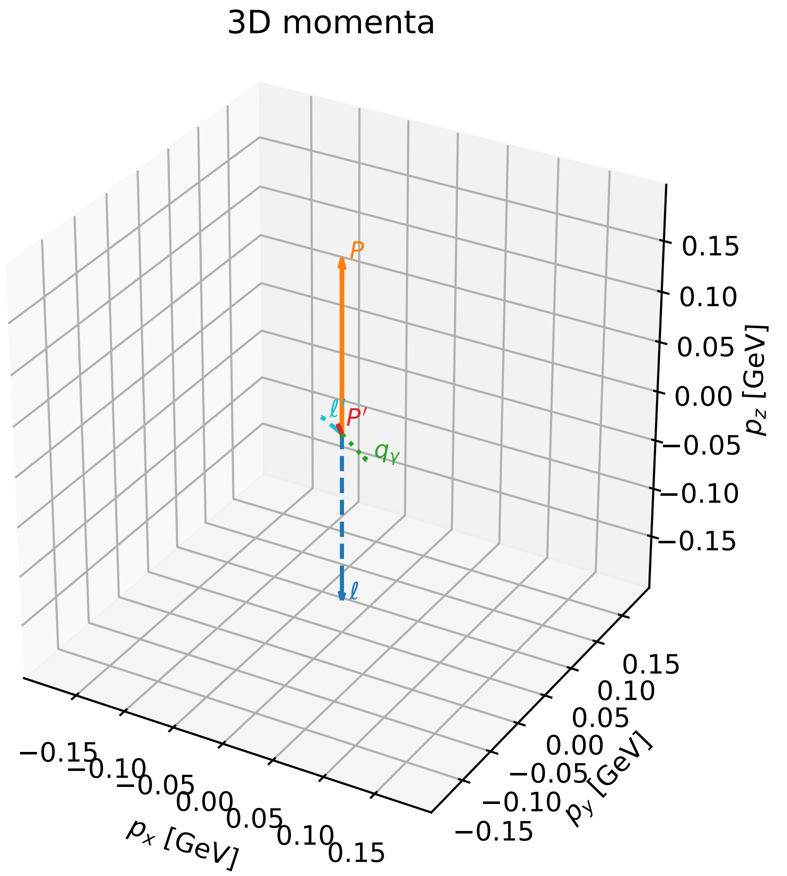

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_1034 D_W=0.0205554
region: polarization_cluster_05_local_minimum_1034
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3264221 rad
incoming lepton: -0.685744|+> +0.727843|->
proton mixing angle: alpha_p=3.0202973 rad
incoming proton: -0.992653|+> +0.120998|->
s=1.22889, $\sqrt{s}$=1.10855, $|\vec{P}|$=0.157432, $|\vec{P}'|$=0.000429062
$\theta_{p'}$=0.813839, θ_γ=1.08325, $\phi_{p'}$=2.64185, ϕ_γ=0.542035
E_γ=0.0852755, $\theta_{\ell'}$=2.06226, $\phi_{\ell'}$=3.68721
Q^2=0.0141793, x_B=0.0814169, t=-0.0245201, W^2=1.03982, y=0.498951
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1574 p_x= 0.0000 p_y= 0.0000 p_z= -0.1574
P E= 0.9511 p_x= 0.0000 p_y= 0.0000 p_z= 0.1574
ℓ' E= 0.0853 p_x= -0.0643 p_y= -0.0390 p_z= -0.0402
P' E= 0.9380 p_x= -0.0003 p_y= 0.0001 p_z= 0.0003
q_γ E= 0.0853 p_x= 0.0645 p_y= 0.0389 p_z= 0.0399
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=6, retained=99.5%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=4.62e+01-1.15e-01i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=3.84e+01-2.57e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-4.18e+01-1.74e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-2.45e+01-1.51e+01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-1.79e+01-9.66e+00i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=1.05e+01-1.44e+01i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P5, local minimum 1034; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour




```
selected local minimum

D_W(local) = 0.020555435
D_W contour = 0.070555435
above global minimum = 0.020539002
8D contour samples = 254

selected phase-space point:
    sqrt_s = 1.108553
theta_p_out = 0.8138388
theta_gamma_out = 1.083248
      qOut = 0.08527547
 phi_p_out = 2.641851
phi_gamma_out = 0.5420352
   alpha_e = 2.326422
   alpha_p = 3.020297
```

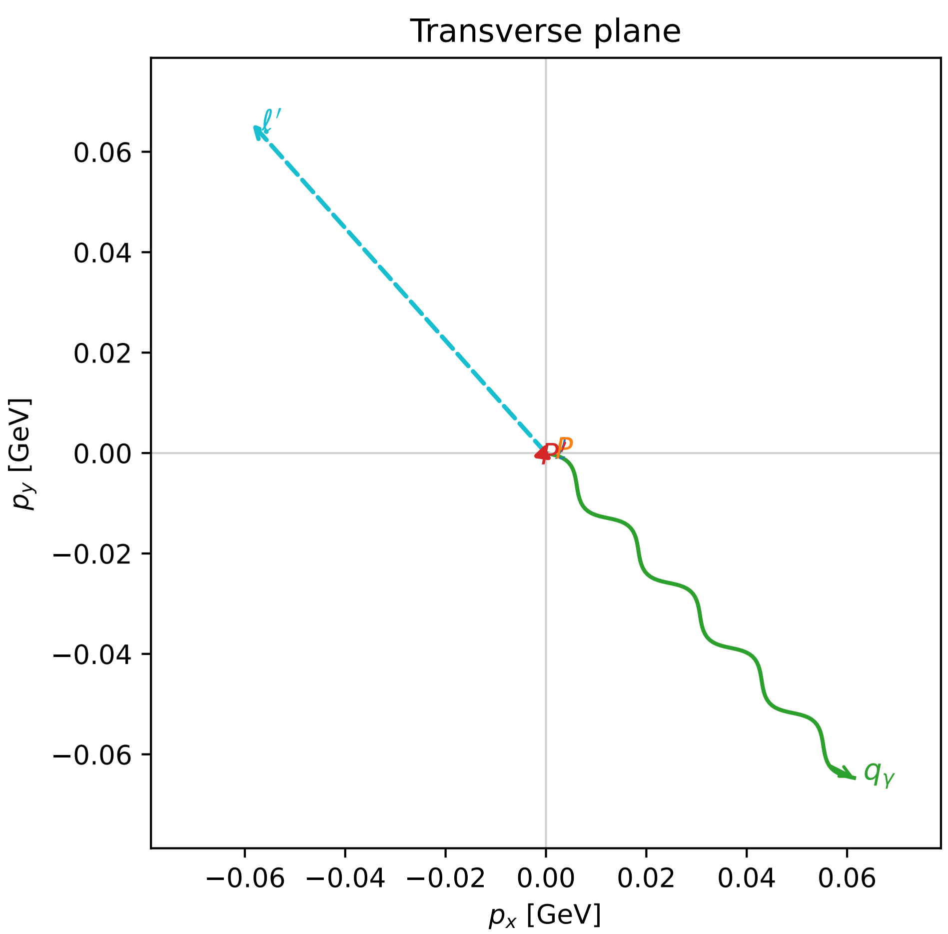
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

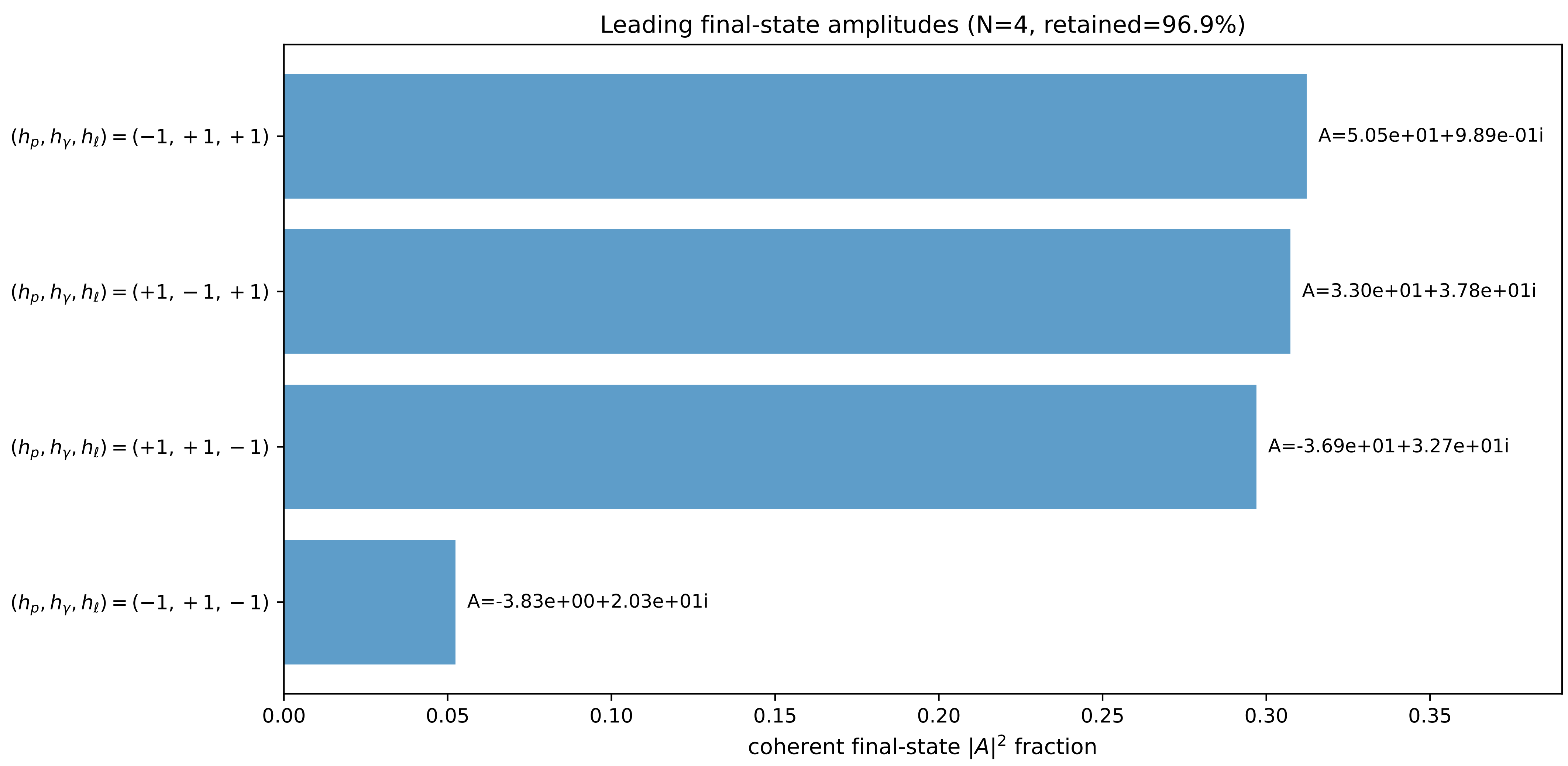


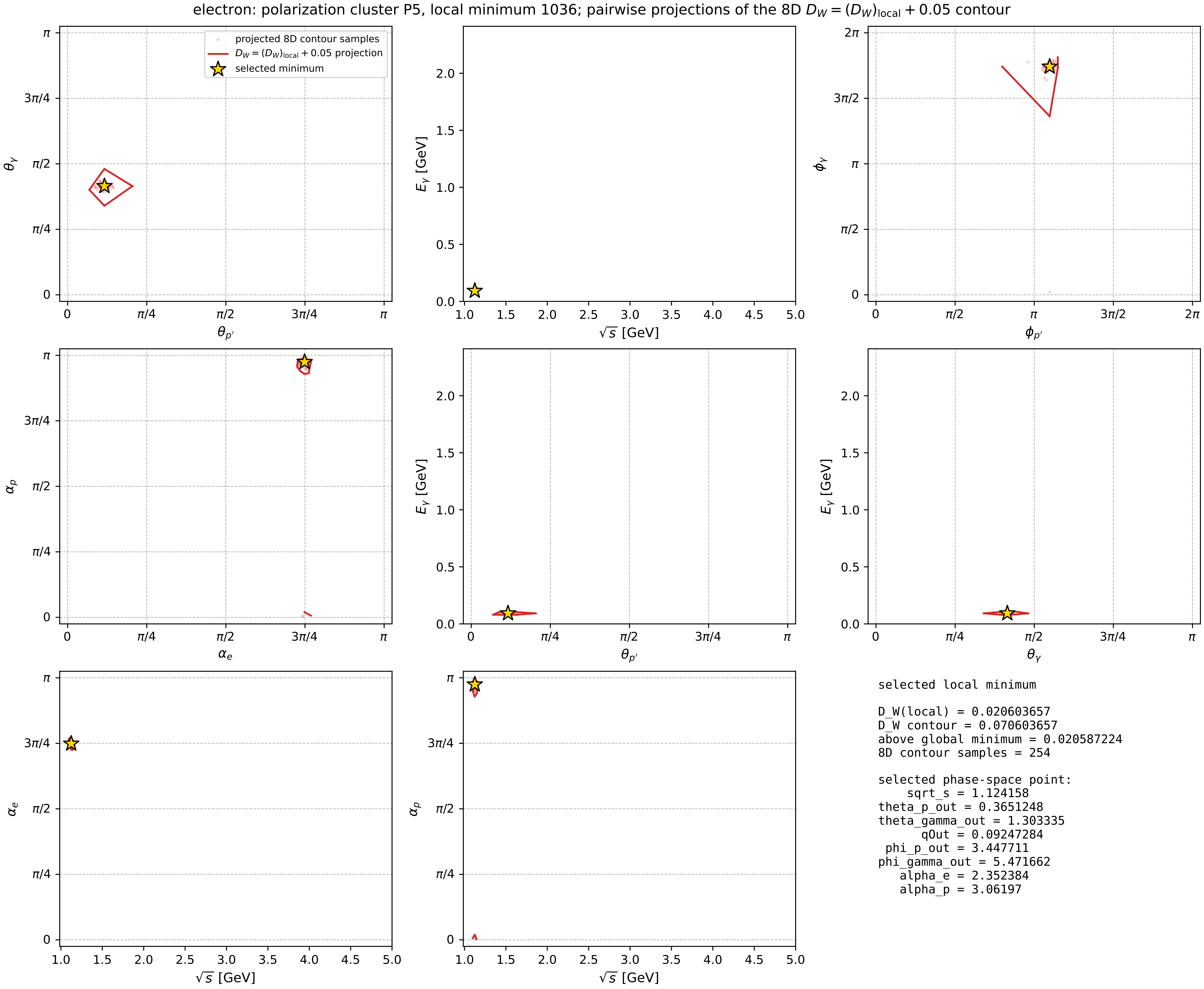
dw_mixing_angles_polarization_05_local_minimum_1036 D_W=0.0206037
region: polarization_cluster_05_local_minimum_1036
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3523837 rad
incoming lepton: -0.704407|+> +0.709796|->
proton mixing angle: alpha_p=3.0619704 rad
incoming proton: -0.996832|+> +0.0795382|->

s=1.26373, $\sqrt{s}$=1.12416, $|\vec{P}|$=0.170743, $|\vec{P}'|$=0.00835788
$\theta_{p'}$=0.365125, θ_γ=1.30333, $\phi_{p'}$=3.44771, ϕ_γ=5.47166
E_γ=0.0924728, $\theta_{\ell'}$=1.92233, $\phi_{\ell'}$=2.29955
Q^2=0.0209675, x_B=0.10791, t=-0.0263208, W^2=1.05318, y=0.506152

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1707 p_x= 0.0000 p_y= 0.0000 p_z= -0.1707
P E= 0.9534 p_x= 0.0000 p_y= 0.0000 p_z= 0.1707
ℓ' E= 0.0936 p_x= -0.0585 p_y= 0.0656 p_z= -0.0322
P' E= 0.9380 p_x= -0.0028 p_y= -0.0009 p_z= 0.0078
q_γ E= 0.0925 p_x= 0.0614 p_y= -0.0647 p_z= 0.0244







electron: polarization cluster P5, local minimum 1036; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.020603657
D_W contour = 0.070603657
above global minimum = 0.020587224
8D contour samples = 254
selected phase-space point:
sqrt_s = 1.124158
theta_p_out = 0.3651248
theta_gamma_out = 1.303335
qOut = 0.09247284
phi_p_out = 3.447711
phi_gamma_out = 5.471662
alpha_e = 2.352384
alpha_p = 3.06197

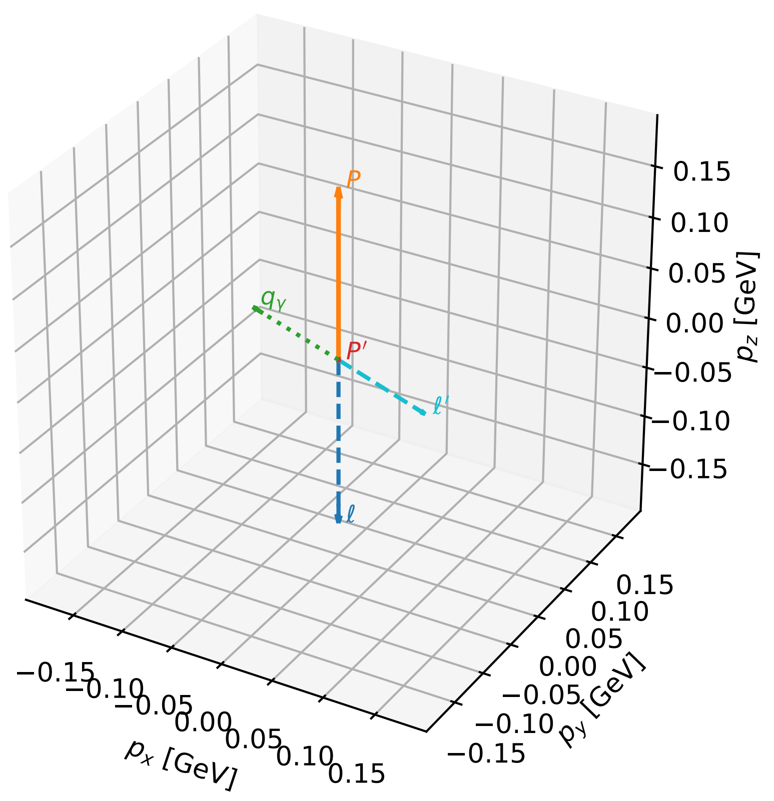
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

```
dw_mixing_angles_polarization_05_local_minimum_1043  D_W=0.021279
region: polarization_cluster_05_local_minimum_1043
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3423273 rad
incoming lepton: -0.697234|+> +0.716844|->
proton mixing angle: alpha_p=0.064289255 rad
incoming proton: 0.997934|+> +0.064245|->
```

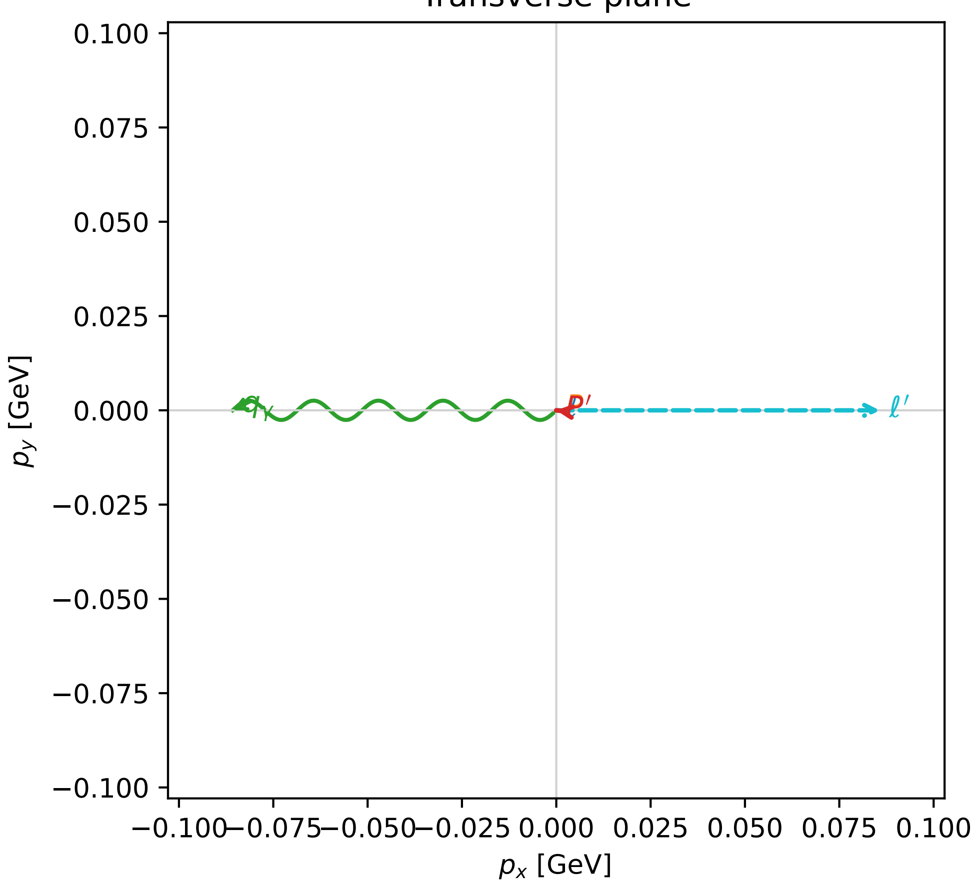
s=1.25165, $\sqrt{s}$=1.11877, $|\vec{P}|$=0.166166, $|\vec{P}'|$=0.000563754
$\theta_{p'}$=0.00159149, θ_γ=1.25216, $\phi_{p'}$=3.02486, ϕ_γ=3.14153
E_γ=0.0902955, $\theta_{\ell'}$=1.89535, $\phi_{\ell'}$=6.28312
Q^2=0.0204796, x_B=0.107877, t=-0.027211, W^2=1.04921, y=0.510595

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1662 p_x= 0.0000 p_y= 0.0000 p_z= -0.1662
P E= 0.9526 p_x= 0.0000 p_y= 0.0000 p_z= 0.1662
ℓ' E= 0.0905 p_x= 0.0858 p_y= -0.0000 p_z= -0.0289
P' E= 0.9380 p_x= -0.0000 p_y= 0.0000 p_z= 0.0006
q_γ E= 0.0903 p_x= -0.0858 p_y= 0.0000 p_z= 0.0283

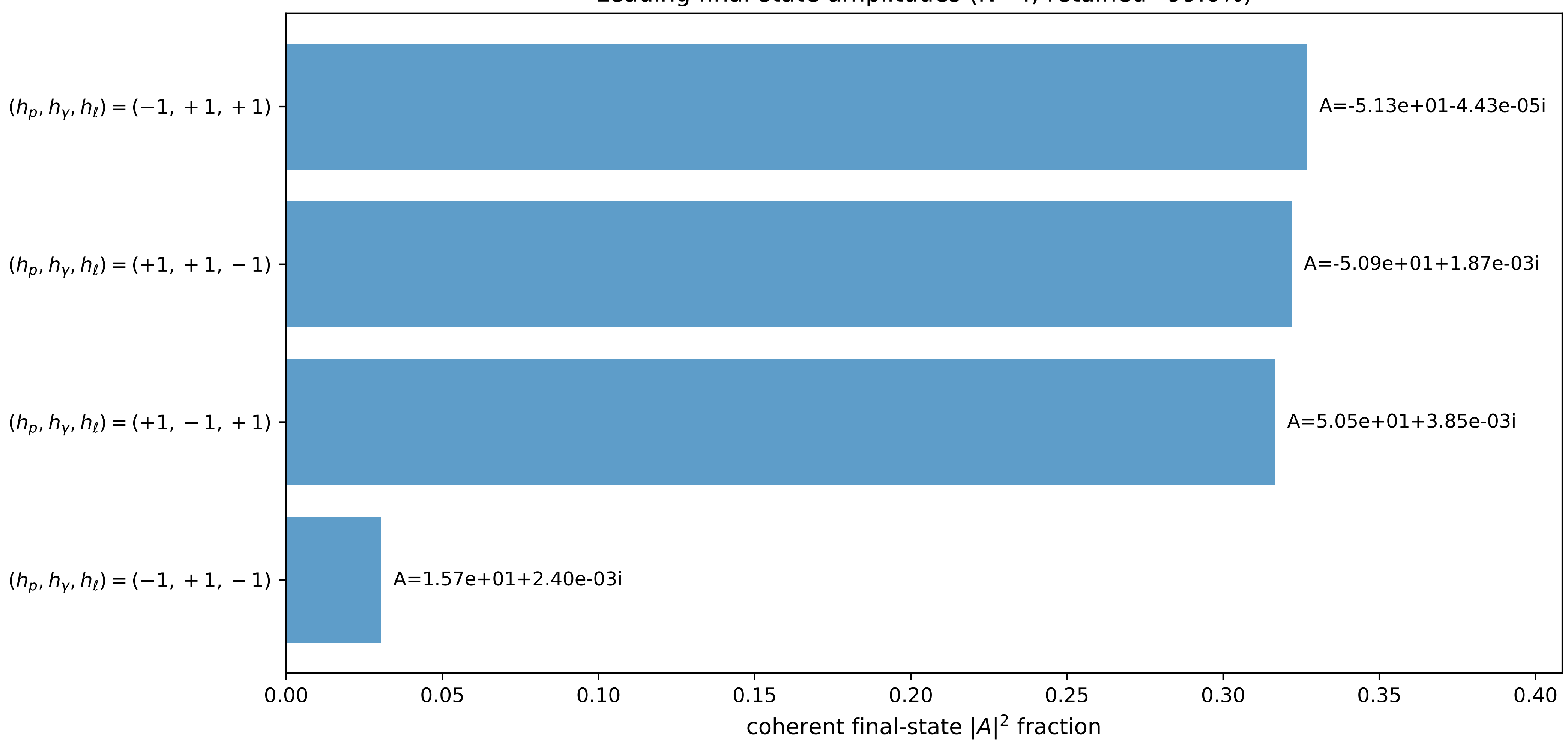
3D momenta



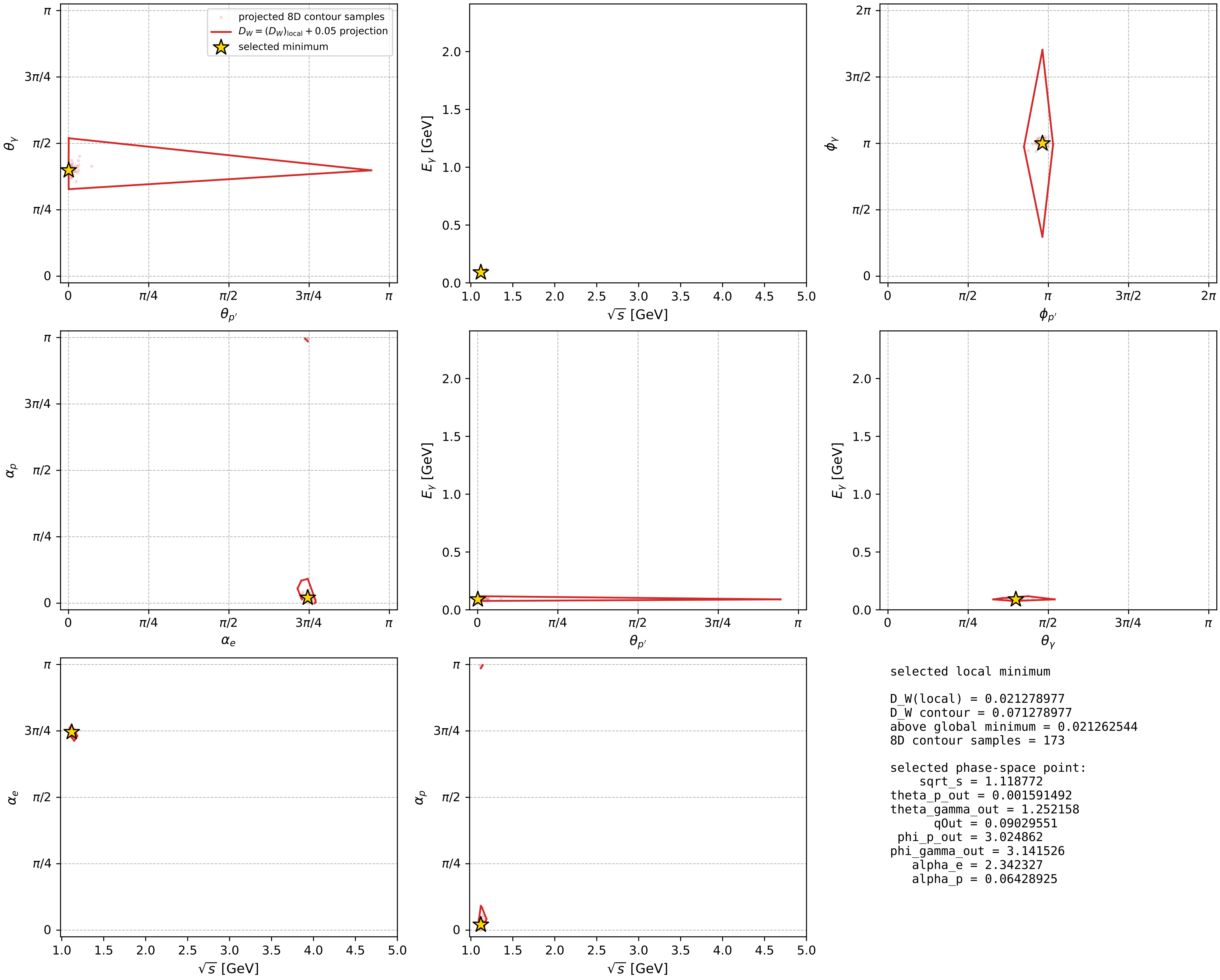
Transverse plane



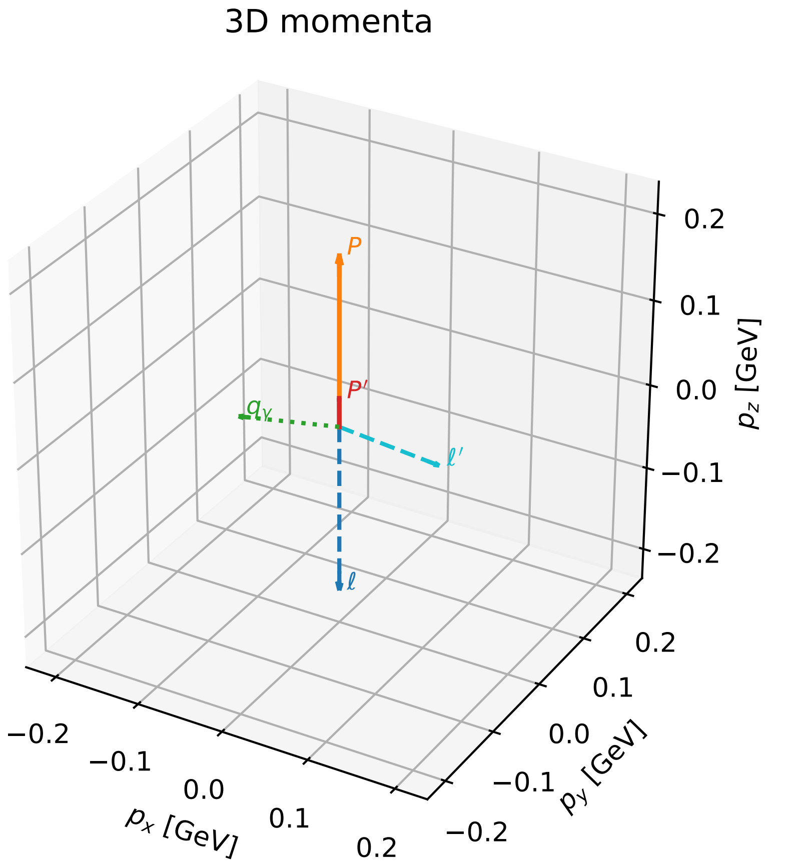
Leading final-state amplitudes (N=4, retained=99.6%)



electron: polarization cluster P5, local minimum 1043; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



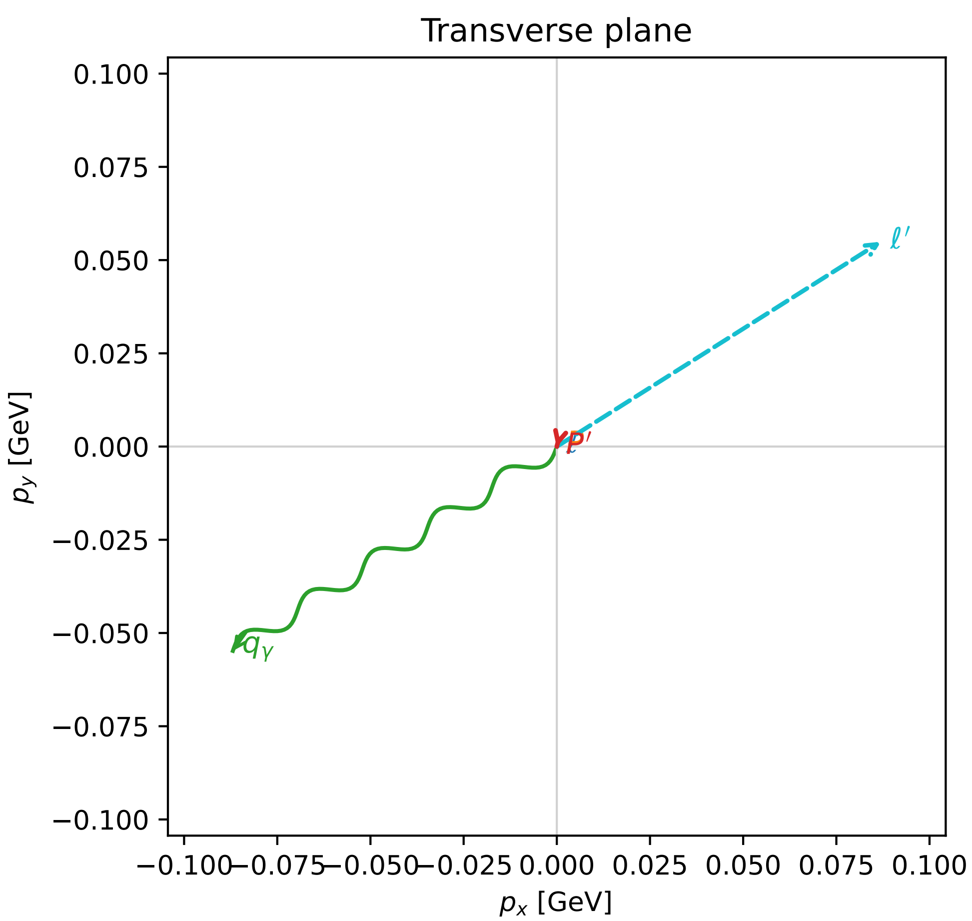
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

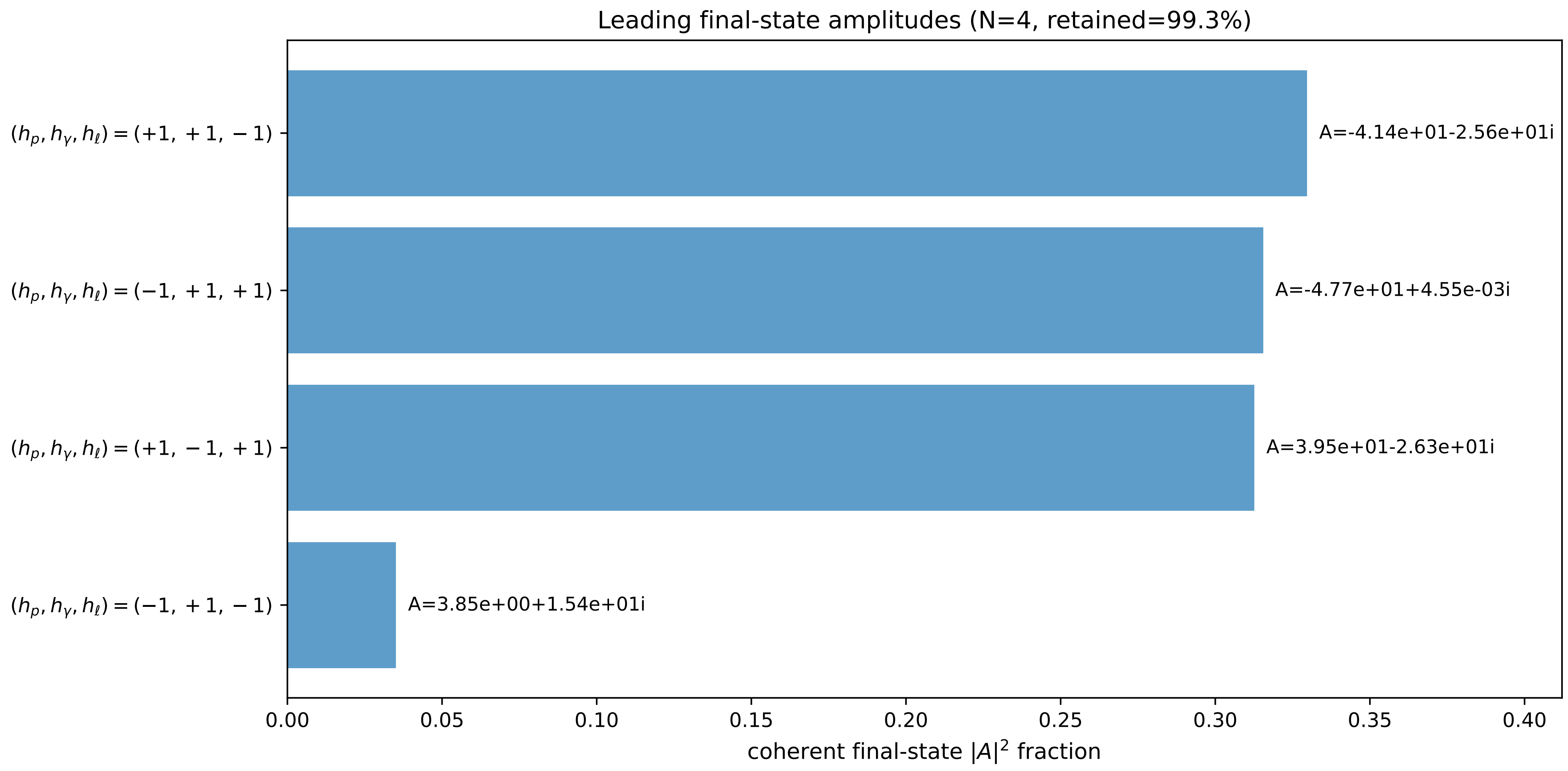


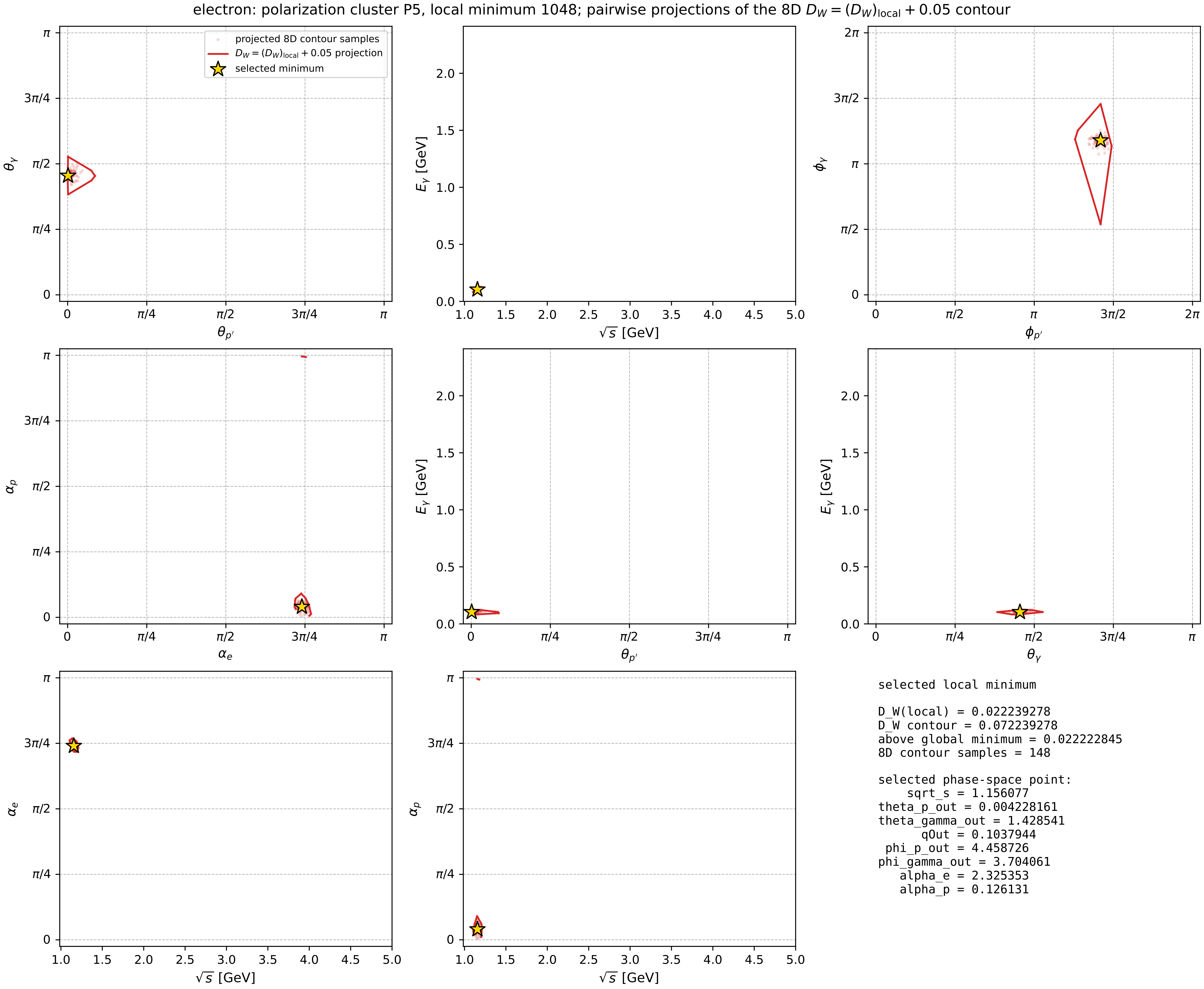

```
dw_mixing_angles_polarization_05_local_minimum_1048  D_W=0.0222393
region: polarization_cluster_05_local_minimum_1048
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3253534 rad
incoming lepton: -0.684966|+> +0.728575|->
proton mixing angle: alpha_p=0.12613102 rad
incoming proton: 0.992056|+> +0.125797|->
```

s=1.33651, $\sqrt{s}$=1.15608, $|\vec{P}|$=0.197508, $|\vec{P}'|$=0.0337011
$\theta_{p'}$=0.00422816, θ_γ=1.42854, $\phi_{p'}$=4.45873, ϕ_γ=3.70406
E_γ=0.103794, $\theta_{\ell'}$=2.01077, $\phi_{\ell'}$=0.563418
Q^2=0.0257786, x_B=0.117384, t=-0.0264342, W^2=1.07367, y=0.480894

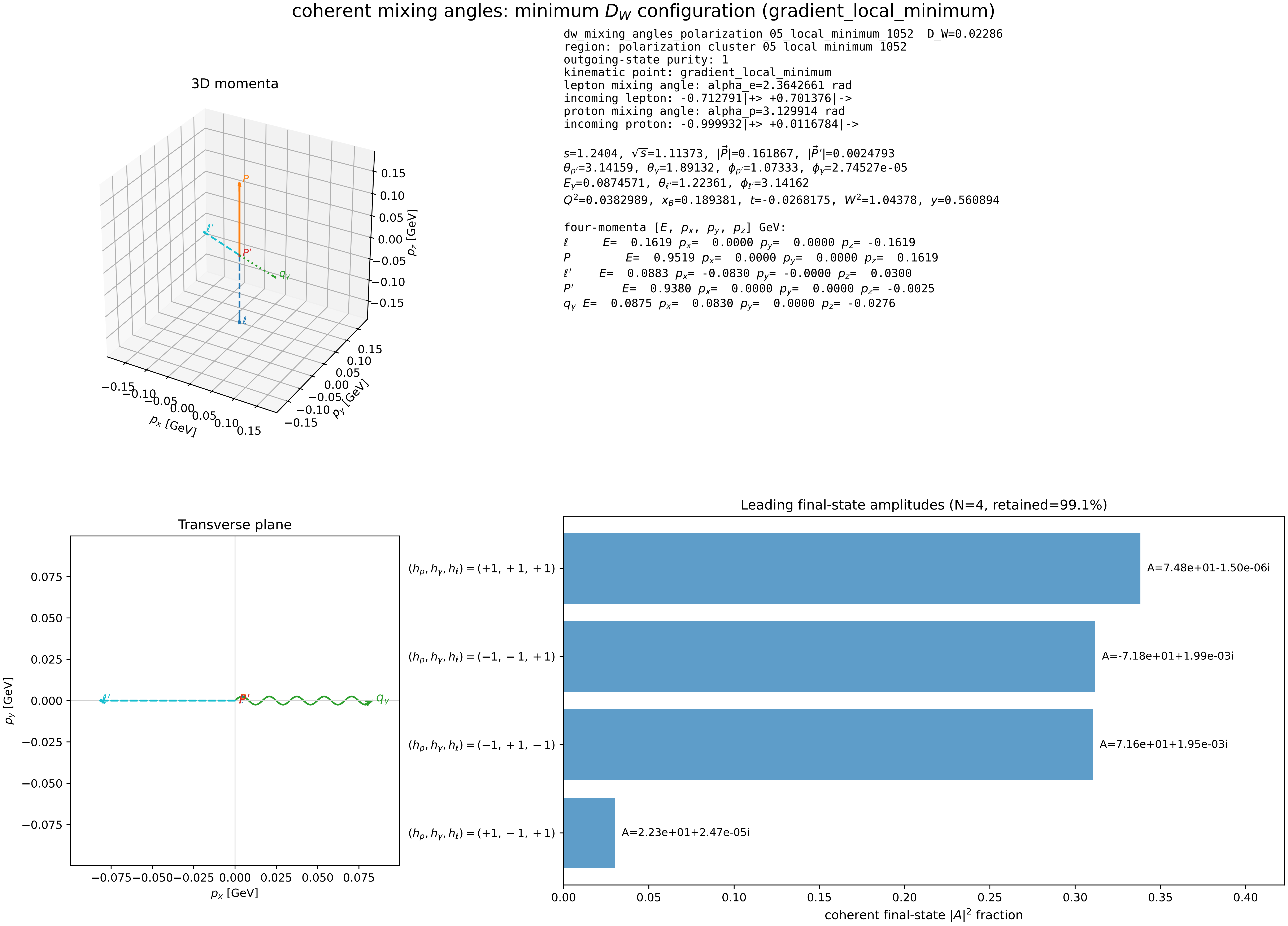
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1975 p_x= 0.0000 p_y= 0.0000 p_z= -0.1975
P E= 0.9586 p_x= 0.0000 p_y= 0.0000 p_z= 0.1975
ℓ' E= 0.1137 p_x= 0.0870 p_y= 0.0549 p_z= -0.0484
P' E= 0.9386 p_x= -0.0000 p_y= -0.0001 p_z= 0.0337
q_γ E= 0.1038 p_x= -0.0869 p_y= -0.0548 p_z= 0.0147



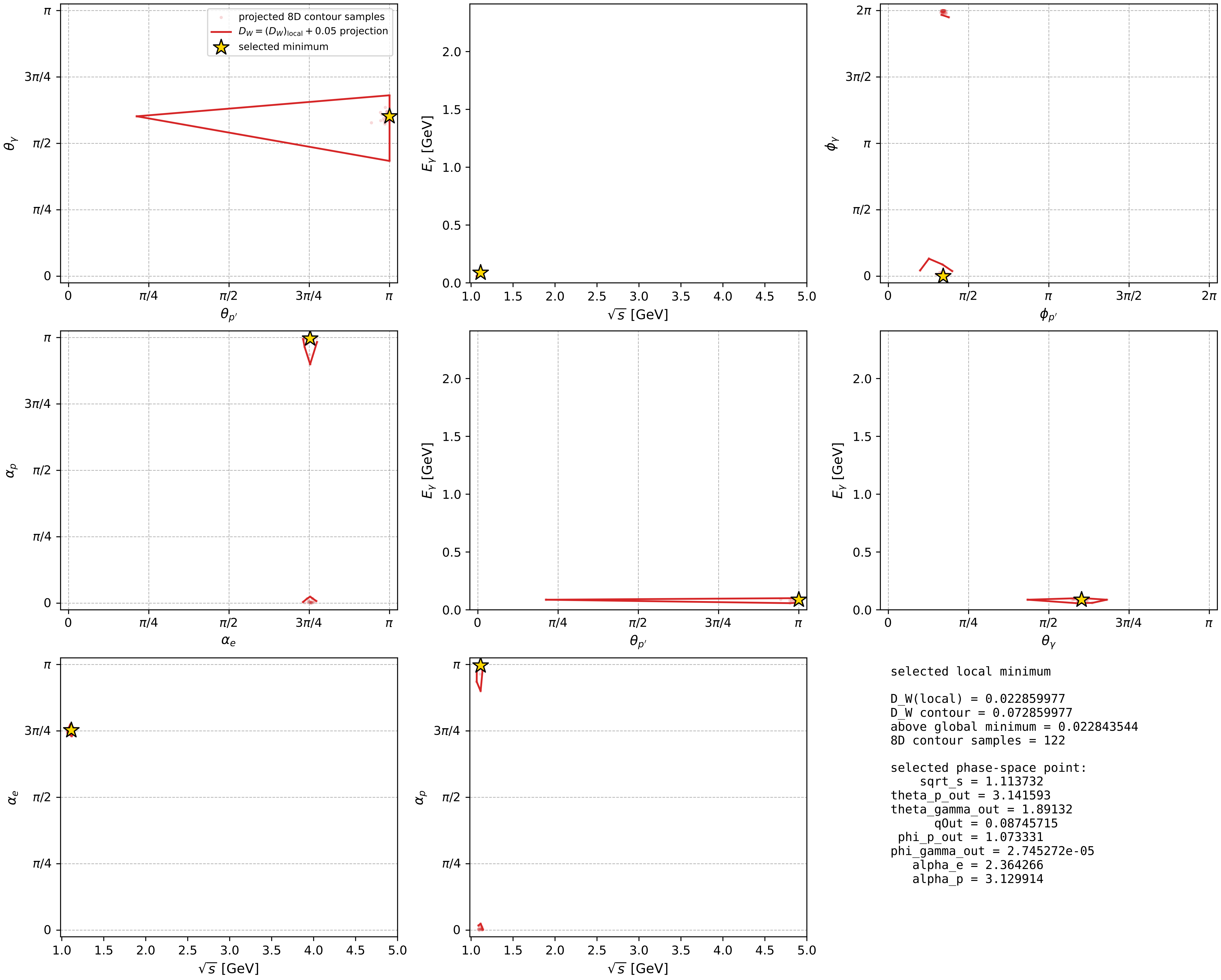




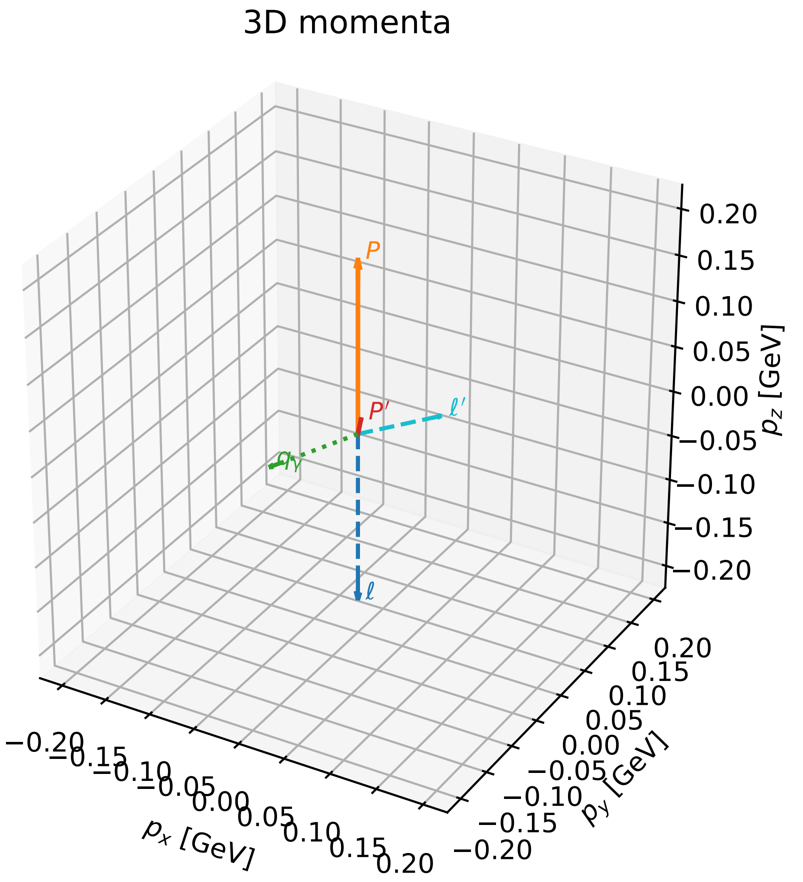
electron: polarization cluster P5, local minimum 1048; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
0
$\pi/4$
$\pi/2$
$3\pi/4$
π
$3\pi/2$
2π
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
selected local minimum
D_W(local) = 0.022239278
D_W contour = 0.072239278
above global minimum = 0.022222845
8D contour samples = 148
selected phase-space point:
sqrt_s = 1.156077
theta_p_out = 0.004228161
theta_gamma_out = 1.428541
qOut = 0.1037944
phi_p_out = 4.458726
phi_gamma_out = 3.704061
alpha_e = 2.325353
alpha_p = 0.126131


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_1052 D_W=0.02286
region: polarization_cluster_05_local_minimum_1052
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3642661 rad
incoming lepton: -0.712791|+> +0.701376|->
proton mixing angle: alpha_p=3.129914 rad
incoming proton: -0.999932|+> +0.0116784|->
s=1.2404, $\sqrt{s}$=1.11373, $|\vec{P}|$=0.161867, $|\vec{P}'|$=0.0024793
$\theta_{p'}$=3.14159, θ_γ=1.89132, $\phi_{p'}$=1.07333, ϕ_γ=2.74527e-05
E_γ=0.0874571, $\theta_{\ell'}$=1.22361, $\phi_{\ell'}$=3.14162
Q^2=0.0382989, x_B=0.189381, t=-0.0268175, W^2=1.04378, y=0.560894
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1619 p_x= 0.0000 p_y= 0.0000 p_z= -0.1619
P E= 0.9519 p_x= 0.0000 p_y= 0.0000 p_z= 0.1619
ℓ' E= 0.0883 p_x= -0.0830 p_y= -0.0000 p_z= 0.0300
P' E= 0.9380 p_x= 0.0000 p_y= 0.0000 p_z= -0.0025
q_γ E= 0.0875 p_x= 0.0830 p_y= 0.0000 p_z= -0.0276
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=4, retained=99.1%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=7.48e+01-1.50e-06i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=-7.18e+01+1.99e-03i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=7.16e+01+1.95e-03i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=2.23e+01+2.47e-05i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P5, local minimum 1052; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

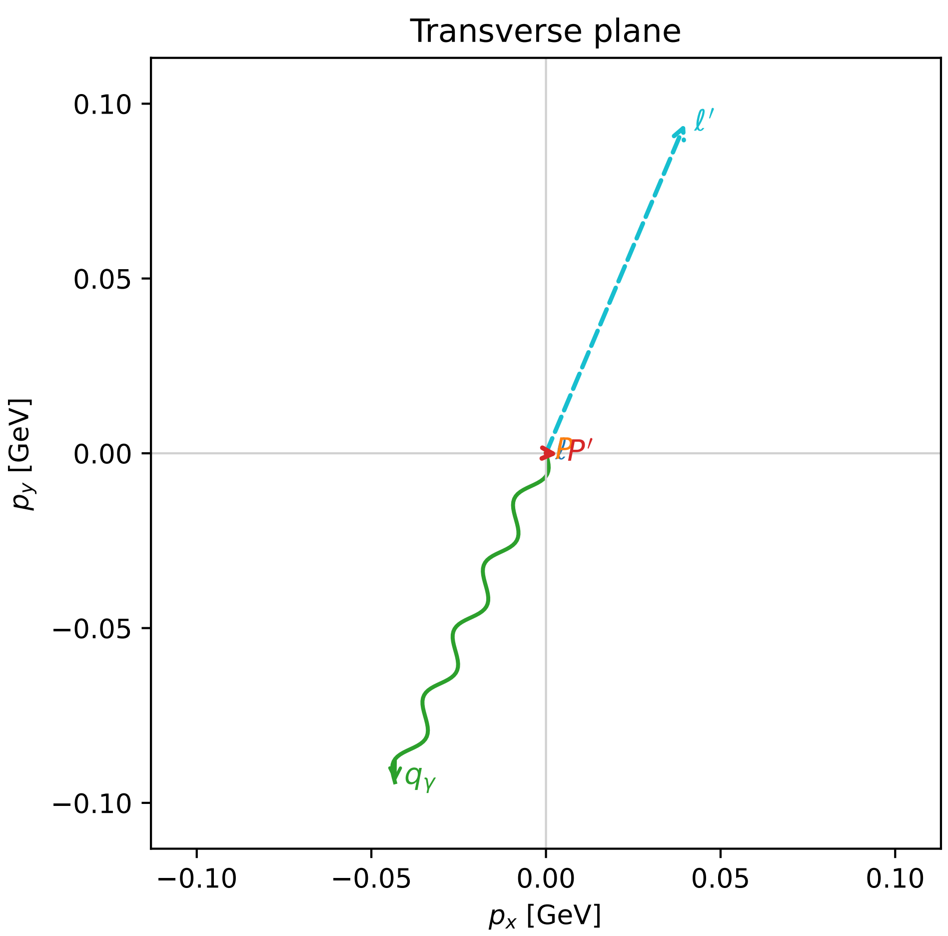


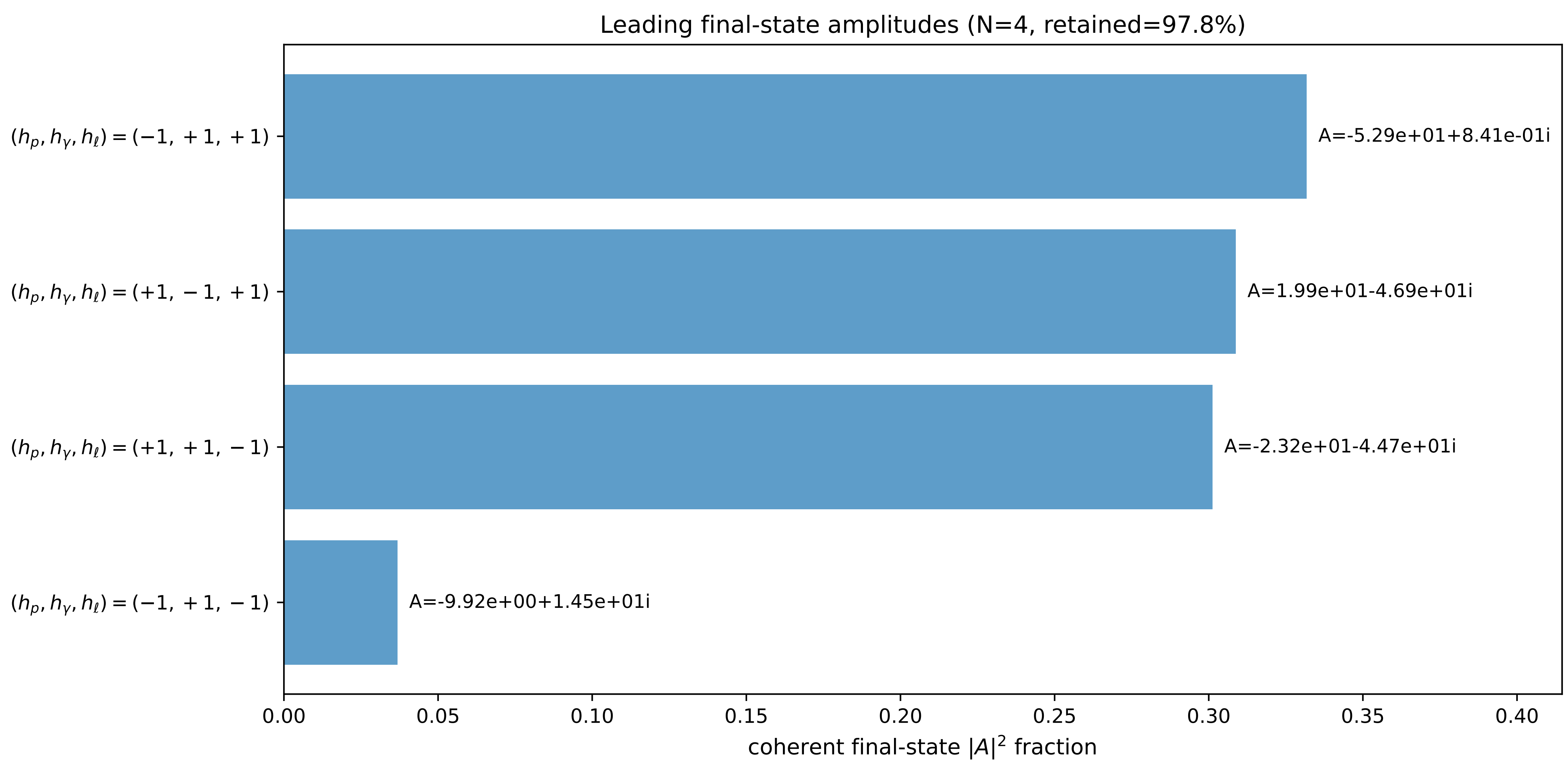

```
dw_mixing_angles_polarization_05_local_minimum_1055  D_W=0.0230509
region: polarization_cluster_05_local_minimum_1055
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3569615 rad
incoming lepton: -0.707649|+> +0.706564|->
proton mixing angle: alpha_p=0.049241873 rad
incoming proton: 0.998788|+> +0.049222|->
```

s=1.31138, $\sqrt{s}$=1.14515, $|\vec{P}|$=0.188416, $|\vec{P}'|$=0.0174277
$\theta_{p'}$=0.198608, θ_γ=1.58974, $\phi_{p'}$=6.2322, ϕ_γ=4.28161
E_γ=0.103561, $\theta_{\ell'}$=1.71754, $\phi_{\ell'}$=1.17123
Q^2=0.0332766, x_B=0.146, t=-0.0290211, W^2=1.07449, y=0.528169

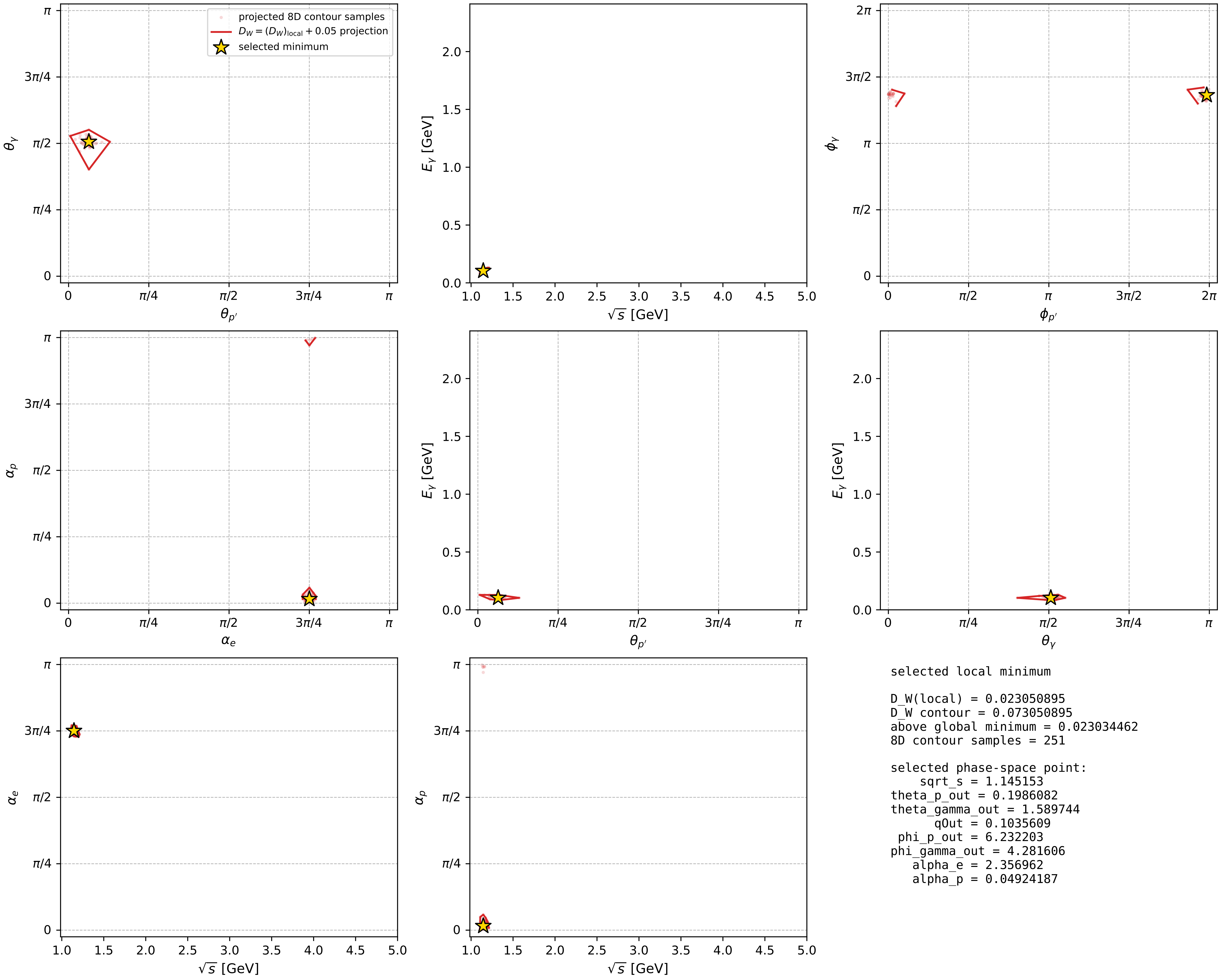
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 0.1884	p_x= 0.0000	p_y= 0.0000	p_z= -0.1884
P	E= 0.9567	p_x= 0.0000	p_y= 0.0000	p_z= 0.1884
ℓ'	E= 0.1034	p_x= 0.0398	p_y= 0.0943	p_z= -0.0151
P'	E= 0.9382	p_x= 0.0034	p_y= -0.0002	p_z= 0.0171
q_γ	E= 0.1036	p_x= -0.0432	p_y= -0.0941	p_z= -0.0020

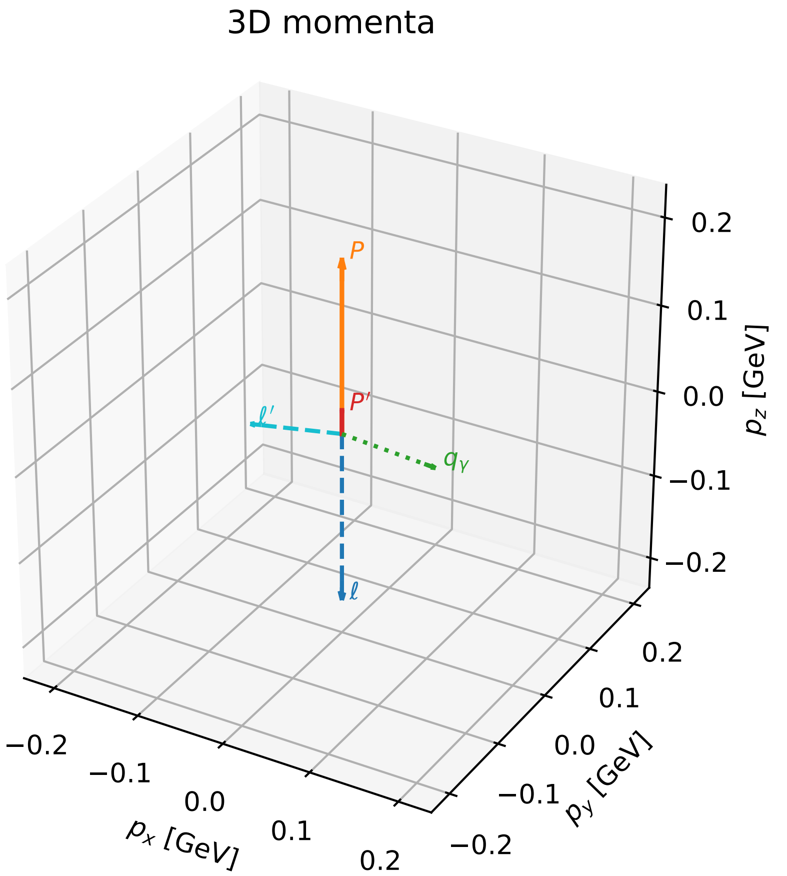




electron: polarization cluster P5, local minimum 1055; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



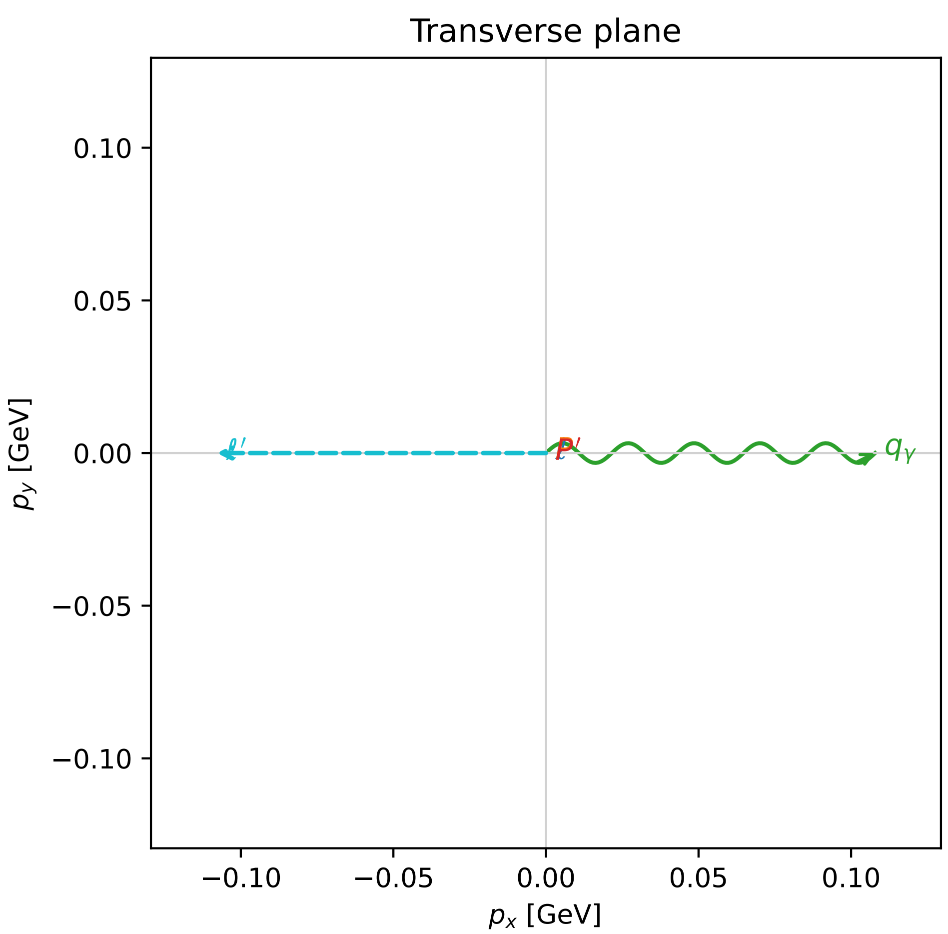
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

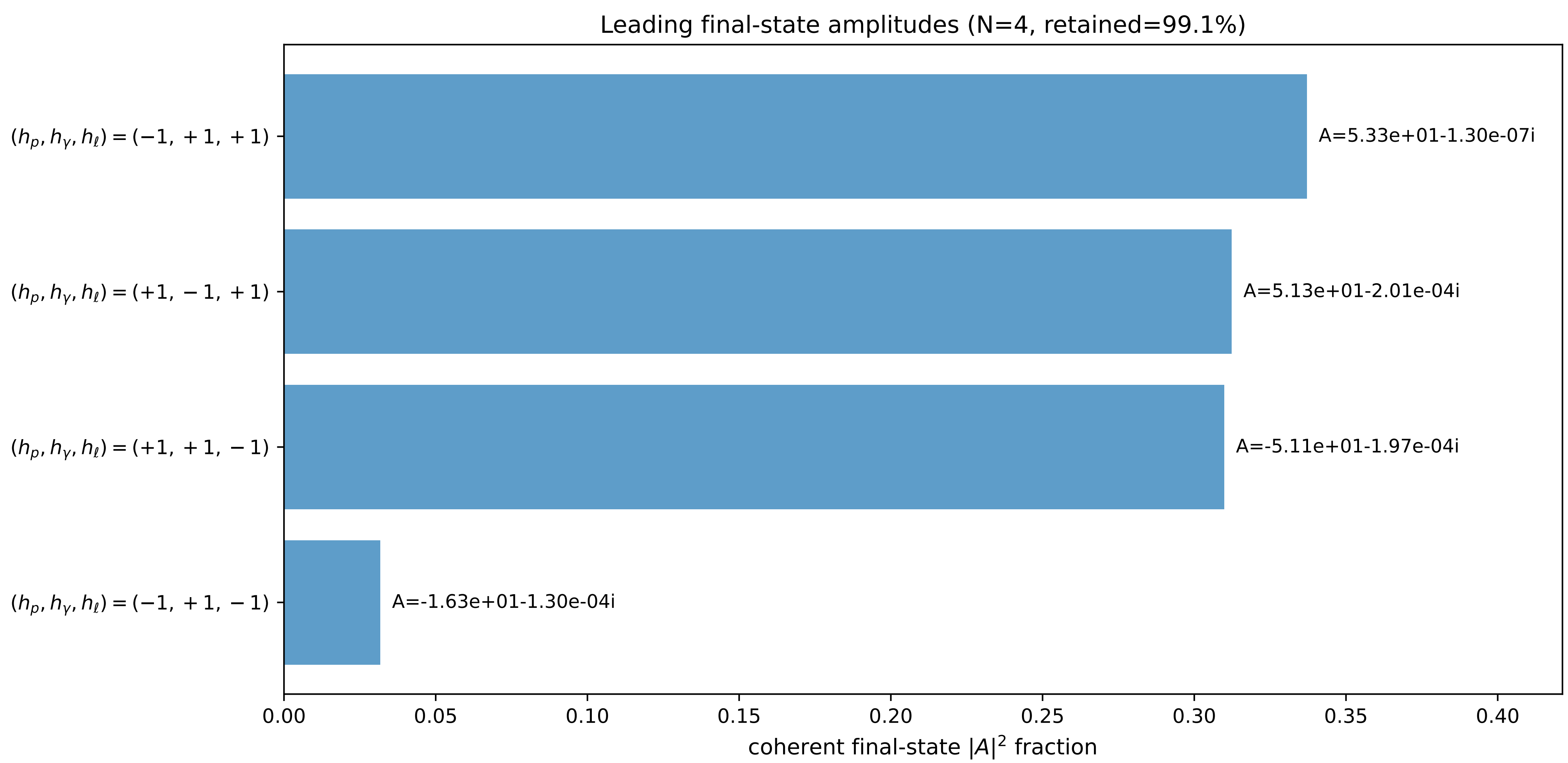


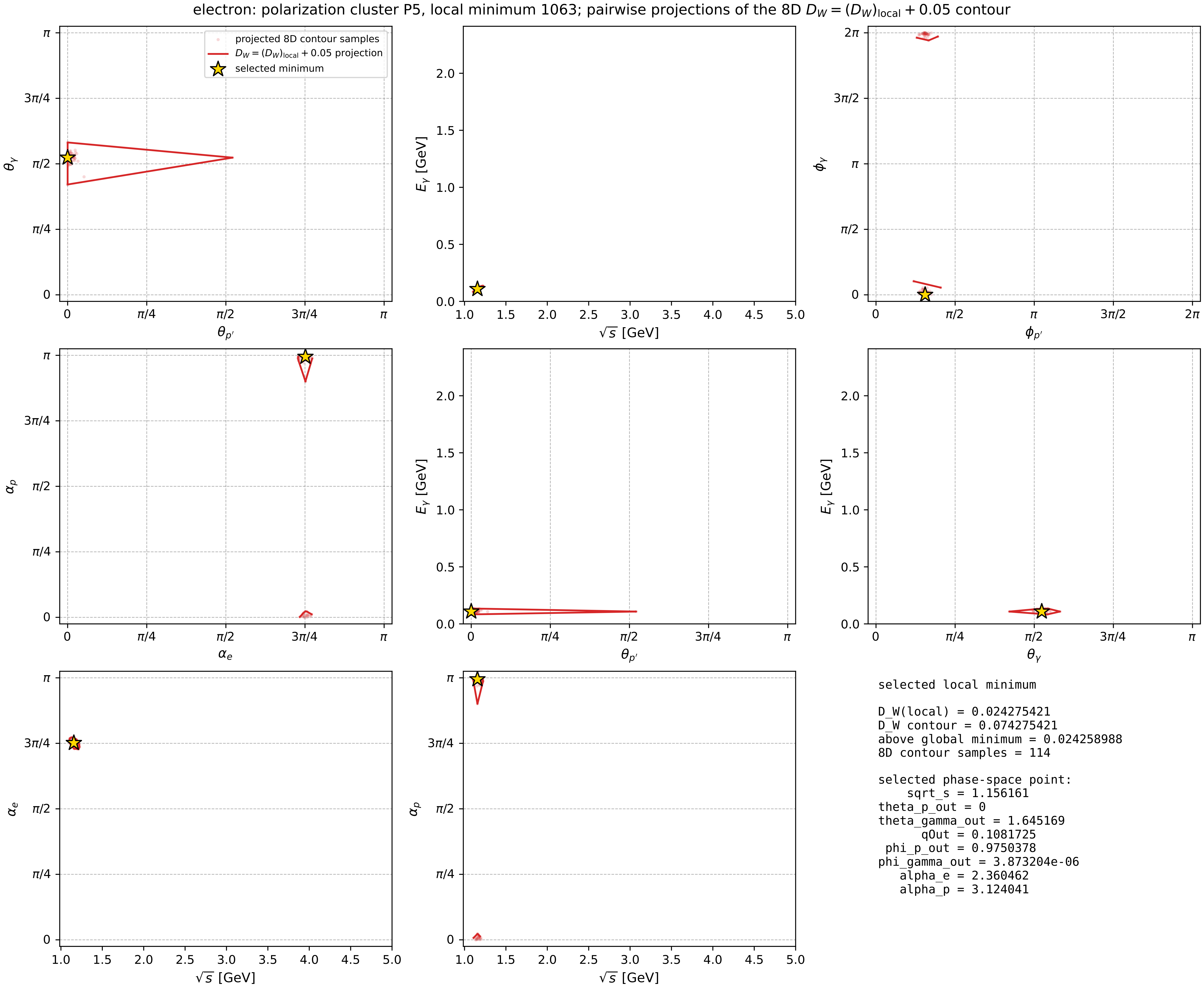

```
dw_mixing_angles_polarization_05_local_minimum_1063  D_W=0.0242754
region: polarization_cluster_05_local_minimum_1063
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3604619 rad
incoming lepton: -0.710118|+> +0.704083|->
proton mixing angle: alpha_p=3.1240407 rad
incoming proton: -0.999846|+> +0.0175511|->
```

s=1.33671, $\sqrt{s}$=1.15616, $|\vec{P}|$=0.197578, $|\vec{P}'|$=0.0273522
$\theta_{p'}$=0, θ_γ=1.64517, $\phi_{p'}$=0.975038, ϕ_γ=3.8732e-06
E_γ=0.108173, $\theta_{\ell'}$=1.74797, $\phi_{\ell'}$=3.1416
Q^2=0.0356726, x_B=0.149176, t=-0.0285694, W^2=1.0833, y=0.523415

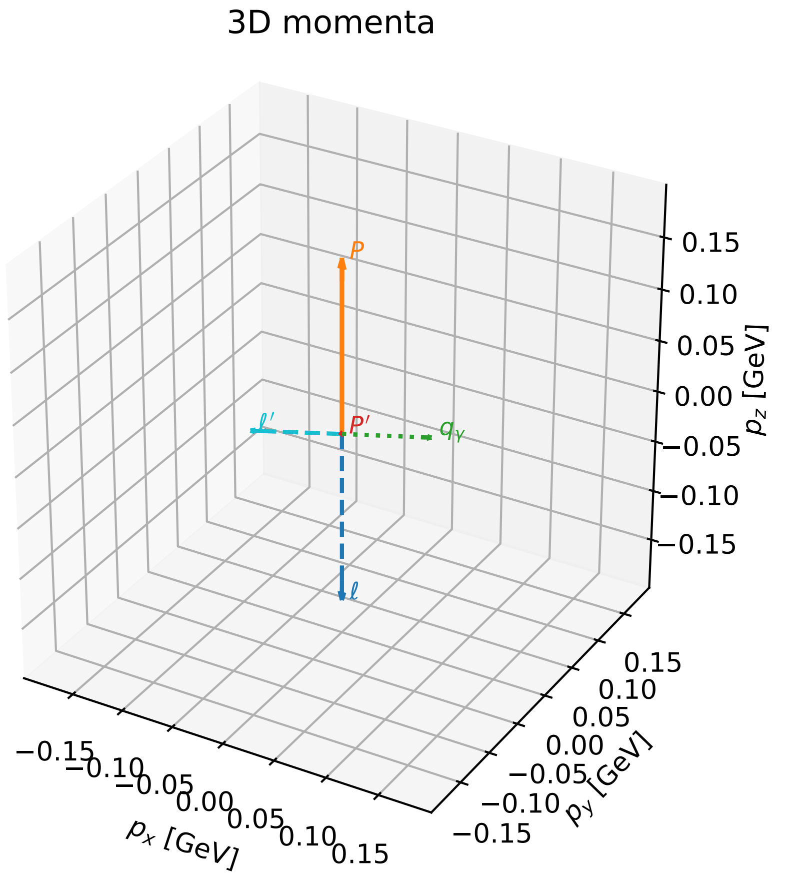
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1976 p_x= 0.0000 p_y= 0.0000 p_z= -0.1976
P E= 0.9586 p_x= 0.0000 p_y= 0.0000 p_z= 0.1976
ℓ' E= 0.1096 p_x= -0.1079 p_y= -0.0000 p_z= -0.0193
P' E= 0.9384 p_x= 0.0000 p_y= 0.0000 p_z= 0.0274
q_γ E= 0.1082 p_x= 0.1079 p_y= 0.0000 p_z= -0.0080






electron: polarization cluster P5, local minimum 1063; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.024275421
D_W contour = 0.074275421
above global minimum = 0.024258988
8D contour samples = 114
selected phase-space point:
sqrt_s = 1.156161
theta_p_out = 0
theta_gamma_out = 1.645169
qOut = 0.1081725
phi_p_out = 0.9750378
phi_gamma_out = 3.873204e-06
alpha_e = 2.360462
alpha_p = 3.124041

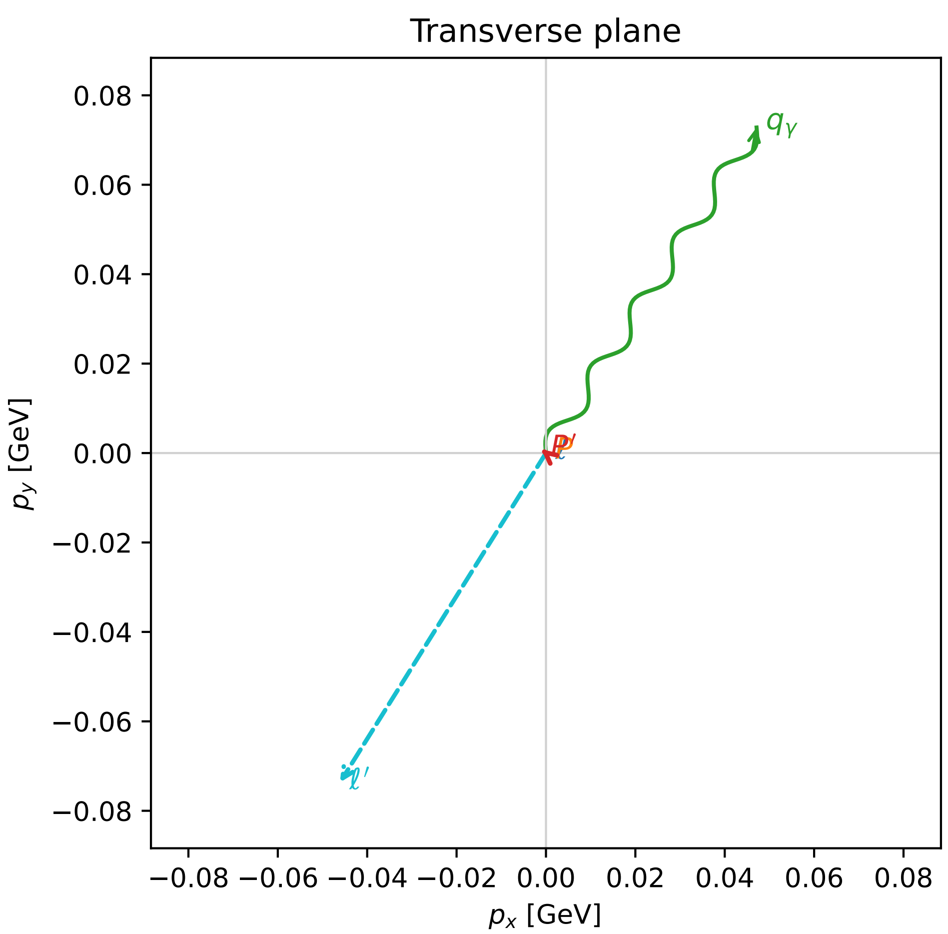
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

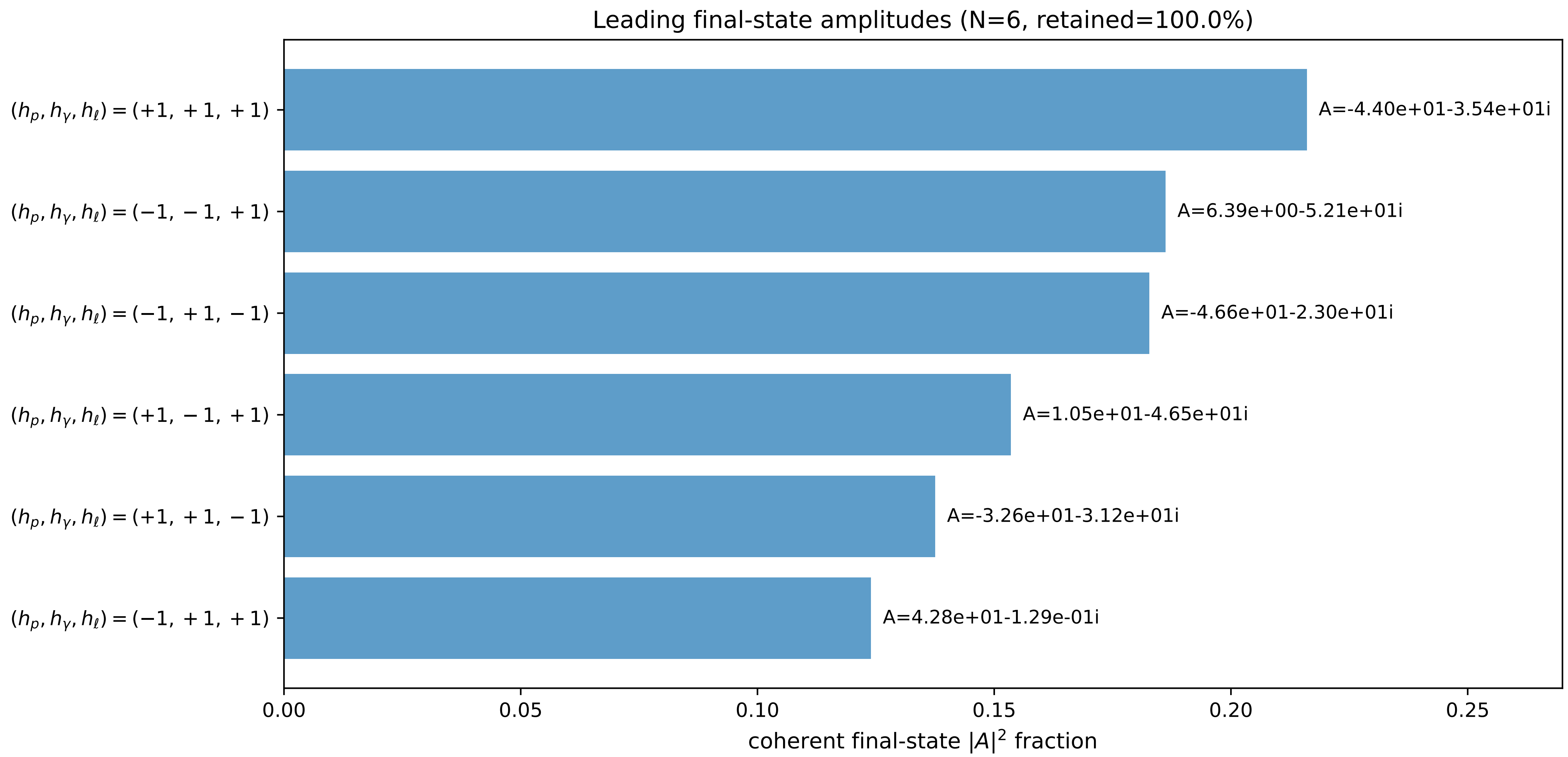


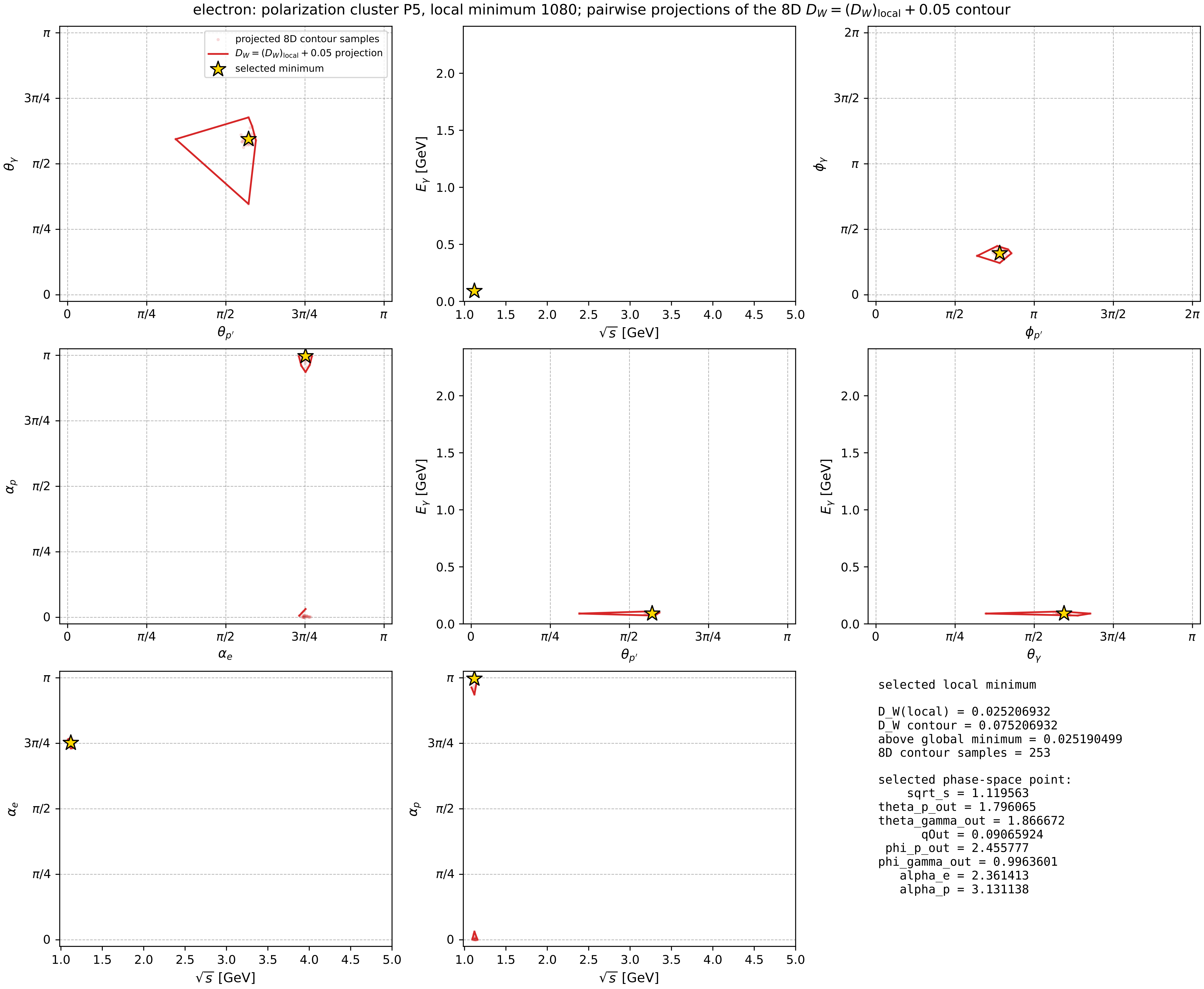
dw_mixing_angles_polarization_05_local_minimum_1080 D_W=0.0252069
region: polarization_cluster_05_local_minimum_1080
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3614126 rad
incoming lepton: -0.710787|+> +0.703407|->
proton mixing angle: alpha_p=3.1311377 rad
incoming proton: -0.999945|+> +0.0104547|->

s=1.25342, $\sqrt{s}$=1.11956, $|\vec{P}|$=0.16684, $|\vec{P}'|$=0.00137372
$\theta_{p'}$=1.79607, θ_γ=1.86667, $\phi_{p'}$=2.45578, ϕ_γ=0.99636
E_γ=0.0906592, $\theta_{\ell'}$=1.2722, $\phi_{\ell'}$=4.15327
Q^2=0.0392549, x_B=0.187563, t=-0.0277231, W^2=1.04988, y=0.560233

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1668 p_x= 0.0000 p_y= 0.0000 p_z= -0.1668
P E= 0.9527 p_x= 0.0000 p_y= 0.0000 p_z= 0.1668
ℓ' E= 0.0909 p_x= -0.0461 p_y= -0.0736 p_z= 0.0267
P' E= 0.9380 p_x= -0.0010 p_y= 0.0008 p_z= -0.0003
q_γ E= 0.0907 p_x= 0.0471 p_y= 0.0728 p_z= -0.0264







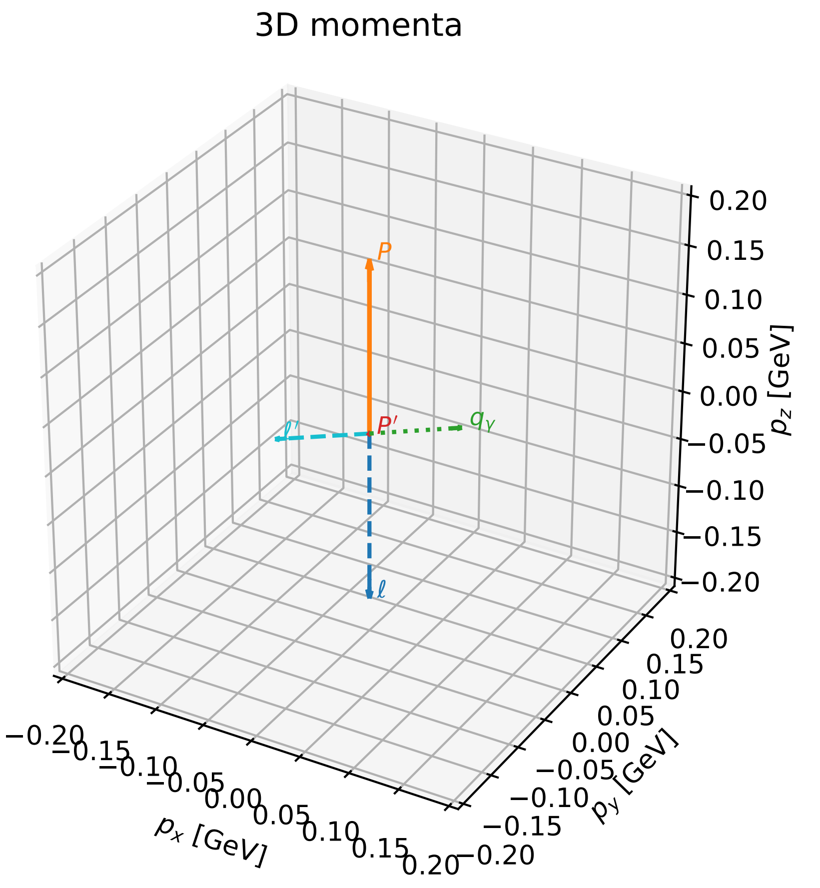
electron: polarization cluster P5, local minimum 1080; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\text{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
selected local minimum
D_W(local) = 0.025206932
D_W contour = 0.075206932
above global minimum = 0.025190499
8D contour samples = 253
selected phase-space point:
sqrt_s = 1.119563
theta_p_out = 1.796065
theta_gamma_out = 1.866672
qOut = 0.09065924
phi_p_out = 2.455777
phi_gamma_out = 0.9963601
alpha_e = 2.361413
alpha_p = 3.131138

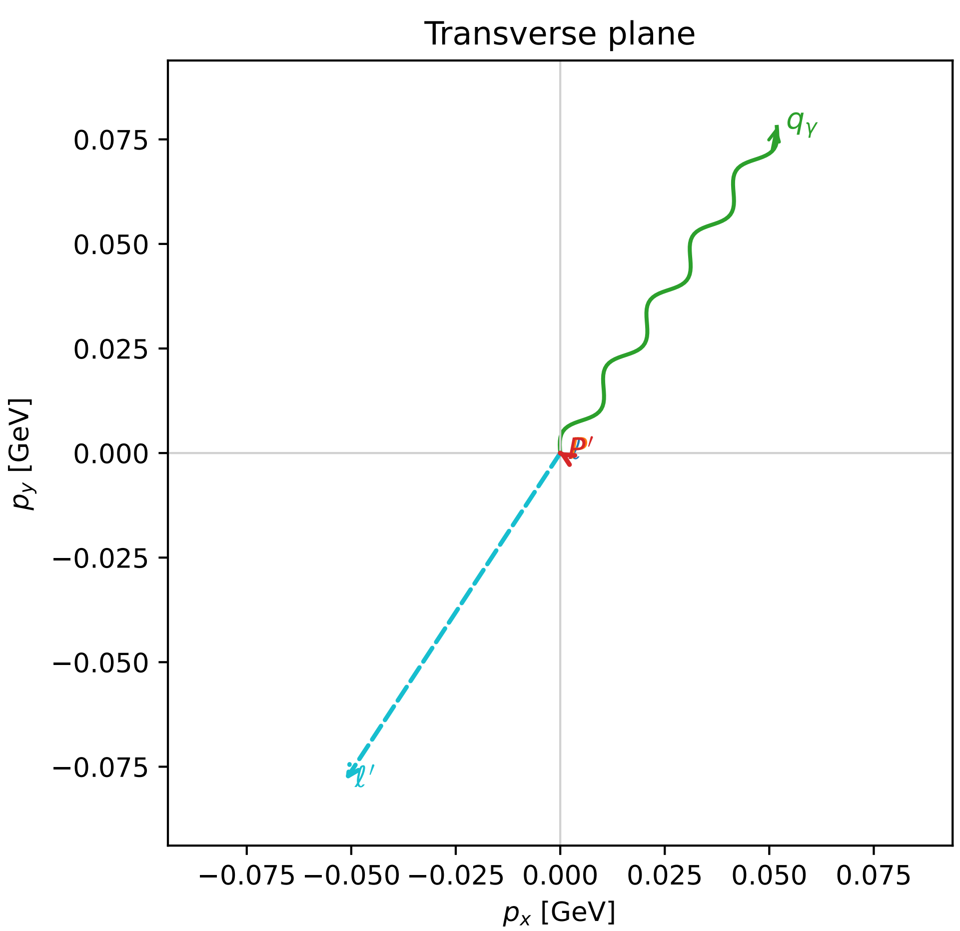
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

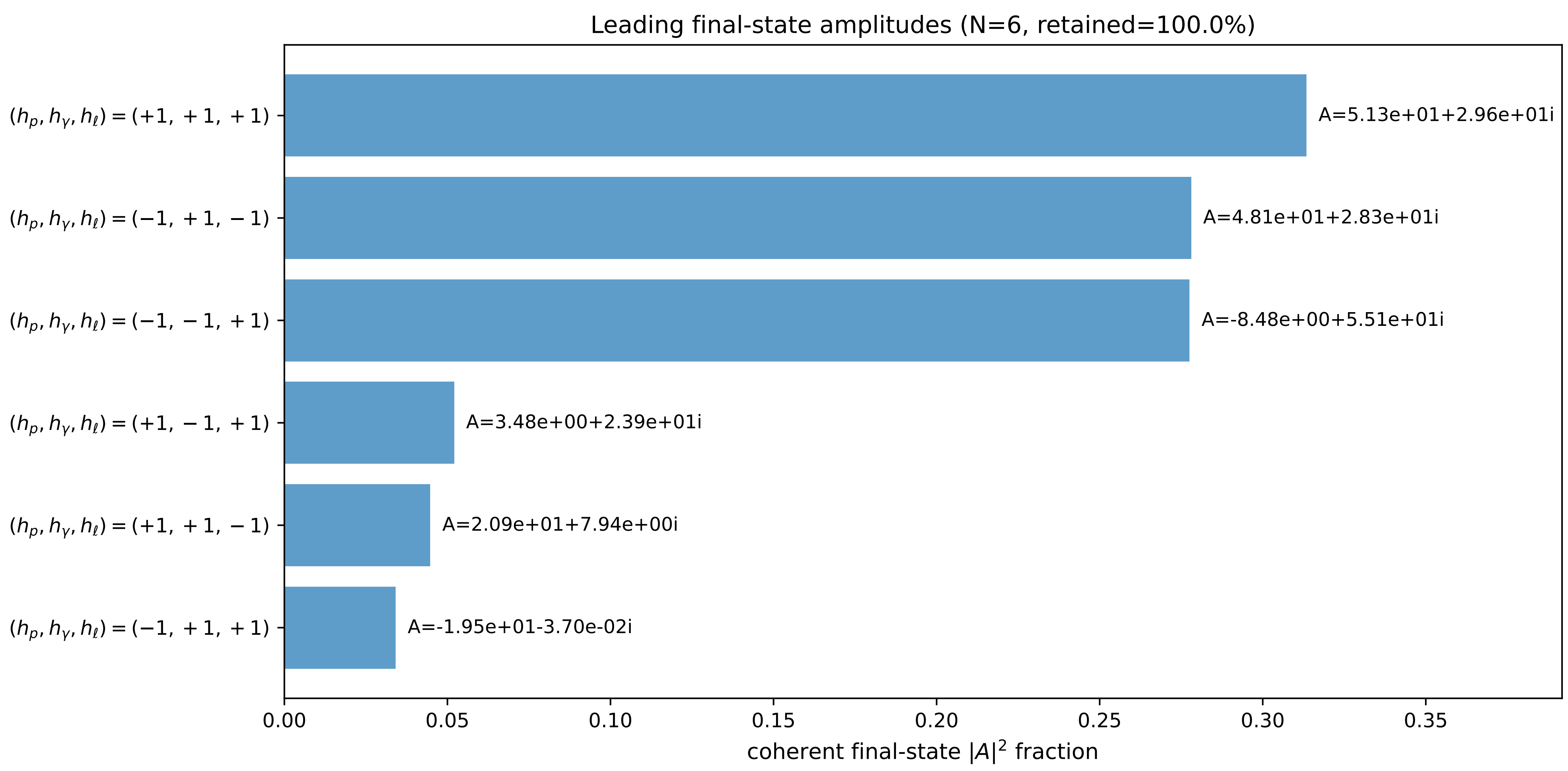
dw_mixing_angles_polarization_05_local_minimum_1085 D_W=0.0256936
region: polarization_cluster_05_local_minimum_1085
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3572648 rad
incoming lepton: -0.707863|+> +0.70635|->
proton mixing angle: alpha_p=6.3323703e-05 rad
incoming proton: 1|+> +6.33237e-05|->

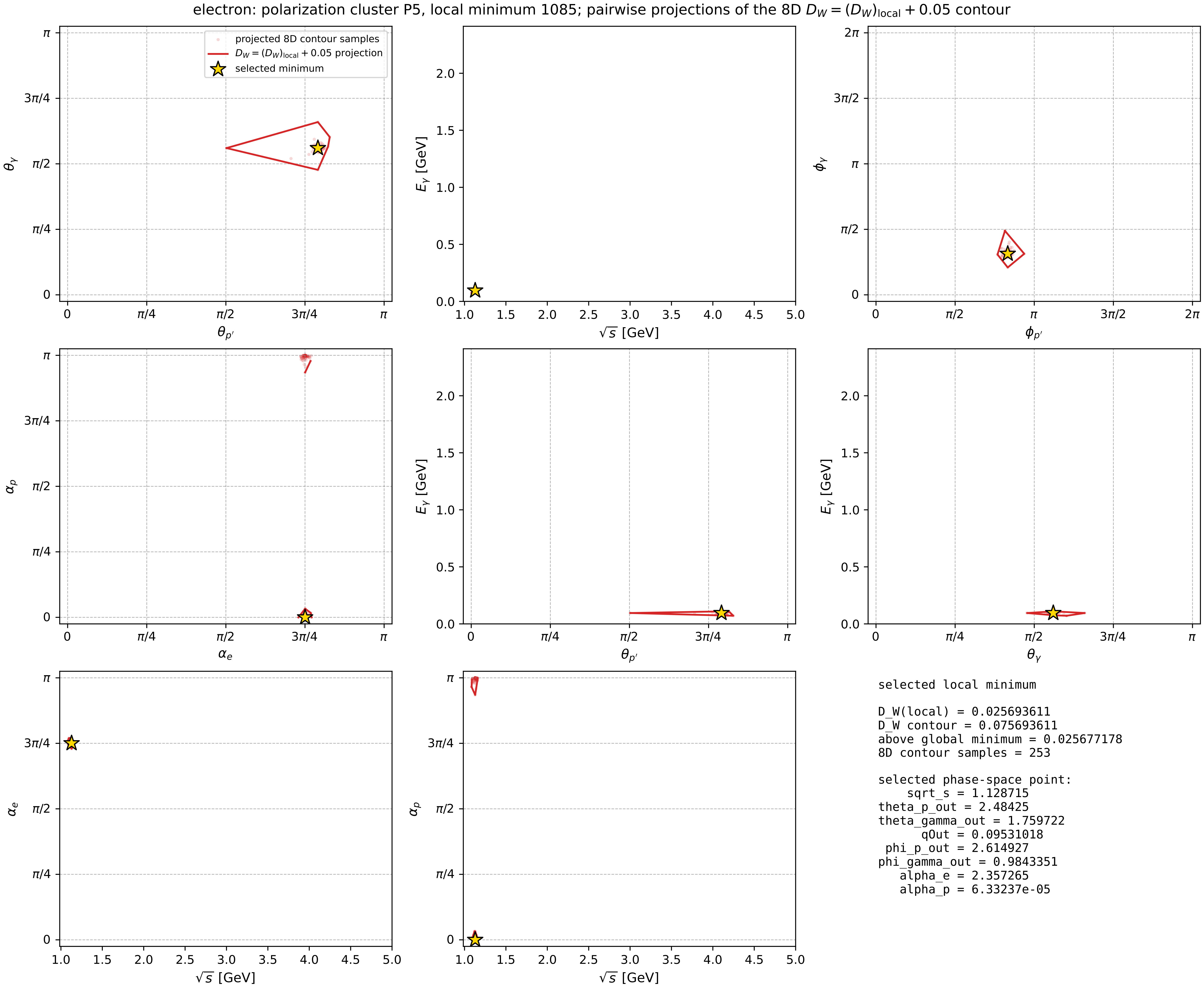
s=1.274, $\sqrt{s}$=1.12871, $|\vec{P}|$=0.174602, $|\vec{P}'|$=0.000794582
$\theta_{p'}$=2.48425, θ_γ=1.75972, $\phi_{p'}$=2.61493, ϕ_γ=0.984335
E_γ=0.0953102, $\theta_{\ell'}$=1.37534, $\phi_{\ell'}$=4.13111
Q^2=0.0397855, x_B=0.182026, t=-0.0304466, W^2=1.05863, y=0.554532

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1746 p_x= 0.0000 p_y= 0.0000 p_z= -0.1746
P E= 0.9541 p_x= 0.0000 p_y= 0.0000 p_z= 0.1746
ℓ' E= 0.0954 p_x= -0.0514 p_y= -0.0782 p_z= 0.0185
P' E= 0.9380 p_x= -0.0004 p_y= 0.0002 p_z= -0.0006
q_γ E= 0.0953 p_x= 0.0518 p_y= 0.0780 p_z= -0.0179

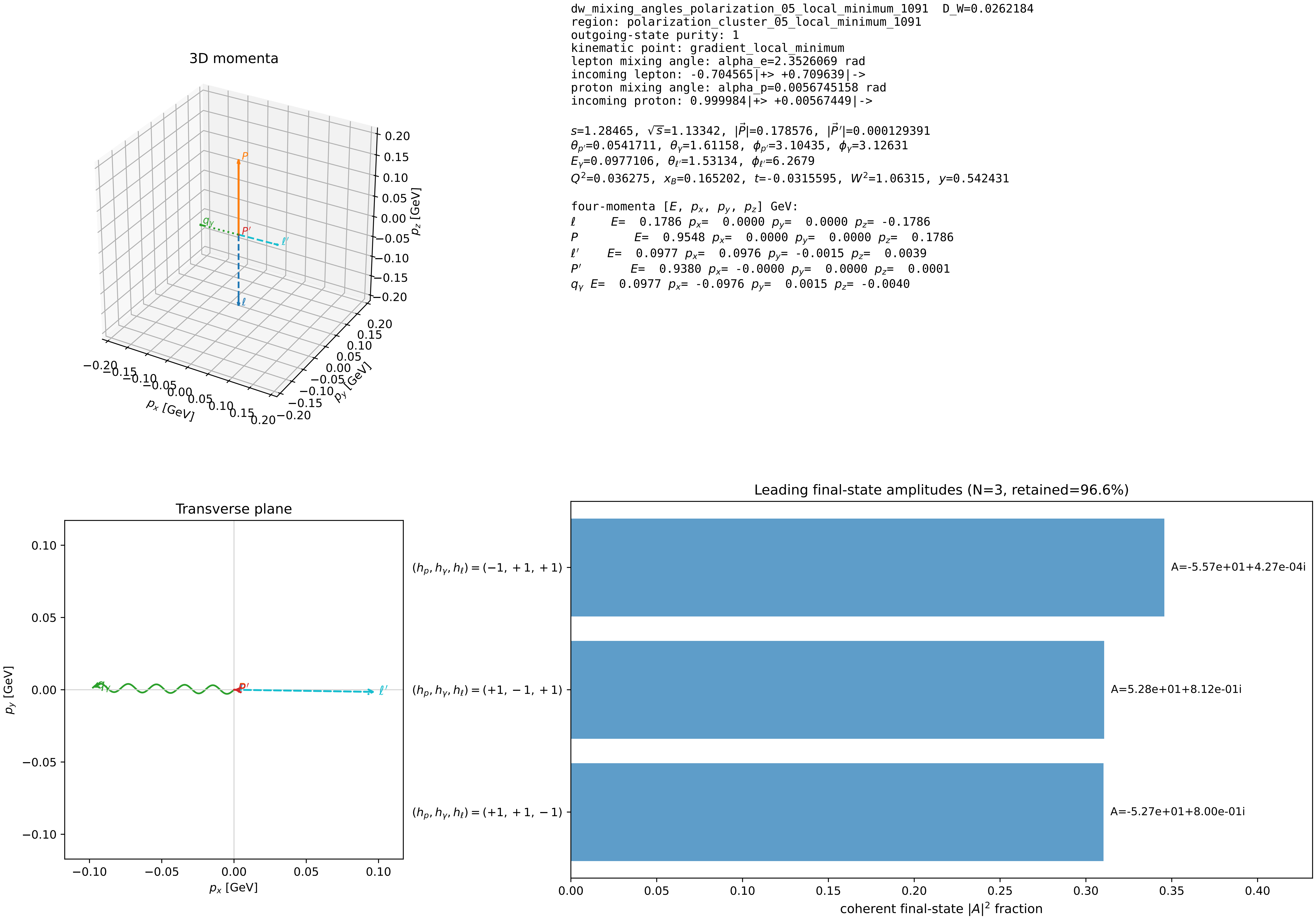


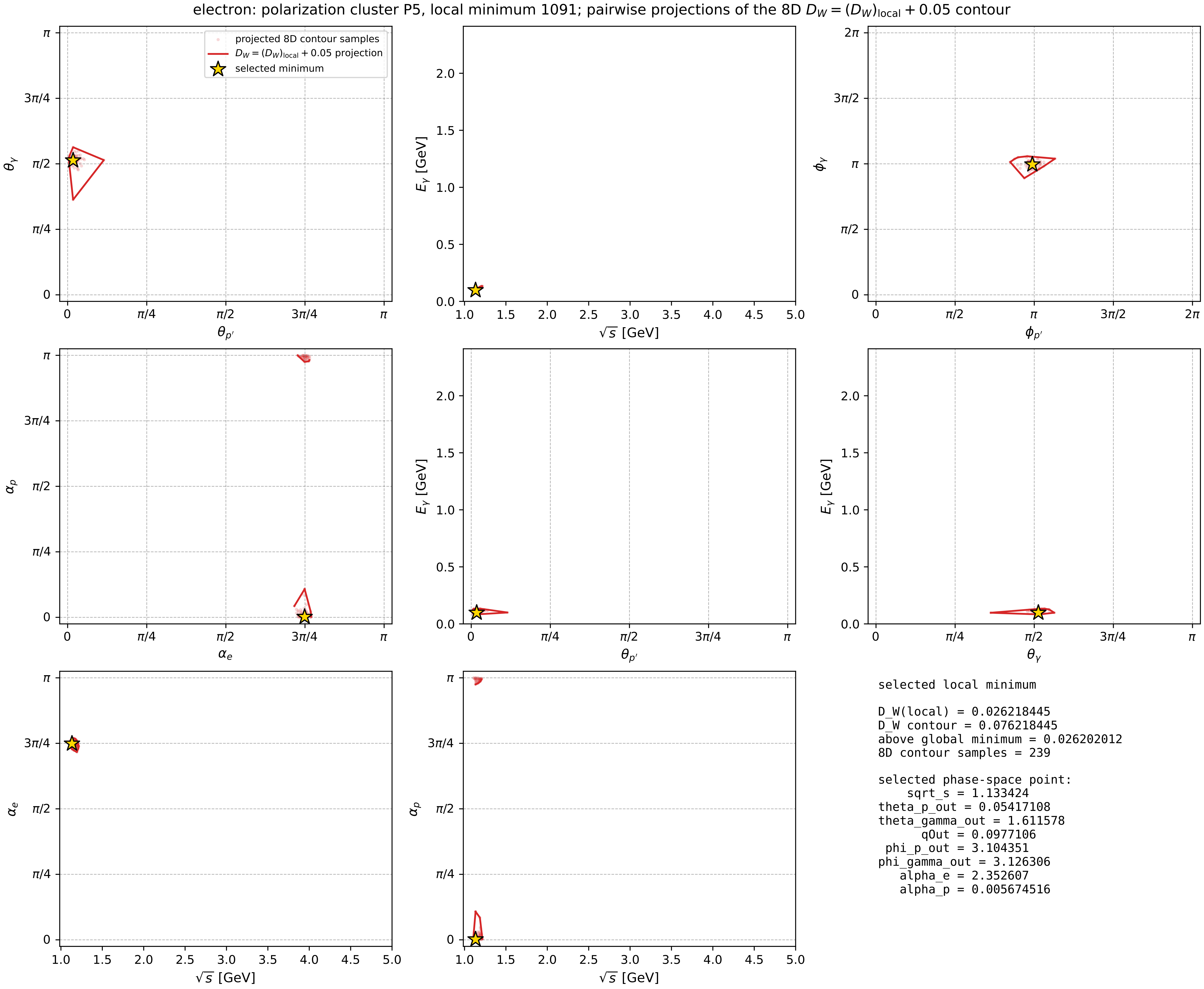


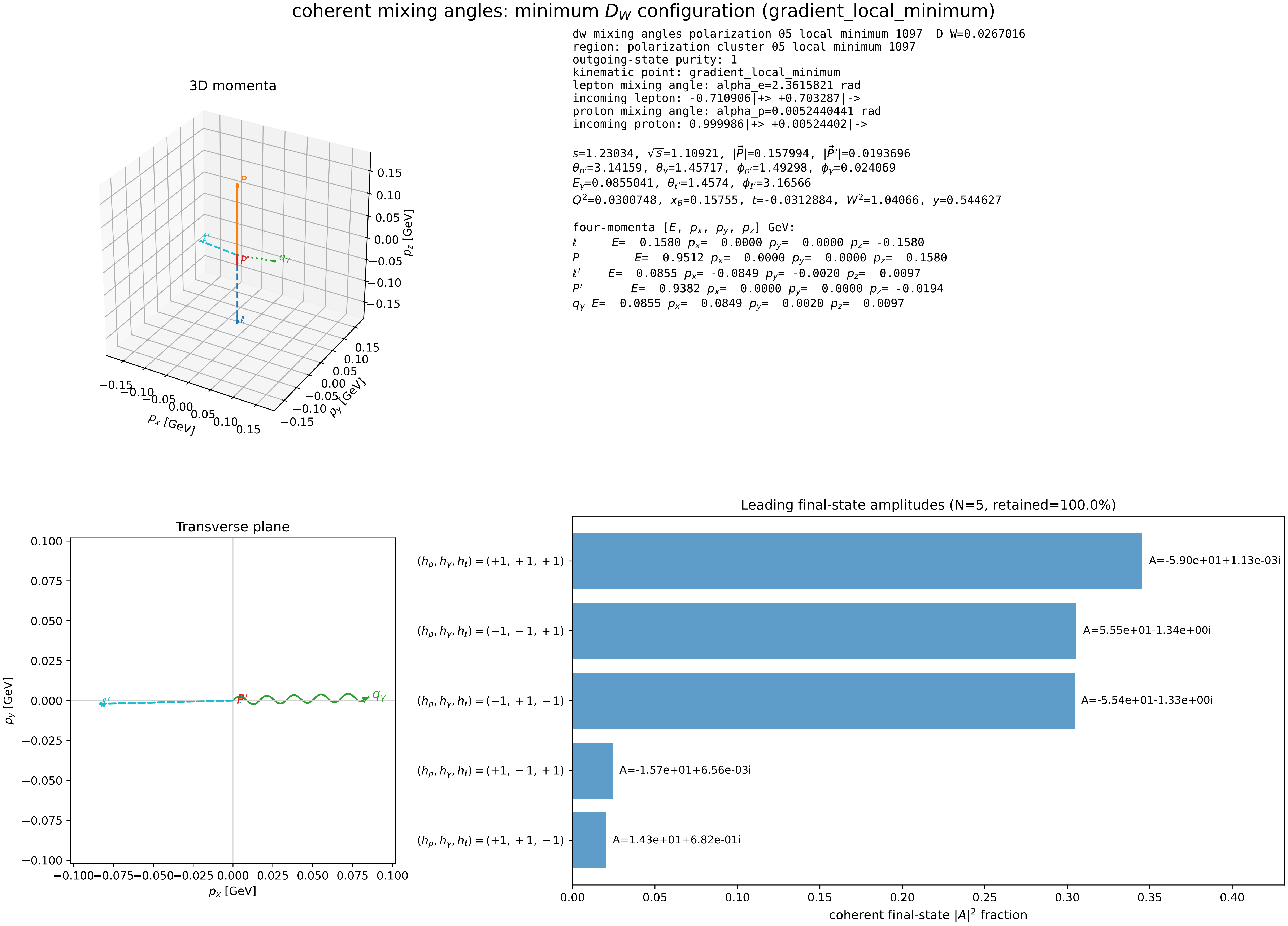



electron: polarization cluster P5, local minimum 1085; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.025693611
D_W contour = 0.075693611
above global minimum = 0.025677178
8D contour samples = 253
selected phase-space point:
sqrt_s = 1.128715
theta_p_out = 2.48425
theta_gamma_out = 1.759722
qOut = 0.09531018
phi_p_out = 2.614927
phi_gamma_out = 0.9843351
alpha_e = 2.357265
alpha_p = 6.33237e-05

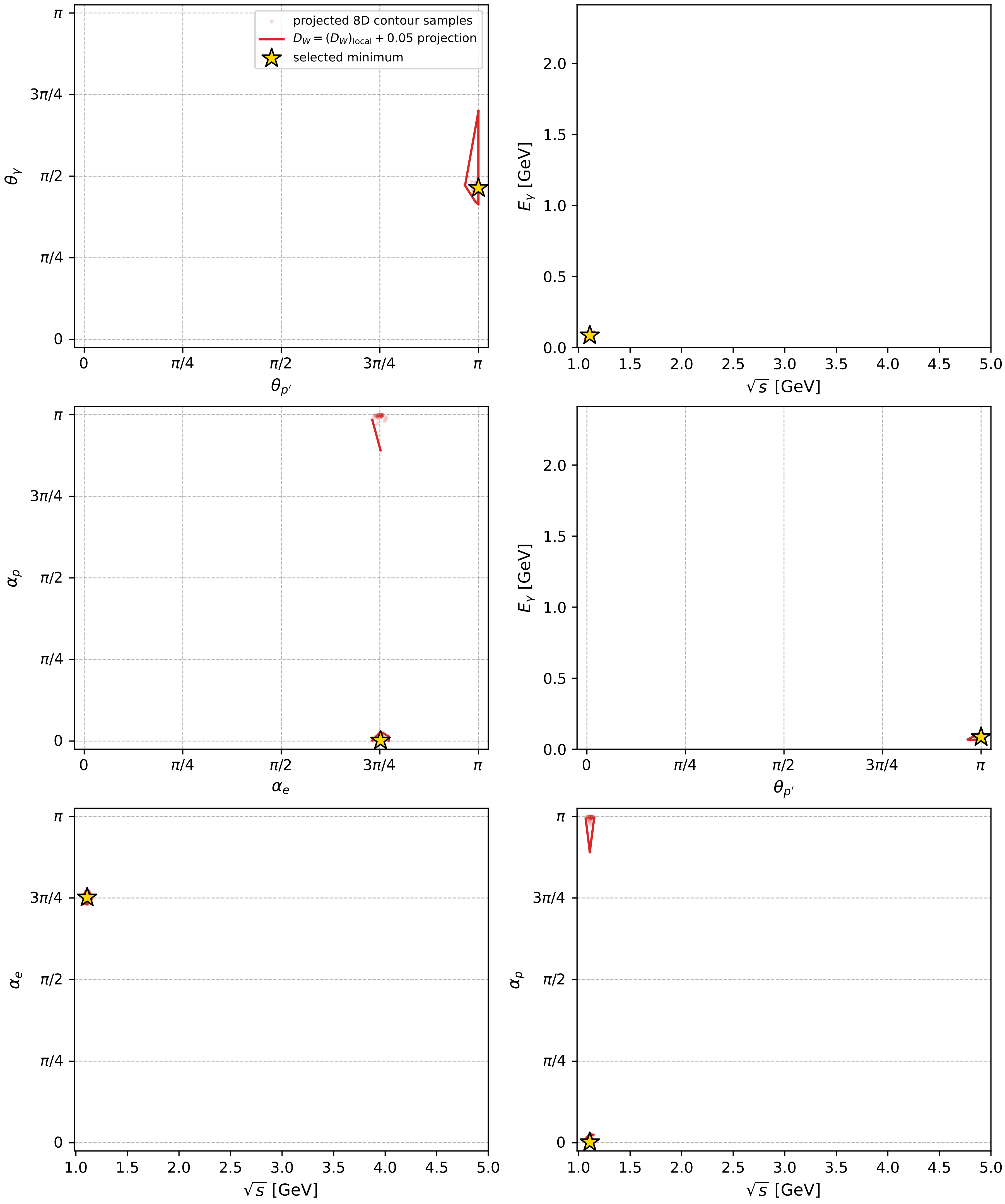
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)




electron: polarization cluster P5, local minimum 1091; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
$\theta_{p'}$
θ_γ
$\sqrt{s}$ [GeV]
E_γ [GeV]
$\phi_{p'}$
ϕ_γ
α_e
α_p
selected local minimum
D_W(local) = 0.026218445
D_W contour = 0.076218445
above global minimum = 0.026202012
8D contour samples = 239
selected phase-space point:
sqrt_s = 1.133424
theta_p_out = 0.05417108
theta_gamma_out = 1.611578
qOut = 0.0977106
phi_p_out = 3.104351
phi_gamma_out = 3.126306
alpha_e = 2.352607
alpha_p = 0.005674516


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_1097 D_W=0.0267016
region: polarization_cluster_05_local_minimum_1097
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3615821 rad
incoming lepton: -0.710906|+> +0.703287|->
proton mixing angle: alpha_p=0.0052440441 rad
incoming proton: 0.999986|+> +0.00524402|->
s=1.23034, $\sqrt{s}$=1.10921, $|\vec{P}|$=0.157994, $|\vec{P}'|$=0.0193696
$\theta_{p'}$=3.14159, θ_γ=1.45717, $\phi_{p'}$=1.49298, ϕ_γ=0.024069
E_γ=0.0855041, $\theta_{\ell'}$=1.4574, $\phi_{\ell'}$=3.16566
Q^2=0.0300748, x_B=0.15755, t=-0.0312884, W^2=1.04066, y=0.544627
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1580 p_x= 0.0000 p_y= 0.0000 p_z= -0.1580
P E= 0.9512 p_x= 0.0000 p_y= 0.0000 p_z= 0.1580
ℓ' E= 0.0855 p_x= -0.0849 p_y= -0.0020 p_z= 0.0097
P' E= 0.9382 p_x= 0.0000 p_y= 0.0000 p_z= -0.0194
q_γ E= 0.0855 p_x= 0.0849 p_y= 0.0020 p_z= 0.0097
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=5, retained=100.0%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, +1)$ A=-5.90e+01+1.13e-03i
$(h_p, h_\gamma, h_\ell) = (-1, -1, +1)$ A=5.55e+01-1.34e+00i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$ A=-5.54e+01-1.33e+00i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=-1.57e+01+6.56e-03i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=1.43e+01+6.82e-01i
coherent final-state $|A|^2$ fraction

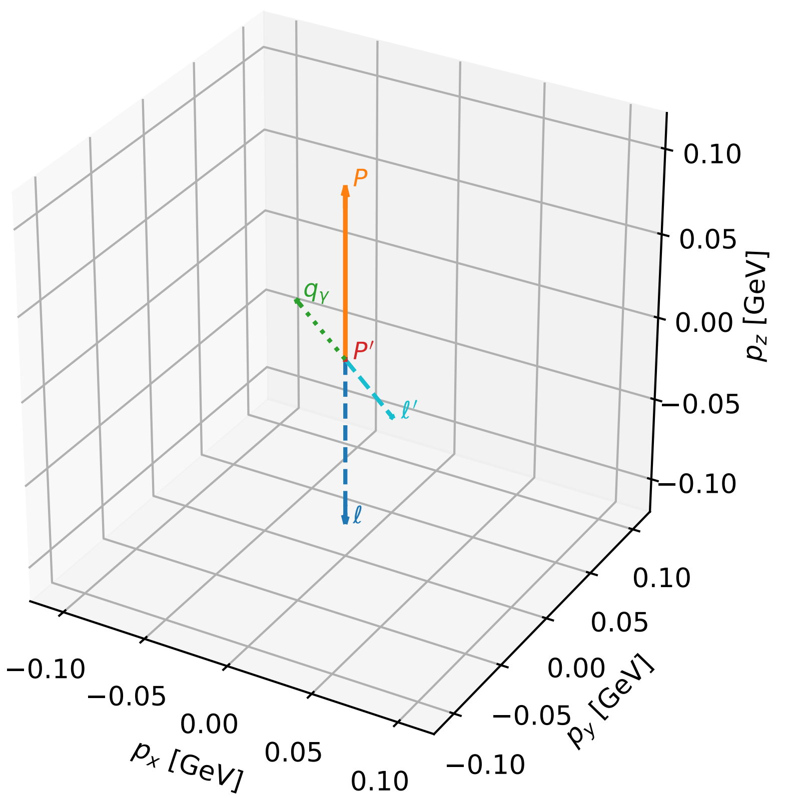
electron: polarization cluster P5, local minimum 1097; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour




```
selected local minimum

D_W(local) = 0.026701643
D_W contour = 0.076701643
above global minimum = 0.02668521
8D contour samples = 131

selected phase-space point:
    sqrt_s = 1.109207
theta_p_out = 3.141593
theta_gamma_out = 1.45717
       qOut = 0.08550408
 phi_p_out = 1.492984
phi_gamma_out = 0.02406903
   alpha_e = 2.361582
   alpha_p = 0.005244044
```

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

3D momenta

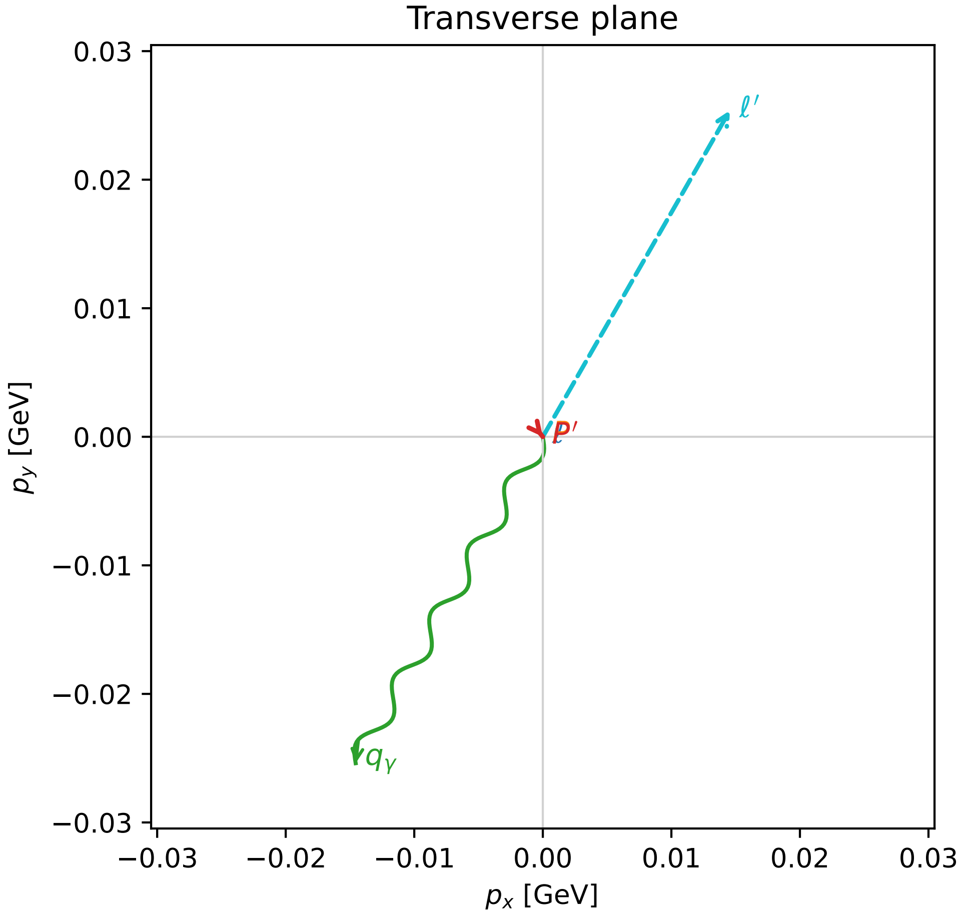


dw_mixing_angles_polarization_05_local_minimum_1099 D_W=0.0268735
region: polarization_cluster_05_local_minimum_1099
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3041987 rad
incoming lepton: -0.669401|+> +0.742901|->
proton mixing angle: alpha_p=0.20746873 rad
incoming proton: 0.978555|+> +0.205984|->

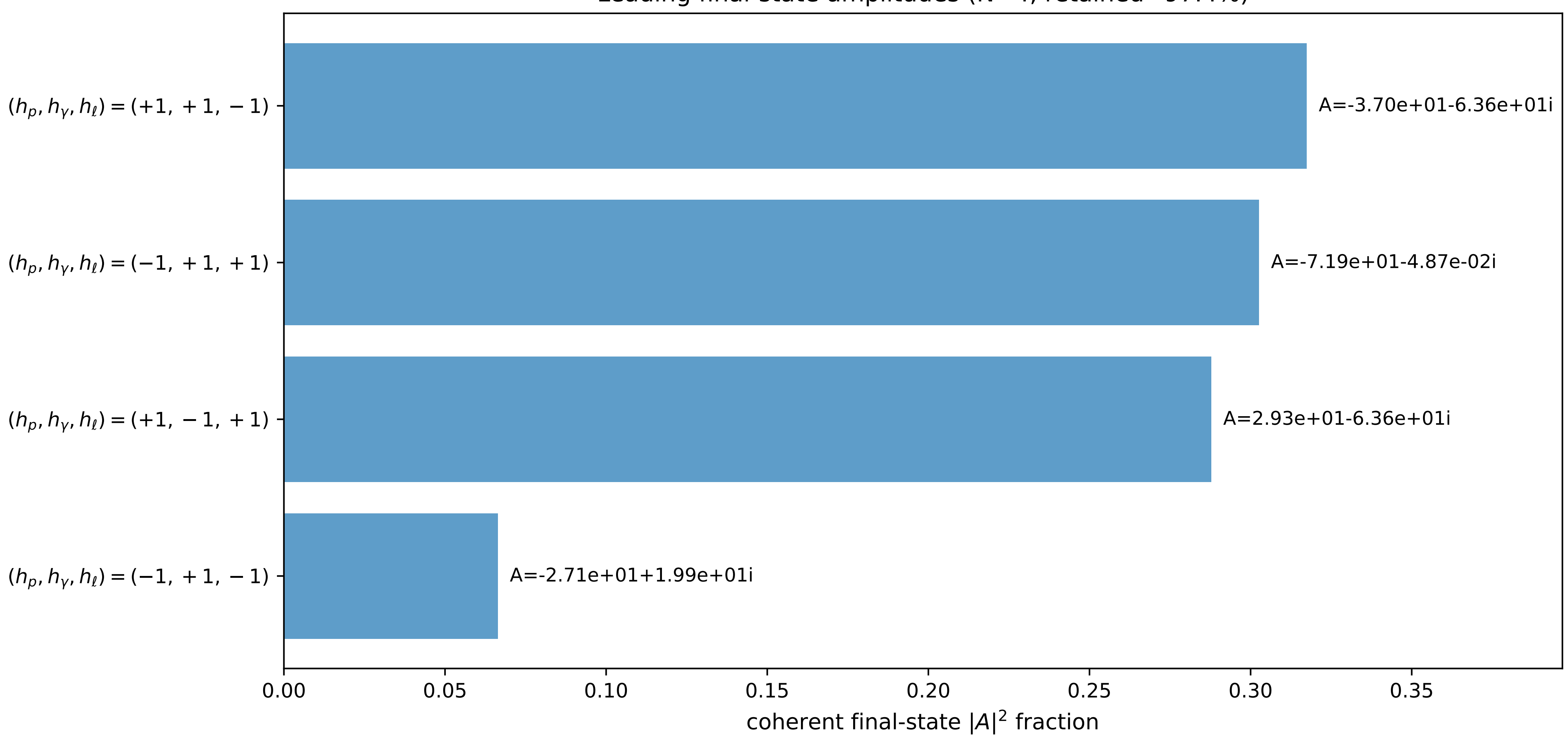
s=1.09018, $\sqrt{s}$=1.04412, $|\vec{P}|$=0.100722, $|\vec{P}'|$=0.000174331
$\theta_{p'}$=0.000153398, θ_γ=0.585264, $\phi_{p'}$=5.38487, ϕ_γ=4.19183
E_γ=0.0529837, $\theta_{\ell'}$=2.55814, $\phi_{\ell'}$=1.05024
Q^2=0.00177067, x_B=0.0175049, t=-0.0100807, W^2=0.979226, y=0.480918

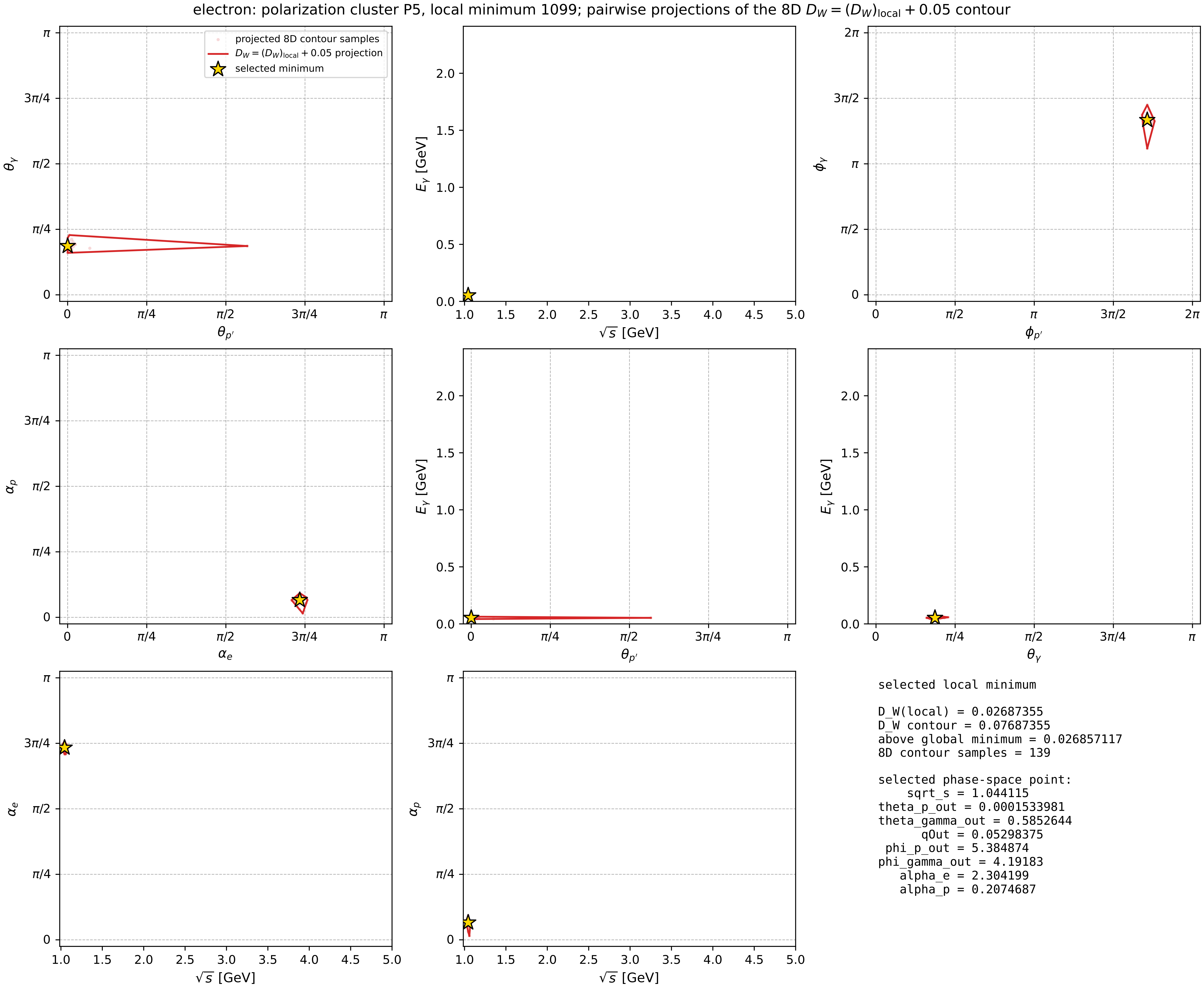
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1007 p_x= 0.0000 p_y= 0.0000 p_z= -0.1007
P E= 0.9434 p_x= 0.0000 p_y= 0.0000 p_z= 0.1007
ℓ' E= 0.0531 p_x= 0.0146 p_y= 0.0254 p_z= -0.0443
P' E= 0.9380 p_x= 0.0000 p_y= -0.0000 p_z= 0.0002
q_γ E= 0.0530 p_x= -0.0146 p_y= -0.0254 p_z= 0.0442

Transverse plane



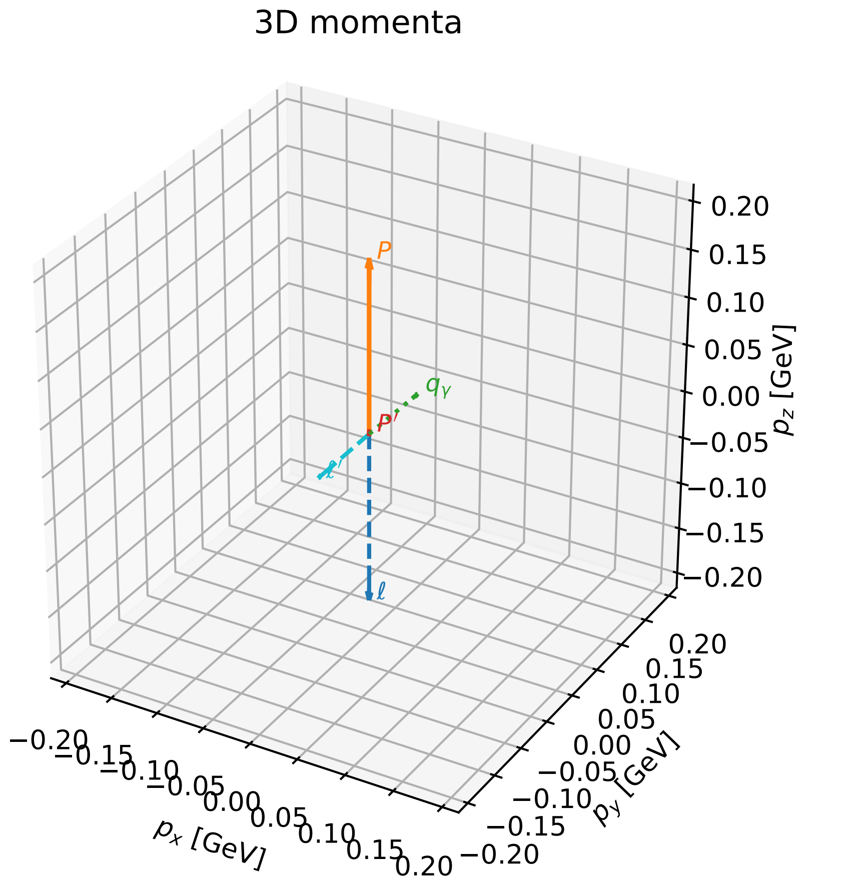
Leading final-state amplitudes (N=4, retained=97.4%)





electron: polarization cluster P5, local minimum 1099; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour
projected 8D contour samples
$D_W = (D_W)_{\mathrm{local}} + 0.05$ projection
selected minimum
θ_γ
$\theta_{p'}$
E_γ [GeV]
$\sqrt{s}$ [GeV]
ϕ_γ
$\phi_{p'}$
α_p
α_e
θ_γ
selected local minimum
D_W(local) = 0.02687355
D_W contour = 0.07687355
above global minimum = 0.026857117
8D contour samples = 139
selected phase-space point:
sqrt_s = 1.044115
theta_p_out = 0.0001533981
theta_gamma_out = 0.5852644
qOut = 0.05298375
phi_p_out = 5.384874
phi_gamma_out = 4.19183
alpha_e = 2.304199
alpha_p = 0.2074687

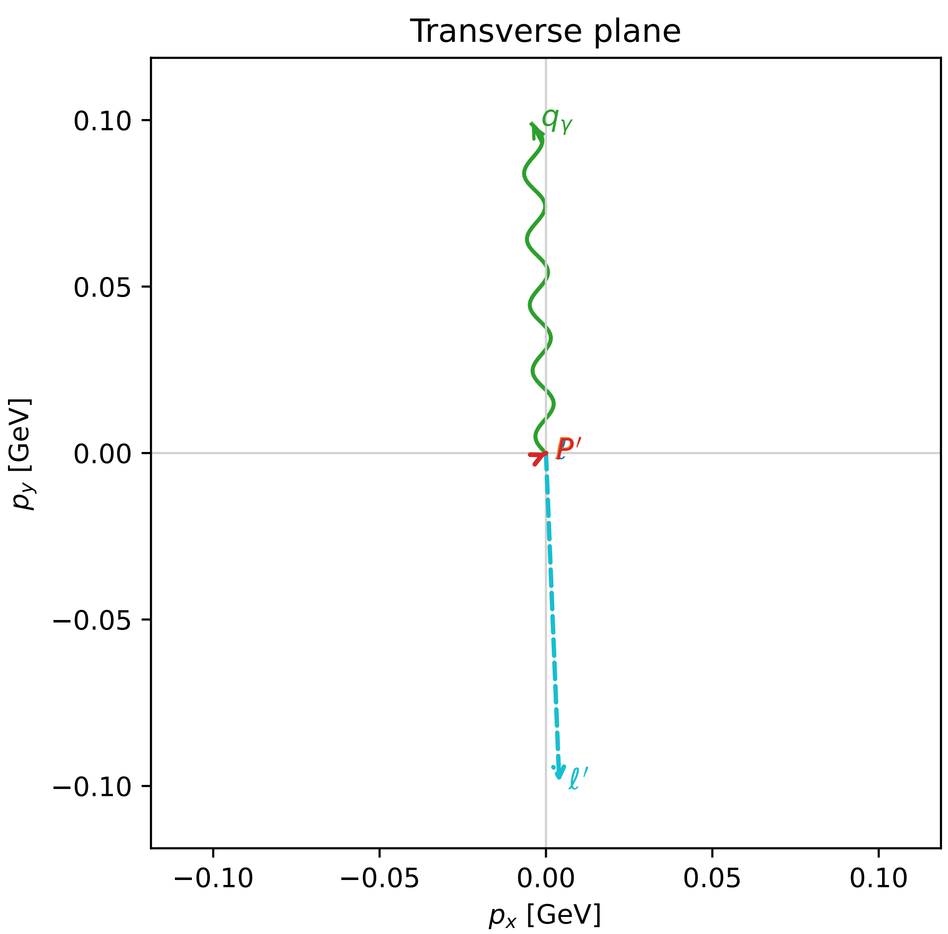
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

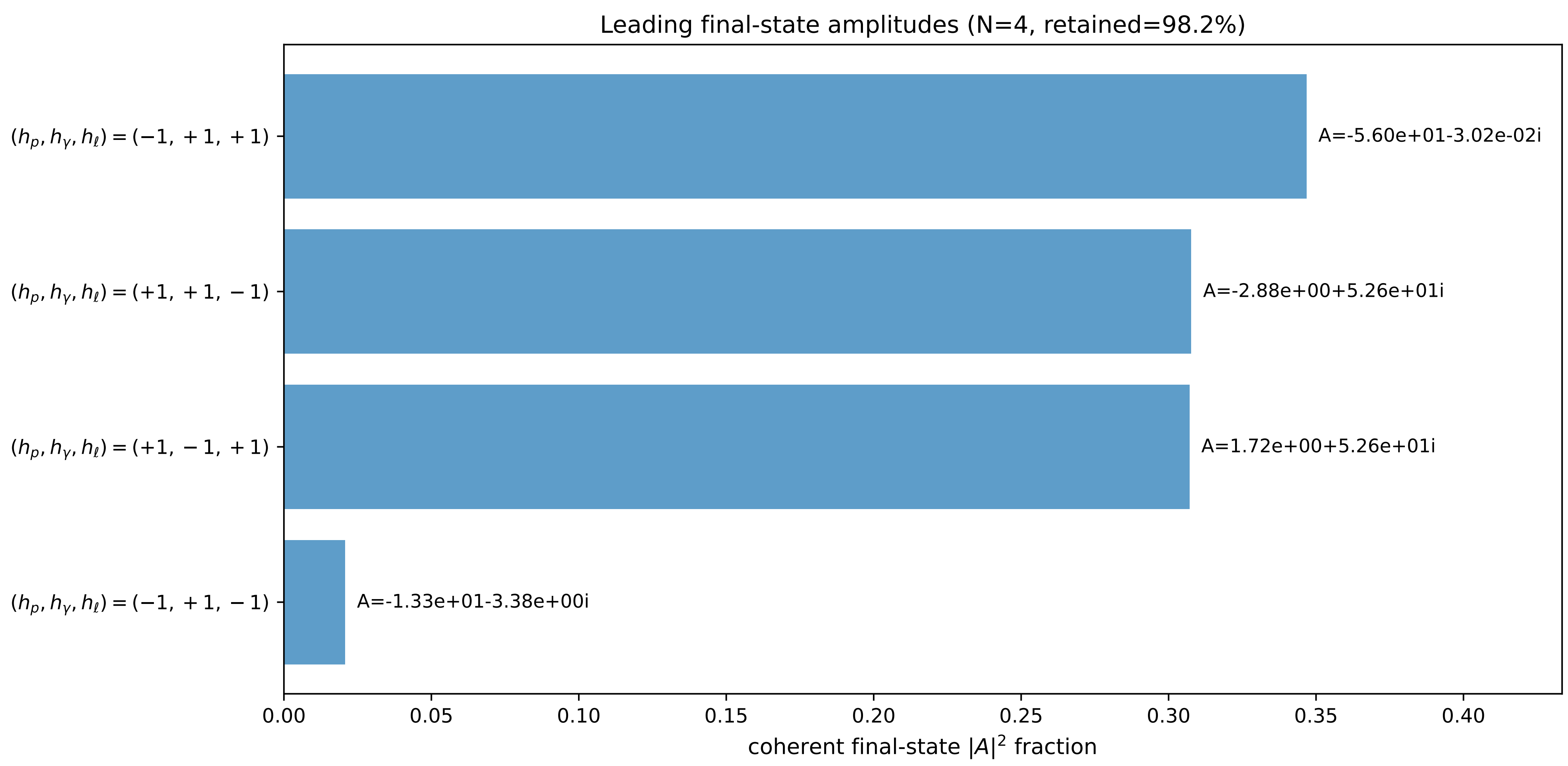


dw_mixing_angles_polarization_05_local_minimum_1100 D_W=0.0271176
region: polarization_cluster_05_local_minimum_1100
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3558261 rad
incoming lepton: -0.706846|+> +0.707367|->
proton mixing angle: alpha_p=0.0068576304 rad
incoming proton: 0.999976|+> +0.00685758|->

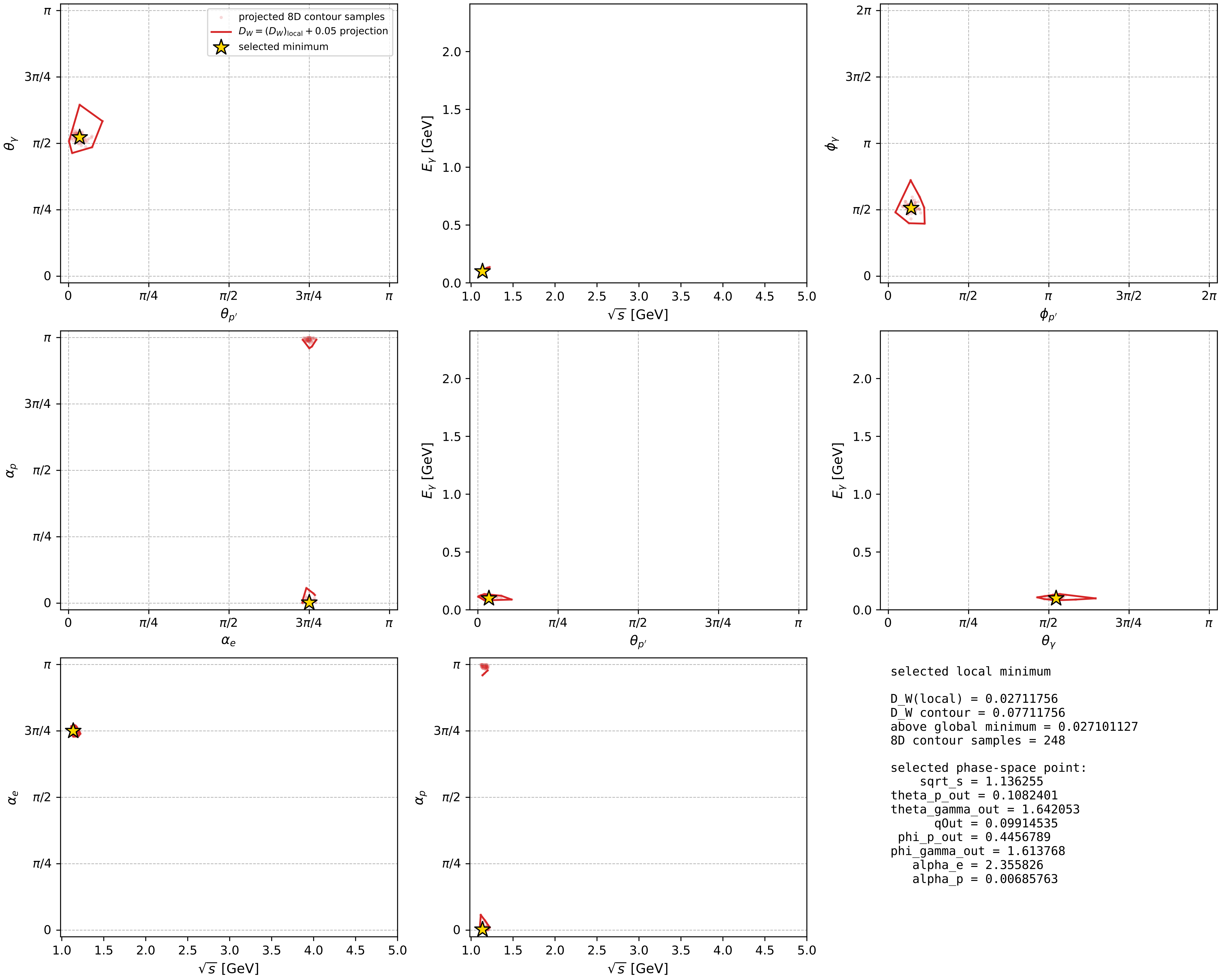
s=1.29108, $\sqrt{s}$=1.13626, $|\vec{P}|$=0.180959, $|\vec{P}'|$=0.0025233
$\theta_{p'}$=0.10824, θ_γ=1.64205, $\phi_{p'}$=0.445679, ϕ_γ=1.61377
E_γ=0.0991454, $\theta_{\ell'}$=1.52487, $\phi_{\ell'}$=4.75283
Q^2=0.0375149, x_B=0.167832, t=-0.0315456, W^2=1.06586, y=0.543553

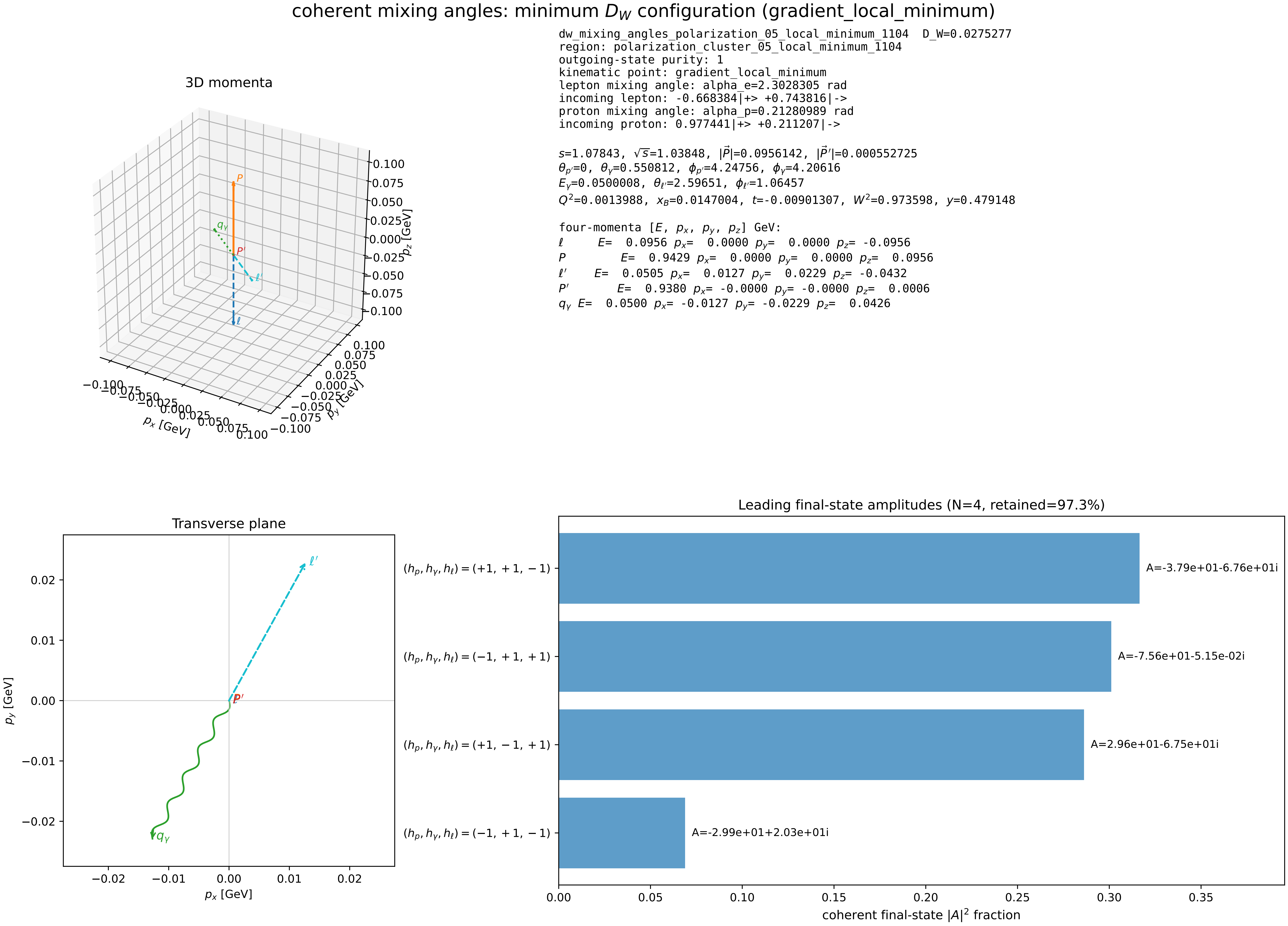
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1810 p_x= 0.0000 p_y= 0.0000 p_z= -0.1810
P E= 0.9553 p_x= 0.0000 p_y= 0.0000 p_z= 0.1810
ℓ' E= 0.0991 p_x= 0.0040 p_y= -0.0989 p_z= 0.0046
P' E= 0.9380 p_x= 0.0002 p_y= 0.0001 p_z= 0.0025
q_γ E= 0.0991 p_x= -0.0042 p_y= 0.0988 p_z= -0.0071



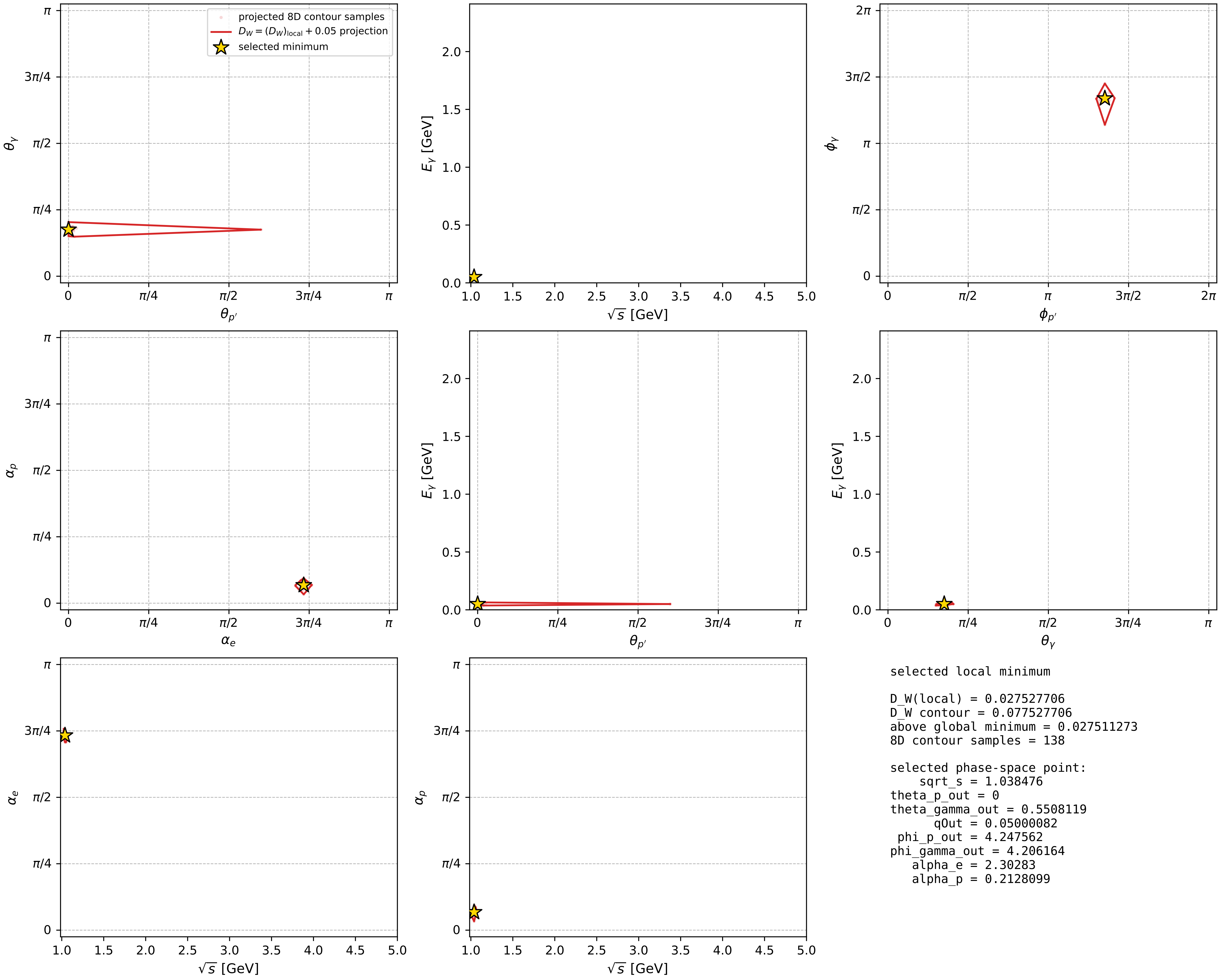


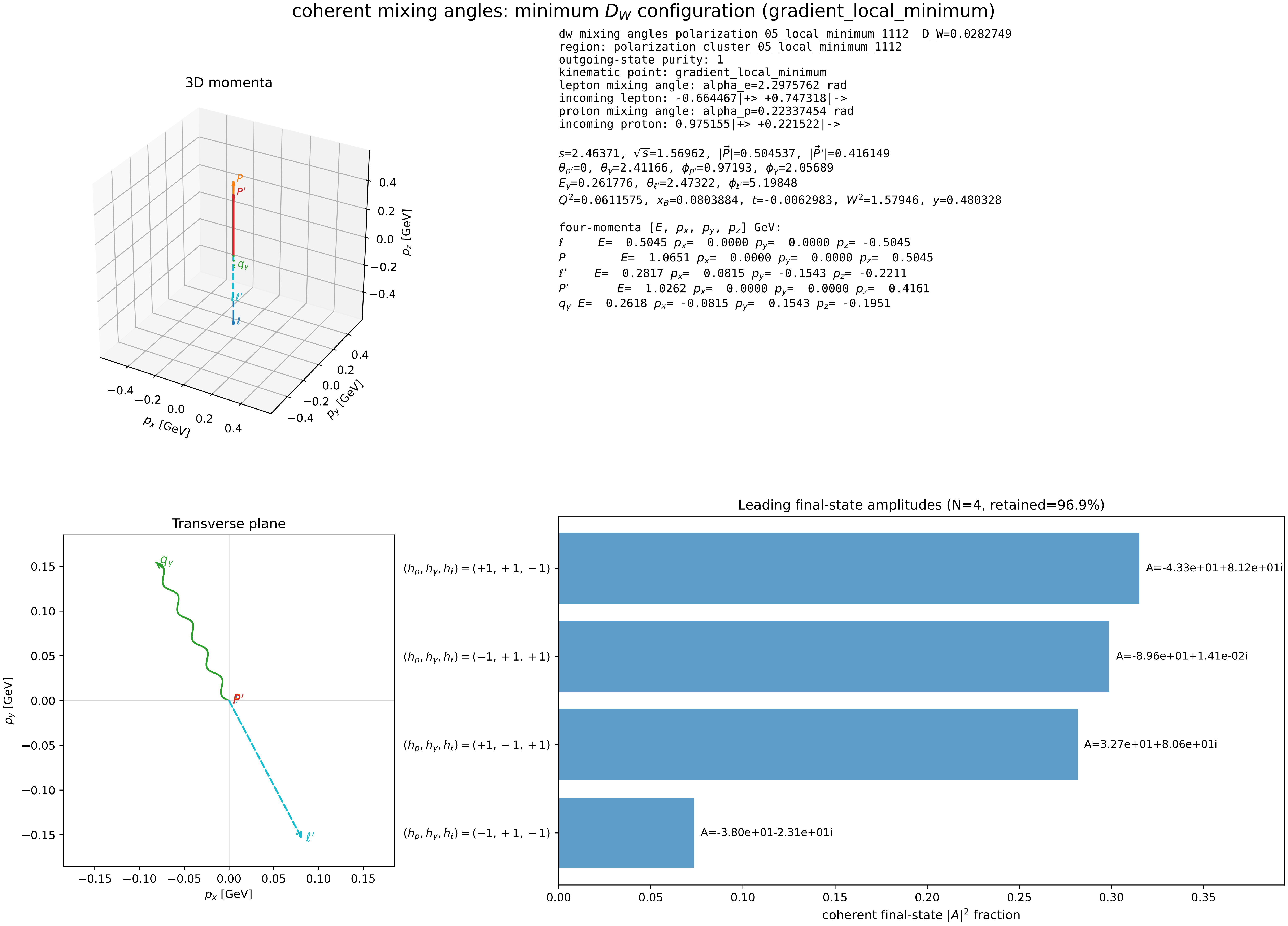
electron: polarization cluster P5, local minimum 1100; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour



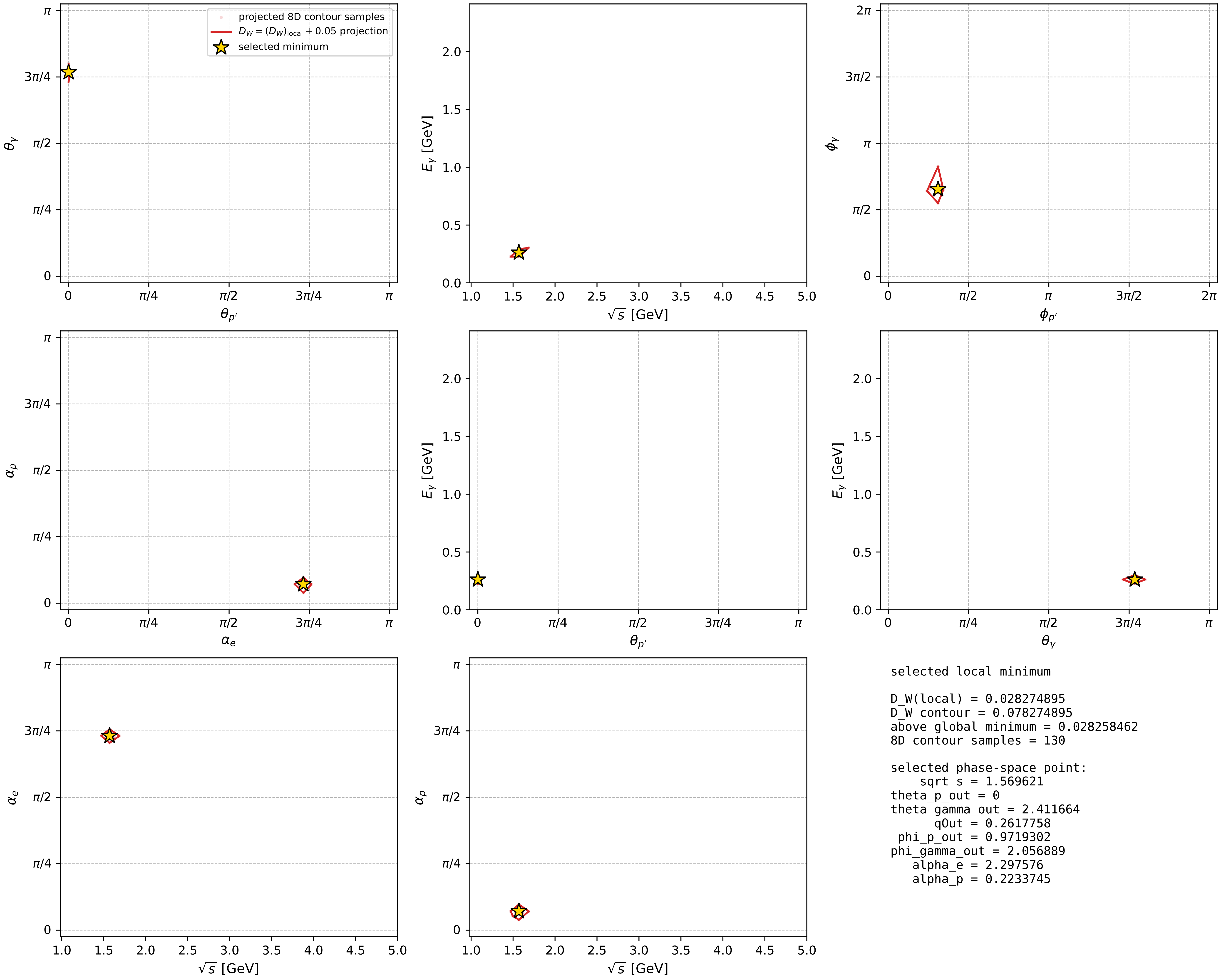

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
P
q_γ
P'
ℓ'
ℓ
p_x [GeV]
p_y [GeV]
p_z [GeV]
dw_mixing_angles_polarization_05_local_minimum_1104 D_W=0.0275277
region: polarization_cluster_05_local_minimum_1104
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3028305 rad
incoming lepton: -0.668384|+> +0.743816|->
proton mixing angle: alpha_p=0.21280989 rad
incoming proton: 0.977441|+> +0.211207|->
s=1.07843, $\sqrt{s}$=1.03848, $|\vec{P}|$=0.0956142, $|\vec{P}'|$=0.000552725
$\theta_{p'}$=0, θ_γ=0.550812, $\phi_{p'}$=4.24756, ϕ_γ=4.20616
E_γ=0.0500008, $\theta_{\ell'}$=2.59651, $\phi_{\ell'}$=1.06457
Q^2=0.0013988, x_B=0.0147004, t=-0.00901307, W^2=0.973598, y=0.479148
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0956 p_x= 0.0000 p_y= 0.0000 p_z= -0.0956
P E= 0.9429 p_x= 0.0000 p_y= 0.0000 p_z= 0.0956
ℓ' E= 0.0505 p_x= 0.0127 p_y= 0.0229 p_z= -0.0432
P' E= 0.9380 p_x= -0.0000 p_y= -0.0000 p_z= 0.0006
q_γ E= 0.0500 p_x= -0.0127 p_y= -0.0229 p_z= 0.0426
Transverse plane
ℓ'
P'
q_γ
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=4, retained=97.3%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=-3.79e+01-6.76e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
A=-7.56e+01-5.15e-02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=2.96e+01-6.75e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=-2.99e+01+2.03e+01i
coherent final-state $|A|^2$ fraction

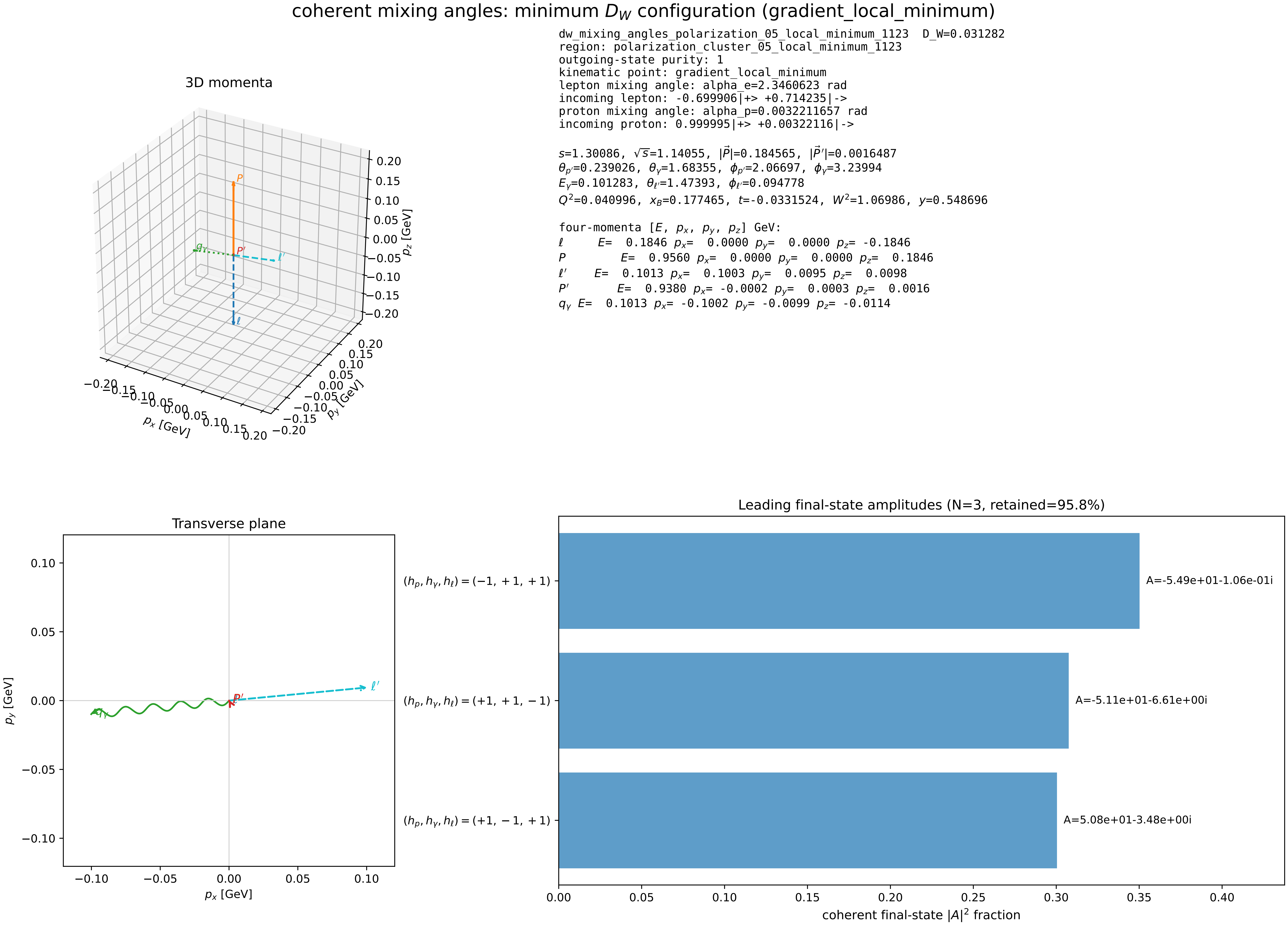
electron: polarization cluster P5, local minimum 1104; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



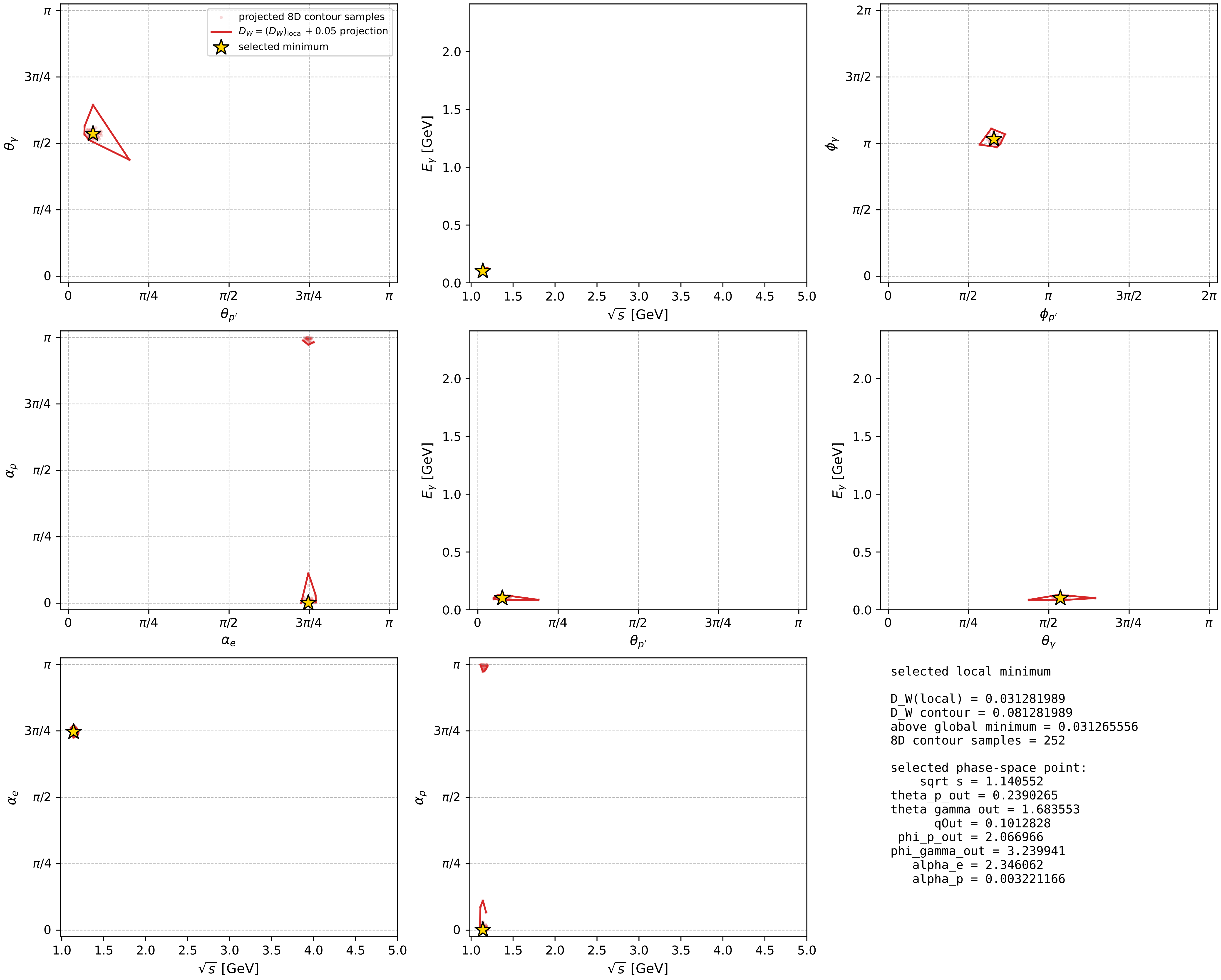

coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
P'
q_γ
ℓ'
ℓ
dw_mixing_angles_polarization_05_local_minimum_1112 D_W=0.0282749
region: polarization_cluster_05_local_minimum_1112
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.2975762 rad
incoming lepton: -0.664467|+> +0.747318|->
proton mixing angle: alpha_p=0.22337454 rad
incoming proton: 0.975155|+> +0.221522|->
s=2.46371, $\sqrt{s}$=1.56962, $|\vec{P}|$=0.504537, $|\vec{P}'|$=0.416149
$\theta_{p'}$=0, θ_γ=2.41166, $\phi_{p'}$=0.97193, ϕ_γ=2.05689
E_γ=0.261776, $\theta_{\ell'}$=2.47322, $\phi_{\ell'}$=5.19848
Q^2=0.0611575, x_B=0.0803884, t=-0.0062983, W^2=1.57946, y=0.480328
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.5045 p_x= 0.0000 p_y= 0.0000 p_z= -0.5045
P E= 1.0651 p_x= 0.0000 p_y= 0.0000 p_z= 0.5045
ℓ' E= 0.2817 p_x= 0.0815 p_y= -0.1543 p_z= -0.2211
P' E= 1.0262 p_x= 0.0000 p_y= 0.0000 p_z= 0.4161
q_γ E= 0.2618 p_x= -0.0815 p_y= 0.1543 p_z= -0.1951
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=4, retained=96.9%)
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$
A=-4.33e+01+8.12e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$
A=-8.96e+01+1.41e-02i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$
A=3.27e+01+8.06e+01i
$(h_p, h_\gamma, h_\ell) = (-1, +1, -1)$
A=-3.80e+01-2.31e+01i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P5, local minimum 1112; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour

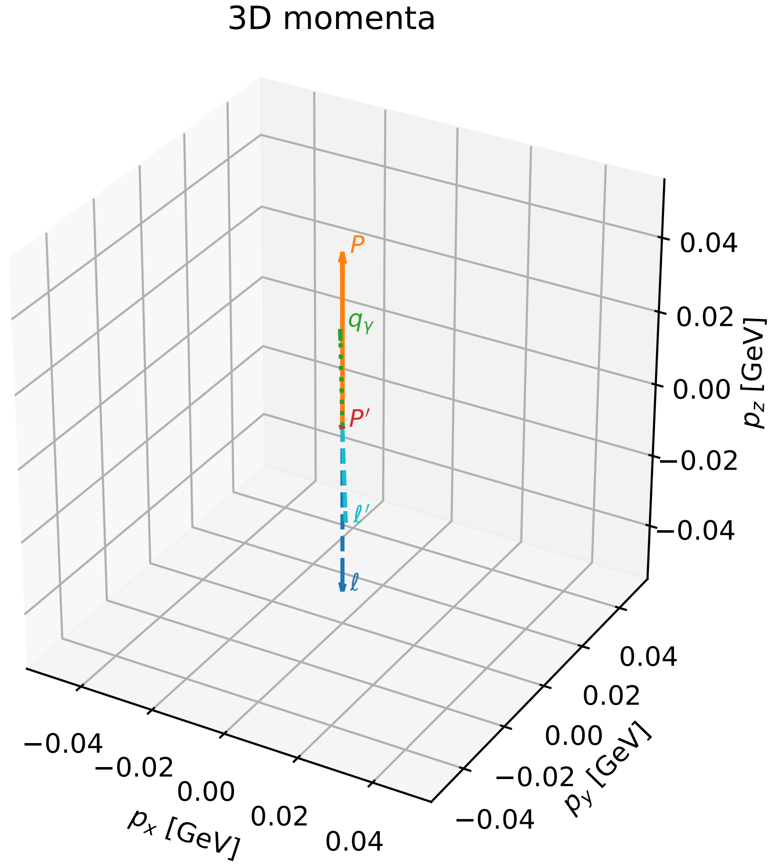



coherent mixing angles: minimum D_W configuration (gradient_local_minimum)
dw_mixing_angles_polarization_05_local_minimum_1123 D_W=0.031282
region: polarization_cluster_05_local_minimum_1123
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3460623 rad
incoming lepton: -0.699906|+> +0.714235|->
proton mixing angle: alpha_p=0.0032211657 rad
incoming proton: 0.999995|+> +0.00322116|->
s=1.30086, $\sqrt{s}$=1.14055, $|\vec{P}|$=0.184565, $|\vec{P}'|$=0.0016487
$\theta_{p'}$=0.239026, θ_γ=1.68355, $\phi_{p'}$=2.06697, ϕ_γ=3.23994
E_γ=0.101283, $\theta_{\ell'}$=1.47393, $\phi_{\ell'}$=0.094778
Q^2=0.040996, x_B=0.177465, t=-0.0331524, W^2=1.06986, y=0.548696
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.1846 p_x= 0.0000 p_y= 0.0000 p_z= -0.1846
P E= 0.9560 p_x= 0.0000 p_y= 0.0000 p_z= 0.1846
ℓ' E= 0.1013 p_x= 0.1003 p_y= 0.0095 p_z= 0.0098
P' E= 0.9380 p_x= -0.0002 p_y= 0.0003 p_z= 0.0016
q_γ E= 0.1013 p_x= -0.1002 p_y= -0.0099 p_z= -0.0114
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
p_x [GeV]
p_y [GeV]
Leading final-state amplitudes (N=3, retained=95.8%)
$(h_p, h_\gamma, h_\ell) = (-1, +1, +1)$ A=-5.49e+01-1.06e-01i
$(h_p, h_\gamma, h_\ell) = (+1, +1, -1)$ A=-5.11e+01-6.61e+00i
$(h_p, h_\gamma, h_\ell) = (+1, -1, +1)$ A=5.08e+01-3.48e+00i
coherent final-state $|A|^2$ fraction

electron: polarization cluster P5, local minimum 1123; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



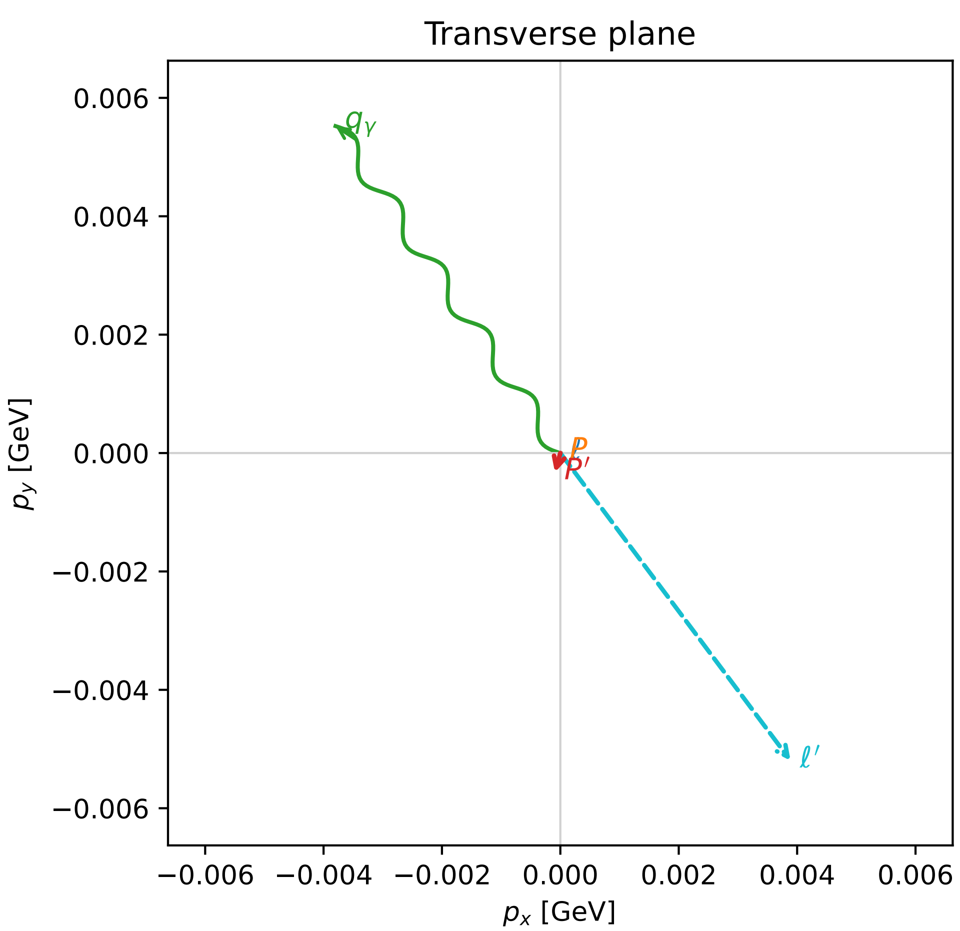
coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

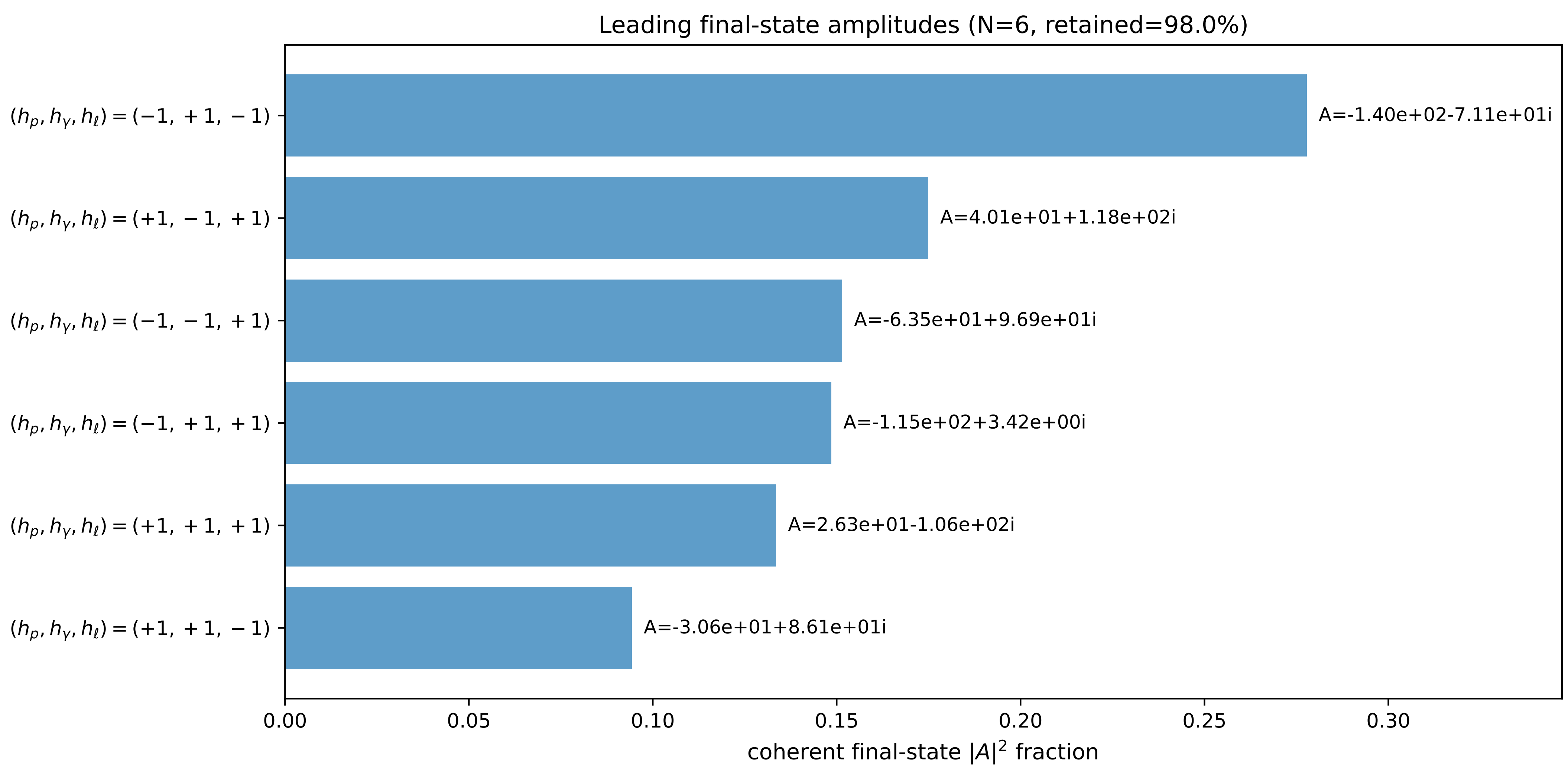



```
dw_mixing_angles_polarization_05_local_minimum_1142  D_W=0.0408868
region: polarization_cluster_05_local_minimum_1142
outgoing-state purity: 1
kinematic point: gradient_local_minimum
lepton mixing angle: alpha_e=2.3155438 rad
incoming lepton: -0.677786|+> +0.735259|->
proton mixing angle: alpha_p=0.22073769 rad
incoming proton: 0.975736|+> +0.218949|->
```

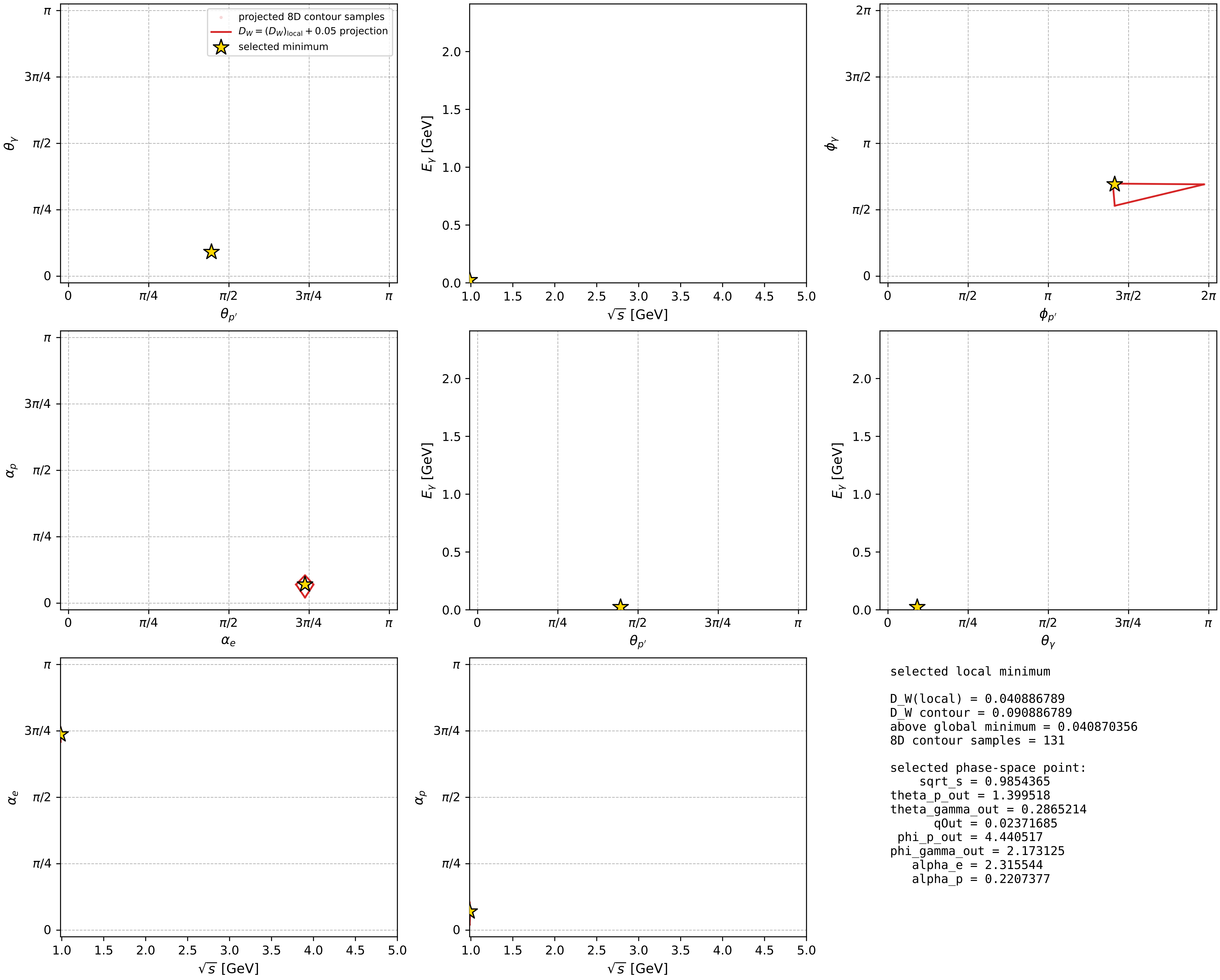
s=0.971085, $\sqrt{s}$=0.985437, $|\vec{P}|$=0.0462921, $|\vec{P}'|$=0.000344821
$\theta_{p'}$=1.39952, θ_γ=0.286521, $\phi_{p'}$=4.44052, ϕ_γ=2.17312
E_γ=0.0237169, $\theta_{\ell'}$=2.86438, $\phi_{\ell'}$=5.35488
Q^2=8.39441e-05, x_B=0.00188313, t=-0.00213633, W^2=0.924337, y=0.488564

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 0.0463 p_x= 0.0000 p_y= 0.0000 p_z= -0.0463
P E= 0.9391 p_x= 0.0000 p_y= 0.0000 p_z= 0.0463
ℓ' E= 0.0237 p_x= 0.0039 p_y= -0.0052 p_z= -0.0228
P' E= 0.9380 p_x= -0.0001 p_y= -0.0003 p_z= 0.0001
q_γ E= 0.0237 p_x= -0.0038 p_y= 0.0055 p_z= 0.0227





electron: polarization cluster P5, local minimum 1142; pairwise projections of the 8D $D_W = (D_W)_{\mathrm{local}} + 0.05$ contour




```
selected local minimum

D_W(local) = 0.040886789
D_W contour = 0.090886789
above global minimum = 0.040870356
8D contour samples = 131

selected phase-space point:
    sqrt_s = 0.9854365
theta_p_out = 1.399518
theta_gamma_out = 0.2865214
       qOut = 0.02371685
 phi_p_out = 4.440517
phi_gamma_out = 2.173125
   alpha_e = 2.315544
   alpha_p = 0.2207377
```